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ANALOG ELECTRONICS

(CIRCUITS and DEVICES)



Dr. D.K. Kaushik

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Dr. D.K. Kaushik presently working as the Head department of Electronics and Computer Science, Dayanand Post-Graduate college, Hisar, Haryana. He obtained his masters degree from Meerut University and Ph.D. in 1981 from Kurukshetra University, Kurukshetra. He has 22 years teaching experience in the subject of Electronics and published more 20 research papers in journals of high repute. He has two patents for indigenous design of electronic thin film thickness monitors, in his credit. He had been the Member board of studies in electronics of Kurukshetra University, Kurukshetra many times. His areas of interest are Semiconductor Electronics, Microprocessor, Computer Architecture and Thin Film Technology.



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ANALOG ELECTRONICS
(CIRCUITS AND DEVICES)

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PREFACE

The book ***Analog Electronics (Circuits and Devices)*** contains eleven chapters with comprehensive material, discussed in a very systematic, elaborative and lucid manner. The author of this book has made sincere efforts in bringing the book very up to date. It has thoroughly been edited by Prof. P. J. George, Senior Most Professor and chairman Electronic Science Department, Kurukshetra University, Kururkshetra.

It will prove to be good text book for polytechnic and B.E./B.Tech. students of all the engineering colleges in India. It will also cater to the needs of the students of B.Sc. (Electronics), U.G.C. sponsored restructured course. The book also covers the syllabi of B.Sc. (IT), M.Sc.(Physics specialization in Electronics), M.Tech (electronics).

The objective of this book is to enable students to understand basic circuits and devices. The discussion on the subject in adequate and after going through the book the students will not only have the fundamental knowledge but will have the overall knowledge.

First three chapters of this book contain the circuit theory and deal the network analysis with d.c. and time varying sources and two port network. Different network theorems, different parameters of Two Port Network and their interconnections, different types of passive filters with their frequency and phase response curves have thoroughly and lucidly been discussed with typical solved problems.

Next two chapters i.e. chapters 4 & 5 deal the physics of semiconductor, different types of semiconductor diodes including Zener diodes, photo diodes and light emitting diodes. The application of diodes such as different types of rectifiers, filters, voltage multipliers, and clipping, clamping and log antilog circuits have also been discussed. Complete theoretical aspects have been considered in these chapters.

Chapters 6 – 8 contain the theory of transistors. These chapters cover physical behaviour of junction transistors their characteristics, base width modulation, models of the transistors along with transistor biasing and thermal stabilization with exhaustive analytical explanation.

Ninth chapter deals the basic theory of Junction Field Effect Transistors and MOS Field Effect Transistors with its model and biasing circuits. Latest theory and analytical treatment have been discussed.

Chapter 10 deals the multistage amplifiers; different types of amplifiers including push pull and power amplifiers.

Theory, construction and working of electronic instruments such as cathode ray oscilloscope, multimeters, digital voltmeters, digital frequency meters etc, have been discussed in the last chapter of the book.

The book has been systematically organized and present form help the students to understand and to have the overall knowledge in the field without having prior background of electronics.

SALIENT FEATURES:

The following are the salient features of the proposed text book:

- The material contained in the book is as per class room lectures. The material is neither too large nor too short.
- Written in the simple language but strong pedagogical approach.
- A large number of simple as well complicated numerical solved problems have been introduced. Some unsolved problems with their answers have also been introduced at the end of each chapter.
- The contents are symmetrically arranged.
- The emphasis has been made on the concept using the proper mathematical derivations wherever necessary.
- It will prove to be good text book for all those who study Electronics. It will help the students preparing for NET/SET competitive examination for Physics and Electronics.

Hisar D.K.Kaushik

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Any constructive comments, suggestions and criticism from the readers will be highly appreciated.

HISAR D. K. KAUSHIK

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1

Network Analysis with d. c. Sources

The components used in electronic circuits may be classified into two categories namely active and passive compo+nents. Active components are those which can perform signal processing functions such as signal generation, rectification and amplification. These components basically are semiconductor diodes, transistors and SCR's etc. Batteries and generators which supply energy, also fall in the category of active components. The passive components are those which can not by themselves perform the above mentioned functions. The basic passive components are resistors, inductors and capacitors. In this chapter the analysis of electric circuits or networks, consisting of d.c. sources as the source of energy and other elements like resistors will be discussed using different methods. In addition different theorems will also be discussed to analyze complicated networks.

Before discussing the methods of analysis of network, it is necessary to give the model of the battery or the generator which supply energy to the network.

1.1 Model for a Battery: To discuss the model for the battery, consider a

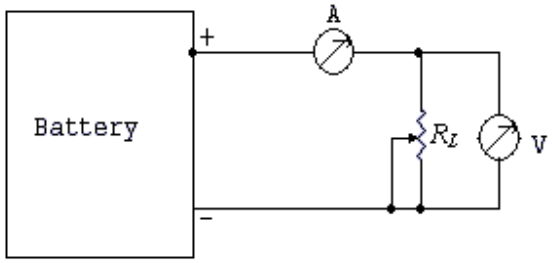


Fig. 1.1

variable load resistance R_L connected to the output terminals of the battery as shown in figure 1.1. Voltage across the load resistance is measured with the help of a voltmeter *V*. The current flowing through the load resistance R_L is measured with the help of an ammeter connected in series with it. By varying the load resistance, the current *I* flowing through and the voltage across the load resistance R_L is measured. A graph is plotted

between the current *I* and voltage *V* as shown in figure (1.2). This is a straight line which cuts at V_0 to the *Y*-axis and at I_0 to the *X*-axis.

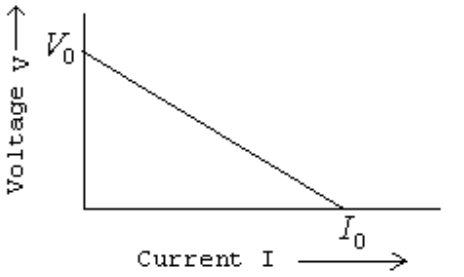


Fig. 1.2

The equation for the straight line is of the form $y = mx + C$,

where $m = -\frac{V_0}{I_0}$, is the slope of the curve

and $C = V_0$, is the intersect to the *Y*-axis.

Thus $V = -\frac{V_0}{I_0} I + V_0$; $\frac{V_0}{I_0}$ has the dimension of resistance represented by R_0.

This equation will be $V = V_0 - R_0 I$ ------ (1.1)

The equivalent circuit of the equation (1.1) is given in figure (1.3).

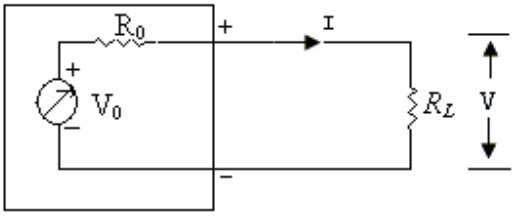


Fig. 1.3

From this circuit, it is clear that the practical battery may be represented by a voltage source V_0 and a resistance R_0 in series with it. The resistance R_0 is known as internal resistance, output resistance or source resistance of the battery.

Now put *I* = 0, i.e., the load resistance is removed from the circuit, then $V = V_0$ and V_0 is called as the open circuit voltage of the source. It is in fact the terminal voltage when no current is drawn from the source. The open circuit voltage when measured with a voltmeter will draw certain amount of current from the source. Thus open circuit voltage should be measured with an ideal voltmeter. Similarly, one more conceptual quantity called the short circuit current may be defined as the current flowing from the battery, when the external terminals of the battery are short circuited. The short circuit

current I_0 will be equal to $\frac{V_0}{I_0}$, since $V = 0$. This current can only be measured with an ideal current meter. The ratio of open circuit voltage V_0 to the short circuit current I_0 is known as internal resistance of the battery.

From the model of the battery the following inferences can be drawn:

(i) If a load resistance R_L is connected to a battery (fig. 1.4), then

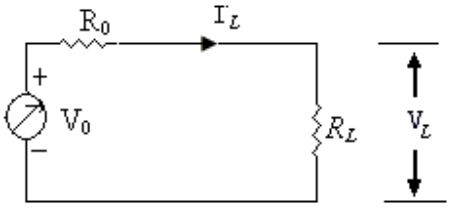


Fig 1.4

the load current I_L is given by:

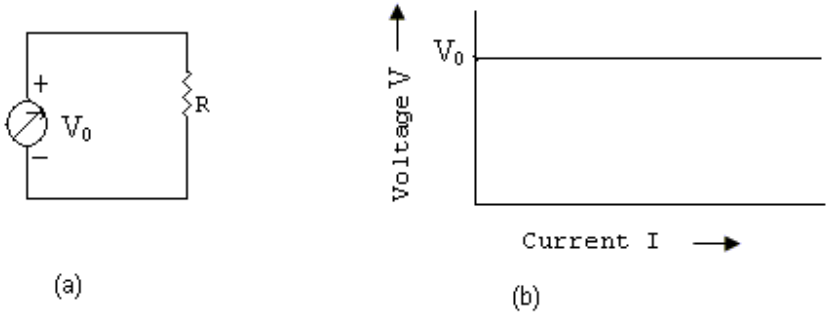
$$I_L = \frac{V_0}{(R_0 + R_L)} \qquad \text{------ (1.2)}$$

The output voltage V_L is given by:

$$V_L = I_L R_L = \frac{V_0 R_L}{(R_0 + R_L)} = \frac{V_0}{(1 + \frac{R_0}{R_L})} \qquad \text{------ (1.3)}$$

From this equation it clear that if $\frac{R_0}{R_L} \to 0$ or $R_0 << R_L$, then $V_L = V_0$; the source will behave like an ideal voltage source. That is the source will said to be a good source if the internal resistance of the source is small enough than the load resistance. The ideal voltage source may be defined as follows:

Ideal voltage source: An ideal voltage source is that source which provides a constant potential difference between its terminals, irrespective of the current drawn from it. An ideal voltage source is represented in figure 1.5(a) and it's V – I relationship in figure 1.5(b).



(a) (b)

Fig. 1.5

(ii) Rewriting the equation (1.2), we get:

$$I_L = \frac{V_0 / R_0}{(1 + \frac{R_L}{R_0})} = \frac{I_0}{(1 + \frac{R_L}{R_0})} \qquad \text{------ (1.4)}$$

It can be understood from this equation that the load current I_L will be equal to the short circuit current, if the $\frac{R_L}{R_0} \to 0$ or $R_L << R_0$; the source will behave like an ideal current source. That is the voltage source will act as a good current source if the source resistance is large enough than the load resistance. The current source is represented as short circuit current I_0 and a source resistance R_0 in parallel with it (fig.1.6). The ideal current source may be defined as follows:

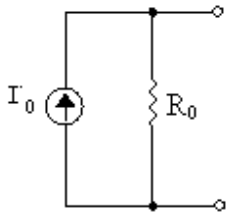


Fig. 1.6

Ideal Current Source: An ideal current source is one which delivers a constant current in the circuit irrespective of the load connected to it. The *V-I* relationship of the ideal current source and its symbolic representation are shown in figure 1.7.

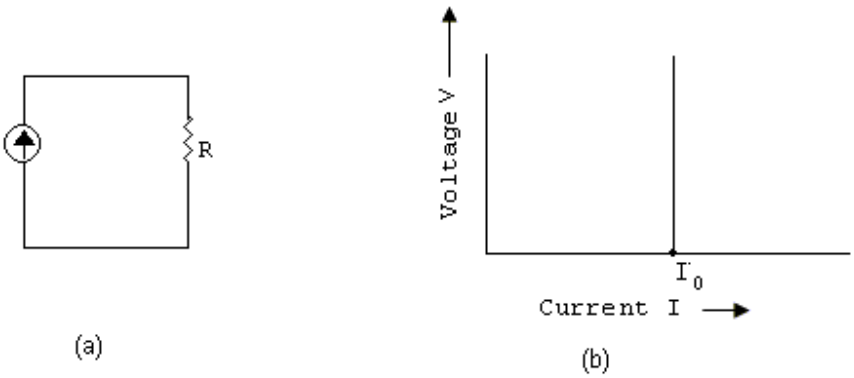


Fig. 1.7

(iii) Sometimes it becomes useful to transform or convert a voltage source into its equivalent current source or vice-versa. We equate the current flowing through the load resistance connected to both the circuits shown in fig. 1.8.

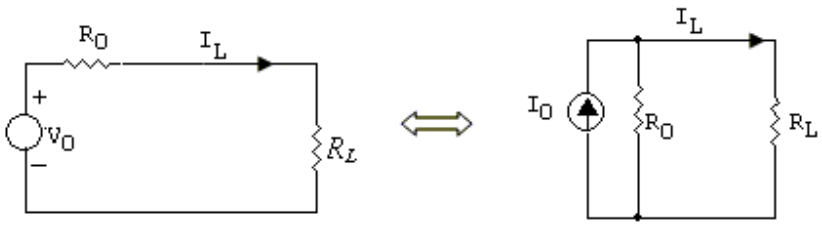


Fig. 1.8

From voltage equivalent circuit $I_L = \dfrac{V_0}{(R_0 + R_L)}$ and

From Current equivalent circuit $I_L = \dfrac{I_0 R_0 R_L}{(R_0 + R_L)}$

Equating these equations we have $V_0 = R_0 I_0$ or $I_0 = \dfrac{V_0}{R_0}$

It is, therefore, concluded that the voltage source V_0 with a series resistance R_0 may be transformed to its equivalent current source I_0 and the resistance R_0 in parallel with it. The value of current source I_0 is given by $I_0 = \dfrac{V_0}{R_0}$. Similarly, a current source I_0 with a resistance in parallel with it may be transformed to voltage source V_0 and a resistance R_0 in series with it. The value of voltage source is given by $V_0 = R_0 I_0$.

Example 1.1 A variable load resistance is connected to the terminals of a battery. When the current flowing in the load is 2*A*, the voltage across the load resistance is 5.8 *volts*; also when the load current is 5*A*, the voltage across the load resistance is 5.5*volts*. All the measurements are made using ideal meters.

Find: (a) Open circuit voltage and source resistance of the battery.

(b) Equivalent current source model of the battery.

Solution: (a) Let V_0 and R_0 are the open circuit voltage and source resistance of the battery respectively. As per statement of the problem:

(i) In the first case $5.8\ volts = 2A\ R_L$ or $R_L = 2.9\ \Omega$;

and $\dfrac{V_0 R_L}{(R_0 + R_L)} = 5.8$ or $\dfrac{V_0.(2.9)}{(R_0 + 2.9)} = 5.8$ or $V_0 = 2R_0 + 5.8$

In the second case $5.5volts = 5\text{A}.\ R_L^{'}$ or $R_L^{'} = 1.1\ \Omega$;

and $\dfrac{V_0 R_L^{'}}{(R_0 + R_L^{'})} = 5.5$ or $\dfrac{V_0.(1.1)}{(R_0 + 1.1)} = 5.5$ or $V_0 = 5R_0 + 5.5$

From these two cases: $V_0 = 6\ volts$ & $R_0 = 0.1\ \Omega$.

(ii) The current source equivalent of the values calculated above is given in figure (1.9). The value of current source $I_0 = 6 \; x \; .1 = 60 \; mA$

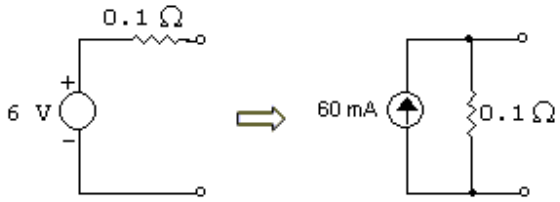


Fig. 1.9

1.2 Network analysis: The analysis of the electric circuits or networks which are formed by interconnecting the sources of electrical energy with other elements like resistances will now be discussed. Here consider the source of electrical energy as d.c. source which does not change with time. Simple circuits may be analyzed using well known Ohm's law. Kirchoff's laws may, however, be used to analyze more complicated circuits. Kirchoff presented two laws namely (i) Kirchoff's Current law (KCL) & (ii) Kirchoff's voltage law (KVL). These laws are the generalization of Ohm's law.

1.2.1 Kirchoff's Current Law (KCL): This law is applicable to any node or junction of electric circuit. The node or junction in an electric circuit is defined as the point where more than two elements meet. This law states that the algebraic sum of currents entering to any node of an electric circuit is zero. The total current entering to a node must be equal to that leaving it. The sign convention for this law is generally assumed that the current entering the node is positive while the current leaving the junction is negative. Mathematically, the law is $\sum I = 0$.

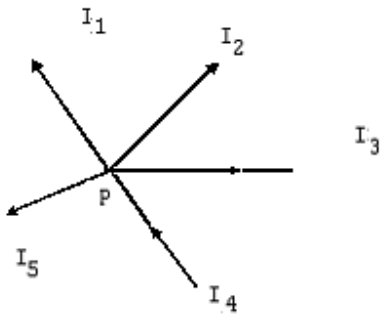


Fig. 1.10

This law may further be illustrated by considering the junction *P* shown in figure 1.10. I_1, I_2 & I_5 are the currents entering the junction which are assumed to be positive while I_3, & I_4 are negative, as these are leaving the junction.

So $$I_1 + I_2 - I_3 - I_4 + I_5 = 0$$

or $$I_1 + I_2 + I_5 = I_3 + I_4$$

Current leaving = Current entering

1.2.2 Kirchoff's Voltage Law (KVL): This law is applicable to a mesh or loop of an electric circuit. A mesh or loop is defined as a closed circuit. The Kirchoff's Voltage law states that the algebraic sum of all the voltage drops in any loop is zero.

The sign convention for applying the KVL to the closed loop is that an arbitrary reference direction of current in the clock wise direction is assumed. The associated reference direction across the resistances is marked positive at the tail of the arrow and negative to head of the arrow. If there is a voltage drop in the circuit, it is assumed to be positive while it is assumed to be negative for the voltage rise in the circuit.

For applying the KVL, we consider a closed circuit given the figure 1.11

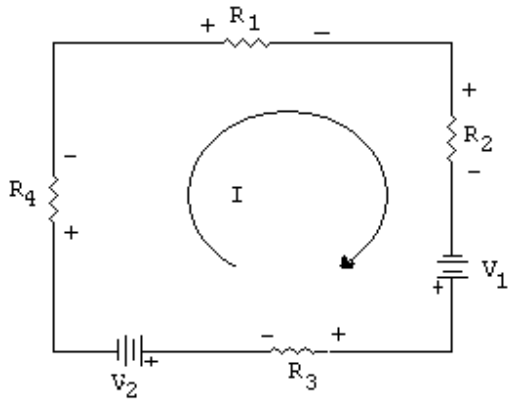


Fig 1.11

From this figure: $R_1 I + R_2 I - V_1 + R_3 I + V_2 + R_4 I = 0$

or $$(R_1 + R_2 + R_3 + R_4) I = V_1 - V_2$$

From this equation it is clear that any unknown quantity may be calculated if rest of the quantities is known.

Example 1.2 A voltmeter having the sensitivity of 20KΩ/V is used to measure the voltage across 50KΩ resistance in the circuit shown in the figure (1.12). The voltmeter is used in 50volts range.

Calculate (a) the reading of the voltmeter,
(b) percentage error in the reading with respect to true value.

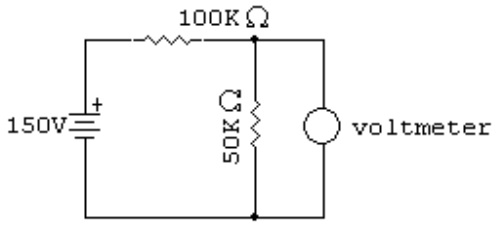


Fig. 1.12

Solution: The true voltage $= \dfrac{150x50K}{(100+50)K} = 50\ volts$

Resistance of the voltmeter in 50volt scale is

$$R_g = 50x20K\ = 1M\Omega$$

When the voltage across 50KΩ resistance is measured, the voltmeter resistance R_g will also come in parallel with 50KΩ resistance. So the voltage will be measured across the parallel combination and not across 50 *K*Ω resistance. Due to which there will be an error.

Reading of the voltmeter V_m will be equal to the voltage across the parallel combination. Resistance of the parallel combination is given by:

$$R_{eq} = \frac{50Kx1M}{(1M+50K)} = 47.6K\Omega$$

Voltmeter reading $V_m = \dfrac{150x47.6}{(100+47.6)} = 48.36\,volts$

% error in the reading $= \dfrac{50-48.36}{50} = 3.28\%$

1.3 Mesh and Node Methods:

The practical or general approach of the KVL and KCL is mesh and node methods. These methods will now be discussed in detail.

1.3.1 Node Method: This method is used to determine the node voltages in the given network. The different nodes are identified in the network and one node is assumed as reference node. All other node voltages are then calculated with respect to the reference node. We find a set of nodal equations, representing one equation for each node. This method may be illustrated by considering a network shown in figure 1.13.

In this network, there are four independent nodes and one reference node. Let V_1, V_2, V_3 & V_4 are the node voltages at four nodes *1*, *2*, *3* & *4* respectively, with respect to reference node. The reference node is grounded (or is at zero potential).

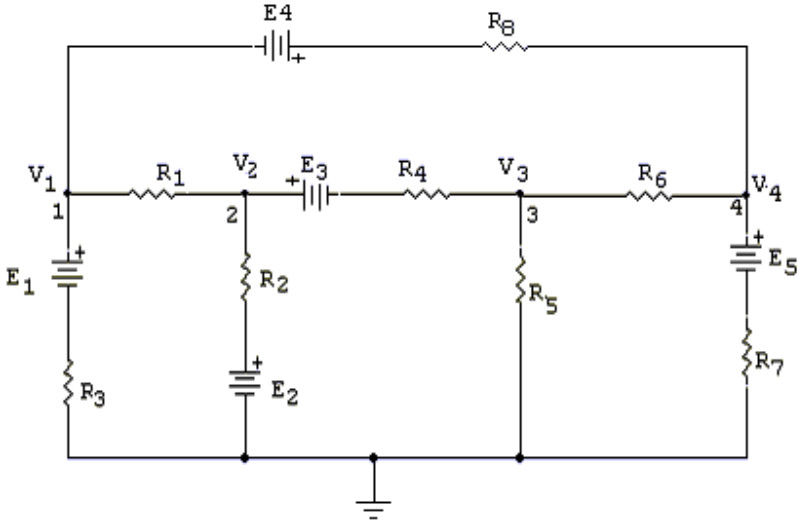


Fig. 1.13

For getting nodal equations, we first of all transform all the voltage sources into current sources as given in figure 1.14. The source *E* with a resistance *R* in series with it is replaced by a current source ($I = \frac{E}{R}$) and the resistance R in parallel with the source. The direction of arrow in the current source will depend upon the sign of voltage source. As it is well known that the conventional current flows from negative to positive inside the voltage source, the direction of arrow in the current source will also represent the inside view of the conventional current in the source.

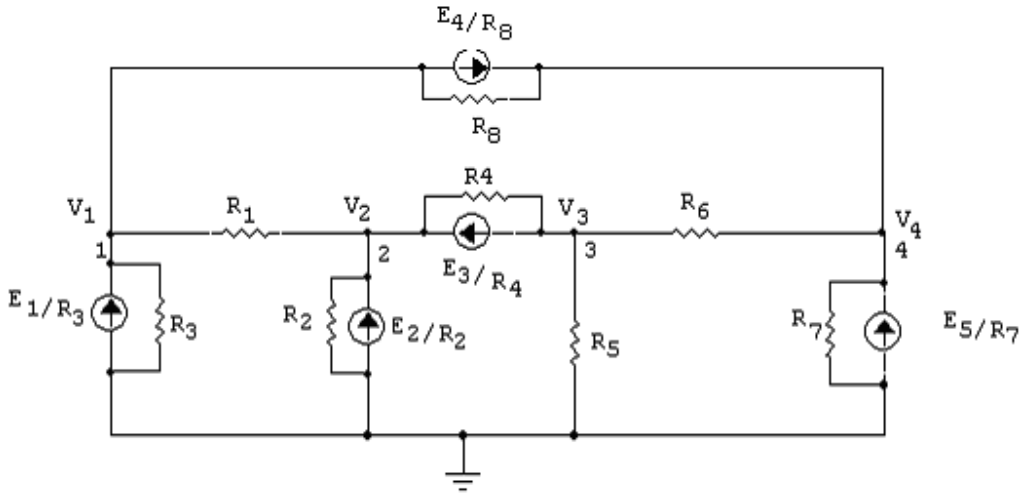


Fig. 1.14

By applying the Kirchoff's current law to each node (fig. 1.14), we may obtain the nodal equations as :

I – Node :
$$\frac{V_1}{R_3} + \frac{V_1 - V_2}{R_1} + \frac{V_1 - V_4}{R_8} = \frac{E_1}{R_3} - \frac{E_4}{R_8}$$

Current leaving the node = current entering the node

or
$$\left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_8}\right)V_1 - \frac{1}{R_1}V_2 - \frac{1}{R_8}V_4 = \frac{E_1}{R_3} - \frac{E_4}{R_8} = I_1 \quad \text{(say)}$$

------ (1.5)

II – Node:
$$\frac{V_2}{R_2} + \frac{V_2 - V_1}{R_1} + \frac{V_2 - V_3}{R_4} = \frac{E_2}{R_2} + \frac{E_3}{R_4}$$

or
$$-\frac{1}{R_1}V_1 + \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}\right)V_2 - \frac{1}{R_4}V_3 = \frac{E_2}{R_2} + \frac{E_3}{R_4} = I_2 \quad \text{(say)}$$

------ (1.6)

III – Node:
$$\frac{V_3}{R_5} + \frac{V_3 - V_2}{R_4} + \frac{V_3 - V_4}{R_6} = -\frac{E_3}{R_4}$$

or $$-\frac{1}{R_4}V_2+\left(\frac{1}{R_4}+\frac{1}{R_5}+\frac{1}{R_6}\right)V_3-\frac{1}{R_6}V_4=-\frac{E_3}{R_4}=I_3 \quad \text{(say)}$$

------ (1.7)

IV – node: $$\frac{V_4}{R_7}+\frac{V_4-V_3}{R_6}+\frac{V_4-V_1}{R_8}=\frac{E_5}{R_7}+\frac{E_4}{R_8}$$

or $$-\frac{1}{R_8}V_1-\frac{1}{R_6}V_3+\left(\frac{1}{R_6}+\frac{1}{R_7}+\frac{1}{R_8}\right)V_4=\frac{E_5}{R_7}+\frac{E_4}{R_8}=I_4 \quad \text{(say)}$$

------ (1.8)

If we concentrate on these equations, we may find some more easy method of writing these nodal equations. For this we rewrite the equation (1.5) as:

$$(G_1+G_3+G_5)V_1-G_1V_2-G_8V_4=I_1 \qquad \text{------ (1.9)}$$

where G are the conductances of their resistance values. This equation (1.9) may further be written in the general form as:

$$G_{11}V_1-G_{12}V_2-G_{13}V_3-G_{14}V_4=I_1$$

G's are the conductances of the corresponding resistances.

G_{11} is the sum of conductances connected to the node 1, which is equal to $\frac{1}{R_1}+\frac{1}{R_3}+\frac{1}{R_8}$ or $(G_1+G_3+G_8)$.

G_{12} is the conductance connected between nodes 1 & 2, which is equal to $\frac{1}{R_1}$ or G_1.

G_{13} is the conductance connected between nodes 1 & 3, which is equal to 0.

G_{14} is the conductance connected between nodes 1 & 4, which is equal to $\frac{1}{R_8}$ or G_8.

The value of current I_1 is the net current entering the node 1 which is equal to $\frac{E_1}{R_3}-\frac{E_3}{R_8}$ as $\frac{E_1}{R_3}$ current is entering the node 1 and $\frac{E_3}{R_8}$ current is leaving the node 1 (hence the – ve sign).

The equations (1.6) to (1.8) may be rewritten in the similar fashion as:

$$-G_{21}V_1+G_{22}V_2-G_{23}V_3-G_{24}V_4=I_2$$
$$-G_{31}V_1-G_{32}V_2+G_{33}V_3-G_{34}V_4=I_3$$
$$-G_{41}V_1-G_{42}V_2-G_{43}V_3+G_{44}V_4=I_4$$

In the matrix form these equations may be written as:

$$\begin{bmatrix} G_{11} & -G_{12} & -G_{13} & -G_{14} \\ -G_{21} & G_{22} & -G_{23} & -G_{24} \\ -G_{31} & -G_{32} & G_{33} & -G_{34} \\ -G_{41} & -G_{42} & -G_{43} & G_{44} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} \quad \text{------ (1.10)}$$

In general G elements of the matrix are defined as:
$G_{ij}\,(i \neq j)$ is the conductance connected between i^{th} and j^{th} node.
$G_{ij}\,(i = j)$ is the sum of conductances connected to the ith or j^{th} node.
I_j is the net amount of current entering the jth node.

$$[G][V] = [I] \quad \text{------ (1.11)}$$

The elements of *G* or *I* matrices are directly obtained from the given problem. It is worthwhile to mention here that the diagonal elements of [G] matrix are positive and off diagonal elements are always negative. This matrix is of 4 *x* 4 orders as there are only 4 nodes. Thus we conclude that a matrix of *n x n* orders will be obtained if there are *n* nodes in the given network. The matrix [G] may be solved for the node voltages by using the Cramer's rule.

Example 1.3 Find the node voltages in the circuit given below.

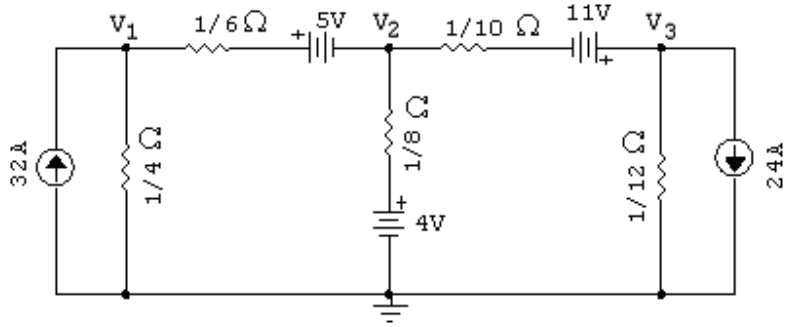


Fig. 1.15

Solution: Transform the voltage sources of the given network in to their equivalent current sources as shown in figure (1.16).

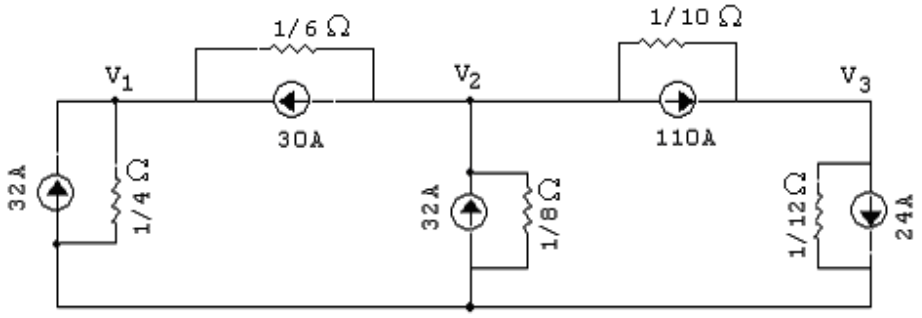


Fig. 1.16

We find the matrix equation of the form: $[G][V] = [I]$

The G and I elements of the two matrices are obtained:

$G_{11} = 4 + 6 = 10\,\text{mhos}$ $G_{12} = 6\,\text{mhos}$ $G_{13} = 0$

$G_{21} = 6\,\text{mhos}$ $G_{22} = 6 + 8 + 10 = 24\,\text{mhos}$ $G_{23} = 10\,\text{mhos}$

$G_{31} = 0$ $G_{32} = 10\,\text{mhos}$ $G_{33} = 10 + 12 = 22\,\text{mhos}$

$I_1 = 32 + 30 = 62A$ $I_2 = -30 + 32 - 110 = -108A$ $I_3 = -24 + 110 = 86A$

So we get the matrix as:

$$\begin{bmatrix} 10 & -6 & 0 \\ -6 & 24 & -10 \\ 0 & -10 & 22 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 62 \\ -108 \\ 86 \end{bmatrix}$$

It can be solved for node voltages using Cramer's rule.

$$\Delta = \begin{vmatrix} 10 & -6 & 0 \\ -6 & 24 & -10 \\ 0 & -10 & 22 \end{vmatrix} = 10[22x24 - 100] + 6[-6x22] = 4280 - 860 = 3490$$

$$V_1 = \frac{\begin{vmatrix} 62 & -6 & 0 \\ -108 & 24 & -10 \\ 86 & -10 & 22 \end{vmatrix}}{\Delta} = \frac{62[22x24 - 100] + 6[(-108)x22 + 86x10]}{3490} = \frac{17440}{3490} = 4.99\,\text{volts}$$

$$V_2 = \frac{\begin{vmatrix} 10 & 62 & 0 \\ -6 & -108 & -10 \\ 0 & 86 & 22 \end{vmatrix}}{\Delta} = \frac{10[(-108)x22 + 860] - 62[(-6)x22]}{3490} = \frac{-6976}{3490} = -1.99\,\text{volts}$$

$$V_3 = \frac{\begin{vmatrix} 10 & -6 & 62 \\ -6 & 24 & -108 \\ 0 & -10 & 86 \end{vmatrix}}{\Delta} = \frac{10[24x86 - 108x10] + 6[(-6)x86] + 62[60]}{3490} = \frac{10464}{3490} = 2.99\,\text{volt}$$

1.3.2 Mesh or Loop Method: Loop or Mesh method is used to find the loop currents in the given network. In the network different loops are first of all identified. The loop currents are assumed to be flowing in the different loops in the clock wise direction (reference direction). KVL is then applied to each loop, to get a set of different equations, the number of which will be equal to the loops present in the network. To discuss this method, consider a circuit shown in the figure (1.17).

In this figure there are four loops 1, 2, 3 & 4 in which I_1, I_2, I_3 & I_4 are assumed to be the loop currents flowing in the clock wise direction as shown figure. Four loop equations are obtained by applying KVL to each loop.

Loop 1: $R_1(I_1 - I_3) + R_2(I_1 - I_2) + E_2 + R_3 I_1 - E_1 = 0$

or $(R_1 + R_2 + R_3)I_1 - R_2 I_2 - R_1 I_3 = E_1 - E_2 = V_1$ (say) ------- (1.12)

Loop 2: $R_2(I_2 - I_1) + E_3 + R_4(I_2 - I_3) + R_5(I_2 - I_4) - E_2 = 0$

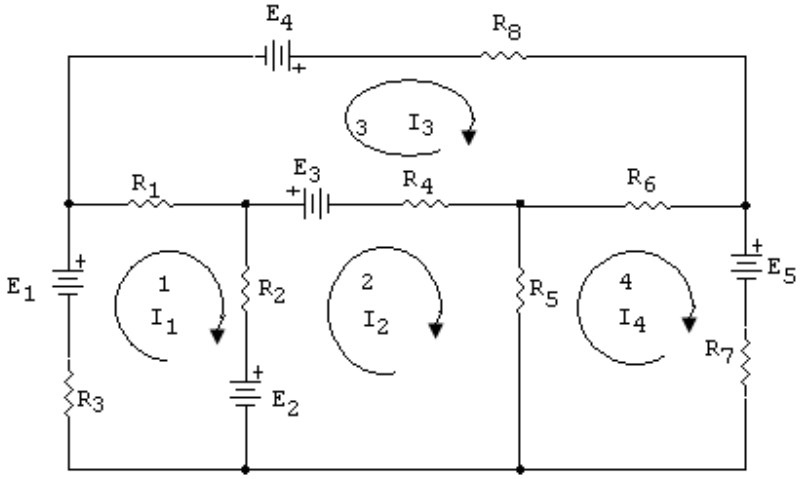


Fig.1.17

or $-R_2 I_1 + (R_2 + R_4 + R_5)I_2 - R_4 I_3 - R_5 I_4 = E_2 - E_3 = V_2$ (say) ------- (1.13)

Loop 3 $-E_4 + R_8 I_3 + R_6(I_3 - I_4) + R_4(I_3 - I_2) - E_3 + R_1(I_3 - I_1) = 0$

or $-R_1 I_1 - R_4 I_2 + (R_1 + R_4 + R_6 + R_8)I_3 - R_6 I_4 = E_3 + E_4 = V_3$ (say) ------(1.14)

loop 4 $R_6(I_4 - I_3) + E_5 + R_7 I_4 + R_5(I_4 - I_2) = 0$

or $-R_5 I_2 - R_6 I_3 + (R_5 + R_6 + R_7)I_4 = -E_5 = V_4$ (say) ------- (1.15)

These equations in the matrix form may be written as:

$$\begin{bmatrix} (R_1+R_2+R_3) & -R_2 & -R_1 & 0 \\ -R_2 & (R_2+R_4+R_5) & -R_4 & -R_5 \\ -R_1 & -R_4 & (R_1+R_4+R_6+R_8) & -R_6 \\ 0 & -R_5 & -R_6 & (R_5+R_6+R_7) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

------ (1.16)

General form of the matrix may be written as:

$$\begin{bmatrix} R_{11} & -R_{12} & -R_{13} & -R_{14} \\ -R_{21} & R_{22} & -R_{23} & -R_{24} \\ -R_{31} & -R_{32} & R_{33} & -R_{34} \\ -R_{41} & -R_{42} & -R_{43} & R_{44} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

------ (1.17)

We define the elements of the matrix in the general form as:
$R_{ij}\,(i \neq j)$ is the resistance common between i^{th} and j^{th} loop.
$R_{ij}\,(I = j)$ is the sum of resistances connected to the ith or j^{th} loop.
V_j is the net voltage rise in j^{th} loop.

$$[R][I] = [V] \qquad \text{------ (1.18)}$$

The elements of R or V matrices are directly obtained from the given problem. The diagonal elements of the [R] matrix are positive and off diagonal elements are negative. The [R] matrix may be solved for the loop currents by using the Cramer's rule.

Example 1.4 Solve for the loop currents in the circuit given below. Values of the resistances in the circuit are given in ohms.

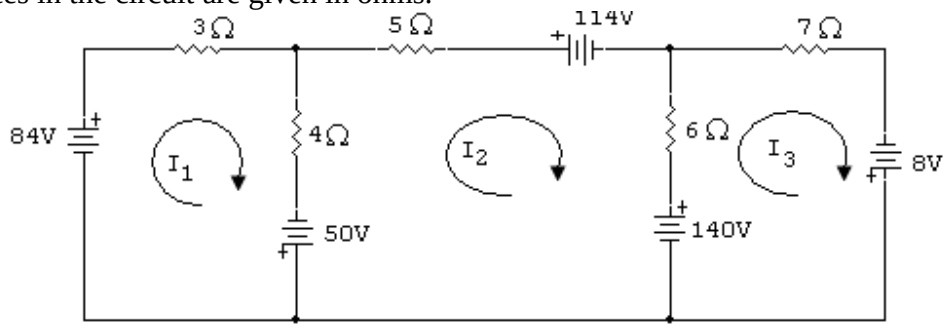


Fig. 1.18

Solution: To find the loop currents we apply the Loop Method and get the matrix of the form [R][I]=[V]

The R's elements of the network are obtained as:

$R_{11} = 3+4 = 7\Omega$ $\quad R_{12} = 4\Omega$ $\quad R_{13} = 0$

$R_{21} = 4\Omega$ $\quad R_{22} = 4+5+6 = 15\Omega$ $\quad R_{23} = 6\Omega$

$R_{31} = 0$ $\quad R_{32} = 6\Omega$ $\quad R_{33} = 6+7 = 13\Omega$

$V_1 = 50+84 = 134V$ $\quad V_2 = -114-140-50 = -304V$ $\quad V_3 = 8+140 = 148V$

We get the matrix as:

$$\begin{bmatrix} 7 & -4 & 0 \\ -4 & 15 & -6 \\ 0 & -6 & 13 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 134 \\ -304 \\ 148 \end{bmatrix} \qquad \text{------ (1.19)}$$

It can be solved for loop currents using Cramer's rule:

$$\Delta = \begin{vmatrix} 7 & -4 & 0 \\ -4 & 15 & -6 \\ 0 & -6 & 13 \end{vmatrix} = 7[15x13-(-6)x(-6)]+4[(-4)x13] = 1113-208 = 905$$

$$I_1 = \frac{\begin{vmatrix} 134 & -4 & 0 \\ -304 & 15 & -6 \\ 148 & -6 & 13 \end{vmatrix}}{\Delta} = \frac{134[15x13-(-6)x(-6)]+4[(-304)x13-(-6)x148]}{905} = \frac{21306-12256}{905} = 10A$$

$$I_2 = \frac{\begin{vmatrix} 7 & 134 & 0 \\ -4 & -304 & -6 \\ 0 & 148 & 13 \end{vmatrix}}{\Delta} = \frac{7[(-304)x13-(-6)x148]-134[(-4)x13]}{905} = \frac{-21448+6968}{905} = -16A$$

$$I_3 = \frac{\begin{vmatrix} 7 & -4 & 134 \\ -4 & 15 & -304 \\ 0 & -6 & 148 \end{vmatrix}}{\Delta} = \frac{7[15x148-(-304)(-6)]+4[(-4)x148]+134[(-4)x(-6)]}{905}$$

$$= \frac{2772-2368+3216}{905} = \frac{3620}{905} = 4A$$

1.4 Network Theorems: General methods of network analysis discussed earlier provide lengthy calculations in complicated networks. It is required to develop some easy methods in solving the network problems. The network theorems, which are applicable for both A.C. and D.C. networks, are helpful in solving the complicated network problems. Some of the important theorems Viz. Superposition Theorem, Thevenin's Theorem, Norton's theorem, Reciprocity Theorem, Millman's Theorem, and Maximum Power Transfer Theorem will be discussed below.

Superposition Theorem: This theorem states that in a network containing impedances and sources (voltage sources and/or current sources), the current flowing at any point is the algebraic sum of the currents that would flow at that point if each source was considered separately, and all other sources were replaced with their internal impedances.

Proof: To prove this theorem, consider a network shown in the figure (1.19). In this circuit we assume that I_1 and I_2 are the two loop currents of the network flowing in the clock wise direction. It is further assumed that I_1' and I_2' are the two currents flowing in these loops when only V_1 voltage source is considered and V_2 is short circuited (Fig. 1.20 a).

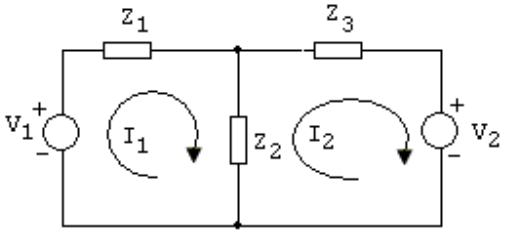


Fig. 1.19

Similarly, $I_1^{"}$ and $I_2^{"}$ are two currents in these two loops when only V_2 voltage source is considered and V_1 is short circuited (Fig. 1.20 b).

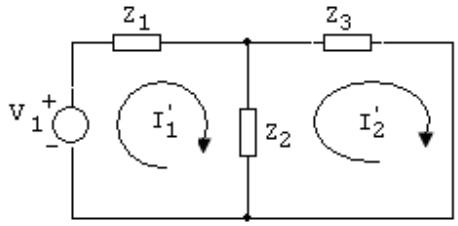
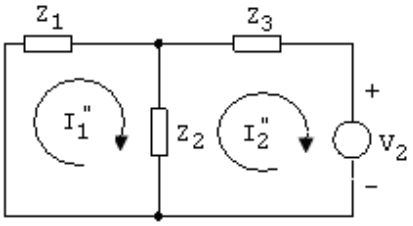


Fig. 1.20(a)



(b)

According to this theorem, we should get:

$$I_1 = I_1^{'} + I_1^{"} \text{ and } \quad I_2 = I_2^{'} + I_2^{"}$$

By applying Loop method in the two meshes (ref. Fig. 1.19), we get:

$$\begin{bmatrix} (Z_1 + Z_2) & -Z_2 \\ -Z_2 & (Z_2 + Z_3) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix} \qquad \text{------ (1.21)}$$

By considering only V_1 source and V_2 is short circuited (putting $V_2 = 0$) in Fig. 1.20(a), we get

$$\begin{bmatrix} (Z_1 + Z_2) & -Z_2 \\ -Z_2 & (Z_2 + Z_3) \end{bmatrix} \begin{bmatrix} I_1^{'} \\ I_2^{'} \end{bmatrix} = \begin{bmatrix} V_1 \\ 0 \end{bmatrix} \qquad \text{------ (1.22)}$$

Now by considering only V_2 source and V_1 is short circuited (putting $V_1 = 0$) Fig. 1.20(b), we get

$$\begin{bmatrix} (Z_1 + Z_2) & -Z_2 \\ -Z_2 & (Z_2 + Z_3) \end{bmatrix} \begin{bmatrix} I_1^{"} \\ I_2^{"} \end{bmatrix} = \begin{bmatrix} 0 \\ -V_2 \end{bmatrix} \qquad \text{------ (1.23)}$$

Since [Z] matrix of both the equations (1.22) & (1.23) are same so we may combine these equations, as:

$$\begin{bmatrix} (Z_1 + Z_2) & -Z_2 \\ -Z_2 & (Z_2 + Z_3) \end{bmatrix} \begin{bmatrix} I_1^{'} + I_1^{"} \\ I_2^{'} + I_2^{"} \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix} \qquad \text{------ (1.24)}$$

Comparing equations (1.21) & (1.24) we get the required result.

$$I_1 = I_1^{'} + I_1^{"} \text{ and } \quad I_2 = I_2^{'} + I_2^{"}$$

Hence the theorem is proved.

Example 1.5 Using Superposition theorems, find the current flowing through each resistances in the following circuit. The values of all resistances in the circuit are given in ohms.

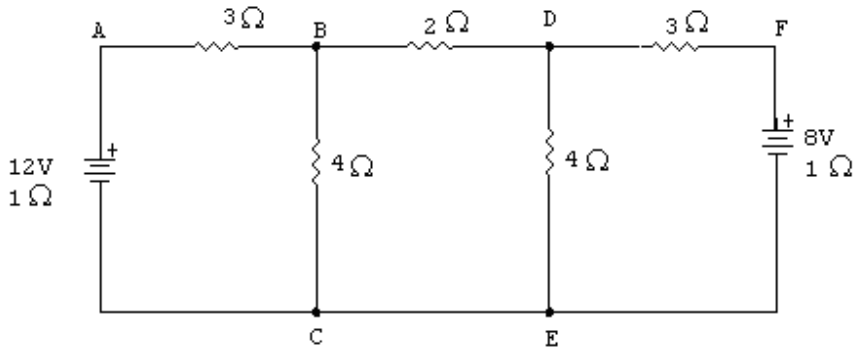


Fig. 1.21

Solution: In this circuit first only 12*V* source is considered and other 8 *V* source is replaced by its internal resistance as shown in the figure 1.22. In this way we find the current following through all the resistances.

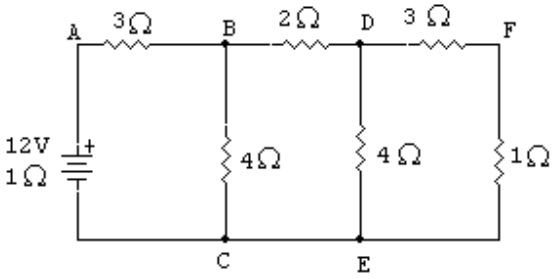


Fig. 1.22

The resistance between B & C points is 2Ω (parallel combination of two 4Ω resistances).

Current flowing through A B branch is $I'_{AB} = \frac{12}{3+1+2} = 2A$ from A to B

Current flowing through BC branch is $I'_{BC} = \frac{I'_{AB}}{2} = 1A$ from B to C

Current flowing through BD branch is $I'_{BD} = I'_{BC} = 1A$ from B to D

Current flowing through DE branch is $I'_{DE} = \frac{I'_{BD}}{2} = 0.5A$ from D to E

Current flowing through DF branch is $I'_{DF} = I'_{DE} = 0.5A$ from D to F

Now other source of 8 volts is considered and 12 volt source is replaced by its internal resistance as shown in the circuit (fig. 1.23). The resistance between *D* & *E* points is 2 Ω (parallel combination of two 4 Ω resistances).

Current flowing through DF branch is $I''_{DF} = \frac{8}{3+1+2} = 1.33A$ from F to D

Current flowing through DE branch is $I''_{DE} = \frac{I''_{DF}}{2} = 0.67A$ from D to E

Current flowing through BD branch is $I''_{BD} = I''_{DE} = 0.67A$ from D to B

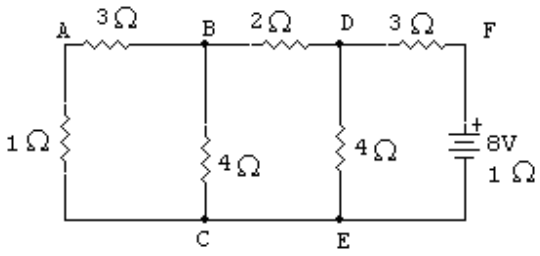


Fig. 1.23

Current flowing through BC branch is $I''_{BC} = \frac{I''_{BD}}{2} = 0.33A$ from B to C

Current flowing through AB branch is $I''_{AB} = I''_{BC} = 0.33A$ from B to A

The required current in AB branch $I_{AB} = I'_{AB} - I''_{AB} = 2 - 0.33 = 1.67A$ (A to B)

The required current in BC branch $I_{BC} = I'_{BC} + I''_{BC} = 1 + 0.33 = 1.33A$ (B to C)

The required current in BD branch $I_{BD} = I'_{BD} - I''_{BD} = 1 - 0.67 = 0.33A$ (B to D)

The required current in DE branch $I_{DE} = I'_{DE} + I''_{DE} = 0.5 + 0.67 = 1.17A$ (D to E)

The required current in DF branch $I_{DF} = I''_{DF} - I'_{DF} = 1.33 - 0.5 = 0.83A$ (F to D)

1.4.2 Thevenin's Theorem: Any network containing impedances and sources (voltage sources and/or current sources) can be replaced with a voltage source V_0 and an impedance Z_0 in series with it. The value of voltage source V_0 is the open circuit voltage at the output terminals of the network, Z_0 is the impedance at the output terminals of the network replacing all the sources with their internal impedances. According to this theorem, any network containing sources and impedances shown in figure 1.24(a) can be replaced by the circuit shown in figure 1.24(b)

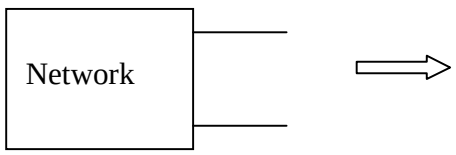


Fig. 1.24(a)

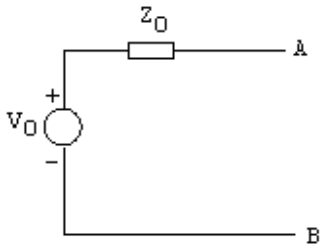


Fig. 1.24(b)

Proof: To establish Thevenin's theorem, a network containing voltage source and impedances is considered, which is shown in figure (1.25a).

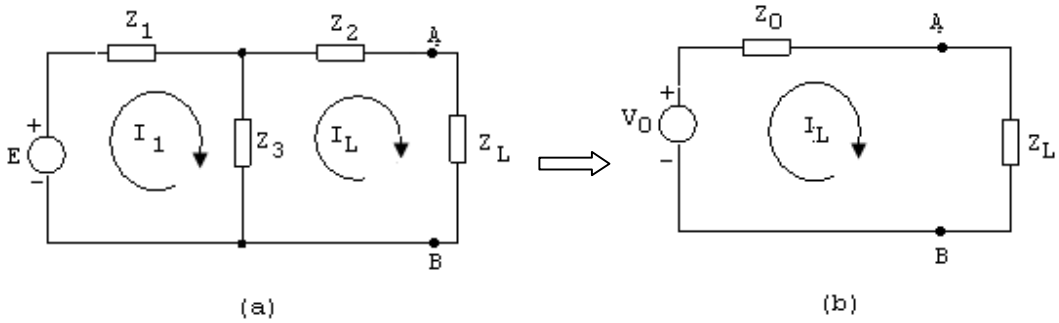


Fig. 1.25

According to this theorem, this network should be equal to a voltage source (V_0) and impedance in series with it, as shown in fig. (1.25b). The load current will be calculated from both the circuits and will be proved that the two currents are equal.

Applying the loop method to the circuit of figure (1.25 a), we get:

$$\begin{bmatrix} (Z_1+Z_3) & -Z_3 \\ -Z_3 & (Z_2+Z_3+Z_L) \end{bmatrix} \begin{bmatrix} I_1 \\ I_L \end{bmatrix} = \begin{bmatrix} E \\ 0 \end{bmatrix} \qquad \text{------ (1.25)}$$

The value of I_L can be calculated from this equation, using Cramer's rule:

$$I_L = \frac{\begin{vmatrix} (Z_1+Z_3) & E \\ -Z_3 & 0 \end{vmatrix}}{\begin{vmatrix} (Z_1+Z_3) & -Z3 \\ -Z_3 & (Z_2+Z_3+Z_L) \end{vmatrix}} = \frac{EZ_3}{(Z_1+Z_3)(Z_2+Z_3+Z_L)-Z_3^2}$$

or $$I_L = \frac{E.Z_3}{Z_1Z_2+Z_1Z_3+Z_1Z_L+Z_2Z_3+Z_3Z_L} \qquad \text{------ (1.26)}$$

We now calculate the value of open circuit voltage V_0 across AB terminals and Thevenin's impedance Z_0 from the figure (1.25a), after removing the load impedance Z_L as:

$$V_0 = \frac{E}{(Z_1+Z_3)}.Z_3$$

and $$Z_0 = Z_2 + Z_1 \| Z_3 \qquad \text{(Short circuiting the voltage source)}$$

$$= Z_2 + \frac{Z_1Z_3}{Z_1+Z_3} = \frac{Z_1Z_2+Z_2Z_3+Z_1Z_3}{Z_1+Z_3}$$

The load current I_L may be obtained from the fig. (1.25b) as:

$$I_L = \frac{V_0}{Z_0+Z_L} = \frac{E.Z_3/(Z_1+Z_3)}{(Z_1Z_2+Z_2Z_3+Z_1Z_3+Z_1Z_L+Z_3Z_L)/(Z_1+Z_3)}$$

$$= \frac{E.Z_3}{Z_1Z_2+Z_1Z_3+Z_1Z_L+Z_2Z_3+Z_3Z_L} \qquad \text{------ (1.27)}$$

From the equations (1.26) & (1.27), we see that the value of I_L is the same, as calculated from both the circuits (Fig. 1.25a & 1.25 b).
Hence the theorem is proved.

Example 1.6 Find the Thevenin's equivalent of the network inside the Box

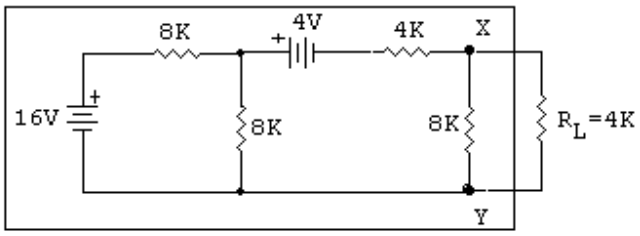


Fig. 1.26

of figure (1.26) and then calculate the current flowing through the load resistance R_L connected between XY terminals.

Solution: We are to find the open circuit voltage across XY terminals and the resistance across these two terminals when the sources have been replaced with their internal resistances. To find the voltage across XY terminals we may use loop method, and get the current flowing through these terminals removing the load resistance. Let I_1 and I_2 are the two current in the two loops as shown in fig. (1.27).

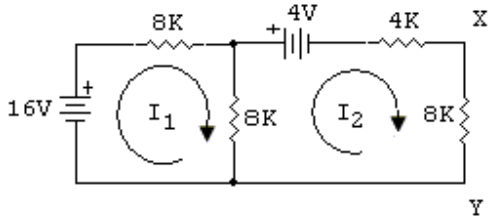


Fig. 1.27

The loop equation is given by: $[R][I]=[V]$

or $$\begin{bmatrix} 16 & -8 \\ -8 & 20 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}=\begin{bmatrix} 16 \\ -4 \end{bmatrix}$$ ------- (1.28)

and $$I_2=\frac{\begin{vmatrix} 16 & 16 \\ -8 & -4 \end{vmatrix}}{\begin{vmatrix} 16 & -8 \\ -8 & 20 \end{vmatrix}}=\frac{[-64+16x8]}{[16x20-64]}=\frac{64}{256}=0.25mA$$

(Resistance values are in Kilo-ohms hence current is in *mA*)

$$V_{XY} = 0.25mA \; x \; 8 \; K\Omega = 2 \; Volts$$

Thevenin's resistance $R_{XY} = [(8K \| 8K) + 4k] \| 8K = 8K \| 8K = 4K\Omega$ (All the sources are shorted).

The Thevenin's equivalent is shown in figure (1.28):

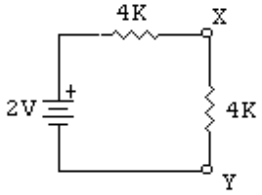


Fig. 1.28

The load current is given by: $I_L = \dfrac{2Volts}{(4K + 4K)} = 0.25mA$.

1.4.3 Norton's Theorem: Any network containing impedances and sources (voltage and / or current sources) can be replaced with a current

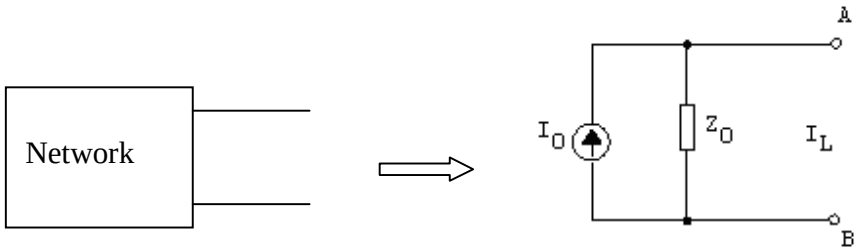


Fig. 1.29

source I_0 and an impedance Z_0 in Parallel with it. The value of current source I_0 is the short circuit current obtained at the output terminals of the network, Z_0 is the impedance at its output terminals replacing all the sources with their internal impedances. According to this theorem, any network containing sources and impedances (ref. figure 1.29a) can be replaced by a circuit shown in fig.(1.29 b).

Proof: To illustrate Norton's theorem, a network containing voltage source and impedances is considered, which is shown in figure (1.30a). As

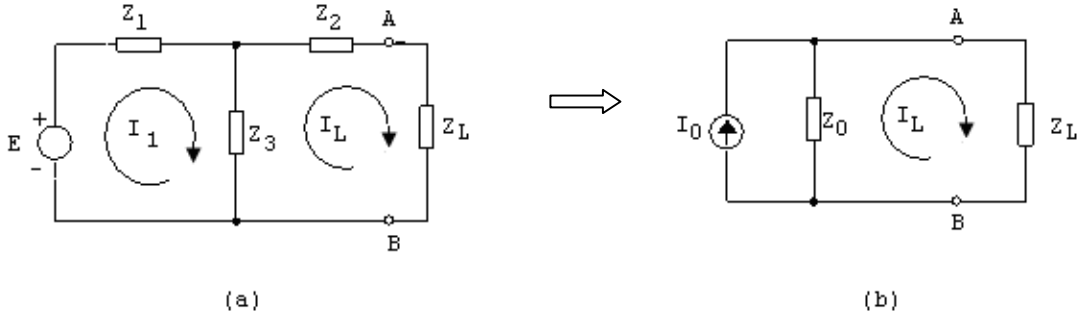


Fig. 1.30

discussed earlier, this network should be equal to a current source (I_0) and impedance (Z_0) in parallel with it (ref. fig. 1.30 b). Norton's theorem will be proved if the load current calculated from both the circuits is equal.
Applying the loop method to the circuit of figure (1.30 a), it is obtained as:

$$\begin{bmatrix} (Z_1+Z_3) & -Z_3 \\ -Z_3 & (Z_2+Z_3+Z_L) \end{bmatrix} \begin{bmatrix} I_1 \\ I_L \end{bmatrix} = \begin{bmatrix} E \\ 0 \end{bmatrix} \qquad \text{------ (1.29)}$$

Using Cramer's rule, the value of I_L can be calculated as:

$$I_L = \frac{\begin{vmatrix} (Z_1+Z_3) & E \\ -Z_3 & 0 \end{vmatrix}}{\begin{vmatrix} (Z_1+Z_3) & -Z3 \\ -Z_3 & (Z_2+Z_3+Z_L) \end{vmatrix}} = \frac{EZ_3}{(Z_1+Z_3)(Z_2+Z_3+Z_L)-Z_3^2}$$

or $$I_L = \frac{E.Z_3}{Z_1Z_2+Z_1Z_3+Z_1Z_L+Z_2Z_3+Z_3Z_L} \qquad \text{------ (1.30)}$$

Now the short circuit current I_0 will be calculated by short circuiting the output terminals as shown in figure (1.31). Loop method may be used to calculate the short circuit current I_0.

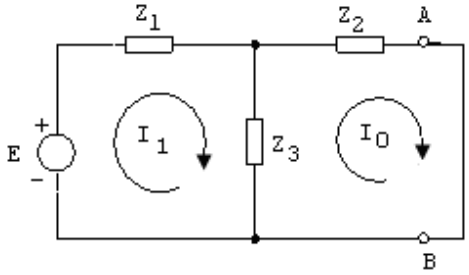


Fig. 1.31

Loop equations are given by:

$$\begin{bmatrix} (Z_1+Z_3) & -Z_3 \\ -Z_3 & (Z_2+Z_3) \end{bmatrix} \begin{bmatrix} I_1 \\ I_0 \end{bmatrix} = \begin{bmatrix} E \\ 0 \end{bmatrix}$$

$$I_0 = \frac{\begin{vmatrix} (Z_1+Z_3) & E \\ -Z_3 & 0 \end{vmatrix}}{\begin{vmatrix} (Z_1+Z_3) & -Z3 \\ -Z_3 & (Z_2+Z_3) \end{vmatrix}} = \frac{EZ_3}{(Z_1+Z_3)(Z_2+Z_3)-Z_3^2}$$

or $$I_0 = \frac{E.Z_3}{Z_1Z_2+Z_1Z_3+Z_1Z_L+Z_2Z_3} \qquad \text{------ (1.31)}$$

The impedance Z_0 is obtained by removing the load impedance in the given network. It is given by:

$$Z_0 = Z_2 + Z_1 \| Z_3$$

$$= Z_2 + \frac{Z_1 Z_3}{Z_1 + Z_3} = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3}{Z_1 + Z_3} \qquad \text{------ (1.32)}$$

The load current can also be calculated using the Norton's equivalent circuit (fig. 1.30 b) as:

$$I_L = \frac{Z_0 . Z_L}{(Z_0 + Z_L)} \frac{I_0}{Z_L} = \frac{Z_0 I_0}{(Z_0 + Z_L)} \qquad \text{------ (1.33)}$$

Putting the values of Z_0 and I_0 from the equations (1.32) & (1.31) in equation (1.33) we get the value of I_L as:

$$I_L = \frac{E.Z_3}{Z_1 Z_2 + Z_1 Z_3 + Z_1 Z_L + Z_2 Z_3 + Z_3 Z_L}$$, which is same as calculated directly from the given network. This proves the Norton's theorem.

Thevenin's and Norton's equivalent circuit of a given network produces same amount of current and voltage in the load impedance. Hence these theorems are the dual of each other. Therefore, either of the two theorems can be applied for the network analysis.

Example 1.7 Draw Norton's equivalent of the network and find the current in 2Ω resistance connected between AB branch in the figure (1.32).

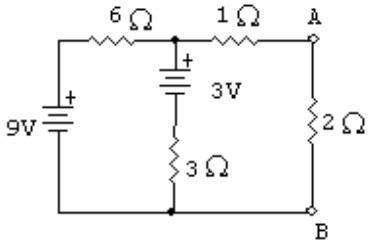


Fig. 1.32

Solution: To draw the Norton's equivalent, the short circuit current in AB branch is obtained by short circuiting these terminals as shown in figure (1.33). In this figure the short circuit current will be I_2 which may be obtained by the loop method.

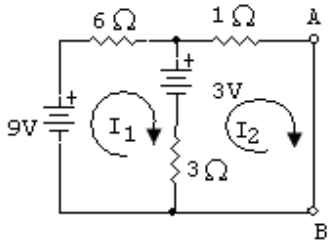


Fig. 1.33

The loop equation is given as: $[R][I]=[I]$

$$\begin{bmatrix} 9 & -3 \\ -3 & 4 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}=\begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

And I_2 is given by: $I_2=\dfrac{\begin{vmatrix} 9 & 6 \\ -3 & 3 \end{vmatrix}}{\begin{vmatrix} 9 & -3 \\ -3 & 4 \end{vmatrix}}=\dfrac{27+18}{36-9}=\dfrac{45}{27}=\dfrac{5}{3}A$

The resistance R_0 is measured at the output terminals, removing the 2Ω resistance between AB branch and short circuiting the voltage sources in the network. $R_0=1+\dfrac{3x6}{3+6}=3\Omega$. The Norton's equivalent is, therefore, shown in figure (1.34).

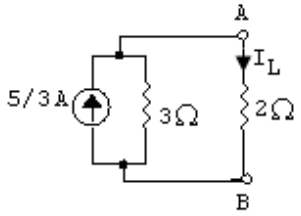


Fig. 1.34

The required current I_L through 2Ω resistance is given by:

$$I_L=\frac{5}{3}.\frac{2x3}{2+3}.\frac{1}{3}=\frac{2}{3}A$$

1.4.4 Reciprocity Theorem: This theorem states that when an ideal voltage source is applied to one loop of the given network of linear impedances, produces a current in the second loop, then the same amount of current will be produced in the first loop of the given network if that ideal voltage source is applied to the second loop.

Proof: To prove this theorem, we consider a network shown in figure (1.35), in

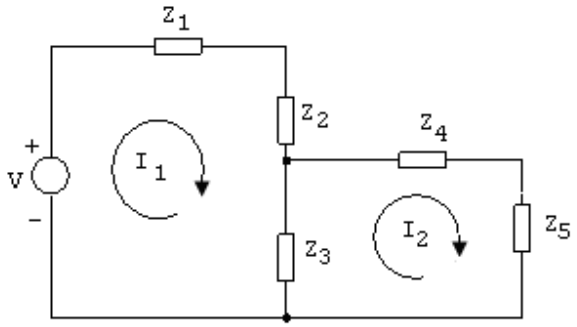


Fig. 1.35

which an ideal voltage source *V* is introduced in loop *1* and current I_2 is produced in loop 2. Again, we introduce the ideal voltage source *V* in the loop *2* and $I_1^{'}$ current is produced in loop 1 as shown in figure (1.36).

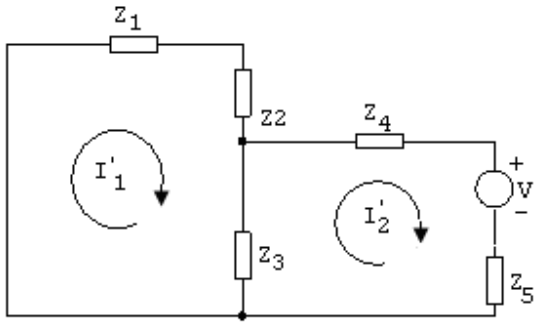


Fig. 1.36

According to this theorem, $|I_2| = |I_1^{'}|$.

To prove this we find the current I_2 from figure (1.35). For this loop method is applied to get the loop equations which are given the matrix form as:

$$\begin{bmatrix} (Z_1+Z_2+Z_3) & -Z_3 \\ -Z_3 & (Z_3+Z_4+Z_5) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V \\ 0 \end{bmatrix} \quad \text{------ (1.34)}$$

The current I_2 is calculated using Cramer's rule as:

$$I_2 = \frac{\begin{vmatrix} (Z_1+Z_2+Z_3) & V \\ -Z_3 & 0 \end{vmatrix}}{\begin{vmatrix} (Z_1+Z_2+Z_3) & -Z3 \\ -Z_3 & (Z_3+Z_4+Z5) \end{vmatrix}} = \frac{Z_3.V}{(Z_1+Z_2+Z_3)(Z_3+Z_4+Z_5)-Z_3^2}$$

------ (1.35)

Similarly, we get the mesh equations in the matrix form from the fig. (1.36)

$$\begin{bmatrix} (Z_1+Z_2+Z_3) & -Z_3 \\ -Z_3 & (Z_3+Z_4+Z_5) \end{bmatrix} \begin{bmatrix} I_1^{'} \\ I_2^{'} \end{bmatrix} = \begin{bmatrix} 0 \\ -V \end{bmatrix} \quad \text{------ (1.36)}$$

The current $I_1^{'}$ is calculated using Cramer's rule as:

$$I_1^{'} = \frac{\begin{vmatrix} 0 & -Z_3 \\ -V & (Z_3+Z_4+Z_5) \end{vmatrix}}{\begin{vmatrix} (Z_1+Z_2+Z_3) & -Z3 \\ -Z_3 & (Z_3+Z_4+Z5) \end{vmatrix}} = \frac{-Z_3.V}{(Z_1+Z_2+Z_3)(Z_3+Z_4+Z_5)-Z_3^2} \quad \text{------ (1.37)}$$

From the equations (1.35) & (1.37), it is clear that $|I_2| = |I_1'|$

Hence the Reciprocity theorem is established.

It follows from Reciprocity theorem that in any linear network containing linear impedances and sources, the ratio of voltage introduced in one loop to the current in any second loop is the same as the ratio of voltage introduced in second loop and the current introduced in first loop, other sources being replaced by their internal impedances.

Example 1.8 Verify the Reciprocity theorem in the circuit shown below. The values of all the resistances in the circuit are given in ohms.

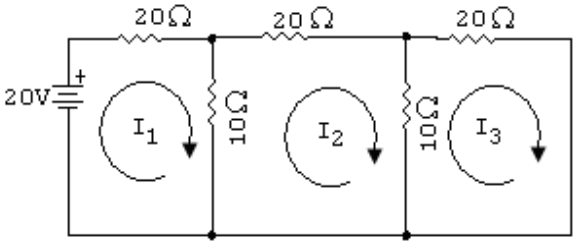


Fig. 1.37

Solution: We find the current I_3 in the third loop using loop method, when 20 *V* source is connected in the first loop.

$$\begin{bmatrix} 30 & -10 & 0 \\ -10 & 40 & -10 \\ 0 & -10 & 30 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix} \quad \text{-------} \quad (1.38)$$

$$I_3 = \frac{\begin{vmatrix} 30 & -10 & 20 \\ -10 & 40 & 0 \\ 0 & -10 & 0 \end{vmatrix}}{\begin{vmatrix} 30 & -10 & 0 \\ -10 & 40 & -10 \\ 0 & -10 & 30 \end{vmatrix}} = \frac{30[0]+10[0]+20[100]}{30[40x30-100]+10[(-10)x30]+0} = \frac{2000}{30000} = \frac{1}{15} = 67mA$$

Now we apply the source in the third loop as given in the figure (1.38) and find the current I_1' in the first loop as:

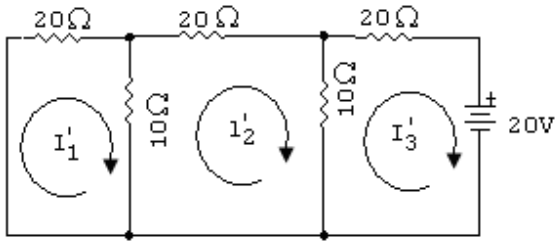


Fig. 1.38

$$I_1^{'} = \frac{\begin{vmatrix} 0 & -10 & 0 \\ 0 & 40 & -10 \\ -20 & -10 & 30 \end{vmatrix}}{\begin{vmatrix} 30 & -10 & 0 \\ -10 & 40 & -10 \\ 0 & -10 & 30 \end{vmatrix}} = \frac{0+10[-200]}{30[40x30-100]+10[(-10)x30]+0} = \frac{-2000}{30000} = \frac{-1}{15} = -67mA$$

Here $|I_3| = |I_1^{'}|$ hence the network is reciprocal.

1.4.5 Millman's Theorem: This theorem states that if several voltage sources in series with impedances are connected as shown in figure (1.39 a), then the equivalent circuit may be represented by a voltage source V and an impedance Z in series with it as shown in figure (1.39 b). The value of voltage source V_m is given by:

$$V_m = \frac{V_1Y_1 + V_2Y_2 + V_3Y_3 + ... + Y_N}{Y_1 + Y_2 + Y_3 + ... + Y_N} = \frac{\sum_{I=1}^{N} V_I Y_I}{\sum_{I=1}^{N} Y_I}$$

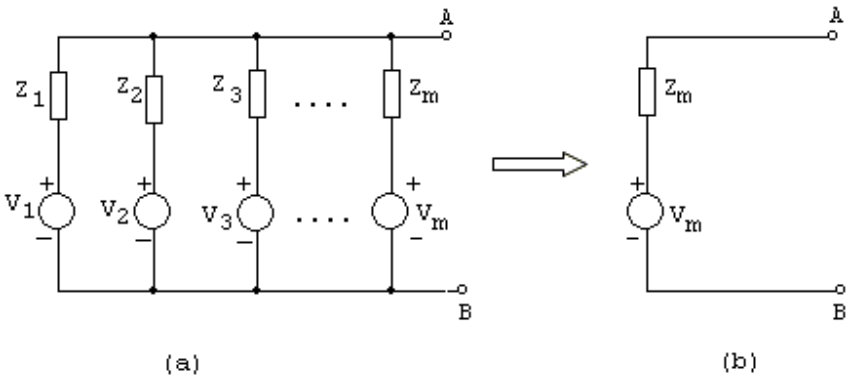


Fig. 1.39

and $Z_m = \frac{1}{Y_1 + Y_2 + Y_3 + ... + Y_N}$, and Y are the admittances ($Z = \frac{1}{Y}$).

Proof: We replace each voltage source with its impedance in series, with current source and its impedance in parallel as shown in figure (1.40 a).

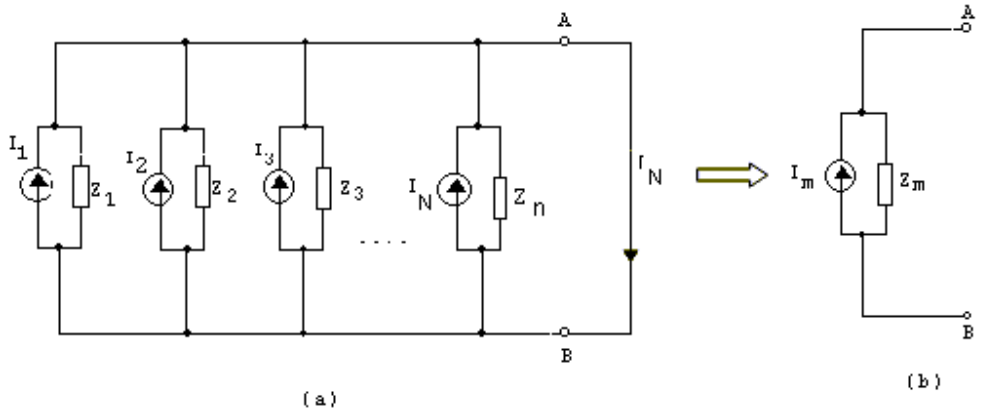


Fig. 1.40

The short circuit current I_m is given by: $I_m = I_1 + I_2 + I_3 + ... + I_n$

$$I_m = \frac{V_1}{Z_1} + \frac{V_2}{Z_2} + \frac{V_3}{Z_3} + ... + \frac{V_N}{Z_N}$$

$$= V_1Y_1 + V_2Y_2 + V_3Y_3 .... + V_NY_N = \sum_{I=1}^{N} V_I Y_I \qquad \text{------- (1.39)}$$

Impedance Z_m at the output terminals when all the sources have been removed and output is open, is given by: $\frac{1}{Z_m} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} + ... + \frac{1}{Z_N}$

or $$Y_m = Y_1 + Y_2 + Y_3 + ... + Y_N = \sum_{I=1}^{N} Y_I$$

or $$Z_m = \frac{1}{Y_m} = \frac{1}{\sum_{I=1}^{N} Y_I} \qquad \text{------- (1.40)}$$

The Norton's equivalent of this network is given in figure (1.40b), which may further be converted to its Thevenin's equivalent. The Thevenin's voltage V_m is given by:

$$V_m = I_m . Z_m = \frac{I_m}{Y_m} = \frac{\sum_{I=1}^{N} V_I Y_I}{\sum_{I=1}^{N} Y_I}$$

and Z_m is given by the equation (1.40). The Thevenin's equivalent will be as shown in figure (1.39 b).

Hence Millman's theorem is proved.

Example 1.9 Calculate the load current I_L In the circuit of figure (1.41), using Millman's theorem.

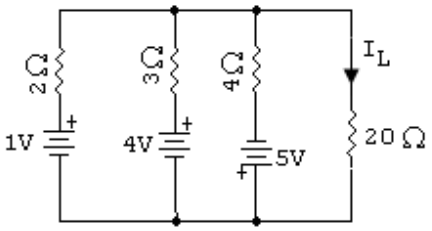


Fig. 1.41

Solution: According to Millman's theorem, the given circuit may be represented by a voltage source *V* and an impedance *Z* in series with it as shown in figure (1.42).

The value of V is given by:

$$V = \frac{V_1Y_1 + V_2Y_2 + V_3Y_3}{Y_1 + Y_2 + Y_3} = \frac{\frac{1}{2} + \frac{4}{3} - \frac{5}{4}}{\frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{\frac{6+16-15}{12}}{\frac{6+4+3}{12}} = \frac{7}{13} volts$$

$$Z = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{\frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{1}{\frac{6+4+3}{12}} = \frac{12}{13}\Omega$$

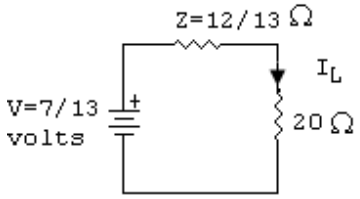


Fig. 1.42

The current I_L is given by: $$I_L = \frac{7/13}{(12/13) + 20} = \frac{7}{273} = 25.7mA$$

1.4.6 Maximum Power Transfer Theorem: This theorem states that when a voltage source is connected to load impedance, then maximum power will be transferred from the voltage source to the load impedance, if load impedance is equal to the complex conjugate of source impedance.

Proof: Let us consider an a.c. voltage source V_S (having Z_S as source impedance) is connected to a load impedance Z_L as shown in figure (1.43).

It is now to be proved that the maximum power will be transferred from source V_S to load impedance Z_S if $$Z_L = Z_S^*$$ ------ (1.41)

We know $Z_S = R_S + jX_S$ and $Z_L = R_L + jX_L$

where R_S & R_L are the resistive parts of Z_S & Z_L respectively and X_S & X_L

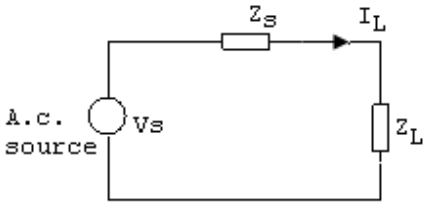


Fig. 1.43

are their corresponding reactive parts.

Load current I_L is given by: $I_L = \dfrac{V_S}{(Z_S + Z_L)} = \dfrac{V_S}{(R_S + R_L) + j(X_S + X_L)}$ ------ (1.42)

Voltage across Z_L is : $V_L = I_l.Z_L = \dfrac{V_{S.}.(R_L + jX_L)}{[R_S + R_L) + j(X_S + X_L)]}$

Power delivered to the load is given by: $P = \text{Re}\,alPart\,(V_{L.}I_L^*)$

$$I_L^* = \frac{V_S}{(R_S + R_L) - j(X_S + X_L)}$$

$$V_L.I_L^* = \frac{V_S^2(R_L + jX_L)}{[(R_S + R_L)^2 + (X_S + X_L)^2]}$$

So $$P = \frac{V_S^2 R_L}{(R_S + R_L)^2 + (X_S + X_L)^2}$$ ------ (1.43)

To find the condition for maximum power delivered to the load, first differentiation of power P is put equal to zero. Here power varies with load resistance R_L and also with load reactance X_L. The variation of power with X_L (keeping R_L constant) is first considered.

We find $\dfrac{dP}{dX_L}$ and put it equal to zero.

$$\frac{dP}{dX_l} = \frac{-2V_S^2 R_L(X_L + X_S)}{\left[(R_L + R_S)^2 + (X_L + X_S)^2\right]^2} = 0$$ ------ (1.44)

This will be equal to zero if $(X_L + X_S) = 0$ or $X_L = -X_S$.

By putting $X_L = -X_S$ in equation (1.44), we get

$$P = \frac{V_S^2.R_L}{(R_S + R_L)^2}$$ ------ (1.45)

Again we consider the variation of power with R_L, here we find $\dfrac{dP}{dR_L}$ and put it equal to

zero as:
$$\frac{dP}{dR_L} = \frac{V_S^2\left[(R_S + R_L)^2 - 2R_L(R_L + R_S)\right]}{(R_L + R_S)^4} = 0$$

or
$$\frac{V_S^2\left[R_S^2 + R_L^2 + 2R_S R_L - 2R_L^2 - 2R_S R_L\right]}{(R_L + R_S)^4} = 0$$

or
$$\frac{V_S^2\left[R_S^2 - R_L^2\right]}{(R_L + R_S)^4} = 0$$

The left hand side of this equation will be zero when either V_S = 0 or $[R_S^2 - R_L^2] = 0$. V_S can not be equal to zero as it is a given voltage source. So $[R_S^2 - R_L^2] = 0$ or $R_S = R_L$.

Thus the power delivered to the load will be maximum when the resistive part of the load impedance are equal to the resistive part of the source impedance; also reactance of the load impedance must be equal but opposite in sign to the reactance of the source impedance.

i.e. $R_L + j\,X_L = R_S - j\,X_S$

or $Z_L = Z_S^*$

In other words one can say that the maximum power will be delivered to the load impedance, when load impedance is equal to the complex conjugate of source impedance. This was to be proved.

It is further interesting to note that if d.c. source V_S having source resistance R_S is considered then the condition for the maximum power will be transferred if $R_S = R_L$ which may be proved in the similar fashion as discussed above.

The expression for maximum power delivered to the load is given by:

$$P_{max} = \frac{V_S^2 R_L}{(2R_L)^2} = \frac{V_S^2}{4R_L}$$

This power is also known as available power.

Example 1.10 Find the value of R_L for which power delivered to it, is maximum in the figure(1.44). Determine the maximum power.

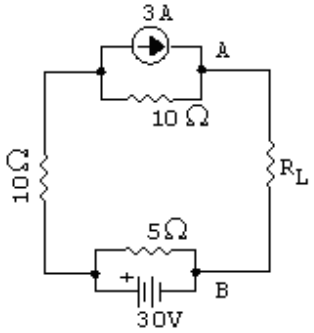


Fig. 1.44

Solution: This circuit is replaced into its Norton's equivalent form. For this find short circuit current by short circuiting the AB terminals as shown in figure (1.45). Short circuit current may be obtained using Superposition theorem.

(i) Consider only current source of 3 *A*, and 30 *V* source is shorted, the short circuit current $I^{'}$ is given by: $I^{'} = \frac{3Ax5}{10} = 1.5A$

(ii) Consider only 30V source, and 3A source is open circuited, we get the short circuit current I" as: $I^{"} = \frac{30V}{10+10} = 1.5A$

Net short circuit current I_0 is given by: $I_0 = I^{'} + I^{"} = 1.5 + 1.5 = 3A$

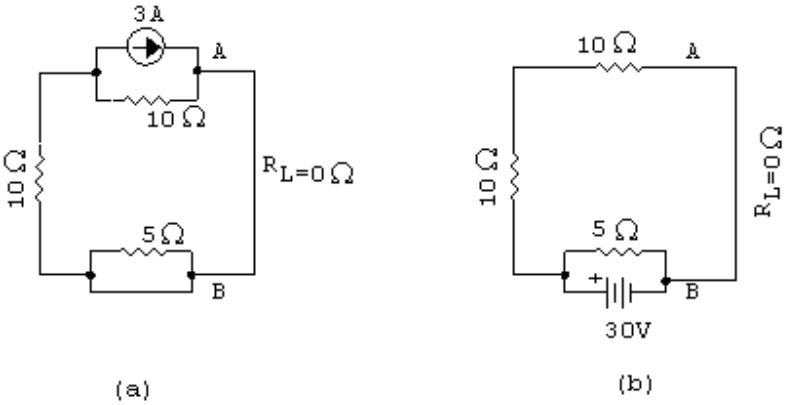


Fig. 1.45

The open circuit resistance R_S (across AB terminals) is obtained by short circuiting the voltage source and open circuiting the current source as:

$$R_S = 10 + 10 = 20\Omega.$$

The circuit may be replaced in Noton's and Thevenin's equivalent forms as shown in figure (1.46).



Fig. 1.46

The value of R_L for maximum power transfer should equal to source resistance R_S. So R_L=20 Ω and maximum power $P_{\max} = \frac{V_0^2}{4R_L} = \frac{(60)^2}{4x20} = 45Watt$

1.4.7 Star – Delta Conversion: Some times network analysis becomes simple if T – network (ref. figure 1.47 a) is converted to $\pi -$ network (ref. figure 1.47 b) and vice - versa. T – Network may be redrawn as a star network

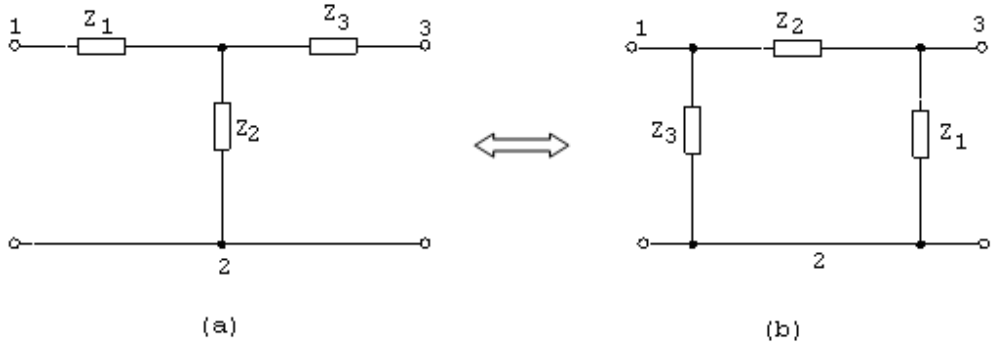
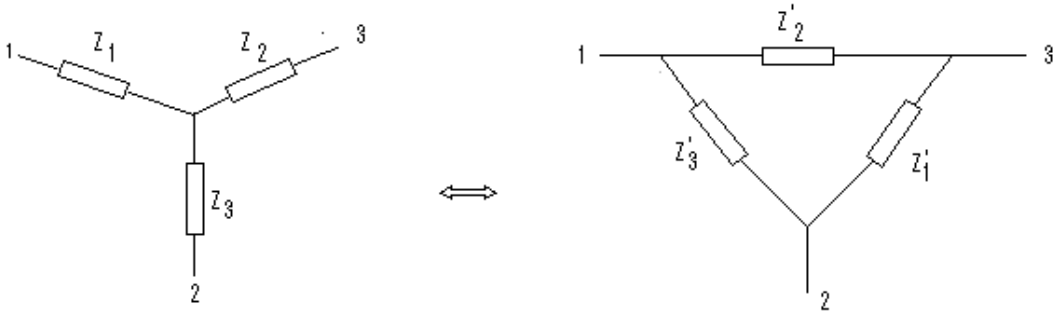


Fig. 1.47

or Y – network and $\pi -$ network may be redrawn as mesh or delta (i.e. Δ) network as shown in figure (1.48). So this conversion is known as star to delta and vice versa (or T to π and vice versa).



(a) (b)

Fig. 1.48

(i) Delta to Star conversion: Consider a delta network and a star network shown in figure (1.48). The two networks will said to be equal if the impedance offered between any two points of one network is equal to impedance offered between the corresponding two points of the other network.

Impedance offered between 1 & 2 terminals of star network (fig. 1.48a) is given by: $Z_{12} = Z_1 + Z_2$

Impedance offered between 1 & 3 terminals of star network (fig. 1.48 a) is given by:

$$Z_{13} = Z_1 + Z_3$$

Impedance offered between 2 & 3 terminals of star network (fig. 1.48a) is given by: $Z_{23} = Z_2 + Z_3$

Similarly, Impedance offered between 1 & 2 terminals of delta network (fig. 1.48 b) is given by: $Z_{12}^{'} = Z_3^{'} \| (Z_1^{'} + Z_2^{'}) = \dfrac{(Z_1^{'} + Z_2^{'})Z_3^{'}}{Z_1^{'} + Z_2^{'} + Z_3^{'}}$

Impedance offered between 1 & 3 terminals of delta network (fig. 1.48 b) is given by: $Z_{13}^{'} = Z_2^{'} \| (Z_1^{'} + Z_3^{'}) = \dfrac{(Z_1^{'} + Z_3^{'})Z_2^{'}}{Z_1^{'} + Z_2^{'} + Z_3^{'}}$

Impedance offered between 2 & 3 terminals of delta network (fig. 1.48 b) is given by: $Z_{23}^{'} = Z_1^{'} \| (Z_2^{'} + Z_3^{'}) = \dfrac{(Z_2^{'} + Z_3^{'})Z_1^{'}}{Z_1^{'} + Z_2^{'} + Z_3^{'}}$

The two networks will be equal if the impedance between two points of one network is equal to impedance between the corresponding two points.

So $$Z_1 + Z_2 = \frac{(Z_1^{'} + Z_2^{'})Z_3^{'}}{Z_1^{'} + Z_2^{'} + Z_3^{'}} \qquad \text{------ (1.46)}$$

$$Z_1 + Z_3 = \frac{(Z_1^{'} + Z_3^{'})Z_2^{'}}{Z_1^{'} + Z_2^{'} + Z_3^{'}} \qquad \text{------ (1.47)}$$

$$Z_2 + Z_3 = \frac{(Z_2^{'} + Z_3^{'})Z_1^{'}}{Z_1^{'} + Z_2^{'} + Z_3^{'}} \qquad \text{----- (1.48)}$$

Adding equations (1.46) & (1.47) and subtracting (1.48) from it we get:

$$2Z_1 = \frac{1}{(Z_1^{'} + Z_2^{'} + Z_3^{'})}\left[Z_1^{'}Z_3^{'} + Z_2^{'}Z_3^{'} + Z_1^{'}Z_2^{'} + Z_2^{'}Z_3^{'} - Z_1^{'}Z_2^{'} - Z_1^{'}Z_3^{'}\right]$$

or $$Z_1 = \frac{1}{(Z_1^{'} + Z_2^{'} + Z_3^{'})}\left[Z_2^{'}Z_3^{'}\right] \qquad \text{------ (1.49)}$$

Similarly, adding equations (1.46) & (1.48) and subtracting (1.47) from it we get: $$Z_2 = \frac{1}{(Z_1^{'} + Z_2^{'} + Z_3^{'})}\left[Z_1^{'}Z_3^{'}\right] \qquad \text{------ (1.50)}$$

Also, adding equations (1.47) & (1.48) and subtracting (1.46) from it, we get:

$$Z_3 = \frac{1}{(Z_1^{'} + Z_2^{'} + Z_3^{'})}\left[Z_1^{'}Z_2^{'}\right] \qquad \text{------ (1.51)}$$

These three equations give the values of impedances of star network in terms of the impedances of delta network. The converted star network of the given delta network may be shown by the dotted lines in figure (1.49).

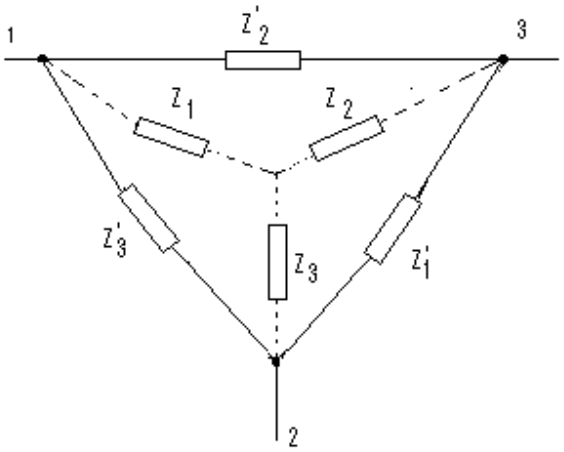


Fig. 1.49

From this figure it is clear that the arms of the star network are obtained by multiplying the impedances of adjacent arms of delta network divided by the sum of all the impedances connected in the delta network.

(ii) Star to Delta Conversion: From the equations (1.49) to (1.51) obtained above, we may get impedances of the Delta network in terms of impedances of star network. This can be done by multiplying the three equations as:

$$Z_1Z_2 + Z_1Z_3 + Z_2Z_3 = \frac{1}{(Z_1' + Z_2' + Z_3')^2}\left[Z_1'Z_2'(Z_3')^2 + Z_1'Z_3'(Z_2')^2 + Z_2'Z_3'(Z_1')^2\right]$$

or
$$= \frac{Z_1'Z_2'Z_3'}{(Z_1' + Z_2' + Z_3')} \qquad \text{------ (1.52)}$$

From equations (1.52) & (1.49) we get :

$$Z_1' = \frac{Z_1Z_2 + Z_1Z_2 + Z_2Z_3}{Z_1}$$

Similarly from equations (1.52) & (1.50) we get:

$$Z_2' = \frac{Z_1Z_2 + Z_1Z_2 + Z_2Z_3}{Z_2}$$

Also from equations (1.52) & (1.51) we get:

$$Z_3' = \frac{Z_1Z_2 + Z_1Z_2 + Z_2Z_3}{Z_3}$$

These three equations give the values of impedances of delta network in terms of the impedances of star network. The converted delta network of the given star network may be shown by the dotted lines in figure (1.50).

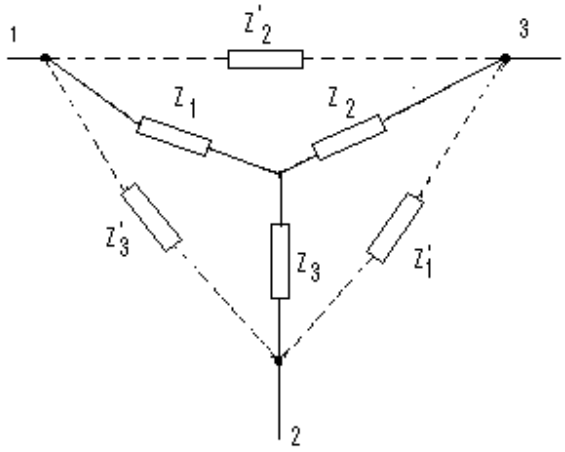


Fig. 1.50

From this figure it is clear that the arms of the delta network are obtained by getting the factor $\sum Z_1 Z_2$ of the star network divided by the impedance of the opposite arm in the star network.

Problems:

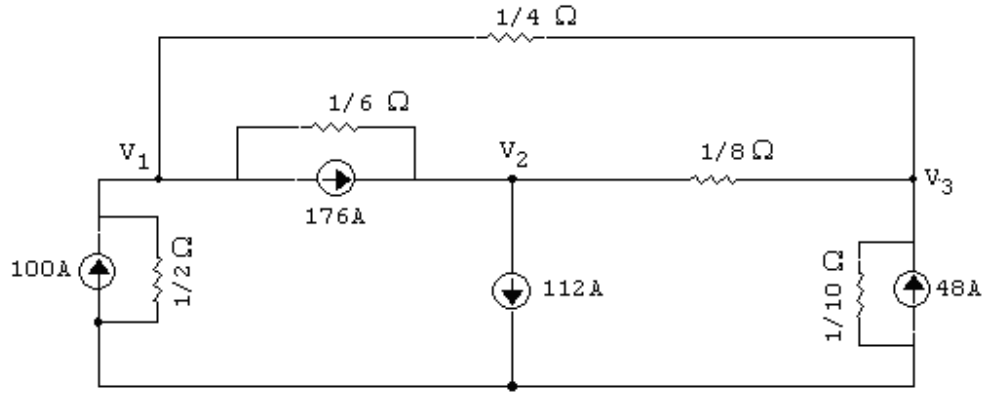
1. State and Explain Kirchoff's laws.
2. Discuss the model for the battery, and show that it is equal to a voltage source and a resistance in series with it. Also explain the terms open circuit voltage and short circuit current.
3. What are ideal voltage source and ideal current source? Prove that for good voltage source the source resistance should be small enough than the load resistance, where as for good current source the source resistance should be larger than load resistance.
4. Discuss in detail, Node method of network analysis by taking a suitable network.
5. Discuss in detail, Loop method of network analysis by taking a suitable network.
6. State and prove Superposition Theorem.
7. State and prove Thevenin's Theorem.
8. State and prove Norton's Theorem.
9. State and prove Reciprocity Theorem.
10. State and prove Millman's Theorem.
11. Discuss the condition for maximum power transfer from an a.c. source to load impedance.
12. Define and compare Thevenin's & Norton's Theorem.
13. Show that a d.c. source having a source resistance connected to load resistance delivers maximum power to the load resistance when source resistance is equal load resistance. Also find the expression for maximum power.
14. Show that the maximum power will be delivered from an a.c. source to the load impedance when load impedance is equal to the complex conjugate of source impedance.

15. How a delta network is converted to star network and vice versa?
16. Explain $\pi - T$ transformation of network.
17. Explain $T - \pi$ transformation of network.
18. A battery has an open circuit voltage of 12 volts and its source resistance as 3Ω. Represent the battery by means of two equivalent circuit elements. Show that these two equivalent circuits draw same amount of current in the load resistance of 9Ω connected to the terminals of the battery.
19. Two equal resistances (each of 1MΩ) in series are connected to the terminals of 75volts source. A multimeter having a sensitivity of 20KΩ/volts is used to measure the voltage across one of the series resistance of 1MΩ. The range of the voltmeter used is 50volts. What will be the reading of the voltmeter?

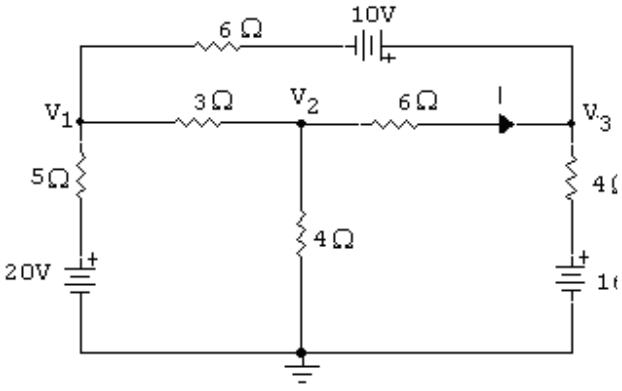
(Ans. 25volts)

20. Solve for the node voltages of the circuit given below.

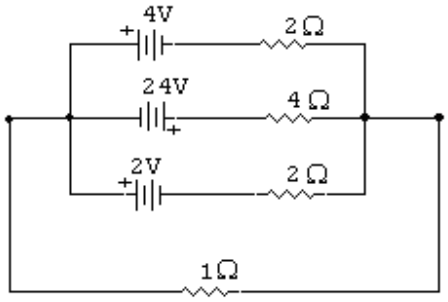
(Ans. $V_1 = -2V, V_2 = 6V, V_3 = 4V$)



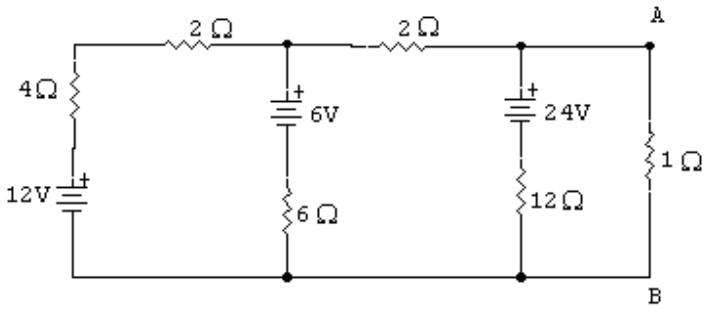
21. Using the node method find the value of current I flowing through 6Ω resistance in the circuit given below.



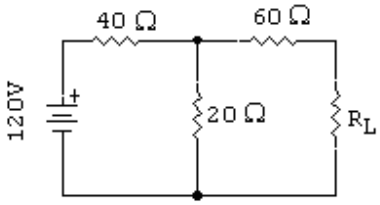
22. Using Millman's theorem, find the value of current flowing through 1Ω resistance in the given circuit. (Ans. 1.33 A)



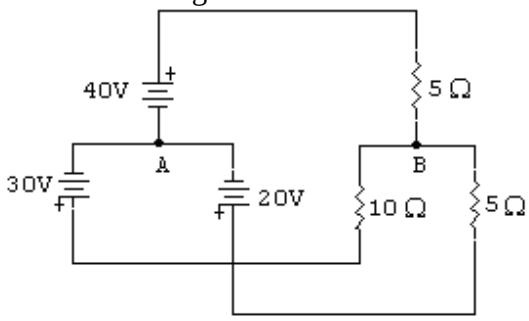
23. Using Norton's theorem, compute current through 1Ω resistance in the given circuit. (Ans. 2.8 A)



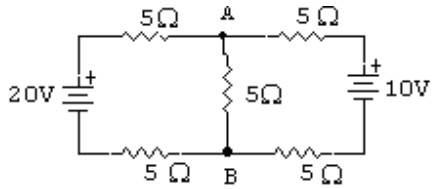
24. In the circuit shown below, find the current flowing through the load resistance R_L of 10Ω. For what value of $R_{L,}$ the power delivered to the load is maximum? Also compute the maximum power. (Ans. 0.48A, 73.3Ω, 5.45watt)



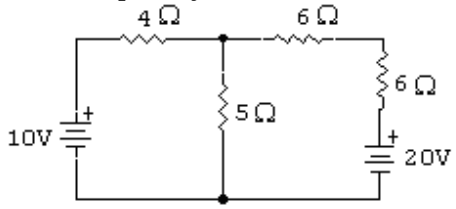
25. Using Millman's theorem find the current through a resistance of 25Ω connected between A & B points in the circuit given below. (Ans. 1A)



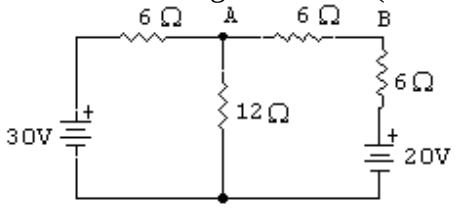
26. Apply Superposition theorem to find the voltage across AB branch in the given circuit. Verify the result using Loop method also. (Ans. 10 volts)



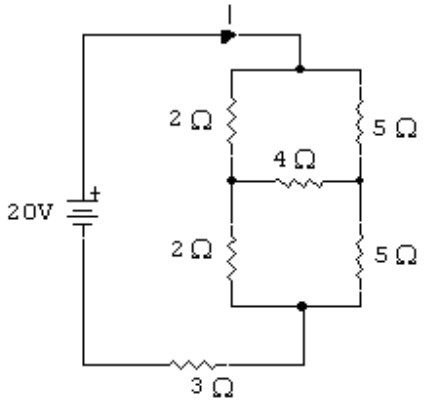
27. Verify the Reciprocity theorem in the circuit shown below.



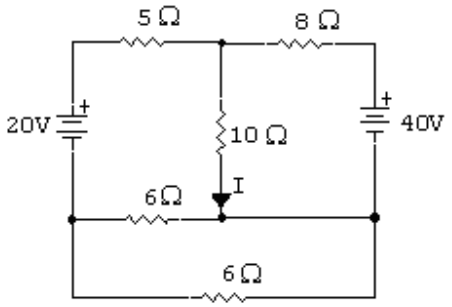
28. Using Superposition theorem, calculate the current through 6Ω resistance in the AB branch in the circuit shown in the figure. (Ans. 7/8A from A to B)



29. Consider the circuit shown below. Determine total impedance of the circuit, current *I* flowing through the circuit, power delivered by the source.
(Ans. 5.86Ω, 3.41A, 68.1Watt)

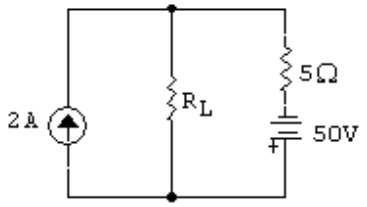


30. Using Superposition theorem find the current I flowing through 10Ω resistance in the given circuit. (Ans. 2.14A)

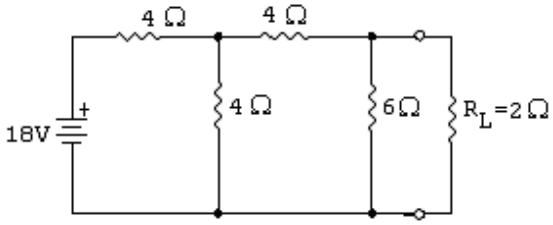


31. Find the value of R_L for which power delivered to it, is maximum as in the figure given below. Determine the maximum power.

(Ans. R_L=5Ω, P_{max}=80Watt)

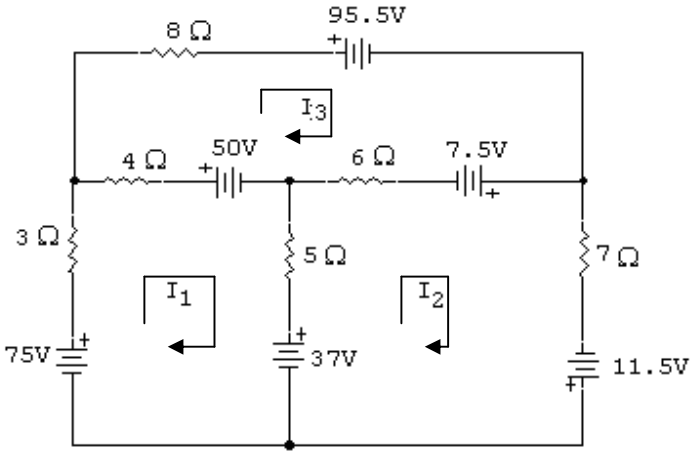


32. Find the current flowing through R_L in the network given below, using Thevenin's theorem. (Ans. 9/5A)



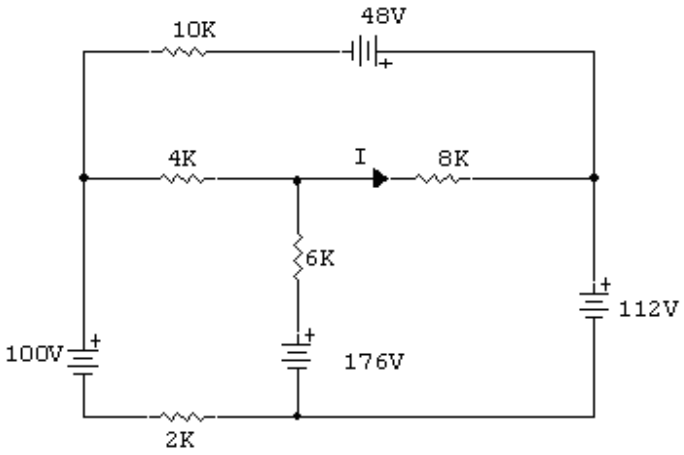
33. Find the loop currents in the circuit given below.

(Ans. $I_1 = -1A, I_2 = 2A, I_3 = -5A$)

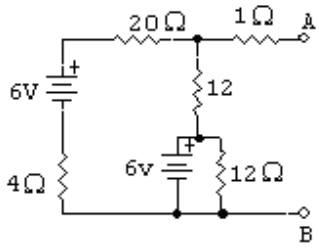


34. In problem 33, if the values of the voltage sources are doubled than show that the loop currents are also doubled.

35. Find the current I flowing through 8KΩ resistance in the circuit shown in the figure. Use loop method to solve the problem. (Ans. 2mA)

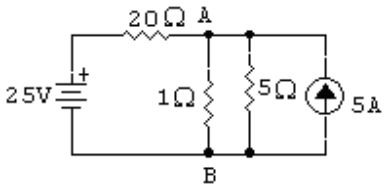


36. Obtain Thevenin's equivalent of the network shown in the figure.

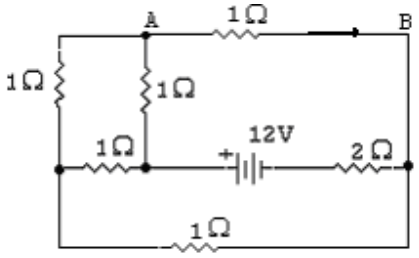


(Ans. $V_0 = 6V$, $R_0 = 9\Omega$)

37. Define Norton's theorem and calculate the current flowing through 1Ω resistance connected between AB terminals of the circuit shown below. (Ans. 5A)



38. Find the current I in AB branch of the circuit shown below.



(Ans. 2A)

2

Two – Port Network

A network contains active and passive elements connected in the form of a circuit. Usually, a network has one pair of terminals for Input and other pair for the output. A pair of Input terminals of the network is called as Input port and the pair of the output terminals is called as the output port. Such a network is called as two port network. If the elements in the network are linear, the network is known as linear two port network. To understand the characteristics or to analyse a linear two port network, consider a black box as shown in figure 2.1. The *1*, *1* terminals of the black box is known as input port and *2, 2* terminals is known as output port.

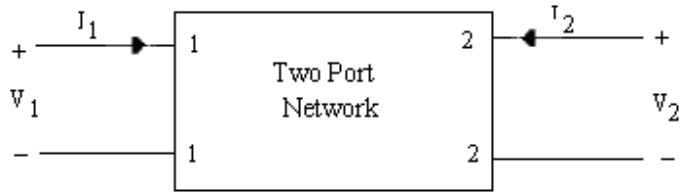


Fig. 2.1

In this network V_1, I_1 are the Input voltage and current; and V_2, I_2 are the output voltage and current. Any pair of variables may be arbitrarily chosen as independent variables, and other variables (dependent variables) may be obtained as a function of independent variables that is dependents variables may assumed to be the functions of independent variables.

2.1 Impedance Parameters: A linear two port network represented by black box is considered, having I_1 and I_2 as independent variables and V_1 and V_2 as dependent variables.

$$V_1 = f_1(I_1, I_2)$$

$$V_2 = f_2(I_1, I_2) \qquad \text{------} \quad (2.1)$$

The changes in the dependent variables may be given by:

$$dV_1 = \frac{\partial V_1}{\partial I_1} dI_1 + \frac{\partial V_1}{\partial I_2} dI_2$$

$$dV_2 = \frac{\partial V_2}{\partial I_1} dI_1 + \frac{\partial V_2}{\partial I_2} dI_2 \text{ ------ (2.2)}$$

The partial derivatives in these equations become constant with operation over linear region of the device curve with constant slope.

The equations may, therefore, be written as:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \text{ ------ (2.3)}$$

In the matrix form it is given by:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \text{ ------ (2.4)}$$

Where Z's are the impedance (resistance for d. c.) parameters, which may be defined as:

$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2 = 0}$, is the input impedance when output is open circuited or open- circuit input impedance.

$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1 = 0}$, is the reverse transfer impedance when input is open circuited or open circuit reverse transfer impedance.

$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2 = 0}$, is the forward transfer impedance when output is open circuited or open circuit forward transfer impedance.

and $Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1 = 0}$, is the output impedance when input is open circuited or open circuit output impedance.

These *Z* parameters also known as open circuit parameters, since in these parameters either input or output is open circuited. The equivalent circuit of the network using *Z*- parameters may be drawn as given below:

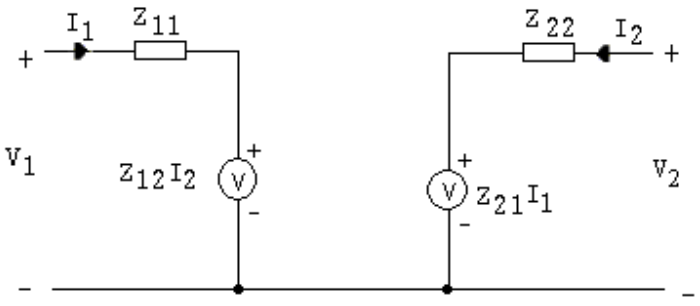


Fig. 2.2

2.2 Admittance Parameters: The admittance parameters of a linear two port network may also be defined in the similar fashion as the impedance parameters discussed above. In the admittance parameters, variables V_1 & V_2 are assumed independent variables and I_1 & I_2 as the dependent variables. The dependent variables I_1 & I_2 may be defined as a linear function of Independent variables V_1 & V_2 as

$$I_1 = f_1(V_1, V_2)$$
$$I_2 = f_2(V_1, V_2) \qquad \text{-------} \quad (2.5)$$

and

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \qquad \text{-------} \quad (2.6)$$

In the matrix form it is given by :

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \qquad \text{------} \quad (2.7)$$

Where Y's are the admittance (conductance for d. c.) parameters, which may be defined as :

$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2 = 0}$$, is the input admittance when output is short circuited or short- circuit input admittance.

$$Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1 = 0}$$, is the reverse transfer admittance when input is short circuited or short circuit reverse transfer admittance.

$$Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2 = 0}$$, is the forward transfer admittance when output is short circuited or short circuit forward transfer admittance.

and $Y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1 = 0}$, is the output admittance when input is short circuited or short circuit output admittance.

These *Y* – parameters are also called as short circuit parameters as given in the network either input or output is shorted. The equivalent circuit of the network using *Y*- parameters is given in figure2.3.

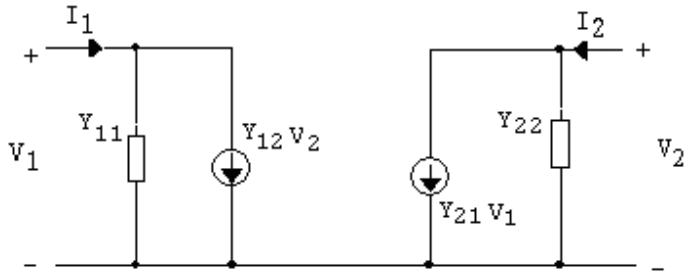


Fig. 2.3

2.3 Hybrid Parameters: Hybrid parameters may be defined by using I_1 & V_2 as Independent variables and V_1 & I_2 as dependent variables. The equations are given:

$$V_1 = H_{11} I_1 + H_{12} V_2$$
$$I_2 = H_{21} I_1 + H_{22} V_2 \qquad \text{--------} \quad (2.8)$$
$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} \qquad \text{-------} \quad (2.9)$$

where $H_{11} = \left.\frac{V_1}{I_1}\right|_{V_2 = 0}$, is known as input impedance when output is shorted, or short circuit input impedance.

$H_{12} = \left.\frac{V_1}{V_2}\right|_{I_1 = 0}$, is known as reverse transfer voltage ratio when input is open circuited or open circuit reverse transfer voltage ratio.

$H_{21} = \left.\frac{I_2}{I_1}\right|_{V_2 = 0}$, is the forward transfer current ratio when input is open circuited or open circuit forward transfer current ratio.

$H_{22} = \left.\frac{I_2}{V_2}\right|_{I_1 = 0}$, is the output admittance when input is open circuited or open circuit output admittance.

The *H*- parameters are known as hybrid parameters as these parameters have the mixed dimensions. The equivalent circuit of the network is given in figure 2.4.

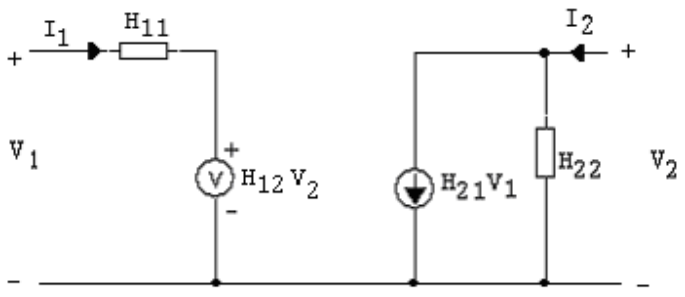


Fig. 2.4

2.4 Inverse hybrid or H' Parameters: In these parameters, V_1 & I_2 are assumed as independent variables and I_1 , V_2 as dependent variables. The dependent equations may be given by:

$$I_1 = H'_{11} V_1 + H'_{12} I_2$$
$$V_2 = H'_{21} V_1 + H'_{22} I_2 \qquad \text{------ (2.10)}$$

$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} H'_{11} & H'_{12} \\ H'_{21} & H'_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} \qquad \text{------ (2.11)}$$

where $H'_{11} = \left.\frac{I_1}{V_1}\right|_{I_2 = 0}$, is the open circuit input admittance.

$H'_{12} = \left.\frac{I_1}{I_2}\right|_{V_1 = 0}$, is the short circuit reverse transfer current ratio.

$H'_{21} = \left.\frac{V_2}{V_1}\right|_{I_2 = 0}$, is the open circuit forward transfer voltage ratio.

$H'_{22} = \left.\frac{V_2}{I_2}\right|_{V_1 = 0}$, is the short circuit output impedance.

The equivalent circuit of the network using H' - parameters are given in figure 2.5.

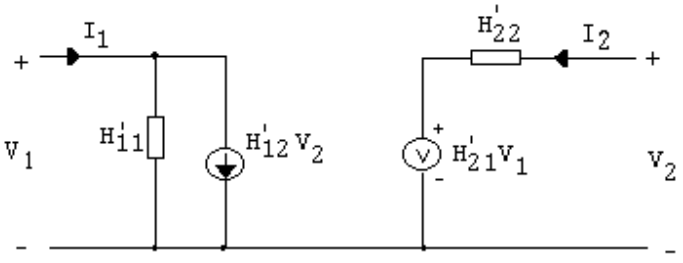


Fig. (2.5)

2.5 Transmission parameters: The Transmission Parameters are obtained by considering the variables of Input port as dependent variables and variables of output port as Independent variables, as given below:

$$V_1 = AV_2 - BI_2$$
$$I_1 = CV_2 - DI_2 \qquad \text{------ (2.12)}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \qquad \text{------ (2.13)}$$

It is customary to choose $-I_2$ in place of I_2 as these parameters are used to find the overall parameters of a cascaded two port network. Transmission parameters are also called *T* or *A, B, C, D,* parameters.

where $A = \left.\frac{V_1}{V_2}\right|_{I_2 = 0}$, is open circuit reverse transfer voltage ratio.

$B = \left.\frac{V_1}{-I_2}\right|_{V_2 = 0}$, is short circuit reverse transfer impedance.

$C = \left.\frac{I_1}{V_2}\right|_{I_2 = 0}$, is open circuit reverse transfer admittance.

$D = \left.\frac{I_1}{-I_2}\right|_{V_2 = 0}$, is short circuit reverse transfer current ratio.

2.6 Inverse Transmission or T'-Parameters: Inverse Transmission or T'-parameters may be defined by using V_1 & $-I_1$ as Independent variables and V_2 & I_2 as dependent variables. The equations are given as:

$$V_2 = A'V_1 - B'I_1$$
$$I_2 = C'V_1 - D'I_1 \qquad \text{------- (2.14)}$$

$$\begin{bmatrix} V_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} V_1 \\ -I_1 \end{bmatrix} \qquad \text{------ (2.15)}$$

where $A' = \left.\frac{V_2}{V_1}\right|_{I_1 = 0}$, is open circuit forward transfer voltage ratio.

$B' = \left.\frac{V_2}{-I_1}\right|_{V_1 = 0}$, is short circuit forward transfer impedance.

$$C' = \left.\frac{I_2}{V_1}\right|_{I_1 = 0}$$, is open circuit forward transfer admittance.

$$D' = \left.\frac{I_2}{-I_1}\right|_{V_1 = 0}$$, is short circuit forward transfer current ratio.

Example 2.1 Find *Z* and *H* Parameters of the Passive *T*-Network given in figure (2.6).

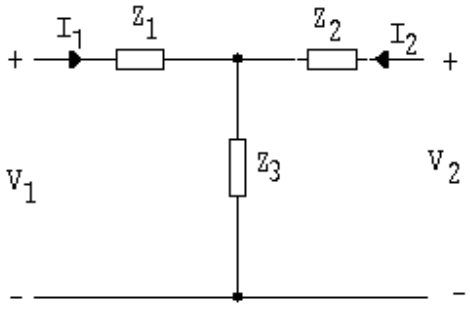


Fig. (2.6)

Solution: 1. Z - Parameters

(i) $Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ Since I_2=0 (output is open circuited),

so $V_1 = I_1(Z_1 + Z_3)$

or $Z_{11} = \frac{V_1}{I_1} = Z_1 + Z_3$

(ii) $Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$ Since I_1=0, the voltage across Z_3 will be equal to V_1

So $V_1 = Z_3 I_2$

or $Z_{12} = \frac{V_1}{I_2} = Z_3$

(iii) $Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$ To find Z_{21}, output port is open circuited (I_2 = 0), the voltage across Z_3 will be equal to V_2 which is given by

$V2 = Z_3 I_1$

or $Z_{21} = \frac{V_2}{I_1} = Z_3$

(iv) $Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1 = 0}$ Z_{22} is obtained by open circuiting the input port .

So $V_2 = I_2 (Z_2 + Z_3)$

$$Z_{22} = \frac{V_2}{I_2} = Z_2 + Z_3$$

2. *H*-Parameters:

(i) $H_{11} = \left.\frac{V_1}{I_1}\right|_{V_2 = 0}$ By short circuiting the output port, the network becomes as shown in the fig. (2.7)

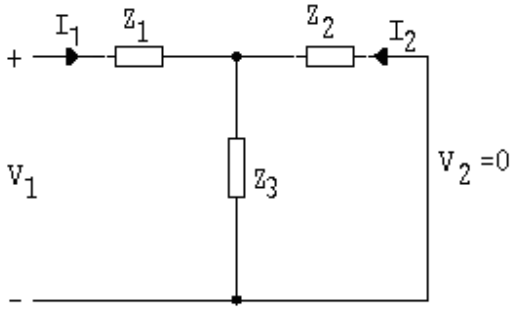


Fig. 2.7

So $$V_1 = I_1 \left[Z_1 + Z_2 \| Z_3 \right]$$

$$H_{11} = \frac{V_1}{I_1} = \frac{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3}{Z_2 + Z_3}$$

(ii) $H_{12} = \left.\frac{V_1}{V_2}\right|_{I_1 = 0}$ Since $I_1 = 0$, so the voltage across Z_3 will be equal to voltage at the input port.

The equations are :

$V1 = Z_3 I_2$ and $V_2 = I_2 (Z_2 + Z_3)$

So $$H_{12} = \frac{V_1}{V_2} = \frac{Z_3}{Z_2 + Z_3}$$

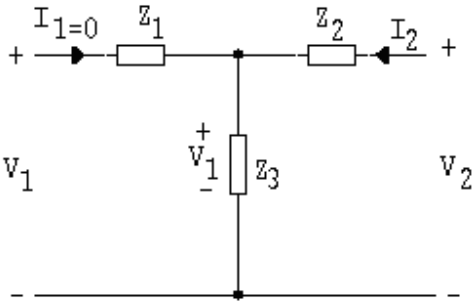


Fig. (2.8)

(iii) $H_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}$ The corresponding diagram is shown below:

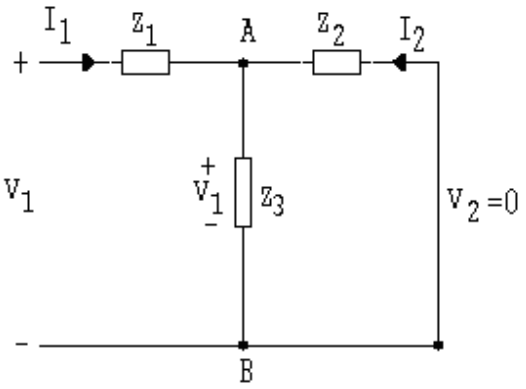


Fig.2.9

Voltage across *AB* points is $V_{AB} = I_1[Z_2 \| Z_3] = \frac{Z_2 Z_3}{Z_2 + Z_3} I_1$

$$I_2 = -\frac{V_{AB}}{Z_2} = -\frac{Z_3 I_1}{(Z_2 + Z_3)}$$

or $$H_{21} = \frac{I_2}{I_1} = -\frac{Z_3}{(Z_2 + Z_3)}$$

(iv) $H_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$ Since I_1=0 (Ref. Fig.2.8)

So $V_2 = I_2 (Z_2+Z_3)$

or $$H_{22} = \frac{I_2}{V_2} = \frac{1}{Z_2 + Z_3}$$

Example. 2.2 Find H' parameters of the given Π - network.

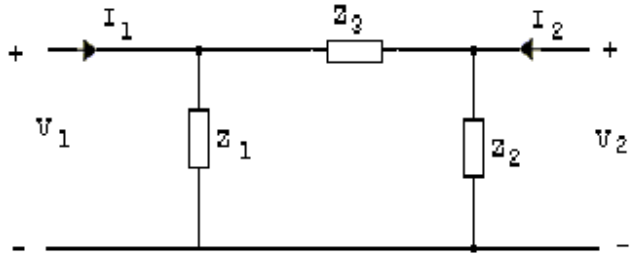


Fig. 2.10

Solution: (i) $H'_{11} = \left.\frac{I_1}{V_1}\right|_{I_2=0}$

The relation between V_1 and I_1 when I_2 = 0, is given by:

$$V_1 = I_1\left[Z_1 \| (Z_2 + Z_3)\right]$$

or $$V_1 = I_1\left[\frac{Z_1(Z_2 + Z_3)}{Z_1 + Z_2 + Z_3}\right]$$

or $$H'_{11} = \frac{I_1}{V_1} = \frac{Z_1 + Z_2 + Z_3}{Z_1Z_2 + Z_1Z_3}$$

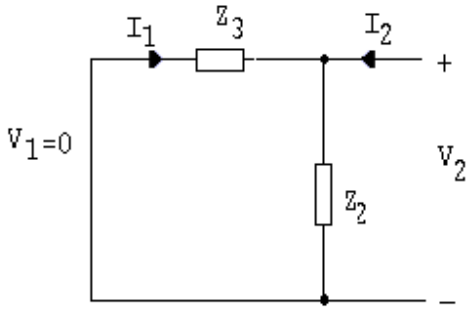


Fig. 2.11

(ii) $$H'_{12} = \left.\frac{I_1}{I_2}\right|_{V_2=0}$$

$$V_2 = I_2\left(\frac{Z_2Z_3}{Z_2 + Z_3}\right) \quad \text{and} \quad I_1 = -\frac{V_2}{Z_3}$$

Combining these two equations, we get

$$I_1 = -\frac{Z_2}{(Z_2 + Z_3)}I_2 \quad \text{or} \quad H'_{12} = \frac{I_1}{I_2} = -\frac{Z_2}{Z_2 + Z_3}$$

(iii) $$H'_{21} = \left.\frac{V_2}{V_1}\right|_{I_2=0}$$ When $I_2 = 0$, the relation between V_1 and V_2 is given by:

$$V_2 = \frac{V_1}{(Z_2 + Z_3)}Z_2 \quad \text{or} \quad H'_{21} = \frac{V_2}{V_1} = \frac{Z_2}{(Z_2 + Z_3)}$$

(iv) $$H'_{22} = \left.\frac{V_2}{I_2}\right|_{V_1=0}$$ The relation between V_2 & I_2 when $V_1 = 0$ is given by (Ref. Fig. 2.11).

$$V_2 = I_2\frac{Z_2Z_3}{Z_2 + Z_3} \quad \text{or} \quad H'_{22} = \frac{V_2}{I_2} = \frac{Z_2Z_3}{Z_2 + Z_3}$$

Example 2.3 Find the transmission Parameters of the network given in figure 2.12. The network is excited by a sinusoidal signal of 10^4 radians /sec.

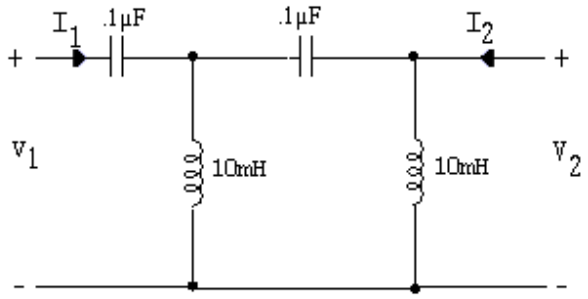


Fig 2.12

Solution: Since the network is excited by a sinusoidal signal of frequency 10^4 radians / sec, the inductive reactance of 10 *mH* inductance is

$$\omega L = 10^4 x 10 x 10^3 = 100\ \Omega$$

and capacitive reactance of .1μF capacitance is

$$\frac{1}{\omega C} = \frac{1}{10^4 x 1 x 10^{-6}} = 100\ \Omega$$

So the network is replaced by the Impedance of 100 Ω each as given in the figure 2.13

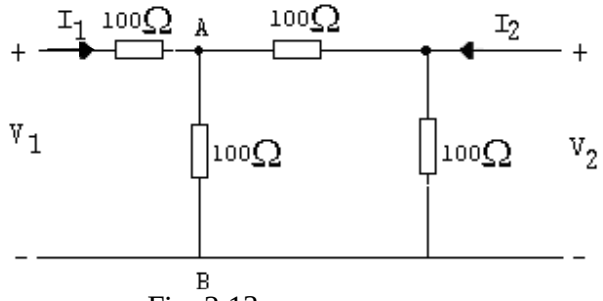


Fig. 2.13

The transmission parameters of this network may be obtained as follows:

(i) $A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$ From the Fig.(2.13) voltage across *AB* terminals is given

as $$V_{AB} = I_1 (100 \parallel 200) = I_1 (\frac{200}{3})$$

and $$V2 = \frac{V_{AB}\, x100}{(100 + 100)} = \frac{V_{AB}}{2}$$

or $$V_2 = I_1 (\frac{100}{3}) \qquad \ldots\ldots\ldots (2.16)$$

also from fig.2.13 $$V_1 = I_1 [100 + 100 \parallel 200] = I_1 \left[100 + \frac{200}{3}\right]$$

or $$V_1 = I_1 (\frac{500}{3}) \qquad \ldots\ldots (2.17)$$

From equations (2.16) & (2.17) we get $V_1 = 5V_2$

or $A = \frac{V_2}{V_2} = 5$

(ii) $B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$ By short circuiting the output terminals we have the network as:

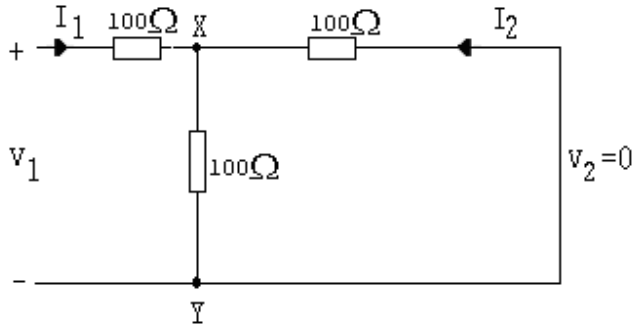


Fig. 2.14

The output current I_2 is given by: $-I_2 = \frac{V_{XY}}{100}$ and

$V_{XY} = I_1(100 \| 100) = I_1(50)$ or $-I_2 = \frac{I_1}{2}$

However $V_1 = I_1[100 + 50] = 150\, I_1$

$V_1 = -150 \times 2\, I_2 = -300\, I_2$

$B = \frac{V_1}{-I_2} = 300\ \Omega$

(iii) $C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$ The relation between I_1 & V_2 when $I_2 = 0$ is given in the calculation of A parameter as $V_2 = I_1 \frac{100}{3}$

so $C = \frac{I_1}{V_2} = \frac{3}{100}$ mhos

(iv) $D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$ The relation between I_1 & I_2 for $V_2 = 0$ is given in the calculation of B parameters of the network shown above as:

$I_2 = -\frac{I_1}{2}$ or $D = \frac{I_1}{-I_2} = 2$

2.7 Transformation of Parameters: For a given linear two port network sometimes one type of parameters is calculated, by considering the suitable independent as well as dependent variables. But for many reasons, other type of parameters also required; which may be calculated

by using the transformation of parameters. The transformation of parameters is just a mathematical transformation, which may be understood by considering an example.

Suppose the Z-parameters of a linear two port network are given and these are to be transformed into its equivalent Y-parameters.

The Z parameters of a linear two port network are given by the equations:

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \qquad \text{------- (2.18)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \qquad \text{-------- (2.19)}$$

The required equations for Y-parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \qquad \text{------ (2.20)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \qquad \text{------ (2.21)}$$

The equations (2.18) and (2.19) may be rewritten as

$$I_1 = \frac{1}{Z_{11}} V_1 - \frac{Z_{12}}{Z_{11}} I_2 \qquad \text{------ (2.22)}$$

$$I_2 = \frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1 \qquad \text{------ (2.23)}$$

From equations (2.22) and (2.23) we get

$$I_1 = \frac{1}{Z_{11}} V_1 - \frac{Z_{12}}{Z_{11}} \left(\frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1 \right)$$

$$I_1 \left(\frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{11} Z_{22}} \right) = \frac{1}{Z_{11}} V_1 - \frac{Z_{12}}{Z_{22} Z_{22}} V_2$$

or $$I_1 = \frac{Z_{22}}{(Z_{11} Z_{22} - Z_{12} Z_{21})} V_1 - \frac{Z_{12}}{Z_{11} Z_{22} - Z_{12} Z_{21}} V_2 \qquad \text{------ (2.24)}$$

$$I_2 = \frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} \left(\frac{1}{Z_{11}} V_1 - \frac{Z_{12}}{Z_{11}} I_2 \right)$$

$$I_2 \left(\frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{11} Z_{22}} \right) = \frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{11} Z_{22}} V_1$$

or $$I_2 = -\frac{Z_{21}}{(Z_{11} Z_{22} - Z_{12} Z_{21})} V_1 + \frac{Z_{11}}{Z_{11} Z_{22} - Z_{12} Z_{21}} V_2 \qquad \text{------ (2.25)}$$

Comparing the coefficients of V_1 and V_2 in equations (2.24) and (2.25) with (2.20) and (2.21) respectively we get

$$Y_{11} = \frac{Z_{22}}{Z_{11} Z_{22} - Z_{12} Z_{21}} \qquad Y_{12} = \frac{Z_{12}}{Z_{11} Z_{22} - Z_{12} Z_{21}}$$

$$Y_{21} = \frac{-Z_{21}}{Z_{11} Z_{22} - Z_{12} Z_{21}} \qquad Y_{22} = \frac{Z_{11}}{Z_{11} Z_{22} - Z_{12} Z_{21}} \qquad \text{------ (2.26)}$$

These are the required parameters in terms of the given Z-parameters.

The transformation of different parameters is shown in the form of a Table given below.

Table 2.1 Transformation of parameters:

	Z	Y	H	$H^{'}$	T	$T^{'}$
$[Z]$	Z_{11} Z_{12} Z_{21} Z_{22}	$\frac{Y_{22}}{\Delta Y}$ $\frac{-Y_{12}}{\Delta Y}$ $\frac{-Y_{21}}{\Delta Y}$ $\frac{Y_{11}}{\Delta Y}$	$\frac{\Delta H}{H_{22}}$ $\frac{H_{12}}{H_{22}}$ $\frac{-H_{21}}{H_{22}}$ $\frac{1}{H_{22}}$	$\frac{1}{H_{11}^{'}}$ $\frac{-H_{12}^{'}}{H_{11}^{'}}$ $\frac{H_{21}^{'}}{H_{11}^{'}}$ $\frac{\Delta H^{'}}{H_{11}^{'}}$	$\frac{A}{C}$ $\frac{\Delta T}{C}$ $\frac{1}{C}$ $\frac{D}{C}$	$\frac{D^{'}}{C^{'}}$ $\frac{1}{C^{'}}$ $\frac{\Delta T^{'}}{C^{'}}$ $\frac{A^{'}}{C^{'}}$
$[Y]$	$\frac{Z_{22}}{\Delta Z}$ $\frac{-Z_{12}}{\Delta Z}$ $\frac{-Z_{21}}{\Delta Z}$ $\frac{Z_{11}}{\Delta Z}$	Y_{11} Y_{12} Y_{21} Y_{22}	$\frac{1}{H_{11}}$ $\frac{-H_{12}}{H_{11}}$ $\frac{H_{21}}{H_{11}}$ $\frac{\Delta H}{H_{11}}$	$\frac{\Delta H^{'}}{H_{22}^{'}}$ $\frac{H_{12}^{'}}{H_{22}^{'}}$ $\frac{-H_{21}^{'}}{H_{22}^{'}}$ $\frac{1}{H_{22}^{'}}$	$\frac{D}{B}$ $\frac{-\Delta T}{B}$ $-\frac{1}{B}$ $\frac{A}{B}$	$\frac{A^{'}}{B^{'}}$ $\frac{-1}{B^{'}}$ $\frac{-\Delta T^{'}}{B^{'}}$ $\frac{D^{'}}{B^{'}}$
$[H]$	$\frac{\Delta Z}{Z_{22}}$ $\frac{Z_{12}}{Z_{22}}$ $\frac{-Z_{21}}{Z_{22}}$ $\frac{1}{Z_{22}}$	$\frac{1}{Y_{11}}$ $\frac{-Y_{12}}{Y_{11}}$ $\frac{Y_{21}}{Y_{11}}$ $\frac{\Delta Y}{Y_{11}}$	H_{11} H_{12} H_{21} H_{22}	$\frac{H_{22}^{'}}{\Delta H^{'}}$ $\frac{-H_{12}^{'}}{\Delta H^{'}}$ $\frac{-H_{21}^{'}}{\Delta H^{'}}$ $\frac{H_{11}^{'}}{\Delta H^{'}}$	$\frac{B}{D}$ $\frac{\Delta T}{D}$ $\frac{-1}{D}$ $\frac{C}{D}$	$\frac{B^{'}}{A^{'}}$ $\frac{1}{A^{'}}$ $\frac{-\Delta T^{'}}{A^{'}}$ $\frac{C^{'}}{A^{'}}$
$[H^{'}]$	$\frac{1}{Z_{11}}$ $\frac{-Z_{12}}{Z_{11}}$ $\frac{Z_{21}}{Z_{11}}$ $\frac{\Delta Z}{Z_{11}}$	$\frac{\Delta Y}{Y_{22}}$ $\frac{Y_{12}}{Y_{22}}$ $\frac{-Y_{21}}{Y_{22}}$ $\frac{1}{Y_{22}}$	$\frac{H_{22}}{\Delta H}$ $\frac{-H_{12}}{\Delta H}$ $\frac{-H_{21}}{\Delta H}$ $\frac{H_{11}}{\Delta H}$	$H_{11}^{'}$ $H_{12}^{'}$ $H_{21}^{'}$ $H_{22}^{'}$	$\frac{C}{A}$ $\frac{-\Delta T}{A}$ $\frac{1}{A}$ $\frac{B}{A}$	$\frac{C^{'}}{D^{'}}$ $\frac{-1}{D^{'}}$ $\frac{\Delta T^{'}}{D^{'}}$ $\frac{B^{'}}{D^{'}}$
$[T]$	$\frac{Z_{11}}{Z_{21}}$ $\frac{\Delta Z}{Z_{21}}$ $\frac{1}{Z_{21}}$ $\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$ $\frac{-1}{Y_{21}}$ $\frac{-\Delta Y}{Y_{21}}$ $\frac{-Y_{11}}{Y_{21}}$	$\frac{-\Delta H}{H_{21}}$ $\frac{-H_{11}}{H_{21}}$ $\frac{-H_{22}}{H_{21}}$ $\frac{-1}{H_{21}}$	$\frac{1}{H_{21}^{'}}$ $\frac{H_{22}^{'}}{H_{21}^{'}}$ $\frac{H_{11}^{'}}{H_{21}^{'}}$ $\frac{\Delta H^{'}}{H_{21}^{'}}$	A B C D	$\frac{D^{'}}{\Delta T^{'}}$ $\frac{B^{'}}{\Delta T^{'}}$ $\frac{C^{'}}{\Delta T^{'}}$ $\frac{A^{'}}{\Delta T^{'}}$
$[T^{'}]$	$\frac{Z_{22}}{Z_{12}}$ $\frac{\Delta Z}{Z_{12}}$ $\frac{1}{Z_{12}}$ $\frac{Z_{11}}{Z_{12}}$	$\frac{-Y_{11}}{Y_{12}}$ $\frac{-1}{Y_{12}}$ $\frac{-\Delta Y}{Y_{12}}$ $\frac{-Y_{22}}{Y_{12}}$	$\frac{1}{H_{12}}$ $\frac{H_{11}}{H_{12}}$ $\frac{H_{22}}{H_{12}}$ $\frac{\Delta H}{H_{12}}$	$\frac{-\Delta H^{'}}{H_{12}^{'}}$ $\frac{-H_{22}^{'}}{H_{12}^{'}}$ $\frac{-H_{11}^{'}}{H_{12}^{'}}$ $\frac{-1}{H_{12}^{'}}$	$\frac{D}{\Delta T}$ $\frac{B}{\Delta T}$ $\frac{C}{\Delta T}$ $\frac{A}{\Delta T}$	$A^{'}$ $B^{'}$ $C^{'}$ $D^{'}$

Where

$$\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21} \qquad \Delta Y = Y_{11}Y_{22} - Y_{12}Y_{21}$$

$$\Delta H = H_{11}H_{22} - H_{12}H_{21} \qquad \Delta H^{'} = H_{11}^{'}H_{22}^{'} - H_{12}^{'}H_{21}^{'}$$

$$\Delta T = AD - BC \qquad \Delta T^{'} = A^{'}D^{'} - B^{'}C^{'}$$

Example 2.4 Determine the *Z*-parameters of the symmetric lattice network given in Fig. (2.15) and then transform them to *Y*-parameters.

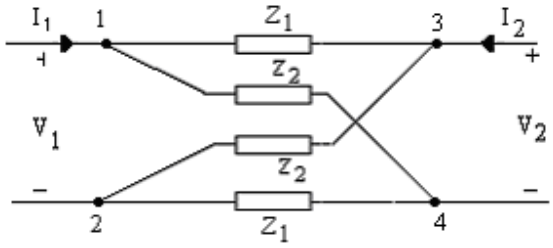


Fig. 2.15

Solution**:**

***Z*-Parameters:** The network shown in figure is symmetric lattice since all the terminals of the network are connected to Z_1 & Z_2 impedances. For simplicity the network may be redrawn as shown in fig. (2.16).

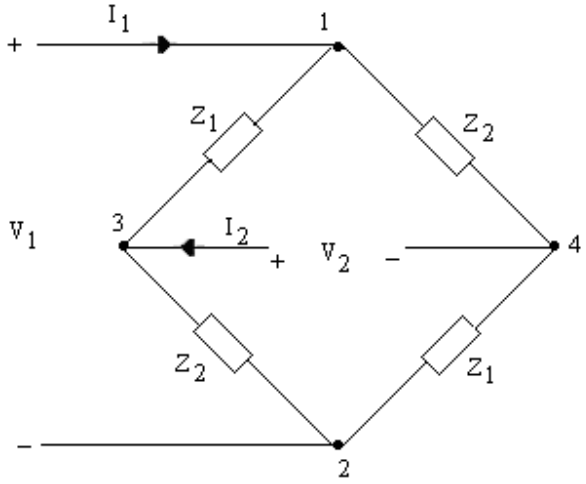


Fig. 2.16

The *Z*-parameters may be calculated as follows:

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$$ Since $I_2 = 0$, the current I_1 is equally distributed in the *1, 3, 2* & *1, 4, 2* branch. i.e. $I_1/2$ current will flow in both the two branches.

So $V_1 = \frac{I_1}{2}(Z_1 + Z_2)$ or $Z_{11} = \frac{V_1}{I_1} = \frac{Z_1 + Z_2}{2}$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1 = 0}$$

Similarly Z_{22} may be calculated by keeping $I_1 = 0$, so the current I_2 will be equally distributed in *3, 1, 4* and *3, 2, 4* branches.

Therefore $V_2 = \frac{I_2}{2}(Z_1 + Z_2)$ or $Z_{22} = \frac{V_2}{I_2} = \frac{Z_1 + Z_2}{2}$

$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$ In this case the distribution of current will be as shown in fig. (2.17).

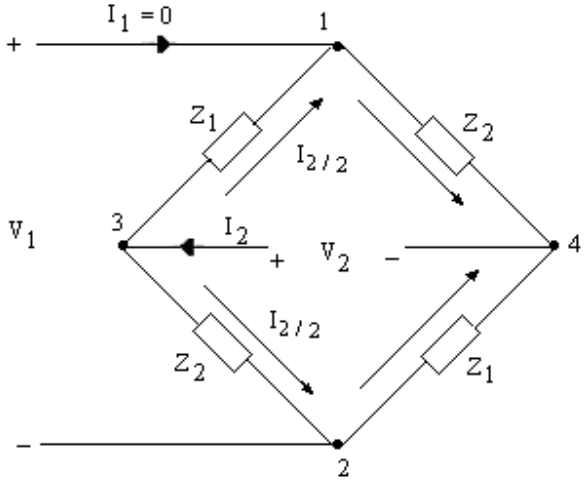


Fig. 2.17

Voltage V_1, which is the algebraic sum of voltages across Z_1 & Z_2, is given by:

$V_1 = -\frac{I_2}{2}Z_1 + \frac{I_2}{2}Z_2$ or $Z_{12} = \frac{V_1}{I_2} = \frac{Z_2 - Z_1}{2}$

$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$ which may in the similar fashion be obtained as:

$V_2 = -\frac{I_1}{2}Z_1 + \frac{I_1}{2}Z_2$ or $Z_{21} = \frac{V_2}{I_1} = \frac{Z_2 - Z_1}{2}$

Y-parameters: We have calculated Z-parameters of the symmetric lattice network, which may be transformed in to Y-parameters. From the table 2.1 *Y* -parameters in terms of *Z*- parameters are given by:

$Y_{11} = \frac{Z_{22}}{\Delta Z}$, $Y_{22} = \frac{Z_{11}}{\Delta Z}$, $Y_{12} = -\frac{Z_{12}}{\Delta Z}$ and $Y_{21} = -\frac{Z_{21}}{\Delta Z}$

Where $\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21}$ putting the values of Z-parameters as calculated above we get $\Delta Z = (\frac{Z_1 + Z_2}{2})^2 - (\frac{Z_2 - Z_1}{2})^2 = \frac{1}{4}\left[Z_1^2 + Z_2^2 + 2Z_1Z_2 - Z_2^2 - Z_1^2 - Z_1Z_2\right]$

$= Z_1Z_2$

So $Y_{11} = \frac{Z_{22}}{\Delta Z} = \frac{(Z_1 + Z_2)/2}{Z_1Z_2} = \frac{Z_1 + Z_2}{2Z_1Z_2} = Y_{22}$ since $Z_{11}=Z_{22}$

$$Y_{12} = -\frac{Z_{12}}{\Delta Z} = -\frac{(Z_2 - Z_1)/2}{Z_1 Z_2} = \frac{Z_1 - Z_2}{2Z_1 Z_2} = Y_{21} \qquad \text{since } Z_{12}=Z_{21}$$

Example 2.5 The Z-parameters of a linear two port network are Z_{11} = 40 Ω, Z_{12} =Z_{21}= 20 Ω and Z_{22} = 30 Ω. Compute the transmission parameters of the network.

Solution: The network equation using *Z*-parameters are given by:

$$V_1 = 40\, I_1 + 20\, I_2 \qquad \text{------ (2.27)}$$

$$V_2 = 20\, I_1 + 30\, I_2 \qquad \text{------ (2.28)}$$

The *T*-parameter equations are given by:

$$V_1 = AV_2 - BI_2 \qquad \text{------ (2.29)}$$

$$I_1 = CV_2 - DI_2 \qquad \text{------ (2.30)}$$

The equations (2.27) & (2.28) may be converted in the form of equations (2.29) & (2.30) respectively.

$$V_1 = \frac{40\,(V_2 - 30\, I_2)}{20} + 20\, I_2$$

or $$V_1 = 2V_2 - 40\, I_2 \qquad \text{------ (2.31)}$$

and $$I_1 = \frac{V_2}{20} - \frac{3}{2} I_2 \qquad \text{------ (2.32)}$$

Comparing eqs. (2.31) & (2.32) with eqs. (2.29) & (2.30) respectively, the required parameters are given as:

$A = 2$ B = 40 Ω C = 1/20 mhos D = 3/2

2.8 Interconnection of Two Port Networks:

Figure (2.18) shows the two networks P_1 and P_2 whose voltage and current variables are given. These two networks may be connected in a number of ways so that the resulting network is also a two port network. The overall parameters of such a two port network may be calculated in different ways.

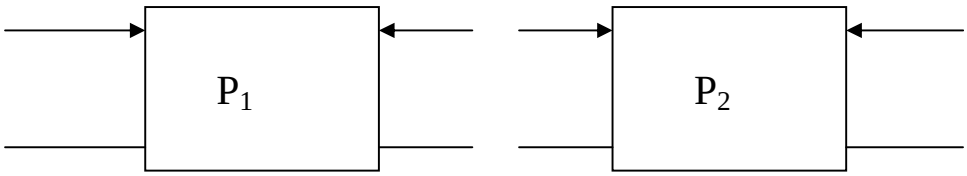


Fig.2.18

(i) Cascade connection of two networks: The two networks are said to be connected in cascaded mode, if the output of one network is connected to the Input of other network as shown in the figure (2.19). The overall parameters of the network may be obtained as follows:

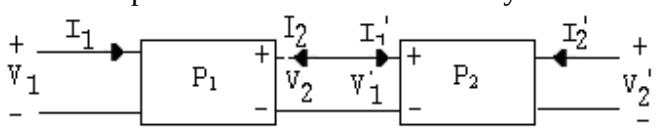


Fig. 2.19

The T-parameter equations of the two networks P_1 and P_2 are given by:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1' \\ I_1' \end{bmatrix} = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix} \qquad \text{------ (2.33)}$$

Since $V_2 = V_1'$ and $I_2 = -I_1'$ so the matrix equation (2.33) may be rewritten as:

$$\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix} \qquad \text{------ (2.34)}$$

From equations (2.33) and (2.34) we get the overall parameters of the cascaded networks:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \qquad \text{------ (2.35)}$$

From this equation it is clear that the overall $[T]$ parameters of the cascaded parameters are obtained by matrix multiplication of the *T*-parameters of the individual networks i.e.

$$[T] = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

(ii) Parallel Connection: - Let us connect the two networks P_1 & P_2 is parallel as shown in the fig. (2.20). The *Y*-parameters of the network P_1 & P_2 are given by the equations:

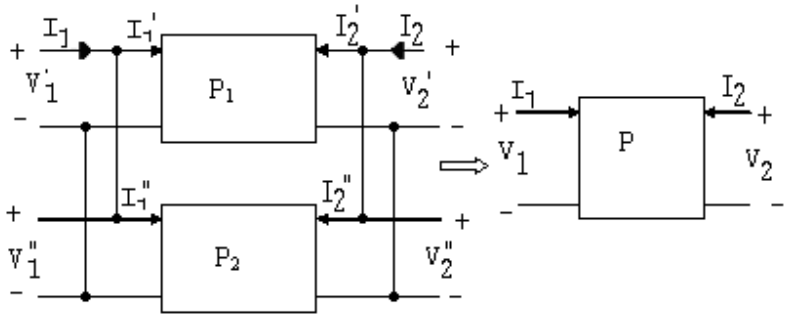


Fig. 2.20

$$I_1' = Y_{11}'V_1' + Y_{12}'V_2' \quad \& \quad I_2' = Y_{21}'V_1' + Y_{22}'V_2' \quad \text{for } P_1 \qquad \text{------ (2.36)}$$

$$I_1'' = Y_{11}''V_1'' + Y_{12}''V_2'' \quad \& \quad I_2'' = Y_{21}''V_1'' + Y_{22}''V_2'' \quad \text{for } P_2 \qquad \text{------ (2.37)}$$

Since inputs of two network are connected in parallel,

so $V_1' = V_1'' = V_1$ (say) $\quad I_1' + I_1'' = I_1$ (say)

also $V_2' = V_2'' = V_2$ (say) $\quad I_2' + I_2'' = I_2$ (say)

Combining equations (2.36) & (2.37) we get:

$$I_1 = I_1' + I_1'' = (Y_{11}' + Y_{11}'')V_1 + (Y_{12}' + Y_{12}'')V_2 \qquad \text{------ (2.38)}$$

$$I_2 = I_2' + I_2'' = (Y_{21}' + Y_{21}'')V_1 + (Y_{22}' + Y_{22}'')V_2 \qquad \text{------ (2.39)}$$

So overall *Y*-parameters of the two networks connected in parallel may be obtained by adding the *Y*-parameters of the individual network i.e.

$$Y_{11} = Y_{11}^{'} + Y_{11}^{''} \qquad Y_{12} = Y_{12}^{'} + Y_{12}^{''}$$
$$Y_{21} = Y_{21}^{'} + Y_{21}^{''} \qquad Y_{22} = Y_{22}^{'} + Y_{22}^{''}$$

(iii) Series connection: In series connection of two port networks, the inputs and outputs of the two different two port networks are connected as shown in fig.(2.21)

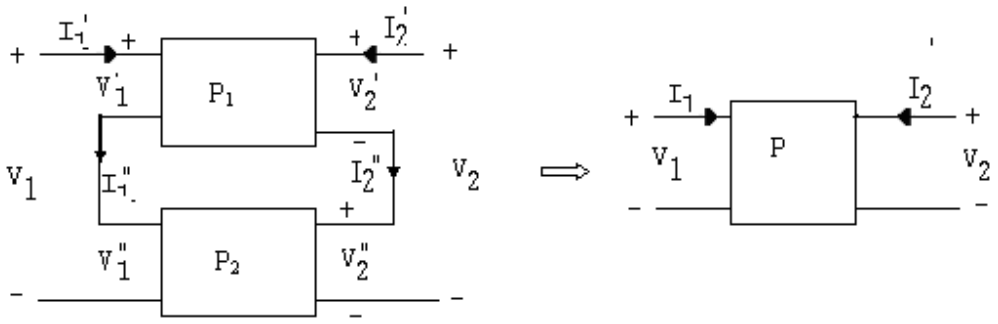


Fig. 2.21

From this figure, we have following relations from the series connection of two port networks (P_1 & P_2)

$$V_1 = V_1^{'} + V_1^{''} \qquad I_1 = I_1^{'} = I_1^{''} \qquad \text{------ (2.40)}$$
$$V_2 = V_2^{'} + V_2^{''} \qquad I_2 = I_2^{'} = I_2^{''} \qquad \text{------ (2.41)}$$

By doing similar calculations, as is in parallel combination discussed above; it can very easily be proved that *Z*-parameters of the series combination of two networks are equal to the sum of corresponding z-parameters of individual networks i.e.

$$Z_{11} = Z_{11}^{'} + Z_{11}^{''} \qquad Z_{12} = Z_{12}^{'} + Z_{12}^{''}$$
$$Z_{21} = Z_{21}^{'} + Z_{21}^{''} \qquad Z_{22} = Z_{22}^{'} + Z_{22}^{''}$$

Example 2.6 Find Y-parameters of the twin T-network of the given figure (2.22). Plot Y_{12} as a function of frequency.

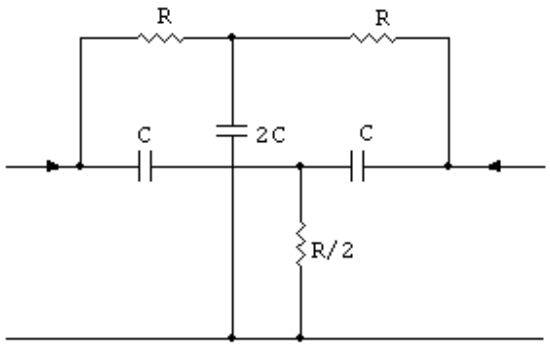


Fig. 2.22

Solution: The given twin T- network is the parallel combination of the two individual T-networks, which is clear from the given circuit shown the fig. (2.23).

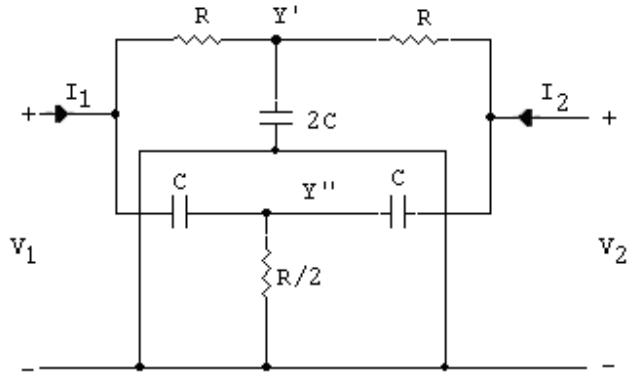


Fig. 2.23

The *Y*-parameters of this *T*- network may be given by:

$$Y_{11} = Y_{11}^{'} + Y_{11}^{''} \qquad Y_{12} = Y_{12}^{'} + Y_{12}^{''}$$
$$Y_{21} = Y_{21}^{'} + Y_{21}^{''} \qquad Y_{22} = Y_{22}^{'} + Y_{22}^{''}$$

Y-parameters of the upper *T*- network may easily be obtained and are given by:

$$Y_{11}^{'} = Y_{22}^{'} = \frac{R + \dfrac{i}{2j\omega C}}{\left[R^2 + \dfrac{R}{2j\omega C} + \dfrac{R}{2j\omega C}\right]} = \frac{(1 + 2j\omega CR)}{2R(1 + i\omega CR)}$$

$$Y_{12}^{'} = Y_{21}^{'} = \frac{-\dfrac{1}{2j\omega C}}{\dfrac{1}{2j\omega C}\left[2R + 2j\omega cR^2\right]} = -\frac{1}{2R(1 + j\omega CR)}$$

Y-parameters of the lower T- network are given by:

$$Y_{11}^{''} = Y_{22}^{''} = \frac{\left[\dfrac{R}{2} + \dfrac{1}{j\omega C}\right]}{\left[-\dfrac{1}{\omega^2 C^2} + \dfrac{R}{2j\omega C} + \dfrac{R}{j\omega C}\right]} = \frac{(2 + j\omega CR)\, j\omega C}{2(1 + j\omega CR)}$$

$$Y_{12}^{''} = Y_{21}^{''} = \frac{-R/2}{\dfrac{2(1 + j\omega CR)}{2j^2\omega^2 C^2}} = \frac{\omega^2 C^2 R^2}{2R(1 + j\omega CR)}$$

The required Y- parameters of the given Twin T- network are given by:

$$Y_{11} = Y_{22} = Y_{11}^{'} + Y_{11}^{''} = \frac{(1+2j\omega CR)}{2R(1+j\omega CR)} + \frac{(2+j\omega CR)j\omega C}{2(1+j\omega CR)}$$

$$= \frac{1+2j\omega CR+2j\omega CR-\omega^2C^2R^2}{2R(1+j\omega CR)}$$

$$= \frac{(1-\omega^2C^2R^2)+4j\omega CR}{2R(1+j\omega CR)}$$

$$Y_{12} = Y_{21} = Y_{12}^{'} + Y_{21}^{''} = -\frac{1}{2R(1+j\omega CR)} + \frac{\omega^2C^2R^2}{2R(1+j\omega CR)}$$

$$= \frac{\omega^2C^2R^2-1}{2R(1+j\omega CR)}$$

If a graph is plotted between Y_{12} as a function of frequency (ω), we get a curve as shown in fig.2.24. The following inference is obtained from this curve.

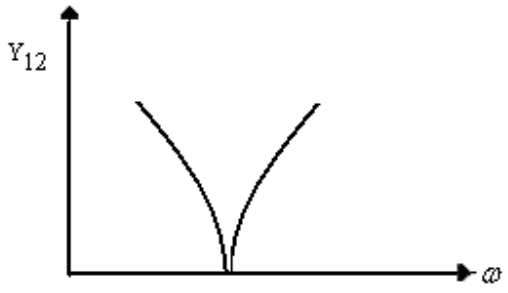


Fig. 2.24

At $\omega^2C^2R^2 = 1$ or $\omega = \frac{1}{RC}$

$Y_{12} = 0$

Y_{12} decreases with ω till $\omega^2C^2R^2 < 1$, but increased with ω when $\omega^2C^2R^2 > 1$. So we may say that the twin *T*- network behaves like a band pass filter.

Example 2.7 Find *Z*- parameters of the given Bridged *T*- network. Draw also its equivalent circuit.

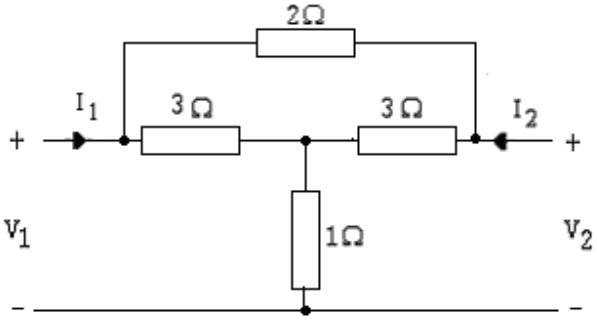


Fig. 2.25

Solution: It is a bridged *T*- network since a resistance (2 Ω) is connected between the Input and Output ports. This network may be redrawn as shown (Fig. 2.26):

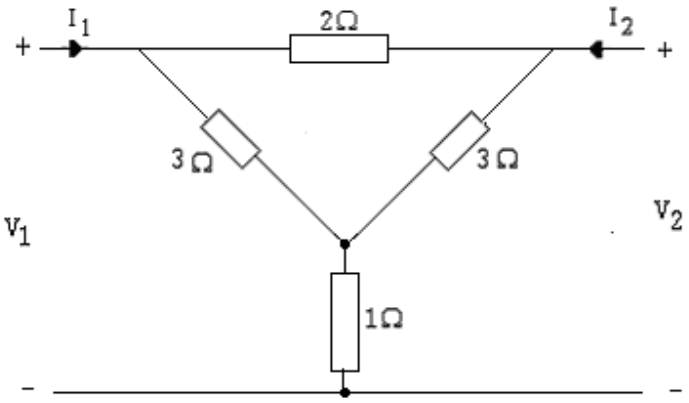


Fig. 2.26

It is clearly seen that this network contains the Delta network, which may converted to its equivalent star network as.

$$R_A = \frac{2x3}{2+3+3} = \frac{6}{8} = \frac{3}{4}\ \Omega\ ; \quad R_B = \frac{3}{4}\ \Omega\ ; \quad R_C = \frac{3x3}{8} = \frac{9}{8}\ \Omega$$

This circuit may further be redrawn as shown in fig. (2.27):

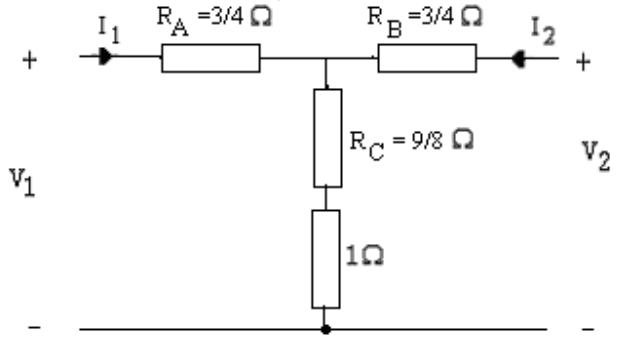


Fig.2.27

The circuit of figure 2.27 may also be reduced as shown in figure 2.28, Z- parameters of this T- network are calculated as:

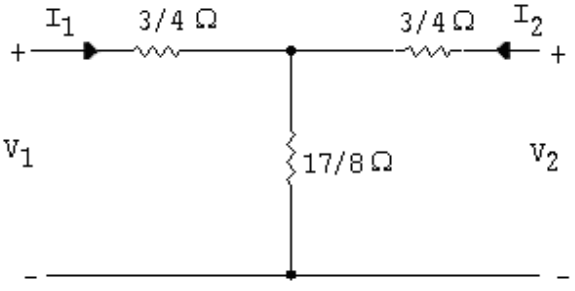


Fig. 2.28

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} \quad \text{i.e.} \quad V_1 = I_1\left(\frac{3}{4} + \frac{17}{8}\right) \quad \text{or} \quad Z_{11} = \frac{V_1}{I_1} = \frac{23}{8}\ \Omega$$

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} \quad \text{i.e.} \quad V_1 = I_2(\frac{17}{8}), \quad \text{or} \quad Z_{12} = \frac{V_1}{I_2} = \frac{17}{8}\,\Omega$$

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} \quad \text{i.e.} \quad V_2 = I_1(\frac{17}{8}), \quad \text{or} \quad Z_{21} = \frac{V_2}{I_1} = \frac{17}{8}\,\Omega$$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} \quad \text{i.e.} \quad V_2 = I_2(\frac{3}{4} + \frac{17}{8}), \text{ or} \quad Z_{22} = \frac{V_2}{I_2} = \frac{23}{8}\,\Omega$$

From the calculated Z- parameters of the given network, the network equations are given as: $V_1 = \frac{23}{8} I_1 + \frac{17}{8} I_2$ & $V_2 = \frac{17}{8} I_1 + \frac{23}{8} I_2$

The equivalent circuit of the given network is (fig. 2.29):

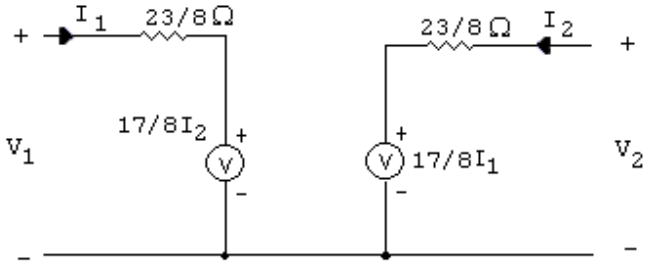


Fig. 2.29

Example 2.8 Find the Z-parameters of the given circuit. All resistance values are in ohms.

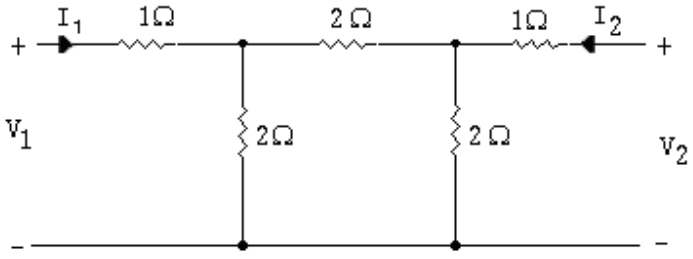


Fig. 2.30

Solution: The given circuit may be redrawn as:

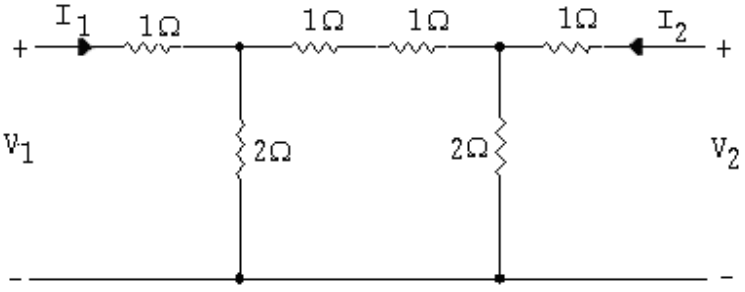


Fig. 2.31

This circuit is a cascaded network of the two T- networks. So we first find the T-parameters of the two networks individually and after matrix multiplication of these *T*-parameters, the overall *T*- parameters of the cascaded network are obtained.
The two networks are identical, so we calculate *T*- parameters of one network (Fig. 2.32) as follows:

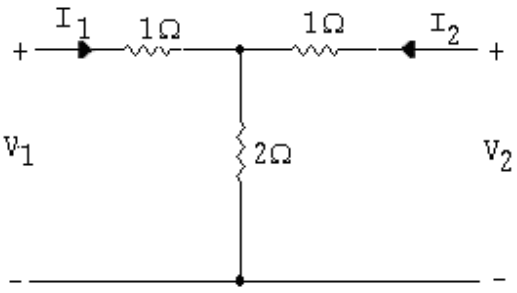


Fig. 2.32

$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$ From the fig.2.32 $V_2 = I_1.2$ and $V_1 = (2+1)I_1$

so $A = \frac{V_1}{V_2} = \frac{3}{2}$

$B = \left.\frac{V_1}{-I_2}\right|_{V_1=0}$ From the fig.2.32 $V_1 = I_1.(1+\frac{1x2}{3}) = \frac{5}{2}I_1$

and $-I_2 = \frac{2}{3}I_1$

so $B = \frac{V_1}{-I_2} = \frac{5}{2}\Omega$

$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$ From the fig.2.32 $V_2 = I_1.2$ or $C = \frac{I_1}{V_2} = \frac{1}{2}$ mhos

$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$ and $-I_2 = \frac{2}{3}I_1$ or $D = \frac{I_1}{-I_2} = \frac{3}{2}$

The overall *T*-parameters of the cascaded network (given network) are obtained by matrix multiplications.

$$\begin{bmatrix} \frac{3}{2} & \frac{5}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix}\begin{bmatrix} \frac{3}{2} & \frac{5}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{7}{2} & \frac{15}{2} \\ \frac{3}{2} & \frac{7}{2} \end{bmatrix}$$

The *T*-parameter equations of the network are therefore given by:

$$V_1 = \frac{7}{2}V_2 - \frac{15}{2}I_2$$

$$I_1 = \frac{3}{2}V_2 - \frac{7}{2}I_2$$

These two equations may be rewritten in the form of *Z*-parameter equations.

$$\frac{3}{2}V_2 = I_1 + \frac{7}{2}I_2 \qquad \text{or} \qquad V_2 = \frac{2}{3}I_1 + \frac{7}{3}I_2$$

$$V_1 = \frac{7}{3}I_1 + \frac{2}{3}I_2$$

Comparing these two equations with *Z*-Parameter equations we get the *Z*-parameters of the given network.

$$Z_{11} = Z_{22} = \frac{7}{3}\Omega \qquad Z_{12} = Z_{21} = \frac{2}{3}\Omega$$

2.9 Dependent sources: So far we have discussed the characteristics of the Passive network having passive elements connected to it. Now the active network having the active elements will be discussed. The active elements used in the network may very likely be transistor, operational amplifiers etc. However, the controlled or dependent Source considered as the basic active element, may be classified as:

(i) Voltage Controlled Voltage Source (VCVS)
(ii) Voltage Controlled Current Source (VCCS)
(iii) Current Controlled Current Source (CCCS)
(iv) Current Controlled Voltage Source (CCVS)

(i) Voltage Controlled Voltage Source (VCVS): It is an ideal voltage source whose voltage is dependent on the input voltage. The network equation may be written by considering following H' or *T*-parameters of the network.

$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \mu & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} \Leftrightarrow \begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\mu} & o \\ o & o \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$I_1 = 0$

& $V_2 = \mu V_1$ Input is open circuited and output is Ideal voltage source, which will depend on the input voltage.

The equivalent network may be drawn as given in Fig. (2.33).

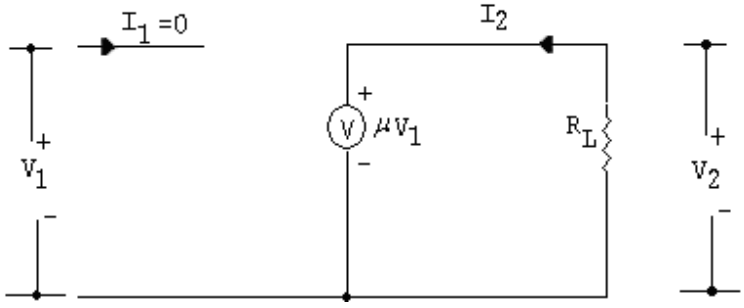


Fig. 2.33

It has input power as zero (since I_1 = 0) and has the finite output power as a load resistance is connected to its output terminals. So there is a power gain and hence it is an active element/device. VCVS may also be called as Voltage Amplifier and μ is called as the amplification factor.

(ii) Voltage Controlled Current Source (VCCS): It is an ideal current source whose current is controlled by the Input voltage. The network equation assuming the *Y*- or *T*-parameters may be given by:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ g_m & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \Leftrightarrow \begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} 0 & -1/g_m \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$I_1 = 0$: Input is open circuited.

$I_2 = g_m V_1$: Output is a current source, whose value may be controlled by input voltage.

The network may be shown as:

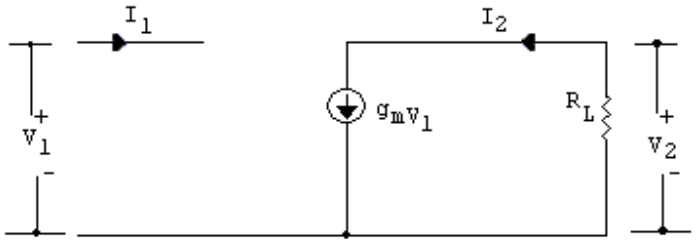


Fig. 2.34

g_m is called as the Transconductance which when multiplied with the input voltage; it gives the magnitude of the Ideal Current Source. It is also the active network since Input Power is zero and out put power is finite as output terminals are connected to some load resistance.

(iii) Current Controlled Current Source (CCCS): The current controlled current source (CCCS) is an ideal current source whose current is controlled by the input current. It is a Current amplifier, whose network equation may be given by the *H*-parameters as:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \alpha & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} \Leftrightarrow \begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1/\alpha \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$V_1 = 0$: Input is short circuited.

$I_2 = \alpha I_1$ Output is the current source, which is controlled by the input current. α is the current gain of the current amplifier. The equivalent circuit of this network is given in fig. (2.35).

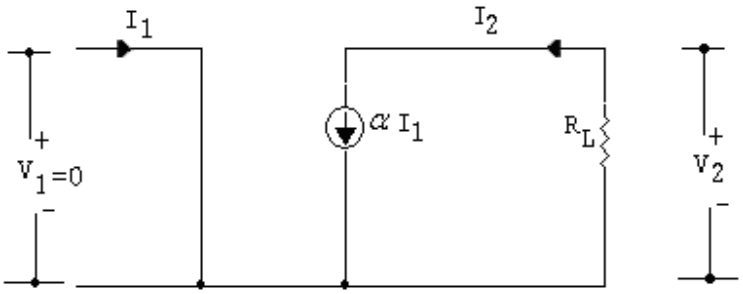


Fig. 2.35

It is also an active network, which may be proved as above.

(iv) Current Controlled Voltage Source (CCVS): It is an ideal voltage source whose voltage is controlled by the Input current source. The network equation of CCVS may be given by using *Z*- parameters as given below:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ r_m & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \Leftrightarrow \begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1/r_m & 0 \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$V_1 = 0$ & $V_2 = r_m I_1$ Input is short circuited and output is a voltage source whose voltage depends on the input current.

The network may be shown as:

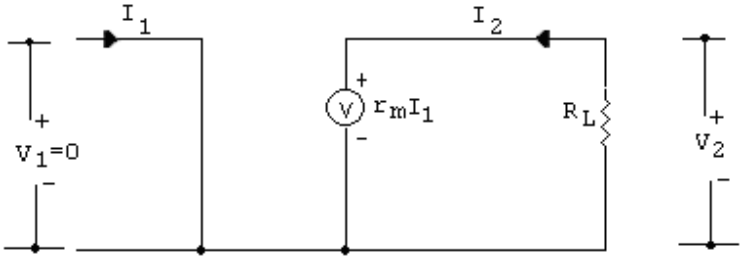


Fig. 2.36

Here r_m is the trans-resistance which when multiplied by the Input current converts into its output voltage. This too is an active element.

2.10 Reciprocity: Passive linear two port network are reciprocal because they exhibit the property of reciprocity. According to reciprocity theorem the ratio $\left.\frac{V_1}{I_2}\right|_{V_2=o} = \left.\frac{V_2}{I_1}\right|_{V_1=0}$

The condition of reciprocity of passive network can be obtained by defining any parameters.

(i) Reciprocity condition for *Z*- parameters: The *Z* - parameters equation of a network (passive) is given by:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \qquad \text{------ (2.42)}$$

Putting $V_2 = 0$ in equation (2.42), it is obtained:

$$I_1 = \frac{-Z_{22}}{Z_{21}} I_2 \qquad \& \qquad V_1 = -\frac{Z_{11}Z_{22}}{Z_{21}} I_2 + Z_{12} I_2$$

$$\frac{V_1}{I_2} = \frac{Z_{12}Z_{21} - Z_{11}Z_{22}}{Z_{21}}$$

Putting $V_1 = 0$ in equation (2.42) it is obtained:

$$I_2 = -\frac{Z_{11}}{Z_{12}} I_1 \qquad \text{or} \qquad V_2 = Z_{21} I_1 - \frac{Z_{11}Z_{22}}{Z_{12}} I_1$$

$$\frac{V_2}{I_1} = \frac{Z_{12}Z_{21} - Z_{11}Z_{22}}{Z_{12}}$$

If the network is reciprocal than $\frac{V_1}{I_2}$ should be equal to $\frac{V_2}{I_1}$, which is possible if $Z_{12} = Z_{21}$. This is the condition of reciprocity.

The condition for reciprocity for the network (passive) represented by other parameters may also be calculated in the similar fashion.

The network is reciprocal if

$Z_{12} = Z_{21}$ in Z –parameters $\quad Y_{12} = Y_{21}$ in Y-parameters

$H_{12} = -H_{21}$ in H-parameters $\quad H'_{12} = -H'_{21}$ in H'-parameters

$AD - BC = 1$ in T-parameters $\quad A'D' - B'C' = 1$ in T' - parameters

The reciprocity theorem, however, in general does not hold well in active network.

2.11 Ideal Transformer: Ideal transformer is one, in which there is no loss of Power, i.e. input power is equal to output power. If n is the turn ratio, then the ratio of Input to the output voltage is given by:

$$\frac{V_1}{V_2} = n \qquad \text{or} \qquad V_1 = n V_2$$

and the ratio of the output current to the Input current is given by:

$$\frac{I_2}{I_1} = -n \qquad \text{or} \qquad I_2 = -n I_1$$

So the *H*-parameters of the ideal transformer may be given by:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$ and its equivalent *T*-parameters are given by

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & 1/n \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

It cannot be represented by the Z or Y-parameters, since input and output resistances of ideal transformer are zero.

2.12 Impedance Converter: Let us consider a two port network terminated by an impedance Z_L to its output port. The equivalent impedance of this network presented at the Input port is given by

$$Z_{in} = (\text{Constant})\, Z_L$$

Such a network is called as the Impedance Converter. So Impedance Converter may be defined as a network whose input impedance is the load impedance terminated at the output port multiplied by constant quantity. The constant quantity is called as converter factor.

Impedance Converter may be classified as:

(i) Positive Impedance Converter (PIC)
(ii) Negative Impedance Converter (NIC)

(i) Positive Impedance Converter (PIC): If the conversion factor discussed above is a +ve quantity, it is called as positive Impedance Converter.

An ideal transformer may be considered as +ve impedance converter. The *H*-parameter matrix of an ideal transformer is given by:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

i.e. $V_1 = nV_2$ and $I_2 = -nI_1$ or $I_1 = -\frac{1}{n} I_2$

$$\frac{V_1}{I_1} = -n^2 \frac{V_2}{I_2} = n^2 Z_l \qquad (\text{since } Z_L = -\frac{V_2}{I_2})$$

$$Z_{in} = n^2 Z_L \qquad (Z_{in} \text{ is the input impedance})$$

n is the turn ratio, n^2 is definitely a +ve quantity. So ideal transformer may be known as +ve Impedance converter.

(ii) Negative Impedance Converter (NIC): If in the Impedance converter, the conversion factor is a negative quantity then it is called as negative impedance converter. It is further of two types:

(i) Voltage Inversion Type –ve Impedance Converter (VNIC)

(ii) Current Inversion type –ve Impedance Converter (CNIC)

VNIC: Consider the *H*-Parameter matrix of a two port network as:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 & -K_1 \\ -K_2 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

i.e. $V_1 = -K_1 V_2$

(Negative sign indicate that the voltage at the output port is inverted)

$$I_2 = -K_2 I_1 \qquad \text{or} \qquad I_1 = -\frac{1}{K_2} I_2$$

$$\frac{V_1}{I_1} = K_1 K_2 \frac{V_2}{I_2} \qquad \text{or} \qquad Z_{in} = -K_1 K_2 Z_L \qquad (\text{since } Z_L = -\frac{V_2}{I_2})$$

It is clear from the above discussion that the load impedance (Z_L) when multiplied by a negative quantity ($-K_1K_2$) is gives us the input impedance. Hence it is known as VNIC.

CNIC: Consider the H-parameter equation of a network given by:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 & K_1 \\ K_2 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$V_1 = K_1 V_2$$

$$I_2 = K_2 I_1 \quad \text{or} \quad I_1 = \frac{1}{K_2} I_2$$

(since normally I_2 should –ve, in this case I_1 & I_2 are of same sign, hence current inversion type network)

$$\frac{V_1}{I_1} = K_1 K_2 \frac{V_2}{I_2} \quad \text{or} \quad Z_{in} = -K_1 K_2 Z_L \quad (\text{since } Z_L = -\frac{V_2}{I_2}).$$

2.13 Gyrator: Gyrator is a two port network, which when terminated a load Impedance Z_L at the output port, hen the Input Impedance is inversely proportional to the load impedance.

i.e. $Z_{in} \propto \frac{1}{Z_L}$ or $z_{in} = K\frac{1}{Z_L}$

where K is called as the Gyrator constant. The parameters of the network are given as;

$$V_1 = -rI_2 \quad \& \quad V_2 = rI_1 \quad \text{or} \quad I_1 = \frac{1}{r} V_2$$

$$\frac{V_1}{I_1} = -r^2 \frac{I_2}{V_2} = r^2 \frac{1}{Z_L} \quad \text{or} \quad Z_{in} = r^2 \frac{1}{Z_L}$$

r^2 is called as the Gyrator Constant. If the load impedance is of capacitive nature ($Z_L = \frac{1}{j\omega C}$) then input impedance will be of inductive nature.

i.e. $Z_{in} = j\,\omega\, C\, r^2$ (since Cr^2 has the dimension of inductance).

Similarly if the load is of inductive nature then the input impedance will be of capacitive nature.

2.14 Cascading of two Gyrators: If two gyrators are cascaded, the overall response of the network (cascaded network) may be obtained by matrix multiplication of T-parameter matrices of individual gyrators.

Consider the T-parameter matrix of individual gyrators:

Gyrator I $\begin{bmatrix} 0 & nr \\ \frac{1}{nr} & 0 \end{bmatrix}$

Gyrator II $\begin{bmatrix} 0 & r \\ \frac{1}{r} & 0 \end{bmatrix}$

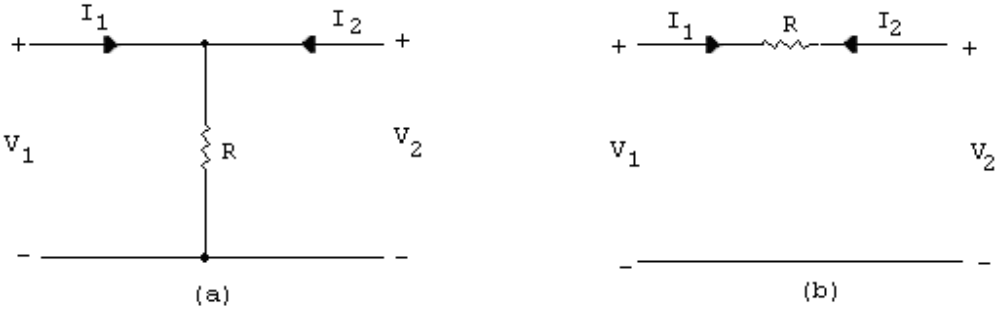
Matrix multiplication of gyrator I and gyrator II

$$\begin{bmatrix} 0 & nr \\ \frac{1}{nr} & 0 \end{bmatrix} \begin{bmatrix} 0 & r \\ \frac{1}{r} & 0 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix}$$

is equal to the T-parameter equation of an ideal transformer. So the network of two cascaded gyrators is equivalent to an ideal transformer.

Problems:

1. What do you understand by Two Port Network? Define H- and T-parameters of a two port network. Find the condition of reciprocity for these parameters.
2. What do you mean by Impedance converter? Explain positive and negative impedance converter. How will you transform the parameters of voltage inversion type negative impedance converter into its equivalent transmission parameters?
3. Find the open- circuit and short circuit parameters of a two port network.
4. Discuss H-parameters of a two port network. Draw its equivalent circuit.
5. Explain the active two port network. What are voltage controlled current source (VCVS) and current controlled voltage source (CCVS)? Show that when CCVS and VCCS are cascaded it behaves like current controlled current source (CCCS).
6. What do you understand by dependent sources? Discuss voltage and current amplifiers in the form of two port network. Show that they are active elements.
7. Discuss voltage control current source (VCCS) and current control voltage source (CCVS). Show that when a VCCS and a CCVS are cascaded, it works as a voltage amplifier.
8. Define gyrator. Show that the two cascaded gyrators can be simulated as an ideal transformer.
9. Define open circuit and short circuit parameters of a two port network. What is transformation of parameters? How can open circuit parameters be converted into its equivalent H-parameters
10. How will you transform H parameters of a passive network into its equivalent Inverse Transmission ($H^{'}$) parameters?
11. What do you understand by dependent sources? By considering a suitable example, explain voltage and current amplifiers.
12. Show that the H-parameters of the two port network will not exist in $Z_{11}=0$.
13. Show that Z-parameters of the series combination of two 'Two Port Networks' are equal to the sum of Z-parameters.
14. Show that when two 'Two Port Networks' are connected in parallel; the Y-parameters of the combined two port network may be obtained by adding the Y-parameters of the individual network.
15. Find Z, Y and H parameters of the given two port networks (a) and (b) shown below.



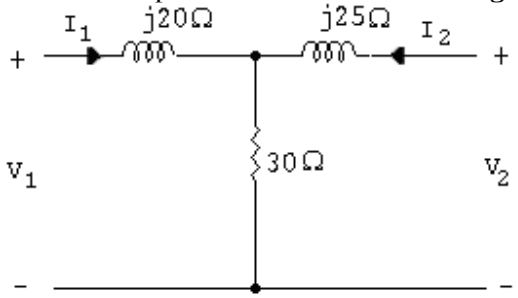
Ans: (a) $Z_{11}= Z_{12} = Z_{21} = Z_{22} = R$, Y – parameters does not exist, $H_{11} = 0$,

$H_{12} = - H_{21} = 1$, $H_{22} = \frac{1}{R}$: (b) Z – parameters does not exist,

$Y_{11}=Y_{22}=\frac{1}{R}$ ℧ , $Y_{12}=Y_{21}=-\frac{1}{R}$ ℧ , H_{11} = R Ω,

H_{12}= - H_{21}=1 , $H_{22}=\frac{1}{R}$ ℧ .

16. Determine Z and Y-parameters of the network given below.



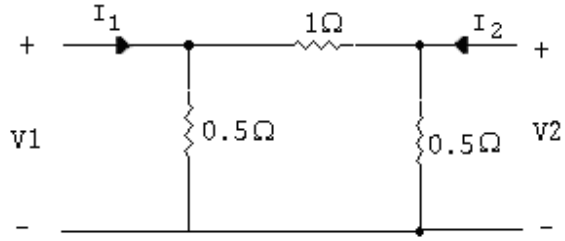
Ans.: Z_{11} =10(3+2j) Ω, Z_{12} =Z_{21} = 30 Ω , Z_{22} = 5(6+j5) Ω

$Y_{11}=\frac{(6+j5)}{10(27j-10)}$ ℧ , Y_{12}=Y_{21}=$\frac{-3}{5(27j-10)}$ ℧ s, Y_{22}=$\frac{(3+2j)}{5(27j-10)}$ ℧

17. A two port network has the following Z-parameters Z_{11}=10 Ω Z_{12}=Z_{21}=5Ω and Z_{22}=12Ω. Calculate the Y-parameters of the network. Also show its equivalent Y-parameter network.

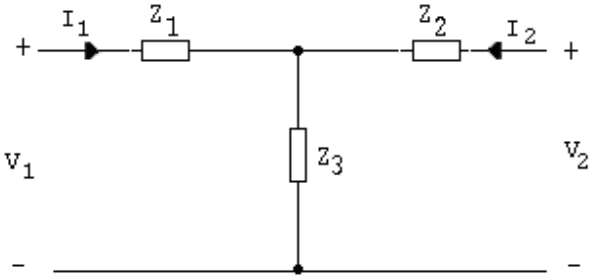
Ans.: Y_{11} = 0.1263 ℧ , Y_{12} = Y_{12}= - 0.053 ℧ , Y_{22} = 0.1053 ℧

18. Find the T-parameters of the network given below. Show that the network is reciprocal.



Ans.: A = D = 3, C = 8 ℧ , B= 1 Ω

19. Find Y, H' ,and T-parameters of the given network.



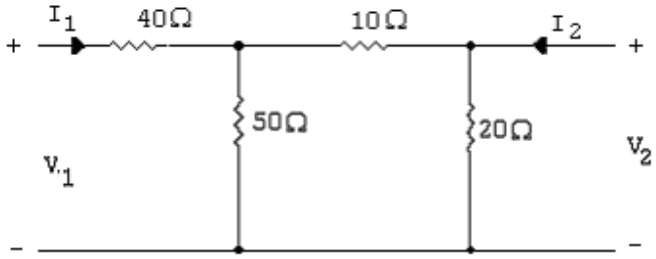
Ans.: $Y_{11}=\frac{Z_2+Z_3}{Z_1Z_2+Z_2Z_3+Z_1Z_3}$ ℧ , $Y_{12}=Y_{21}=\frac{-Z_3}{Z_1Z_2+Z_2Z_3+Z_1Z_3}$ ℧

$$Y_{22} = \frac{Z_1 + Z_3}{Z_1Z_2 + Z_2Z_3 + Z_1Z_3} \ \Omega \ : H'_{11} = \frac{1}{Z_1 + Z_3} \ \Omega \ ,$$

$$H'_{22} = \frac{Z_1Z_2 + Z_2Z_3 + Z_1Z_3}{Z_1 + Z_3} \Omega \ , \ H'_{12} = \frac{-Z_3}{Z_1 + Z_3} = -H'_{21} \ , \ A = \frac{Z3}{Z_1 + Z_3} \ ,$$

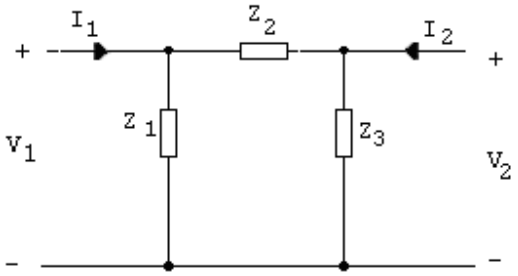
$$B = \frac{Z_1Z_2 + Z_1Z_3 + Z_2Z_3}{Z_3} \ \Omega \ , \ C = \frac{1}{Z_3} \ \Omega \ , \ D = \frac{Z_3}{Z_2 + Z_3}$$

20. Find H-parameters of the network.



Ans.: $H_{11} = 48.33 \ \Omega$, $H_{12} = 0.83 = - H_{21}$, $H_{22} = 0.067$ Ω

21. Find H, Z, Y and T- parameters of the given Π network.



Ans.: $H_{11} = \frac{Z_1Z_2}{Z_1 + Z_2} \Omega \ , \ H12 = \frac{Z_1}{Z_1 + Z_2} = -H_{21} \ , \ H_{22} = \frac{Z_1 + Z_2 + Z_3}{(Z_1 + Z_2)Z_3}$ Ω

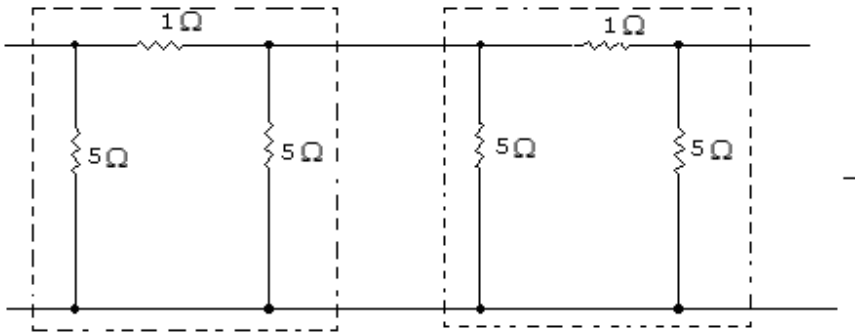
$$Z_{11} = \frac{Z_1(Z_2 + Z_3)}{Z_1 + Z_2 + Z_3} \ \Omega \ , \ Z_{12} = Z_{21} = \frac{Z_1Z_3}{Z_1 + Z_2 + Z_3} \Omega,$$

$$Z_{22} = \frac{Z_3(Z_1 + Z_2)}{Z_1 + Z_2 + Z_3} \ \Omega: \ A = \frac{Z_2 + Z_3}{Z_3} \ , \ B = Z_2 \ \Omega \ ,$$

$$C = \frac{Z_1 + Z_2 + Z_3}{Z_1Z_3} \ \Omega \ , \ D = \frac{Z_1 + Z_2}{Z_1} \ : Y_{11} = \frac{Z_1 + Z_2}{Z_1Z_2} \ \Omega \ ,$$

$$Y_{12} = Y_{21} = \frac{-1}{Z_2} \ \Omega \ , \ Y_{22} = \frac{Z_{21} + Z_3}{Z_2Z_3} \ \Omega \ .$$

22. Two identical π - networks are cascaded together as shown in the figure given below. Find the H – parameters of this cascaded network. Show that this network is reciprocal.

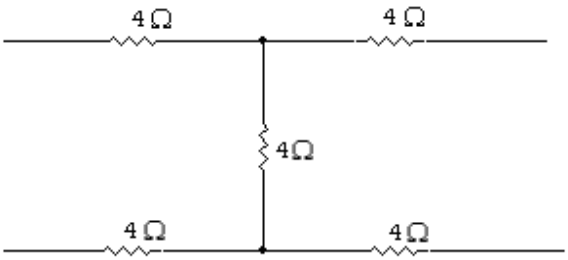


Network 1 Network 2

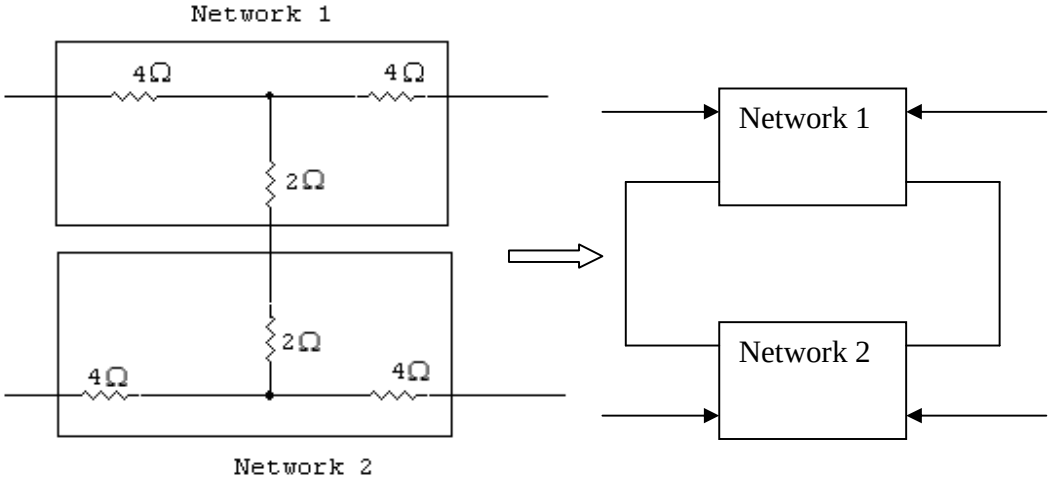
(Hint: First find the A, B, C, D parameters of the one given π- network (identical). Overall T - parameters of the cascaded network are obtained by matrix multiplication of the individual π - network. These parameters may further be transformed to H – parameters.)

Ans.: $H_{11} = \frac{60}{47}\Omega$, $H_{12} = -H_{21} = \frac{25}{47}$, $H_{22} = \frac{132}{47}$ ℧

23. Find $T^{'}$ – parameters of the given network.



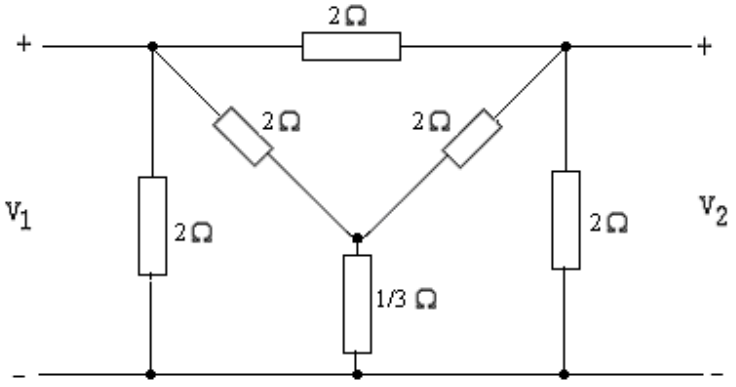
(Hint: This circuit is a series combination of two identical T- networks shown in the figure given below.)



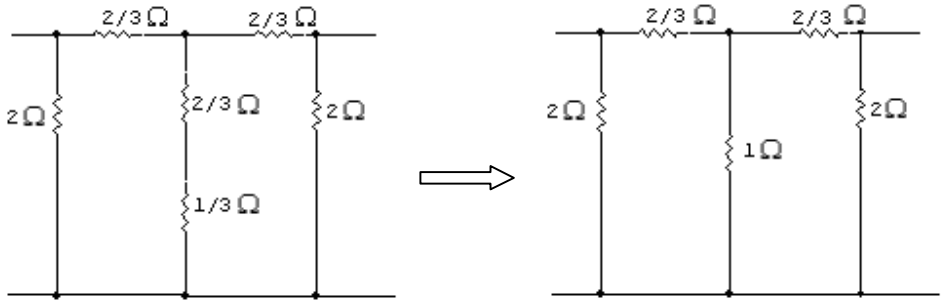
Find the Z – parameters of the individual T-network; by adding the Z-parameters of the two networks, the overall Z-parameters of the given network are obtained, which may further be transformed to $T^{'}$ – parameters.)

Ans.: $A^{'} = 3 = D^{'}$, $B^{'} = 32\ \Omega$, $C^{'} = \frac{1}{4}$ ℧ .

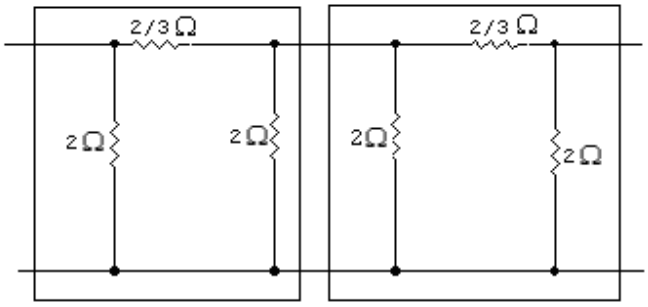
24. Find T - parameters of the network shown below.



Hint: The network may be redrawn using the delta to star conversion of upper delta network as:



This network may further be shown as a cascaded network of two identical π – networks as:



Network 1 Network 2

Ans.: $A = \frac{23}{9} = D$, $B = \frac{16}{9}\Omega$, $C = \frac{28}{9}$ ℧ .

3

Networks with time Varying Sources

So far the analysis of networks containing only batteries and resistances has been discussed. In this chapter the network with time varying sources, resistances and other elements like inductances capacitances and transformers, etc. will also be discussed. The time varying sources are generally of three types, viz., periodic, aperiodic and random. Sine, Square and Triangular waves are periodic, since it repeats after a fixed interval of time. Periodic waves are generally used in the electronic circuits and are of our great interest. A pulse is aperiodic and noise is of random nature.

3.1 Fourier series: Any function *f(t)* can be expanded using Fourier series, which is expressed as the summation of sinusoidal (sine, cosine or both) terms as given by:

$$f(t) = A_0 + \sum_{n=1}^{\infty} A_n Sin\ (n\omega t) + \sum_{n=1}^{\infty} B_n \cos(\ n\omega t) \qquad \text{-------- (3.1)}$$

where

$$A_0 = \frac{1}{T}\int_0^T f(t).dt \qquad \text{------ (3.2)}$$

$$A_n = \frac{2}{T}\int_0^T f(t).Sin(n\omega t).dt \qquad \text{------ (3.3)}$$

$$B_n = \frac{2}{T}\int_0^T f(t).Cos(n\omega t).dt \qquad \text{------ (3.4)}$$

Example 3.1 Find the first four coefficients of the half wave rectified output, using Fourier series expansion. Characteristics of the wave are:

$$f(t) = E_m Sin\omega t \qquad \text{when } 0 \le t \le T/2$$
$$f(t) = 0 \qquad T/2 \le t \le T$$

Solution: The coefficients of Fourier series of the half wave rectified output wave are given as:

$$A_0 = \frac{1}{T}\int_0^T E_m Sin(\omega t).dt = \frac{1}{T}\left[\int_0^{T/2} E_m Sin(\omega t)dt + \int_{T/2}^{T}(0).dt\right]$$

$$= \frac{E_m}{\pi} \qquad \text{------ (3.5)}$$

$$A_n = \frac{2}{T}\left[\int_0^{T/2} E_m Sin(\omega t).Sin(n\omega t).dt\right] \qquad \text{------ (3.6)}$$

$$= \frac{2E_m}{T}\int_0^{T/2}[Cos(n-1)\omega t - Cos(n+1)\omega t]dt$$

$$= \frac{2E_m\omega}{2\pi}\left[\frac{Sin(n-1)\omega t}{(n-1)\omega} - \frac{Sin(n+1)\omega t}{(n+1)\omega}\right]_0^{T/2}$$

$$= \frac{E_m}{\pi}\left[\frac{Sin(n-1)\pi}{(n-1)} - \frac{Sin(n+1)\pi}{(n+1)}\right]$$

$= 0$ (for all values of n, except for $n = 1$)

For $n = 1$ denominator of the above equation will be inderminent. So we calculate A_1 by putting $n = 1$ directly in equation (3.6) as:

$$A_1 = \frac{2}{T}\int_0^{T/2} E_m Sin\,\omega t.Sin\,\omega t.dt$$

$$= \frac{2E_m}{T}\int_0^{T/2} Sin^2\omega t.dt = \frac{2E_m}{2T}\int_0^{T/2}[1 - Cos2\omega t]dt$$

$$= \frac{E_m}{T}\left[\frac{T}{2} - \left[\frac{Sin2\omega t}{2\omega}\right]_0^{T/2}\right] = \frac{E_m}{T}\left[\frac{T}{2} - 0\right]$$

$$= \frac{E_m}{2} \qquad \text{------ (3.7)}$$

B_n may be calculated as:

$$B_n = \frac{2}{T}\int_0^{T/2} E_m Sin(\omega.t).Cos(n\omega.t).dt$$

$$= \frac{2E_m}{T}\int_0^{T/2}\left[\frac{-Sin(n-1)\omega.t + Sin(n+1)\omega.t}{2}\right]dt$$

$$= \frac{E_m\omega}{2\pi}\left[\frac{Cos(n-1)\omega t}{(n-1)\omega} - \frac{Cos(n+1)\omega t}{(n+1)\omega}\right]_0^{T/2}$$

$$= \frac{E_m}{2\pi}\left[\frac{Cos(n-1)\pi}{(n-1)} - \frac{Cos(n+1)\pi}{(n+1)} - \frac{1}{(n-1)} + \frac{1}{(n+1)}\right]$$

If n is odd, then $\qquad B_n = \frac{E_m}{2\pi}\left[\frac{1}{(n-1)} - \frac{1}{(n+1)} - \frac{1}{(n-1)} + \frac{1}{(n+1)}\right] = 0$ ------ (3.8)

If n is even, then

$$B_n = \frac{E_m}{2\pi}\left[-\frac{1}{(n-1)}+\frac{1}{(n+1)}-\frac{1}{(n-1)}+\frac{1}{(n+1)}\right]$$

$$= \frac{2E_m}{2\pi}\left[-\frac{1}{(n-1)}+\frac{1}{(n+1)}\right]$$

$$= \frac{E_m}{\pi}\left[\frac{-n-1+n-1}{(n^2-1)}\right] = -\frac{2E_m}{(n^2-1)\pi} \qquad \text{------- (3.9)}$$

(where n is even i.e. n = 2,4,6,8…..)

So by putting the values of *A's* and *B's* in equation 3.1 we get the required function for half wave rectified output as:

$$E = \frac{E_m}{\pi}+\frac{E_m}{2}Sin\omega t-\frac{2E_m}{3\pi}Cos2\omega t-\frac{2E_m}{15\pi}Cos4\omega t-\frac{2E_m}{35\pi}Cos6\omega t..... \qquad \text{------- (3.10)}$$

Example 3.2 Consider a full wave rectified signal of peak value E_m and period T. Find its coefficients in the Fourier series expansion.

Solution: The coefficients of Fourier series of the full wave rectified output wave are given as:

$$A_0 = \frac{1}{T}\int_0^T E_m Sin(n\omega t).dt = \frac{1}{T}\left[\int_0^{T/2} E_m Sin(\omega t)dt + \int_{T/2}^{T}(-)E_m Sin(\omega.t)dt\right]$$

$$= \frac{2E_m}{\pi} \qquad \text{------ (3.11)}$$

$$A_n = \frac{2}{T}\left[\int_0^{T/2} E_m Sin(\omega t).Sin(n\omega t).dt + \int_{T/2}^{T} E_m Sin(\omega t).Sin(n\omega t).dt\right]$$

$$= \frac{4E_m}{T}\left[\int_0^{T/2} Sin(\omega t).Sin(n\omega t).dt\right]$$

$$= \frac{4E_m}{2T}\int_0^{T/2}\left[Cos(n-1)\omega t - Cos(n+1)\omega t\right]dt$$

$$= \frac{2E_m\omega}{2\pi}\left[\frac{Cos\,(n-1)\omega t}{(n-1)\omega}-\frac{Cos\,(n+1)\omega t}{(n+1)\omega}\right]_0^{T/2}$$

$$= \frac{E_m}{\pi}\left[\frac{Sin(n-1)\pi - Sin0}{(n-1)}-\frac{Sin(n+1)\pi + Sin0}{(n+1)}\right]$$

$$= \frac{E_m}{\pi}\left[\frac{Sin(n-1)\pi}{(n-1)}-\frac{Sin(n+1)\pi}{(n+1)}\right] = 0 \qquad \text{------ (3.12)}$$

$$B_n = \frac{2x2}{T}\int_0^{T/2} E_m Sin(\omega.t).Cos(n\omega.t).dt$$

$$= \frac{4E_m}{T}\int_0^{T/2}\left[\frac{-Sin(n-1)\omega.t + Sin(n+1)\omega.t}{2}\right]dt$$

$$= \frac{2E_m\omega}{2\pi}\left[\frac{Cos(n-1)\omega t}{(n-1)\omega} - \frac{Cos(n+1)\omega t}{(n+1)\omega}\right]_0^{T/2}$$

$$= \frac{E_m}{\pi}\left[\frac{Cos(n-1)\pi}{(n-1)} - \frac{Cos(n+1)\pi}{(n+1)} - \frac{1}{(n-1)} + \frac{1}{(n+1)}\right]$$

If *n* is odd, then $B_n = \frac{E_m}{\pi}\left[\frac{1}{(n-1)} - \frac{1}{(n+1)} - \frac{1}{(n-1)} + \frac{1}{(n+1)}\right] = 0$ ----- (3.13)

If *n* is even, then

$$B_n = \frac{E_m}{\pi}\left[-\frac{1}{(n-1)} + \frac{1}{(n+1)} - \frac{1}{(n-1)} + \frac{1}{(n+1)}\right]$$

$$= \frac{2E_m}{2\pi}\left[-\frac{1}{(n-1)} + \frac{1}{(n+1)}\right]$$

$$= \frac{2E_m}{\pi}\left[\frac{-n-1+n-1}{(n^2-1)}\right] = -\frac{4E_m}{(n^2-1)\pi}$$ ------- (3.14)

(where *n* is even i.e. *n* = 2, 4, 6, 8…..)

So by putting the values of A's and B's in equation 3.1, the required function for full wave rectified output is obtained as:

$$E = \frac{2E_m}{\pi} - \frac{4E_m}{3\pi}Cos2\omega t - \frac{4E_m}{15\pi}Cos4\omega t - \frac{4E_m}{35\pi}Cos6\omega t..... \quad \text{------- (3.15)}$$

Example 3.3 Consider a symmetrical triangular wave of peak value E_m and period T as shown in figure (3.1). Find its coefficients in the Fourier series.

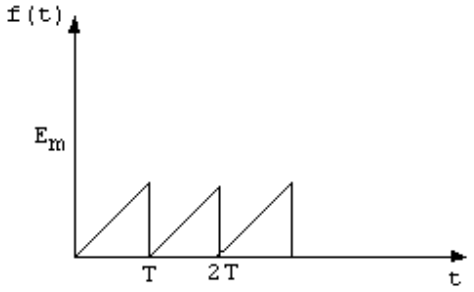


Fig. 3.1

Solution: let $f(t) = X + Y.t$ at $t = 0$ $f(t) = 0$ So $X = 0$

at $t = T$ $f(t) = E_m$

$E_m = Y\,T$ or $Y = (E_m)/T$

$f(t) = (\frac{E_m}{T}).t$ is the equation of the triangular wave. So the coefficients of Fourier series of this wave are given as:

$$A_0 = \frac{1}{T}\int_0^T \frac{E_m}{T}.t.dt = \frac{E_m}{T^2}\left[\frac{T^2}{2}\right] = \frac{E_m}{2} \qquad \text{------ (3.16)}$$

$$A_n = \frac{2}{T}\left[\int_0^T \frac{E_m}{T}.t.Sin(n\omega t).dt\right]$$

$$= \frac{2E_m}{T^2}\int_0^T t.Sin(n\omega t).dt$$

$$= \frac{2Em}{T^2}\left[\frac{-tCos(n\omega t)}{n\omega} + \int_0^T \frac{1.Cos(n\omega t)}{n\omega}.dt\right]$$

$$= \frac{2E_m}{T^2}\left[\frac{-tCos(n\omega t)}{n\omega} + \frac{Sin(n\omega t)}{n^2\omega^2}\right]_0^T$$

$$= \frac{2E_m}{T^2}\left[\frac{-TCos(2n\pi)}{n\omega} + \frac{Sin(2n\pi)}{n^2\omega^2}\right]$$

$$= -\frac{2E_m}{T^2}\left[\frac{TCos(2n\pi)}{n\omega}\right] = -\frac{2E_m}{n\omega}Cos2n\pi$$

$$= -\frac{E_m}{n\pi} \quad \text{(since } Cos\ 2n\pi = 1 \text{ for all values of } n\text{)}$$

$$B_n = \frac{2}{T}\left[\int_0^T \frac{E_m}{T}.t.Cos(n\omega t).dt\right]$$

$$= \frac{2E_m}{T^2}\left[\frac{tSin(n\omega t)}{n\omega} - \int_0^T \frac{1.Sin(n\omega t)}{n\omega}.dt\right]$$

$$= \frac{2E_m}{T^2}\left[\frac{tSin(n\omega t)}{n\omega} + \frac{Cos(n\omega t)}{n^2\omega^2}\right]_0^T$$

$$= \frac{2E_m}{T^2}\left[\frac{TSin(2n\pi)}{n\omega} + \frac{Cos(2n\pi)}{n^2\omega^2} - \frac{1}{n^2\omega^2}\right]$$

$$= \frac{2E_m}{n^2\omega^2T^2}\left[Cos(2n\pi)-1\right] = \frac{2E_m}{2n^2\pi^2}\left[1-1\right] = 0 \qquad \text{------ (3.17)}$$

So by putting the values of *A's* and *B's* in equation 3.1, the required Fourier series of the triangular wave is as: $E = \frac{E_m}{2} - \frac{E_m}{\pi}Sin\omega t - \frac{E_m}{2\pi}Sin2\omega t - \frac{E_m}{3\pi}Sin3\omega t.....$

Example 3.4 Find the Fourier series expansion of the square wave shown in figure (3.2).

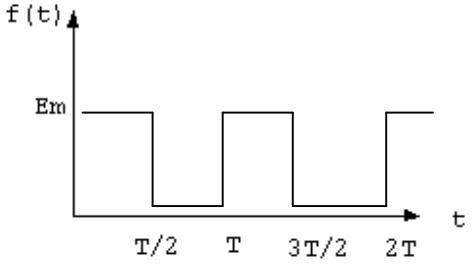


Fig. 3.2

Solution: The coefficients of Fourier series of the square wave are defined as:

$$A_0 = \frac{1}{T}\int_0^T f(t).dt = \frac{1}{T}\left[\int_0^{T/2} E_m dt + \int_{T/2}^{T}(0)dt\right]$$

$$= \frac{E_m}{T}\left[\frac{T}{2}\right] = \frac{E_m}{2}$$

$$A_n = \frac{2}{T}\left[\int_0^{T/2} E_m.Sin(n\omega t).dt + \int_{T/2}^{T}(0).Sin(n\omega t).dt\right] = \frac{2E_m}{n\omega T}\left[-Cos(n\omega t)\right]_0^{T/2}$$

$$= (\frac{2E_m}{n\omega T})\left[1-Cos(n\omega T/2)\right]$$

$$= (\frac{2E_m}{n\pi})[1-Cos(n\pi)] = 0 \text{ for even values of } n$$

$$= (\frac{2E_m}{n\pi}) \qquad \text{for odd values of } n$$

$$B_n = \frac{2}{T}\left[\int_0^{T/2} E_m.Cos(n\omega.t).dt + \int_{T/2}^{T}(0)Cos(n\omega t).dt\right]$$

$$= (\frac{2E_m}{n\omega T})[Sin(n\omega t)]_o^{T/2} = (\frac{2E_m}{n\omega T})Sin(\frac{n\omega T}{2}) = \frac{E_m}{n\pi}Sin(n\pi) = 0 \quad \text{for all values of n.}$$

So by putting the values of *A's* and *B's* in this equation, the required Fourier series of the Square wave is given as:

$$f(t) = \frac{E_m}{2} + \frac{2E_m}{\pi} Sin(\omega t) + \frac{2E_m}{3\pi} Sin(3\omega t) + \frac{2E_m}{5\pi} Sin(5\omega t) + \frac{2E_m}{7\pi} Sin(7\omega t) + ...$$

3.2 Sinusoidal Signal applied to different elements: When an A.C. signal $E(t) = E_m Sin\omega t$ is applied to a simple resistance, according to Ohm's Law one may obtain the instantaneous value of the current flowing through the circuit as:

$$I(t) = \frac{E(t)}{R} = \frac{E_m}{R} Sin\ \omega t$$
$$= I_m\ Sin\ \omega t \quad (I_m \text{ is the peak value of current})$$

Thus the current wave form is the exact replica of the voltage, i.e., both are in the same phase. But when an A.C. source is applied to any combination of resistance, inductance and capacitance, we get the current and voltage are having certain phase difference. Phase relation between the current flowing through and voltage applied across the circuit may best be understood if the Impedances and reactances are represented on the complex plain.

When an A.C. signal $E(t) = E_m Sin\ \omega t$ is applied across an inductance L, the instantaneous value of the current flowing through the circuit is given by:

$$I(t) = \frac{1}{L}\int E(t).dt = \frac{1}{L}\int (E_m \sin\ \omega t).dt$$
$$= \frac{E_m}{L}.\frac{-\cos\ \omega t}{\omega} = \left(\frac{E_m}{\omega L}\right)\sin\left(\omega t - \frac{\pi}{2}\right) \qquad \text{------ (3.18)}$$

It is clear from this equation that the current is lagging by an angle of 90 ^{o} with the voltage. The quantity $\left(\frac{E_m}{\omega L}\right)$ is known as the peak value of the current. It is customary to represent the magnitude as well as the phase relation between the voltage and the current in a graphical form. Such a graphical form is known as phasor diagram.

The phasor diagram is generally represented on the complex plain, as shown in figure (3.3). When a quantity coincides with the *X*-axis (real axis), it is known as real quantity. The quantity on *X*-axis when multiplied by j ($(j = \sqrt{-1})$, makes it purely imaginary quantity. It rotates by an angle of 90° in the anti-clock wise direction. However, when the real quantity is divided by j or multiplied by $(-j)$ it rotates in the clock wise direction by 90°.

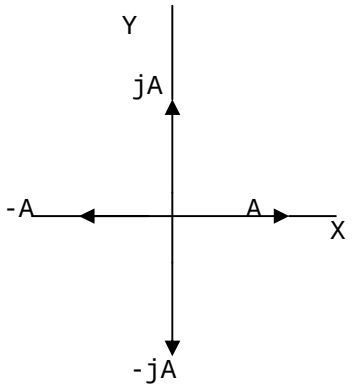


Fig. 3.3

From the equation (3.18), it is clear that the voltage *E*, when divided by ωL, the correct magnitude of the current is obtained. In order to rotate the phase by – 90 ^{o} the magnitude of current is further divided by *j* as:

$$I = \frac{E}{j\omega L} \qquad \text{and} \qquad \angle I = \angle E - \frac{\pi}{2}$$

The imaginary quantity ($j\omega L$) is called the inductive reactance of the inductance *L*.

The current is lagging behind the voltage by an angle of 90° as shown in figure (3.4). This is the phase relation between the current and voltage.

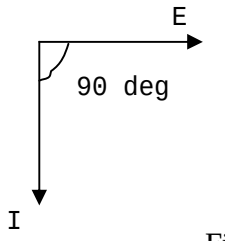


Fig. 3.4

Similarly the reactance of a capacitance is $\frac{1}{j\omega C}$ or $\frac{-j}{\omega C}$, which is also an imaginary quantity represented on the imaginary axis. The current flowing through the capacitance *C* when an A.C. signal of voltage *E* is applied across the capacitance is given by:

$$I = \frac{E}{(1/j\omega C)} = j\omega CE$$

or $$\angle I = \angle E + \frac{\pi}{2}$$

The current is 90° ahead by the voltage as is shown in figure (3.5).

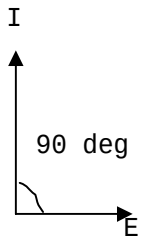


Fig. 3.5

The impedance of a series combination of a resistance *R* and an inductance *L* is given by $Z = R + j\omega L$ or $|Z| = \sqrt{R^2 + (\omega L)^2}$

The phase angle θ the impedance makes with the resistive component *R* is given by: $\theta = \tan^{-1}(\frac{\omega L}{R})$

The impedance Z is written as: $Z = |Z|.\angle \tan^{-1}(\frac{\omega L}{R})$

The current flowing through this combination is given by:

$$I = \frac{E}{Z} = \frac{E}{R + j\omega L} \quad \text{or} \quad \angle I = \angle E - \tan^{-1}(\frac{\omega L}{R})$$

The current is lagging behind with the voltage by an angle $\tan^{-1}(\frac{\omega L}{R})$.

The impedance of a series combination of a resistance *R* and an capacitance *C* is given by $Z = R + \frac{1}{j\omega C}$ or $|Z| = \sqrt{R^2 + (1/\omega C)^2}$

The phase angle θ the impedance makes with the resistive component *R* is given by: $\theta = \tan^{-1}(\frac{1}{\omega CR})$

The impedance *Z* is written as: $Z = |Z|.\angle \tan^{-1}(\frac{1}{\omega CR})$

The current flowing through this combination is given by:

$$I = \frac{E}{Z} = \frac{E}{R + (1/\ j\omega C)} = \frac{E.(j\omega C)}{1 + j\omega CR} \quad \text{or} \quad \angle I = \angle E + \frac{\pi}{2} - \tan^{-1}(\omega CR)$$

The phase difference between current and voltage is $\frac{\pi}{2} - \tan^{-1}(\omega CR)$.

Example 3.5 An A.C. signal of 20 volts and frequency 200Hz as applied to a circuit consisting of 10 *mH* inductance and 10 Ω resistance in series with it. Find the magnitude and phase of the current.

Solution: The impedance *Z* of series combination is given by:

$$Z = R + j2.\pi f.L = 10 + jx2x3.14x200x10x10^{-3} = 10 + jx12.56$$

$$|Z| = \sqrt{(10)^2 + (12.56)^2} = 16\Omega \quad \text{and} \quad \theta = \tan^{-1}(\frac{12.56}{10}) = 51.47^0$$

Current $|I| = \frac{20}{16} = 1.25amp$

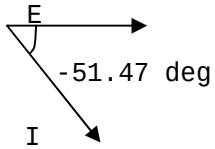


Fig. 3.6

And phase difference $\varphi = \angle I - \angle V = -51.47^0$

Example 3.6 Consider a series R – C circuit having R = 1.5$K\Omega$ and C = 0.2μF is excited by a sinusoidal signal of 20 volts and frequency 2 *KHz*. Find the magnitude and phase of the current.

Solution: The impedance of the R – C circuit is given by:

$$Z = R - \frac{j}{\omega C} = 1500 - \frac{j}{2x3.14x2000x0.2x10^{-6}}$$

$$= 1500 - \frac{j10^4}{25.12} = 1500 - j398.1$$

$$|Z| = \sqrt{(1500)^2 + (398.1)^2} = 1552\Omega$$

$$\theta = -\tan^{-1}(\frac{398.1}{1500}) = -14.86^0$$

So $Z = 1552\Omega\angle -14.86^0$

The current is given by: $I = \frac{20}{1552\angle -14.86^0} = 12.9mA\angle 14.86^0$

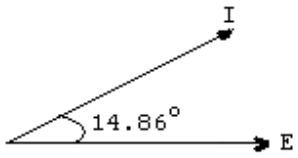


Fig. 3.7

The phasor diagram is given in figure 3.7.

Example 3.7 Consider a sinusoidal signal of peak value of 20 volts with frequency of 2000 *radians/sec* is applied to a circuit shown in figure (3.8). L=20 *mH*, C_1=C_2=0.2 μF, R=100 Ω . Calculate the total current I and the currents I_1 & I_2 in the two branches.

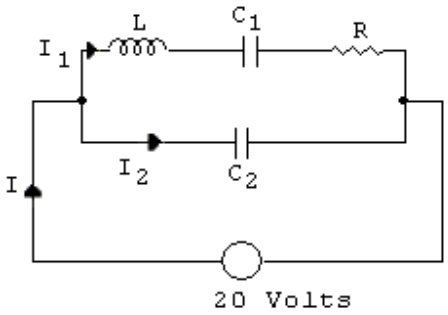


Fig. 3.8

Solution: Impedance Z_1 of the series branch is given by:

$$Z_1 = R + j(\omega L - \frac{1}{\omega C_1}) = 100 + j(2000x20x10^{-3} - \frac{1}{2000x0.2x10^{-6}})$$

$$= 100 + j(40 - \frac{10^4}{4}) = 100 - j(2460) = 2.46K\Omega\angle -87.67^0$$

Current I_1 in this series branch is given by:

$$I_1 = \frac{20}{Z_1} = \frac{20}{2.46K\Omega\angle -87.67^0} = 8.13mA\angle 87.67^0$$

Impedance of the capacitor branch $Z_2 = \frac{-j}{\omega C_2} = \frac{-j}{2000x0.2x10^{-6}} = \frac{-j10^4}{4}$

$$= -2500\,j = 2.5K\Omega\angle -\pi/2$$

Current I_2 in this branch is given by:

$$I_2 = \frac{20}{2.5K\Omega\angle -\pi/2} = 8mA\angle \pi/2$$

Admittance of Z_1 & Z_2 impedances is given by:

$$Y = \frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} = \frac{1}{2.46K\Omega\angle -87.67^0} + \frac{1}{2.5K\Omega\angle -\pi/2}$$

$$= 0.00041\angle 87.67^0 + 0.0004\angle \pi/2$$
$$= [0.000017 + j0.00081] + [j0.0004] = 0.000017 + j0.00081$$
$$= 0.00081\angle 88.8^0$$

Total impedance $Z = \frac{1}{0.00081\angle 88.8^0} = 1.23K\Omega.\angle -88.8^0$

Total current $I = \frac{20}{1.23K\Omega.\angle -88.8^0} = 16.1mA\angle 88.8^0$

3.3 R – L Low pass filter: Consider a circuit shown in figure (3.9), in which an A.C. signal *E* is applied across the series combination of resistance *R* and Inductance *L*.

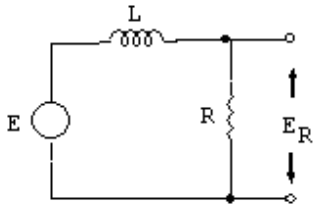


Fig. 3.9

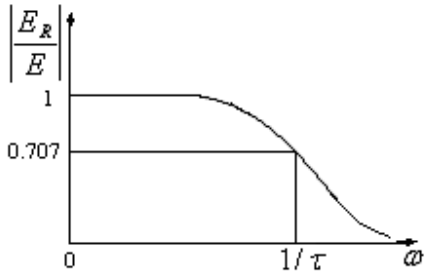
The output is taken across the resistance R, which is given by:

$$E_R = \frac{E.R}{(R + j\omega L)} = \frac{E}{(1 + \frac{j\omega L}{R})} \qquad \text{------ (3.19)}$$

The quantity $\frac{L}{R}$ has the dimension of time and is represented by τ.

So $$|E_R| = \frac{|E|}{\sqrt{(1+\omega^2\tau^2)}} \qquad \text{or} \qquad \frac{|E_R|}{|E|} = \frac{1}{\sqrt{(1+\omega^2\tau^2)}} \qquad \text{------ (3.20)}$$

Now a graph between $\left|\frac{E_R}{E}\right|$ and the frequency ω is plotted as shown in figure (3.10). This graph is known as the frequency response curve of the circuit.



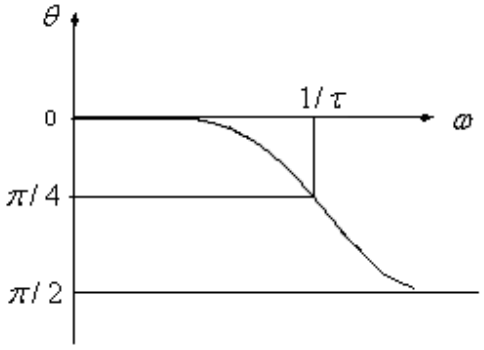


Fig. 3.10 (a) (b)

From this graph it is clear that if the input frequency is low enough then the output of the circuit will be almost the same as the input signal; and if the frequency of the input signal is high enough then the output is attenuated i.e. only a small amount of input reaches at the output. Hence the circuit behaves like a low pass filter. In other words we may say that this circuit is suitable to separate the signal of low frequency from a mixer of signals of low and high frequencies. The best use of this circuit is to detect a signal of low frequency from the high frequency noise. Its behaviour may also be explained if the reactance of the inductance is considered. It offers low reactance at low frequency and acts as an on switch, thus allows the input signal to pass to the output. However, it offers high reactance for the high frequency acting as an open switch

resulting thereby an attenuation of the input signal. In the equation (3.20), if $\omega = \frac{1}{\tau} = \frac{R}{L}$, then $\left|\frac{E_R}{E}\right|$ will be equal to $\frac{1}{\sqrt{2}}$ or 0.707, which is known as cutoff frequency ω_0 or $-3db$ (decibel)[1] point.

The voltage gain in *db* is defined as:

$$\text{Voltage gain (in } db) = 20 \log_{10}(A_V)$$

If $A_V = \frac{1}{\sqrt{2}}$ then $20\log_{10}(A_V) = 20\log_{10}(\frac{1}{\sqrt{2}}) = -\frac{20}{2}\log_{10} 2 = -3db$.

The cutoff frequency is given by: $\omega_0 = \frac{1}{\tau} = \frac{R}{L}$ *radians/sec* or $f_0 = \frac{1}{2\pi}.\frac{R}{L}$ *Hz.*

Putting the value of ω_0 or f_0 in equation (3.20), it is obtained:

$$\frac{|E_R|}{|E|} = \frac{1}{\sqrt{[1+(\omega/\omega_0)^2]}} = \frac{1}{\sqrt{[1+(f/f_0)^2]}}$$

So beyond cutoff frequency ω_0 or $-3db$ point, the gain (ratio of the output signal to the input signal) of the circuit decreases and becomes zero as $\omega \to \infty$. From this equation it is clear that when the frequency is increased by 10 fold beyond cutoff ($\omega/\omega_0 = 10$), the gain decreases 10 times.

So $$\frac{|E_R|}{|E|} = \frac{1}{\sqrt{[1+(\omega/\omega_0)^2]}} = \frac{1}{\sqrt{1+100}} \cong \frac{1}{10} \text{ or } 20db$$

As $20\log_{10}(\frac{1}{10}) = -20\log_{10} 10 = -20db$); or when the frequency is increased by two fold ($\omega/\omega_0 = 2$) the gain decreases 2 times ($\frac{|E_R|}{|E|} = \frac{1}{\sqrt{[1+(\omega/\omega_0)^2]}} = \frac{1}{\sqrt{1+4}} \cong \frac{1}{2}$) or $6db$ ($20\log_{10}(\frac{1}{2}) = -20\log_{10} 2 = -6db$). In other words the gain rolls off beyond cutoff at the rate of $20db/decade$ or $6db/octave$; octave signifies a two fold increase in frequency.

The phase relation between the input and the output may be obtained from the equation (3.19) as: $\angle E_R = \angle E - \tan^{-1}(\frac{\omega L}{R})$ or the phase difference between the output and input is given by:

---

[1] Fro more details please see appendix – I.

$$\theta = -\tan^{-1}(\frac{\omega L}{R}) = -\tan^{-1}(\omega/\omega_0)$$

The graph plotted between θ and ω, known as phase response curve is shown in figure (3.10b). From this curve it is clear that the phase difference is equal to ($-\frac{\pi}{4}$) at the cutoff frequency.

3.4 R – C Low pass filter: Low pass filter can also be designed using the series combination of resistance R and capacitance C as shown in figure (3.11).

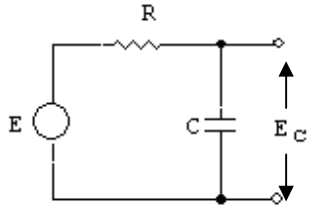


Fig. 3.11

The output voltage is taken across the capacitance *C*, which is given by:

$$E_C = \frac{E.(1/j\omega C)}{(R + \frac{1}{j\omega C})} = \frac{E}{(1 + j\omega CR)}$$

The value of *CR* has the dimension of time and is represented by τ.

So $\frac{E_C}{E} = \frac{1}{(1 + j\omega\tau)}$ or $\left|\frac{E_C}{E}\right| = \frac{1}{\sqrt{(1+\omega^2\tau^2)}}$ ------ (3.21)

This equation is identical with equation (3.20), hence frequency response and phase response curves will be the same as that of R-L low pass filter. This circuit is therefore called as R – C low pass filter. The cut off frequency is given by:

$\omega_0 = \frac{1}{\tau} = \frac{1}{CR}$ radians/sec or $f_0 = \frac{1}{2\pi CR}$ Hz.

The phase difference between the output and input is given by:

$$\theta = -\tan^{-1}(\omega\tau) = -\tan^{-1}(\omega CR) = -\tan^{-1}(\omega/\omega_0)$$

So at $\omega = \omega_0$ $\theta = (-\pi/4)$.

Example 3.8 Find the value of capacitance in the *RC* low pass filter to obtain the cut off frequency of 1.5 *KHz*. The value of the resistance is given as *R* = 2 *K*Ω.

Solution: The cut off frequency is given by $f_0 = \frac{1}{2\pi RC}$

or $C = \frac{1}{2\pi R f_0} = \frac{1}{2\,x\,3.14\,x1500\;\;x\,2000} = \frac{10^{-6}}{18.84} = .05\,\mu F$

3.5 R – L High pass filter: A series combination of a resistance and an Inductance L can also act as High pass filter if the output voltage is taken across inductance in place of resistance as shown in figure (3.12).

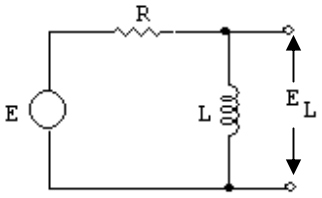


Fig. 3.12

The output voltage across the inductance *L* is given by:

$$E_L = \frac{E.(j\omega L)}{(R + j\omega L)} = \frac{E(\frac{j\omega L}{R})}{(1 + \frac{j\omega L}{R})} \qquad \text{or} \qquad \frac{E_L}{E} = \frac{(j\omega\tau)}{(1 + j\omega\tau)} \qquad \text{------ (3.22)}$$

as *L/R* has the dimension of time denoted by τ.

So $$|E_L| = \frac{|E|.(\omega\tau)}{\sqrt{(1+\omega^2\tau^2)}} \qquad \text{or} \qquad \frac{|E_L|}{|E|} = \frac{\omega\tau}{\sqrt{(1+\omega^2\tau^2)}} \qquad \text{------ (3.23)}$$

Now a graph between $\left|\frac{E_L}{E}\right|$ and the frequency ω is plotted as shown in figure (3.13a). From this graph it is clear that if the input frequency is high enough then the output of the circuit will be almost the same as the input signal; and if the frequency of the input signal is low enough then the output is attenuated i.e. only a small amount of input reaches at the output. Hence the circuit behaves like a high pass filter. This circuit can be used to detect a signal of high frequency from the low frequency noise. Its behaviour may also be explained if the reactance of the inductance is considered. It offers low reactance at low frequency and acts as an on switch, thus the output across the inductance is almost negligibly small. However, it offers high reactance for the high frequency, the output works like an open circuited and the output is same as the input.

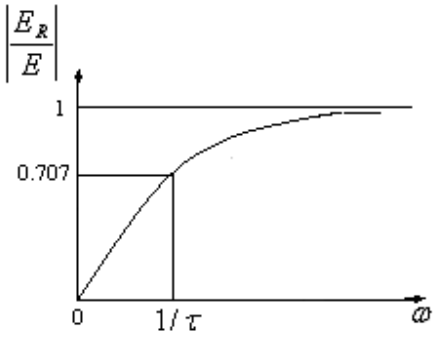
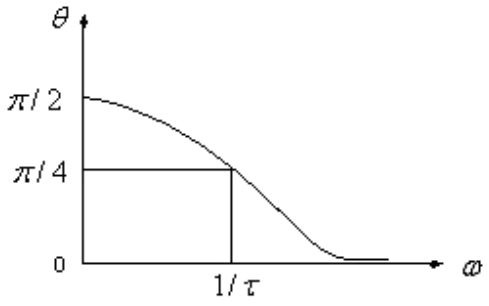


Fig. 3.13 (a)



(b)

In the equation (3.23), if we put $\omega = \frac{1}{\tau} = \frac{R}{L}$ then $\left|\frac{E_L}{E}\right|$ will be equal to $\frac{1}{\sqrt{2}}$ or 0.707, which is known as higher cutoff frequency ω_0 or – 3db point. It can be shown that the gain of this circuit increases at the rate of 20db/decade below the cut off frequency. At the cut off frequency $\left|\frac{E_L}{E}\right|$ is given by:

$$\frac{|E_L|}{|E|} = \frac{(\omega/\omega_0)}{\sqrt{[1+(\omega/\omega_0)^2]}} = \frac{(f/f_0)}{\sqrt{[1+(f/f_0)^2]}} \qquad \text{------- (3.24)}$$

The phase relation between the input and the output may be obtained from the equation (3.19) as: $\angle E_L = \angle E + \frac{\pi}{2} - \tan^{-1}(\frac{\omega L}{R})$ or the phase difference between the output and input is given by:

$$\theta = \frac{\pi}{2} - \tan^{-1}(\frac{\omega L}{R}) = \frac{\pi}{2} - \tan^{-1}(\omega/\omega_0) = \frac{\pi}{2} - \tan^{-1}(f/f_0)$$

The graph plotted between θ and ω, known as phase response curve is shown in figure (3.13 b). From this curve it is clear that the phase difference is equal to $\frac{\pi}{4}$ at the cutoff frequency.

3.6 R – C High pass filter: High pass filter may also be designed using the series combination of resistance R and capacitance C as shown in figure (3.14).

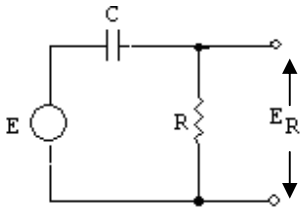


Fig. 3.14

The output voltage is taken across the resistance R, which is given by:

$$E_R = \frac{E.R}{(R + \frac{1}{j\omega C})} = \frac{E.j\omega CR}{(1 + j\omega CR)}$$

The value of CR has the dimension of time and is represented by τ.

So $$\frac{E_R}{E} = \frac{j\omega\tau}{(1 + j\omega\tau)} \quad \text{or} \quad \left|\frac{E_R}{E}\right| = \frac{\omega\tau}{\sqrt{(1+\omega^2\tau^2)}} \qquad \text{------ (3.25)}$$

This equation is identical with equation (3.23), hence frequency response and phase response curves will be the same as that of R-L High pass filter. This circuit is,

therefore, called as R – C High pass filter. The cut off frequency is given by: $\omega_0 = \frac{1}{\tau} = \frac{1}{CR}$ radians/sec or $f_0 = \frac{1}{2\pi CR}$ Hz.

The phase difference between the output and input is given by:

$$\theta = \frac{\pi}{2} - \tan^{-1}(\omega\tau) = \frac{\pi}{2} - \tan^{-1}(\omega/\omega_0) = \frac{\pi}{2} - \tan^{-1}(f/f_0)$$

3.7 Band Pass Filter or Series R – L – C Circuit: Series combination of R, L, & C elements connected to an A.C. signal as shown in figure (3.15) behaves as Band pass filter.

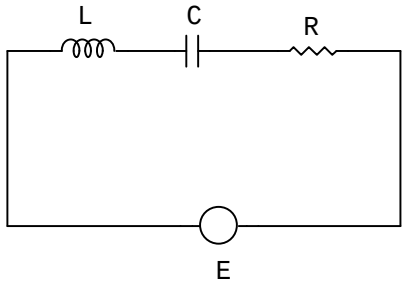


Fig. 3.15

The impedance of the circuit is given by:

$$Z = R + j\omega L + \frac{1}{j\omega C} = R + j(\omega L - \frac{1}{\omega C}) \qquad \text{------- (3.26)}$$

The series R-L-C circuit is said to resonant if the current and voltage are in the same phase. This is possible when the impedance becomes purely resistive or reactive components are zero.

i.e. $\omega L - \frac{1}{\omega C} = 0$ or $\omega L = \frac{1}{\omega C}$

$\omega^2 = \frac{1}{LC}$ or $\omega = \frac{1}{\sqrt{LC}}$

This frequency is known as resonance frequency denoted by $\omega_0 = \frac{1}{\sqrt{LC}}$ or $f_0 = \frac{1}{2\pi\sqrt{LC}}$. So one may say that at resonance frequency the impedance is minimum (resistive), consequently the current in the circuit is maximum, i.e. $I_{max} = \frac{E}{R}$. At frequency lower than the resonance frequency, the capacitive reactance is large compared

to the inductive reactance ($\frac{1}{\omega C} > \omega L$) and thus total reactance is capacitive in nature. At frequency higher than the resonance frequency, inductive reactance is large compared to the capacitive reactance ($\omega L > \frac{1}{\omega C}$) and the circuit is inductive. The current flowing through the series R – L –C circuit is given by:

$$I = \frac{E}{R + j(\omega L - \frac{1}{\omega C})} = \frac{E.j\omega C}{(1-\omega^2 LC) + j\omega CR} \qquad \text{------ (3.27)}$$

$$|I| = \frac{|E|\omega C}{\sqrt{(1-\omega^2 LC)^2 + \omega^2 C^2 R^2}} \qquad \text{------ (3.28)}$$

The phase relation between voltage and current is given by:

$$\theta = \angle I - \angle E = \frac{\pi}{2} - \tan^{-1}\left[\frac{\omega CR}{(1-\omega^2 LC)}\right] \qquad \text{------ (3.29)}$$

The frequency and phase response curve are shown in figure (3.16).

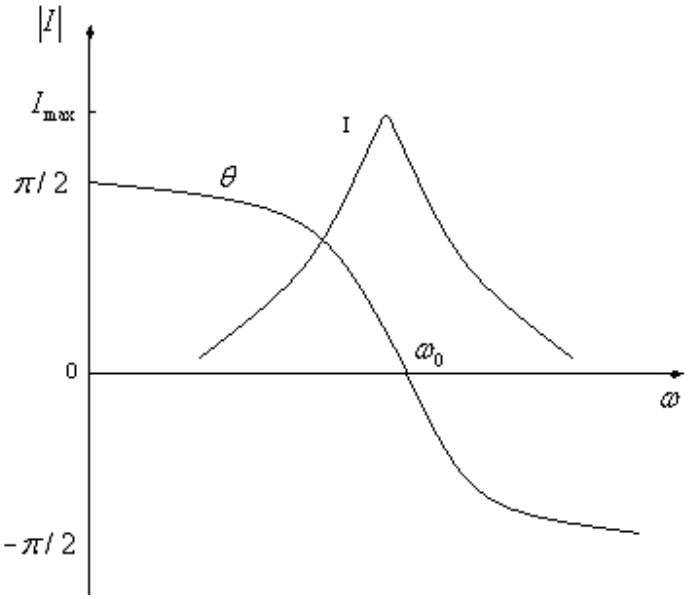


Fig. 3.16

From this figure it is clear that the current is maximum at the resonant frequency ω_0, it decreases in the similar fashion on either side of the resonant frequency ω_0. The circuit therefore, acts as band pass filter if we consider the voltage as the input and the current as the output. When the frequency is equal to ω_1 or ω_2, then the current is $\frac{I_{max}}{\sqrt{2}} = \frac{E}{\sqrt{2}R}$.

Thus $$\frac{E}{\sqrt{R^2+\left[\omega L-\frac{1}{\omega C}\right]^2}}=\frac{E}{\sqrt{2}R} \qquad \text{------ (3.30)}$$

or $$\sqrt{R^2+\left[\omega L-\frac{1}{\omega C}\right]^2}=\sqrt{2}R \qquad \text{------ (3.31)}$$

or $$R^2+\left[\omega L-\frac{1}{\omega C}\right]^2=2R^2$$

or $$\left[\omega L-\frac{1}{\omega C}\right]^2=R^2$$

or $$\left[\omega L-\frac{1}{\omega C}\right]=\pm R \qquad \text{------ (3.32)}$$

since $\omega_2 > \omega_1$, then $$\left[\omega_1 L-\frac{1}{\omega_1 C}\right]=-R \qquad \text{------ (3.33)}$$

and $$\left[\omega_2 L-\frac{1}{\omega_2 C}\right]=R \qquad \text{------ (3.34)}$$

Adding equations (3.33) & (3.34) we get $$(\omega_1+\omega_2)L-\frac{1}{C}\left(\frac{\omega_1+\omega_2}{\omega_1\omega_2}\right)=0$$

or $$\omega_1\omega_2=\frac{1}{LC}$$

Subtracting equation (3.33) from (3.34), one may get:

$$(\omega_2-\omega_1)L+\frac{1}{C}\left(\frac{\omega_2-\omega_1}{\omega_1\omega_2}\right)=2R$$

or $$(\omega_2-\omega_1)\left(L+\frac{1}{C\omega_1\omega_2}\right)=2R$$

or $$(\omega_2-\omega_1).2L=2R \qquad \text{or} \quad (\omega_2-\omega_1).=\frac{R}{L} \qquad \text{------ (3.35)}$$

$\omega_2-\omega_1$ is known as the Band Width ($\Delta\omega$) and is the band of frequency which lies between two points of either side of resonant frequency where the current falls to $\frac{1}{\sqrt{2}}$ of its resonant value.

The Q-factor of this series circuit is $$Q=\frac{\omega_0 L}{R} \qquad \text{------ (3.36)}$$

From equations (3.35) &(3.36), we have $$Q=\frac{\omega_0}{(\omega_2-\omega_1)}$$

or $Q = \frac{\omega_0}{\Delta\omega}$ or $Q = \frac{f_0}{\Delta f}$ ------ (3.37)

From this equation it is clear that greater is the value of quality factor Q, smaller is the bandwidth and more sharper is the resonance.

3.8 Band Rejection Filter or Parallel R – L – C Circuit: The parallel combination inductance L having a small leakage resistance R and capacitance C connected to an A.C. signal as shown in figure (3.17), forms a band rejection filter.

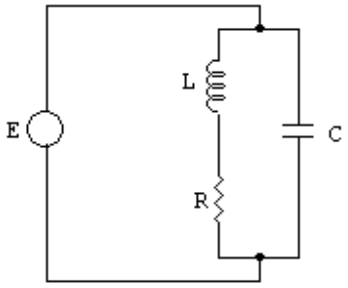


Fig. 3.17

The impedance of this circuit is given by:

$$Z = \frac{(R + j\omega L).(\frac{1}{j\omega C})}{\left[R + j\omega L + \frac{1}{j\omega C}\right]} = \frac{(R + j\omega L)}{\left[(1 - \omega^2 LC) + j\omega RC\right]} \quad \text{------- (3.38)}$$

The inductance L may assume to be of high quality factor, so $\omega L >> R$ and the impedance is approximated as:

$$Z = \frac{j\omega L}{\left[(1 - \omega^2 LC) + j\omega RC\right]} \quad \text{------ (3.39)}$$

$$|Z| = \frac{\omega L}{\sqrt{\left[(1 - \omega^2 LC)^2 + \omega^2 R^2 C^2\right]}} \quad \text{------ (3.40)}$$

The impedance of the circuit will be maximum and purely resistive if:

$1 - \omega^2 LC = 0$ or $\omega = \frac{1}{\sqrt{LC}} = \omega_0$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{------ (3.41)}$$

The circuit is said to be in resonance and ω_0 is called as resonance frequency. The impedance at resonance frequency is L/CR which is infinite for ideal inductance.

The current flowing through the parallel circuit will be small at resonance frequency and it increases on either side of the resonance frequency as shown in figure (3.18).

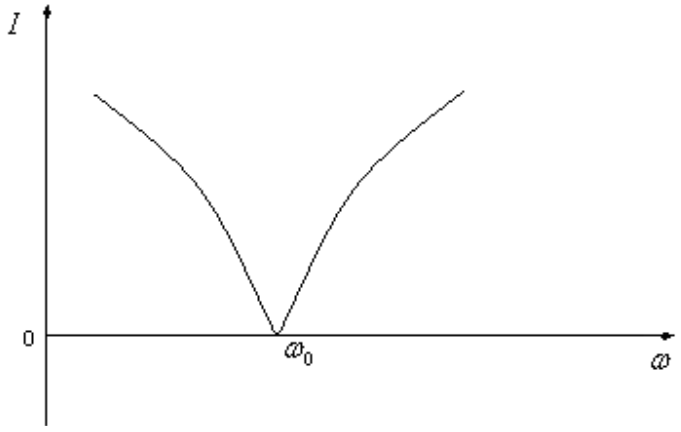


Fig. 3.18

It is clear from this figure that the circuit eliminates a particular band of frequency hence the name band rejection filter. At two frequencies ω_1 and ω_2 on either side of the resonance frequency, the magnitude of the impedance is $(1/\sqrt{2})$ times the impedance at resonance frequency.

i.e. $$\frac{\omega L}{\sqrt{(1-\omega^2 LC)^2 + \omega^2 C^2 R^2}} = \frac{L}{\sqrt{2}CR}$$

or $$\sqrt{(1-\omega^2 LC)^2 + \omega^2 C^2 R^2} = \sqrt{2}\omega CR$$

or $$(1-\omega^2 LC)^2 + \omega^2 C^2 R^2 = 2\omega^2 C^2 R^2$$

or $$1-\omega^2 LC = \omega CR \qquad \text{------ (3.42)}$$

This quadratic equation may be solved for ω which will have two roots given by:

$$\omega_1 = \left[\frac{1}{\sqrt{LC}}\right] - \frac{R}{2L} \qquad \text{and} \quad \omega_2 = \left[\frac{1}{\sqrt{LC}}\right] + \frac{R}{2L}$$

The band width $$\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L}$$

The quality factor Q of the circuit is given by

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0 L}{R} = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} = \sqrt{\frac{L}{C}} \cdot \frac{1}{R}$$

Example 3.9 Calculate the Q-factor and the Band width of the Band Rejection filter (Parallel Resonant Circuit). Given L = 200 *mH*, R = 20 Ω and C = 100 *pf*.

Solution: The resonance frequency f_0 is given by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2x3.14x\sqrt{200x10^{-3}\,x100x10^{-12}}} = \frac{1}{6.28x\sqrt{20x10^{-12}}}$$

$$= \frac{10^6}{6.25x4.47} = 35.6KHz$$

The quality factor Q is given by: $Q = \frac{\omega_0 L}{R} = \frac{2x3.14x35.6x10^3\,x200x10^{-3}}{20}$

$$= 20x3.14x35.6 = 2240$$

Band width BW $BW = \frac{f_0}{Q} = \frac{35.6x10^3}{2240} = 15.9Hz$

3.9 Transient Response: when an inductance or a capacitor is connected to a d.c. source, the inductance or capacitor starts charging and the voltage or current is build up gradually in the element. Ultimately it attains the steady state. Now when the source is switched off, the current or voltage stored in the elements gradually deceases. The process of gradually increasing and decreasing of the current or voltage in the inductance or capacitance is known as the transient response of the circuit. We shall now study the transient response of R – C and R – L circuits.

3.9.1 Transient Response of R – C circuit: Let us consider a circuit consisting of a resistance R and a capacitance C connected to a d.c. source through a SPDT switch as shown in figure (3.19). The switch has two positions such that when it is thrown to position *A* as shown in figure (3.19a) the capacitor starts charging through *R* with the source *E*. When the switch is thrown to position *B* as shown in figure (3.19b) the capacitor starts discharging.

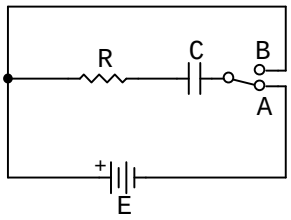
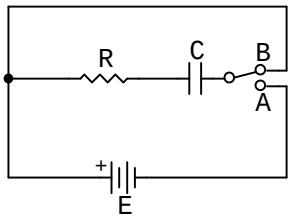


Fig. 3.19 (a)



(b)

(i) Charging of Capacitance: Initially the switch is thrown to position *A*, the capacitor will start charging through *R*. The current *I* flowing through the circuit is given by: $I = \frac{dq}{dt}$

Applying the KVL to the circuit we get:

$$RI + \frac{q}{C} = E \quad \text{or} \quad R\frac{dq}{dt} + \frac{q}{C} = E$$

or $R\frac{dq}{dt} = E - \frac{q}{C}$ or $dt = \frac{Rdq}{E - \frac{q}{C}} = -RC\frac{-(1/C)dq}{E - \frac{q}{C}}$

Integrating on both sides $\int dt = -RC\int \frac{-(1/C)dq}{E - \frac{q}{C}}$

or $t = -RC\ln(E - \frac{q}{C}) + A$ ------ (3.43)

where A is a constant of integration, its value may be obtained form the initial conditions.
At time $t = 0$, $q = 0$ so $A = RC\ln E$ ------ (3.44)

From equation (3.43) and (3.44), $t = -RC\ln(E - \frac{q}{C}) + RC\ln E$

or $t = -RC\ln\left(\frac{E - \frac{q}{C}}{E}\right)$ or $-\frac{t}{RC} = \ln\left(\frac{E - \frac{q}{C}}{E}\right)$

or $\left(\frac{E - \frac{q}{C}}{E}\right) = e^{-t/RC}$ or $q = CE(1 - e^{-t/RC}) = q_0(1 - e^{-t/RC})$

(Where q_0 is the maximum charge, the capacitor can attain; $q_0 = CE$)

The current I is given by: $I = \frac{dq}{dt} = \frac{d}{dt}\left[q_0(1 - e^{-t/RC})\right]$

$= \frac{q_0}{RC}.e^{-t/RC} = \frac{E}{R}e^{-t/RC} = I_0 e^{-t/RC}$ ------ (3.45)

where $I_0 = \frac{E}{R}$, is the maximum or initial value of current.

At time $t = 0$, $I = I_0$

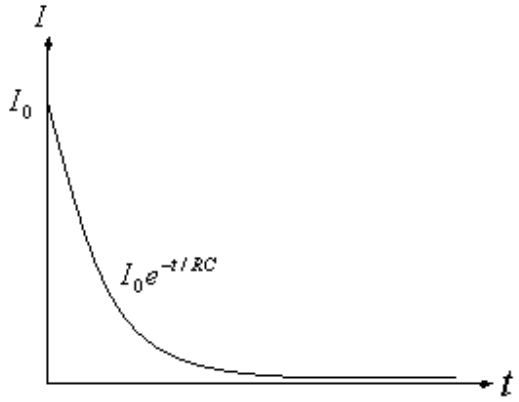


Fig. 3.20

and at $t = \infty$, $I = 0$. Figure (3.20) shows the transient response of R – C circuit, the current during the charging of capacitor *C*.

(ii) Discharging of Capacitance: After charging the capacitor, the switch is thrown to position *B* as shown in figure (3.19b). The capacitor will start discharging through resistance *R*.

Now applying the KVL to the circuit we get: $RI + \frac{q}{C} = 0$

or $R\frac{dq}{dt} + \frac{q}{C} = 0$ (since $I = \frac{dq}{dt}$)

or $\frac{dq}{q} = \frac{-dt}{RC}$ or $\int \frac{dq}{q} = \int \frac{-dt}{RC}$ ------ (3.46)

or $\ln q = \frac{-t}{RC} + B$ ------ (3.47)

(Where *B* is a constant of Integration which may be obtained from the initial conditions)

At $t = 0$, $q = q_0$ and thus $B = \ln q_0$ ------ (3.48)

From equations (3.47) & (3.48) it is obtained : $\ln q = \frac{-t}{RC} + \ln q_0$

$\ln \frac{q}{q_0} = \frac{-t}{RC}$ or $q = q_0 e^{-t/RC}$ ------- (3.49)

The current I is given by: $I = \frac{dq}{dt} = \frac{d}{dt}\left[q_0 . e^{-t/RC}\right] = \frac{-E}{R} e^{-t/RC}$

or $I = -I_0 e^{-t/RC}$ ------- (3.50)

where $I_0 = E/R$ is the initial value of current, at $t = 0$, $I = -I_0$ and at $t = \infty$, $I = 0$. Figure (3.21) shows the transient response of R – C circuit, the current during the discharging of capacitor *C*.

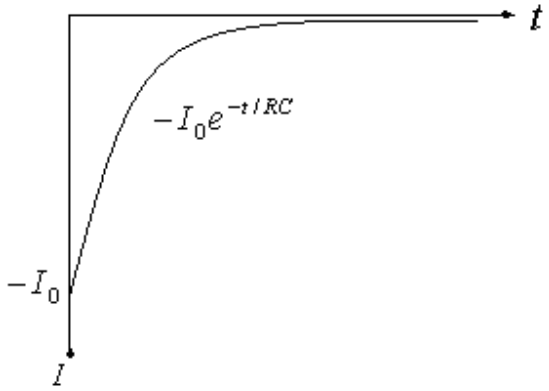


Fig. 3.21

3.9.2 Transient Response of R – L circuit: Let us consider a circuit consisting of a resistance *R* and an inductance *L* connected to a d.c. source through a SPDT switch as shown in figure (3.22). The switch has two positions such that when it is thrown to position *A* as shown in figure 3.22 a,

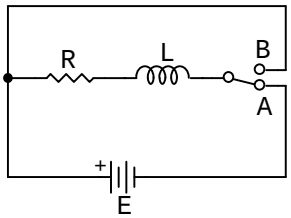
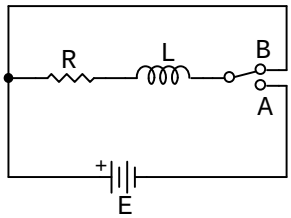


Fig. 3.22 (a)



(b)

the source *E* gets connected to the series combination of resistance and inductance *L*. When the switch is thrown to position *B* as shown in figure (3.22b), the current induced in the coil decays through the resistance.

(i) Rising of current: Initially the switch is thrown to position *A*, the current stats to flow and a flux is induced in the coil. The induced e.m.f. in the coil opposes the rising of the current in the coil. The current *I(t)* thus in the coil attain its steady value gradually. Applying KVL to the circuit we get:

$$L\frac{dI(t)}{dt} + RI(t) = E$$

or $$L\frac{dI}{dt} = E - RI \qquad \text{or} \qquad \frac{dI}{E - RI} = \frac{dt}{L} \qquad \text{------ (3.51)}$$

or $$\int \frac{dI}{E - RI} = \int \frac{dt}{L} \text{ or } \qquad -\frac{1}{R}\ln(E - RI) = -\frac{t}{L} + A$$

where *A* is a constant of Integration which may be obtained from the initial conditions;

at t = 0, I = 0 and $A = \frac{-1}{R}\ln E$ ------ (3.52)

From equations (3.51) and (3.52) we get:

$$-\frac{1}{R}\ln(E - RI) = -\frac{t}{L} - \frac{1}{R}\ln E \qquad \text{or} \qquad \ln\frac{(E - RI)}{E} = -\frac{R}{L}t$$

or $$\frac{RI}{E} = 1 - e^{-\frac{R}{L}t} \qquad \text{or} \qquad I = \frac{E}{R}\left(1 - e^{-(\frac{R}{L})t}\right)$$

$$I = I_0\left(1 - e^{-(\frac{R}{L})t}\right) \qquad \text{------ (3.53)}$$

I_0 is the maximum value of current which is equal to $\frac{E}{R}$. The rising of the current in the inductance is shown in figure (3.23).

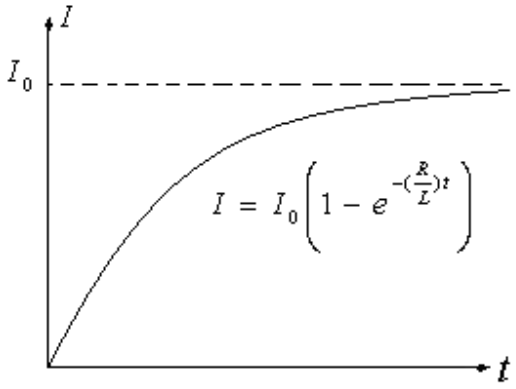


Fig. 3.23

(ii) Decay of current: After attaining the maximum value of current $I_0 = \frac{E}{R}$, the switch is thrown to position B as shown in figure (3.22 b). The current *I* in the circuit will decay from maximum value to zero. The equation (3.51) may be written as:

$$L\frac{dI(t)}{dt} + RI(t) = 0$$

or $L\frac{dI}{dt} = -RI$ or $\frac{dI}{I} = -\frac{R}{L}dt$ ------ (3.54)

or $\int\frac{dI}{I} = -\int\frac{R}{L}dt$ or $\ln I = -\frac{R}{L}t + B$

B is a constant of Integration which may be obtained from the initial conditions; at $t = 0$, $I = I_0$ and $B = \ln I_0$ ------ (3.55)

or $\ln I = -\frac{R}{L}t + \ln I_0$ or $\ln\frac{I}{I_0} = -\frac{R}{L}t$

or $I = I_0 e^{-(\frac{R}{L})t}$ ------ (3.56)

The decay of current is shown in figure (3.24).

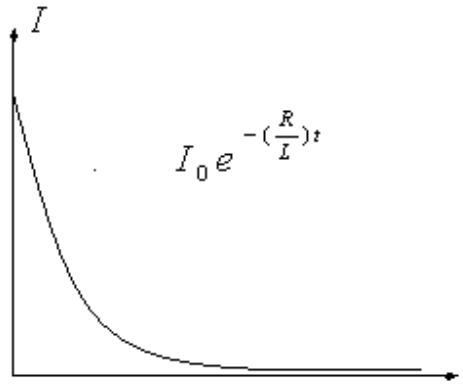


Fig. 3.24

Example 3.10 An inductance of 10 Henry and resistance 20 Ω is connected to d.c. signal of 20 volts. Calculate (i) steady current in the circuit, (ii) the rate of change of current at the instant of closing the circuit and (iii) time to get the 90% of the steady current.

Solution: The current in the R – L circuit is given by:

$$I = I_0\left(1 - e^{-(\frac{R}{L})t}\right)$$

(i) The steady state current $I_0 = \frac{E}{R} = \frac{20}{20} = 1 amp.$

(ii) $$\frac{dI}{dt} = \frac{R}{L}e^{-(R/L)t} \text{ and } \left.\frac{dI}{dt}\right|_{t=0} = \frac{R}{L}.e^{-0} = \frac{20}{10} = 2\,amp \text{ / sec}$$

(iii) R/L =20/10 =2

$$0.9I_0 = I_0\left(1 - e^{-(\frac{R}{L})t}\right)$$

$0.9 = 1 - e^{-2t}$ or $e^{-2t} = 0.1$

$e^{2t} = 10$ or $2t = \ln(10)$

$$t = \frac{\ln(10)}{2} = \frac{2.30}{2} = 1.15 Sec$$

3.10 Differentiating and Integrating Circuit: Differentiation and Integration with respect to time are the important operations of signal processing. Capacitance and Inductance may perform these operations if we define input and output variables properly. Capacitance is generally used for these operations as practical inductors have some resistance. We shall, therefore, discuss R – C differentiating and integrating circuits.

3.10.1 R – C Differentiating Circuit: Consider a capacitance *C* is charged with a voltage source E_I, then the current flowing through the capacitance is given by:

$$I = C\left(\frac{dE_I}{dt}\right) \qquad \text{------ (3.57)}$$

This equation clearly indicates that the current forms the differentiation of

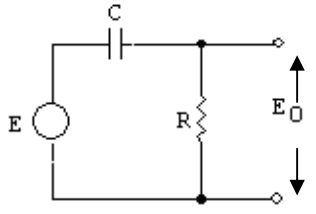


Fig. 3.25

To measure the current a resistance *R* is connected in series with the capacitance C as shown in figure (3.25). The voltage across the resistance is proportional to the

differentiation of the input signal which can be given by:

$$E_0 = RI = RC\left(\frac{dE_I}{dt}\right) \qquad \text{------ (3.58)}$$

This circuit may thus be called as the differentiating circuit. The R – C Differentiating Circuit is identical with the R – C high pass filter.

3.10.2 R – C Integrating Circuit: When a current *I* is passed through a capacitance, then the voltage across the capacitance E_0 is be given by:

$$E_0 = \frac{1}{C}\int Idt \qquad \text{------ (3.59)}$$

This voltage is proportional to the integration of the input current *I*. Now we consider the R – C network as shown in figure (3.26a), a signal *E*

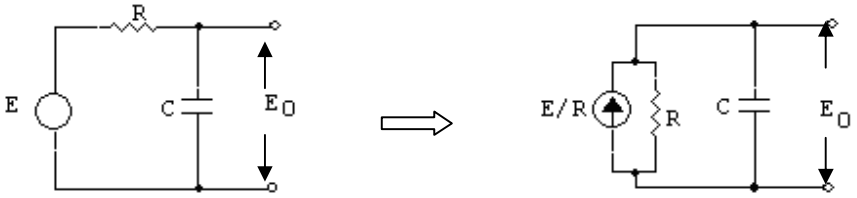


Fig. 3.26(a) (b)

is applied to the input terminals and voltage E_0 is measured across the capacitance *C*. The input signal *E* is converted to the current source *I* (conversion of Thevenin's equivalent to Norton's equivalent). The current *I* is given by: $I = E / R$

So $$E_0 = \frac{1}{C}\int(\frac{E}{R})dt = \frac{1}{RC}\int Edt \qquad \text{------ (3.60)}$$

The equation (3.60) clearly indicates that the output voltage across the capacitance is proportional to the input voltage *E*. Hence this circuit is called as an Integrator. The integrator circuit is identical to the R – C low pass filter.

Problems:

1. Prove that the Fourier Series expansion of the half wave rectified output is given by:

   $$E = \frac{E_m}{\pi} + \frac{E_m}{2}Sin\omega t - \frac{2E_m}{3\pi}Cos2\omega t - \frac{2E_m}{15\pi}Cos4\omega t - \frac{2E_m}{35\pi}Cos6\omega t.....$$

   where $f(t) = E_m Sin\omega t$ when $0 \le t \le T/2$

   $f(t) = 0$ when $(T/2) \le t \le T$
2. Prove that the Fourier Series expansion of the Full wave rectified output is given by:

   $$E = \frac{2E_m}{\pi} - \frac{4E_m}{3\pi}Cos2\omega t - \frac{4E_m}{15\pi}Cos4\omega t - \frac{4E_m}{35\pi}Cos6\omega t.....$$

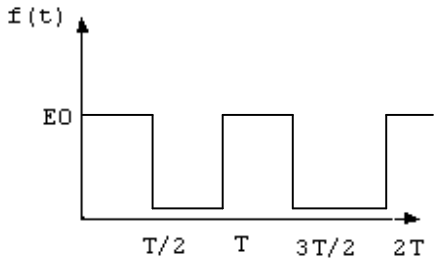
   where $f(t) = E_m Sin\omega t$ when $0 \le t \le T/2$

$$f(t) = -E_m Sin\omega t \quad \text{when} \quad (T/2) \le t \le T$$

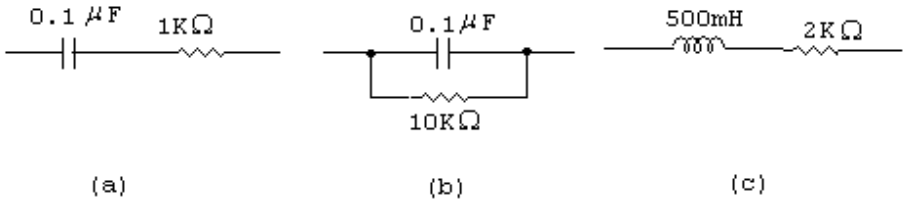
3. Consider a symmetrical triangular wave of peak value E_m and period T. Show that the Fourier Series expansion of the wave is given by:

$$E = \frac{E_m}{2} - \frac{E_m}{\pi} Sin\ \omega t - \frac{E_m}{2\pi} Sin\ 2\omega t - \frac{E_m}{3\pi} Sin\ 3\omega t .....$$

4. Find the Fourier series expansion of the square wave shown in figure given below.



5. Discuss (i) R – L low pass filter, (ii) R – C Low pass filter. Draw the frequency and phase response curves of these filters. Find the expression of cut-off frequency in each case.
6. Discuss (i) R – L High pass filter, (ii) R – C High pass filter. Draw the frequency and phase response curves of these filters. Find the expression of cut-off frequency in each case.
7. Explain the working of R – C integrator circuit.
8. Explain the working of R – C differentiator circuit.
9. Show that the output across the resistance in an R – C circuit is the differential of the input signal.
10. Show that the output across the capacitance in an R – C circuit is the integration of the input signal.
11. Discuss the transient response of an R – C circuit.
12. Discuss the transient response of an R – L circuit.
13. Show that the series R – L – C circuit behaves as a band pass filter. Draw the frequency and phase response curve of this series circuit. Find the expression for the band width also.
14. Show that the parallel R – L – C circuit behaves as a band rejection filter. Draw its frequency and phase response curve. Find the expression for the band width also.
15. Find the impedances of the following network at 1 KHz frequency.



Ans.(a) $1.88K\Omega\angle -57.9^0$, (b) $8.48K\Omega\angle -32.13^0$, (c) $3.72K\Omega\angle 57.5^0$

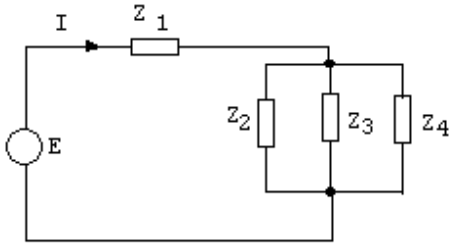
16. Three impedances $Z_1 = 125\angle -38^0$, $Z_2 = 213.5\angle 62^0$ and $Z_3 = 83\angle 54.5^0$ are connected in series. What is the total impedance of the circuit?

Ans. $305\Omega\angle -36^0$

17. A series R L circuit is excited by an sinusoidal voltage of magnitude 20volts and frequency 1.5KHz. If R = 2KΩ and L = 200 mH. Find the magnitude and phase of the current flowing through the circuit.

Ans. $7.2mA\angle -43.29^0$

18. Consider the circuit shown in figure given below, in which $Z_1 = 14.4\angle 45^0$, $Z_2 = 25\angle 53.1^0$, $Z_3 = 5\angle -53.1^0$ & $Z_4 = 10\angle 36.9^0$. Find (i) the impedance of the circuit, (ii) total amount of current drawn if the signal E is of 10 volts source, (iii) phase of the current I with respect to the applied voltage.



Ans.: $16.6\Omega\angle 31.87^0$, $0.6amp$, phase angle -31.87^0

19. A series RLC circuit is excited by a sinusoidal signal E =E_m Sin ωt. R= 20Ω , L = 20mH , C=0.2μF, E_m = 20 volts and ω = $2x10^4$ radians/sec. Find the magnitude and phase of the current.

Ans. $I = 53.3mA\angle -86.9^0$

20. An inductor of 200 mH and resistance 2Ω is connected to a d.c. source. Calculate the time in which the current attains half of its steady current after making the circuit.

Ans. 0.0693 Sec

21. A series LCR circuit has the following parameters; Quality factor Q = 120, C= 200 μF , L = 150 μH. Calculate the Band width.

Ans.7.7 KHz

4

Physics of Semiconductors

In this chapter the physical behaviour of semiconductors and semiconductor diodes has been discussed. Semiconductors have some useful properties and thus extensively being used in electronics. Semiconductor devices such as diodes, transistors, integrated circuits etc have brought a revolution in the modern world.

4.1 Semiconductors: The elements whose resistivity or conductivity lies in between that of the conductors and insulators are known as semiconductors. The resistivity of good conductor (such as copper) is approximately $1.7x10^{-8}\,\Omega m$ and that of insulator (Glass) is $9x10^{11}\,\Omega m$. Germanium, Silicon, Selenium, Carbon etc has the resistivity $10^{-4}\,\Omega m$ to $10.5\,\Omega m$, which is quite high as compared to conductor (Copper) and is quite low as compared to insulator (glass). Hence these are known as semiconductors. The resistivity of the element is not the only factor which decides if the particular material is semiconductor. There are certain alloys which behave like the semiconductors i.e. their resistivity lies within the range of semiconductors. The alloys like Cadmium Sulphide (CdS), Lead Sulphide (PbS), Gallium arsenide (GaAs), etc are also semiconductors which are being used to fabricate the solid state devices.

Following are few of the essential characteristics of the semiconductors:

(i) Semiconductors have negative temperature co-efficient of resistance that is, its resistivity decreases with the increase in temperature.

(ii) The conductivity of the semiconductors can extensively be increased if the additional impurities of suitable metals are introduced in it.

The difference in behaviour of conductors, semiconductors and insulators can best be explained on the basis of energy band structure of solids. The range of energies possessed by an electron in a solid is known as band. The electron in the outer most orbit of an atom is known as valence electron. The energy band occupied by the valence electron is called as valence band. This band may be completely or partially filled. The valence band is the highest occupied band. The electrons which have left the valence band are called the conduction electrons and are weakly bound to the nucleus. The band occupied by these electrons is called the conduction band. Thus in the conduction band the electrons can move freely and will be responsible for the current flow. The conduction band is separated from the valence band by certain energy gap which has no allowed energy levels. This gap between the valence and the conduction band is known

as bad gap or forbidden gap. Figure (4.1) shows the energy band picture of insulators, conductors and semiconductors.

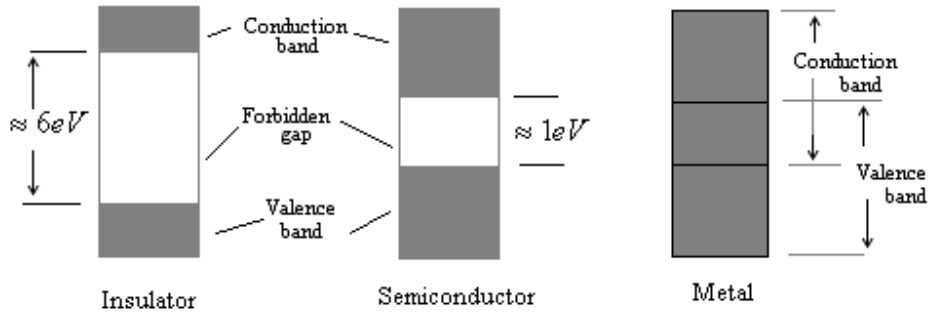


Fig. 4.1

It is clear from this figure that in case of insulators, there is a large forbidden gap between the valence and conduction bands. This band gap is of the order of several electron volts. In this case the valence band is full while the conduction band is empty. Therefore, a very high electric field is required for the electrons to move from valence band to conduction band. That is why the electrical conductivity of insulators is very poor.

In case of conductors the valence and conduction band overlap each other. Due to this overlapping a large number of free electrons are available in the conduction band and constitute an electric current. The conductors show the positive temperature coefficient of resistance i.e. the resistivity of the conductors increases with the increase of the temperature. This is because that the electrons are already in the conduction band and when the temperature of the conductors is increased; the electrons in the conduction band become thermally agitated and their energy is wasted in colliding with the other electrons in the conduction band.

The forbidden gap in case of semiconductors is of the order of one eV. Hence at room temperature some of the electrons in the valence band will have sufficient energy to jump from valence band to conduction band. In this way the conductivity of the semiconductors will be more than the insulators and smaller than the conductors. The semiconductors have almost full valence band and partially filled conduction band. The most commonly used semiconductors are germanium and silicon. The band gap energy of the germanium is 0.7 eV and 1.12 eV for silicon. The semiconductors show negative temperature coefficient of resistance; as the temperature is increased the more number of electrons will get into the conduction band which results the flow of current.

The semiconductors may be classified in to two categories:

(i) Intrinsic Semiconductors
(ii) Extrinsic or Doped Semiconductors

4.1.1 Intrinsic Semiconductors: Extremely pure form of a semiconductor is known as Intrinsic Semiconductor. The most commonly used semiconductors are Germanium and Silicon, which lies in the IV group of the periodic table. The atomic number of Germanium is 32, so it has 32 electrons, 2 electrons in the first orbit, 8 in the second orbit and 4 in the outer most orbit. Similarly the atomic number of Silicon is 14, and has 14 electrons; 2 electrons are in the first orbit, 8 electrons in the second orbit and 4 electrons in the outer most orbit. Both Germanium and Silicon have the crystalline structure.

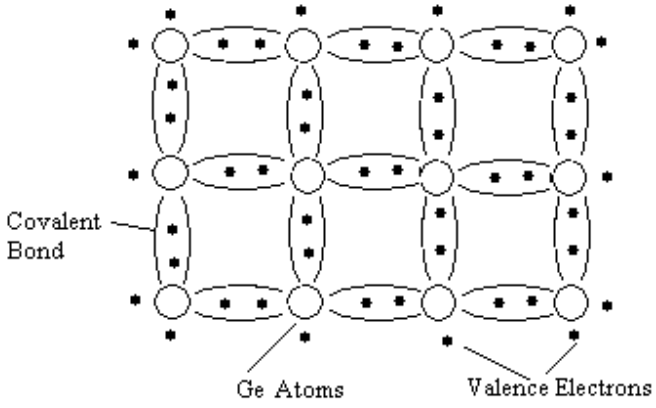


Fig. 4.2

Figure (4.2) shows that germanium (or silicon) atoms are held through covalent bonds. Each of the four electrons in a germanium (or silicon) atoms are shared by the valence electrons of four adjacent germanium (or silicon) atoms. The covalent bonds thus provide the binding force between the neighbouring atoms. The outer most orbits of atoms seem to be complete having 8 (shared) electrons. Consequently, the pure semiconductors (Ge or Si) behave as perfect insulator at 0^0 K. When the temperature is increased, some of the covalent bond is broken, the electrons are released and move to the conduction band. The empty space is left behind in the valence band. This empty space is known as hole and has the charge equal to that of an electron but opposite in sign. Thus each broken bond creates an electron-hole pair. The holes move through the crystal lattice in the random fashion as the free electrons and contribute to the current flow when an electric field is applied. Both these carriers drift in opposite directions giving rise to conventional current in the direction of flow of holes or in the direction opposite to the flow of electrons.

It is worthwhile to mention the following points with regards to the intrinsic semiconductor:

(1) When a covalent bond breaks an electron – hole pair is created. So in an intrinsic semiconductor the number of electrons and number of holes are equal.

(2) In the semiconductors the current flow is due to both the charge carriers unlike in the case of metals in which the current flow is only due electrons.

(3) At any finite temperature, some bonds break which result the generation of electron – hole pairs and some bonds may be reforming. The process of remaking the bond is known as recombination. At a given temperature, an equilibrium will be setup between the generation of electron – hole pairs and recombination of them. The hole concentration *p* must be equal the electron concentration *n*, so that $n = p = n_i$, where n_i is called the concentration of electron – hole pairs in the intrinsic or pure semiconductor.

(4) The carrier concentration in a semiconductor at a temperature is given by:

$$n_i^2 = A_0 T^3 e^{-Eg/KT} \qquad \text{------ (4.1)}$$

where A_0 is a constant for a given material, E_g is the band gap energy, *K* is the Boltzman constant and T is the temperature in Kelvin.

4.1.2 Extrinsic or Doped Semiconductors: If a small amount of impurity (one atom in 10^8 atoms) is added to the intrinsic semiconductor, it significantly increases the conductivity of the semiconductor. The semiconductor thus formed is known as extrinsic or doped semiconductor. The extrinsic semiconductors may further be classified in two categories depending upon the type of impurity being added to the intrinsic semiconductors.

(i) N – type semiconductor

(ii) P – type semiconductor

N – type semiconductor: When the atoms of V group (pentavalent) such as antimony, phosphorous or arsenic, having 5 electrons in its outer most orbit, are introduced as impurity in the intrinsic semiconductor, impurity atoms will displace few of the host atoms of the intrinsic semiconductor and form an N – type semiconductor. In the crystal structure four valence electrons of the impurity atoms will form the covalent bonds with four atoms of the intrinsic semiconductor and one electron finds no space in the covalent bond and remains free to move randomly in the crystal lattice. The impurity thus is called donor type impurity since each impurity atom donates a free electron to the crystal lattice. The semiconductor formed by introducing the donor type of impurity is called as N – type semiconductor. Figure (4.3) shows the crystal lattice of germanium with pentavalent impurity.

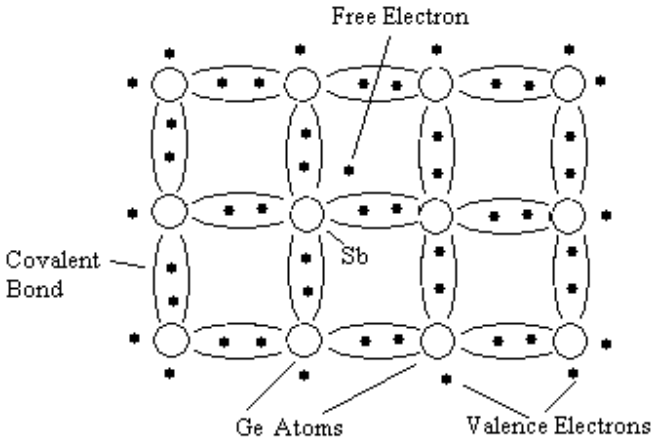


Fig. 4.3

Figure (4.4) shows the energy band diagram of N – type semiconductor. The addition of donor impurities in the intrinsic semiconductor introduces the allowable energy level just below the bottom of the conduction band. The gap between this level and the conduction band is very small ~ 0.01 eV in germanium (0.05 eV in Silicon); therefore, at room temperature almost all the donor electrons get into the conduction band. This increases the free electron concentration *n* in the crystal. This also reduces the hole concentration *p* due to recombination, as the rate of recombination of electrons with hole is increased. However, the product *np* remains constant. In this way the number of electrons in the crystal is more than the number of holes, i.e., the electrons are the majority charge carriers and holes are the minority charge carriers.

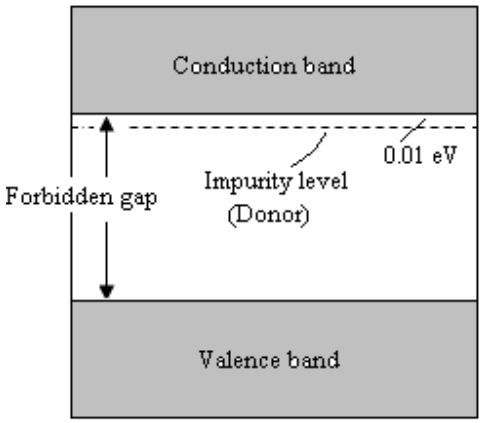


Fig. 4.4

P – type semiconductor: If on the contrary**,** the atoms of III group (trivalent) such as boron, gallium or indium, having 3 electrons in its outer most orbits, are introduced as impurity in the intrinsic semiconductor, impurity atoms will displace the atoms of the

intrinsic semiconductor and forms P – type semiconductor. In the crystal structure three valence electrons of the impurity atoms will form the covalent bonds with three atoms of the intrinsic semiconductor and the fourth covalent bond will be incomplete resulting thereby, the deficiency of an electron which constitutes a hole. The impurity thus is called the acceptor type impurity since it has the tendency to accept an electron in forming the covalent bond. The semiconductor formed by introducing the acceptor type of impurity is called as P – type semiconductor. Figure (4.5) shows the crystal lattice of germanium with trivalent impurity.

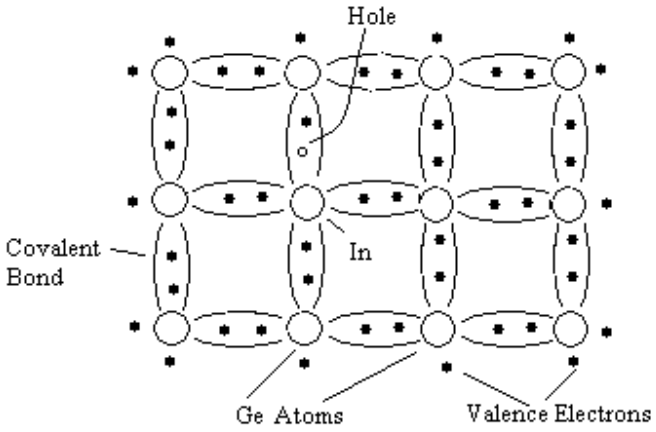


Fig. 4.5

Figure (4.6) shows the energy band diagram of P – type semiconductor. The addition of acceptor impurities in the intrinsic semiconductor introduces the allowable energy level just above the top of the valence band. A very small amount of energy is needed for an electron to leave the valence band and occupy the acceptor level. This increases the hole concentration *p* in the crystal. In this way the number of holes in the crystal is more than the number of electrons, i.e., the holes are the majority charge carriers and electrons are the minority charge carriers in P – type semiconductor. The product np, however, remains constant.

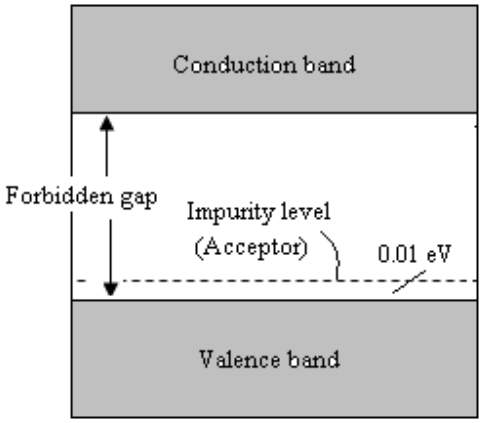


Fig. 4.6

4.2 Effect of Temperature on extrinsic Semiconductors: To discuss the effect of temperature on the extrinsic semiconductor, we consider an N – type semiconductor. At room temperature the electrons are the majority charge carriers and holes are the minority charge carriers. On increasing the temperature of the semiconductor, more electron hole pairs will be created. This will increase the population of holes in the semiconductor. A temperature may reach when number of broken covalent bonds is very large such that the number of electrons is almost equal to number of holes. The extrinsic semiconductor will then behave like an intrinsic semiconductor. Similarly it can be proved that the P – type semiconductor too will behave as intrinsic semiconductor at higher temperature.

4.3 Concentration of Holes and Electrons in Extrinsic Semiconductor: From the above discussion it is clear that the introduction of donor impurities in intrinsic semiconductor decreases the number of holes below that in the intrinsic semiconductor. Similarly the introduction of acceptor impurities in intrinsic semiconductor decreases the number of free electrons below that in the intrinsic semiconductor. A theoretical analysis reveals that under thermal equilibrium, the product of the free electron and hole concentration is constant. It is independent of the doping concentration. This relationship is called the law of mass action and is given by:

$$np = n_i^2 \qquad \text{------ (4.2)}$$

where n_i is the intrinsic concentration which is a function of temperature given by equation (4.1).

The densities of *n* and *p* may further be inter-related using the law of electrical neutrality. In an N – type semiconductor, each donor atom donates a free electron which moves randomly in the crystal lattice. The donor atom becomes a positive ion fixed in the lattice. Similarly in a P – Type semiconductor, each acceptor atom accepts one electron and becomes a negative ion. The hole created due to the acceptance of electron moves freely in the crystal lattice. Since the crystal as a whole must be electrically neutral, the positive and negative charge must be equal:

$$N_D + p = N_A + n \qquad \text{------ (4.3)}$$

Consider an N – type semiconductor having no acceptor atom ($N_A = 0$) and $n >> p$ (number of free electrons is much larger than the number of holes), equation (4.3) becomes: $n \approx N_D$ ------ (4.4)

In an N – type semiconductor, the number of free electrons is approximately equal to the density of donor atoms.

Similarly it can be proved that the number of holes in P – type semiconductor is equal to the density of acceptor atoms.

i.e., $p \approx N_A$ ------ (4.5)

To be more specific, the subscript *n* or *p* is introduced to the concentrations of free electrons *n* and holes *p*, to show whether these concentrations belong to N – type or P – type semiconductors

i.e. n_n and p_n show respectively the electron and hole concentrations in *N* – type semiconductor.

and n_p and p_p show respectively the electron and hole concentrations in P – type semiconductor.

Using these conventions, in N – type semiconductor $n_n = N_D$ and $p_n = (\frac{n_i^2}{N_D})$, since $n_n p_n = n_i^2$.

Similarly in P – type semiconductor $p_p = N_A$ and $n_p = (\frac{n_i^2}{N_A})$, since $n_p p_p = n_i^2$.

4.4 Currents in semiconductors: Two types of currents flow in the semiconductors (i) diffusion current & (ii) drift current.

(i) Diffusion Current: The diffusion phenomenon also comes into play in the semiconductors, due to which the charge carriers may diffuse from higher concentration to lower concentration. The movement of charge carriers will constitute the current, which is known as diffusion current. Consider a semiconductor in which carrier concentration is not uniform. Let the hole concentration p decreases with the increase of x. The holes will diffuse in the positive direction of x – axis which results the flow of diffusion current in the semiconductor. The current density for holes J_p will be proportional to the concentration gradient for holes ($\frac{\partial p}{\partial x}$), which is given by:

$$J_p = -qD_p(\frac{\partial p}{\partial x}) \qquad \text{------- (4.6)}$$

where q is electronic charge and D_p is the proportionality constant known as diffusion constant for holes. The negative sign in this equation indicates that the concentration gradient decreases with increase of x.

The equation for current density for electrons J_n may also be given in the similar fashion if the variation of electron concentration is considered in the semiconductor crystal.

$$J_n = qD_n(\frac{\partial n}{\partial x}) \qquad \text{------- (4.7)}$$

D_n is called as the diffusion constant for electrons and ($\frac{\partial n}{\partial x}$) is concentration gradient for electrons.

(ii) Drift Current: The other type of current is the drift current that would flow due to the movements of charge carriers in an applied electric field across the semiconductor. It is well known that if the field is not too large then the velocity v attained by the charge carriers (drift velocity) will be proportional to the applied electric field E i.e. $v \propto \mu E$ or $v = \mu E$. μ is called the mobility of the charge carriers, which may be defined as the velocity attained by the charge carriers in an unit electric field.

Let v_n and v_p are the drift velocities of electrons and holes respectively in the electric field E applied across the semiconductor, which are given by:

$$v_n = \mu_n E$$

and $$v_p = \mu_p E$$

where μ_n and μ_p are the mobility for electrons and holes respectively. Because of the drift velocities of electrons and holes in the semiconductor, the current called as the Drift current will flow. The drift current density for electrons J_n and holes J_p are given by:

$$J_n = nqv_n = nq\mu_n E \qquad \text{------ (4.8)}$$

$$J_p = pqv_p = pq\mu_p E \qquad \text{------ (4.9)}$$

where n and p are number of electrons and holes respectively. The total current density J is given by:

$$J = J_n + J_p = q(n\mu_n + p\mu_p)E$$

The overall conductivity of the semiconductor containing electrons and holes is given by: $$\sigma = \frac{J}{E} = q(n\mu_n + p\mu_p)$$

For intrinsic semiconductors $n = p = n_i$, so the conductivity of the intrinsic semiconductor is given by :

$$\sigma_{\text{int}} = qn_i(\mu_n + \mu_p)$$

Conductivity of the N – Type semiconductor is :

$$\sigma_{N-type} = qn\mu_n = q\mu_n N_D$$

N_D is the density of ionized donor atoms.

Conductivity of the P – Type semiconductor is :

$$\sigma_{P-type} = qp\mu_p = q\mu_p N_A$$

N_A is the density of ionized acceptor atoms.

The total current densities for electrons and holes are given by:

$$J_{Tn} = q(n\mu_n E + D_n \frac{\partial n}{\partial x})$$

and $$J_{Tp} = q(p\mu_p E - D_p \frac{\partial p}{\partial x}) \qquad \text{------- (4.10)}$$

4.5 Properties of Ge and Si: Some important properties of Ge and Si semiconductors are shown in form of table given below:

Table 4.1

Property	Symbols	Units	Ge	Si
Atomic No.	Z		32	14
Atomic Weight	W		72.6	28.1
Density	g	Kg/m^3	$5.32x10^3$	$2.33x10^3$
Atomic Concentration		atoms/ m^3	$4.4x10^{28}$	$5x10^{28}$
Dielectric constant	ε_r		16	12
Band gap at 0^0 K	E_{go}	eV	0.785	1.21
Band gap at 300^0 K	E_g	eV	0.72	1.1
Intrinsic carriers	n_i	$Carriers/m^3$	$2.5x10^{19}$	$1.5x10^{16}$
Intrinsic resistivity	ρ	Ohm-m	0.45	2300
Mobility of electrons	μ_n	m^2/sec-volt	0.38	0.13
Mobility of holes	μ_p	m^2/sec-volt	0.18	0.05
Diff. const. for electrons	D_n	m^2/sec	$9.9x10^{-3}$	$3.4x10^{-3}$
Diff. const. for holes	D_p	m^2/sec	$4.7x10^{-3}$	$1.3x10^{-3}$

It may be mentioned here that the band gap (E_g) of Ge or Si varies with temperature given by:

$Eg(T) = 1.21 - 3.60\ x10^{-4}.T$ for Si

$Eg(T) = 0.785 - 2.23\ x10^{-4}.T$ for Ge

where T is the temperature in *Kelvin.*

Example 4.1 (a) Find the resistivity of the intrinsic germanium crystal at 300^0 K. (b) What will be the resistivity of this germanium crystal if the donor impurity of 1 atom per 10^8 germanium atoms is introduced? Given that n_i = 2.5 $x10^{19}$ atoms/m^3, μ_n = 0.38 m^2/volt-sec, μ_p = 0.18 m^2/volt-sec, Atomic conc. of *Ge* = $4.4x10^{28}$ atoms/m^3 .

Solution: (a) The conductivity of the intrinsic germanium is given by:

$$\sigma_{\text{int}} = qn_i(\mu_n + \mu_p) = 1.6x10^{-19}\,x2.5x10^{19}(0.38+0.18)$$
$$= 1.6x2.5x5.6 = 2.24 \quad \text{mhos/m}$$

The resistivity of the intrinsic germanium is given by:

$$\rho = \frac{1}{\sigma} = \frac{1}{2.24} = 0.45 \quad \Omega\text{-}m$$

(b) $$N_D = \frac{4.4x10^{28}}{10^8} = 4.4x10^{20}$$

$$\sigma_{N-type} = q\mu_n N_D = 1.6\,x10^{19}\,x0.38\,x4.4\,x10^{20}$$
$$= 26.752 \quad mhos/m$$

$$\rho_{N-type} = \frac{1}{\sigma_{N-type}} = \frac{1}{26.752} = 0.037 \quad \Omega\text{-}m$$

Example 4.2 (a) Pure germanium crystal has the resistivity of 0.45 Ω-m. How much donor impurity should be added to Ge crystal so that its resistivity decreases to 10% of the original value? (b) Find the values of n and p in this N – type Ge crystal. Given that μ_n = 0.38 m^2/volt-sec, n_i = 2.5 x10^{19} atoms/m^3

Solution: (a) $\rho = \frac{1}{\sigma} = 0.045$ and $\sigma = q\mu_n N_D$

So $\frac{1}{0.045} = 1.6x10^{-19} x0.38 xN_D$

or $N_D = \frac{1}{0.045x1.6x10^{-19}x0.38} = \frac{10^{19}}{0.274}$

$= 3.65x10^{20}$ atoms/m^3

(b) $n \approx N_D = 3.65x10^{20}$ atoms/m^3

$p = \frac{n_i^2}{N_D} = \frac{(2.5x10^{19})^2}{3.65x10^{20}}$

$= \frac{6.25x10^{18}}{3.65} = 1.71x10^{18}$ atoms/m^3

Example 4.3 An N – type Ge crystal is .02m long and has a cross section of 0.002m x 0.002 m. A current of 10 mA flows through the crystal, when a 2 volt battery is applied across it. Find (a) doping concentration N_D and (b) Drift velocity of the charge carriers. (μ_n = 0.38 m^2/volt-sec)

Solution: (a) Resistance R of the Ge crystal is: $R = \frac{2volt}{10mA} = 200\Omega$

L = 0.02 m area A = 0.0002x0.0002 = 4x10^{-8} m^2

Resistivity of the crystal $\rho = \frac{RA}{L} = \frac{200x4x10^{-8}}{.02} = 4x10^{-4}$ Ω-m

Conductivity $\sigma = \frac{1}{\rho} = \frac{1}{4x10^{-4}} = 2500$ mhos/m

Since the resistivity of the crystal is very large so it is heavily doped crystal. The approximate formula may be used: $\sigma_{N-type} = qn\mu_n = q\mu_n N_D$

so $N_D = \frac{\sigma}{q\mu_n} = \frac{2500}{1.6x10^{-19}x0.38} = 4.11x10^{22}$ atoms/m^3

(b) Electric field $E = \frac{Volts}{length} = \frac{2}{0.02} = 100$ volts/m

Drift velocity of charge carriers

$v = \mu_n E = 0.38x100 = 38$ m/sec

4.6 P–N Junction Diode: The properties of *N* – type and *P* – Type semiconductors have been studied in the preceding sections. In this section the behaviour of the combination of the two types of extrinsic semiconductors will be discussed. If a *P* – type semiconductor in combined with the *N*– Type semiconductor such that the crystal structure is continuous, then a P-N junction is formed. The device thus formed is called a *P-N* junction diode or semiconductor diode. A useful *P – N* junction cannot be produced by simply placing the two semiconductors together or by welding etc., because it gives rise to discontinuous crystal structure. Special fabrication techniques are adopted to form a *P – N* junction diode.

To understand the working of a *P – N* junction diode, let us consider a hypothetical case where a *P* – type semiconductor is brought in physical contact with the *N* – type semiconductor to form a junction diode as shown in figure (4.7). Both the *P* – and *N* – type semiconductors are separately electrically neutral. On the left side of the junction, the P – type crystal consists of the majority charge carrier holes and immobile acceptor ions. The acceptor ion is indicated by negative sign, because after the acceptor atom accepts an electron it becomes a negative ion. Similarly on the left side of the junction, the *N* – type semiconductor consists of majority charge carriers as electrons and the immobile donor ions. The donor ion is represented by the positive sign, because after a donor atom donates an electron, it becomes a positive ion. In addition, the *P* – type crystal also has the electrons as the minority charge carriers and *N* – type semiconductor has the holes as minority charge carriers.

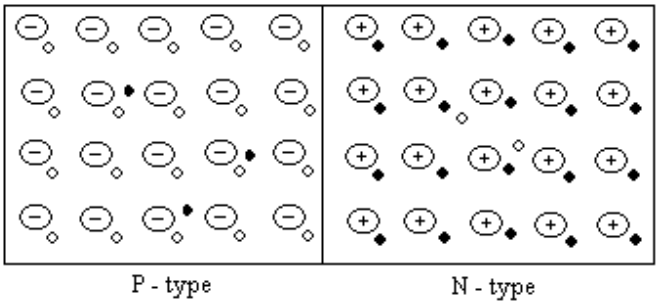


Fig. 4.7

A concentration gradient of holes will exist across the junction as the concentration of holes decreases from *P* – region to *N* – region. Similarly concentration gradient of electrons across the junction will also exist as the concentration of electrons decreases from *N* – region to *P* – region.

Because of the concentration gradients, holes will diffuse from *P* – region to *N* – region and electrons will diffuse from *N* – region to *P* – region. During the diffusion of holes from *P* to *N*, and diffusion of free electrons from *N* to *P*, some holes and some free electrons recombine. Each recombination leads the elimination of an electron – hole pair. In this process, the acceptor ions of *P* – region and donor ions of *N* – region in the neighbourhood of the junction are left uncompensated.

In the neighbourhood of the junction, the region which contains only uncompensated acceptor and donor ions, is called the depletion region, space charge

region or the transition region (fig. 4.8). The depletion region is devoid of mobile charge carriers. An electric field will develop across the junction due to these uncompensated ions, which will create a potential barrier or potential hill at the junction. The approximate value of the potential barrier is 0.3 *V* for *Ge* and 0.7 *V* for *Si* at room temperature. Further diffusion of the majority charge carriers from *P* to *N* and vice-versa is discouraged by the potential barrier. The total recombination of electron – hole pairs is, therefore, not possible. There will be some majority carriers in both the regions having sufficient energy to surmount the potential hill and may cross the junction. However, the potential barrier will encourage the drifting of minority charge carriers from one region to other. An equilibrium known as thermal equilibrium will be set up in the junction such that the drift of minority carriers across the junction is counter balanced by the diffusion of same number of majority carriers across the junction. It may be noted that no external battery has been applied across the junction so far or the junction is open circuited.

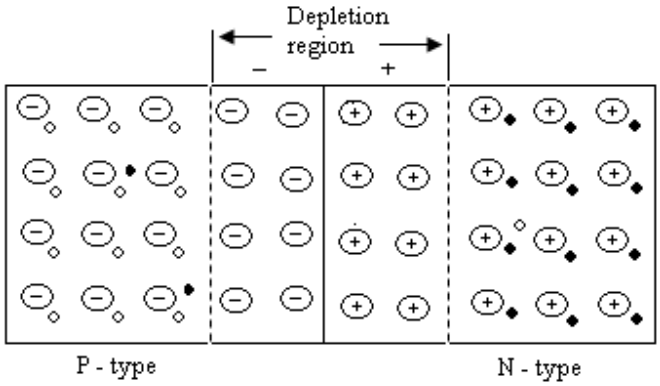


Fig. 4.8

Now we connect a battery such that positive terminal of the battery is connected to *P* – side of the diode and negative terminal of the battery to the *N* – side as shown in figure (4.9). The diode in this condition is said to be biased in forward bias. The reduction in the built-in potential is due to the applied voltage forcing more electrons into the n-type region and more holes into the p-type region, thus covering some of the fixed charges and narrowing the depletion layer. Since the total uncovered charge is reduced, the built-in potential must be lower. Remembering that the built-in potential opposes the flow of majority carriers across the junction, a reduction in the potential makes it easier for holes in the p-type region to cross the junction and for electrons in the n-type region to cross the junction in the opposite direction.

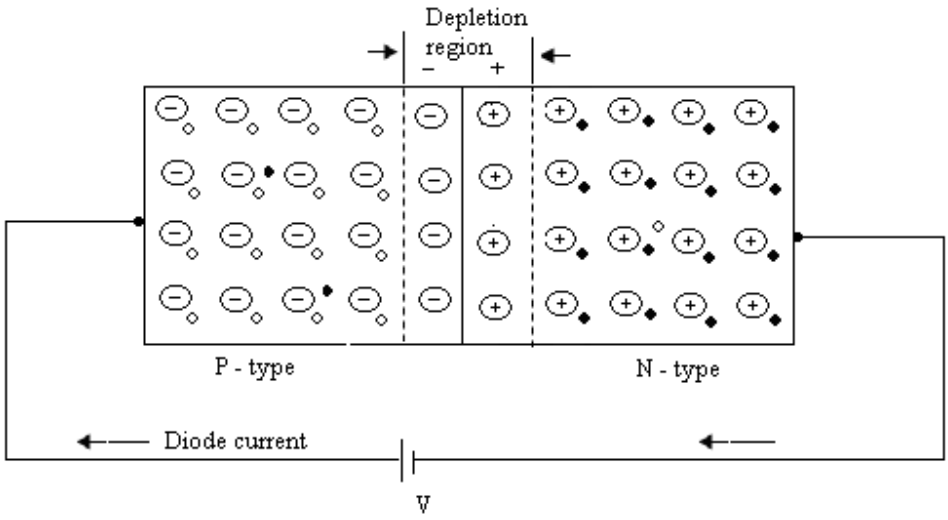


Fig. 4.9

In the *N* region where the electrons combine with the equal number of holes from the *p* region, then equal number of electrons arrives from the negative terminal of the battery and enters the *N* region to replace electrons lost by combination with holes near the junction. These electrons move towards the junction at the left, where these electrons combine with the new holes. This process thus goes on and large amount of current flows.

If the terminals of the battery connected to the diode is reversed that is positive terminal of the battery is connected to the *N* region and negative terminal to the *P* region, then the diode is said to be biased in the reverse bias. Figure (4.10a) shows the biasing of the diode in the reverse direction. In this condition majority charge carrier holes in *P* region are attracted towards the negative terminal of the battery. The majority charge carrier electrons in *N* region are attracted towards the positive terminal of the battery. Thus these majority charge carriers move away from the junction. This action widens the depletion region and increases the potential hill. The increased barrier potential almost completely stops the flow of majority charge carriers from one region to other. However, it will help the minority charge carriers to flow across the junction. The current that is possible to flow in the reverse bias is due to the minority charge carriers only. The minority charge carriers are minority in number hence a very small amount of current (in the range of 10^{-6} to 10^{-9} amperes) flows in this bias. The generation of minority charge carriers depends upon the temperature. So at the fixed temperature the reverse current is almost constant and is independent of potential of the external battery.

The symbolic representation of the *P N* junction diode is shown in the figure (4.10b). It is a two terminal device, the *P* side of which is known as anode and N side is called as cathode. The arrow in the symbol represents the direction of the conventional current to flow when the diode is in forward bias.

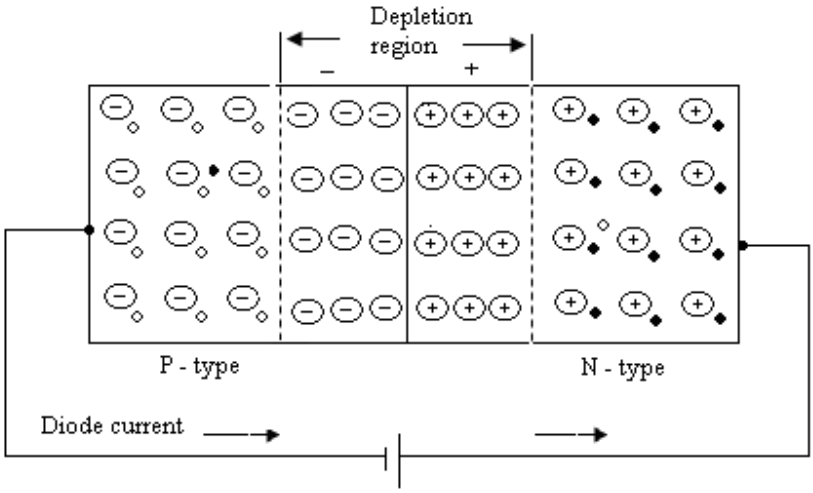


Fig 4.10 (a)

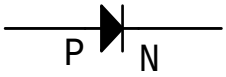


Fig. 4.10(b)

The V – I characteristic curve of the diode both in the forward and reverse bias is shown in figure (4.11). In the forward bias practically a very small current flows until the applied voltage is equal barrier potential of the diode (0.3 *V* for Ge and 0.7 *V* for Si). When there is small increase in the applied voltage beyond the barrier potential, a sharp increase in the current is observed. The voltage at which the current starts to increase rapidly is known as cut – in or knee voltage represented by V_γ. The effect of potential barrier is more pronounced when applied voltage is less then the barrier potential, hence a very small current. When the applied voltage is greater than the potential barrier of the diode, the effect of potential hill is nullified and current flows due to majority charge carriers, hence the large current.

In the reverse bias a very small amount of current flows due to the minority charge carriers. This current is known as reverse saturation current or leakage current. When the reverse voltage is increased beyond certain limit known as break down voltage, the minority carriers drifting across the junction acquire enough energy to ionize the atoms in the depletion region and produce more charge carriers; resulting a large current to flow. This region of the characteristics is called the avalanche break down. Ordinary diodes are never used in this break down region, since the excessive current in the diode rises its temperature to a very high value and thus the diode may be damaged. Special types of diodes known as Zener diodes are designed which work in the break down region only. These diodes are discussed in section 4.12 of this chapter.

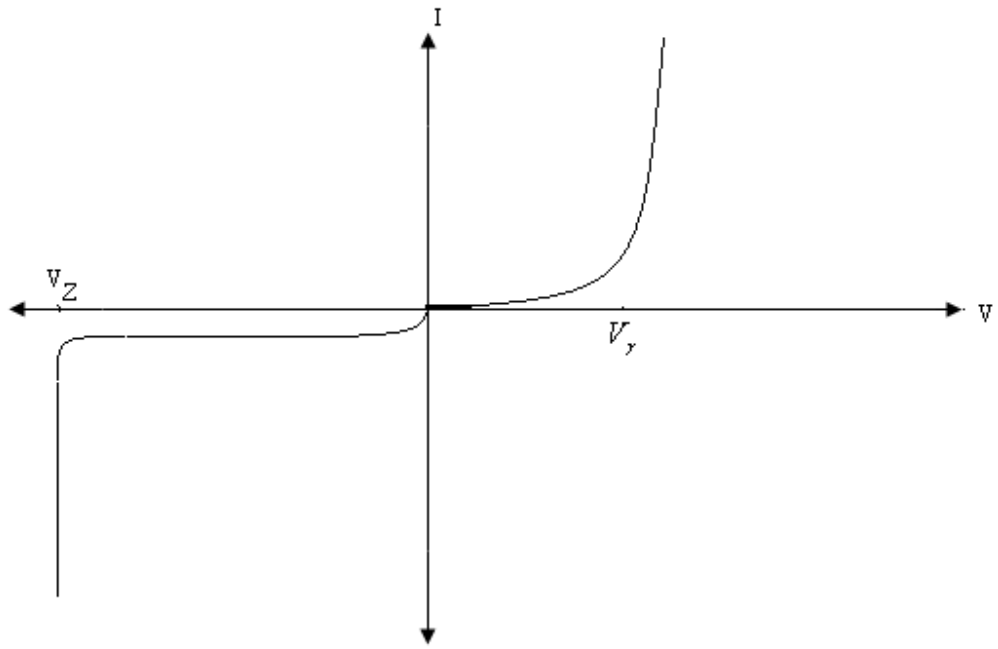


Fig. 4.11

The $V - I$ characteristics of the junction diode may also be expressed in the form of a equation given by:

$$I = I_s \left[e^{(qV/\eta KT)} - 1 \right] \qquad \text{------ (4.11)}$$

Where I_s is called the reverse saturation of the diode, K is the Boltzmann constant and its value is 1.38×10^{-23} J/K, q is the electron charge which is equal to 1.6×10^{-19} Coulomb, T is temperature in ^0K, η is a constant whose value depends on the material of the diode and the quality of the junction, its value is 1 for Ge and 1 to 2 for Si.

The term (KT/q) has the dimension of voltage known as thermal voltage is denoted by V_T. The value of V_T is given by $V_T = \dfrac{KT}{q} \approx \dfrac{T}{11{,}600}$

At room temperature (T = 300 ^{0}K) V_T = 0.026 volt = 26 mV.

The equation (4.11) may be rewritten as:

$$I = I_s \left[e^{(V/\eta V_T)} - 1 \right] \qquad \text{------ (4.12)}$$

From this equation it is clear that when the diode is in forward bias and $V > V_T$, the value of $e^{(V/\eta V_T)} >> 1$, the equation (4.12) may be approximated as:

$$I = I_s \left[e^{(V/\eta V_T)} \right] \qquad \text{------ (4.13)}$$

The current, therefore, rises exponentially; this verifies the nature of the V – I characteristics of the diode in the forward direction.

When the diode is in reverse bias, the voltage V will be negative. The equation (4.12) will be written as:

$$I = I_s\left[\frac{1}{e^{V/\eta V_T}} - 1\right] \qquad \text{------ (4.14)}$$

For large value of V, $I \approx -I_s$ i.e. the reverse saturation current will be independent of *V*.

Example 4.4 The saturation current density of a P – N junction Ge diode is 220 mA/m^2 at 300 ^{0}K. Find the voltage that would have to be applied to cause a forward current density of (i) 10^3 *A/m*^{2} and (ii) 10^4 *A/m*^{2} to flow.

Solution: we know $\frac{I}{A} = \frac{I_s}{A}\left[e^{(V/\eta V_T)} - 1\right]$

or $J = J_s\left[e^{(V/\eta V_T)} - 1\right] = J_s\left[e^{\frac{11600\ xV}{300}} - 1\right]$

η = 1 for Ge , V_T = (T/11600) and T = 300

or $\frac{J}{J_s} \equiv e^{38.67xV}$

(i) case $J = 10^3$ *A/m*^{2} $J_S = 220$ *mA/m*^{2} = 0.22 *A/m*^{2}

$$e^{38.67xV} = \frac{100000}{22} = 4545.45$$

$$V = \frac{\ln(4545.45)}{38.67} = \frac{8.42}{38.67} = 0.217 \text{ volts}$$

(ii) case $J = 10^4$ A/m^2 $J_s = 220$ *mA/m*^{2} = 0.22 *A/m*^{2}

$$e^{38.67xV} = \frac{1000000}{22} = 45454.5$$

$$V = \frac{\ln(45454.5)}{38.67} = \frac{10.72}{38.67} = 0.277 \text{ volts}$$

Example 4.5 A silicon diode operates at a forward voltage of 0.4 Volt. Calculate the factor by which the current is multiplied when the temperature is increased from 27^{0}C to 125^{0}C.

Solution: V_{T1} at 27^{0}C is given by:

$$V_{T1} = \frac{T}{11600} = \frac{(273+27)}{11600} = \frac{300}{11600} = 0.026 Volts$$

V_{T2} at 125^{0}C is given by:

$$V_{T2} = \frac{T}{11600} = \frac{(273+125)}{11600} = \frac{398}{11600} = 0.034 Volts$$

Now $I_1 = I_s(e^{\frac{V}{\eta V_{T1}}} - 1) = I_s(e^{\frac{0.4}{2x0.026}} - 1) = I_s(e^{7.69} - 1) = 2185.4 I_s$

$$I_2 = I_s(e^{\frac{V}{\eta V_{T1}}} - 1) = I_s(e^{\frac{0.4}{2x0.034}} - 1) = I_s(e^{5.88} - 1) = 356.81 I_s$$

$$\text{Multiplication factor} = \frac{I_1}{I_2} = \frac{2185.4 I_s}{356.81 I_s} = 6.12$$

4.7 Temperature Dependence of Reverse Saturation current of the Diode: The reverse saturation current of the diode is very much dependent on temperature and very important factor to be studied. Its theoretical variation with temperature is approximately given by:

$$I_s = K' T^m e^{-\left(\frac{qV_{g0}}{\eta KT}\right)} = K' T^m e^{-\left(\frac{V_{g0}}{\eta V_T}\right)} \qquad \text{------- (4.15)}$$

Where K' is a constant, qV_{g0} is the forbidden gap energy in Joules:

$\eta = 1$	$m = 2$	$V_{g0} = 0.785\ V$	for Ge
$\eta = 2$	$m = 1.5$	$V_{g0} = 1.12\ V$	for Si

Taking logarithms on both sides of the equation (4.15) and differentiating, we get :

$$\ln I_s = \ln K' + m \ln T - \frac{qV_{g0}}{\eta KT}$$

$$\frac{1}{I_s}\left(\frac{dI_s}{dT}\right) = \frac{m}{T} + \frac{qV_{g0}}{\eta K} \cdot \frac{1}{T^2}$$

$$\frac{dI_s}{dT} = \frac{I_s}{T}\left[m + \frac{qV_{g0}}{\eta KT}\right] \qquad \text{------ (4.16)}$$

From this equation it can be shown that I_s varies with temperature by 8% per degree Celsius for Si and 11% per degree Celsius for Ge. The experimental data is slightly different from this theoretical variation. The experimental variation is about 7% per degree Celsius in temperature for both Ge and Si. Since $(1.07)^{10} \cong 2.0$, it is therefore concluded that for every 10 ^{0}C rise in temperature the reverse saturation approximately doubles. This statement can be put in the form of a equation given by:

$$I_{s2} = I_{s1} . 2^{\frac{(T_2 - T_1)}{10}}$$

Where I_{s2} and I_{s1} are the values of I_s at temperatures T_2 and T_1 respectively. The reason for the discrepancy between the theoretical and experimental results is that, in a physical diode there is a component of reverse saturation current I_s due to leakage over the surface that has not been taken into account. This component of I_s is independent of temperature. The diode may assume to be shunted with a resistance R as shown in figure (4.12). The component of reverse saturation current over the surface is given by:

$$I_2 = I_s - I_1$$

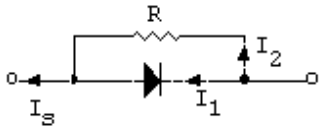


Fig. 4.12

Example 4.6 Calculate theoretically the factor by which the reverse saturation current of a Ge diode be multiplied when the temperature of the diode is increased from 27 to 75^{0}C.

Solution: I_s at 27^0C (300^0K) is given by:

$$I_{s1} = K^{'} x(300)^2 e^{-\left(\frac{11600x0.785}{300}\right)} = K^{'} x(300)^2 e^{-\left(\frac{9106}{300}\right)}$$

I_s at 75^0C (348^0K) is given by:

$$I_{s2} = K^{'} x(348)^2 e^{-\left(\frac{11600x0.785}{348}\right)} = K^{'} x(348)^2 e^{-\left(\frac{9106}{348}\right)}$$

Multiplication factor $$= \frac{I_{s2}}{I_{s1}} = \left(\frac{348}{300}\right)^2 \frac{e^{\left(\frac{9106}{300}\right)}}{e^{\left(\frac{9106}{348}\right)}} = 1.3456 \frac{e^{30.35}}{e^{26.167}}$$

$$= 88.19$$

Example 4.7 It is predicted that, for Ge diode the reverse saturation I_s should increase at the rate of 11% per degree Celsius rise in temperature. Experimentally it was found that in the diode the reverse saturation current is 5 μA at a reverse voltage of 10 volts, and it increases by 7% per degree Celsius rise in temperature. What is the value of leakage resistance shunting the diode?

Solution: From the figure (4.12)

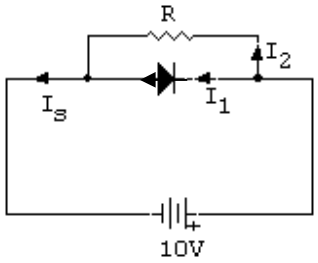


Fig.4.13

$$I_2 = I_s - I_1 \quad \text{or} \quad I_1 + I_2 = 5x10^{-6}$$

According to the problem I_s is increased by 7% so I_s = 1.07x5 μA

And I_1 is increased by 11 % so $I_1 = 1.11xI_1$

$$I_2 = 1.07 \; x \; 5x10^{-6} - 1.11 \; x \; I_1$$

$$= 5.35 \times 10^{-6} - 1.11(5x10^{-6} - I_2)$$

or $$= (0.2/.11) \; x \; 10^{-6} = 1.82 \; \mu A$$

Therefore $$R = \frac{10volt}{1.82\mu A} = 5.5 \; M\Omega$$

4.8 Diode Resistance: The characteristic curve of the diode is non-linear so the diode is called as the non-linear device. Its resistance value changes at different point of the curve. The two types of resistances for the diode may be defined namely: (i) D.C. or Static Resistance (ii) A.C or Dynamic resistance

D.C. or Static Resistance: It is defined as the ratio of the voltage and current at any point on the characteristic curve of the diode. This resistance is equal to the reciprocal of the slope of the line joining the operating point to the origin. The static resistance varies with voltage and current and is not a useful parameter. The forward resistance (static) R_f of the diode is very small and reverse resistance R_r is very high.

A.C. or Dynamic Resistance: The very useful parameter for the diode is the dynamic or A.C. or incremental resistance, which is defined as the reciprocal of the slope of the V – I characteristic curve i.e. $r_{a.c} \cong \frac{dV}{dI}$. The dynamic resistance is not constant but depends upon the operating voltage.

The diode equation is given by: $I = I_s \left[e^{(V/\eta V_T)} - 1 \right]$

Differentiating this equation with V, we get:

$$\frac{dI}{dV} = \frac{I_s . e^{(V/\eta V_T)}}{\eta V_T} = \left(\frac{I + I_S}{\eta V_T} \right) \qquad \text{------ (4.17)}$$

$\frac{dI}{dV}$ is called the incremental conductance denoted by g. The reciprocal of g is called the incremental resistance or a.c. resistance denoted by $r_{a.c.,}$ which is given by:

$$r_{a.c.} = \left(\frac{\eta V_T}{I + I_s} \right) \approx \frac{\eta V_T}{I} \qquad \text{------ (4.18)}$$

As I_s very small comparative to forward current I, hence neglected.

Example 4.8 Find the static and dynamic resistance of a P – N junction Ge diode for an applied voltage of 0.2 Volt. Temperature = 27^0 C and reverse saturation current is 2 μA.

Solution: We know $I = I_s \left[e^{(V/\eta V_T)} - 1 \right]$

η =1 for Ge, I_s = 2x10^{-6} A, V_T = T/11600 and T = 273+27 =300

$$I = 2x10^{-6} \left[e^{\frac{11600 \; xV}{300}} - 1 \right] = 2x10^{-6} \left[e^{38.67 \; x \; 0.2} - 1 \right]$$

or

$$= 2x10^{-6} x (2284 \;\; .72 \;\; - 1)$$

$$= 0.004567 = 4.567mA$$

Static resistance $r_{d.c.} = \frac{V}{I} = \frac{0.2x10^3}{4.567} = 43.8\Omega$

Dynamic resistance $r_{a.c.} = \left(\frac{V_T}{I + I_s} \right) \approx \frac{V_T}{I} = \frac{25.8mV}{4.4567mA}$

$= 5.7\Omega$

4.9 Ideal Diode:

The P – N junction diode has been discussed in detail. It offers a low resistance in forward bias (i.e. R_f is low) and offers high resistance in reverse bias

(i.e. R_r is high). The perfect or ideal diode may, however, be defined as a diode which offers zero resistance to the current flow in forward bias, the current in the diode is limited by the applied potential. On the other hand, it offers infinite resistance to the current flow in the reverse bias or the current is zero. The ideal diode is purely an imaginary quantity and practically such diodes are not available. These diodes behave as ON switch in the forward bias and OFF switch in the reverse bias as shown in figure (4.14).

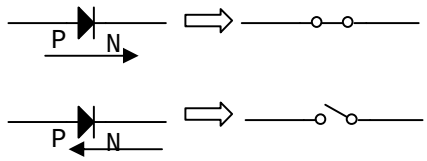


Fig. 4.14

The characteristic curve of the ideal diode is shown in figure (4.15), which coincides with V and I axes. ($I > 0$ for $V = 0$, and $I = 0$ for $V < 0$)

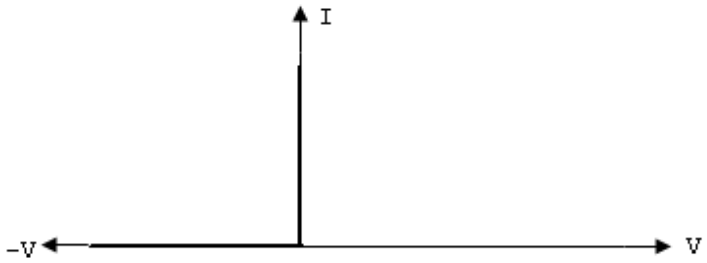


Fig. 4.15

4.10 Circuit Model for Junction Diode: It is well known that a voltage drop across the diode is observed in the forward bias. This voltage drop is about 0.7 V for Si diode and 0.3 V for Ge diode for a forward current of 1mA. Thus if we are dealing with voltage like hundreds of volts in the circuit containing the diodes, this voltage drop of about 0.7 V or 0.3 V may assumed to be negligibly small. For the approximate calculations in the circuit the diodes may be assumed as the ideal diodes discussed above. On the contrary if the applied voltage is comparable to the forward voltage drop of the diodes, then junction diodes may not be approximated as ideal diodes but a battery of knee voltage V_γ is assumed to be connected in series with the ideal diode as shown in figure (4.16). This model is known as the piecewise linear model or the approximate model of the junction diode.

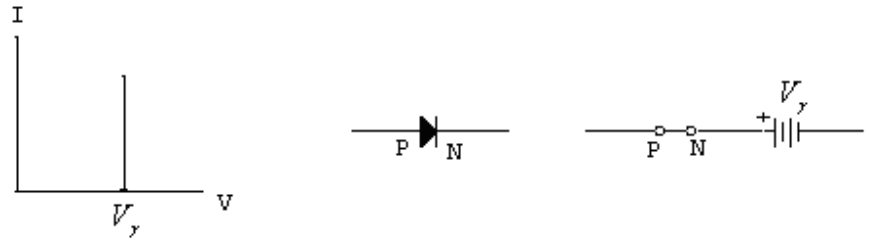


Fig. 4.16

For more accurate calculations, we use the model shown in figure (4.17), in which a small resistance r is also included in the model. The non-linear V – I relationship of the junction diode is here approximated a straight line of slope (*1/r*).

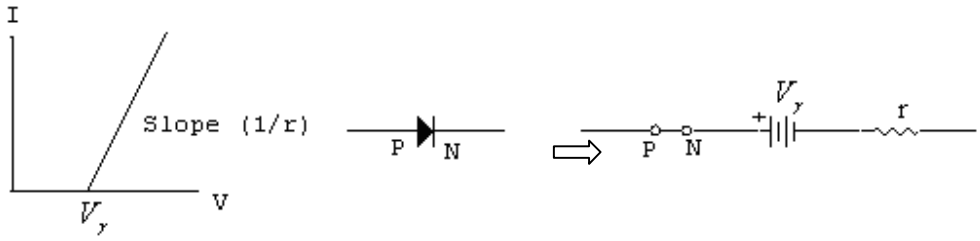


Fig. 4.17

Example 4.9 In the figure (4.18) the diode is reverse biased. (a) If the diode is ideal what is the voltage across it? (b) If the diode has a reverse current of 0.25 mA at – 50 Volts, what is the voltage across the diode?

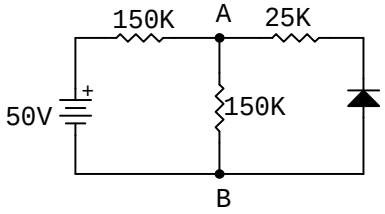


Fig. 4.18

Solution: The Thevenin's equivalent of this circuit is given by (fig. 4.19):

$$V_{AB} = \frac{50Vx150K}{(150+150)K} = 25V$$

$$R_{AB} = \frac{150Kx150K}{(150+150)K} = 75K\Omega$$

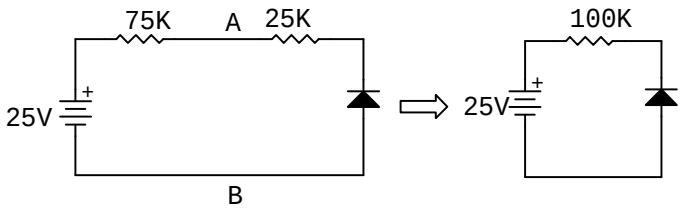


Fig. 4.19

(a) In the first case when the diode is ideal (behaves as an open switch), the voltage across the diode will be equal to the open circuit voltage i.e. 25 Volts.

(b) In the second case diode resistance in reverse bias is

$$R_r = \frac{50}{0.25x10^{-3}} = 200K\Omega$$

Now the diode can be replaced by a resistance of 200 KΩ as shown in figure (4.20).

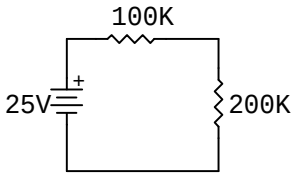


Fig. 4.20

The voltage across the diode (200 $K\Omega$ resistance) is given by:

$$Vo = \frac{25Vx200K}{300K} = \frac{50}{3} = 16.67 \ volts$$

Example 4.10 The Si diode in the given figure (4.21) has a current of $2\mu A$ for a reverse voltage of 100 volts. If the diode is approximated by a battery of 0.7 volt, what is the output wave form across $1M\Omega$ resistance?

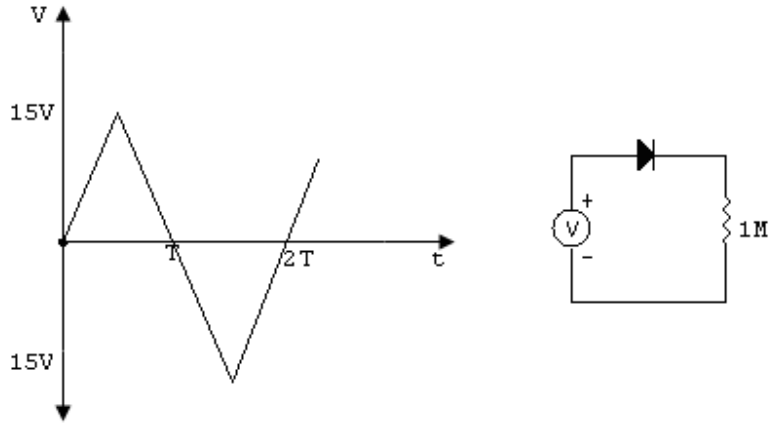


Fig. 4.21

Solution: In the forward bias, the diode will have the voltage drop of 0.7 Volt so the output will increase only up to 15 – 0.7 =14.3 volt.

In the reverse bias, the diode will behave like a resistance of (100Volt/2μA) 50 MΩ. So voltage across 1 MΩ resistance is $\frac{15Vx1M}{(50+1)M} = 0.29$ volt.

In the negative half cycle of the input wave, output will increase only up to 0.29 volts. The output wave form is, therefore, given in figure (4.22).

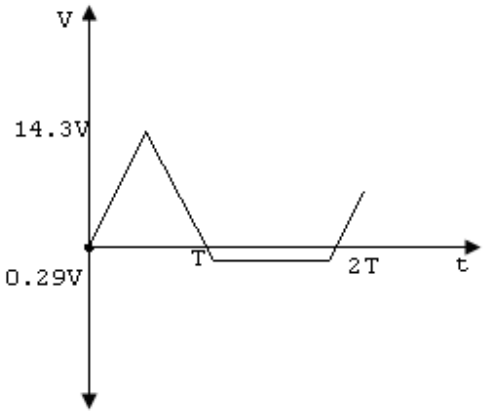


Fig.4.22

4.11 Junction Capacitances: Two types of Junction capacitances are observed in the P N junction diode. These capacitances are:

(i) Transition or Depletion region capacitance (C_T) and

(ii) Diffusion capacitance or Storage capacitance (C_D).

Both types of capacitances are present in forward and reverse bias region. But the transition capacitance is more effective in reverse bias and diffusion capacitance is more effective in forward bias. These capacitances are very small in magnitude so they are sensitive at high frequencies. At low frequencies their reactance are very high hence behave like an open circuit, but at high frequencies their reactance are finite and introduce impedances in the circuit.

(i) Transition Capacitance: In the reverse bias there is a depletion region that behaves essentially like an insulator between the layers of opposite charges. So it will behave like a parallel plate capacitor whose value is given by: $C = \frac{\varepsilon A}{W}$, where ε is the permittivity of the dielectric (insulator), A is the area of the plates, W is the distance between the parallel plates (or width of the depletion region). As is well known the width of the depletion region increases with the increase of the reverse bias; a decrease in the transition capacitance with the increase of the reverse bias will be observed. This property of the diode is used in the construction of special types of diode called Varactor diodes or Varicaps.

(ii) Diffusion Capacitance: The diffusion capacitance C_D is observed in the junction diode when it is in forward bias. It is caused by the injected charge stored on both sides of the junction just outside the space charge region. It is basically the rate of change of injected charge with voltage. The increased level of current will result the increased level of diffusion capacitance.

The capacitive effect of the junction diode is represented by a capacitance in parallel with the ideal diode as shown in figure (4.23).

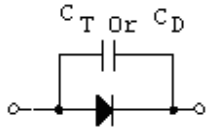


Fig. 4.23

4.12 Zener Diodes: Zener diodes also called as Avalanche diodes are the heavily doped P – N junction diodes which work in the reverse bias and operate in the break down region. Ordinary diodes are never used in the break down region, since the excessive current in the diode rises its temperature to a very high value and hence the diode may be damaged. It has been observed that when the reverse voltage of the diode is increased beyond certain limit then two types of break downs are possible. (i) Avalanche Break Down; & (ii) Zener Break Down.

(i) Avalanche Break Down: When the reverse voltage of the heavily doped junction diode reaches a certain limit, the electric field is quite intense and the minority carriers in this field are accelerated and get sufficient velocity to collide with the ions in the depletion region resulting thereby the liberation of electron – hole pairs. These librated carriers also gain enough velocities to dislodge other electron – hole pairs. This cumulative process is referred to avalanche multiplication. This leads a large reverse current and the diode is said to be in the avalanche breakdown region.

(ii) Zener Break Down: The Zener break down does not occur by the collision of carriers with semiconductor ions as in the case of avalanche breakdown, but the breaking of covalent bonds occurs by the strong field set up in the depletion region due to the applied reverse bias. Thus a large reverse current is produced.

The V – I characteristic of a Zener diode is shown in figure (4.24). It is clear from this curve that in forward bias its characteristics are the same as that of the ordinary diode. In the reverse bias, a constant reverse current flows until the break down voltage is reached. This breakdown voltage is known as Zener voltage (V_z). The Zener voltage may be different for different Zener diodes depending upon the amount of doping in the diode. A heavily doped diode has a narrow depletion region and the breakdown occurs at low voltage and hence the low Zener voltage. For lightly doped diodes the breakdown occurs at high voltage. The Zener diodes having the break down voltage from 1.8 volts to 200 volts are available with power rating of ¼ watt to 200 watts. In high voltage Zener diodes the avalanche multiplication is more pronounced and in the low voltage diodes Zener

breakdown is effective. However, in the diodes of Zener voltages between 5 volts to 8 volts both types of breakdown occur.

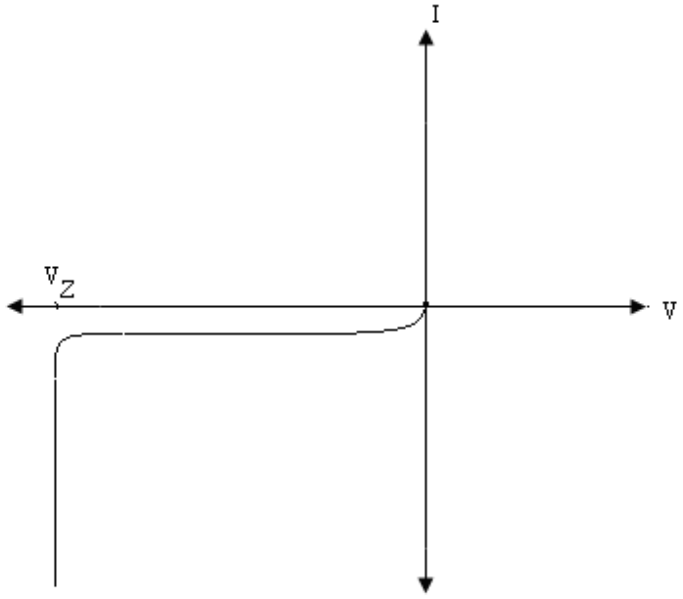


Fig. 4.24

The Zener diodes are used as voltage regulator i.e. a constant voltage equal to the Zener voltage may be obtained with these diodes. Its symbolic representation is the same as that of ordinary diode with the only difference that the bar is replaced by a symbol *Z* (fig.4.25 a). Its equivalent circuit may be re[resented by a battery equal to Zener potential (fig. 4.25 b). The accurate equivalent circuit includes a small dynamic resistance in series with the battery, as shown in figure (4.25 c).

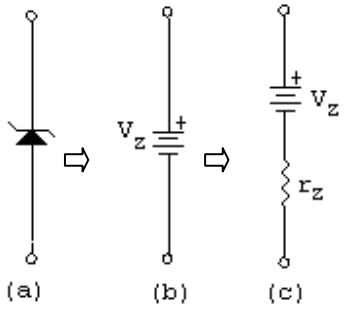


Fig. 4.25

4.13 Light Emitting Diodes: Light emitting diodes (LED's) are specially doped P N junction diodes, which emit light when a proper forward bias is applied across the diodes. In the forward biasing of the diodes the electrons from N – type semiconductor move across the junction and enter P – type semiconductor. These electrons combine with holes. This recombination requires that the energy possessed by the free electrons be transferred to some other form. In all semiconductor P N junctions some of this energy

will be given off as heat and some in the form of light energy. In Si and Ge diodes this energy is given off more in the form of heat and very negligible amount of energy is transferred in the form of light. However, the energy is given up in the form of visible lights than the heat if the diodes are designed by the Gallium Phosphide or Gallium Arsenide Phosphide semiconductor materials. The LED's fabricated by Gallium Phosphide produces visible red light and Gallium Arsenide Phosphide semiconductor produces visible green light. Infrared LED's are designed by Gallium Arsenide semiconductors which emit invisible infrared light in forward bias. The LED is represented by the simple diode in addition emission of light is also shown (fig.4.26). The LED's are used as display devices and are available in different shape and size.

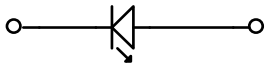


Fig. 4.26

4.14 Photodiodes: If the radiations are allowed to fall on the junction of a P – N diode, the reverse current varies linearly with the radiations. The device which exhibits this property is known as Photodiode. This device is embedded in a plastic cover having a window for the light to fall on the junction as shown in the figure (4.27). It is well known

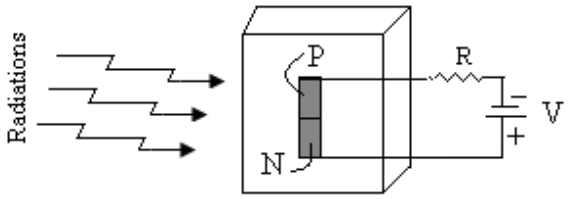


Fig. 4.27

that when the diode is in reverse bias, then reverse saturation current flows in the diode which is generated due to the minority charge carriers. When the light energy incident on the junction, generation of minority carriers are increased thereby increasing the reverse current. The dark current of the photodiode is that current which is obtained when no photons are incident on the junction. Figure (4.28) shows the characteristics of photodiode for different intensity of light given in foot candles.

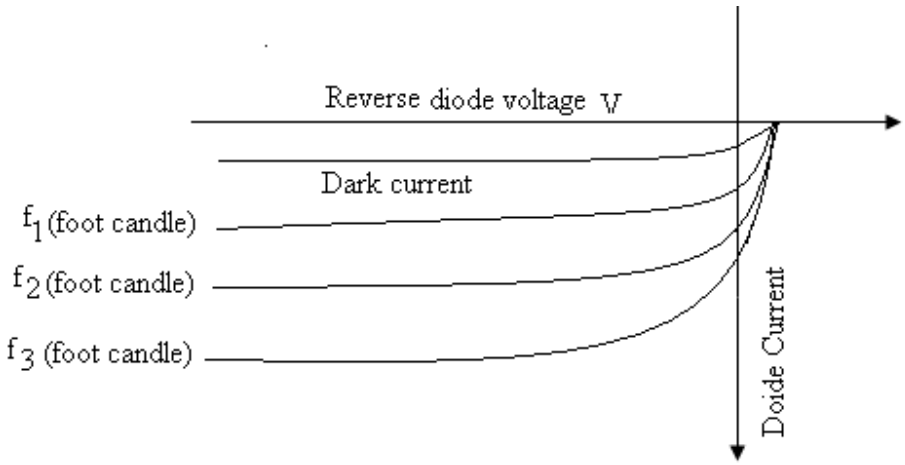


Fig. 4.28

Problems:

1. What is the difference between metal, insulator and semiconductor? Discuss on the basis of energy band diagram.
2. What is a semiconductor? Distinguish between intrinsic and extrinsic semiconductor?
3. What do you understand by intrinsic and extrinsic semiconductors? Give the energy band description of semiconductors and discuss the effect of temperature on it.
4. What are N type and P type semiconductors? Discuss the effect of temperature on the extrinsic semiconductor.
5. Discuss the currents in extrinsic semiconductors. Find the expression for the total current density in the extrinsic semiconductor.
6. What is drift current? Derive an expression for drift current density and conductivity in a semiconductor.
7. What is diffusion current? Drive the expression for diffusion current density for holes and electrons.
8. What is a junction diode? Explain the working of a P – N junction diode under forward and reverse biasing. Draw the V – I characteristic curve of the junction diode.
9. What are P and N type semiconductor? How they are used to form the junction diode. Discuss how the depletion region is formed in a junction diode.
10. Discuss the effect of temperature on the reverse saturation current of the junction diode.
11. What do you understand by static and dynamic resistance of a junction diode? Find the expression for the dynamic resistance.
12. What is the difference between an ideal diode and a practical diode? Discuss the circuit models for the junction diode.
13. Write short notes on the following:
    (i) Junction capacitances

(ii) Photodiode
(iii) Light emitting diode

14. Discuss the physical mechanism of breakdowns in a junction diode. Explain the working and characteristics of Zener diode.

15. (a) Show that the resistivity of the intrinsic Si crystal at 300^0 K is 0.45Ω-m (b) what will be the resistivity of this Si crystal if the donor impurity of 1 atom per 10^8 Si atoms is introduced? Given that $n_i = 1.5 \times 10^{16}$ atoms/m^3, $\mu_n = 0.13$ m^2/volt-sec, $\mu p = 0.05$ m^2/volt-sec, Atomic conc. of Si = 5.0×10^{28} atoms/m^3.

Ans. 0.037 Ω-m

16. Find the density of impurity atoms that must be added to intrinsic Si wafer to convert it to (i) 0.2 Ω-m P – type Silicon.
(ii) 0.2 Ω-m N – type Silicon.
Given for Si $\mu_n = 0.13$ m^2/volt-sec, $\mu p = 0.05$ m^2/volt-sec.

Ans. $N_A=6.25 \times 10^{20}/m^3$, $N_D=2.4 \times 10^{20}/m^3$

17. Calculate the density of donor atoms to be added to an intrinsic semiconductor to produce N – type semiconductor of 4500 mhos/m. Given $\mu_n = 0.385$ m^2/volt-sec.

Ans. $N_D=73 \times 10^{21}/m^3$

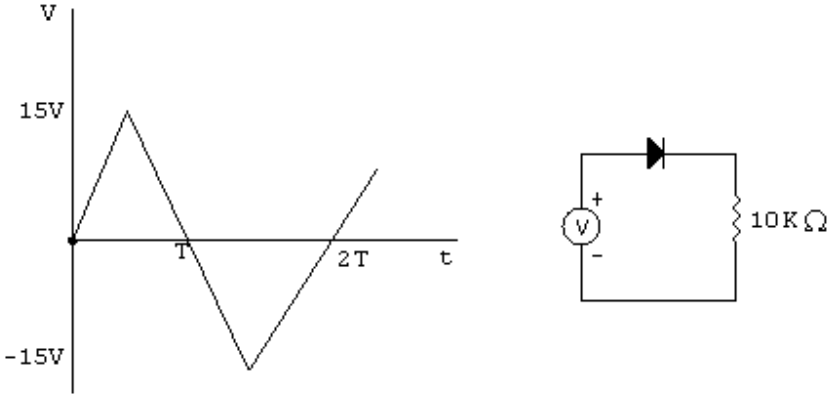
18. Mobilities of of electrons and holes in intrinsic Ge crystal at room temperature are 0.38 m2/volt-sec and 0.18 m^2/volt-sec respectively. If the densities of electrons and holes are 2.5×10^{19} /m^3 and $1.5 \times 10^{16}/m^3$ respectively. Calculate electrical conductivity and resistivity of Ge.

Ans. 1.35 mhos/m , 0.74 ohm-m

19. A forward biased P N junction diode requires 1 volt to pass a current of 300 mA. The same junction requires 200 volts to pass 20 μA current in reverse bias. Find the forward and reverse resistance of the diode.

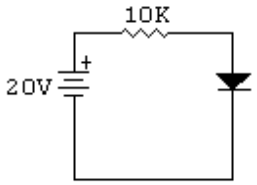
Ans. $R_f = 3.33$ Ω, $R_r = 10$ MΩ

20. Sketch the output voltage across 10 KΩ resistance of the circuit shown in the figure given below. Use ideal diode approximation.

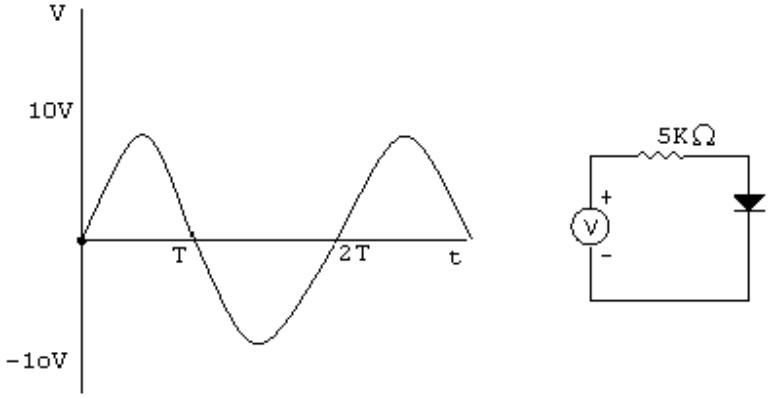


21. A forward biased Ge diode is connected as shown in figure given below. Find the voltage across the diode if the reverse saturation current is 10 μA.

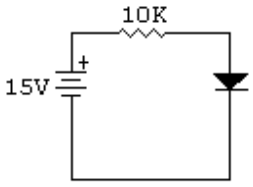
Ans. 0.138 Volt



22. Sketch the output voltage across the diode of the circuit shown in the figure given below. Use ideal diode approximation.

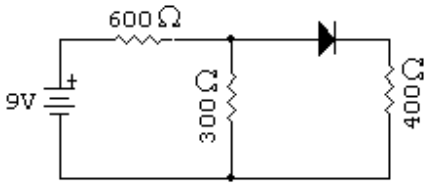


23. In the figure given below the diode is made of Ge and has a forward current of 10 mA at 1 volt. (a) How much current flows if the diode is considered ideal? (b) How much current flows if the diode is considered by a battery and a resistance in series with it?



Ans. (a)1.5 mA, (b) 1.4 mA

24. A Si diode is used in the circuit shown in the figure given below. Determine the voltage across 400 Ω resistance, if the diode is approximated by a battery of potential 0.7 volt.



Ans. 1.53 volt

25. A Si diode at room temperature of 27 ^{0}C, conduct 1 mA current at 0.7 volt. Now the voltage is increased to 0.9 volt. Calculate the diode current and the reverse saturation current. Given $\eta = 2$.

Ans. 0.046 A, 1.4 nA

26. A Ge diode operates at a forward voltage of 0.25 Volt. Calculate the factor by which the current is multiplied when the temperature is increased from 27^0C to 150^0C. Ans. 17.8

27. For what voltage will the reverse current in a Ge diode attains a value of 90% of its saturation value at 300 ^{0}K? What will now be the current if I_s = 10μA and the forward voltage of the magnitude calculated above is applied?
Ans.: – 60 mV, 91 μA

28. (a) Calculate theoretically the factor by which the reverse saturation current of a Ge diode be multiplied when the temperature of the diode is increased from 25 to 85^0C.
(b) Repeat the calculations of part (a) for Si diode whose temperature is increased from 25 to 150^0C. Ans.: 244.73 , 1739

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5 Applications of Diodes

In the forgoing chapter of this book, the details of the semiconductors and different types of diodes have been discussed. The applications of the semiconductor diodes such as rectifiers, filters circuits, voltage multiplier circuits, clipping, clamping Log antilog circuits will be discussed in this chapter. Zener diode as the voltage regulator circuit will also be discussed in this chapter.

5.1 Rectifier Circuits: Many electronic equipments work with d.c. power supply. The line voltage available from the power plugs is a.c. voltage (220 Volts & 50Hz frequency). A circuit which converts a.c. voltage to d.c. voltage is required. Such a circuit is called the rectifier circuit. The nonlinear elements like vacuum diodes or semiconductor diodes may be used for the rectification of the supply. There are two types of rectifier circuits (i) Half Wave Rectifier & (ii) Full Wave Rectifier. In the following articles these circuits will be discussed in detail using the semiconductor diodes.

5.1.1 Half Wave Rectifier : Consider a circuit shown in the figure (5.1). In this circuit a sinusoidal voltage $E_s = E_m \sin \omega.t$, obtained from the transformer, is applied to a series combination of the diode D_1 and a load resistance R_L. E_m is the peak voltage of the signal and $\omega = 2\pi f$ is the frequency of the a.c. mains in radians and f is the mains frequency in Hz. The transformer used here may be step up or step down transformer which is chosen according to the required d.c. supply. If the a.c. mains supply (220 volts, 50 Hz) is to be rectified, then it may directly be applied to the series combination of diode and load resistance.

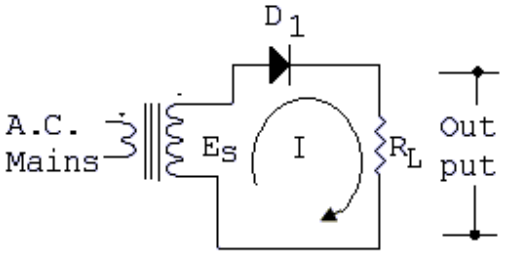


Fig. 5.1

To discuss its operation, assume that the peak value of voltage E_m is large enough than the cut in voltage of the diode, so that the diode behaves like an ideal diode. During the positive half cycle of the input wave E_s, the diode is in forward bias and works as a closed switch. Therefore, applied voltage appears across the load resistance R_L. However, during the negative half cycle of the input wave, the diode is in reverse bias and it acts as an open switch. In this case, no current flows through the circuit and voltage across the load resistance R_L is zero. The input, output waves of the circuit is shown in the fig. (5.2).

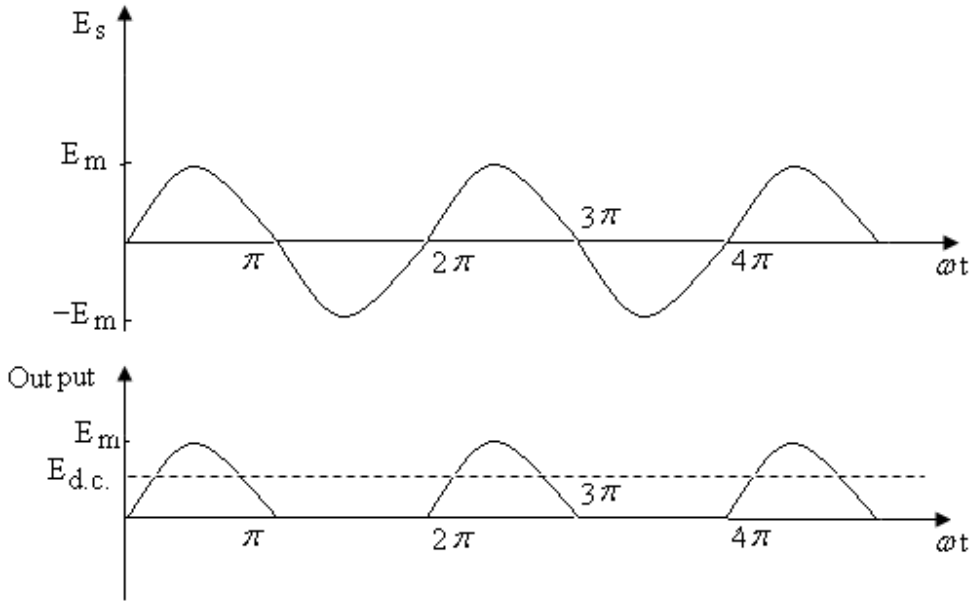


Fig. 5.2

From this figure it is clear that the current in the load resistance flows only during positive half cycle of the input wave and is zero during the negative half cycle, i.e., the current is unidirectional. Hence the circuit is called a half wave rectifier.

Analysis: The input a.c. voltage is given by $E_s = E_m \sin \omega.t$

Since the current in the load resistance is unidirectional, so we take the average value of the voltage across the resistance. It may be given as:

$$E_{d.c.} = \frac{1}{T}\int_0^{T/2} E_m \sin \omega.t.dt \qquad = \frac{E_m}{\pi} \qquad \text{since } T = \frac{2\pi}{\omega}$$

The d.c. current flowing through the load resistance R_L is given by:

$$I_{d.c.} = \frac{E_{d.c.}}{R_L} = \frac{E_m}{\pi R_L}$$

$$= \frac{I_m}{\pi} \qquad \text{(where } I_m = \frac{E_m}{R_L} \text{ is the peak current)}$$

The output voltage across the load resistance may be given by the Fourier analysis as:

$$E_{out} = \frac{E_m}{\pi} + \frac{E_m}{2}\sin \omega t - \frac{2E_m}{3\pi}\cos 2\omega t - \frac{2E_m}{15\pi}\cos 4\omega t - \frac{2Em}{31\pi}\cos 6\omega t - .....$$

------ (5.1)

From equation (5.1), it may be seen that the first term of the equation is the same as $E_{d.c.}$ as calculated above. In addition to this term there are other terms of frequency ω and its higher harmonics. So the output voltage across the load resistance has the required voltage ($E_{d.c}$) and other unwanted components called the ripple.

The root mean square value of the output voltage may be given by:

$$E_{r.m.s.}^2 = \frac{1}{T}\int_0^{T/2} E_s^2 .dt$$

$$= \frac{1}{T}\int_0^{T/2} E_m^2 \sin^2 \omega t.dt \quad = \frac{E_m^2}{4}$$

or $$E_{r.m.s.} = \frac{E_m}{2}$$ ------ (5.2)

The total power delivered to the load resistance is:

$$P_{in} = \frac{E_{r.m.s.}^2}{R_L} = \frac{E_m^2}{4R_L}$$ ------ (5.3)

The d.c. power delivered to the load is:

$$P_{d.c.} = \frac{E_{d.c.}^2}{R_L} = \frac{E_m^2}{\pi^2 R_L} \qquad (\text{Putting } E_{d.c.} = \frac{E_m}{\pi})$$

Therefore, power delivered in load resistance due to unwanted components i.e. ripple may be given as:

$$P_{ripple} = P_{in} - P_{d.c.}$$

$$= \frac{E_{r.m.s.}^2}{R_L} - \frac{E_{d.c.}^2}{R_L}$$ ------ (5.4)

The P_{ripple} is equal to $\frac{E_{a.c.}^2}{R_L}$. The equation (5.4) may be rewritten as:

$$\frac{E_{a.c.}^2}{R_L} = \frac{E_{r.m.s.}^2}{R_L} - \frac{E_{d.c.}^2}{R_L} \quad \text{or} \quad \gamma = \frac{E_{a.c.}}{E_{d.c.}} = \sqrt{\left(\frac{E_{r.m.s.}^2}{E_{d.c.}^2} - 1\right)}$$

Where γ is the ripple factor and is defined as the ratio of the a.c. components to the d.c. components at the output.

Putting the values of $E_{d.c.}$ and $E_{r.m.s.}$ respectively we get:

$$\gamma = \sqrt{\left(\frac{\pi^2}{4} - 1\right)} = 1.21$$

From this equation, it is clear that γ is more than 1, which indicates that at the output, the unwanted a.c. components (ripple) are more than the wanted d.c. components. Hence the half wave rectifier being a poor circuit for rectification is not of much practical use.

Rectifier Efficiency: It is useful to define the quantity called the rectifier efficiency, which is defined as:

Rectifier Efficiency η = (d.c. power input to load) / (Input power delivered to load)

$$= \frac{P_{d.c.}}{P_{in}} x100\ \%$$

$$= \frac{(E_m^2 / \pi^2 R_L)}{(E_m^2 / 4R_L)} x100 \quad = \frac{4}{\pi^2} x100 \quad = 40.6\ \%$$

This means, for half wave rectifier, under ideal conditions, only 40.6 % of the a.c. input power is converted into d.c. power in the load resistance.

Example 5.1 A 12 volt a.c. from the secondary of a transformer is applied to the input of a half wave rectifier circuit having 10KΩ load resistance. If the diode is ideal, find:

(i) Peak value of the a.c. signal

(ii) D.C. output voltage

(iii) Peak value of the current through the load resistance

(iv) Average value of the current through the load resistance

(v) PIV of the diode

Solution: Given $E_{r.m.s.}$ =12 Volts , R_L = 10 KΩ

(i) $E_m = \sqrt{2} E_{r.m.s.} = 1.414x12 = 16.97 volts$

(ii) $E_{d.c} = \frac{E_m}{\pi}. = \frac{16.97}{3.14} = 5.4 volts$

(iii) $I_m = \frac{E_m}{R_L} = \frac{16.97}{10000} = 1.7 mA$

(iv) $I_{d.c}. = \frac{E_{d.c.}}{R_L} = \frac{5.4}{10000} = 540 \mu A$

(v) PIV of the diode = E_m = 16.97 Volts

Example 5.2 A P – N junction diode having forward resistance R_f = 25Ω, is used in half wave rectifier circuit. The input applied signal is given by $Es = 25Sin(100\pi t)$. If the load resistance R_L is 500Ω, then calculate I_m, $I_{r.m.s.}$, $E_{d.c.}$, d.c. current following through the load resistance and the rectifier efficiency.

Solution: The peak value of the input a.c. is given E_m = 25 Volts

$$I_m = \frac{E_m}{R_f + R_L} = \frac{25}{(25 + 500)} = 0.0476\, A = 47.6 mA$$

$$I_{r.m.s.} = \frac{I_m}{2} = \frac{47.6}{2} = 23.8 mA$$

$$E_{d.c.} = \frac{E_m x R_L}{(R_f + R_L)\pi} = \frac{25x500}{525x3.14} = 7.58 volts$$

$$I_{d.c.} = \frac{E_{d.c.}}{R_L} = \frac{7.85}{500} = 15.17 mA$$

Rectifier Efficiency $\eta = \frac{P_{d.c.}}{P_{in}} x100$

$$P_{d.c.} = I_{d.c.}^2 R_L = (15.17)^2 500 = 0.115 watt$$

$$P_{in} = I_{r.m.s.}^2 x(R_L + R_f) = (23.8)^2 x525 = 0.2974 watt$$

$$\eta = \frac{P_{d.c.}}{P_{in}} = \frac{0.115}{0.2974} x100 = 38.67\%$$

5.1.2 Full Wave Rectifier: Figure (5.3) shows the circuit diagram of full wave rectifier. It consists of a centre tapped transformer and two diodes D_1 and D_2 in addition to the load resistance R_L. The centre tapped transformer converts a sinusoidal signal into two equal sinusoidal signals which are 180^0 out of phase. In order to understand the operation of this circuit, we consider the diodes are ideal. The two signals E_{S1} and E_{S2} obtained from the transformer are being applied to the two diodes D_1 and D_2. During first half cycle of the input wave (E_S), point A is positive and B is negative with respect to. the common terminal. The diode D_1, therefore, conducts and D_2 is in reverse bias. This causes I_{b1} current to flow through the diode D_1 and load resistance R_L. During the other half cycle, diode D_2 conducts and diode D_1 is in reverse bias. This caused I_{b2} current to flow through the diode D_2 and load resistance R_L.

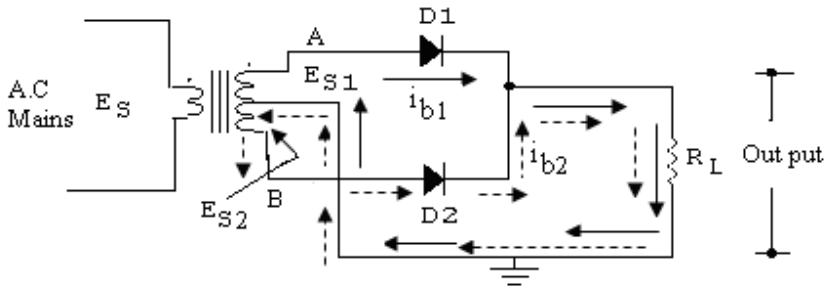


Fig 5.3

Fig. (5.4) shows the input and output waves of the full wave rectifier circuit,

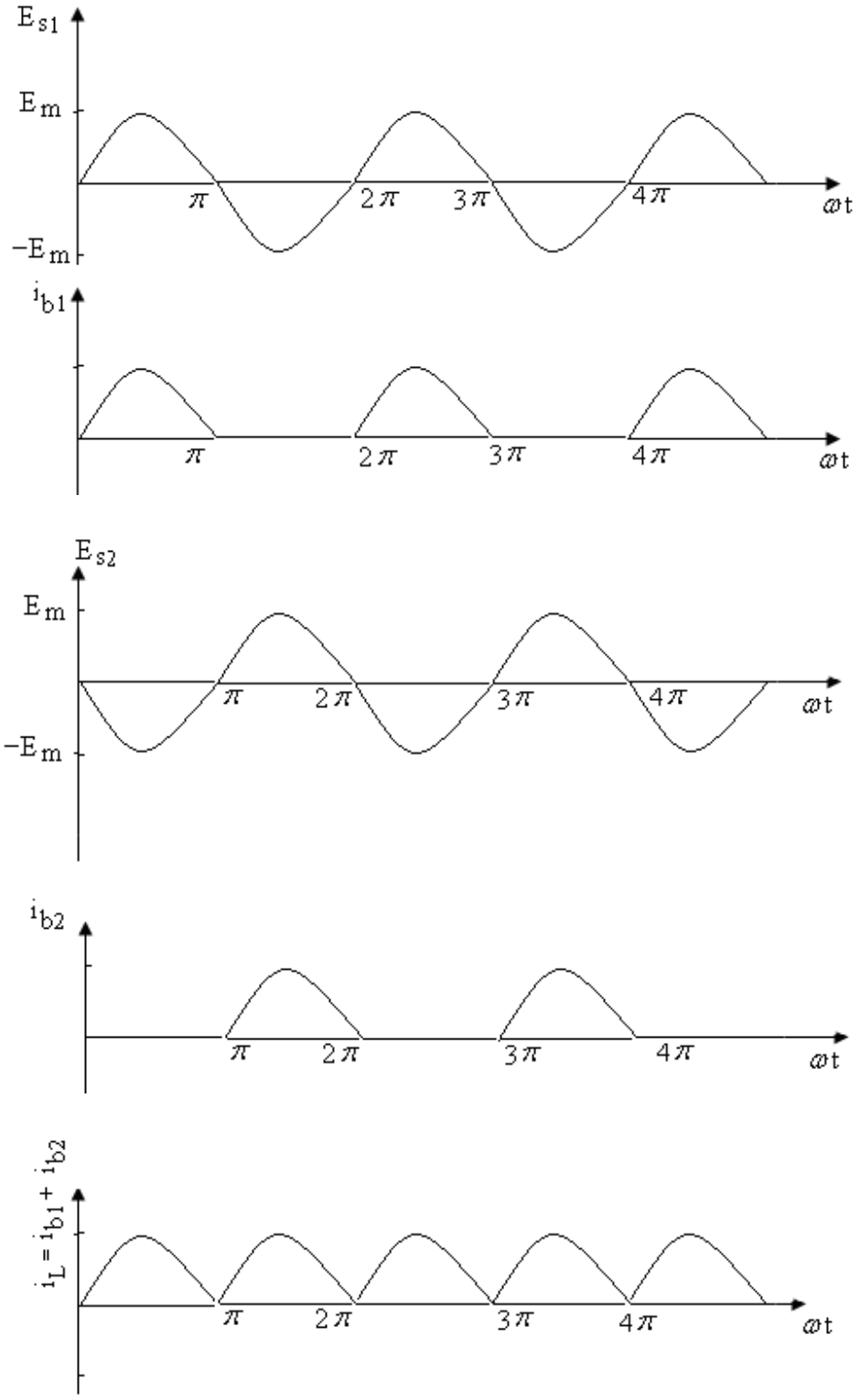


Fig. 5.4

From these wave shapes, it is clear that the direction of current in the load resistance is same for both the half cycles of the output wave. Hence, the circuit is called as full wave rectifier.

Analysis: In this circuit the centre tapped terminal of the transformer is used as the common terminal of two voltages E_{S1} and E_{S2} which are 180 out of phase.

Let $E_{S1} = -E_{S1} = E_m Sin\,\omega t$

The average value (d.c.) of the voltage across the load resistance for the full wave rectifier is given by:

$$E_{d.c.} = \frac{1}{T}\left[\int_0^{T/2} E_{S1}\,dt + \int_{T/2}^{T} E_{S2}\,dt\right]$$

$$= \frac{1}{T}\left[\int_0^{T/2} E_m \sin\omega t.dt - \int_{T/2}^{T} E_m \sin\omega t.dt\right]$$

which may be simplified as $= \frac{2E_m}{\pi}$.

The r.m.s. value of the voltage across the load resistance is given by:

$$E^2_{r.m.s.} = \frac{1}{T}\left[\int_0^{T/2} E^2_{S1}.dt + \int_{T/2}^{T} E^2_{S2}.dt\right]$$

$$= \frac{1}{T}\left[\int_0^{T/2} E^2_m \sin^2\omega t.dt + \int_{T/2}^{T} E^2_m \sin^2\omega t.dt\right] \quad = \frac{E^2_m}{2}$$

or $E_{r.m.s.} = \frac{E_m}{\sqrt{2}}$

From the Fourier analysis the voltage across the load resistance is given by:

$$E_L = \frac{2E_m}{\pi} - \frac{4E_m}{3\pi}\cos 2\omega.t - \frac{4E_m}{15\pi}\cos 4\omega.t - \frac{4E_m}{31\pi}\cos 6\omega t - \ldots \quad \text{------ (5.5)}$$

It is clear from this equation that first term of the output voltage across the load resistance is the same as that of $E_{d.c.}$, calculated above. Further the lowest frequency term (ripple) in the full wave rectifier circuit is twice the frequency of supply signal.

The ripple factor γ is calculated as:

$$\gamma = \sqrt{\left(\frac{E_{r.m.s.}^2}{E_{d.c.}^2} - 1\right)} = \sqrt{\left(\frac{E_m^2/2}{4E_m^2/\pi^2} - 1\right)}$$

$$= \sqrt{\left(\frac{\pi^2}{8} - 1\right)} = 0.48 \qquad \text{------ (5.6)}$$

It clearly indicates that γ being less that unity has the d.c. components (desired) in the output more than the a.c. components (undesired).

Rectifier Efficiency: This is calculated as:

$$\eta = \frac{P_{d.c.}}{P_{in}} x100 \ \%$$

$$= \frac{(4E_m^2/\pi^2 R_L)}{(E_m^2/2R_L)} x100 \quad = \frac{8}{\pi^2} x100 \quad = 81.2\ \%$$

Thus for full wave rectifier 81.2 % of the a.c. input power is converted to d.c. output power.

5.2 Peak Inverse Voltage (PIV): Peak inverse voltage of a rectifier diode in a circuit is the maximum reverse voltage across the diode. For the rectifier circuit PIV of the diodes used should be less than the break down voltage of the diode. It can be seen from the circuit of half wave rectifier that PIV of the diode used is equal to peak voltage of the applied input a.c. i.e. $+E_m$.. But in case of full wave rectifier PIV of each diode used is twice the peak voltage of the applied a.c. ($+2E_m$). It is clear from the figure (5.5) of the full wave rectifier that during +ve half cycle of the input wave, the diode is in forward bias and the voltage at the point C may thus reaches to $+E_m$ and at point B it is $-E_m$. So the total reverse voltage across the diode D_2 is $2E_m$. Similarly it can be seen that during –ve half cycle PIV of the diode is also $+2E_m$. From the above discussion it is clear that in full wave rectifier circuit, we should use the diodes whose break down voltage in greater than +2Em (i.e. double the peak value of the input a.c. voltage).

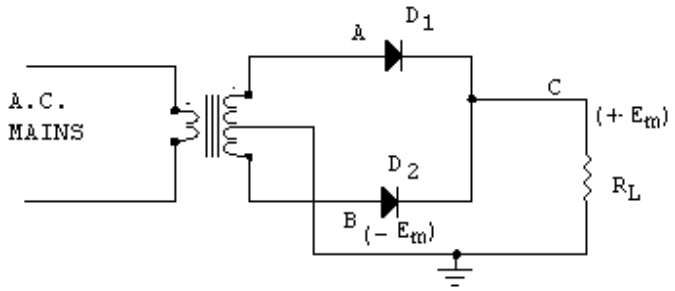


Fig. 5.5

Disadvantage of Full Wave Rectifier: The full wave rectifier circuit has the following disadvantages:

(i) As discussed above, the PIV of the diode used in the circuit is $+2E_m$. So such diodes whose break down voltage is greater than $2E_m$ is to be chosen for this purpose.

(ii) In this circuit, especially designed transformer called centre tapped transformer is to be used. To rectify the a.c. mains voltage a center tapped transformer of 220-0-220 volts is to be used. However, in case of half wave rectifier circuit, transformer is not to be used if an a.c. main is to be rectified.

(iii) Since centre terminal of the centre tapped transformer is used as a common terminal (or ground terminal) –ve d.c. output voltage may not be obtained by simply reversing the output terminals; but will be obtained by reversing the diode connections.

Example 5.3 A full wave rectifier circuit has the input signal $E_s = 100Sin(100\pi.t)$, R_L = 900 Ω and each diode has the forward resistance of 100 Ω. Calculate

(i) Peak value of the current through the load resistance,
(ii) D.C. load current,
(iii) D.C. output voltage
(iv) Rectifier Efficiency

Solution: Given $E_m = 100$ Volts, $R_L = 900\ \Omega$, $R_f = 100\Omega$

(i) $$I_m = \frac{E_m}{R_L + R_f} = \frac{100}{900+100} = 100mA$$

(ii) $$I_{d.c.} = \frac{2I_m}{\pi} = \frac{2x100}{3.14} = 63.7mA$$

(iii) $$E_{d.c.} = I_{d.c.}R_L = 63.7mAx900 = 57.3 \text{ Volts}$$

(iv) Rectifier Efficiency $\eta = \frac{P_{d.c.}}{P_{in}} x100$

$$P_{d.c.} = I_{d.c.}^2 R_L = (63.7)^2 900 = 3.65watt$$

$$P_{in} = I_{r.m.s.}^2 x(R_L + R_f) = \frac{I_m^2}{\sqrt{2}\sqrt{2}} x1000 = \frac{100x100x1000x10^{-6}}{2} = 5watt$$

$$\eta = \frac{P_{d.c.}}{P_{in}} = \frac{3.65}{5} x100 = 73\%$$

Example 5.4 A 15 – 0 – 15 volts transformer is used for full wave rectifier circuit. Each diode has a forward resistance of 10Ω. The load resistance is 6000 Ω. Find $E_{d.c.}$, $I_{d.c.}$, $I_{r.m.s.}$, and rectifier efficiency η.

Solution: $R_f = 10\ \Omega$, $R_L = 600\ \Omega$, $E_{r.m.s.} = 15$ Volts

$$E_m = 15x\sqrt{2} = 21.21volts$$

$$E_{d.c.} = \frac{2E_m}{(R_f + R_L)}.R_L = \frac{2x21.21x600}{(10+600)} = 41.72volts$$

$$I_{d.c.} = \frac{E_{d.c.}}{R_L} = \frac{41.72}{600} = 69.5mA$$

$$I_{r.m.s.} = \frac{E_m}{\sqrt{2}(R_f + R_L)} = \frac{21.21}{1.414x610} = 24.6mA$$

$$\eta = \frac{0.812}{(R_f + R_L)}.R_L = \frac{0.812}{(1+\frac{R_f}{R_L})}$$

$$= \frac{0.812}{(1+\frac{20}{600})}x100 = \frac{0.812x30x100}{31} = 78.6\%$$

5.3 Bridge Rectifier: The disadvantages that we have in case of full wave rectifier will be removed in the Bridge Rectifier. The basic circuit diagram of bridge rectifier is shown in Fig. (5.6). This circuit has four diodes connected in the four arms of a bridge, hence the name bridge rectifier. To understand the working of this circuit let us consider that during +ve half cycle of the input E_S, the point A is +ve with respect to B. In this condition, diodes D_1 and D_3 conducts and diodes D_2 and D_4 are in reverse bias. The current flows in the load resistance R_L, in the direction of arrow as shown in fig. (5.6a). During the next half cycle the point B is +ve with respect to A and thus diodes D_2 and D_4 conducts and diodes D_1 and D_3 goes in reverse bias. This also causes the flow of current through the load resistance R_L in the same direction which is shown by dotted lines in the figure 5.6(b).

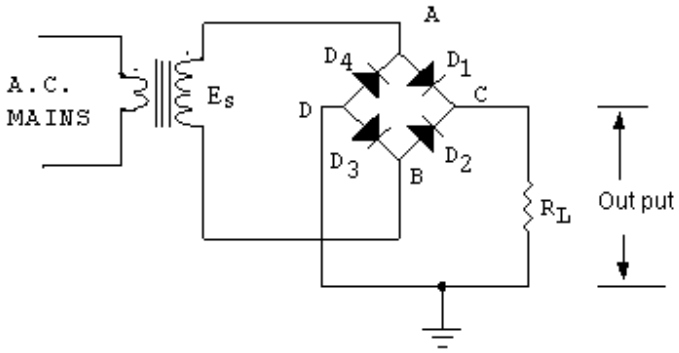


Fig. 5.6

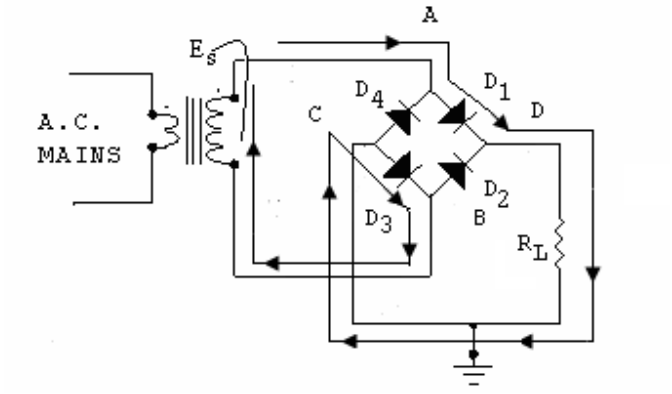


Fig. 5.6(a)

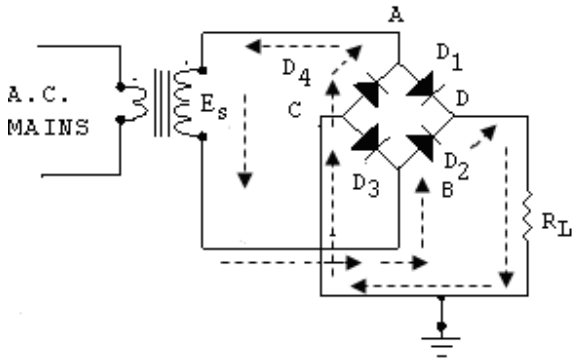


Fig. 5.6(b)

The output wave form of the bridge rectifier is essentially the same as in the case of full wave rectifier.

Advantages of Bridge Rectifier:

(i) No especially designed center tapped transformer is required in the bridge rectifier circuit.

(ii) There is no common terminal (common between input and output), so –ve supply may be obtained just by reversing the output terminals.

(iii) The peak inverse voltage across any of the four diodes in the bridge rectifier circuit is equal to E_m and not $+2E_m$ as in the case of full wave rectifier.

(iv) For the same output voltage, the transformer secondary line – to – line voltage, in the bridge rectifier should be one half of that used for full wave rectifier. Thus bridge rectifier supply large amount of d.c. power.

Example 5.5 The forward resistance of each diode used in bridge rectifier is 10 Ω. The a.c. voltage used to the input is 230 volts and load current is 1K Ω. Find $E_{d.c.}$, $I_{d.c.}$, I_m, a.c. current to the load $I_{r.m.s.}$ and PIV of the diodes used in the circuit.

Solution: $E_{r.m.s.} = 230$ Volts $R_f = 10\Omega$ $R_L = 1000\ \Omega$

$$E_m = 230\sqrt{2} = 325.22 \text{ Volts}$$

$$E_{d.c.} = \frac{2E_m x R_L}{(2R_f + R_L)\pi} = \frac{2x325.22x1000}{(20+1000)x3.14} = 203.1 \text{ Volts}$$

(Here $2R_f$ is used in place of R_f as two diodes go in forward bias at a time)

$$I_{d.c.} = \frac{E_{d.c.}}{R_L} = \frac{203.1}{1000} = 203.1mA$$

$$I_m = \frac{E_m}{(2R_f + R_L)} = \frac{325.22}{(20+1000)} = 319mA$$

PIV of the diodes = E_m = 325.22 Volts

5.4 Filter Circuits: In the preceding section we have studied the rectifier circuits. The rectified output contains a large amount of unwanted a.c. components (ripples) in addition to the d.c. voltage. The ripple can be eliminated or considerably reduced by using a filter circuit between the output of the rectifier and the load resistance. We shall now study the different commonly used filter circuits.

5.4.1 Half Wave Rectifier with Shunt Capacitor Filter: Figure (5.7) shows the circuit diagram of a half wave rectifier with shunt capacitor filter. In this circuit a capacitor C is connected in parallel (in shunt) with R_L hence it is called shunt capacitor filter. The operation of this circuit may be explained as follows:

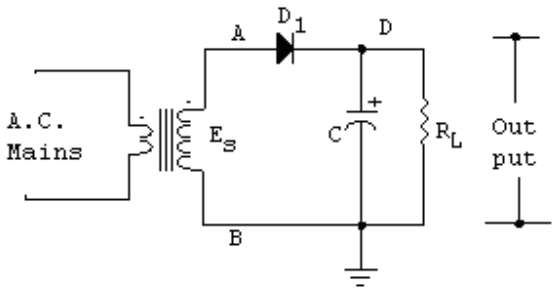


Fig. 5.7

During the voltage rise of +ve half cycle of the input wave E_s, the diode conducts and the output follows as the input voltage at point A. The capacitor C starts charging and thus it charges to the maximum of peak voltage E_m of the input wave E_s. Further during the decrease of the input wave from E_m to zero of the same half cycle, the diode D is in reverse bias. The output seems to be disconnected from the point A and the capacitor C will now start discharging through the load resistance R_L. The R_LC time constant is chosen large enough compared to half the time period of the input wave, so that the capacitor C is not completely discharged. As soon as the input voltage again becomes greater than the output voltage (during the next +ve half cycle) the diode again conducts and capacitor C starts charging further. In this way the cycle repeats. The input and output wave with shunt capacitor filter is shown in Fig. (5.8).

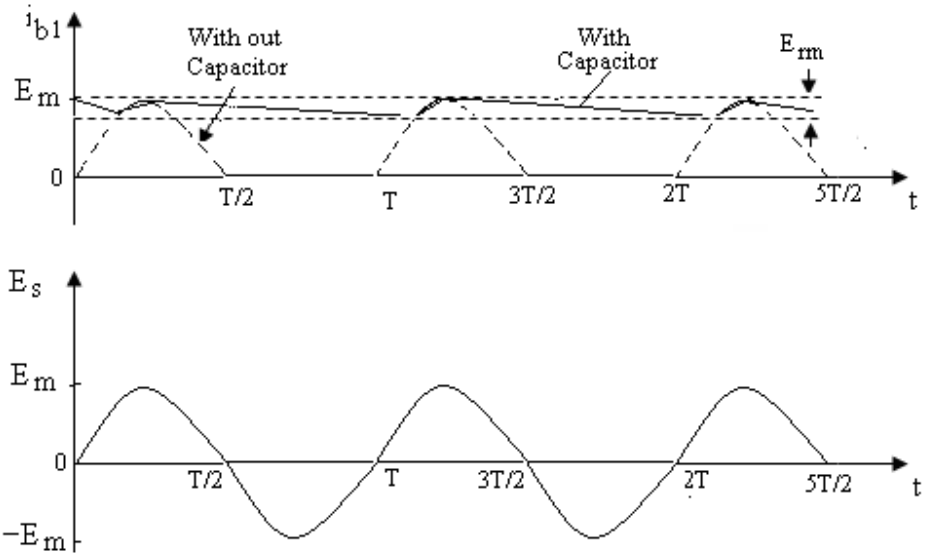


Fig. 5.8

The discharging path will depend on the R_LC time constant. Since it is large, the discharging path will therefore, seem to be almost linear.

Calculation of ripple factor: Given input signal is $E_S = E_m \sin \omega.t$

The total change in the output (ripple voltage) is shown in Fig. (5.8). The average value of this triangular wave is approximately half the total change in the output wave (E_R), which is given by:

$$E_{d.c.} = E_m - \frac{E_R}{2} \quad . \quad \text{----- (5.7)}$$

For the good filtering action the discharge period T_2 should be very much greater than the charging period T_1 i.e.

$T_2 >> T_1 \approx T$ (say) where T is the time period of the one complete cycle.

For simplicity of analysis the ripple voltage is approximated the exact triangular wave as shown in Fig.(5.9).

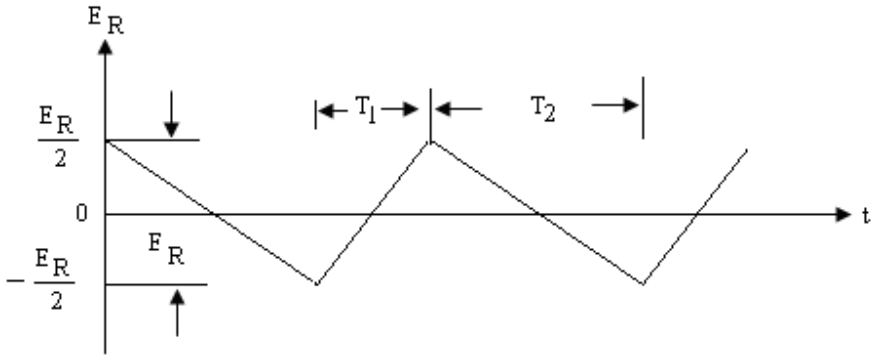


Fig. 5.9

The equation of the Triangular wave is given by:

$$E = A_1 + A_2.t \qquad \text{------ (5.8)}$$

where A_1 and A_2 are constants which may be obtained from the boundary conditions:

$$E = \frac{E_R}{2} \quad \text{at} \quad t = o \quad \text{and} \quad E = -\frac{E_R}{2} \quad \text{at} \quad t = T \qquad \text{------(5.9)}$$

so we get from equations (5.8) & (5.9) $\quad A_1 = \frac{E_R}{2} \quad \& \quad A_2 = -\frac{E_R}{T}$

The wave equation (5.8) will become $\quad E = \frac{E_R}{2} - \frac{E_R}{T}.t$

The r.m.s. value of the ripple voltage is given by:

$$E^2_{r.m.s.} = \frac{1}{T}\int_0^T E^2 dt$$

$$= \frac{E_R^2}{T}\int_0^T \left[\frac{1}{2} - \frac{t}{T}\right]^2 .dt$$

$$= \frac{E_R^2}{T}\int_0^T (\frac{1}{4} + \frac{t^2}{T^2} - \frac{t}{T}).dt \qquad = \frac{E_R^2}{12}$$

or $\quad E_{r.m.s.} = \frac{E_R}{2\sqrt{3}} \qquad \text{------ (5.10)}$

During the discharge period T_2, the capacitor loses the charge $I_{d.c.}.T_2$ at a constant rate. Hence the change in capacitor voltage E_R is calculated as:

E_R = (Loss of charge)/(capacity of the capacitor)

$$= \frac{I_{d.c.}.T_2}{C} \qquad \text{------ (5.11)}$$

It has already been assumed that $T_2 = T = \frac{1}{f}$ for good filtering action. Where f is the frequency of the wave in Hz.

So $\quad E_R = \frac{I_{d.c.}}{f.C} \qquad \text{------ (5.12)}$

From the equations (5.7) & (5.12) we get :

$$E_{d.c.} = E_m - \frac{I_{d.c.}}{2\,f.C}$$

The term $\frac{1}{2.f.C}$ has the dimension of resistance, which represents the source resistance of d.c. supply $E_{d.c.}$; and may assumed to be an open circuit voltage. For having the low ripple voltage and ensure good voltage regulation the capacitor C must be large enough.
The ripple factor γ is given by:

$\gamma =$ (r.m.s. value of a.c. output)/(d.c. output voltage)

$$= \frac{E_R}{2\sqrt{3}.E_{d.c.}} \qquad \left(\text{since } E_R = \frac{I_{d.c.}}{f.C}\right)$$

$$= \frac{1}{2\sqrt{3}.f.C.R_L} \qquad \left(\text{where } \frac{E_{d.c.}}{I_{d.c.}} = R_L\right) \qquad \text{----- (5.13)}$$

From this equation it is clear that we may get small ripple content at high load resistance. At no load ($R_L \to \infty$) the output voltage will be equal to E_m, the open circuit voltage of the d.c. supply; and we thus say that this circuit behaves like a peak detector. This filter is suitable for low current.

The PIV of the diode used in half wave rectifier with shunt capacitor filter is approximately $2E_m$ since the point D (in fig. 5.7), the cathode of the diode is at about $+E_m$ potential and anode goes to $-E_m$ with respect to common point. Hence the total reverse voltage across the diode is $2E_m$.

Example 5.6 A transformer whose secondary winding is rated at 12 volts r.m.s., is used for half rectifier with shunt capacitor filter. If the value of capacitor C = 100 μF and load resistance R_L = 1KΩ and frequency f = 50 Hz, determine $E_{d.c.}$ and $I_{d.c.}$, peak to peak ripple voltage at the output, peak forward current in the diode. How do the above change if C is increased to 1000 μF? What should be voltage ratings of the capacitor, if 25% variation is allowed in the input a.c. voltage.

Solution: C = 100 μF R_L = 1000 Ω $E_{r.m.s.}$ =12 volts f = 50 Hz

$$E_m = 12\sqrt{2} = 12x1.414 = 16.97 \text{ Volts}, \qquad I_{d.c.} = \frac{E_{d.c.}}{R_L}$$

$$E_{d.c.} = E_m - \frac{I_{d.c.}}{2f.C} = E_m - \frac{E_{d.c.}}{2f.C.R_L}$$

or $$E_{d.c.}\left(1+\frac{1}{2f.C.R_L}\right) = E_m \qquad \text{and} \qquad E_{d.c.} = \frac{E_m}{\left(1+\frac{1}{2.f.C.R_L}\right)}$$

$$Ed.c. = \frac{16.97}{1+\frac{1}{2x50x100x10^{-6}x1000}} = \frac{16.97}{1.1} = 15.43 \text{ Volts}$$

$$I_{d.c.} = \frac{E_{d.c.}}{R_L} = \frac{15.43}{1000} = 15.43mA$$

Peak to peak ripple voltage at the output E_R is given by:

$$E_R = \frac{I_{d.c.}}{f.C} = \frac{15.43\ x10^{-3}}{50\ x100\ x10^{-6}} = 3.086 \text{ Volts}$$

Peak forward current in the diode $I_m = \frac{E_m}{R_L} = \frac{16.97}{1000} = 16.97mA$

Now the value of C is increased to 1000 μF so put C = 1000 μF in the above calculations, we get:

$$Ed.c. = \frac{16.97}{1 + \frac{1}{2x50x1000x10^{-6}x1000}} = \frac{16.97}{1.1} = 16.8 \text{ Volts}$$

$$I_{d.c.} = \frac{E_{d.c.}}{R_L} = \frac{16.8}{1000} = 16.8mA$$

$$E_R = \frac{I_{d.c.}}{f.C} = \frac{16.8x10^{-3}}{50\ x1000\ \ x10^{-6}} = 0.336 \text{ Volts}$$

$$I_m = \frac{E_m}{R_L} = \frac{16.97}{1000} = 16.97mA$$

If 25% increase in the a.c. signal then peak voltage also increases in the same ratio i.e.

$E_m = 16.97 + 0.25x16.97 = 21.2$ Volts

So the voltage rating of the capacitor should be greater than 21.2 Volts may be taken of 25 Volts or 50 volts.

5.4.2 Full Wave Rectifier with Shunt Capacitor Filter: The circuit diagram of this filter is shown in figure (5.10). The working of this circuit is similar to that of the half wave rectifier with shunt capacitor filter; the only difference between the two is that the circuit works for the half cycles. The input – output wave shaped is given in figure (5.11).

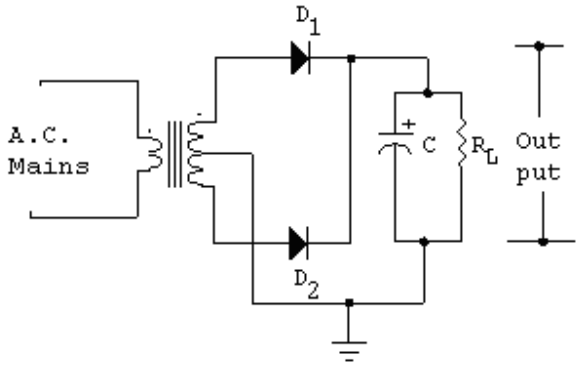


Fig. 5.10

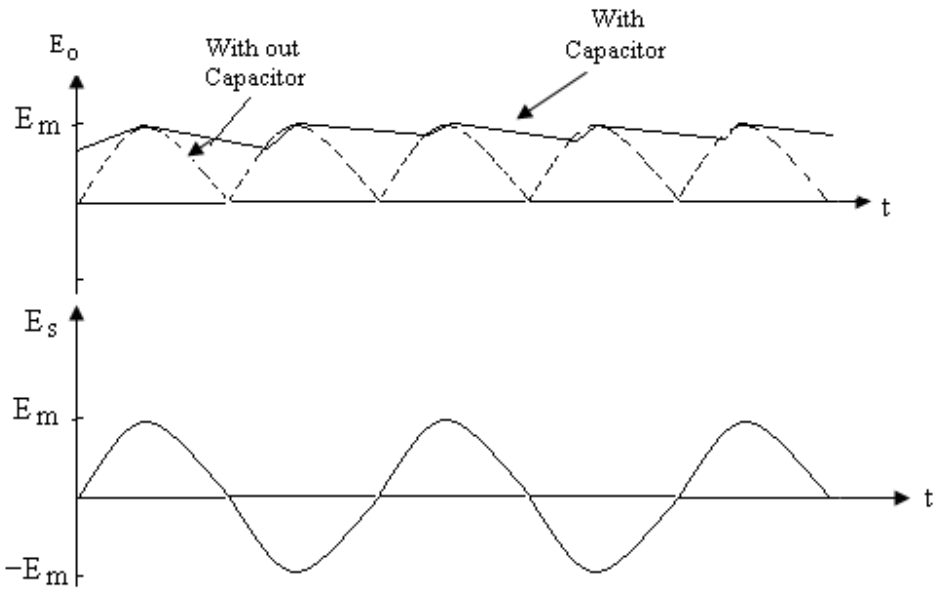


Fig. 5.11

The ripple factor for the full wave rectifier with shunt capacitor filter may also be calculated in the similar fashion as for half wave case simply by replacing $T_2 \approx T/2$. Since in this case, the discharge period T_2 must be greater than T/2. So T_2 is assumed to be equal to T/2.
The ripple factor γ is given by:

$$\gamma = \frac{1}{4\sqrt{3}.f.C.R_L} \qquad \text{------ (5.14)}$$

5.4.3 Percentage Regulation: The percentage regulation of a d.c. power supply may be defined as:
Percentage regulation = (**No load voltage – Full load voltage) / Full load voltage**
If the supply designed by the rectifier and filter, then it can be represented by the model of the battery, i. e., a voltage source E_i (open circuit voltage) and a source resistance R_0 in series with it as given in figure (5.12).

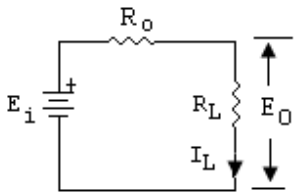


Fig. 5.12

From this figure we get $E_i = R_0 I_L + E_0$ and $E_0 = R_L I_L$
or $E_i - E_0 = R_0 I_L$

E_I = No load voltage E_0 = Full load voltage

So Percentage regulation = $\frac{R_0 I_L}{R_L I_L} x100\% = \frac{R_0}{R_L} x100\%$

Example 5.7 A full wave rectifier is supplied with an a.c. signal of 50 – 0 – 50 volts r.m.s. and 50 Hz frequency. A 100 μF capacitor serves as a filter and takes 300 mA load current. What is the d.c. load voltage? Also find the ripple factor.

Solution: $E_{r.m.s.}$ = 50 Volts, f = 50 Hz C = 100 μF $I_{d.c.}$ = 300 mA

$$E_m = 50x\sqrt{2} = 50x1.414 = 70.7 \text{ Volts}$$

For full wave rectifier with shunt capacitor filter is given by:

$$E_{d.c.} = E_m - \frac{I_{d.c.}}{4\ f\ .C}$$

$$= 70.7 - \frac{300x10^{-3}}{4x50x100x10^{-6}} = 70.7 - 15 = 55.7 \text{ Volts}$$

$$R_L = \frac{E_{d.c.}}{I_{d.c.}} = \frac{55.7}{300x10^{-3}} = 185.7\Omega$$

$$\gamma = \frac{1}{4\sqrt{3}.f.C.R_L} = \frac{1}{4x1.73x50x100x10^{-6}x185.7}$$

$$= 0.156 = 15.6\%$$

Example 5.8 It is required to design full wave rectifier with shunt capacitor filter which is capable of supplying 20 volts d.c. at no load. The regulation of this supply is required to be less than 10% for a full load current of 1 ampere. The maximum ripple is to be less than 3 volts (peak to peak). Find (i) the required secondary rating of the transformer,
(ii) the value and voltage rating of the capacitor,
(iii) the peak forward current and PIV ratings of the diodes.

Solution: No load voltage $E_{d.c.}$ = 20 Volts % regulation = 10%

Percentage regulation = (**No load voltage – Full load voltage) x100/ Full load voltage**

10 = (20 – Full load voltage)x100/Full load voltage

or **Full load voltage = 18.18 volts**

The full load voltage should be greater than 18.18 volts say 18.5 Volts

Ripple voltage E_R = 3 Volts (peak to peak)

$$E_R = \frac{I_{d.c.}}{2.f.C} = \frac{1amp}{2x50\ xC} = 3volts$$

or $$C = \frac{1}{300} = 0.003333F = 3333\mu F$$

$$R_L = \frac{E_{d.c.}}{I_{d.c.}} = \frac{18.5}{1amp} = 18.5\Omega$$

We know $$I_{d.c.}R_L = E_m - \frac{I_{d.c.}}{4\ f\ .C}$$

$$I_{d.c.} = \frac{E_m}{(R_L + \frac{1}{4.f.C})} \qquad \text{or} \qquad E_m = I_{d.c.}(R_L + \frac{1}{4.f.C})$$

$$Em = 1(18.5 + \frac{300}{4x50x1}) = 20 \text{ volts}$$

$$Er.m.s. = \frac{Em}{\sqrt{2}} = \frac{20}{1.414} = 14.14 \text{ volts} = 15 \text{ (say)}$$

(i) So secondary of the transformer should be rated as 15 – 0 – 15 volts and current 1amp.

(ii) The value of capacitor C= 3333 μF and voltage ratings should greater than 20 volts.

(iii) Peak forward current $I_m = \frac{E_m}{R_L} = \frac{20}{18.5} = 1.08amp$

PIV of the diodes = $2E_m$= 40 volts

5.4.4 Series Inductor Filter: The circuit diagram of the series inductor filter is shown in figure (5.13). Here an inductance is connected in series with the load resistance. Therefore, it is known as series inductor filter. The signal E_I has the d.c. components along with the higher harmonics of input frequency ω. E_I is given by:

$$E_I = \frac{2E_m}{\pi} - \frac{4E_m}{3\pi}\cos 2\omega.t - \frac{4E_m}{15\pi}\cos 4\omega.t - .....$$

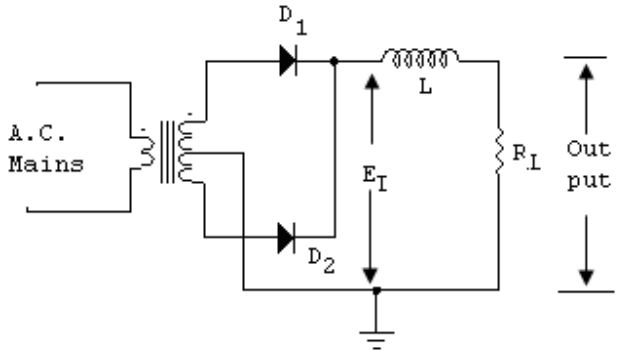


Fig. 5.13

Inductance L is connected to eliminate or reduce the ripple content at the output. Out of the ripple contents, the second harmonic term $\frac{4E_m}{3\pi}.\cos \omega.t$ is the only effective term. Other terms may be neglected. The inductance offers zero reactance for d.c. components and thus d.c. voltage appears at the output terminals. However, it offers very high reactance for the second and other higher harmonic terms. Thus the inductance attenuates the a.c. components to appear at the output terminals.
Now the ripple for this type of filter is calculated as below:

Peak value of the a.c. components to appear at the output is

$$E_{a.c.} = \frac{4E_m}{3\pi}.\frac{R_L}{(R_L + X_L)}) \qquad \text{where} \quad X_L = 2\,j\omega.L\,, \text{ is the inductive}$$

reactance at second harmonic frequency 2ω . The root mean square value of this ripple content is :

$$E_{r.m.s.} = \frac{4E_m}{3\sqrt{2}\pi}\frac{R_L}{(R_L + X_L)}$$

$$\left|E_{r.m.s.}\right| = \frac{4E_m}{3\sqrt{2}\pi}\frac{R_L}{\sqrt{\left[R_L + 4\omega^2 L^2\right]}} = \frac{4E_m}{3\sqrt{2}\pi\sqrt{\left(1 + \frac{4\omega^2 L^2}{R_L^2}\right)}}$$

If $\frac{4\omega^2 L^2}{R_L^2} >> 1$ then $\left|E_{r.m.s.}\right| = \frac{2E_m.R_L}{3.\sqrt{2}\pi.\omega.L}$

$$E_{d.c.} = \frac{2E_m}{\pi}$$

The ripple factor $\gamma = \frac{E_{r.m.s.}}{E_{d.c.}} = \frac{2E_m R_L / 3\sqrt{2}\pi\omega}{\frac{2E_m}{\pi}} = \frac{1}{3\sqrt{2}}.\frac{R_L}{\omega.L}$ ------ (5.15)

From this equation, it is clear that the ripple factor depends on both the load resistance and the magnitude of the inductances. The ripple factor decrease with the increase of inductance; and also it is smaller for smaller value of R_L. That is the inductor filter is suitable for higher value of load current.

Example 5.9 A full wave rectifier with series inductor filter has the load resistance of 1KΩ and inductor of 25 Henry. The peak value of the applied a.c. signal is 50 volts and frequency 50Hz. Calculate d.c. output voltage, d.c. current to the load resistance and the ripple factor. Diodes used are ideal.

Solution: L = 25 Henry E_m = 50 Volts R_L = 1000 Ω f = 50 Hz

$$E_{d.c.} = \frac{2E_m}{\pi} = \frac{2x50}{3.14} = 31.85 \text{ Volts}$$

$$I_{d.c}. = \frac{E_{d.c.}}{R_L} = \frac{31.85}{1000} = 31.85mA$$

Ripple factor $\gamma = \frac{1}{3\sqrt{2}}.\frac{R_L}{\omega.L} = \frac{R_L}{3\sqrt{2}x2x\pi xf} = \frac{1000}{3x1.414x2x3.14x50} = 0.75$

5.4.5 L –Section (or L-C) Filter: It has been discussed earlier that the shunt capacitor filter is used for small value of load currents and inductor filter is suitable where we wish to draw large amount of currents. The series combination of inductor and

capacitor used for the filtering action is suitable for all values of the currents. Figure (5.14) shows the circuit diagram of L-section or L-C filter with full wave rectifier.

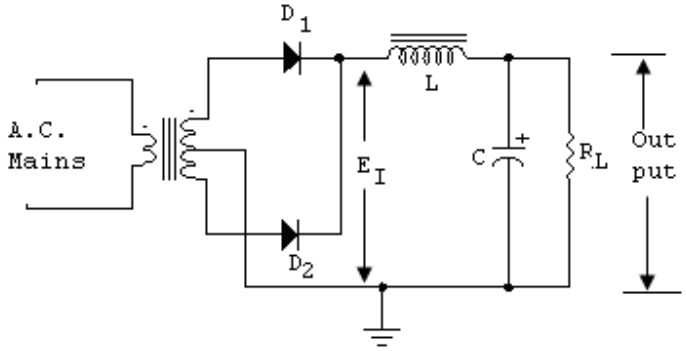


Fig. 5.14

In this circuit, the values of inductance L and capacitance C are chosen so that X_L (inductive reactance) is much greater than X_C (capacitive reactance) at the ripple frequency. In this way the inductor attenuates the ripple and capacitor bypasses it.
The voltage E_I available at the input of the L-section filter is given by:

$$E_I = \frac{2E_m}{\pi} - \frac{4E_m}{3\pi}\cos 2\omega.t - \frac{4E_m}{15\pi}\cos 4\omega.t - .....$$

Higher Harmonic terms (having frequency more than 2ω) may be neglected. The effective ripple frequency may be assumed as 2ω.

$$E_{d.c.} = \frac{2E_m}{\pi}$$

The a.c. components (ripple content) at the output terminals may be given by:

$$E_{a.c.} = \frac{4E_m.(X_C \| R_L)}{3\pi.(X_L + X_C \| R_L)}\cos 2\omega.t$$

It is further assumed that X_L (inductive reactance at 2ω frequency) is large enough than the parallel combination of load resistance R_L and X_C (capacitive reactance at 2ω ripple frequency) also $R_L >> X_C$.

So $$Ea.c. = \frac{4E_m.X_C}{3\pi.X_L}\cos 2\omega.t .$$

The r.m.s. value of a.c. component is $$E_{r.m.s.} = \frac{4E_m}{3\sqrt{2}\pi}\frac{X_C}{X_L}$$

Putting $X_C = \frac{1}{2\omega.C}$ (capacitive reactance at 2ω ripple frequency)

and $X_L = 2\omega L$ (inductive reactance at 2ω ripple frequency)

we get $$E_{r.m.s.} = \frac{4E_m}{3\sqrt{2}\pi\omega^2.2\omega L.2\omega C} = \frac{E_m}{3\sqrt{2}\pi\omega^2 L.C}$$

The ripple factor γ is given as: $\gamma = \frac{E_{r.m.s}}{E_{d.c.}} = \frac{E_m / (3\sqrt{2}\pi\omega^2 LC)}{2E_m/\pi}$

or $\gamma = \frac{1}{6\sqrt{2}\omega^2 L.C}$ ------ (5.16)

It is clear from this equation that the ripple factor is independent of load resistance. So this filter circuit is suitable for all values of currents.

5.4.6 π -Section Filter: A more common filter is the π -Section Filter shown in figure (5.15). It has a capacitor filter followed by L-section filter and is used to provide

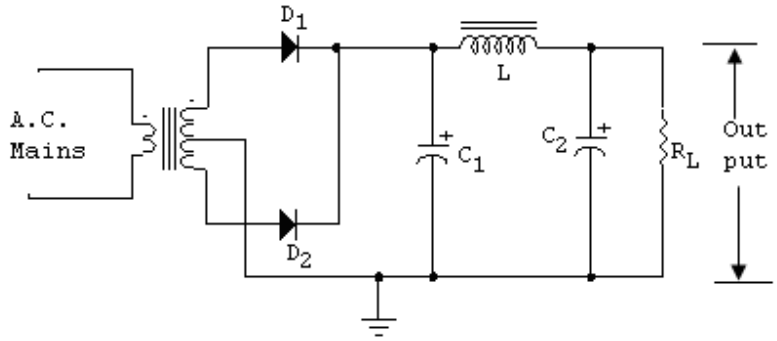


Fig. 5.15

higher d.c. output voltage with low ripple. It used for medium load currents. The capacitors C_1, C_2 and inductance L form a $\pi-$type network, hence it is known as π - Section filter. Here the capacitor C_1 bypasses the ripple frequency, the inductance L attenuates the ripple and C_2 further bypasses it. So it is assumed that pure d.c. is available at the output.

The r.m.s. value of a.c. components (ripple content) across the capacitor C_1 is given by:

$$E_{r.m.s.} = \frac{I_{d.c.}}{4\sqrt{3}fC_1} = \frac{\pi . I_{d.c.}}{2\sqrt{3}\omega C_1} \quad \text{where } \omega = 2\pi f$$

The r.m.s. value of the a.c. voltage across the parallel combination of C_2 and R_L may be calculated as:

$$E'_{r.m.s.} = \frac{E_{r.m.s.}(X_{C_2} \| R_L)}{(X_L + X_{C_2} \| R_L)} \cong \frac{I_{d.c.}\pi . X_{C_2}}{2\sqrt{3}\omega C_1 X_L} \quad \text{(since } X_L >> X_{C_2} \text{ and } R_L >> X_{C_2})$$

Putting $X_C = \frac{1}{2\omega.C}$ (capacitive reactance at 2 ω ripple frequency)

and $X_L = 2\omega L$ (inductive reactance at 2 ω ripple frequency)

We get $E'_{r.m.s.} \cong \frac{\pi . I_{d.c.}}{8\sqrt{3}\omega^3 C_1 C_2 L}$

d.c. voltage across the load resistance $E_{d.c.} = I_{d.c.} R_L$

The ripple factor is given by:

$$\gamma = \frac{E'_{r.m.s.}}{E_{d.c.}} = \frac{\pi . I_{d.c.}}{8\sqrt{3}\omega^3 . C_1 . C_2 . L . I_{d.c.} R_L}$$

or $$\gamma = \frac{\pi}{8\sqrt{3}\omega^3 . C_1 . C_2 . L . R_L} \qquad \text{------ (5.17)}$$

the ripple factor is inversely proportional to the load resistance.

We can, of course, use more than one section of filters depending upon the extent of ripple reduction as shown in figure (5.16). Each section acts like a voltage divider; the overall attenuation equals the product of the individual attenuation.

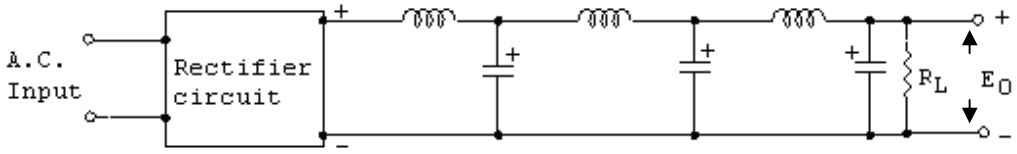


Fig. 5.16

Example 5.10 A full wave rectifier with $\pi -$ section filter has the following circuit elements. $C_1 = C_2 = 20$ μF, L = 20H, $R_L = 5K\Omega$, peak value of the a.c. signal 200 volts and frequency = 50 Hz. Find the ripple factor of this circuit.

Solution: The ripple factor γ is given by $\gamma = \frac{\pi}{8\sqrt{3}\omega^3 . C_1 . C_2 . L . R_L}$

or $$\gamma = \frac{3.14}{8x\sqrt{3}x(2x3.14x50)^3 x20x20x10^{-6} x10^{-6} x20x5000}$$
$$= 0.002 = 0.02\%$$

5.5 Voltage Multiplier Circuits: Some times high voltage / low current d.c. supply is required in electronic circuits. Such a supply is needed for accelerating the electrons in a cathode ray tube of CRO. Voltage multiplier circuits are used to design such a supply, it gives a d.c. output which is a multiple of the peak value of the input a.c. signal applied to the circuit.

5.5.1 Half Wave Voltage Doubler: As the name indicates it gives d.c. output which is almost double the peak value of the input signal. It is also known as the cascade voltage doubler. The basic circuit diagram of the half wave voltage doubler is shown in figure (5.17a).

It works as follows. During the negative half cycle of the input sinusoidal wave, diode D_1 conducts and the capacitor C_1 charges to the maximum of the peak value of the input wave (E_m) with the polarity of the capacitor as shown in figure (5.17). The diode D_2 will be in the reverse bias. Now during the positive half cycle of the input wave, diode D_1 will

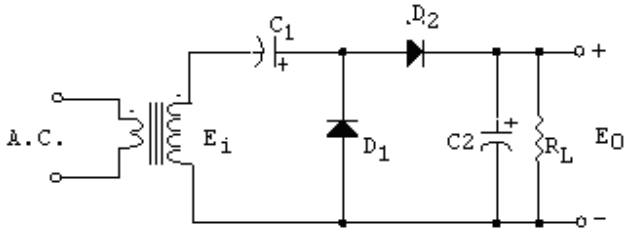


Fig. 5.17

be in reverse bias and diode D_2 conducts as the voltage across the diode D_1 is positive and its magnitude will be approximately equal to double the peak value of the input sinusoidal wave. The capacitor C_2 will charge to steady voltage $+2E_m$ for the number of cycles of operations. So the output voltage E_0 which is equal to the double of the peak value of the input will be available across the load resistance R_L. Hence this circuit is known as voltage doubler. The PIV ratings of the diodes used in this circuit, is $2E_m$. Half wave voltage doubler has very poor regulation and its ripple content is also high as ripple frequency is equal to the frequency of the input signal.

5.5.2 Full Wave Voltage Doubler: The basic circuit diagram of the full wave voltage doubler is shown in figure (5.18). It works as follows.

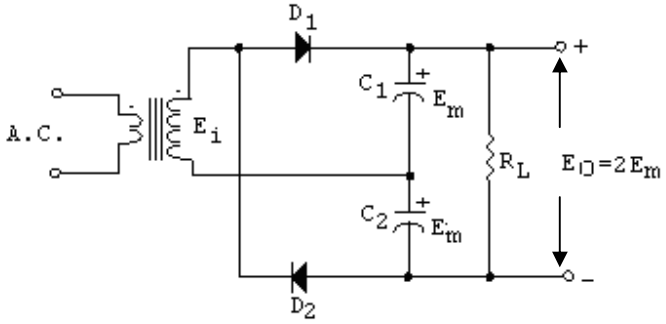


Fig. 5.18

During the first positive half cycle of the input sinusoidal signal of the wave the diode D_1 will conduct and capacitor C_1 charges to the maximum of the peak value E_m of the wave, with the polarity of the capacitor C_1. In this case, the diode D_2 will be in reverse bias. Now during the next negative half cycle of the input wave, diode D_1 will be in reverse bias and diode D_2 will conduct and capacitor C2 will charge to E_m. The total voltage across the load resistance R_L will be just double of the peak value of the input wave i.e. $+2E_m$. The PIV ratings of the diodes used in this circuit, is $2E_m$. The ripple frequency in this circuit is equal to twice the frequency of the input signal (second harmonic terms).

5.5.3 Half Wave Voltage Multiplier: The half wave voltage doubler discussed above can be extended to obtain any multiple of the peak value of the input wave i.e. $3E_m$, $4E_m$, $5E_m$, $6E_m$ etc. The circuit thus obtained may be called as voltage multiplier which is shown in figure (5.19).

The circuit may be explained on the same lines as the half wave voltage doubler. The voltage obtained across A & B terminals is equal to $2E_m$, and the voltage across A &

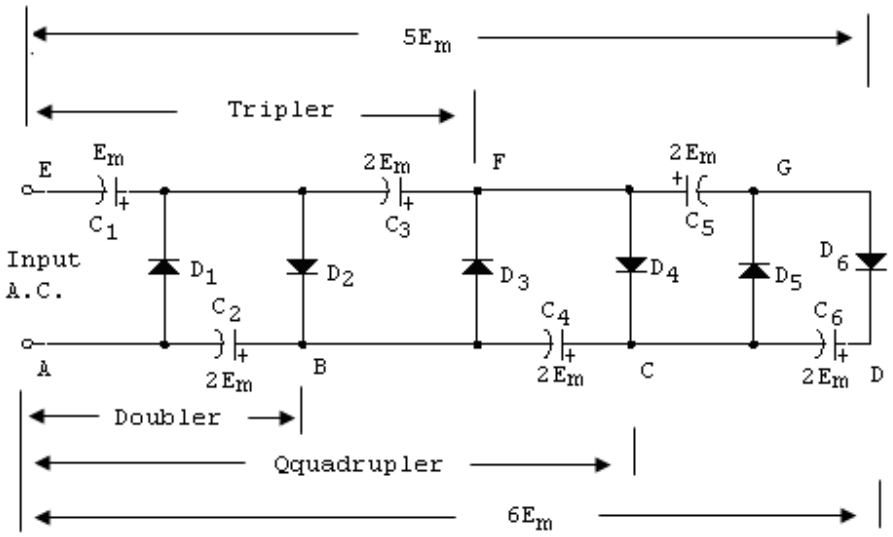


Fig. 5.19

C terminals is equal to 4Em (Quadrupler), and across A & D is 6Em. The voltage across the terminals E & F is $3E_m$ (Tripler), and the voltage across E & G terminals is $5E_m$.

Example 5.11 What will be voltage at the output of half wave voltage doubler, if the voltage at the secondary of the transformer is 25 volts? What will be PIV of the diodes used?

Solution: $E_{r.m.s.} = 25$ volts,

Peak value $E_m = \sqrt{2}xE_{r.m.s.} = 1.414x25 = 35.35$ volts

Output voltage of the doubler = $2.E_m = 2x35.35 = 70.70$ volts

PIV of the diodes used = $2.E_m = 70.70$ volts

5.6 Clipper Circuits: The clipping circuits clip off the unwanted portion of the signal without distorting the remaining part of the applied signal. The clipping circuits basically are of two types (i) Series Clipper and (ii) Shunt Clipper. In series clippers the diodes are connected in series with the load resistance and hence the name series clipper. In shunt clippers the diodes are connected in parallel or in shunt with the load resistance. These clippers may further be classified as (i) unbiased and (ii) biased clippers. In unbiased clippers no additional battery is applied to the diodes. However, in case of biased clippers, the diodes are biased with the additional battery of desired magnitude.

The classification of clipper circuits is shown in figure (5.20). Now we shall discuss each of these circuits in detail.

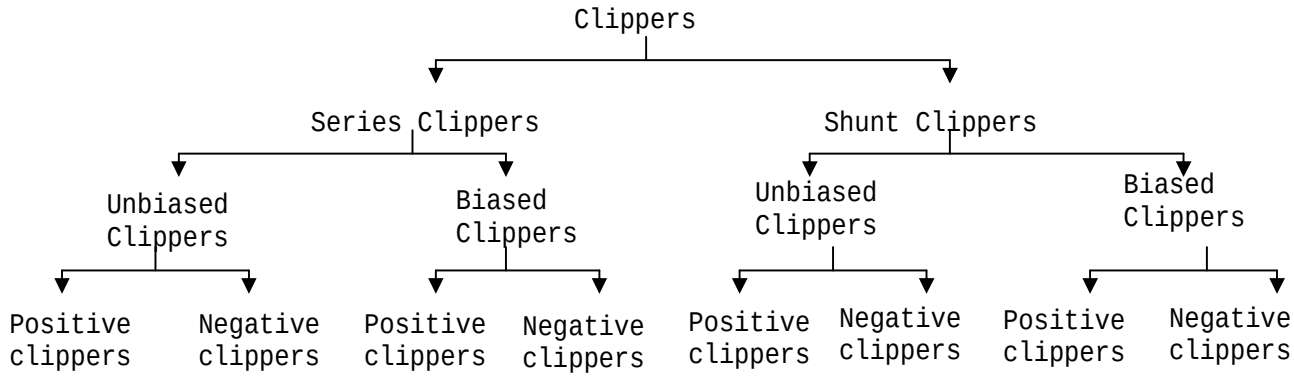


Fig. 5.20

5.6.1 Unbiased Positive Series Clipper: This circuit is basically the half wave rectifier as shown in figure (5.21). If the input wave sine, triangular or square wave is applied to the input of the circuit, we get zero output for the positive half cycle of the input wave; and for the negative half cycle output is the same as the input. This is because that the diode goes in reverse bias during positive half cycle of the input wave and during the negative half cycle the diode is forward bias and output follows the input. The input output wave shapes are shown in figure (5.22). From the wave shapes it is clear that this circuit clip off the positive half cycle of the input wave hence the name positive clipper.

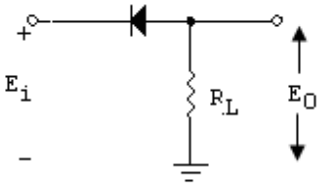


Fig. 5.21

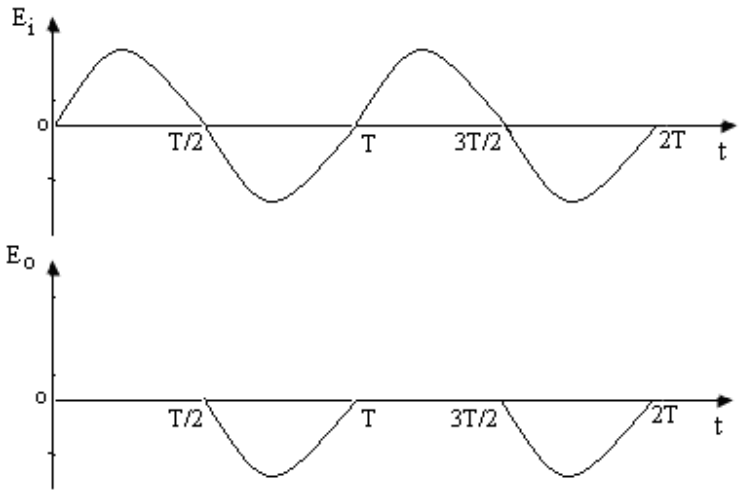


Fig. 5.22

5.6.2 Unbiased Negative Series Clipper: If the connections of the diode are reversed as shown in figure (5.23), then the circuit is known as negative unbiased series clipper circuit. It clips off the negative half cycle of the input wave; and the output is the same as the input for negative half cycle. The input output wave shapes are shown in figure (5.24).

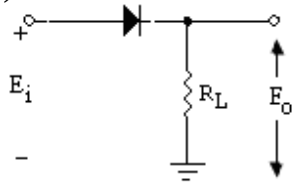


Fig. 5.23

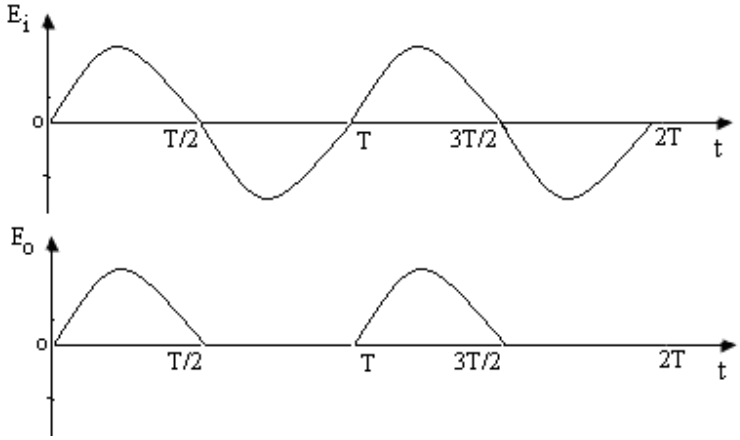


Fig 5.24

5.6.3 Biased Positive Series Clipper: The circuit diagram of this clipper is shown in figure (5.25). To explain the working of the circuit let us assume the diode and battery

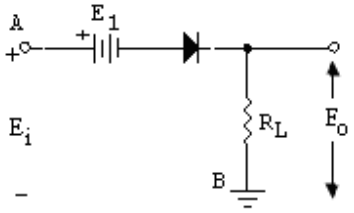


Fig. 5.25

are ideal. During the positive half cycle of the input, the point A is positive with respect to point B and the diode will remain in reverse bias till the cathode of the diode is more negative than E_1. During the next half cycle of the input wave the diode will remain in reverse bias and the output E_0 is zero. So when the input is positive and beyond E_1, the output follows the input; and when the input is between zero and E_1 then the output E_0 is zero. It is, therefore, concluded that this circuit has clipped off the portion between zero and $+E_1$ of the input wave. Hence the name biased positive clipper. The input output wave shapes are given in figure (5.26). It is worth while to mention that the input should be greater than the magnitude of the battery otherwise the clipping will not take place.

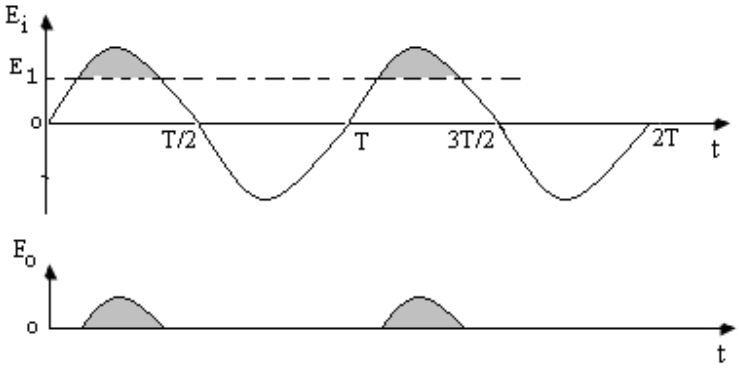


Fig. 5.26

5.6.4 Biased Negative Series Clipper: The circuit diagram of this clipper is shown in figure (5.27). To explain the working of the circuit let us assume the diode and

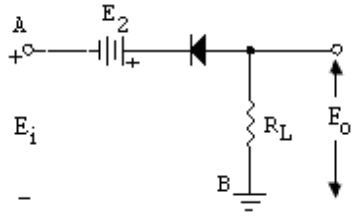


Fig. 5.27

battery are ideal. During the positive half cycle of the input, the point A is positive with respect to point B and the diode will be in reverse bias. During the next half cycle of the input wave the diode will remain in reverse bias till the anode of the diode is more positive than E_2 and the output E_0 will follow the input beyond E_2. So when the input is positive, the output is zero and when the input is negative and beyond E_1 then the output follows the input. It is, therefore, concluded that this circuit has clipped off the portion between zero and $– E_2$ of the input wave. Hence the name biased negative clipper. The input output wave shapes are given in figure (5.28). It is worth while to mention that the input should be greater than the magnitude of the battery otherwise the clipping will not take place.

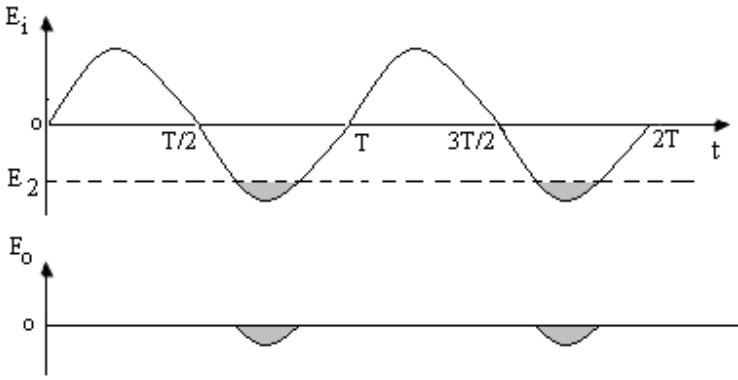


Fig. 5.28

Further if both the circuits of figures (5.25) & (5.27) are combined as shown in figure (5.29), then the clipper will be both sided biased clipper.

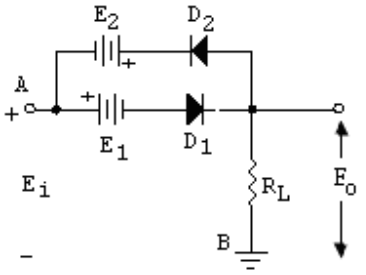


Fig. 5.29

It is very easy to explain the circuit on the similar lines as discussed above. In this circuit the diode D_1 conducts when input is positive and beyond E_1. The output thus follows the input, clipping off the portion between 0 and E_1. The diode D_2 will conduct when the input is negative and beyond E_2. The output follows the input, clipping off the portion between 0 and – E_2. The input output wave shapes are shown in figure (5.30).

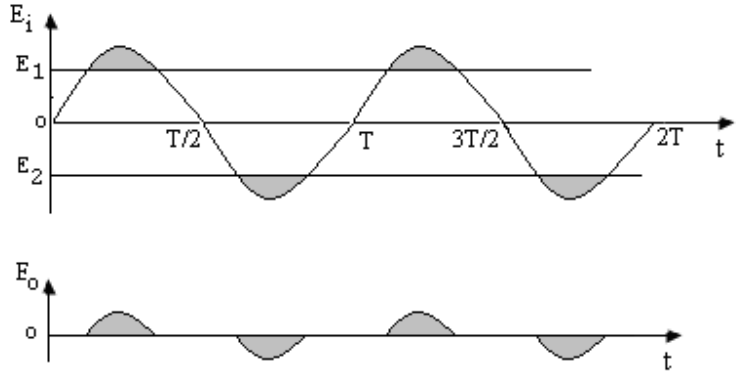


Fig. 5.30

In the clipping circuits discussed above, we have considered the ideal diodes and ideal batteries. Practically it is sometimes important to consider the knee voltage or cut – in voltage V_γ of the diode, its resistance r_d and also the internal resistance of the battery r_b. In this situation the V_γ potential will be added with the voltage of the battery that is clipping voltage will be $E_1 + V_\gamma$ and $E_2 + V_\gamma$ in double sided series clipper. The transfer characteristics which is the input output relationship, will have less than unity slope at the points $(E_1 + V_\gamma)$ and $-(E_2 + V_\gamma)$. The value of slope is given by $\frac{R_L}{R_L + r_d + r_b}$. This will result a distortion in the out put wave forms.

Example 5.12 Draw the transfer characteristics of the series clipper circuit shown in the figure (5.31a). The values of the circuit parameters are given as: E_1 =1.5 V, E_2 = 2.0 V, R_L = 200 Ω, diode resistance of each diode r_d=20 Ω and internal resistance of each battery r_b = 10 Ω. The cut – in voltage of each diode V_γ = 0.7 V.

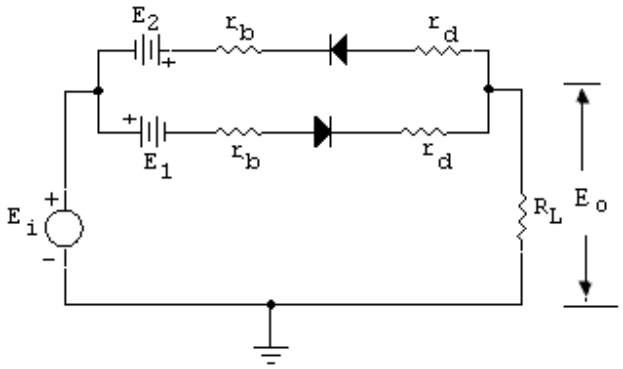


Fig. 5.31a

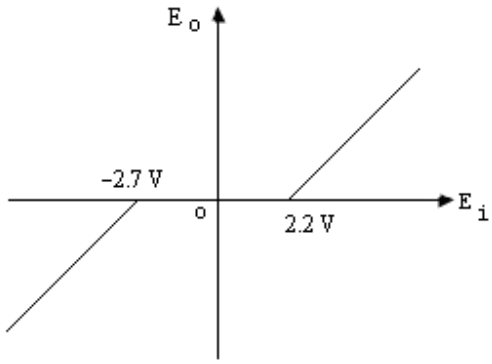


Fig. 5.31b

Solution: The break points are $(E_1 + V_\gamma) = 1.5 + 0.7 = 2.2$ V

$-\ (E_2 + V_\gamma) = 2.0 + 0.7 = -2.7$ V

The slope at these points is $\dfrac{R_L}{R_L + r_d + r_b} = \dfrac{200}{200 + 20 + 10} = 0.87$

The transfer characteristics are shown in figure (5.31b).

5.6.5 Unbiased Positive Shunt Clipper:

The circuit diagram for unbiased

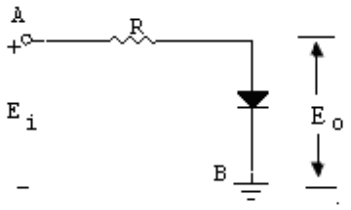


Fig. 5.32

positive shunt clipper is shown in figure (5.32). Its working may very easily be explained as follows. During the positive half cycle of the input wave, the diode is forward biased and will act as an on switch. The output will be zero for the positive half cycle. However,

for the negative half cycle diode is in reverse bias and will act as open switch. The output will therefore, follows the input. So during positive half cycle no output is obtained clipping off this cycle of the input wave. The output is taken in parallel with the diode hence called unbiased positive shunt clipper. The input output wave shapes are shown in figure (5.33).

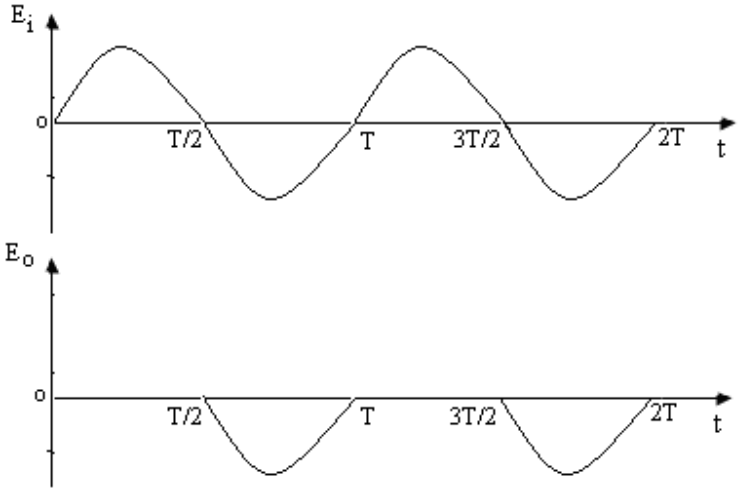


Fig. 5.33

5.6.6 Unbiased Negative Shunt Clipper: In this circuit the connections of the diode are reversed as shown in figure (5.34a). This circuit clip off the negative half cycle of the input wave and positive half cycle is obtained without any distortion. The input output wave shapes are shown in figure (5.34 b)

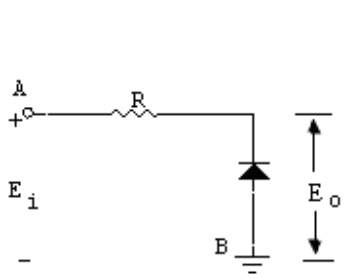


Fig. 5.34 a

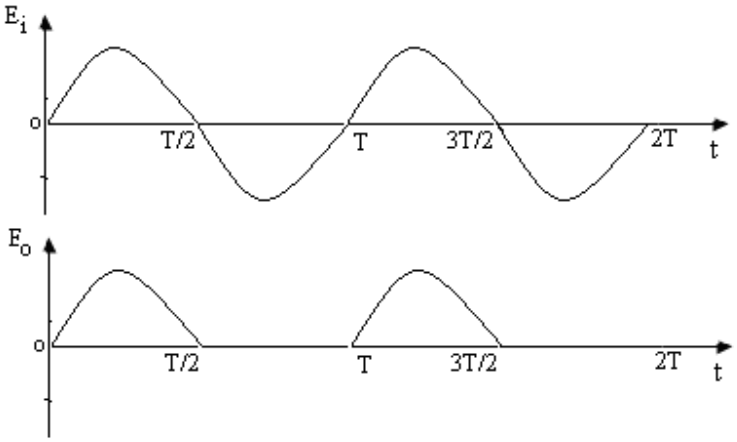


Fig. 5.34 b

5.6.7 Biased Positive Shunt Clipper: The circuit diagram of the biased positive

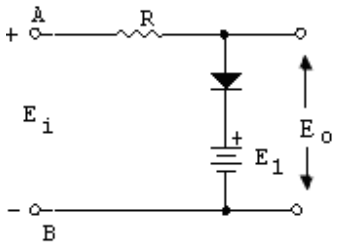


Fig. 5.35 a

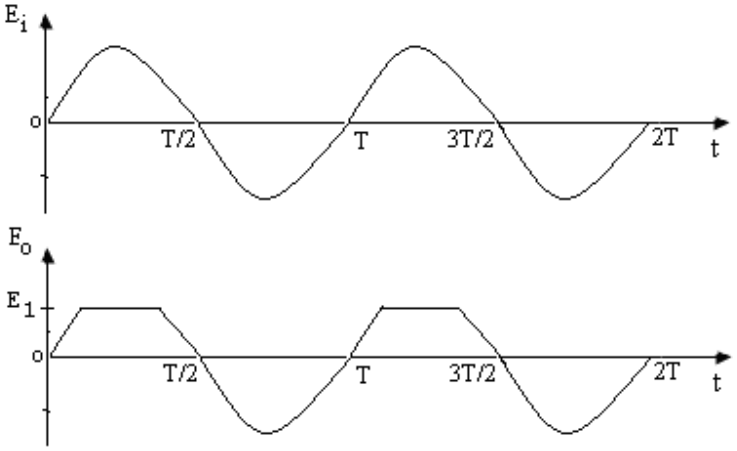


Fig. 5.35b

clipper is shown in figure (5.35a). It work as follows, during the positive half cycle of the input wave, the diode will be in reverse bias and behaves as an open switch till input is less than or equal to the magnitude of the battery E_1. In this condition, output follows the input. When the input becomes more positive than E1, diode conducts and works as an on switch. The output will equal to E_1. During the next half cycle the diode will be in reverse bias and output will follow the input. So it is concluded that the portion of the input beyond E1, in positive half cycle is clipped off. The input output wave shapes is shown in figure (5.35 b).

5.6.8 Biased Negative Shunt Clipper: The circuit diagram of the biased negative clipper is shown in figure (5.36 a). In this circuit the connections of the diode and

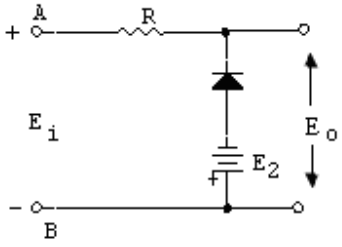


Fig. 5.36 a

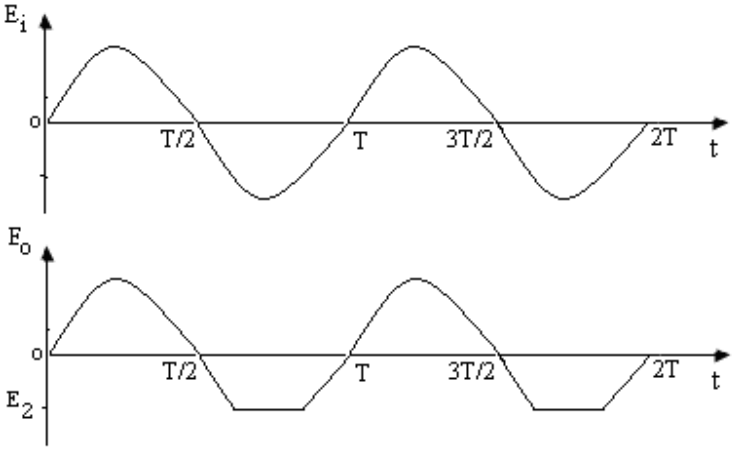


Fig. 5.36 b

battery are reversed. It work as follows, during the positive half cycle of the input wave, the diode will be in reverse bias and behaves as an open switch and the output follows the input. During the next half cycle of the input wave the diode will be in reverse bias till the input is less negative than E_2.When the input becomes more negative than E_2, diode conducts and works as an on switch. The output will equal to E_2. So it is concluded that the portion of the input beyond E_2, in negative half cycle is clipped off. The input output wave shapes is shown in figure (5.36 b).

Further both the circuits of figures (5.35 a) & (5.36 a) may be combined to get both sided

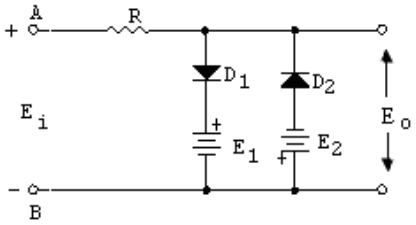


Fig. 5.37 a

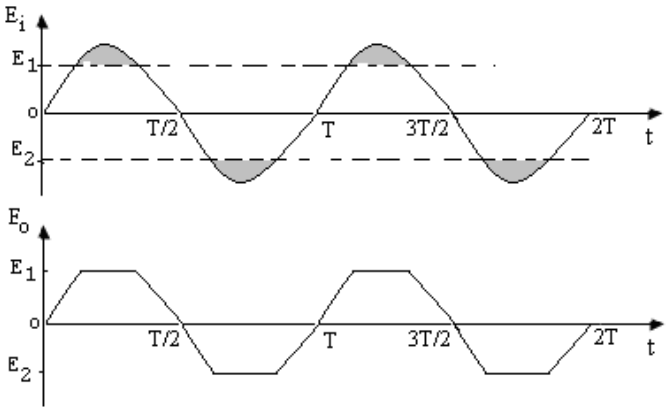


Fig. 5.37 b

biased shunt clipper as shown in figure (5.37 a). It may be explained that the portion beyond E1 in the positive half cycle of the input, is clipped off and in the negative half cycle the portion beyond E_2 is clipped off. The input output wave shapes are shown in figure (5.37 b).

Example 5.13: **T**he diodes connected in the circuit of figure (5.37 a) are not ideal but have some finite forward resistance r_d. Draw its transfer characteristics and the input output wave forms.

Solution: When the input signal is lying between E_1 and E_2, both the diodes will be in reverse bias and the slope will be unity at the origin. When the input is beyond E_1 or E_2, the slope at these points E_1 and E_2 will not be zero but will have the slope equal to $\frac{r_d}{R+r_d}$ which is small positive. So the output will be slightly distorted at the points E_1 & E_2. The transfer characteristics and its input output wave shapes are given in figure (5.38)

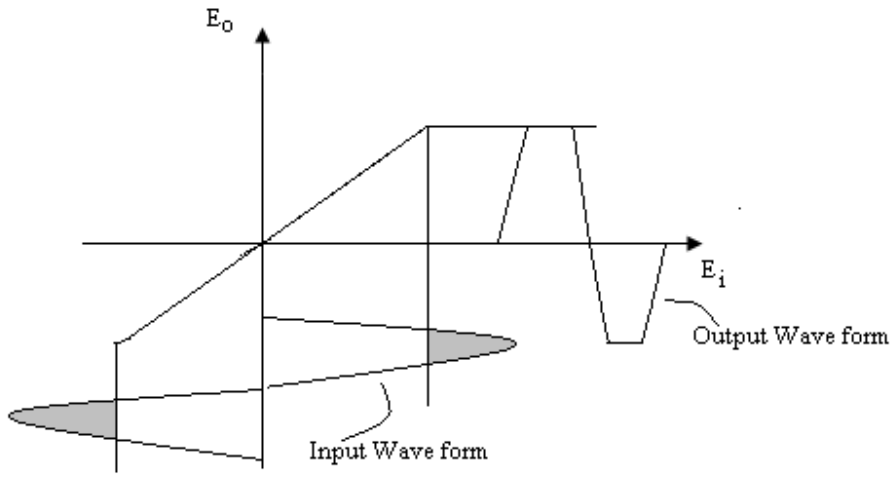


Fig. 5.38

5.7 Clamping Circuits: The clamping circuit is used to clamp a signal to a desired d.c. level. The sinusoidal, square or triangular signal normally swings symmetrically about the X- axis with equal magnitude on both sides as shown in figure (5.39). The signal in this condition is said to have the zero average value or its d.c. level is zero. If the signal is lifted upward to touch the negative peak points of the signal to X- axis, the signal is said to have positively clamped at zero d.c. level. If on the contrary the signal is lifted downward to touch the positive peak points of the signal to X- axis, the signal is said to have negatively clamped at zero d.c. level (ref fig. 5.39). The signal may further be lifted upward or down ward to a desired positive or negative voltage (E) as shown in figure (5.39). In this condition the signal is said to have clamped at positive E volts or negative E volts.

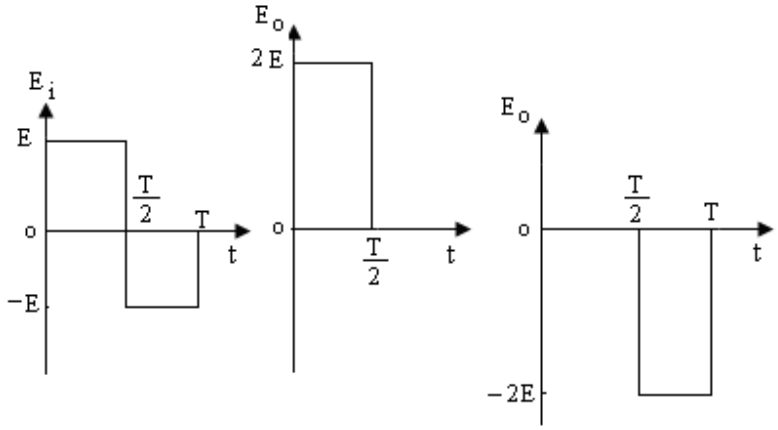


Fig. 5.39

The clamping circuit shown in figure (5.40a) consists of a diode, a capacitance and a resistance. To the input of this circuit we apply a signal whose d.c. level is zero i.e. which swings both sides of X – axis symmetrically. During the first negative half cycle of the input wave E_i, diode conducts and it behaves as a closed switch. The capacitor C

charges to the peak value of the negative swing (E_N) with the polarity shown in figure (5.40a). The values of the capacitor C and resistance R are so chosen so that the RC time constant is large enough so that the capacitor is not immediately discharged. Now during the next positive half cycle of the input wave, diode is in reverse bias and behaves as an open switch. The voltage at the output will be equal to the magnitude (E_N+E_P).The cycle repeats and we get the output as shown in figure (5.40b). It is clear from this figure that the output wave touches the negative peak of the signal to the X-axis hence it is positively clamped at zero d.c. level.

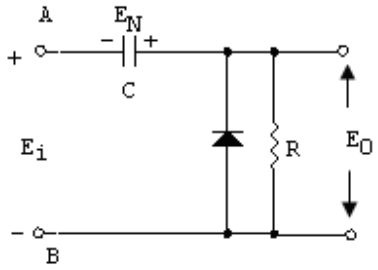


Fig. 5.40 a

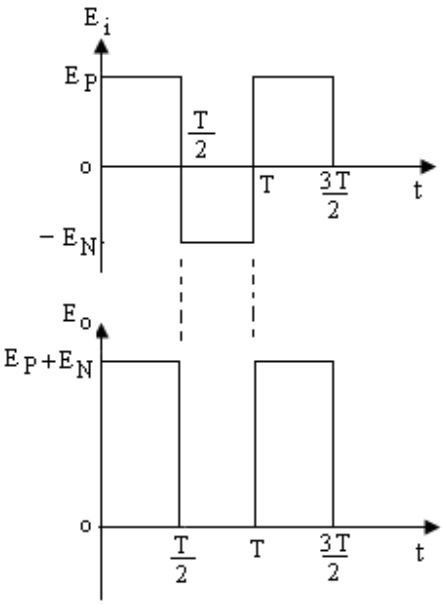


Fig. 5.40 b

If the diode connections are reversed as shown in figure (5.41a), the circuit may easily be explained that the signal is negatively clamped at zero d.c. level (ref. 5.41 b).

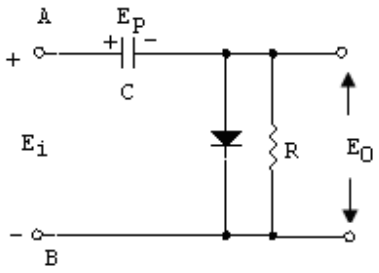


Fig. 5.41 a

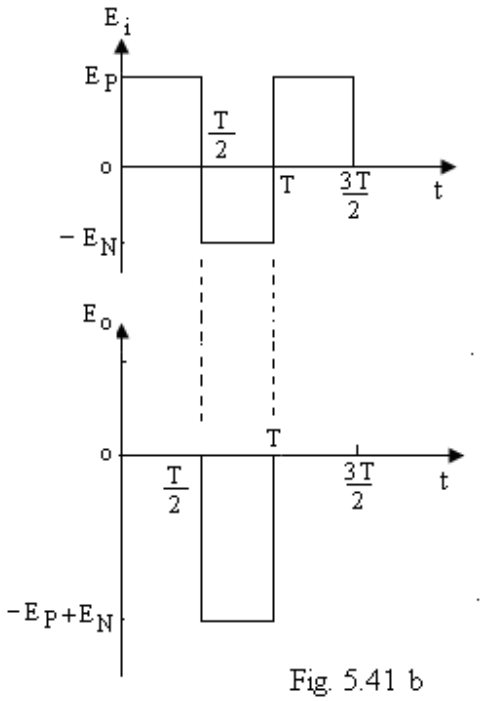


Fig. 5.41 b

Now to clamp the signal at some positive E volt, then a battery of E volt may be introduced is series with the diode as shown in figure (5.42 a). Input output signals are shown in figure (5.42 b).

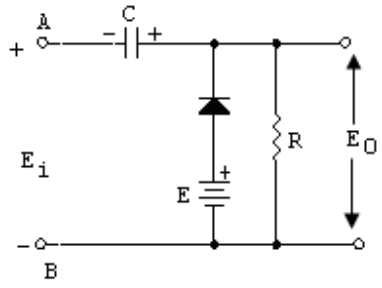


Fig. 5.42 a

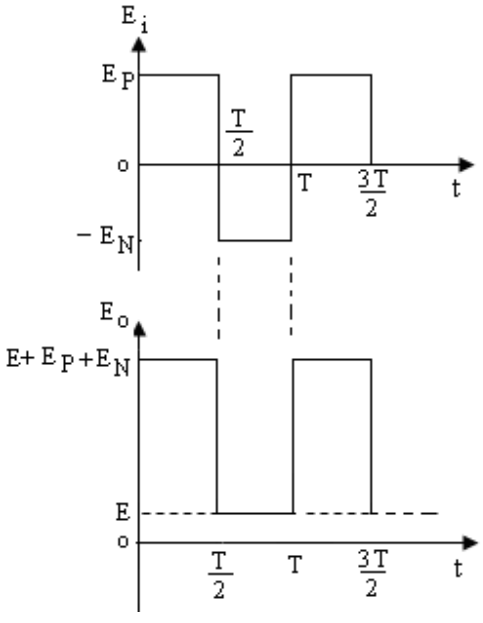


Fig. 5.42 b

Similarly, to clamp the signal at some negative E volt, then a battery of E volt may be introduced in series with the diode as shown in figure (5.43 a). Input output signals are shown in figure (5.43 b).

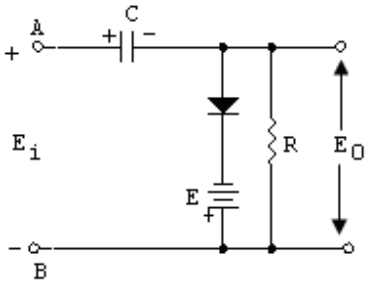


Fig. 5.43 a

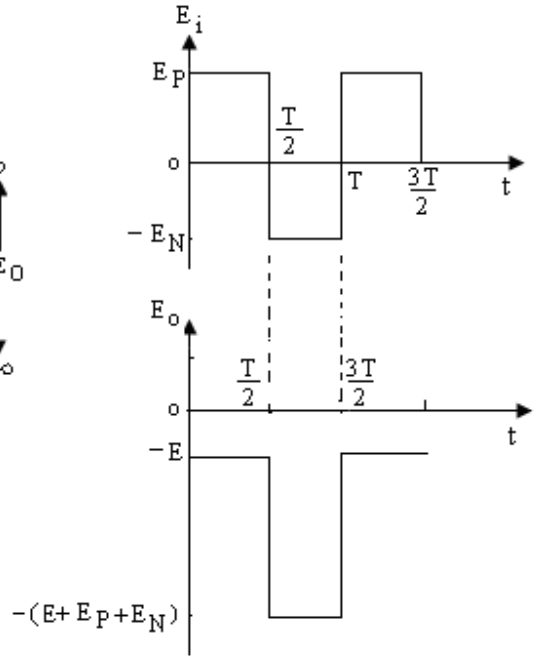


Fig. 5.43 b

5.8 Log and anti log circuit: The signal processing operations such as logarithm and antilogarithm can be performed using the p – n junction diode characteristics. As it is well known that the current flowing through the diode is given by

$$I = I_s(e^{V/V_T} - 1) \qquad \text{------- (5.18)}$$

where V_T is the thermal voltage and I_s is the reverse saturation current which are constants at the particular temperature and semiconductor material. If V >> VT, this equation will of the form

$$I = I_s e^{V/V_T} \qquad \text{------ (5.19)}$$

i.e. if a voltage V is applied across a diode as shown in figure (5.44a), the current flowing through the diode will be proportional to the antilog function

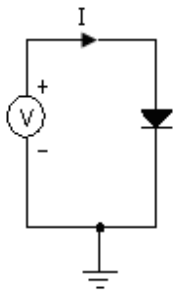


Fig. 5.44 a

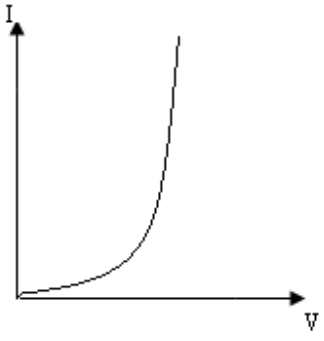


Fig. 5.44 b

of this voltage V. This circuit may be called as the antilog circuit. Figure (5.43b) shows the antilog function of the input voltage V.
Taking the log on both sides of equation (5.19), we get:

$$\ln\left(\frac{I}{I_s}\right)=\frac{V}{V_T} \qquad \text{or} \qquad V=V_T \ln\left(\frac{I}{Is}\right) \qquad \text{------ (5.20)}$$

from this equation it is clear that if a known current I from a current source

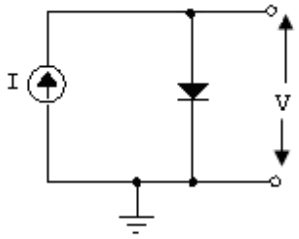


Fig. 5.45a

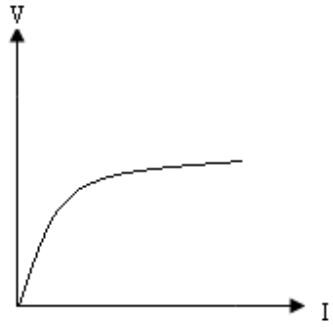


Fig. 5.45 b

is passed through the diode (fig. 5.45 a), then the voltage across the diode will be proportional to the logarithm of current I. hence this circuit may be called as log circuit. The logarithm function is shown in figure (5.45 b).

5.9 Zener Diode as Voltage Regulator: The heavily doped P N junction diodes which work in reverse bias and operate in the break down region, known as Zener diodes have already been discussed in the preceding chapter. Here we shall discuss its most common application as a voltage regulator which provides a constant voltage from a source whose voltage may vary over a sufficient range. A simple voltage regulator circuit using Zener diode is shown in figure (5.46).
In this circuit a series combination of Zener diode and a resistance R_s, is connected across the unregulated supply E_i. The regulated output voltage E_0 is available across the parallel combination of Zener diode and the load resistance R_L. The polarity of the Zener diode is such that it is biased in reverse bias. The Zener diodes of different break down voltages

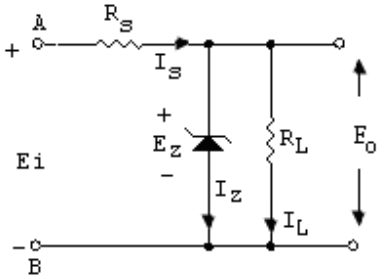


Fig. 5.46

are available in the market, so the diode of required break down voltage E_z is chosen in the circuit. If the circuit elements are such that the voltage across the Zener diode is less than the Zener break down voltage E_z, the diode will behave like an off switch.

The output voltage E_0 will be given by $E_0 = \frac{E_i R_L}{R_s + R_L}$. The Zener diodes are never used in this state.

If the voltage across the Zener diode is more than or equal to the Zener break down voltage E_z, the diode is in on state and acts as voltage source of voltage E_z. The output voltage E_0 will be equal to E_z. This is the required state for the voltage regulation.

The current drawn I_s from the unregulated supply is given by $I_s = I_z + I_L$

or $I_z = I_s - I_L$ where $I_L = \frac{E_i}{R_L}$ and $I_s = \frac{E_i - E_0}{R_s}$

The power dissipation of the Zener diode is given by: $P_z = E_z.I_z$.

As discussed above the Zener diode is used in the break down region. The output voltage will be equal to the Zener voltage. This circuit is commonly used as voltage regulator or as a fixed reference source. The regulator circuit gives the constant voltage irrespective of the change in the load resistance R_L or the change in the unregulated supply E_i. We shall now find the minimum value of load resistance R_L and the minimum value of the E_i. If the load resistance R_L or unregulated supply is too small the diode will not be in the break down region or the Zener will be off.

We know $E_0 = E_z = \frac{E_i x R_L}{(R_s + R_L)}$

or $(R_s + R_L)E_z = E_i R_L$

or $R_L(E_i - E_z) = R_s E_z$

or $R_{L\min} = \frac{R_s E_z}{(E_i - E_z)}$,

so any load resistance greater than this value will ensure the Zener diode is in the break down region or on state.

The maximum value of R_L may also be calculated. $I_{L\max} = \frac{E_0}{R_L} = \frac{E_z}{R_{L\max}}$

or $I_{L\max} = \frac{E_i - E_z}{R_s} = \frac{E_s}{R_s} = I_s$

The Zener current is minimum, as $I_s = I_z + I_L$.

I_{Lmin} may also be given by: $I_{L\min} = I_s - I_{z\max}$

so $R_{L\max} = \frac{E_z}{I_{L\max}}$

The minimum value of the unregulated supply is to be calculated as follows:

We know $E_0 = E_z = \frac{E_i x R_L}{(R_s + R_L)}$

and $E_{i\min} = \frac{(R_L + R_s)xE_z}{R_L}$

The maximum value of E_i is limited by the maximum Zener current I_{zmax}. Since

$I_{zmax} = I_s - I_L$.

Now $E_i = I.R_s + E_0$

As $E_0 = E_z$ is constant, the input voltage will be maximum when Is is maximum, so $E_{imax} = I_{smax}.R_s + E_z$

It is worth while to mention the following points regarding the Zener regulator circuits.

1. More than one Zener diodes may be connected in series, and the output voltage will be sum of all the break down voltages of the Zener diodes. Let three Zener diodes Z_1, Z_2, Z_3 are connected in series as shown in figure (5.47), whose values are $E_{z1} = 6$ volts, $E_{z2} = 12$ volts, $E_{z3} = 15$

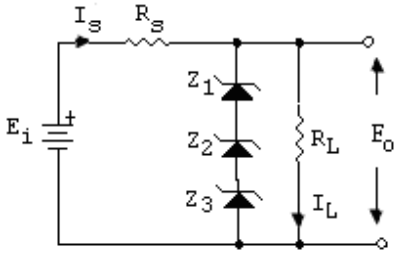


Fig. 5.47

volts. The output voltage will be 33 volts, provided the circuit parameters are properly chosen, so that the voltage across the load resistance is more than 33 volts, when diodes are assumed in off state.

2. The Zener diodes are never connected in parallel, as in the break down region the Zener diodes behave like ideal voltage source.
3. The Zener diodes may be connected in series back to back as shown in figure (5.48a). This circuit will behave like the shunt clipper circuit. Input

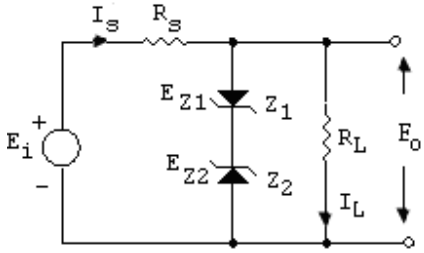


Fig. 5.48 a

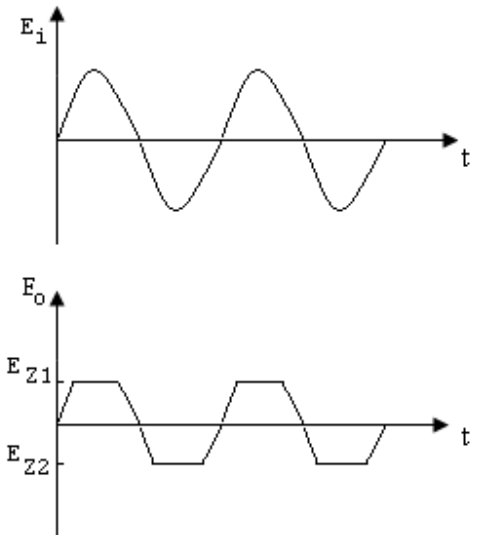


Fig. 5.48 b

signal connected to this circuit is a varying signal whose peak value should be greater than break down voltage of each diode. During the positive half cycle of the input signal the diode Z_1 will be in forward and will work like an ordinary diode. The Zener diode Z_2 will be in reverse bias till the input is less than its break down voltage, in this condition the

output follows the input. But when the input is greater than the break down voltage of Z_2, a constant voltage equal to E_{z2} will be maintained across the output. Thus it clips off the portion of the input signal beyond E_{z2}. Similarly it can be shown that this circuit clips off the portion of the negative half cycle beyond E_{z1}, as the process is reverse in this case. The input output wave shapes are shown in figure (5.48 b).

Example 5.14 The Zener diode regulator circuit shown in figure (5.49) has the following parameters. E_i = 20 volts, R_s = 1KΩ, E_z = 8.2 volts and R_L = 2KΩ. Find E_0, E_s, I_z. If the maximum wattage of the Zener diode is 35 mW, then suggest whether this diode will work or not.

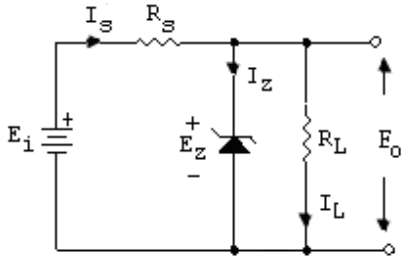


Fig. 5.49

Solution: If the Zener diode is in the off state, the voltage across the load resistance R_L is given by: $$E = \frac{20Vx2K\Omega}{(1+2)K\Omega} = 13.33Volts$$

This voltage is greater than the break down voltage of the Zener diode, the diode will go in the on state and out put voltage will be equal to the E_z.

So $E_0 = E_z = 8.2$ Volts

and voltage across R_s is $E_s = 20 - 8.2 = 11.8$ volts

The current following through the resistance R_s is $$I_s = \frac{11.8Volts}{1K\Omega} = 11.8mA$$

The current following through the resistance R_L is $$I_L = \frac{8.2Volts}{2K\Omega} = 4.1mA$$

The current following through the Zener diode is

$$I_z = I_s - I_L = 11.8 - 4.1 = 7.7mA$$

Power of the Zener diode $P_z = E_z \times I_z = 7.7$ mA x 8.2 volts = 63.14 mW

This is the required power of the Zener diode. So we have to use the Zener diode whose maximum power is greater than 63.14 mW. But in the given problem, the required power of the Zener diode is less than the given power 35 mW. So the Zener diode of 35 mW power will not work. The Zener diode of higher than 63.14 mW should be used.

Example 5.15 The Zener diode regulator circuit shown in figure (5.50) has the following parameters. E_i = 35 volts, R_s = 1.2 KΩ, E_z = 12 volts and maximum Zener current is 10mA. Find the range of R_L so that E_0 remains to be constant to 12 volts.

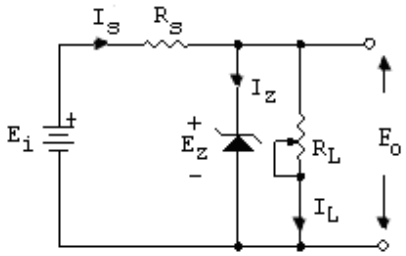


Fig. 5.50

Solution: The minimum value of R_L is given by:

$$R_{L\min} = \frac{R_s E_z}{(E_i - E_z)} = \frac{1200x12}{35-12} = 626\Omega$$

The current flowing through the resistance R_s is then given by:

$$I_s = \frac{E_i - E_z}{R_s} = \frac{(35-12)volts}{1.2K\Omega} = 19.2mA$$

The minimum value of I_L is $I_{L\min.} = I_s - I_{z\max} = 19.2 - 10 = 9.2mA$

The maximum value of R_L is $R_{L\max} = \frac{E_z}{I_{L\min.}} = \frac{12volts}{9.2mA} = 1.3K\Omega$

Problems:

1. Draw a circuit diagram of a full wave rectifier along with a series inductor (choke) filter and obtain a formula for the ripple factor.
2. Give the circuit diagram and describe the working of a full wave rectifier with shunt capacitor filter. Why for very low ripple the output current should be very small?
3. Explain, giving the circuit diagram, the working of a full wave rectifier. Obtain the expressions for average, r.m.s. value of the output voltage, efficiency and ripple factor.
4. Drive the expressions for (a) average and (b) r.m.s. value of the output voltage of a half wave rectifier. Find the expression of rectifier efficiency and ripple factor for half wave rectifier.
5. Describe the circuit diagram and explain the working of half wave rectifier. Explain the ripple voltage and ripple factor for the same.
6. Give the circuit diagram of a full wave rectifier with an L-Section filter and explain its working. Find the expression for the ripple factor.
7. Give the circuit diagram of a full wave rectifier with π- section filter and explain its working. Find the expression for the ripple factor.
8. What do you mean by the peak inverse voltage of the diode? Show that when a capacitor is connected across the load resistance of a half wave rectifier circuit, and then the peak inverse voltage of the diode is approximately twice the peak voltage of the input signal.

9. Discuss the bridge rectifier circuit with neat circuit diagram. What are its merits and demerits? Show that the PIV of the diodes used in bridge rectifier is equal to the peak value of the input signal.
10. Discuss the circuit of the full wave rectifier and explain its working. Mention its advantages and disadvantages. Show that PIV of the diodes used in this circuit is equal to twice the peak value of the input signal.
11. Making the suitable approximations, determine the d.c. output voltage as well as ripple factor of the half wave rectifier with shunt capacitor filter.
12. Making the suitable approximations, determine the d.c. output voltage as well as ripple factor of the full wave rectifier with shunt capacitor filter. Show that its value is approximately half the value calculated for half wave rectifier.
13. What do you mean by clipping circuits? Mention various types of clipping circuits. Discuss double sided biased series clipper.
14. Discuss various types of shunt clipper with the help of suitable circuit diagrams.
15. Discuss half wave voltage doubler with a suitable diagram. Show that the PIV of the diodes used in this circuit is twice the peak value of the input signal.
16. Discuss the voltage multiplier circuit.
17. Discuss full wave voltage doubler with suitable circuit diagram. What are its advantages?
18. What do you understand by Clamper? Discuss a clamper circuit with suitable circuit diagram.
19. What is Zener diode? Explain how the Zener diode is used as a voltage regulator.
20. Prove that r.m.s. value of the triangular ripple wave is $\frac{E_R}{2\sqrt{3}}$, where E_R is the peak value of the ripple.
21. A 200 volts a.c. signal of frequency 50 Hz is applied to the half wave rectifier circuit having load resistance 500 Ω. If the diode used in the circuit is ideal then find d.c. output voltage across the load resistance, d.c. and r.m.s. current flowing through the load resistance.

    Ans. 90 Volts, 180 mA, 141.4 mA
22. Repeat the problem 21, assuming the diode has the forward resistance of 20 Ω.

    Ans. 86.6 volts, 173.2 mA, 141.4 mA
23. A full wave rectifier is to supply 30 volts d.c. across the load resistance of 1KΩ. Find (a) r.m.s. value voltage of the transformer, (b) peak diode current and (c) power rating of the transformer.

    Ans. 33 – 0 – 33 volts transformer, 33mA, 1.1 watt
24. A 20 – 0 – 20 volts transformer is used for full wave rectifier circuit. Each diode has a forward resistance of 20Ω. The load resistance is 2KΩ. Find $E_{d.c.}$, $I_{d.c.}$, $I_{r.m.s.}$, and rectifier efficiency η.

    Ans. 17.8 volts, 8.9mA, 9.9 mA, 80.4%
25. A center tap transformer having 20 volts r.m.s. on each side of the center terminal is used in full wave rectifier with shunt capacitor filter. If frequency of the input a.c. is 50Hz, the load resistance is 300 Ω and capacity of the shunt capacitor is 470μF, find the d.c. load voltage and the ripple factor.

    Ans. 27.3 volts, 2.1%

26. A full wave rectifier with shunt capacitor filter has the following parameters. $E_{d.c.}$ = 50 volts, $I_{d.c.}$ =277mA, $I_{r.m.s.}$ = 289 mA and f = 50Hz. Find the value of ripple factor and the capacity of the capacitor.

Ans. 27.6%, 58μF

27. A full wave rectifier with shunt capacitor filter is to supply a 200Ω load with 150 mA current. What should be voltage rating of the transformer at 50 Hz frequency, to have a ripple factor of 0.04?

Ans. 23 – 0 – 23 V

28. A full wave rectifier circuit is connected to a π - section filter, having two 100μF capacitors and a 5 H inductor. Find the value of ripple content if the frequency of the a.c. mains is 50 Hz and load connected at the output is 150Ω.

Ans. 0.098%

29. Find the maximum and minimum values of Zener current in a Zener regulator circuit. Given R_s = 3.5KΩ, R_L = 5KΩ, E_z = 15 Volts and the input varies from 35 volts to 55 volts. Ans. 8.43mA, 2.7mA

30. If in Zener diode regulator circuit, R_s = 1KΩ, E_z = 12 Volts, R_L=3KΩ and E_i = 30 volts, find the output voltage, the voltage across the series resistance R_s and also the Zener current.

Ans. 12 volts, 18 volts, 14mA

6
Junction Transistors

W. Schockley, J. Bardeen and W. H. Brattain of Bell Telephone Laboratories, U.S.A. in 1948 invented an important semiconductor device named as transistor which is now most commonly being used in electronic instruments. The junction transistor is also called Bipolar Junction Transistor (BJT) because the current flows in it due to both types of charge carriers electrons and holes. Transistors have the advantages that its size is very small, works on low applied voltage and capable of producing large amplifications. Physical behaviour of transistors including V – I characteristics will be studied in this chapter.

6.1 The Transistor: Junction transistors are made up of two PN junctions of Si or Ge, forming PNP or NPN transistors. The PNP transistor is formed by sandwiching a thin N – layer between two P – layers. Similarly an NPN transistor is formed by sandwiching a thin P – layer between two N – layers. These two types of transistors with their symbols are shown in figure (6.1). The three regions of the transistor are known as emitter, base and collector. The emitter is heavily doped, base region is thin and very lightly or sparingly doped and the collector is moderately doped. Three currents flow in the transistor namely emitter current I_E, base current I_B and collector current I_C. Conventionally, it is assumed that all the three currents are entering the junctions irrespective of the types of the transistor. That is whether the transistor is PNP or NPN, the current entering the junction is positive and leaving the junction is negative. If the actual current in either type of transistors is entering the junction, the current is taken as positive otherwise negative.

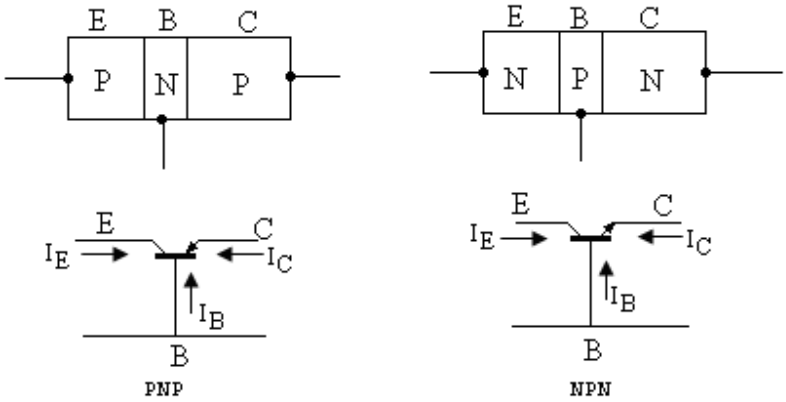


Fig. 6.1

From the figure (6.1) $I_E + I_B + I_C = 0$

The symbol V_{EB} represents the potential across the emitter base junction, V_{CB} the potential across collector base junction and V_{CE} the potential across collector and emitter junction. Normally the transistor is biased in the active region, in which emitter base junction is kept in forward bias and collector base junction in the reverse bias. Input circuit has low resistance as the emitter base junction is forward biased and output circuit has high resistance as collector base junction is biased in the reverse bias. The transistor therefore, transfers the resistor from input circuit to output circuit, hence the name Transistor (**Trans**fer + Res**istor** = **Transistor**). The working of both the two types of transistors is the same with the difference that the role of electrons and holes are reversed in these transistors.

6.1.1 Minority Carrier Concentration in a Transistor: Let us consider a PNP transistor which is open circuited so that all the transistor currents are zero. Further it is assumed that the transistor is having perfectly symmetrical junctions. The symmetrical junctions we mean that the emitter and collector junctions are having identical physical dimensions and doping concentrations. The barrier height at the emitter junction J_E and collector junction J_C will be equal as shown in figure 6.2 (a). The narrow space charge regions at the junctions are neglected.

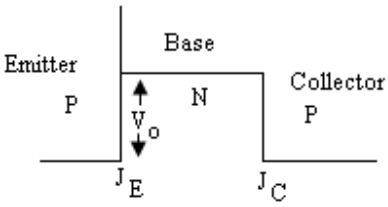


Fig. 6.2 a

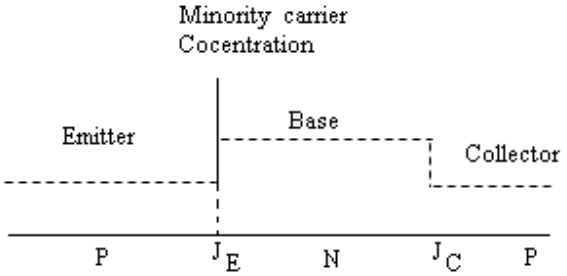


Fig. 6.2 b

The majority charge carrier concentration in each region will be equal to the corresponding doping level.

i.e., $p = N_A$ in p – type semiconductor,

and $n = N_D$ in n – type semiconductor.

The minority carrier concentration can be calculated from the law of mass action

$$np = n_i^2$$

Thus the number of electrons (minority carriers) in p type semiconductor or emitter is given by:

$$n_p = \frac{n_i^2}{N_A} = n_{po} \quad ,$$

the suffix p represents the p type semiconductor and n_{po} represents the minority carrier electron in p type emitter in thermal equilibrium. N_A is the acceptor concentration in p type emitter.

Similarly the number of holes (minority carriers) in n type semiconductor or base is given by: $p_n = \frac{n_i^2}{N_D} = p_{no}$,

p_{no} represents the minority carrier holes in n type base in thermal equilibrium. N_D is the donor concentration in the *n* type base.

Generally in a transistor base is lightly doped $N_A >> N_D$,

this implies $p_{no} >> n_{po}$

The no. of minority charge carriers (electrons) in *p* – type collector can also be given as: $n_{po} = \frac{n_i^2}{N_A^{'}}$ where $N_A^{'}$ is the acceptor concentration in the *p* type collector. The minority carrier concentrations in the three regions in thermal equilibrium state are therefore, depicted in figure 6.2(b). These concentrations will vary with the application of biasing voltages to the transistor junctions.

6.2 The Transistor in active region: Internal physical behaviour of the transistor will be discussed by considering the transistor (PNP) in the active region i.e. emitter base junction is biased in forward bias and collector base junction is biased in reverse bias, which is shown in figure 6.3(a). The variation of potential barrier with biasing of the transistor is shown in figure 6.3(b). The dotted lines show this variation in open circuit condition of the transistor; the solid lines, however, show this variation after the application of biasing voltages. The space charge regions at the emitter junction J_E and at the collector junction J_C are not shown, as it is assumed to be negligibly small. The forward biasing of the emitter junction lowers barrier potential V_0 by an amount equal to the magnitude of the applied emitter base voltage $|V_{EB}|$, where as the reverse biasing of the collector junction increases the barrier potential by the amount equal to the magnitude of the applied collector base voltage $|V_{CB}|$ as depicted in figure 6.3(b). The lowering of the emitter base potential barrier permits the injection of minority carrier holes of the emitter region to the base region; and the minority carrier electrons of the base region are injected in to the emitter region.

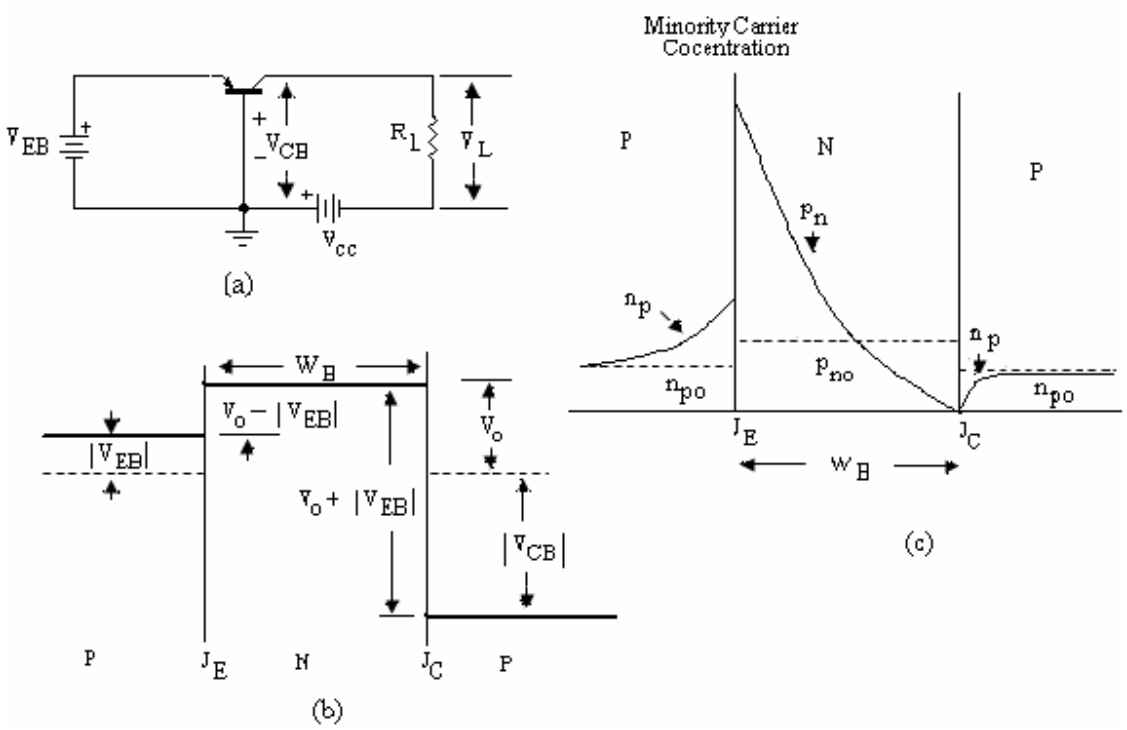


Fig. 6.3

The minority carrier electrons in the emitter (n_p) rises exponentially from thermal equilibrium concentration n_{p0} to $n_p(0)$ at the emitter junction J_E (where $x = 0$), given by:

$$p_n(0) = p_{n0} e^{\frac{V_{EB}}{V_T}} \quad \text{------ (6.1)}$$

The minority carrier holes in the base (n_p) also rises exponentially from thermal equilibrium concentration p_{n0} to $p_n(0)$ at the emitter junction J_E (where $x = 0$), given by:

$$n_p(0) = n_{p0} e^{\frac{V_{EB}}{V_T}} \quad \text{------ (6.2)}$$

as $p_{n0} >> n_{p0}$, so $p_n(o) >> n_p(0)$.

Similarly minority carrier holes in the base p_n and electrons in the collector n_p , near the collector junction J_C (where $x = W$) are given by:

$$p_n(w) = p_{n0} e^{\frac{V_{CB}}{V_T}} \quad \text{------ (6.3)}$$

and

$$n_p(w) = n_{p0} e^{\frac{V_{CB}}{V_T}} \quad \text{------ (6.4)}$$

As V_{CB} is negative and $|V_{CB}| >> V_T$ so $p_n(w) = n_p(w) = 0$. These variations of the minority carriers are depicted by solid lines in figure 6.3(c).

The excess holes in the base diffuse from emitter base junction J_E to collector base junction J_C across the base where the electric field intensity E is zero. The collector base

junction is reverse biased so the electric field intensity E at the junction J_C will be positive and large as ($E = -dV/dx$). Due to which the holes are accelerated across the junction. In other words, the holes which reach the junction J_C fall down the potential barrier, and are therefore collected by the collector.

6.3 Current Components in a Transistor: We shall find components of currents in a transistor, which is biased in the active region i.e. the emitter base junction is biased in the forward bias and collector base junction is biased in the reverse bias. Due to the forward biasing of the emitter base junction, majority carrier holes are injected into base region and majority carrier electrons of the base region move into the emitter region. This means emitter current I_E consists of two types of current (i) hole current I_{pE} constituted by the movement of holes from emitter to the base, and (ii) electron current I_{nE} constituted by the movements of electrons from the base region to the emitter region. The forward biasing of the emitter junction increases the population of holes in the base region which will finally produce the collector current. So to have the collector current proportional to the emitter current, the emitter current should only be constituted by the hole current I_{pE} (i.e. $I_E \approx I_{pE}$) or electron current I_{nE} should be negligibly small. This is possible when the emitter is heavily doped and base is very lightly doped, which is normally being done in the transistors.

We assume that the injection of electrons into the base region is a low level injection. Hence the minority carrier current I_{pE} is a hole diffusion current into the base. Its value is proportional to the concentration gradient of holes at J_E which is given by:

$$I_{pE} = -qD_p A\frac{dp_n}{dx} \qquad \text{------ (6.5)}$$

Where D_p is the diffusion constant for holes, A is the area of cross section and q is the charge of an electron.

Similarly I_{nE} , electron diffusion current is proportional to the concentration gradient of electrons and is given by:

$$I_{nE} = -qD_n A\frac{dn_p}{dx} \qquad \text{------ (6.6)}$$

where D_n is the diffusion constant for electrons. Thus I_{nE} is proportional to the slope of electron concentration n_p at the junction J_E.

The total emitter current is given by $I_E = I_{pE} + I_{nE}$ ------ (6.7)

The directions of these currents are shown in figure (6.4), which are all positive flowing from emitter to base as these are conventional currents. The holes on crossing the emitter junction diffuse through the base region. In this process, some of the holes combine with the majority carrier electrons in the base region. In this way the number of holes reaching the collector junction will be less than the number of holes emitted at the emitter base junction. To reduce the possibility of this recombination in the base region, the width of the base region is made to be extremely small. Let I_{pC1} is the hole current at collector junction J_C , which is due to the holes reaching this junction through the base region. The current ($I_{pE} - I_{pC1}$) is the recombination current which leaves the base as shown in figure

(6.4). The electrons enter the base region from the external circuit through the base lead to supply those charges which have been lost by recombination with the holes injected into base across J_E.

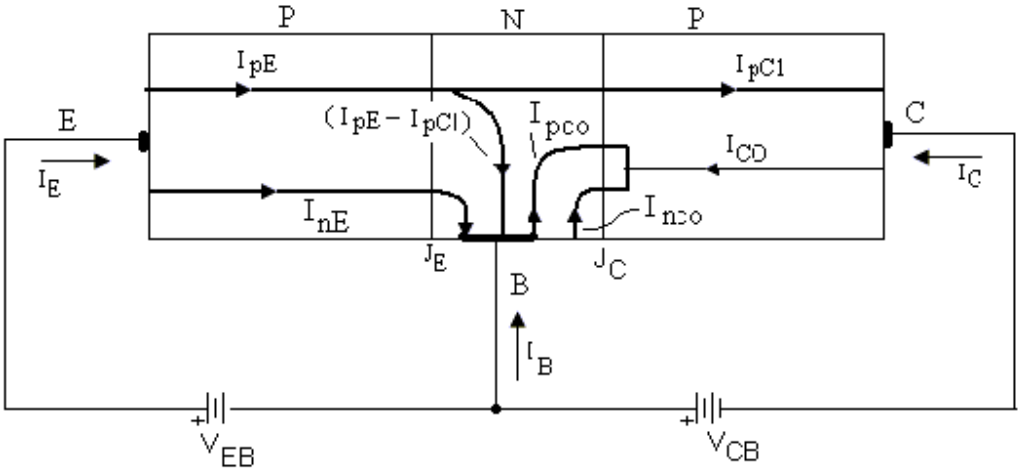


Fig. 6.4

It is further assumed for the time being, that the emitter base junction is open circuited so that I_E and I_{pC1} are equal to zero. Under this condition, the collector current I_C will be equal to the reverse saturation current I_{C0} of the base collector junction which is working as a simple reverse bias diode. This reverse saturation current I_{C0} will be the sum of two currents I_{Cno} and I_{Cpo}. The I_{Cno} is the current due to the movement of minority charge carrier electrons across J_C from collector (p type) to base (n type), while I_{Cpo} is the current due to the movement of minority charge carrier holes across J_C from base (*n* type) to collector (*p* type). Thus it may be written as :

$$-I_{Co} = I_{nCo} + I_{pCo} \qquad \text{------ (6.8)}$$

(The minus sign is chosen arbitrarily so that I_C and I_{C0} will have the same sign.)

We now return to our original situation that emitter base junction is biased in the forward direction. Under this condition $I_E \neq 0$ and collector current I_C is given by:

$$I_C = I_{Co} - I_{pC}$$

or
$$I_C = I_{Co} - \alpha I_E \qquad \text{------ (6.9)}$$

where α is defined as the fraction of the total emitter current I_E that represents the holes reaching the collector after traveling from the emitter through base. Since I_{C0} is small enough so α is the ratio of the collector current to the emitter current

In a PNP transistor, I_E is positive and both I_C and I_{Co} are negative, which indicates the actual collector current is opposite to the assumed direction of the collector current shown in figure (6.4). However, in NPN transistor these currents are reversed.

We wish to find out the generalized transistor equation, as the equation (6.9) represents the collector current when the transistor is in the active region and also this current is independent of the collector voltage. To obtain such a generalized equation for

the collector current, which should be valid not only when the collector junction is reversed biased but also valid for any voltage across the J_C . We replace I_{Co} of equation (6.9) by the diode current equation.

The diode current is given by: $$I = I_s(e^{\frac{V}{V_T}} - 1)$$ ------ (6.10)

Here I_s is replaced by (– I_{Co}) and V by V_C, the voltage across junction from collector to base. The equation (6.9) will become as:

$$I_C = -\alpha I_E + I_{C0}\left(1 - e^{\frac{V_C}{V_T}}\right)$$ ------ (6.11)

If V_C is negative and larger than V_T, the equation (6.11) reduces to:

$I_C = -\alpha I_E + I_{Co}$, which is the equation (6.9).

6.4 Base - Width Modulation or The Early Effect: Consider a PNP transistor in active region whose emitter base junction is biased in forward direction and collector base junction is biased in reverse direction. The reverse biasing of the collector base junction in a transistor acts as a sink of minority charge carriers injected into the base region from the emitter region. It is well known that the width of the depletion region or the space charge region near the junction is large when the junction is in inverse bias and this width increases with the increase of reverse bias. So the width of the depletion region W is large at the collector junction J_C as compared to this value at the emitter junction J_E. Since the emitter base junction is in forward bias so the width of the depletion region is negligibly small at the emitter junction J_E. Since the doping in the base is ordinarily substantially smaller than that into the collector, so the penetration of the transition region in the base near the collector junction J_C will be larger than that into the collector. Hence, the whole of the depletion region may assume to be in the base region, as shown in figure (6.5a). Let W_b is the metallurgical base width and W is the width of the depletion layer. Now as the collector voltage is increased, the space charge layer takes up more of the metallurgical base width W_b, and as a result , the effective base width $W_b^{'}$ is decreased and is given by: $W_b^{'} = W_b - W$. This effect given by J.M.Early, is known as base narrowing, base width modulation or the Early effect.

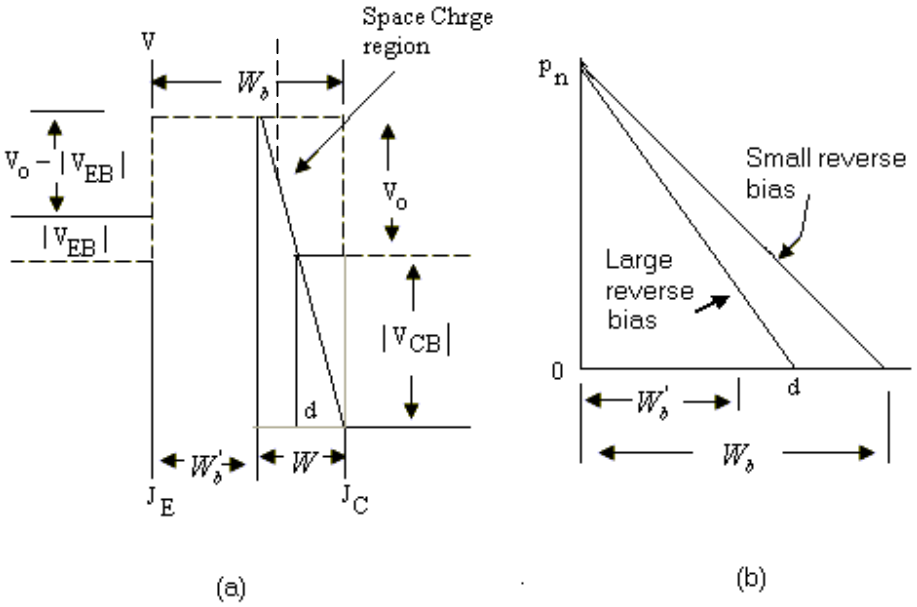


Fig. 6.5

The modulation of the effective base width $W_b^{'}$ with the increase in the collector voltage has three consequences:

1. The possibility of the recombination within the base is reduced, thereby increasing the collector current. Hence α increases with the increase of $|V_{CB}|$.
2. The concentration gradient of the minority charge carriers in increased within the base region with the increase of $|V_{CB}|$ and consequently hole current density J_E and then I_E increases (Ref. figure 6.5b).
3. For extremely large reverse bias at the collector base junction, the width of the deletion region becomes too large consequently the effective base width reduces to zero, causing voltage break down in the transistor. This breakdown phenomenon is called as 'punch through' or 'reach through'.

6.5 The Transistor as an Amplifier: It is well known that the transistor has three terminals namely emitter, base and the collector; hence it is called a three terminal device. While using the transistor, one terminal is used as a reference or common terminal between input and output which is usually grounded. So the transistor may be used in three configurations, namely common base (CB), common emitter (CE) and common collector (CC) configurations. If the base is common between input and output terminals, the transistor is called in common base configuration. Similarly common emitter and common collector configurations may be defined. In any of the three configurations, transistor can provide the power gain (voltage amplification or current amplification) in the circuit. The transistor is, therefore, said to work as an amplifier.

To show that the transistor can work as an amplifier, we consider a transistor in common base configuration shown in figure (6.6), where a load resistance R_L

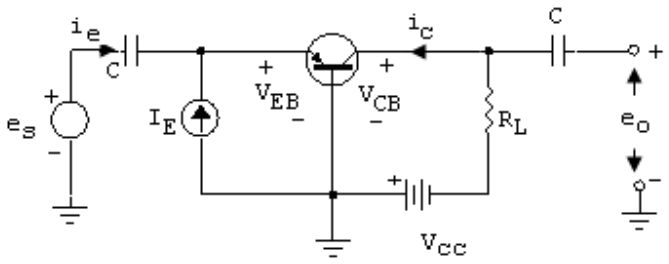


Fig. 6.6

is connected in series with the collector supply voltage V_{CC}. The transistor is biased in active or linear operating region, such that it remains in active region through out the change in the emitter current due to the application of the input a.c. signal e_s. Thus the d.c. biasing of the emitter base junction is shown only by the current source I_E. The small signal e_s is connected across the emitter base junction, through a coupling capacitor C_C for the effective a.c. coupling. Since the emitter base junction is in forward bias, its incremental resistance r_e is very small of the order of 25Ω. The variation in emitter current due to e_s a.c. signal is represented by i_e, which is given by:

$$i_e = \frac{e_s}{r_e} \qquad \text{------ (6.12)}$$

The corresponding change in collector current i_c will be give by: $i_c = -\alpha . i_e$ The value of α is nearly equal to unity. This collector current will flow in the load resistance R_L which is practically kept as high resistance since the reverse biased CB junction has a high output resistance. The output voltage is given by:

$$e_o = -i_c R_L = \alpha . i_e . R_L \qquad \text{------ (6.13)}$$

From equations (6.12) & (6.13) we get : $e_o = \dfrac{\alpha . e_s . R_L}{r_e}$

or $$A_V = \frac{e_o}{e_s} = \frac{\alpha . R_L}{r_e}$$

A_v is the voltage gain and its value will be of the order of 400 if R_L is chosen the practical value, as 10$K\Omega$. The current gain α is less than or equal to unity. The circuit, therefore, will have finite power gain, which is α times the voltage gain. Thus the transistor gives the power gain by transferring current from the low resistance emitter circuit to the high resistance collector circuit, which signifies the name of the transistor as the transfer of resistance. The transistor may also be said to work as amplifier.

Example 6.1 A transistor having $\alpha = 0.96$ is connected in common base configuration with load resistance R_L = 5 $K\Omega$. If the incremental resistance of emitter base junction is 60 Ω, find the values of current gain, voltage gain and power gain of the amplifier.
Solution: The current gain of the amplifier in CB configuration is given by:

$$A_I = \frac{|I_C|}{|I_E|} = \alpha = 0.96$$

The voltage gain A_V is given by : $A_V = \dfrac{\alpha R_L}{r_e} = \dfrac{(0.96)(5000)}{60} = 80$

The power gain = $A_V.A_I$ = 80x0.96= 76.8

6.6 Transistor Characteristics in Common Base Configuration: Consider the PNP transistor in common base (CB) configuration, as shown in figure (6.7). In this circuit we have two variable voltages namely V_{EB} and

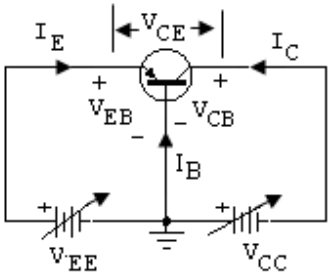


Fig. 6.7

V_{CB} and two currents I_E and I_C. Graphs may be plotted by choosing two out of four variables as dependent variables and other two as independent variables. It is customary to choose input current I_E and output voltage V_{CB} as independent variables; and input voltage V_{EB} and output current I_C as the dependent variables. Thus the dependent variables as the function of independent variables are given by:

$$V_{EB} = f_1(V_{CB}, I_E) \qquad \text{------ (6.14)}$$

$$I_C = f_2(V_{CB}, I_E) \qquad \text{------ (6.15)}$$

6.6.1 Input characteristics: Figure (6.8) shows the plot of emitter to base voltage V_{EB} versus emitter current I_E for different collector to base voltage V_{CB}. The set of these curves are known as the static input characteristic curves or static emitter characteristics.

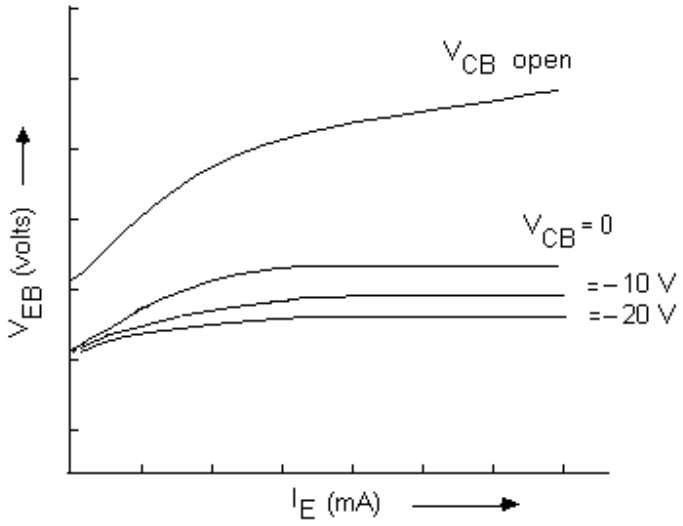


Fig. 6.8

The input characteristics shown in figure (6.8) represent simply the forward characteristics of the diode formed between the emitter and base region, for various collector voltages. It may be noted that there exists a cut in, or threshold voltage V_γ, blow which the emitter current is negligibly small. In general this voltage is approximately 0.1 volt for Ge transistors, and 0.5 volt for Si transistors. From the characteristic curves it is clear that there is an increasing trend in the emitter current for the increase in the magnitude of the collector voltage for a constant V_{EB}. This can be explained on the basis of Early effect that when the magnitude of the collector voltage increases, the effective base width decreases causing thereby an increase in the concentration gradient of minority carriers in the base region due to which emitter current increases.

6.6.2 Output characteristics: Figure (6.9) shows the plot of collector to base voltage V_{CB} versus collector current I_C for different emitter current I_E. The set of these curves are known as the static output characteristic curves or static collector characteristics. On the basis of biasing conditions of the two junctions, these output characteristics may be divided in to three regions namely active region, saturation region and cutoff region.

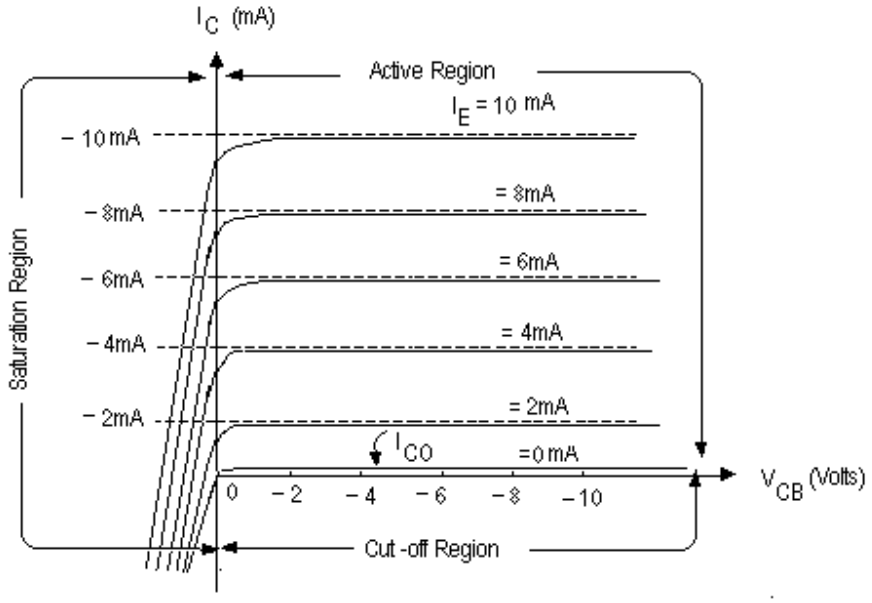


Fig. 6.9

Active region: In the active region the collector base junction is biased in the reverse bias whereas the emitter base junction is biased in the forward direction. From the output characteristics, it is clear that when $I_E = 0$, the collector current is very small and is equal to the reverse saturation I_{CO} of the diode formed by the collector base junction. The emitter current I_E flowing the emitter circuit due to the forward biasing of the emitter base junction causes a fraction of I_E to flow as collector current given by $I_C = -\alpha I_E + I_{Co.}$ In the active region, the collector current depends only on the emitter current and is independent of V_{CB}. However, because of the early effect, there is only a small increase in the collector current (approximately 0.5%) with the increase in the magnitude of the V_{CB}.

Since $\alpha \leq 1$, magnitude of the collector current is slightly less than that of the emitter current.

Saturation region: In the saturation region both the emitter base and collector base junctions of the transistor are in forward bias. This region exists to the left side of the ordinate, $V_{CB} = 0$, and above $I_E = 0$ in the output characteristics shown in figure (6.9). In this region, near $V_{CB} \approx 0$ the magnitude of the collector current decreases to the bottom of the curve and it is known that the bottoming has taken place. Actually, V_{CB} is slightly positive in this region, this forward biasing of the collector junction leads a large change in the collector current. So there is an exponential rise in the collector current for the small increase in V_{CB} given by diode equation. This is the reason of the bottoming. The collector current may even become positive if there is a large increase in the forward bias of the collector voltage.

Cutoff Region: The collector current I_C is very small roughly equal to I_{C0}, for $I_E = 0$ in the output characteristics shown in figure (6.9), so this curve is slightly above the V_{CB} axis and passes through the origin. The region below $I_E = 0$ is known as the cutoff region. In this region both emitter and collector junctions are in reverse bias. In the cutoff region the collector current $I_C \approx 0$. So the transistor in the cutoff region is used in switching circuits as the OFF state. Conversely, the ON state in the switching circuits is represented by the saturation region of the transistor.

6.7 Transistor Characteristics in Common Emitter Configuration: The common emitter configuration is most commonly used in the transistor circuits. Figure (6.10) shows the PNP transistor in common emitter configuration in which the emitter is common or grounded.

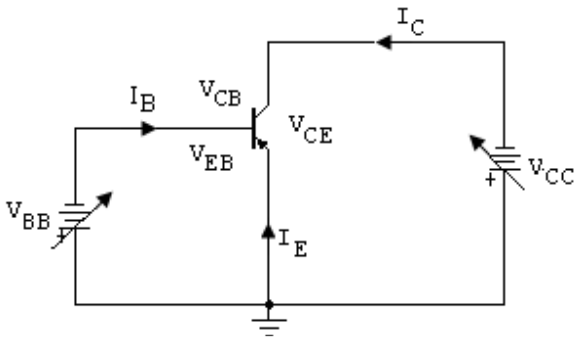


Fig. 6.10

In this configuration too we choose input current I_B and output voltage V_{CE} as independent variables; and input voltage V_{BE} and output current I_C as the dependent variables. Thus the dependent variables as the function of independent variables are given by:

$$V_{BE} = f_1(V_{CE}, I_B) \qquad \text{------ (6.16)}$$

$$I_C = f_2(V_{CE}, I_B) \qquad \text{------ (6.17)}$$

6.7.1 Input characteristics: Figure (6.11) shows the plot of base current I_B versus base to emitter voltage V_{BE} for different collector to emitter voltage V_{CE}. The set of these curves are known as the static input characteristics for CE configuration. For collector

shorted with emitter (i.e. the curve indicated by $V_{CE} = 0$), the emitter is forward bias and the curve is essentially the forward bias diode characteristic curve. If V_{BE} becomes zero then both emitter and collector junctions are short circuited, resulting thereby the base current zero. It is clear from these input characteristic curves that there is a decrease in magnitude of base current with the increase in magnitude of V_{CE} for constant V_{BE}. In general, if V_{BE} is held constant there is an increase in magnitude of the V_{CB} with the increase in V_{CE} , as $V_{CE} = V_{CB} + V_{BE}$. As per early effect, the increase in V_{CB} results the decrease in effective base width $W_b^{'}$. There will be less chances of recombination in the base region and thus base current decreases. It may be mentioned here that the break away point (the point where the current breaks away from zero current) in the transistor characteristic curves lies in the range of 0.5 to 0.6 volt for Si transistor and 0.1 to 0.2 for Ge transistor.

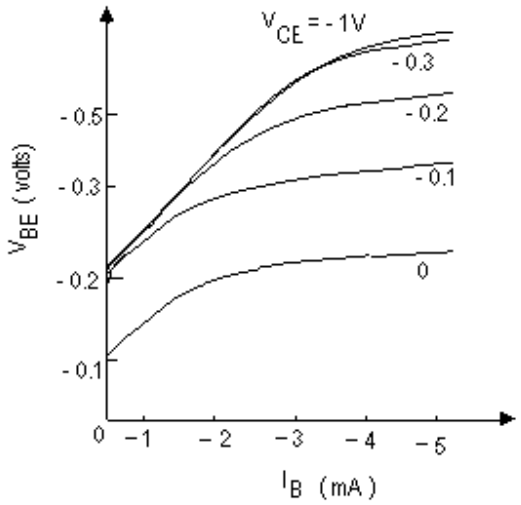


Fig. 6.11

6.7.2 Output characteristics: Figure (6.12) shows the plot of collector to emitter voltage V_{CE} versus collector current I_C for different base current I_B. The set of these curves are known as the static output characteristic curves for CE configuration. On the basis of biasing conditions of the two junctions, these output characteristics may be divided in to three regions namely active region, saturation region and cutoff region.

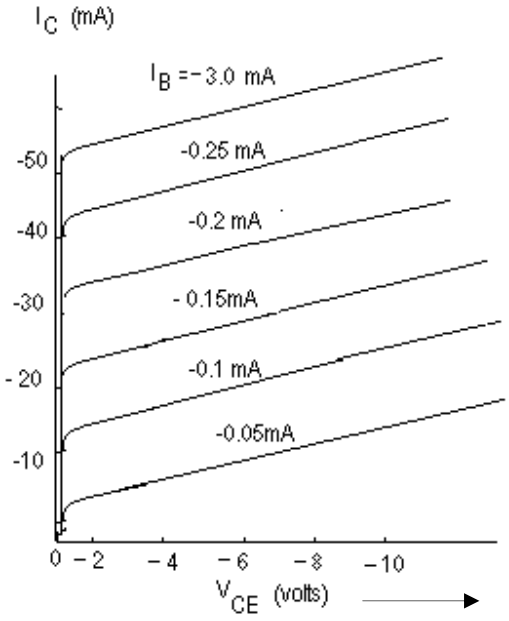


Fig. 6.12

Active region: In the active region the collector junction is biased in the reverse bias whereas the emitter junction is biased in the forward direction. The region above $I_B = 0$, to the right side of the I_C axis in figure (6.12) where V_{CE} is few tenths of a volt, is known as active region. The transistor to be used as an amplifying device, it should be operated in this active region.

As already discussed, the sum of the three transistor currents is zero i.e.

$I_E + I_B + I_C = 0$ ------ (6.18)

The vale of I_E from equation (6.9) is given by:

$$I_E = -\frac{I_C}{\alpha} + \frac{I_{C0}}{\alpha} \qquad \text{------ (6.19)}$$

From equations (6.18) & (6.19) we get:

$$I_B + I_C - \frac{I_C}{\alpha} + \frac{I_{C0}}{\alpha} = 0$$

or $$I_C\left(\frac{\alpha - 1}{\alpha}\right) + \frac{I_{C0}}{\alpha} = -I_B$$

or $$I_C = \left(\frac{\alpha}{1-\alpha}\right) I_B + \left(\frac{1}{1-\alpha}\right) I_{C0} \qquad \text{------ (6.20)}$$

or $I_C = \beta . I_B + (1 + \beta) . I_{C0}$ ------(6.21) where

$\beta = \left(\frac{\alpha}{1-\alpha}\right)$. Since $I_{C0} << I_B$ so $I_C \approx \beta . I_B$, in the active region. The factor β is called as the current gain defined as the ratio of the collector current to the base current.

If α were perfectly constant, I_C would be independent of V_{CE} (equation 6.20) and the characteristics should have been horizontal. These curves are not horizontal but the large variations in the characteristics are observed, which may be explained on the basis of Early Effect. According to the Early Effect α increases with the increase of V_{CE} (as V_{BE} is small so $V_{CE} \approx V_{CB}$). There is only about 0.5% increase in α when V_{CE} increases few volts. The numerical data, however, reveals that there is about 34% increase in β with 0.5% increase in α (as $\beta = \frac{\alpha}{1-\alpha}$). In view of the fact that $I_C = \beta I_B$, this leads a large increase in the magnitude of collector current. It clearly indicates that the characteristic curves are having some slope rather being horizontal.

CE Cutoff Region: In a transistor, cutoff refers to the condition where the collector current is zero or it is very small, approximately equal to the reverse saturation current I_{C0}. The transistor may, therefore, be in the open circuit condition. The cutoff in the figure (6.12) occurs at the intersection of load line with the base current I_B =0. If I_B is equal to zero then the collector current I_C will be given by (using equation 6.20):

$$I_C = -I_E = \frac{I_{C0}}{(1-\alpha)} \equiv I_{CE0} \qquad \text{------ (6.22)}$$

The symbol I_{CE0} is the collector current with collector junction reverse biased and base open circuited. This value of collector current in cutoff region is quite large for Ge transistors as α is generally equal to 0.9 (Ge transistor) and thus

$$I_C \approx \frac{I_{C0}}{1-0.9} \approx 10 I_{C0}$$

i.e. the collector current is approximately ten times to that of I_{C0} for Ge transistor. In order to satisfy the condition for cutoff, the collector current is to be reduced so that it becomes equal to I_{C0}. To obtain the collector current to be equal to I_{C0}, I_E should be made equal to zero as $I_C = I_{Co} - \alpha I_E$. It is found that if a reverse biasing voltage of the order of 0.1 volt is applied across the emitter junction, then Ge transistor will be in cutoff. However, in Si transistors, cutoff occurs at V_{BE} = 0. It is concluded from the above discussions that the cutoff means I_E = 0, $I_C = I_{C0}$, $I_B = -I_C$ and V_{BE} is a reverse voltage whose magnitude is of the order of 0.1 volt for Ge transistor and 0 volt for Si transistor (ref. fig. 6.13).

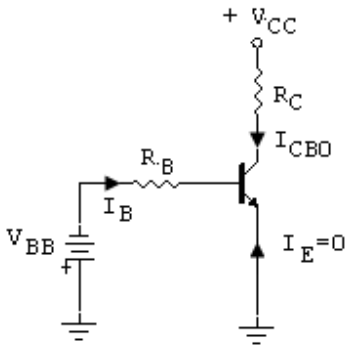


Fig 6.13

The collector current, when the emitter current is zero is represented by I_{CB0}. This current is larger than I_{C0} because (i) a large leakage current also flows across the surface not through the junction which is proportional to the voltage across the junction; and (ii) the new carriers may be generated by collision in the transition region of the collector junction leading to breakdown due to avalanche multiplication. At room temperature the I_{CB0} is of the order of a few microamperes for Ge and a few nano amperes for Si transistor. Since the current I_{CB0} approximately doubles with the 10^0C rise in the temperature as in the case of junction diode. Hence Si transistor can be used up to 200 ^{0}C and the Ge transistor can be used only up to 100^0C.

CE Saturation Region: In the saturation region both emitter and collector junctions are in forward bias. The saturation region is very close to zero voltage axis, where the bottoming is taking place i.e. the region where all the curves merge and rapidly fall to the origin. It has been observed that the region to the left of 0.3 V for Si (0.1 V for Ge) is the saturation region. To explain the saturation region in more details, the characteristic curves in between 0 to – 0.5 volt has been expanded as shown in figure 6.14. From this figure, we find that V_{CE} and I_C do not respond appreciably to the variations in the base current, once the base current exceeds the value –0.15 mA i.e. these curves are independent of base current and look like approximately the straight lines. The inverse of the slope of the curve in this region is known as common emitter saturation resistance R_{CS}, R_{CES} or $R_{CE(Sat)}$ given by the ratio $V_{CE(Sat)} / I_C$.

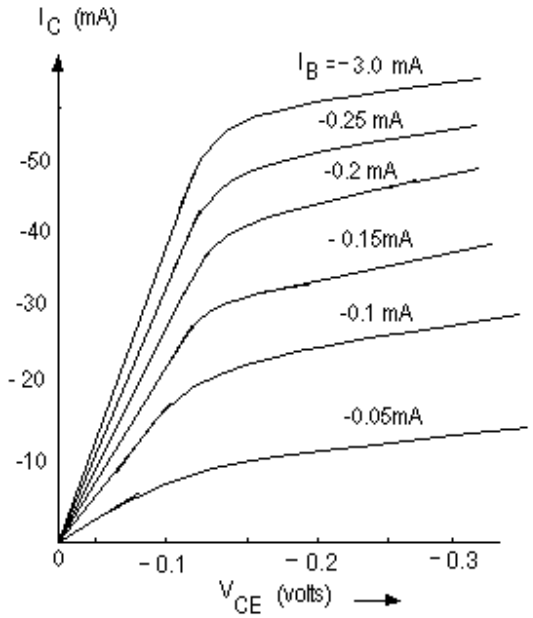


Fig. 6.14

6.8 Common Emitter Current Gain: Three different definitions of the current gain of a transistor will be discussed here:

(i) **Large Signal Current gain β: T**his current gain has earlier been defined as $\beta = \left(\frac{\alpha}{1-\alpha}\right)$. However, in equation (6.21) we replace I_{C0} with I_{CB0} we get,

$$I_C = \beta.I_B + (1+\beta).I_{CB0} \qquad \text{------ (6.23)}$$

or $$\beta = \frac{I_C - I_{CB0}}{I_B - (-I_{CB0})} \qquad \text{------- (6.24)}$$

The common emitter cutoff region is defined by $I_E = 0$, $I_C = I_{CB0}$ and $I_B = -\ I_{CB0}$. The numerator of this equation (6.24) is the increment in the collector current where as denominator is the increment in the base current from the currents of the cut-off region. So β may be defined as the ratio of collector current increment to the base current increment. Thus β represents the large signal current gain of a CE transistor.

(ii) **D.C. Current Gain h_{FE} :** The current gain or the d.c. forward transfer current gain or the β d.c. denoted by $\beta_{d.c.}$ or h_{FE} is defined as:

$$\beta_{d.c.} \equiv \frac{I_C}{I_B} = h_{FE}. \qquad \text{------- (6.25)}$$

(iii) **Small Signal Current gain:** The quantity β' is defined as the ratio of collector current gain ΔI_C to the base current increment ΔI_B at the operating point with the fixed V_{CE} . Thus

$$\beta' \equiv \left.\frac{\partial I_C}{\partial I_B}\right|_{V_{CE}=constant} = h_{fe} \qquad \text{------- (6.26)}$$

β' is used for the analysis of small signal amplification hence known as small signal current gain. If β is independent of the magnitude of the current then differentiating equation (6.23) w.r.t. I_C we get:

$$(I_{CB0} + I_B)\frac{\partial \beta}{\partial I_C} + \beta \frac{\partial I_B}{\partial I_C} = 1$$

or $$\frac{\beta}{\beta'} = \left(1 - (I_{CB0} + I_B)\frac{\partial \beta}{\partial I_C}\right)$$

or $$\beta = \beta'\left(1 - (I_{CB0} + I_B)\frac{\partial \beta}{\partial I_C}\right)$$

or $$h_{fe} = \frac{h_{FE}}{\left(1 - (I_{CB0} + I_B)\frac{\partial \beta}{\partial I_C}\right)} \qquad \text{------(6.24)}$$

as $\beta = h_{FE}$ and $\beta' = h_{fe}$.

It has been observed that over the entire range of I_C , h_{fe} differs from h_{FE} by less than 20%.

It may, however, be noted this equation (6.24) is true only for the active region. In saturation region, $h_{fe} \to 0$ as $\Delta I_C \to 0$ for a small increment ΔI_B.

6.9 Common Collector Configuration: The common collector configuration as will be discussed in the next chapter is mainly used for impedance matching since it has high input impedance and low output impedance. The CC configuration is shown in figure

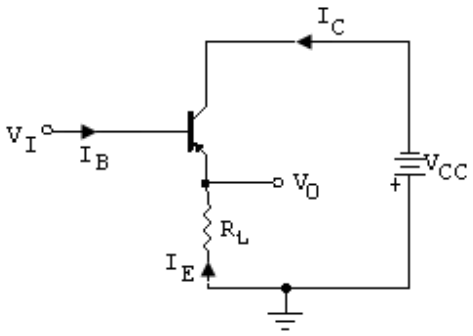


Fig. 6.15

(6.15), in which the load resistance connected between emitter and ground rather than collector and ground. This circuit seems to be similar to the common emitter configuration. From the deign point of view, there is no need for a set of common collector characteristics to choose the parameter of the amplifier. For all practical purposes, the output characteristics of the CC configuration are the same as for the CE configuration. For the CC configuration the output characteristics are a plot of I_E versus V_{EC} for the different range of base current I_B. The input current is, therefore, the same both the two types of the configurations. The horizontal voltage axis for CC configuration is obtained by simply changing the sign of the collector to emitter voltage of the CE characteristics. Since $I_C = \alpha I_E$ and $\alpha \approx 1$, so there will not be a remarkable change in the vertical axis.

Typical junction voltages: The typical junction voltages of a NPN transistor at 25 ^{0}C are given in volts, in the form of a table :

Transistor	$V_{BE,cutoff}$	$V_{BE,cutin}$	$V_{BE,active}$	$V_{BE,Sat}$	$V_{CE,Sat.}$
Si	0	0.5	0.7	0.8	0.2
Ge	-0.1	0.1	0.2	0.3	0.1

For a PNP transistor the sign of all the entries are reversed. It should be remembered that the transistor to be in the active region, the collector current should be equal to $\beta.I_B$ (neglecting I_{C0}); and in the saturation region the base current should be greater than or equal to $\frac{I_C}{\beta}$.

Example 6.2 (i) Find the values of I_B and I_C in the circuit given below (figure 6.16), knowing in which region the transistor is working. The Si transistor with β = 150 and I_{C0} = 22 nA is used in the circuit.
(b) Repeat part (i) with R_B = 370 KΩ .

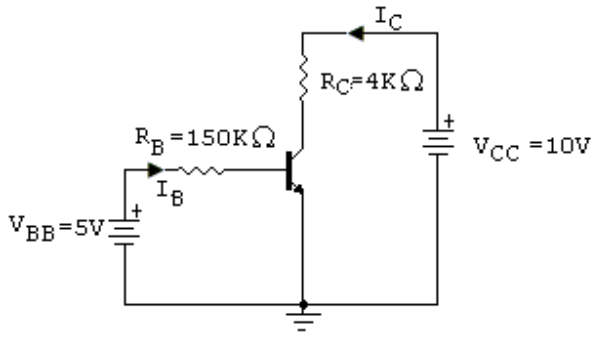


Fig. 6.16

Solution: (i) First of all we find the region in which the transistor is working, whether it is in cutoff, active or saturation region. Since the emitter base junction is in forward bias, it is therefore, clear that the transistor is not in the cutoff region. The transistor may either be in the active or in the saturation region.

Let us assume that the transistor is in the active region. Applying the KVL to the input circuit, we get:

$$V_{BB} = R_B I_B + V_{BE,active}$$

or $5\ V = 150\ K.\ I_B + 0.7\ V$

or $$I_B = \frac{5-0.7}{150} mA = 28.7 \mu A$$

and $$I_C = \beta . I_B = 150 x 28.7 \mu A = 4.3 mA$$ neglecting I_{C0}.

Applying KVL to the output circuit we get:

$$V_{CC} = R_C I_C + V_{CE}$$

or $V_{CE} = 10\ V - (4K)(4.3mA) = -7.2\ Volts$

But $V_{CE} = V_{CB} + V_{BE,active}$

or $V_{CB} = -7.2 - 0.7 = -7.9$

The transistor used in this circuit is NPN, so negative value of V_{CB} means collector is negative w.r.t. base; and the collector base junction is in forward bias. The transistor is therefore, not in the active region and our assumption is wrong. The transistor in now said to be biased in the saturation region.

Now applying KVL to the input circuit we get:

$$V_{BB} = R_B I_B + V_{BE,\ Sat}$$

or $5\ V = 150\ K.\ I_B + 0.8\ V$

or $$I_B = \frac{5-0.8}{150} mA = 28 \mu A$$

Applying KVL to the output circuit we get:

$$V_{CC} = R_C I_C + V_{CE,\ Sat}$$

or $$I_C = \frac{10.0-0.2}{4K} = \frac{9.8}{4} mA = 2.45 mA$$

We check the condition of saturation ($I_B \geq \frac{I_C}{\beta}$), which is true as 28 μA > 1.63 μA. The required values of I_C and I_B are 2.45 *mA* and 28 μA, and the transistor in saturation.

(ii) Let us assume that the transistor is in the active region. Applying the KVL to the input circuit, we get:

$$V_{BB} = R_B I_B + V_{BE,\ active}$$

or $\quad 5\ V = 370\ K.\ I_B + 0.7\ V$

or $\quad I_B = \frac{5 - 0.7}{370} mA = 11.62 \mu A$

and $\quad I_C = \beta . I_B = 150 x 11.62 \mu A = 1.74 mA$ neglecting I_{C0}.

Applying KVL to the output circuit we get:

$$V_{CC} = R_C I_C + V_{CE}$$

or $\quad V_{CE} = 10\ V - 4Kx1.74mA = 3.04\ Volts$

But $\quad V_{CE} = V_{CB} + V_{BE,active}$

or $\quad V_{CB} = 3.04 - 0.7 = 2.34\ volts$

The positive value of V_{CB} means the collector is positive w.r.t. base, CB junction is in reverse bias. The transistor is therefore, in the active region, which verifies our assumption; and the required values I_B and I_C are *11.62* μA and *1.74 mA* respectively.

Example 6.3 For the circuit shown in figure (6.17), assume β = 100. Find

(i) if the Si transistor is in cutoff, saturation or in active region,
(ii) output voltage V_0, and
(iii) minimum value for R_E for which the transistor operates in the active region.

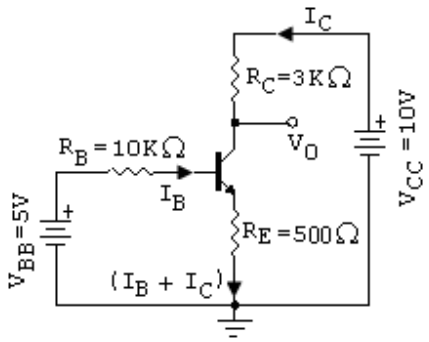


Fig. 6.17

Solution: (i) Suppose the transistor is in saturation region. Applying the KVL to the input and output circuits we get:

$V_{BB} = R_B I_B + V_{BE,\ Sat} + R_E(I_B + I_C)$

$5V = 10K.I_B + 0.8V + 0.5K(I_B + I_C)$

$4.2V = 10.5K.I_B + 0.5K.I_C$ ------ (6.25)

$V_{CC} = R_C I_C + V_{CE,\ Sat} + R_E(I_B + I_C)$

$10V = 3K.I_C + 0.2V + 0.5K(I_B + I_C)$

$9.8V = 3.5K.I_C + 0.5K.I_B$ ------ (6.26)

Solving the equations (6.25) & (6.26) for I_B and I_C we get:

$I_B = 0.28\ mA$ and $I_C = 2.76\ mA$

Since $I_B \geq \frac{I_C}{\beta}$ ($0.28\ mA > 0.0276\ mA$), hence the transistor is in saturation.

(ii) $V_0 = V_{CC} - R_C I_C = 10V - (3K)(2.76mA)$
$= 1.72$ volts

(iii) For the transistor to operate in the active region, $V_{BE,\ active} = 0.7\ V$ and V_{BC} should be in the reverse bias (greater than zero say 0.1V). So the transistor to operate in the active region V_{CE} should be at least equal to 0.8 V (as $V_{CE} = V_{BE} + V_{CB}$). Also in the active region $I_C = \beta . I_B$, (neglecting I_{C0}).

Applying KVL to the input circuit we get:

$V_{BB} = R_B.I_B + V_{BE,\ Active} + R_E(1+\beta)I_B$
$5V = (10K)I_B + 0.7V + R_E(101)I_B$
$4.3V = (10K)I_B + R_E(101)I_B$ ------ (6.27)

Now applying KVL to the output circuit we get:

$V_{CC} = (3K)(\beta I_B) + VCE + R_E(1+\beta)I_B$
$10V = (3K)(100\ I_B) + 0.8V + R_E(101)I_B$
$9.2V = (300K)I_B + R_E(101)I_B$ ------ (6.28)

Solving the equations (6.27)&(6.28) we get $R_E = 2.42\ K\Omega$.
This is the minimum value of R_E for the transistor to operate in the active region.

Example 6.4 Find the value of the resistance *R* connected between collector and base of a Si transistor with $\beta = 49$, shown in figure (6.18). Neglect the reverse saturation collector current of the transistor. Transistor is operated in the active region.

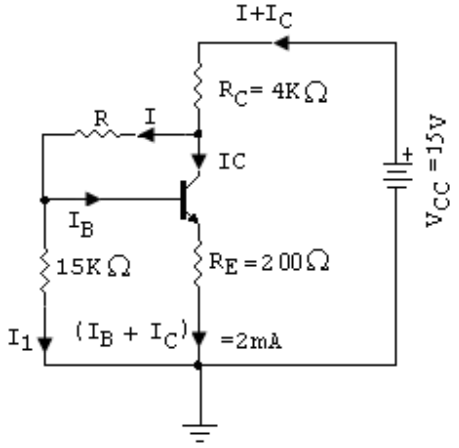


Fig. 6.18

Solution: Since $V_{BE} = 0.7V$, it indicates that the transistor is in the active region. So $I_C = \beta I_B$ and $I_C + I_B = 2mA$ (Given)

$I_B = (2/50)mA = 0.04\ mA$ and $I_C = (49)x(0.04) = 1.96\ mA$

Voltage across $15K\Omega$ resistance $V_i = V_{BE,\ Active} + (0.2K)\ x\ (2mA)$
$= 0.7 + 0.4 = 1.1\ V$

Current flowing through 15KΩ resistance $I_1 = \frac{1.1V}{15K\Omega} = 0.0733mA$

Current through resistance R is $I = I_1 + I_B = 0.0733 + 0.04 = 0.1133\ mA$

Applying KVL to the input circuit we get:

$V_{CC} = R_C(I + I_C) + R.I + V_i$

or $15V = (4K)(0.1133+1.96)mA + R(0.1133mA) + 1.1V$

or $R = 49.49\ K\Omega$

Example 6.5 A Si transistor with β = 100 is connected as shown in figure (6.19) Find the minimum value of the collector resistance R_C for which the transistor remains in saturation.

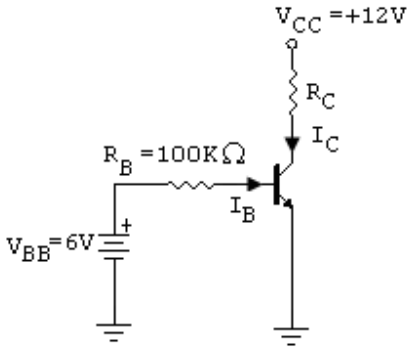


Fig. 6.19

Solution: To have the transistor in saturation region, $V_{BE} = V_{BE,Sat} = 0.8\ V$ and $V_{CE} = V_{CE,Sat.} = 0.2\ V$.

Now applying the KVL to the input circuit we have:

$V_{BB} = R_B I_B + V_{BE,Sat}$

$6V = (100K)(I_B) + 0.8V$

or $I_B = \frac{(6-0.2)V}{100K\Omega} = 0.052mA$

Similarly applying KVL to the input circuit we have:

$V_{CC} = R_C I_C + V_{CE,Sat}$

$12V = R_C I_C + 0.2$

or $I_C = \frac{(12-0.2)V}{R_C} = \frac{11.8V}{R_C}$

For the transistor to be in the saturation region :

$\beta . I_B \geq I_C$

$(100)x(0.052mA) \geq \frac{11.8}{R_C}$

or $R_C \geq \frac{11.8V}{5.2\ mA}$

or $R_C \geq 2.27K\Omega$

So the required minimum value of $R_C = 2.27\ K\Omega$.

Example 6.6 A Si transistor with β = 30 is used in the circuit shown in figure (6.20). $I_{CBO} = 10\ nA$ at $25\ ^0C$.

Find (i) V_0 for $V_i = 12\ V$ and show that the transistor in saturation for $R_1 = 15K$,

(ii) the minimum value of R_1 for which the transistor is in active region for $V_i = 12\ V$,

(iii) V_0 for $R_1 = 15\ K\Omega$ and $V_i = 1\ V$ and show that the transistor in cut-off, and

(iv) the maximum temperature at which transistor remains in cut-off for $R_1 = 15\ K\Omega$ and $V_i = 1\ V$.

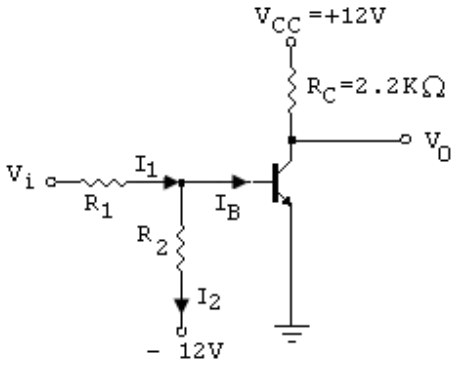


Fig. 6.20

Solution: (i) Let the transistor is in saturation, $V_{BE,\ Sat} = 0.8$ V

$$I_1 = \frac{(Vi - 0.8)V}{15K\Omega} = \frac{(12 - 0.8)V}{15K\Omega} = 0.75mA$$

$$I_2 = \frac{(12.0 + 0.8)V}{100K\Omega} = \frac{12.8V}{100K\Omega} = 0.128mA$$

$$I_B = I_1 - I_2 = 0.75 - 0.128 = 0.622\ mA$$

Applying KVL to the output circuit, we get:

$$V_{CC} = R_C I_C + V_{CE,\ Sat}$$

$$12V = (2.2\text{K})I_C + 0.2$$

or
$$I_C = \frac{(12 - 0.2)V}{2.2K\Omega} = \frac{11.8}{2.2K\Omega} = 5.36mA$$

$$\frac{I_C}{\beta} = \frac{5.36mA}{30} = 0.179mA$$

Since $I_B > (I_C/\beta)$ hence the transistor is in saturation.

So $V_0 = V_{CE,\ Sat} = 0.2\ V$

(ii) Let the transistor is in active region, $V_{BE,\ active} = 0.7\ V$

$$I_1 = \frac{(Vi - 0.7)V}{R_1} = \frac{(12 - 0.7)V}{R_1} = \frac{11.3V}{R_1}$$

$$I_2 = \frac{(12.0 + 0.7)V}{100K\Omega} = \frac{12.7V}{100K\Omega} = 0.127mA$$

$$I_B = I_1 - I_2 = \frac{11.3V}{R_1} - 0.127mA$$

In the active region $I_C = \beta\, I_B$

So $I_C = 30(\frac{11.3V}{R_1} - 0.127mA) = \frac{339}{R_1} - 3.81mA$

From the output circuit $V_{CE} = V_{CC} - I_C R_C = 12V - (\frac{339}{R_1} - 3.81mA)(2.2K\Omega)$

$$= 20.38 - \frac{745.8}{R_1}$$

The transistor will be in the active region if $V_{CB} \le -0.5V$ which implies:

$$V_{CB} = V_{CE} - V_{BE} = 20.38 - \frac{745.8}{R_1} - 0.7V = 19.68 - \frac{745.8}{R_1}$$

or $19.68 - \frac{745.8}{R_1} \le -0.5V$

or $\frac{745.8}{R_1} \le 20.18V$

or $R_1 \ge 36.95K\Omega$ So minimum value of $R_1 = 36.95K\Omega$

(iii) We find V_{BE} by applying Superposition theorem as:

$$V_{BE} = \frac{(1V)(100K\Omega)}{(100+15)K\Omega} + \frac{(12V)(15K\Omega)}{(100+15)K\Omega} = \frac{100}{115} - \frac{180}{115} = 0.896 - 1.565$$

$$= -0.7V$$

This implies the transistor is in cut-off region. So $V_0 = V_{CC} = 12V$

(iv) For the transistor to be in the cut-off region, $V_{BE} = 0V$ and the base current will be equal to I_{CBO} which is given by:

$$I_{CBO} = I_2 - I_1 = \frac{12V}{100K\Omega} - \frac{1V}{15K\Omega} = 0.12mA - 0.67mA = 0.053mA$$

The I_{CBO} varies with temperature as:

$$I_{CBO}(T) = I_{CBO}(T_0) x 2^{(T-T_0)/10}$$

As $T_0 = 25\ ^0C$ so $0.053mA = (10x10^{-6}mA)x2^{(T-25)/10}$

$$2^{(T-25)/10} = 5300$$

$$\{(T-25)/10\}\log 2 = \log 5300$$

$$(T-25)/10 = 12.37$$

$$T = 148.7\ ^0C$$

This is the maximum temperature for which the transistor remains in cut-off.

Example 6.7 (i) The reverse saturation current of the Ge diode connected in the circuit shown in figure (6.21), is $2\ \mu A$ at $25\ ^0C$ and increases by a factor of two for every $10\ ^0C$ rise in temperature. If $V_{BB} = 5V$, find the maximum allowable value of the resistance RB if the transistor is to remain in cut-off at a temperature of $75\ ^0C$.

(ii) If $V_{BB} = 1\ V$ and $R_B = 50\ K\Omega$, how high may the temperature increase before the transistor comes out of cut-off.

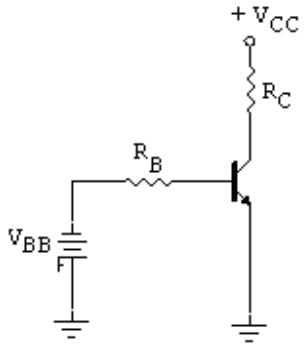


Fig. 6.21

Solution: (i) For the Ge transistor to be in the cut-off region, $V_{BE} = -\ 0.1V$
The I_{CO} varies with temperature as:

$$I_{CO}(T) = I_{CO}(T_0)x2^{(T-T_0)/10}$$

Where $T = 75\ ^0C$ $T_0 = 25\ ^0C$ and $I_{C0}\ (T_0) = 2\ \mu A$

So $$I_{CO}(75) = (2\mu A)x2^{(75-25)/10} = (2\mu A)(2^5) = 64\mu A$$

In the cut-off region $I_{C0}\ (T_0)$ current will flow as the base current, so

$$V_{BB} - V_{BE,Cutoff} = R_B\ I_{C0}(T)$$
$$5V - 0.1V = R_B(64\mu A)$$

or $$R_B = 76.5\ K\Omega$$

(ii) $$V_{BB} - V_{BE,Cutoff} = R_B\ I_{C0}\ (T)$$
$$V_{BB} = 1.0\ V,\ R_B = 50\ K\Omega$$

The transistor will come out of cut-off region when V_{BE} is slightly more than – 0.1V so

$$1 - 0.1 = (50\ K\Omega)\ I_{C0}(T)$$
$$I_{C0}(T) = 18\ \mu A$$
$$18\mu A = (2\mu A)x2^{(T-T_0)/10}$$
$$2^{(T-T_0)/10} = 9\mu A \qquad \text{or} \qquad (T - T_0) = 31.7^0C$$
$$T = 31.7 + 25 = 56.7\ ^0C$$

6.10 Ebers – Moll Model of a Transistor: We have already studied the equation which shows the dependence of currents in a transistor upon the junction voltages given by equation (6.11) as:

$$I_C = -\alpha I_E - I_{C0}\left(e^{\frac{V_C}{V_T}} - 1\right) \qquad \text{------ (6.29)}$$

For a physical transistor, the emitter and collector junctions are quite alike, except a little difference in the electrical conductivities of each layer due to their doping levels. Theoretically one may think of using a transistor in an inverted mode, i.e. the role of emitter junction and the collector junction are interchanged. Such an arrangement might not give as effective results as from its normal mode. So we write two current equations

one for the normal mode of operation, replacing α by α_N (current gain in normal operation):

$$I_C = -\alpha_N I_E - I_{C0}\left(e^{\frac{V_C}{V_T}} - 1\right) \qquad \text{------ (6.30)}$$

and other for inverted mode of operation, replacing α by α_I (current in inverted operation):

$$I_E = -\alpha_I I_C - I_{E0}\left(e^{\frac{V_E}{V_T}} - 1\right) \qquad \text{------ (6.31)}$$

In the inverted mode of operation I_E has been replaced by I_C and vice versa, V_C by V_E and I_{C0} by $I_{EO,}$ as the role of emitter junction and collector junctions are interchanged. I_{E0} is the emitter junction reverse saturation current. These two equations are defined in figure (6.22) for PNP transistor. For either transistor (PNP or NPN), a positive value of current means that positive charge flows into the junction and a positive V_E or V_C means that the corresponding junction is in forward bias. A resistance $r_{bb'}$, known as base spreading resistance is supposed to be connected between a point on active base region *B'* and the base terminal *B* as shown in figure (6.22). The base spreading resistance is the d.c. ohmic resistance of the semiconductor material between the two points. The voltage drop across collector to base terminal differs from V_C by the potential drop across the base spreading resistance $r_{bb'}$ i.e. $V_{CB} = V_C - I_B r_{bb'}$

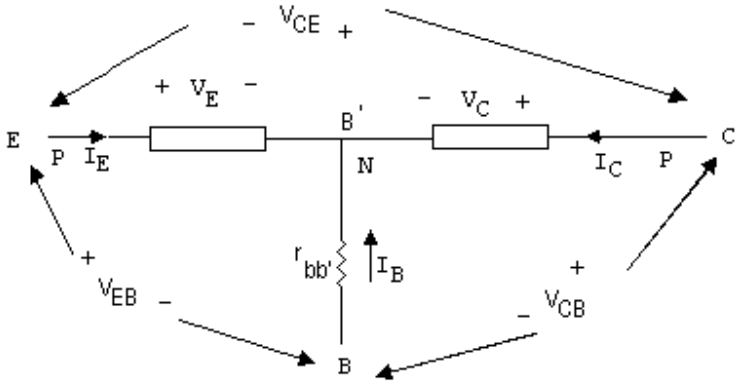
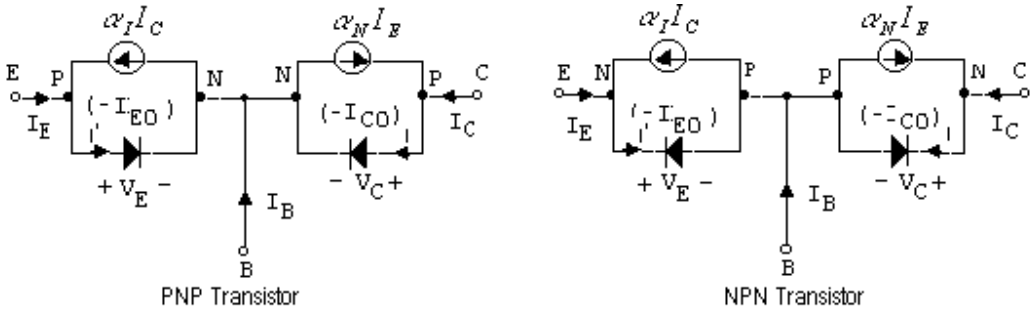


Fig. 6.22

J.J.Ebers and J.L.Moll of Bell Telephone in 1954 proposed a simple transistor model based on the equations (6.30 & 6.31). This model, known as Ebers – Moll model, is shown in figure (6.23a) for a PNP transistor and in figure (6.23b) for NPN transistor.



(a) (b)

Fig. 6.23

This model consists of two diodes placed back to back with reverse saturation currents $-I_{C0}$ and $-I_{E0}$ and two dependent current sources shunting the diodes. For NPN transistor the direction of the diodes are reversed. The corresponding equations for NPN transistor will be same as given in equations (6.30 & 6.31), with the difference that V_E will be replaced by $-V_E$ and V_C by $-V_C$, as V_E and V_C make the diodes in reverse bias. It should be noted that the base spreading resistance has been neglected from the figure (6.22) and thus eliminating the difference between I_{C0} and I_{CB0}.

The model which have been developed in this section are characterized by four parameters α_N, α_I, I_{C0} and I_{E0}. However, these parameters are not independent, but are related by the relation:

$$\alpha_N . I_{E0} = \alpha_I I_{C0} \qquad \text{------ (6.32)}$$

This reciprocity condition shows that at least three parameters out of these four parameters are sufficient to find to characterize the static V – I relationship of a transistor.

If the dependent sources from the figure (6.23) are eliminated, then transistor may be represented simply by two diodes placed back to back. This is possible if $\alpha_N = \alpha_I = 0$. The current gain α will be zero if the base width is made much larger than the diffusion length of the injected minority charge carriers in the base region, all the minority carriers will recombine in the base and none will survive to reach the collector. In this condition the transistor action ceases. We therefore, get the conclusion that it is impossible to construct a transistor by simply connecting two diodes back to back.

Example 6.8 Find the explicit expressions for I_C and I_E in terms of V_C and V_E from equations (6.30) and (6.31).

Solution: Rewriting the equations (6.30) and (6.31) in the following form:

$$I_C + \alpha_N I_E = -I_{C0}\left(e^{\frac{V_C}{V_T}} - 1\right) \qquad \text{------ (6.30)}$$

$$I_E + \alpha_I I_C = -I_{E0}\left(e^{\frac{V_E}{V_T}} - 1\right) \qquad \text{------ (6.31)}$$

Putting the value of I_E from equation (6.31) in equation (6.33) we get:

$$I_C + \alpha_N \left[-\alpha_I I_C - I_{E0}(e^{\frac{V_E}{V_T}} - 1) \right] = -I_{C0}(e^{\frac{V_C}{V_T}} - 1)$$

or $$I_C(1-\alpha_N\alpha_I) - \alpha_N I_{E0}(e^{\frac{V_E}{V_T}} - 1) = -I_{C0}(e^{\frac{V_C}{V_T}} - 1)$$

or $$I_C = \frac{\alpha_N I_{E0}(e^{\frac{V_E}{V_T}} - 1)}{(1-\alpha_N\alpha_I)} - \frac{I_{C0}(e^{\frac{V_C}{V_T}} - 1)}{(1-\alpha_N\alpha_I)} \qquad \text{------ (6.35)}$$

Similarly by putting the value of I_C from equation (6.30) in equation (6.34) we get:

$$I_E = \frac{\alpha_I I_{C0}(e^{\frac{V_C}{V_T}} - 1)}{(1-\alpha_N\alpha_I)} - \frac{I_{E0}(e^{\frac{V_E}{V_T}} - 1)}{(1-\alpha_N\alpha_I)} \qquad \text{------ (6.36)}$$

These are the required expressions.

Example 6.9 (i) A transistor is operating in the cut-off region with both the emitter and collector junctions reversed biased by at least a few tenths of a volt. Prove that the currents are given by:

$$I_E = \frac{I_{E0}(1-\alpha_N)}{1-\alpha_N\alpha_I}$$

$$I_C = \frac{I_{C0}(1-\alpha_I)}{1-\alpha_N\alpha_I}$$

(ii) Prove that the emitter junction voltage required just to produce cut-off ($I_E = 0$ and the collector is reversed biased) is:

$$V_E = V_T \ln(1-\alpha_N)$$

Solution: (i) In the equations (6.35) and (6.36), the values of $e^{(V_C/V_T)}$ and $e^{(V_E/V_T)}$ may be put equal to zero as V_C and V_E are reversed biased and are larger than V_T. So we get :

$$I_E = \frac{-\alpha_I I_{C0}}{(1-\alpha_N\alpha_I)} + \frac{I_{E0}}{(1-\alpha_N\alpha_I)}$$

As $\alpha_N I_{E0} = \alpha_I I_{C0}$, above equation may be modified as :

$$= \frac{I_{E0} - \alpha_N I_{E0}}{(1-\alpha_N\alpha_I)} = \frac{I_{E0}(1-\alpha_N)}{(1-\alpha_N\alpha_I)} \qquad \text{------ (6.37)}$$

Similarly we may prove $$I_C = \frac{I_{C0}(1-\alpha_I)}{1-\alpha_N\alpha_I} \qquad \text{------ (6.38)}$$

(ii) We know $$I_E = -\alpha_I I_C - I_{E0}\left(e^{\frac{V_E}{V_T}} - 1\right)$$

$$I_C = -\alpha_N I_E - I_{C0}\left(e^{\frac{V_C}{V_T}} - 1\right)$$

Put $I_E = 0$ and $e^{(V_C/V_T)} \approx 0$ as $|V_C| >> 1$ in the above equations we get:

$$0 = -\alpha_I I_C - I_{E0}\left(e^{\frac{V_E}{V_T}} - 1\right)$$

or $$\left(1 - e^{\frac{V_E}{V_T}}\right) = \frac{\alpha_I I_C}{I_{E0}} \qquad \text{------ (6.39)}$$

and $I_C = I_{C0}$

So $$\left(1 - e^{\frac{V_E}{V_T}}\right) = \frac{\alpha_I I_{C0}}{I_{E0}} = \alpha_N \qquad \text{as} \quad \alpha_N I_{E0} = \alpha_I I_{C0}$$

or $V_E = V_T \ln(1 - \alpha_N)$ Proved.

Example 6.10 Find both collector and emitter current for a transistor when emitter and collector junctions are reversed biased. Given $I_{C0} = 5\ \mu A$, $I_{E0} = 3.57\ \mu A$ and $\alpha_N = 0.98$.

Solution: We know $\alpha_N I_{E0} = \alpha_I I_{C0}$ so $\alpha_I = \frac{\alpha_N I_{E0}}{I_{C0}} = \frac{(0.98)(3.57\mu A)}{5\mu A} = 0.7$

From equation (6.35) we get:

$$I_C = \frac{I_{C0}(1 - \alpha_I)}{1 - \alpha_N \alpha_I} = \frac{(5\mu A)(1 - 0.7)}{(1 - 0.98\ x\ 0.7)} = 0.48\ \mu A$$

From equation (6.34) we get:

$$I_E = \frac{I_{E0}(1 - \alpha_N)}{(1 - \alpha_N \alpha_I)} = \frac{(3.57\mu A)(1 - 0.98)}{(1 - 0.98 x 0.7)} = 0.23\mu A$$

Example 6.11 Prove that the junction voltages in terms of the transistor currents are given by: $$V_C = V_T . \ln\left(1 - \frac{I_C + \alpha_N I_E}{I_{C0}}\right)$$

$$V_E = V_T . \ln\left(1 - \frac{I_E + \alpha_I I_C}{I_{E0}}\right)$$

Solution: The transistor currents are given by:

$$I_C = -\alpha_N I_E - I_{C0}\left(e^{\frac{V_C}{V_T}} - 1\right) \qquad \text{------ (6.40)}$$

$$I_E = -\alpha_I I_C - I_{E0}\left(e^{\frac{V_E}{V_T}} - 1\right) \qquad \text{------ (6.41)}$$

From equation (6.40) we have: $$e^{\frac{V_C}{V_T}} = \left(1 - \frac{I_C + \alpha_N I_E}{I_{C0}}\right)$$

or $$V_C = V_T\left(1 - \frac{I_C + \alpha_N I_E}{I_{C0}}\right) \qquad \text{----- (6.42)}$$

Similarly from equation (6.41), it may be proved that:

$$V_E = V_T.\ln\left(1-\frac{I_E+\alpha_I I_C}{I_{E0}}\right) \qquad \text{------ (6.43)}$$

Example 6.12 (i) Show that the exact expression for the CE output characteristics of a PNP transistor is:

$$V_{CE} = V_T.\ln(\frac{\alpha_I}{\alpha_N}) + V_T.\ln.\left(\frac{I_{C0}+\alpha_N I_B - I_C(1-\alpha_N)}{I_{E0}+I_B+I_C(1-\alpha_I)}\right)$$

(ii) If $I_B >> I_{E0}$ and $I_B >> I_{C0}/\alpha_N$ then

$$V_{CE} = V_T.\ln\left(\frac{1-\frac{1}{\beta_N}.\frac{I_C}{I_B}}{\frac{1}{\alpha_I}+\frac{1}{\beta_I}\frac{I_C}{I_B}}\right)$$

Where $\beta_I = \dfrac{\alpha_I}{1-\alpha_I}$ and $\beta_N = \dfrac{\alpha_N}{1-\alpha_N}$

Solution: (i) For the PNP transistor $V_{CE} = V_C - V_E$, putting the values of V_C and V_E from equations (6.42) & (6.43) we have:

$$V_{CE} = V_C - V_E = V_T\left[\ln\left(1-\frac{I_C+\alpha_N I_E}{I_{C0}}\right)-\ln\left(1-\frac{I_E+\alpha_I I_C}{I_{E0}}\right)\right]$$

$$= V_T \ln\left[\left(\frac{I_{C0}-I_C-\alpha_N I_E}{I_{E0}-I_E-\alpha_I I_C}\right)\left(\frac{I_{E0}}{I_{C0}}\right)\right] \qquad \text{Further put} \quad -I_E = I_B + I_C$$

$$= V_T \ln\left[\left(\frac{I_{C0}-I_C+\alpha_N I_C+\alpha_N I_B}{I_{E0}+I_B+I_C-\alpha_I I_C}\right)\left(\frac{I_{E0}}{I_{C0}}\right)\right]$$

$$= V_T \ln\left[\left(\frac{I_{C0}-(1-\alpha_N)I_C+\alpha_N I_B}{I_{E0}+I_B+I_C(1-\alpha_I)}\right)\left(\frac{I_{E0}}{I_{C0}}\right)\right] \quad \text{Put} \quad \frac{I_{E0}}{I_{C0}} = \frac{\alpha_I}{\alpha_N}$$

We get $V_{CE} = V_T.\ln(\dfrac{\alpha_I}{\alpha_N}) + V_T.\ln.\left(\dfrac{I_{C0}+\alpha_N I_B - I_C(1-\alpha_N)}{I_{E0}+I_B+I_C(1-\alpha_I)}\right)$ ------ (6.44)

(ii) Equation (6.41) may be rewritten as:

$$V_{CE} = V_T \ln\left[\left(\frac{(\frac{I_{C0}}{\alpha_N}+I_B)\alpha_N-(1-\alpha_N)I_C}{I_{E0}+I_B+I_C(1-\alpha_I)}\right)\left(\frac{I_{E0}}{I_{C0}}\right)\right]$$

Neglecting $\dfrac{I_{C0}}{\alpha_N}$ and I_{E0} as $I_B >> I_{E0}$ and $I_B >> I_{C0}/\alpha_N$ we get

$$V_{CE} = V_T \ln\left[\left(\frac{I_B\alpha_N-(1-\alpha_N)I_C}{I_B+I_C(1-\alpha_I)}\right)\left(\frac{\alpha_I}{\alpha_N}\right)\right]$$

or $$V_{CE} = V_T \ln\left[\left(\frac{1-\left\{\frac{1-\alpha_N}{\alpha_N}\right\}\frac{I_C}{I_B}}{\frac{1}{\alpha_I}+\left\{\frac{1-\alpha_I}{\alpha_I}\right\}\frac{I_C}{I_B}}\right)\left(\frac{\alpha_I\alpha_N}{\alpha_N\alpha_I}\right)\right]$$

Putting $\beta_I = \frac{\alpha_I}{1-\alpha_I}$ and $\beta_N = \frac{\alpha_N}{1-\alpha_N}$ we get:

$$V_{CE} = V_T.\ln\left(\frac{1-\frac{1}{\beta_N}.\frac{I_C}{I_B}}{\frac{1}{\alpha_I}+\frac{1}{\beta_I}\frac{I_C}{I_B}}\right)$$

6.11 Maximum Voltage Rating: The limiting ratings given by the manufacturers for a transistor are maximum current, maximum voltage and maximum power dissipation. The power dissipation in the junctions of a transistor is important since excessive heating can either damage the transistor completely or alter its characteristics considerably. It has been observed that even if the rated dissipation of a transistor is not exceeded, there is an upper limit to the maximum allowable collector junction voltage. Since at high voltage, there is a possibility of voltage break down in the transistor. Two types of breakdowns are possible (i) Avalanche breakdown and (ii) Punch through.

(i) Avalanche Breakdown: This is a simple collector diode breakdown, which is due to the avalanche multiplication, similar to one that occurs in a junction diode. It is well known that the collector current increases sharply at a well defined breakdown voltage. For CE configuration, there is a strong influence of carrier multiplication and the breakdown voltage in this case is significantly smaller than CB configuration. Let BV_{CBO} is the breakdown voltage for CB configuration for the condition $I_E = 0$ (with emitter is open); and BV_{CEO} is the breakdown voltage for CE configuration for the condition $I_B = 0$ (with base is open). Let M is the factor by which the current entering the collector depletion region is multiplied, to obtain the collector I_C in each case. The current I_C is then given by the relation:

$$I_C = -(\alpha.I_E + I_{C0})M \qquad \text{------ (6.45)}$$

The factor M is given by empirical relation as:

$$M \cong \frac{1}{1-(V_{CB}/BV_{CBO})^n} \qquad \text{------ (6.46)}$$

The value of n is found to be in the range of about 2 to 10 to all the transistors.

From the equation (6.45), it is clear that in case of CB if $I_E = 0$, then I_C is simply $-MI_{CO}$, which defines the breakdown of a simple junction. In CE configuration the situation is somewhat complicated. If $I_B = 0$ then I_C will be equal to I_E and thus equation (6.45) becomes:

$$I_C = -\frac{MI_{C0}}{(1 - M\alpha)} \qquad \text{------- (6.47)}$$

It is clear from the equation (6.47) that the collector current increases indefinitely when $M.\alpha$ approaches unity. Since α is close to unity in most transistors, M need only be slightly greater than unity. This equation then approaches to breakdown. Avalanche multiplication thus dominates the current in a CE transistor well below the breakdown voltage of the collector junction in CB configuration.

(ii) Punch through: The second breakdown that usually takes place in the transistor is known as Punch through or Reach through. This phenomenon occurs because of Early Effect. As per this effect when the reverse bias on the collector junction in increased far enough, it is possible to decrease the effective base width W_b, to the extent that the collector depletion region essentially fills the entire base region (fig. 6.24). That is the effective base width is zero. In this punch through condition holes are swept directly from the emitter region to the collector, and the transistor action is lost. Punch through is a breakdown effect which is generally avoided in the circuit design.

The punch through is controlled by the basic transistor design parameters such as base conductivity and base width and not by the circuit configuration so it takes place at a fixed collector base voltage. However, the avalanche multiplication not only depends on the collector voltage but also upon the circuit in which it is used. In a particular transistor, the maximum allowable voltage limit is determined by punch through or the avalanche breakdown, whichever occurs earlier.

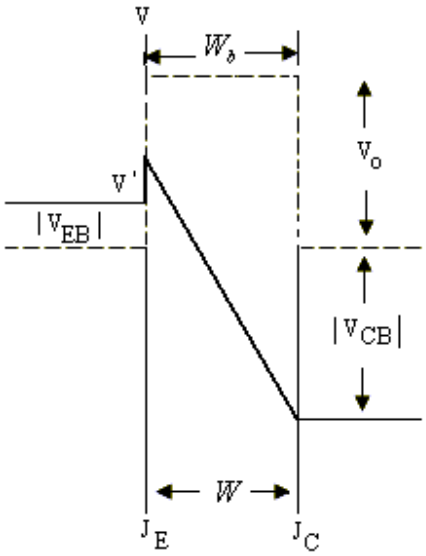
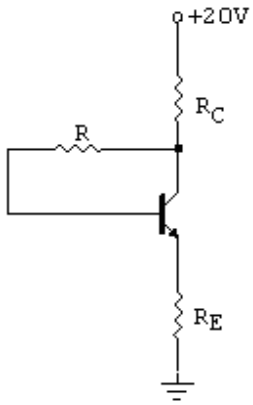


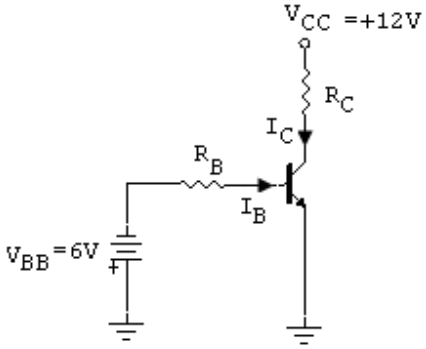
Fig. 6.24

Problems:

1. Explain Base – width modulation (The Early Effect) with the aid of plot of potential and minority carrier concentration in the base region.
2. Sketch the output static characteristic curves of a PNP transistor in CB configuration. Explain the shapes of these curves qualitatively in active, cut-off and saturation region.
3. Sketch the input static characteristic curves of a PNP transistor in CB configuration. Explain the shapes of these curves qualitatively.
4. Draw the output static characteristic curves of a PNP transistor in CE configuration. Explain the shapes of the curves qualitatively in active, cut-off and saturation region.
5. Draw the input static characteristic curves of a PNP transistor in CE configuration. Explain the shapes of these curves qualitatively.
6. Discuss the Early Effect with its three consequences.
7. For a PNP transistor biased in the active region, plot the variations in potential and minority carrier concentration in each emitter, base and collector region. Explain the shapes of these plots.
8. Write the Ebers and Moll equations. Draw the circuit model which satisfies these equations.
9. Draw the Ebers and Moll model of a transistor. Explain a transistor can not be represented by two diodes connected back to back.
10. Discuss two possible sources of break down in a transistor as the collector to emitter voltage is increased.
11. Discuss maximum voltage ratings of a transistor.
12. Prove that the transistor can be used as an amplifier.
13. Discuss qualitatively different current components in a PNP transistor. Deduce the equation $I_C = -\alpha I_E + I_{C0}\left(1 - e^{\frac{V_C}{V_T}}\right)$, where symbols have their usual meanings.
14. Define the current gain α and β. Derive the relation between them.
15. Discuss the phenomenon of punch trough and reach through in a transistor.
16. Define the following regions in a transistor (a) active (b) saturation and (c) cut-off.
17. Discuss the variations in potential and minority carrier concentrations in each section of open circuited symmetrical PNP transistor.
18. Discuss the different current components in a PNP transistor. Deduce an expression for the collector current I_C. Define each symbol in this equation. Finally deduce the generalized expression for I_C so that it is valid even if the transistor is not operating in its active region.
19. Define the three current gain β, $\beta_{d.c.}$ (h_{FE}),and β' (h_{fe}) in CE transistor. Derive the relation between h_{FE} and h_{fe}.
20. Find the value of resistance R in the circuit given below in which Si transistor with $\beta = 49$ is used. $V_{CC} = 20$ V, $V_{CE} = 5.8$ V, $R_E = 0.2\ K\Omega$, $R_C = 8.2\ K\Omega$. Neglect reverse saturation current. (Ans. 159.4 KΩ)



21. Find the values of R_B and R_C in the circuit given below, so that collector current of 12mA flows in the Si transistor with $\beta = 80$. $V_{CE} = 4$ V. Neglect reverse saturation current. (Ans. 667Ω, 35KΩ)



7

The Transistors at Low Frequencies

In the preceding chapter the static characteristics of a transistor have been discussed. The use of transistor as amplifier will be discussed in this chapter. Transistor can be used in large signal or small signal operation. The graphical approach may be studied for the large signal behaviour. However, for small signal, the transistor is operated in the active region. Thus small signal linear model of the transistor will be derived in the active region for the analysis of the transistor behaviour. The *h* – parameter model of the transistor is generally used to characterise a transistor which can further be used easily to explain the amplifier characteristics in CE, CB and CC configuration.

7.1 Low Frequency *h* – Parameter Model of a Transistor: There are three modes of transistor namely CE, CB and CC configurations. In all three configurations one terminal is assumed as common between input and output terminals. One pair of terminal is used as the input port and other pair of terminals as output port. So any transistor configuration can be represented by a two port network. In CE configuration base and emitter terminals are treated as input port and collector & emitter terminals as the output port. In common base configuration emitter & base are known as the input port and collector & base are known as the output port. In common collector configuration input port consists of base and collector; and output port is emitter and collector.

We consider the common emitter configuration since it is most commonly used amplifier. In figure (7.1a) the transistor in common emitter configuration is shown with their terminal variables. We choose i_b and v_{ce} as independent variables and v_{be} and i_c as the dependent variables. We may write the *h* – parameter equations as discussed in section 2.3.

$$v_{be} = h_{11} i_b + h_{12} v_{ce} \qquad \text{------ (7.1)}$$

$$i_c = h_{21} i_b + h_{22} v_{ce} \qquad \text{------ (7.2)}$$

We define:

$h_{11} = (v_{be}/i_b)|_{v_{ce}=0}$ as the input resistance with output shorted, it is represented by h_{ie}

$h_{12} = (v_{be}/v_{ce})|_{i_b=0}$ as the reverse transfer voltage ratio with input open, it is represented by h_{re}.

$h_{21} = (i_c/i_b)|_{v_{ce}=0}$ as the forward transfer current ratio with output shorted, it is represented by h_{fe}.

$h_{22} = (i_c/v_{ce})|_{i_b=0}$ as the output conductance with input open, it is represented by h_{oe}.

In all these parameters the subscript *e* is used for common emitter configuration. For common base or common collector configuration the subscript *e* will be replaced with *b* or *c* respectively.

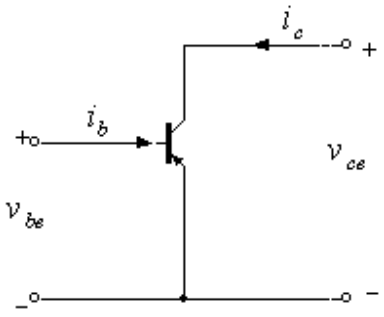


Fig. 7.1a

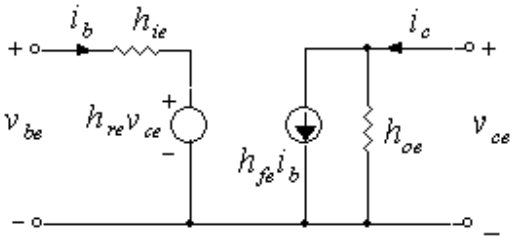


Fig. 7.1b

Thus the common emitter *h* – parameter equations in the form of the traditional symbols are given as:

$$v_{be} = h_{ie} i_b + h_{re} v_{ce} \qquad \text{------ (7.3)}$$

$$i_c = h_{fe} i_b + h_{oe} v_{ce} \qquad \text{------ (7.4)}$$

The equivalent circuit for these two equations may be drawn as shown in figure (7.1b), which can be verified by applying KVL to the input circuit and KCL to the output circuit. Thus the circuit of figure (7.1b) forms the model of the transistor which is known as *h* – Parameter Model (or hybrid model) of a Transistor in common emitter configuration.

The *h* – parameter model of a transistor in CB configuration is shown in figure (7.2). The terminal variables are chosen properly and *h* – parameters have the suffix *b* in place of *e*. The h – parameter equations in this configuration may be written as:

$$v_{eb} = h_{ib} i_e + h_{rb} v_{cb} \quad \text{------ (7.5)}$$

$$i_c = h_{fb} i_e + h_{ob} v_{cb} \quad \text{------ (7.6)}$$

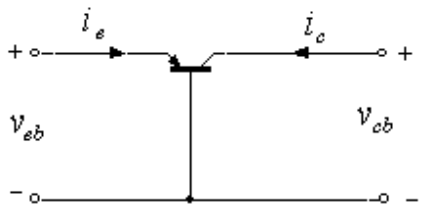


Fig. 7.2a

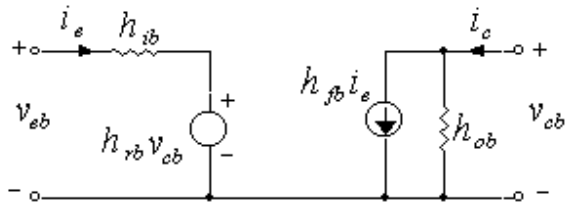


Fig.7.2b

Similarly, h – parameter equations in CC configuration may be given below and its equivalent model is shown in figure (7.3)

$$v_{bc} = h_{ic} i_b + h_{rc} v_{ec} \quad \text{------ (7.7)}$$

$$i_e = h_{fc} i_b + h_{oc} v_{ec} \quad \text{------ (7.8)}$$

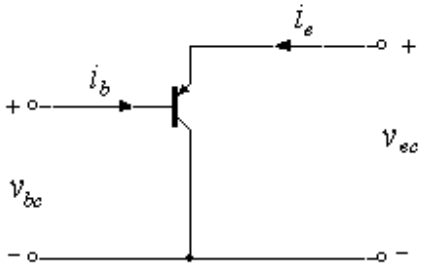


Fig. 7.3a

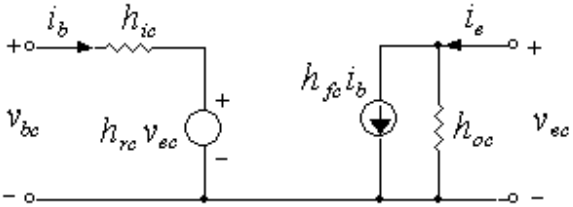


Fig.7.3b

The Hybrid models and the equations discussed above are valid for NPN as well as for PNP transistors.

7.2 Determination of h – parameters: Although the *h* – parameters are directly measured in practice with suitable circuit arrangement, but these parameters can also be estimated from the static characteristics of the transistor in any configuration. Figures (7.4a) & (7.4b) show the output and input characteristics respectively of an NPN transistor in CE configuration. In the output characteristic we choose the operating point Q for an ideal current bias flowing in the base and ideal collector to emitter voltage. The point *P* in the input characteristic corresponds to the values of base current and V_{BE} for a particular V_{CE}. Considering two points P_1 and P_2 lying in the horizontal line (base current is constant), we find the difference ΔV_{BE} as ($V_{BE,}$ at P_2 – $V_{BE,}$ at P_1) and the difference ΔV_{CE} corresponding to these two points in the input characteristics. The ratio of ΔV_{BE} and ΔV_{CE} will be the estimate of h_{re}. On the curve of given V_{CE} (the curve on which the

point *P* lies) the two more points A and B are chosen, the ratio of ΔV_{BE} and ΔI_B at these two points gives h_{ie}. On the output characteristics of figure (7.4a) two points Q_1 and Q_2 are chosen for the fixed value of V_{CE}. The ratio of ΔI_C and ΔI_B, corresponding to these two points gives h_{fe}. The ratio of ΔI_C and ΔV_{CE} on the curve of fixed I_B (the curve on which the point Q lies) will give the estimate of h_{oe}.

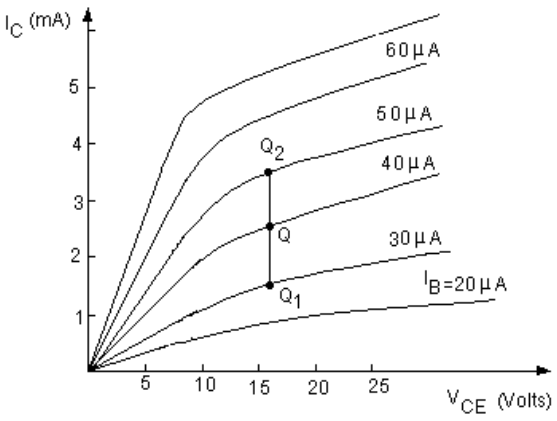


Fig. 7.4(a)

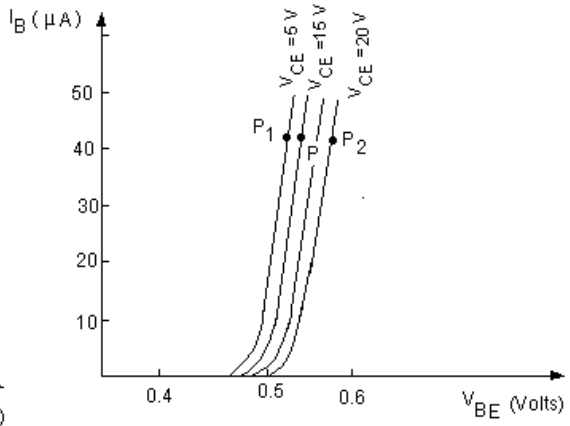


Fig. 7.4(b)

The method described above for the determination of *h* – parameters in CE configuration may also be used for evaluating these parameters in CB and CC configuration using their corresponding static characteristics.

7.3 Conversion of *h* – Parameters in Three Configurations: Sometimes *h* – parameters in one configuration is given and we are required to use the transistor in other configuration, so it becomes necessary to have the parameters in other configuration. For this one has to transform the parameters from one configuration to other configuration. Procedure for conversion of *h* – parameters from one configuration to other configuration is given as follows:

Step 1: Write the set of *h* – parameter equations in the given configuration.

Step 2: Draw the circuit model in the given configuration.

Step 3: Write the set of *h* – parameter equations in the required configuration.

Step 4: Redraw the circuit model in the required configuration.

Step 5: Using the circuit model obtained in step 4, we write some possible equations. By doing some manipulation in these equations, a set of two equations in the formats of required configuration is obtained.

Step 6: Comparing the equations (in step 5) with the *h* – parameter equations in the required configuration (in step 3), the required *h* – parameters are obtained in terms of the parameters of given configuration.

Examples 7.1 Convert the CB *h* – parameters in to CC *h* – parameters.
Solution: We write *h* – parameter equations in CB configuration as:

$$v_{eb} = h_{ib} i_e + h_{rb} v_{cb} \quad \text{------ (7.9)}$$

$$i_c = h_{fb} i_e + h_{ob} v_{cb} \quad \text{------ (7.10)}$$

We draw the *h* – parameter model in CB configuration (Fig. 7.5):

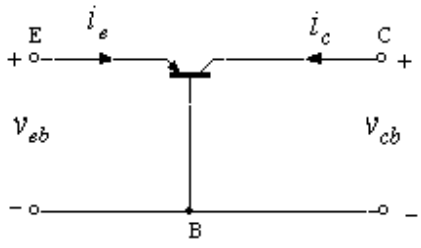


Fig. 7.5a

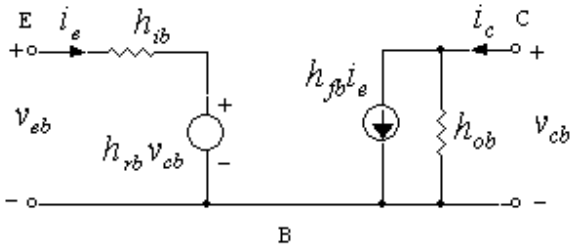


Fig.7.5b

We write *h* – parameter equations in CC configuration as:

$$v_{bc} = h_{ic} i_b + h_{rc} v_{ec} \quad \text{------ (7.11)}$$

$$i_e = h_{fc} i_b + h_{oc} v_{ec} \quad \text{------ (7.12)}$$

We redraw circuit model of fig. 7.5 in the CC configuration as (Fig. 7.6):

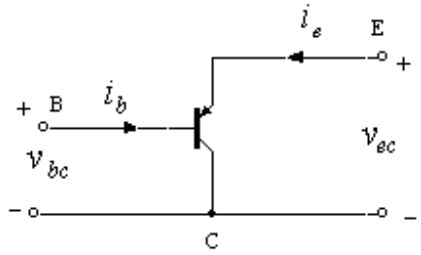


Fig. 7.6a

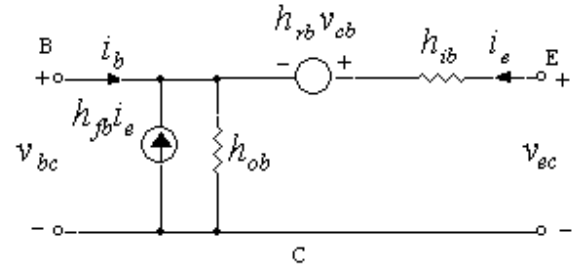


Fig.7.6b

From the figures (7.6 b) we get the following equations:

$$i_b + h_{fb} i_e + i_e - h_{ob} v_{bc} = 0$$

or $$i_b + (1 + h_{fb}) i_e - h_{ob} v_{bc} = 0 \quad \text{------ (7.13)}$$

$$v_{ec} = h_{ib} i_e + h_{rb} v_{cb} + v_{bc}$$

or $$v_{ec} = h_{ib} i_e + (1 - h_{rb}) v_{bc} \quad \text{------ (7.14)}$$

From equation (7.13) we get the value of i_e as:

$$i_e = \frac{h_{ob} v_{bc}}{(1 + h_{fb})} - \frac{i_b}{(1 + h_{fb})} \qquad \text{------ (7.15)}$$

From equations (7.14) & (7.15) we get:

$$v_{ec} = v_{bc}(1 - h_{rb}) + h_{ib}\left(\frac{h_{ob} v_{bc}}{(1 + h_{fb})} - \frac{i_b}{(1 + h_{fb})}\right)$$

or $$v_{bc} = \frac{h_{ib}}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]} i_b + \frac{(1 + h_{fb})}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]} v_{ec}$$

------ (7.17)

Putting the value of v_{bc} from equation (7.17) in equation (7.15) and doing some manipulation we get:

$$i_e = \frac{-(1 - h_{rb})}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]} i_b + \frac{h_{ob}}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]} v_{ec}$$

------ (7.18)

These are the two equations in the required configuration, the parameters in this configuration may be obtained by comparing these two equations with equations (7.11 & 7.12) we get:

$$h_{ic} = \frac{h_{ib}}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]} \qquad h_{rc} = \frac{(1 + h_{fb})}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]}$$

$$h_{fc} = \frac{-(1 - h_{rb})}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]} \qquad h_{oc} = \frac{h_{ob}}{[h_{ib} h_{ob} + (1 - h_{rb})(1 + h_{fb})]}$$

Example 7.2 Find the CC *h* – parameters in terms of CE *h* – parameters.

Solution: We write *h* – parameter equations in CE configuration as:

$$v_{be} = h_{ie} i_b + h_{re} v_{ce} \qquad \text{------ (7.19)}$$

$$i_c = h_{fe} i_b + h_{oe} v_{ce} \qquad \text{------ (7.20)}$$

We draw the *h* – parameter model in CE configuration:

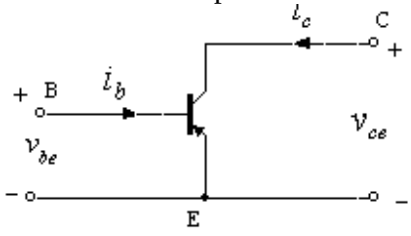


Fig. 7.7a

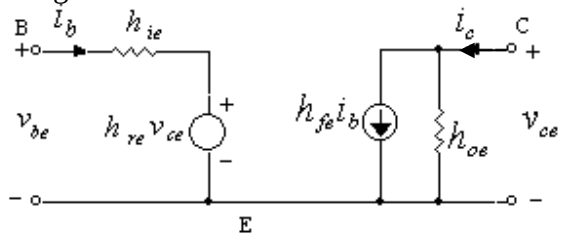


Fig.7.7b

We write *h* – parameter equations in CC configuration as:

$$v_{bc} = h_{ic} i_b + h_{rc} v_{ec} \qquad \text{------ (7.21)}$$

$$i_e = h_{fc} i_b + h_{oc} v_{ec} \qquad \text{------ (7.22)}$$

We redraw circuit model of fig. 7.7 in the CC configuration as:

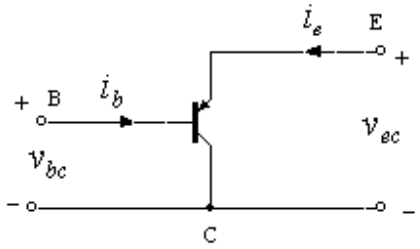


Fig. 7.8a

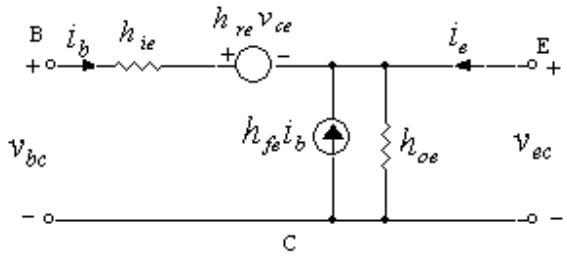


Fig.7.8b

From the figures (7.8 b) we get the following equations:

$$v_{bc} = h_{ie} i_b + h_{re} v_{ce} + v_{ec}$$

or $$v_{bc} = h_{ie} i_b + v_{ec} (1 - h_{re}) \qquad \text{------ (7.23)}$$

$$i_b + h_{fe} i_b + i_e - h_{oe} v_{ec} = 0$$

or $$(1 + h_{fe}) i_b + i_e - h_{oe} v_{ec} = 0 \qquad \text{------ (7.24)}$$

Comparing the equations (7.23) & (7.24) with equations (7.21) & (7.22) respectively, we get CC *h* – parameters in CE parameters.

$$h_{ic} = h_{ie} \qquad h_{rc} = (1 - h_{re})$$

$$h_{fc} = -(1 + h_{fe}) \qquad h_{oc} = h_{oe}$$

Examples 7.3 Convert the CC *h* – parameters in to CB *h* – parameters.

Solution: We write *h* – parameter equations in CC configuration as:

$$v_{bc} = h_{ic} i_b + h_{rc} v_{ec} \qquad \text{------ (7.25)}$$

$$i_e = h_{fc} i_b + h_{oc} v_{ec} \qquad \text{------ (7.26)}$$

We draw the *h* – parameter model in CC configuration:

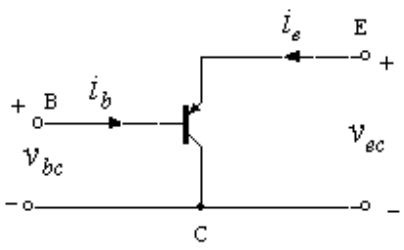


Fig. 7.9a

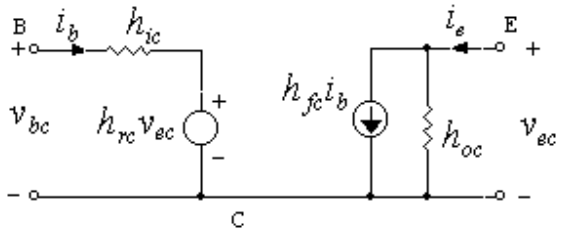


Fig.7.9b

We write h – parameter equations in CB configuration as:

$$v_{eb} = h_{ib} i_e + h_{rb} v_{cb} \qquad \text{------ (7.27)}$$

$$i_c = h_{fb} i_e + h_{ob} v_{cb} \qquad \text{------ (7.28)}$$

We redraw circuit model of fig. 7.9 in the CB configuration as:

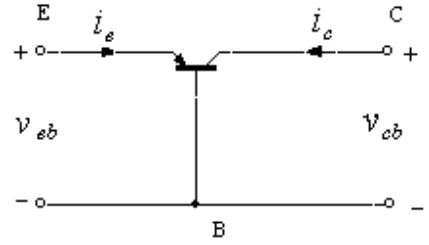


Fig. 7.10a

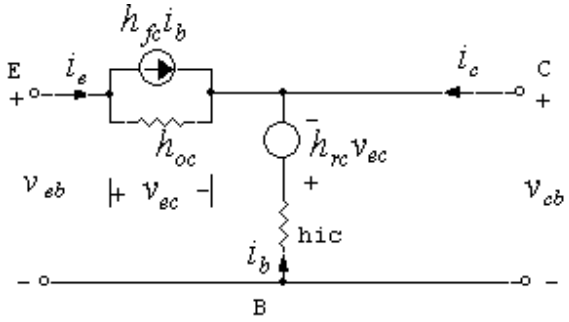


Fig.7.10b

From the figures (7.10 b) we get the following equations:

$$v_{eb} = v_{ec} - h_{rc} v_{ec} - h_{ic} i_b$$

or
$$v_{eb} = v_{ec}(1 - h_{rc}) - h_{ic} i_b \qquad \text{------ (7.29)}$$

$$i_e = h_{fc} i_b + h_{oc} v_{ec} \qquad \text{------ (7.30)}$$

$$v_{eb} = v_{ec} + v_{cb} \qquad \text{------ (7.31)}$$

$$i_e + i_b + i_c = 0 \qquad \text{------ (7.32)}$$

Eliminate the values of i_b and v_{ec} from equation (7.29) using equations (7.30) & (7.31) we get:

$$v_{eb} = \frac{h_{ic}}{[h_{ic} h_{oc} - h_{rc} h_{fc}]} i_e + \frac{(1 - h_{rc}) h_{fc} + h_{ic} h_{oc}}{[h_{ic} h_{oc} - h_{rc} h_{fc}]} v_{cb} \qquad \text{------ (7.33)}$$

Using the equations (7.30) to (7.33) and doing some manipulations we get:

$$i_c = \frac{h_{rc}(1 + h_{fc}) - h_{ic} h_{oc}}{[h_{ic} h_{oc} - h_{rc} h_{fc}]} i_e + \frac{h_{oc}}{[h_{ic} h_{oc} - h_{rc} h_{fc}]} v_{cb} \qquad \text{------ (7.34)}$$

Comparing the equations (7.33) & (7.34) with equations (7.27) & (7.28) respectively, we get CB *h* – parameters in CC parameters.

$$h_{ib} = \frac{h_{ic}}{[h_{ic}h_{oc} - h_{rc}h_{fc}]} \qquad h_{rb} = \frac{(1 - h_{rc})h_{fc} + h_{ic}h_{oc}}{[h_{ic}h_{oc} - h_{rc}h_{fc}]}$$

$$h_{fb} = \frac{h_{rc}(1 + h_{fc}) - h_{ic}h_{oc}}{[h_{ic}h_{oc} - h_{rc}h_{fc}]} \qquad h_{ob} = \frac{h_{oc}}{[h_{ic}h_{oc} - h_{rc}h_{fc}]}$$

Conversions for h – parameters in three configurations are given in table 7.1.

Table 7.1

CB *h* – parameters	
In terms of CE parameters	In terms of CC parameters
$h_{ib} = \frac{h_{ie}}{(1+h_{fe})(1-h_{re}) + h_{ie}h_{oe}}$ $h_{rb} = \frac{h_{ie}h_{oe} - h_{re}(1+h_{fe})}{(1+h_{fe})(1-h_{re}) + h_{ie}h_{oe}}$ $h_{fb} = \frac{-h_{fe}(1-h_{re}) - h_{ie}h_{oe}}{(1+h_{fe})(1-h_{re}) + h_{ie}h_{oe}}$ $h_{ob} = \frac{h_{oe}}{(1+h_{fe})(1-h_{re}) + h_{ie}h_{oe}}$	$h_{ib} = \frac{h_{ic}}{h_{ic}h_{oc} - h_{fc}h_{rc}}$ $h_{rb} = \frac{h_{fc}(1-h_{rc}) + h_{ic}h_{oc}}{h_{ic}h_{oc} - h_{fc}h_{rc}}$ $h_{fb} = \frac{h_{rc}(1+h_{fc}) - h_{ic}h_{oc}}{h_{ic}h_{oc} - h_{fc}h_{rc}}$ $h_{ob} = \frac{h_{oc}}{h_{ic}h_{oc} - h_{fc}h_{rc}}$
CE *h* – parameters	
In terms of CB parameters	In terms of CC parameters
$h_{ie} = \frac{h_{ib}}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}}$ $h_{re} = \frac{h_{ib}h_{ob} - h_{rb}(1+h_{fb})}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}}$ $h_{fe} = \frac{-h_{fb}(1-h_{rb}) - h_{ib}h_{ob}}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}}$ $h_{oe} = \frac{h_{ob}}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}}$	$h_{ie} = h_{ic}$ $h_{re} = (1 - h_{rc})$ $h_{fe} = -(1 + h_{fc})$ $h_{oe} = h_{oc}$
CC *h* – parameters	
In terms of CB parameters	In terms of CE parameters
$h_{ic} = \frac{h_{ib}}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}}$	$h_{ic} = h_{ie}$

$h_{rc} = \dfrac{1+h_{fb}}{(1+h_{fb})(1-h_{rb})+h_{ib}h_{ob}}$ $h_{fc} = \dfrac{-(1-h_{rb})}{(1+h_{fb})(1-h_{rb})+h_{ib}h_{ob}}$ $h_{oc} = \dfrac{h_{ob}}{(1+h_{fb})(1-h_{rb})+h_{ib}h_{ob}}$	$h_{rc} = (1-h_{re})$ $h_{fc} = -(1+h_{fe})$ $h_{oc} = h_{oe}$

The h – parameters of actual transistors are such that some approximation will greatly simplify the conversion formulas with reasonable accuracy. Generally following approximations are made.

$h_i h_o << 1$ $\quad h_{rc} \cong 1$ $\quad h_{rb} << 1$ $\quad$ and $\quad h_{re} << 1$

Approximate formulas for the three configurations are given in table 7.2.

Table 7.2

CB h – parameters	
In terms of CE parameters	In terms of CC parameters
$h_{ib} = \dfrac{h_{ie}}{1+h_{fe}}$ $h_{rb} = \dfrac{h_{ie}h_{oe}}{1+h_{fe}} - h_{re}$ $h_{fb} = \dfrac{-h_{fe}}{1+h_{fe}}$ $h_{ob} = \dfrac{h_{oe}}{1+h_{fe}}$	$h_{ib} = -\dfrac{h_{ic}}{h_{fc}}$ $h_{rb} = h_{rc} - \dfrac{h_{ic}h_{oc}}{h_{fc}} - 1$ $h_{fb} = -\dfrac{1+h_{fc}}{h_{fc}}$ $h_{ob} = -\dfrac{h_{oc}}{h_{fc}}$
CE h – parameters	
In terms of CB parameters	In terms of CC parameters
$h_{ie} = \dfrac{h_{ib}}{1+h_{fb}}$ $h_{re} = \dfrac{h_{ib}h_{ob}}{1+h_{fb}} - h_{ib}$ $h_{fe} = \dfrac{-h_{fb}}{1+h_{fb}}$ $h_{oe} = \dfrac{h_{ob}}{1+h_{fb}}$	$h_{ie} = h_{ic}$ $h_{re} = (1-h_{rc})$ $h_{fe} = -(1+h_{fc})$ $h_{oe} = h_{oc}$
CC h – parameters	
In terms of CB parameters	In terms of CE parameters

$h_{ic} = \dfrac{h_{ib}}{1+h_{fb}}$	$h_{ic} = h_{ie}$
$h_{rc} = 1$	$h_{rc} = 1$
$h_{fc} = \dfrac{-1}{1+h_{fb}}$	$h_{fc} = -(1 + h_{fe})$
$h_{oc} = \dfrac{h_{ob}}{1+h_{fb}}$	$h_{oc} = h_{oe}$

7.4 An Analysis of Transistor Amplifier: To analyse the transistor amplifier, we consider a transistor in any of the three configurations. To the input of the amplifier, a signal source V_s having R_s as the source resistance is applied, and a load impedance Z_L is connected to its output. The transistor is however biased properly. The amplifier is replaced by a two port network (active) as shown in figure (7.11) without bothering about in which mode the transistor is used. The quantities of our interest, regarding the transistor amplifier are current gain $A_I = \dfrac{I_L}{I_1} = \dfrac{-I_2}{I_1}$, voltage gain $A_V = \dfrac{V_2}{V_1}$, input impedance $Z_I = \dfrac{V_1}{I_1}$ and output impedance $Z_O = \dfrac{V_2}{I_2}$. To find these quantities we represent the transistor into its equivalent small signal hybrid model as shown in figure (7.12). Here we are considering the general case of the amplifier, so no second suffix to

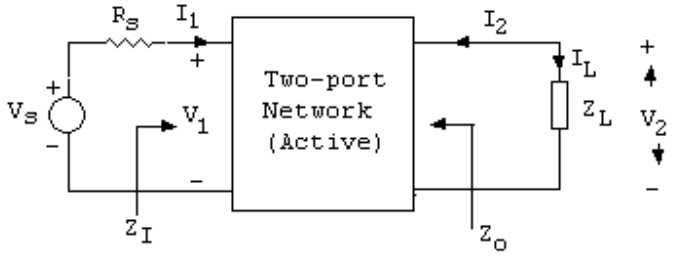


Fig. 7.11

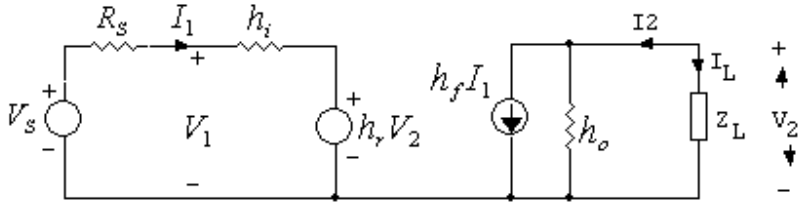


Fig. 7.12

the *h* – parameters are introduced. Later on, we will introduce the suffix e, b, or c for CE, CB and CC configurations respectively, as the case may be. We shall now calculate these quantities as given below.

(i) Current Gain A_I : The current gain A_I is defined as the ration of output current to the input current as $A_I = \frac{I_L}{I_1} = \frac{-I_2}{I_1}$

From the circuit of figure (7.12), we have :

$$I_2 = h_f I_1 + h_o V_2 \qquad \text{------ (7.35)}$$

Putting $V_2 = I_L Z_L = -I_2 Z_L$ in the above equation we get:

$$I_2 = h_f I_1 - h_o I_2 Z_L$$

or $$I_2(1 + h_o Z_L) = h_f I_1$$

or $$A_I = \frac{-I_2}{I_1} = \frac{-h_f}{(1 + h_o Z_L)} \qquad \text{------ (7.36)}$$

Note that the current gain of the amplifier depends on the h – parameters of the transistor as well as on the load impedance Z_L. The maximum current gain of the amplifier is $|h_f|$ which is obtained when $h_o Z_L \to 0$.

(ii) Input Impedance Z_I: The impedance looking into the input terminals of the amplifier is known as input impedance $Z_I = \frac{V_1}{I_1}$.

From the input circuit of figure (7.12), we have

$$V_1 = h_i I_1 + h_r V_2$$

or $$Z_I = \frac{V_1}{I_1} = \frac{(h_i I_1 + h_r V_2)}{I_1} = h_i + \frac{h_r V_2}{I_1}$$

Putting $V_2 = -I_2 Z_L$ in the above equation we get:

$$Z_I = h_i - \frac{h_r I_2 Z_L}{I_1} = h_i + h_r A_I Z_L$$

Substituting the value of A_I from equation (7.36), we have

$$Z_I = h_i - \frac{h_r h_f Z_L}{(1 + h_o Z_L)} = h_i - \frac{h_f h_r}{(Y_L + h_o)} \qquad \text{------ (7.37)}$$

Where $Y_L = (1/Z_L)$, is the load admittance. It is clear from the above equation that the input impedance is not only a function of *h* – parameters of the transistor but also depends on the load impedance.

(iii) Voltage Gain A_V : The voltage gain A_V is defined as $A_V = \frac{V_2}{V_1}$ which is given by:

$$A_V = \frac{-I_2 Z_L}{V_1} = \frac{A_I Z_L}{Z_I} \qquad \text{------ (7.38)}$$

(iv) Output Impedance Z_o : By definition the output impedance is $Z_O = \frac{V_2}{I_2}$, with source replaced by its internal resistance R_s and $Z_L = \infty$.
From equation (7.17) we have the output admittance Y_o

$$Y_o = \frac{I_2}{V_2} = h_f \frac{I_1}{V_2} + h_o \qquad \text{------ (7.39)}$$

From figure (7.12), with $V_s = 0$ we have

$$R_s I_1 + h_i I_1 + h_r V_2 = 0$$

or

$$\frac{I_1}{V_2} = -\frac{h_r}{h_i + R_s} \qquad \text{------ (7.40)}$$

Putting the value of $\frac{I_1}{V_2}$ from equation (7.40) to equation (7.39), we get

$$Y_o = h_o - \frac{h_f h_r}{(h_i + R_s)} \qquad \text{------ (7.41)}$$

Note that output impedance is a function of the source resistance in addition to the h – parameters of the transistor. Further if source resistance is resistive than the output admittance will be real (conductance). If the output impedance of the amplifier Z_L is included than output impedance will be parallel combination of Z_L and Z_o.

In addition to the formulas derived above, it is important to find the overall voltage gain and overall current gain also.

(v) Overall Voltage Gain A_{vs} : The overall voltage gain A_{vs}, taking into account the resistance R_s of the source is given by:

$$A_{VS} = \frac{V_2}{V_S} = \frac{V_2}{V_1} . \frac{V_1}{V_S} = A_V . \frac{V_1}{V_S}$$

From figure (7.12),

$$V_1 = \frac{V_S}{R_S + Z_I} . Z_I$$

Then

$$A_{VS} = \frac{A_V}{R_S + Z_I} . Z_I = \frac{A_I Z_L}{R_S + Z_I} \qquad \text{------ (7.42)}$$

Note that if $R_S = 0$, then $A_{VS} = A_V$.

(vi) Overall Current Gain A_{IS} : If the input source is a current source I_S in parallel with the source resistance R_S (i.e. Norton's equivalent of the voltage source), then the overall current gain A_{IS} is given by:

$$A_{IS} = -\frac{I_2}{I_S} = \frac{-I_2}{I_1}.\frac{I_1}{I_S} = A_I \frac{I_1}{I_S}$$

Using the Norton's equivalent of the voltage source V_S in the figure (7.12), we get:

$$I_1 = \frac{R_S}{R_S + Z_I} I_S$$

Then $$A_{IS} = \frac{A_I R_S}{R_S + Z_I}$$

Note that if $R_S = 0$, then $A_{IS} = A_I$. Also we have

$$A_{VS} = \frac{A_{IS} Z_L}{R_S}$$

The formulas derived above are summerised in table 7.3.

Table 7.3

$A_I = \frac{-h_f}{(1+h_o Z_L)}$	$Y_o = h_o - \frac{h_f h_r}{(h_i + R_s)} = \frac{1}{Z_0}$
$Z_I = h_i + h_r A_I Z_L$	$A_{VS} = \frac{A_V}{R_S + Z_I}.Z_I = \frac{A_I Z_L}{R_S + Z_I}$
$A_V = \frac{A_I Z_L}{Z_I}$	$A_{IS} = \frac{A_I R_S}{R_S + Z_I}$

Example 7.4 Prove for any single transistor amplifier

$$Z_I = \frac{h_i}{1 - h_r A_V}$$

Solution: We know $Z_I = h_i + h_r A_I Z_L$ ------- (7.43)

and $$A_V = \frac{A_I Z_L}{Z_I}$$ ------ (7.44)

From the equations (7.43) & (7.44) we have:

$$Z_I = h_i + h_r A_V Z_I$$

$$Z_I (1 - A_V Z_I) = h_i$$

or $$Z_I = \frac{h_i}{1 - h_r A_V}$$ Proved

Example 7.5 Prove that the output admittance of the transistor may be given in the following form:

$$Y_o = h_o \left(\frac{R_s + Z_{I\infty}}{R_s + Z_{I0}} \right)$$

where $Z_{I\infty} = Z_I$ for $Z_L = \infty$ and $Z_{Io} = Z_I$ for $Z_L = 0$.

Solution: We know that

$$Y_o = h_o - \frac{h_f h_r}{(h_i + R_s)}$$ ------ (7.45)

$$Z_I = h_i + h_r A_I Z_L$$ ------ (7.46)

$$A_I = \frac{-h_f}{(1 + h_o Z_L)}$$ ------ (7.47)

From equations (7.46) & (7.47) have:

$$Z_I = h_i - \frac{h_f h_r Z_L}{(1 + h_o Z_L)} = h_i - \frac{h_f h_r}{(h_o + \frac{1}{Z_L})}$$

For $Z_L = 0$ $Z_I = Z_{I0}$ so $Z_I = h_i$

For $Z_L = \infty$ $Z_I = Z_{I\infty}$ so $Z_{I\infty} = h_i - \frac{h_f h_r}{h_o}$

Now from eq. (7.45) $$Y_o = \frac{h_o (h_i + R_s) - h_f h_r}{(h_i + R_s)}$$

or $$Y_o = \frac{h_o \left[h_i + R_s - \frac{h_f h_r}{h_o} \right]}{(h_i + R_s)}$$

or $$Y_o = h_o\left(\frac{R_s + Z_{I\infty}}{R_s + Z_{I0}}\right)$$ proved

Example 7.6 For the circuit shown in figure (7.13), verify that the modified h – parameters (indicated by h') are given by:

(i) $h'_{ie} \approx h_{ie} + \dfrac{(1+h_{fe})R_e}{(1+h_{oe}R_e)}$ (ii) $h'_{re} = \dfrac{h_{re} + h_{oe}R_e}{(1+h_{oe}R_e)}$

(iii) $h'_{fe} = \dfrac{h_{fe} - h_{oe}R_e}{(1+h_{oe}R_e)}$ (iv) $h'_{oe} = \dfrac{h_{oe}}{(1+h_{oe}R_e)}$

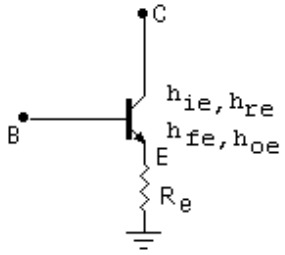


Fig. 7.13

Solution: The h – parameter model of the circuit is shown in figure (7.14)

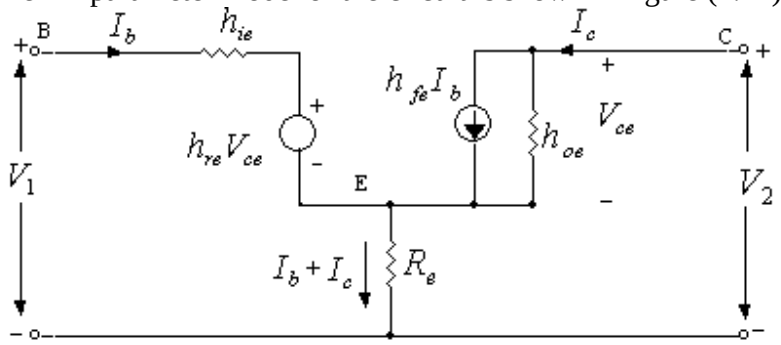


Fig. 7.14

The h – parameter equations of this network is given as :

$$V_1 = h'_{ie}I_b + h'_{re}V_2$$

$$I_C = h'_{fe}I_b + h'_{oe}V_2$$

where $$h'_{ie} = \left.\frac{V_1}{I_b}\right|_{V_2=0} \qquad h'_{re} = \left.\frac{V_1}{V_2}\right|_{I_b=0}$$

$$h'_{fe} = \left.\frac{I_c}{I_b}\right|_{V_2=0} \qquad h'_{oe} = \left.\frac{I_c}{V_2}\right|_{I_b=0}$$

For $$h_{fe}^{'} = \left.\frac{I_c}{I_b}\right|_{V_2=0}$$

Applying the KVL & KCL to the input and output circuits we get:

$$V_1 = h_{ie} I_b + h_{re} V_{ce} + I_b R_e + I_c R_e \quad \text{------ (7.48)}$$

$$V_2 = V_{ce} + R_e (I_b + I_c) \quad \text{------ (7.49)}$$

$$I_c = h_{fe} I_b + h_{oe} V_{ce} \quad \text{------ (7.50)}$$

From Eq. (7.50) $$V_{ce} = \frac{I_c - h_{fe} I_b}{h_{oe}} \quad \text{------ (7.51)}$$

Since $V_2 = 0$ So eq. (7.49) reduces to

$$V_{ce} = -R_e (I_b + I_c) \quad \text{or} \quad I_c = -\frac{V_{ce} + R_e I_b}{R_e} \quad \text{------ (7.52)}$$

From eqs. (7.51) & (7.52) we have

$$\frac{I_c - h_{fe} I_b}{h_{oe}} = -(I_b + I_c) R_e$$

or $$I_c \left(\frac{1}{h_{oe}} + R_e\right) = -I_b R_e + \frac{h_{fe} I_b}{h_{oe}}$$

or $$\frac{I_c}{I_b} = \frac{h_{fe} - R_e h_{oe}}{(1 + R_e h_{oe})}$$

or $$h_{fe}^{'} = \frac{I_c}{I_b} = \frac{h_{fe} - R_e h_{oe}}{(1 + R_e h_{oe})} \quad \text{------- (7.53)}$$

Note that part (iii) is proved.

For $$h_{ie}^{'} = \left.\frac{V_1}{I_b}\right|_{V_2=0}$$

From equations (7.48) & (7.53) we have:

$$V_1 = h_{ie} I_b + h_{re} V_{ce} + I_b R_e + R_e \left(\frac{h_{fe} - R_e h_{oe}}{1 + R_e h_{oe}}\right) I_b$$

or $$V_1 = I_b \left(h_{ie} + R_e \left\{ \frac{1 + R_e h_{oe} + h_{fe} - R_e h_{oe}}{(1 + R_e h_{oe})} \right\} \right) + h_{re} V_{ce}$$

or $$V_1 = I_b \left(h_{ie} + R_e \left\{ \frac{1 + h_{fe}}{(1 + R_e h_{oe})} \right\} \right) + h_{re} V_{ce}$$

Neglecting $V_{ce}h_{re}$, we have

$$V_1 \approx I_b\left(h_{ie} + R_e\left\{\frac{1+h_{fe}}{(1+R_e h_{oe})}\right\}\right)$$

or $$h_{ie}^{'} = \frac{V_1}{I_b} \approx \left(h_{ie} + R_e\left\{\frac{1+h_{fe}}{(1+R_e h_{oe})}\right\}\right) \quad \text{------ (7.54)}$$

Note that part (i) is proved.

For $$h_{re}^{'} = \left.\frac{V_1}{V_2}\right|_{I_b=0}$$

Putting $I_b = 0$ in Equations (7.48) to (7.50) we have:

$$V_1 = h_{re}V_{ce} + I_c R_e \quad \text{------ (7.55)}$$

$$V_2 = V_{ce} + R_e I_c \quad \text{------ (7.56)}$$

$$I_c = h_{oe}V_{ce} \quad \text{------ (7.57)}$$

From eqs. (7.55) to (7.57)

$$V_1 = h_{re}V_{ce} + R_e h_{oe}V_{ce}$$

$$V_1 = (h_{re} + R_e h_{oe})V_{ce}$$

and $$V_2 = V_{ce}(1+R_e h_{oe}) \quad \text{------ (7.58)}$$

or $$h_{re}^{'} = \frac{V_1}{V_2} = \frac{h_{re} + h_{oe}R_e}{(1+h_{oe}R_e)} \quad \text{------ (7.59)}$$

Note that part (ii) is proved.

For $$h_{oe}^{'} = \left.\frac{I_c}{V_2}\right|_{I_b=0}$$

From eqs. (7.57) & (7.55)

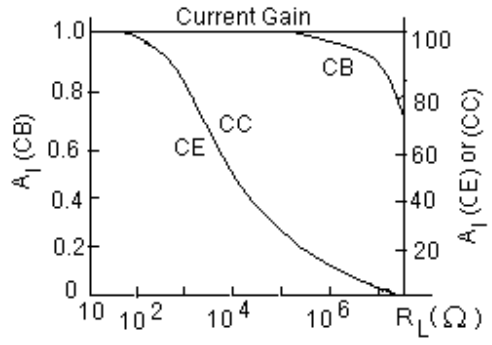
$$h_{oe}^{'} = \frac{I_c}{V_2} = \frac{h_{oe}}{(1+h_{oe}R_e)} \quad \text{----- (7.60)}$$

Part (iv) is proved**.**

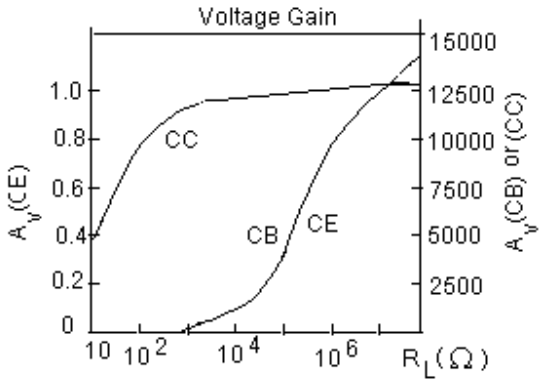
7.5 Comparison of Transistor Amplifier Configurations: Typical values of *h* – parameters of a transistor in the three configurations are given in the Table 7.4. Using these values of *h* – parameters in the formulas given in table 7.4, the current gain, voltage gain, input impedances were calculated as functions of load impedances; and output impedance as the function of source resistance for the three configurations. The variations of current gain, voltage gain, input impedance and output impedance as functions of load impedance and source resistance are shown in figure 7.15.

Table 7.4

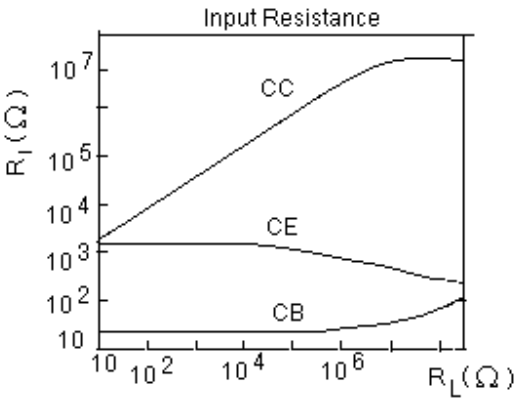
Parameter	Common Emitter CE	Common Base CB	Common Collector CC
h_i	h_{ie} = 2600 Ω	h_{ib} = 25.8 Ω	h_{ic} = 2600 Ω
h_f	h_{fe} = 100	h_{fb} = – 0.99	h_{fc} = – 101
h_r	h_{re} = 0.62x10 $^{-4}$	h_{rb} =0.67x10 $^{-4}$	h_{rc} = 1
h_o	h_{oe} = 5 μ-mhos	h_{ob} = 0.05 μ-mhos	h_{oc} = 5 μ-mhos



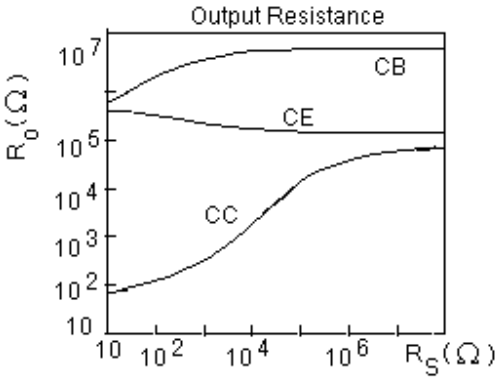
(a)



(b)



(c)



(d)

Fig. 7.15

On the careful study of these curves, the characteristics of the three configurations may be summarised as shown in table 7.5.

Table 7.5

Quantity	CB	CE	CC
Current gain (A_I)	≈ 1	High	High
Voltage gain (A_V)	High	High	≈ 1

Input impedance (Z_I)	Low	Medium	High
Output impedance (Z_o)	High	Medium	Low

From this table, we get the following inferences regarding the three transistor configurations.

1. In case of CE amplifier both voltage gain and current gain are high. Generally speaking, we get high power gain also. The input and output impedance are of medium range, so the CE amplifier may be called a general purpose amplifier. The middle stages of a multistage amplifier are usually designed using this configuration.

2. The CC configuration has the high input impedance and low output impedance; its voltage gain is nearly equal to unity. So such an amplifier is used as a buffer amplifier between a source of high source resistance (poor source) and a low impedance load. Generally a source of high source resistance does not deliver much current to the load impedance as load impedance loads the source. The common collector configuration overcomes this difficulty as high input impedance of this configuration may properly be matched with the source resistance; thus it gives the output signal of almost the same magnitude as the input having the low output impedance. The low output impedance of CC amplifier will be properly matched with the low impedance load. The common collector configuration thus transforms the source of high output impedance to a source of low output impedance having the signal of almost same magnitude.

3. The common base configuration has the current gain of nearly unity, its input impedance is low and output impedance is high. So CB configuration may be used as buffer amplifier between a current source of low source resistance and high load impedance. It therefore transforms the source of low output impedance to a source of high output impedance having the current of almost same magnitude.

7.6 Miller's Theorem: This theorem states that an impedance Z shunting an active network having a known voltage gain A_V may be eliminated by connecting impedance $Z_1 = \frac{Z}{(1-A_V)}$ and another impedance $Z_2 = \frac{ZA_V}{(A_V - 1)}$ across the input and the output terminals respectively. This is illustrated in figure (7.16).

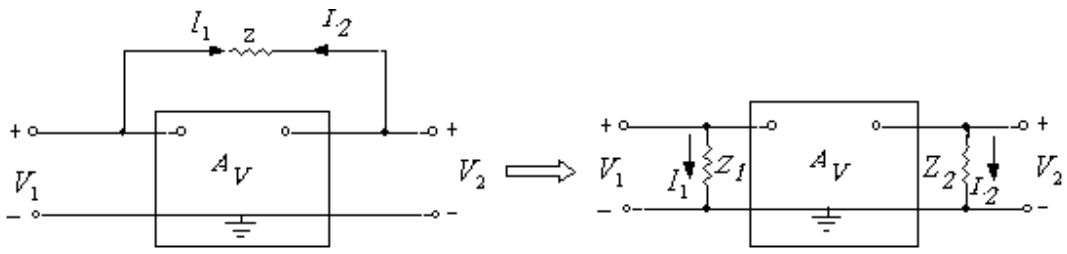


Fig. 7.16(a) Fig. 7.16(b)

This can be proved by considering fig.(7.16a). The current I_1 from this figure is given by:

$$I_1 = \frac{V_1 - V_2}{Z} \quad \text{------ (7.61)}$$

and the voltage V_2 is given by: $V_2 = A_V . V_1$ ------ (7.62)

From equations (7.61) & (7.62) we have:

$$I_1 = \frac{V_1 - A_V V_1}{Z} = \frac{V_1(1 - A_V)}{Z} \quad \text{----- (7.63)}$$

From figure (7.16 b) the current I_1 is given by $I_1 = \frac{V_1}{Z_1}$ ----- (7.64)

From the equations (7.63) and (7.64), it is clear that the two networks given in figures (7.16a) & (7.16b) will be identical when $Z_1 = \frac{Z}{(1 - A_V)}$

Similarly, from fig.(7.16a):

$$I_2 = \frac{V_2 - V_1}{Z} \quad \text{and} \quad V_1 = . \frac{V_2}{A_V}$$

or $$I_2 = \frac{V_2 - \frac{V_2}{A_V}}{Z} = \frac{V_2(1 - \frac{1}{A_V})}{Z} \quad \text{------ (7.65)}$$

From fig. (7.16b) $I_2 = \frac{V_2}{Z_2}$ ------ (7.66)

From the equations (7.65) and (7.66), it is clear that the two networks given in figures (7.16a) & (7.16b) will be identical when $Z_2 = \frac{ZA_V}{(A_V - 1)}$

Hence the theorem is proved.

7.7 Dual of Miller's Theorem: It states that an impedance *Z*, in series with both the input and the output terminals of an active network having a known current source A_I may be eliminated by adding impedance $Z_1 = Z(1 - A_I)$ and another impedance $Z_2 = \frac{Z(A_I - 1)}{A_I}$ in series with the input and the output terminals respectively of the active network. This may be illustrated in figure (7.17).

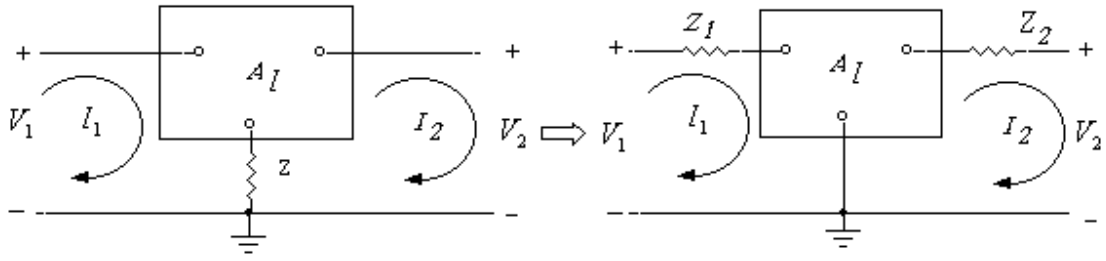


Fig. 7.17(a) Fig. 7.17(b)

From figure (7.17a) the voltage across impedance Z is

$$Z(I_1 + I_2) = Z(I_1 - A_I I_1) = ZI_1(1 - A_I) \qquad \text{------ (7.67)}$$

where $A_I = -\dfrac{I_2}{I_1}$

From figure (7.17b) the voltage across impedance Z_1 is $Z_1 I_1$

These two voltages will be equal when $Z_1 = Z(1 - A_I)$

In the similar fashion it can be proved that $Z_2 = \dfrac{Z(A_I - 1)}{A_I}$

7.8 The Emitter Follower: As discussed above that the common collector amplifier has very high input impedance and very low output impedance and its voltage gain is almost unity. So this amplifier is used for impedance transformation. The practical circuit of the common collector amplifier is shown in figure (7.18), in which the collector is

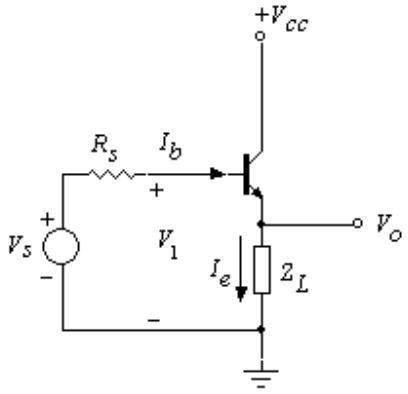


Fig. 7.18

directly connected to the supply voltage and a load impedance Z_L is connected to the emitter, and the output is taken across the load impedance. This circuit seems more as the common emitter amplifier than the common collector amplifier, but this is the common collector amplifier. Infect when the a.c. analysis of a network is made the d.c. source is assumed to be short circuited, this leads the collector to be shorted to the ground. Hence it is a common collector amplifier. The common collector amplifier is also known as the emitter follower, because its voltage is gain almost unity, any change in the base voltage will appear across the load impedance. The emitter (output) follows the input signal and hence the name emitter follower.

The expressions for current gain A_I, input impedance Z_I, voltage gain A_V and the output impedance Z_o are to be calculated. While calculating these expressions the common collector parameters of the transistor are to be used, but some times the common emitter *h* – parameters are given. So conversions of CE *h* – parameters to CC h parameters are to be made as shown in table 7.1. These expressions are therefore given below in terms of both common collector and common emitter h – parameters.

Current Gain $$A_I = \frac{-I_e}{I_b} = \frac{-h_{fc}}{1+h_{oc}Z_L} = \frac{1+h_{fe}}{1+h_{oe}Z_L}$$

Input Impedance $$Z_I = \frac{V_1}{I_b} = h_{ic} + h_{rc}A_I Z_L = h_{ie} + A_I Z_L$$

Voltage Gain $$A_V = \frac{V_o}{V_1} = \frac{A_I Z_L}{Z_I} = 1 - \frac{h_{ie}}{Z_I}$$

The voltage gain is slightly less than one and no phase reversal between input and output signal.

Output Admittance $$Y_o = h_{oc} - \frac{h_{rc}h_{fc}}{h_{ic}+R_s} = h_{oe} + \frac{1+h_{fe}}{h_{ie}+R_s}$$

Example 7.7 A CE transistor amplifier is driven by a signal source of source resistance 500 Ω. The load impedance is 2 KΩ. The h –parameters values are given in the table 7.4. Calculate the current gain, input impedance, voltage gain and output impedance of the amplifier. Find the overall voltage gain and current gain also.

Solution:

$$A_I = \frac{-h_{fe}}{(1+h_{oe}Z_L)} = \frac{-100}{(1+5x10^{-6}x2x10^3)} = \frac{-100}{1.01} = -99$$

$$Z_I = h_{ie} + h_{re}A_I Z_L = 2600 + 0.62x10^{-4}x(-99)(2x10^3)$$

$$= 2600 - 12.276 = 2587.7\Omega = 2.59K\Omega$$

$$A_V = A_I\frac{Z_L}{Z_I} = (-99)\frac{2K\Omega}{2.59K\Omega} = -76.45$$

$$Y_o = h_{oe} - \frac{h_{fe}h_{re}}{(h_{ie}+R_s)} = 5x10^{-6} - \frac{(100)(.62x10^{-4})}{(2600+500)}$$

$$= 5x10^{-6} - 2x10^{-6} = 3x10^{-6}\,mhos$$

$$Z_o = \frac{1}{Y_o} = \frac{1}{3x10^{-6}} = 333.33K\Omega$$

$$A_{VS} = \frac{A_I Z_L}{R_S + Z_I} = \frac{(-99)(2x10^3)}{(500+2587.7)} = -64.125$$

$$A_{IS} = \frac{A_I R_S}{R_S + Z_I} = \frac{(-99)(500)}{(500+2587.7)} = -16.03$$

Example 7.8 A CB transistor amplifier is driven by a signal source of source resistance 500 Ω. The load impedance is 2 KΩ. The h –parameters values are given in the table 7.4. Calculate the current gain, input impedance, voltage gain and output impedance of the amplifier. Find the overall voltage gain and current gain also.

Solution:

$$A_I = \frac{-h_{fb}}{(1+h_{ob}Z_L)} = \frac{-(-.99)}{(1+0.05x10^{-6}x2x10^3)} \approx 0.99$$

$$Z_I = h_{ib} + h_{rb}A_I Z_L = 25.8 + 0.67x10^{-4}x(0.99)(2x10^3)$$

$$= 25.8 - 0.66 = 25.73\Omega$$

$$A_V = A_I \frac{Z_L}{Z_I} = (0.99)\frac{2K\Omega}{25.73\Omega} = 76.95$$

$$Y_o = h_{ob} - \frac{h_{fb}h_{rb}}{(h_{ib} + R_s)} = 0.05\,x10^{-6} - \frac{(-0.99)(.67\,x10^{-4})}{(25.8 + 500)}$$

$$= 0.05\,x10^{-6} + .126\,x10^{-6} = 0.176\,x10^{-6}\,mhos$$

$$Z_o = \frac{1}{Y_o} = \frac{1}{0.176x10^{-6}} = 5.68M\Omega$$

$$A_{VS} = \frac{A_I Z_L}{R_S + Z_I} = \frac{(0.99)(2x10^3)}{(500 + 25.73)} = 3.77$$

$$A_{IS} = \frac{A_I R_S}{R_S + Z_I} = \frac{(0.99)(500)}{(500 + 25.73)} = 0.942$$

Example 7.9 Design a single stage buffer amplifier which is to be used with a transducer having a source resistance of 12 KΩ. The amplifier should have input impedance greater than 100 KΩ when connected to the load impedance of 1 KΩ. The h – parameters of the transistor used are given in the table 7.4. Calculate the current gain, overall current gain, input impedance, voltage gain, overall voltage gain and output impedance of the amplifier thus designed.

Solution: From the above problem it is clear that the amplifier to be designed should have high input impedance (greater than 100 KΩ) and low output impedance (lower than 1 KΩ). The CC amplifier has such properties, so we are to design the CC amplifier. Now we calculate the various gain and input and output impedance as follows:

$$A_I = \frac{-h_{fc}}{(1 + h_{oc}Z_L)} = \frac{-(-101)}{(1 + 5x10^{-6}\,x1x10^3)} = \frac{101}{1.005} = 100.5$$

$$Z_I = h_{ic} + h_{rc}A_I Z_L = 2600 + 1x(100.5)(1x10^3)$$

$= 2600+ = 2587.7\Omega = 103.1K\Omega$, which is greater than 100 KΩ.

$$A_V = A_I \frac{Z_L}{Z_I} = 100.5\frac{1K\Omega}{103.1K\Omega} = 0.975$$

$$Y_o = h_{oc} - \frac{h_{fc}h_{rc}}{(h_{ic} + R_s)} = 5\,x10^{-6} - \frac{(-101)\,x1}{(2600 + 12000)}$$

$$= 5\,x10^{-6} + 0.0069 = 6.9\,x10^{-3}\,mhos$$

$$Z_o = \frac{1}{Y_o} = \frac{1}{6.9x10^{-3}} = 144.9\Omega$$, which is less than 1 KΩ

$$A_{VS} = \frac{A_I Z_L}{R_S + Z_I} = \frac{(100.5)(1x10^3)}{(12K + 103.1K)} = 0.87$$

$$A_{IS} = \frac{A_I R_S}{R_S + Z_I} = \frac{(100.5)(12\,K)}{(12\,K + 103.1K)} = 10.48$$

7.9 Cascaded Transistor Amplifier: We have studied that the three modes of the transistor are used to design the single stage amplifiers. But sometimes single stage amplifier does not provide the amplification up to the desired level or the input/ output impedances do not properly match. In such cases it becomes necessary to cascade the transistor amplifiers. The cascading we mean that the output of the first stage is connected to the input of the second stage. More than two stages may be connected as per our requirement. Several types of cascaded stages may be used for different purposes. A few of them are being discussed here.

CE – CC Cascaded Amplifier: Figure (7.19) shows the CE –CC cascaded amplifier, in which first stage is the common emitter stage and the second stage is the common

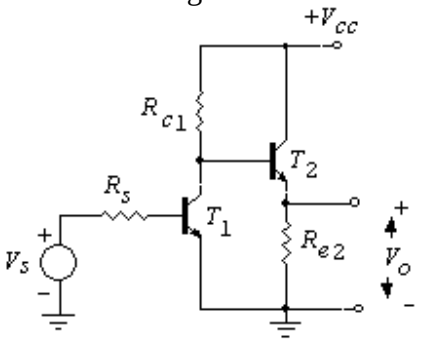


Fig. 7.19 a

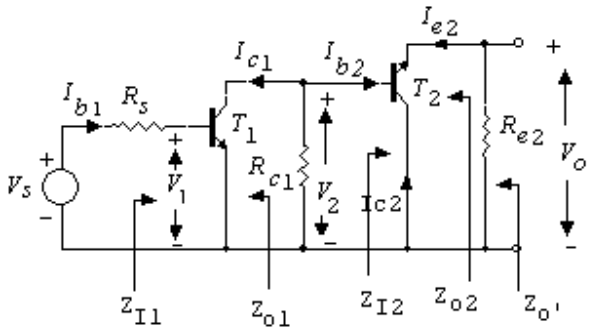


Fig. 7.19 b

collector stage. We will find the expressions of various gains and input & output impedances of the cascaded amplifier. The current gain, input impedance and voltage gain of the second stage are calculated first and subsequently of first stage, since load resistance is required in the calculations of these quantities. However, the output impedance of the first stage is calculated first followed by the output impedance of the second stage.

Analysis of the second stage: The current gain of the second stage which is the common collector stage is given by (ref. table 7.3):

$$A_{I2} = -\frac{I_{c2}}{I_{b2}} = \frac{-h_{fc}}{1+h_{oc}R_{c2}}$$

The input impedance is $Z_{I2} = h_{ic} + h_{rc}A_{I2}R_{c2}$

The voltage gain of this stage is given by:

$$A_{V2} = \frac{V_o}{V_2} = A_{I2}\frac{R_{c2}}{Z_{I2}}$$

Analysis of first stage: For the first stage the effective load resistance will be the parallel combination of R_{c1} and Z_{I2} given by

$$R_{L1} = \frac{R_{c1}Z_{I2}}{R_{c1}+Z_{I2}}$$

The current gain of this stage is

$$A_{I1} = -\frac{I_{c1}}{I_{b1}} = \frac{-h_{fe}}{1 + h_{oe} R_{L1}}$$

The input impedance of the first stage which is also the input impedance of the cascaded amplifier is given by:

$$Z_{I1} = h_{ie} + h_{re} A_{I1} R_{L1}$$

The voltage gain of this stage is

$$A_{V1} = \frac{V_2}{V_1} = A_{I1} \frac{R_{L1}}{Z_{I1}}$$

The output admittance $Y_o = (1/Z_{o1})$ of the first stage is given by

$$Y_{o1} = \left(h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s} \right)$$

The output impedance of this stage, taking into account the resistance R_{c1}, will be the parallel combination of Z_{o1} and R_{c1}.

i.e. $Z'_{o1} = Z_{o1} \| R_{c1}$

The output admittance of the second stage can be calculated by considering the effective source resistance R_{s2} as the parallel combination of Z_{o1} and $R_{c1.}$

$$Y_{o2} = \left(h_{oc} - \frac{h_{fc} h_{rc}}{h_{ic} + R_{s2}} \right)$$

The output impedance of the cascaded amplifier will be the parallel combination of Z_{o2} and R_{c2}

i.e. $Z_0 = Z_{o2} \| R_{c2}$

Now the total (overall) current gain of both stages is

$$A_I = -\frac{I_{c2}}{I_{b1}} = -\frac{I_{c2}}{I_{b2}} \frac{I_{b2}}{I_{c1}} \frac{I_{c1}}{I_{b1}} = -A_{I2}.A_{I1} \frac{I_{b2}}{I_{c1}}$$

From figure (7.19) we have

$$I_{b2} = (-I_{c1}) \left(\frac{R_{c1} Z_{I2}}{Z_{I2} + R_{c1}} \right) \left(\frac{1}{Z_{I2}} \right)$$

or $$\frac{I_{b2}}{I_{c1}} = -\left(\frac{R_{c1}}{Z_{I2} + R_{c1}} \right)$$

Hence $$A_I = A_{I2}.A_{I1} \left(\frac{R_{c1}}{Z_{I2} + R_{c1}} \right)$$

The voltage gain of the cascaded amplifier is

$$A_V = \frac{V_o}{V_1} = \frac{V_o}{V_2} \frac{V_2}{V_1} = A_{V2} A_{V1}$$

The overall voltage gain of the cascaded amplifier by considering the source resistance also, is given by:

$$A_{Vs} = \frac{V_o}{V_s} = A_V \frac{Z_{I1}}{Z_{I1} + R_s}$$

7.10 Simplified Common – Emitter Hybrid Model: In the preceding sections the exact hybrid models of the transistors were used for getting the expressions of various gains and input & output impedances of different transistor amplifiers. However, in most cases the approximate hybrid model of the transistor will give acceptable results with in the specified limits, thus tedious and lengthy calculations may be avoided. The CE amplifier is most frequently used amplifier, so we shall get the simplified h – parameter model in this configuration which will further be used in CB and CC configuration circuits.

The exact model in CE configuration is redrawn in figure (7.20).

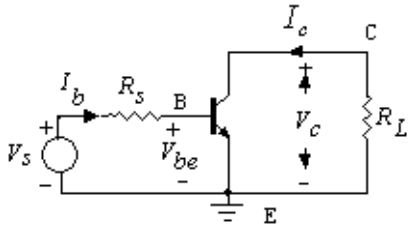


Fig. 7.20 a

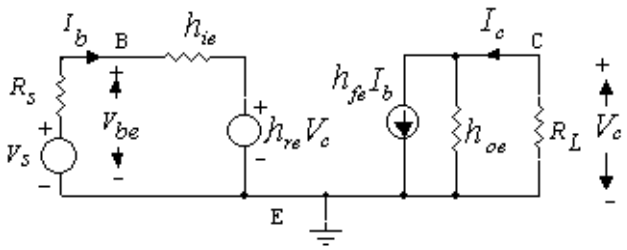


Fig. 7.20 b

Out of four *h* – parameters only two parameters h_{ie} and h_{fe} are sufficient to use in the simplified model in CE configuration and other two parameters may be neglected provided the load resistance R_L is small enough in comparison with ($1/h_{oe}$), so that the parallel combination R_L and $1/h_{oe}$ is approximately equal to R_L. We may, therefore, omit h_{oe} from the exact model shown in figure (7.20 b). In this condition the collector current will be approximately equal to $h_{fe}I_b$. The magnitude of the voltage source ($h_{re}V_c$) in the emitter circuit will be equal to $h_{re}h_{fe}I_bR_L$ since $V_c = -h_{fe}I_bR_L$. In most of the transistor parameters $h_{re}h_{fe} \cong 0.01$, so the voltage $h_{re}V_c$ may be neglected in comparison with the voltage drop ($h_{ie}I_b$) across h_{ie} provided R_L is small enough. The simplified model may therefore be given in figure (7.21). It has been observed that the errors in using this simplified model will not be more than 10% if $h_{oe}R_L < 0.1$.

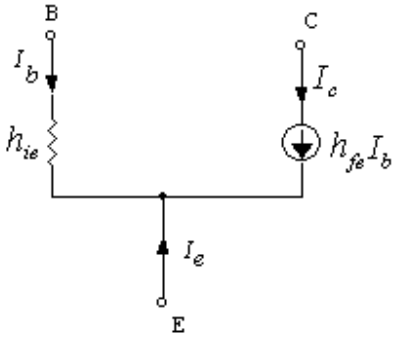


Fig. 7.21

We shall now calculate various gains and input and output impedances in all the three configurations using this simplified model.

7.10.1 Simplified Calculation for the Common Emitter Configuration: The simplified model in CE configuration is shown in figure (7.22).

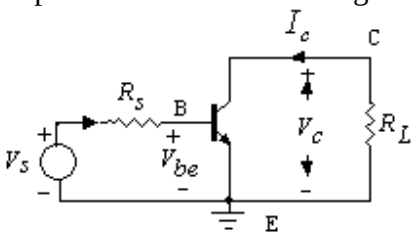


Fig. 7.22 a

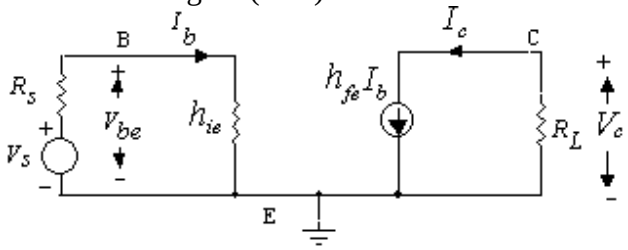


Fig. 7.22 b

(i) **Current Gain**: From the figure (7.22 b) we have

$$A_I = \frac{-I_c}{I_b} \cong -\frac{h_{fe} I_b}{I_b} = -h_{fe} \qquad \text{------ (7.68)}$$

(ii) **Input impedance**: The input impedance Z_I is given by:

$$Z_I = h_{ie} + h_{re} A_I R_L$$

which may be put in the form

$$Z_I = h_{ie}\left[1 - \frac{h_{re} h_{fe}}{h_{ie} h_{oe}} \frac{|A_I|}{h_{fe}} h_{oe} R_L\right] \qquad \text{----- (7.69)}$$

The value of the quantity $\left[\frac{h_{re} h_{fe}}{h_{ie} h_{oe}}\right]$ is approximately 0.5 for the typical h – parameter values of the transistor.

Thus, if $h_{oe} R_L < 0.1$ than $Z_I \cong h_{ie}$.

(iii) **Voltage Gain**:

$$A_V = \frac{A_I R_L}{Z_I} \cong -\frac{h_{fe} R_L}{h_{ie}} \qquad \text{------(7.70)}$$

(iv) Output Impedance: From Fig. 7.22 b, if V_c is the applied voltage at the output, we get $Z_o = \frac{V_c}{I_c}$ with $V_s = 0$.

Since $h_{fe}I_b$ reduces to zero then $Z_o = \infty$. The true value of Z_o depends upon the source resistance and lies between 40 to 80 KΩ.

Output impedance Z_o' taking into account the load resistance R_L will be approximately equal to R_L as it is the parallel combination of output resistance ($Z_o = \infty$) and R_L.

7.10..2 Simplified Calculation for the Common Base Configuration: **T**he simplified model of CE configuration can be used to get approximate model in CB configuration as shown in figure (7.23) in which base is grounded and the collector is connected to the ground through the load resistance R_L.

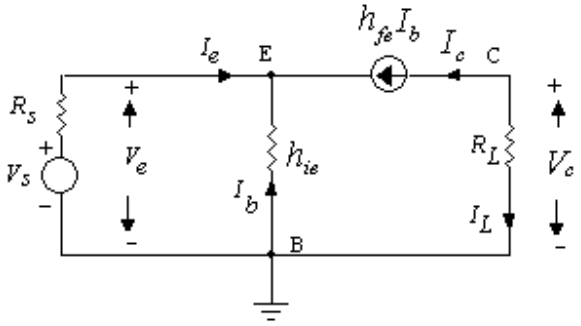


Fig. 7.23

(i) **Current Gain**: From the figure (7.23) we have

$$A_I = \frac{I_L}{I_e} = \frac{-I_c}{I_b}$$

$$\cong \frac{-h_{fe}I_b}{-I_b(1+h_{fe})} = \frac{h_{fe}}{1+h_{fe}}$$

(ii) **Input impedance:** $Z_I = \frac{V_e}{I_e} \cong \frac{-h_{ie}I_b}{-(1+h_{fe})I_b} = \frac{h_{ie}}{1+h_{fe}}$

(iii) **Voltage Gain**: $A_v = A_I \frac{R_L}{Z_I} \cong \frac{h_{fe}}{1+h_{fe}} \frac{(1+h_{fe})}{h_{ie}} R_L$

$$= \frac{h_{fe}}{h_{ie}} R_L$$

(iv) Output Impedance: $Z_o = \frac{V_c}{I_c} = \infty$ as $V_s = 0$

The output resistance taking into account R_L will be equal to R_L.

7.10.3 Simplified Calculation for the Common Collector Configuration: **T**he simplified model of CE configuration can be used to get approximate model in CC

configuration as shown in figure (7.24) in which collector is grounded and the load resistance R_L is connected between emitter and the ground.

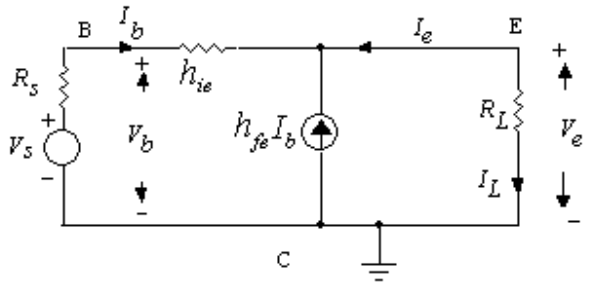


Fig. 7.24

(i) **Current Gain**: From the figure (7.24) we have

$$A_I = \frac{I_L}{I_b} = \frac{-I_e}{I_b}$$

$$\cong \frac{(1+h_{fe})I_b}{I_b} = 1+h_{fe}$$

(ii) **Input impedance:** $$Z_I = \frac{V_b}{I_b} \cong \frac{h_{ie}I_b + (1+h_{fe})I_b R_L}{I_b}$$

$$= h_{ie} + (1+h_{fe})R_L$$

(iii) **Voltage Gain**: $$A_v = A_I \frac{R_L}{Z_I} \cong \frac{(1+h_{fe})R_L}{h_{ie} + (1+h_{fe}R_L)}$$

$$= \frac{h_{ie} + (1+h_{fe})R_L - h_{ie}}{h_{ie} + (1+h_{fe}R_L)} = 1 - \frac{h_{ie}}{Z_I}$$

(iv) Output Impedance: From the figure (7.24), the open circuit output voltage is V_s while the short circuit output current is

$$I = (1+h_{fe})I_b = \frac{(1+h_{fe})V_s}{R_s + h_{ie}}.$$

So the output impedance is given by: $$Z_o = \frac{V_s}{I} = \frac{R_s + h_{ie}}{1+h_{fe}}$$

The output resistance taking into account R_L will be equal to the parallel combination of R_L and Z_o calculated above.

The approximate formulas for different quantities obtained using the simplified model of the transistor are given in table 7.6 for the three configurations.

Table 7.6

Quantity	CE	CB	CC
A_I	$-h_{fe}$	$\frac{h_{fe}}{1+h_{ie}}$	$1+h_{fe}$
Z_I	h_{ie}	$\frac{h_{ie}}{1+h_{fe}}$	$h_{ie}+(1+h_{fe})R_L$
A_v	$-\frac{h_{fe}R_L}{h_{ie}}$	$\frac{h_{fe}}{h_{ie}}R_L$	$1-\frac{h_{ie}}{Z_I}$
Z_o	∞	∞	$\frac{R_s+h_{ie}}{1+h_{fe}}$
$Z_o^{'}$	R_L	R_L	$Z_o \Vert R_L$

7.11 CC – CC Cascaded Amplifier (Darlington pair): As is well known that the common collector amplifier is used where we need input impedance to be large enough and output impedance to be very small. Some times one stage of such an amplifier does not provide much degree of buffering (ratio of Z_I / Z_o), it becomes necessary to cascade two transistors, each in CC configuration as shown in figure (7.25). A pair of transistors connected like this is known as a Darlington pair. Analysis of this circuit will be done as in the previous case of cascaded amplifier.

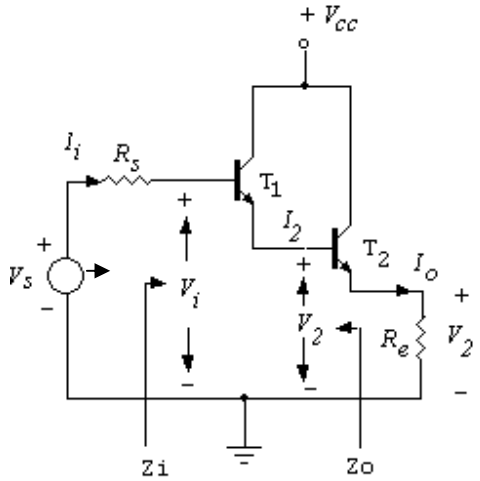


Fig. 7.25

The current gain of the transistor T_2 is calculated first, assuming $h_{oe}R_e \le 0.1$ and $h_{fe}R_e >> h_{ie}$. The relations given in table 7.6 may be used for calculating the current gain and input impedance of the second stage.

$$A_{I2} = \frac{I_o}{I_2} \approx 1+h_{fe} \qquad \text{and} \qquad Z_{I2} \approx (1+hfe)R_e$$

We shall find the current gain of the first stage by considering the exact formula of the current gain as $h_{oe}Z_{I2} \le 0.1$ requirement is not meet out since Z_{I2} is the effective load resistance for the transistor T_1.

Thus
$$A_{I1} = \frac{I_2}{I_i} = \frac{1+h_{fe}}{1+h_{oe}Z_{I2}}$$
$$= \frac{1+h_{fe}}{1+h_{oe}(1+h_{fe})R_e}$$
$$\approx \frac{1+h_{fe}}{1+h_{oe}h_{fe}R_e} \qquad \text{as } h_{oe}R_e \le 0.1.$$

The overall current gain of the Darlington pair is given by:
$$A_I = \frac{I_o}{I_i} = \frac{I_o}{I_2}\frac{I_2}{I_i} = A_{I2}A_{I1}$$

or
$$A_I = \frac{(1+h_{fe})^2}{1+h_{oe}h_{fe}R_e}$$

The input impedance of T1, which will be the overall input impedance of the cascaded circuit, is given by:
$$Z_{I1} = h_{ie} + A_{I1}R_{i2} \approx \frac{(1+h_{fe})^2 R_e}{1+h_{oe}h_{fe}R_e}$$

The voltage gain of the first stage is given by the expression,
$$A_{V1} = 1 - \frac{h_{ie}}{Z_{I1}}$$

and
$$A_{V2} \approx 1 - \frac{h_{ie}}{A_{I1}Z_{I2}}$$

The overall voltage gain of the Darlington emitter follower is
$$A_V = A_{V1}.A_{V2} = \left(1 - \frac{h_{ie}}{Z_{I2}}\right)\left(1 - \frac{h_{ie}}{A_{I1}Z_{I2}}\right)$$
$$\approx 1 - \frac{h_{ie}}{A_{I1}Z_{I2}} - \frac{h_{ie}}{Z_{I2}}$$

But $A_{I1}Z_{I2} >> h_{ie}$, hence $A_V \approx \left(1 - \frac{h_{ie}}{Z_{I2}}\right)$

It is clear from this equation that the voltage gain of the Darlington emitter follower is approximately the same as that of the conventional emitter follower of transistor T_2 and is very close to unity.

The output impedance Zo1 of the first stage is given by:
$$Z_{o1} = \frac{R_s + h_{ie}}{1+h_{fe}}$$

This output impedance works as the source resistance for the second stage of the circuit. Hence the output impedance of the second stage, which will be the overall output impedance of the circuit given by:

$$Z_{o2} \approx \frac{\frac{R_s + h_{ie}}{1+h_{fe}} + hie}{1+h_{fe}} \approx \frac{R_s + h_{ie}}{(1+h_{fe})^2} + \frac{h_{ie}}{1+h_{fe}}$$

Thus the output impedance of this Darlington pair circuit is smaller than that of single stage emitter follower.

From the above calculations of the Darlington emitter follower circuit, it is concluded that:

(i) Its overall current gain is high.

(ii) Its input impedance is higher than that of the single stage emitter follower.

(iii) Its voltage gain close to unity and approximately the same as that of the single stage emitter follower.

(iv) Its output impedance is smaller than that of the single stage.

The major drawback of this Darlington pair of transistors is that the leakage current of the first stage of the amplifier is amplified by the second stage. Hence the overall leakage current is high and is not suitable to use more than three transistors in cascade in emitter follower mode.

Example 7.10 Show that the overall h parameters of the two – stage cascaded amplifier shown in figure (7.26)are

(i) $h_{11} = h_{11}^{'} - \frac{h_{12}^{'} h_{21}^{'}}{1+h_{22}^{'} h_{11}^{''}} h_{11}^{''}$ (ii) $h_{12} = \frac{h_{12}^{'} h_{12}^{''}}{1+h_{22}^{'} h_{11}^{''}}$

(iii) $h_{21} = -\frac{h_{21}^{'} h_{21}^{''}}{1+h_{22}^{'} h_{11}^{''}}$ (iv) $h_{22} = h_{22}^{''} - \frac{h_{12}^{''} h_{21}^{''}}{1+h_{22}^{'} h_{11}^{''}} h_{22}^{'}$

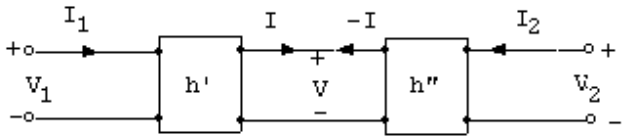


Fig. 7.26

Solution: For the two individual networks shown in figure (7.26 h – parameter equations are given

$$V_1 = h_{11}^{'} I_1 + h_{12}^{'} V \quad \text{------ (7.71)}$$

$$I = h_{21}^{'} I_1 + h_{22}^{'} V \quad \text{------ (7.72)}$$

and $$V = -h_{11}^{''} I + h_{12}^{''} V_2 \quad \text{------ (7.73)}$$

$$I_2 = -h_{21}^{''} I + h_{22}^{''} V_2 \quad \text{------ (7.74)}$$

The h - parameter equations of the cascaded network in the required form are given

$$V_1 = h_{11} I_1 + h_{12} V_2 \qquad \text{------ (7.75)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \qquad \text{------ (7.76)}$$

Using the equations (7.71) to (7.74), we get the two equations of the form given in equations (7.75) & (7.76).

Putting the value of I from equation (7.72) in equations (7.73) & (7.74) we have

$$V = -h_{11}^{''}(h_{21}^{'} I_1 + h_{22}^{'} V) + h_{12}^{''} V_2$$

or $$V(1 + h_{11}^{''} h_{22}^{'}) = -h_{11}^{''} h_{21}^{'} I_1 + h_{12}^{''} V_2$$

or $$V = -\frac{h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})} I_1 + \frac{h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} V_2 \qquad \text{------- (7.77)}$$

and $$I_2 = -h_{21}^{''}(h_{21}^{''} I_1 + h_{22}^{'} V) + h_{22}^{''} V_2$$

or $$I_2 = -h_{21}^{''} h_{21}^{''} I_1 - h_{21}^{''} h_{22}^{'} V + h_{22}^{''} V_2 \qquad \text{------ (7.78)}$$

Put the value of V from equation (7.77) in equations (7.71) & (7.78) we have

$$V_1 = h_{11}^{'} I_1 + h_{12}^{'}\left(-\frac{h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})} I_1 + \frac{h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} V_2\right)$$

or $$V_1 = \left(h_{11}^{'} - \frac{h_{12}^{'} h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})}\right) I_1 + \frac{h_{12}^{'} h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} V_2$$

or $$V_1 = \left(\frac{h_{11}^{'} + h_{11}^{'} h_{11}^{''} h_{22}^{'} - h_{12}^{'} h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})}\right) I_1 + \frac{h_{12}^{'} h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} V_2$$

or $$V_1 = \left(h_{11}^{'} - \frac{h_{12}^{'} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})} h_{11}^{''}\right) I_1 + \frac{h_{12}^{'} h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} V_2 \qquad \text{------ (7.79)}$$

and $$I_2 = -h_{21}^{''} h_{21}^{'} I_1 - h_{21}^{''} h_{22}^{'}\left(-\frac{h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})} I_1 + \frac{h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} V_2\right) + h_{22}^{''} V_2$$

or $$I_2 = \left(-h_{21}^{''} h_{21}^{'} + \frac{h_{21}^{''} h_{22}^{'} h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})}\right) I_1 + \left(h_{22}^{''} - \frac{h_{21}^{''} h_{22}^{'} h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})}\right) V_2$$

or

$$I_2 = \left(\frac{-h_{21}^{''} h_{21}^{'} - h_{21}^{''} h_{21}^{'} h_{11}^{''} h_{22}^{'} + h_{21}^{''} h_{22}^{'} h_{11}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})}\right) I_1 + \left(\frac{h_{22}^{''} + h_{11}^{''} h_{22}^{'} h_{22}^{''} - h_{21}^{''} h_{22}^{'} h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})}\right) V_2$$

or $$I_2 = \left(\frac{-h_{21}^{''} h_{21}^{'}}{(1 + h_{11}^{''} h_{22}^{'})}\right) I_1 + \left(h_{22}^{''} - \frac{h_{21}^{''} h_{12}^{''}}{(1 + h_{11}^{''} h_{22}^{'})} h_{22}^{'}\right) V_2 \qquad \text{------ (7.80)}$$

Comparing the equations (7.79) & (7.80) with equations (7.75) & (7.76), we get the required result.

(i) $h_{11} = h_{11}^{'} - \dfrac{h_{12}^{'} h_{21}^{'}}{1 + h_{22}^{'} h_{11}^{''}} h_{11}^{''}$ (ii) $h_{12} = \dfrac{h_{12}^{'} h_{12}^{''}}{1 + h_{22}^{'} h_{11}^{''}}$

(iii) $h_{21} = -\dfrac{h_{21}^{'} h_{21}^{''}}{1 + h_{22}^{'} h_{11}^{''}}$ (iv) $h_{22} = h_{22}^{''} - \dfrac{h_{12}^{''} h_{21}^{''}}{1 + h_{22}^{'} h_{11}^{''}} h_{22}^{'}$

Example 7.11 Show that the overall *h* –parameters for the composite transistors illustrated in figure (7.27), are:

(i) $h_{ie} = h_{ie1} + \dfrac{(1 - h_{re1})(1 + h_{fe1}) h_{ie2}}{1 + h_{oe1} h_{ie2}}$

(ii) $h_{fe} = h_{fe1} + \dfrac{(h_{fe2} - h_{oe1} h_{ie2})(1 + h_{fe1})}{1 + h_{oe1} h_{ie2}}$

(iii) $h_{oe} = h_{oe2} + \dfrac{(1 + h_{fe2})(1 - h_{re2}) h_{oe1}}{1 + h_{oe1} h_{ie2}}$

(iv) $h_{re} = h_{re2} + \dfrac{(h_{ie2} hoe1 + h_{re1})(1 - h_{re2})}{1 + h_{oe1} h_{ie2}}$

(v) Obtain the numerical values for the *h* – parameters of the composite transistor by assuming transistors T_1 and T_2 as identical and using $h_{ie} = 1100\Omega$, $h_{re} = 2.5x10^{-4}$, $h_{fe} = 50$ and $h_{oe} = 24\mu A/volts$.

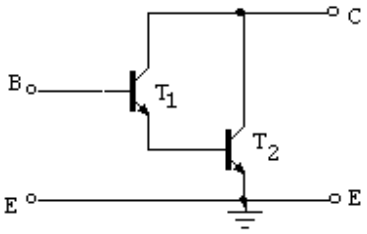


Fig. 7.27

Solution: The transistor T_1 is in CC configuration and T_2 is in CE configuration. The h – parameters of the composite configuration may be obtained using the results of example 7.10.

(i) $h_{ie} = h_{ic1} - \dfrac{h_{rc1} h_{fc1}}{1 + h_{oc1} h_{ie2}} h_{ie2}$

Put CC *h* –parameters in the form of CE *h* – parameters as

$h_{ic1} = h_{ie1}$ $h_{fc1} = -(1 + h_{fe1})$

$$h_{rc1} = (1 - h_{re1}) \qquad h_{oc1} = h_{oe1}$$

$$h_{ie} = h_{ie1} - \frac{(1 - h_{re1})(1 + h_{fe1})}{1 + h_{oe1} h_{ie2}} h_{ie2}$$

part (i) proved.

(ii) $$h_{fe} = -\frac{h_{fc1} h_{fe2}}{1 + h_{oc1} h_{ie2}} = \frac{(1 + h_{fe1}) h_{fe2}}{1 + h_{oe1} h_{ie2}}$$

$$= \frac{h_{fe2} + h_{fe1} h_{fe2} - h_{oe1} h_{ie2} h_{fe1} + h_{oe1} h_{ie2} - h_{oe1} h_{ie2} + h_{oe1} h_{ie2} h_{fe1}}{1 + h_{oe1} h_{ie2}}$$

$$= h_{fe1} + \frac{(1 + h_{fe1}) h_{fe2} - h_{oe1} h_{ie2} (1 + h_{fe1})}{1 + h_{oe1} h_{ie2}}$$

(iii) $$h_{re} = \frac{h_{rc1} h_{re2}}{1 + h_{oc1} h_{ie2}} = \frac{(1 - h_{re1}) h_{re2}}{1 + h_{oe1} h_{ie2}}$$

Problems:

1. Define *h* – parameters of a transistor. Draw the *h* – parameter model of a transistor.
2. Discuss how are *h* – parameters deduced from the input output characteristics curves of the transistor.
3. Discuss the method of converting *h* – parameters from one configuration to other configuration.
4. Draw the circuit of transistor in common emitter configuration and give its *h* – parameter model.
5. Draw the circuit of transistor in common base configuration and give its *h* – parameter model.
6. Draw the circuit of transistor in common collector configuration and give its *h* – parameter model.
7. Find CE *h* – parameters in terms of CC configuration.
8. Find CC *h* – parameters in terms of CE configuration.
9. Find CE *h* – parameters in terms of CB configuration.
10. Find CB *h* – parameters in terms of CE configuration.
11. Find CB *h* – parameters in terms of CC configuration.
12. Find CC *h* – parameters in terms of CB configuration.
13. Discuss the method of converting *h* – parameter from one configuration to other configuration.
14. Explain how to obtain h_{fe} and h_{re} from the output characteristics of the transistor.
15. Prove that $h_{ic} = h_{ie}$, $h_{rc} = (1 - h_{re})$, $h_{fc} = -(1 + h_{fe})$, $h_{oc} = h_{oe}$.
16. Prove that $h_{ie} = h_{ic}, h_{re} = (1 - h_{rc})$, $h_{fe} = -(1 + h_{fc})$, $h_{oe} = h_{oc}$.
17. Prove the following expressions:

$$h_{ic} = \frac{h_{ib}}{[h_{ib}h_{ob} + (1-h_{rb})(1+h_{fb})]} \qquad h_{rc} = \frac{(1+h_{fb})}{[h_{ib}h_{ob} + (1-h_{rb})(1+h_{fb})]}$$

18. Prove the following expressions:

$$h_{fc} = \frac{-(1-h_{rb})}{[h_{ib}h_{ob} + (1-h_{rb})(1+h_{fb})]} \qquad h_{oc} = \frac{h_{ob}}{[h_{ib}h_{ob} + (1-h_{rb})(1+h_{fb})]}$$

19. (i) Prove the following expressions

$$h_{ib} = \frac{h_{ie}}{(1+h_{fe})(1-h_{re}) + h_{ie}h_{oe}} \quad h_{rb} = \frac{h_{ie}h_{oe} - h_{re}(1+h_{fe})}{(1+h_{fe})(1-h_{re}) + h_{ie}h_{oe}}$$

(ii) From these exact formulas deduce the approximate formulas given below:

$$h_{ib} = \frac{h_{ie}}{1+h_{fe}} \qquad h_{rb} = \frac{h_{ie}h_{oe}}{1+h_{fe}} - h_{re}$$

20. (i) Prove the following expressions

$$h_{fe} = \frac{-h_{fb}(1-h_{rb}) - h_{ib}h_{ob}}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}} \qquad h_{oe} = \frac{h_{ob}}{(1+h_{fb})(1-h_{rb}) + h_{ib}h_{ob}}$$

(ii) From these exact formulas prove the following expressions

$$h_{fe} = \frac{-h_{fb}}{1+h_{fb}} \qquad h_{oe} = \frac{h_{ob}}{1+h_{fb}}$$

21. By drawing the variations of voltage gain, current gain and input impedance with load impedance and output impedance with source resistance for the three configuration of the transistor, find the characteristics of the transistor amplifier in the three configurations.
22. Discuss the comparative study of the transistor amplifier in the three configurations.
23. Draw the circuit model of a transistor amplifier (general amplifier configuration). Using this model find the expressions of various gains, input and output impedances.
24. Drive the expressions of A_I and Z_I of a transistor amplifier in terms of load impedance.
25. State and prove Miller theorem and its dual.
26. Using the approximate model *h* – parameter model, obtain the expression for a CE circuit for A_I, Z_I, A_V, and Z_o.
27. Using the approximate model *h* – parameter model, obtain the expression for the emitter follower circuit for A_I, Z_I, A_V, and Z_o. What are the advantages of the emitter follower circuit?
28. Using the simplified model of the CE configuration obtain the expressions of various gain and input output impedances of the CB configuration.
29. Using the simplified model of the CE configuration obtain the expressions of various gain and input output impedances of the CC configuration.
30. Using the simplified model of the CE configuration obtain the expressions of various gain and input output impedances of this configuration.
31. What do you understand by the cascading of the transistor amplifiers? Why cascading is done? Draw the circuit of CE – CC cascaded amplifiers. Find the

expressions of the overall voltage gain, current gain and input / output impedances of this cascaded amplifier.

32. Draw a Darlington emitter follower circuit and find the expressions of the overall voltage gain, current gain and input / output impedances of this Darlington emitter follower circuit.

8

Transistor Biasing and Thermal Stabilization

In this chapter we shall discuss the different biasing methods of transistor, to operate the transistor in the linear operating region of the characteristics. The operating point shifts with temperature as the transistor parameters are dependent on temperature. The temperature compensated techniques will also be discussed in this chapter in addition to the study of the stability factor of different biasing methods.

8.1 The Operating Point: It is well known that the CE amplifier is most commonly used amplifier and whenever the general amplification of the signal is needed this amplifier configuration is used. So we shall discuss the operating point and other factor by considering the CE amplifier. Figure 8.1(a) shows a simple circuit of CE

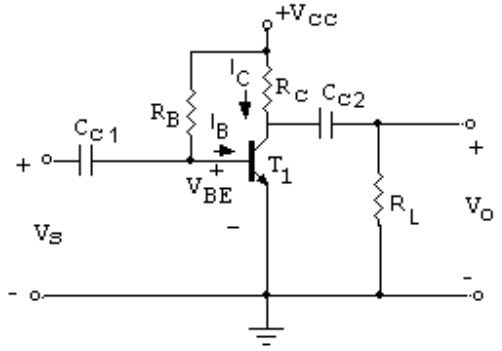


Fig. 8.1(a)

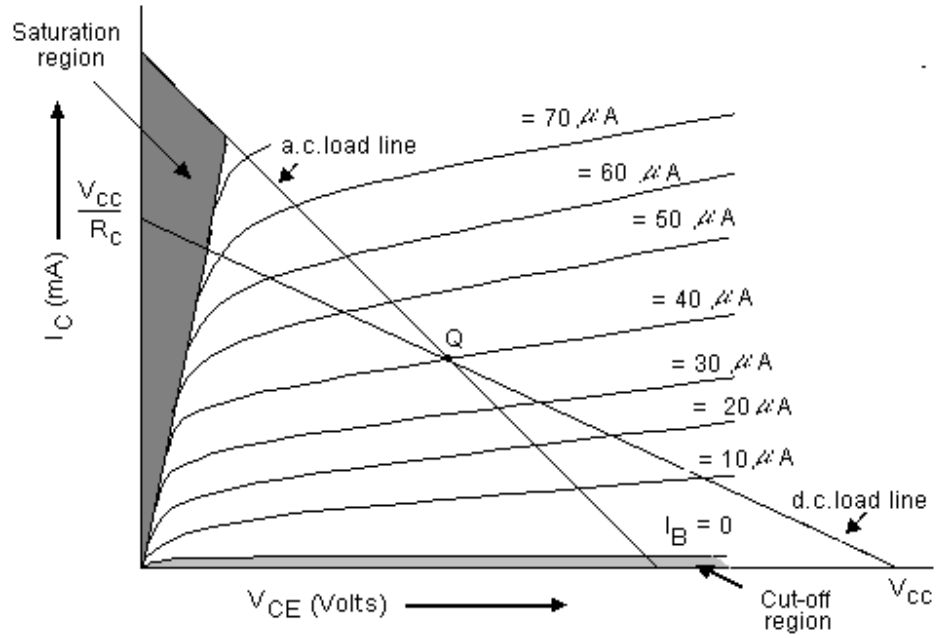


Fig. 8.1(b)

amplifier. In this circuit V_{CC} and R_C are fixed. The values of collector current I_C and collector to emitter voltage $V_{C\,E}$ are dependent on the values of R_B. The signal to be amplified from the source V_s is applied to the base of the transistor through the coupling capacitor C_{C1} and the output is taken from the collector through the other coupling capacitor $C_{C\,2}$. These two coupling capacitors offer low reactances to the signal frequency thus the signal is passed through them and d.c. voltage is blocked. Figure 8.1(b) shows a set of output characteristics of a transistor.

Applying KVL to the collector circuit (fig. 8.1a) under d.c. condition the coupling capacitor $C_{C\,2}$ will act as an open circuit and $R_L = \infty$, it is obtained

$$V_{CC} = R_C I_C + V_{CE}$$

or
$$I_C = (-\frac{1}{R_C}).V_{CC} + \frac{V_{CC}}{R_C}$$

The above equation is a straight line equation of the form $y = mx + C$, having the slope $(-\frac{1}{R_C})$ and intercept on y- axis as $\frac{V_{CC}}{R_C}$.

A straight line drawn between $\frac{V_{CC}}{R_C}$ and V_{CC} on the characteristic curves of the transistor will have the slope $(-\frac{1}{R_C})$ as shown in figure 8.1(b). This straight on the characteristic curves of the transistor is known as the d.c or static load line.

The point of intersection of the load line and characteristic curve of the particular base current is known as quiescent operating point or simply the operating point. The operating point may be chosen anywhere on the load line, which will be decided by the base current. For the proper amplification of the signal the transistor should be operated in the centre of the load line.

If on the other hand $R_L \neq \infty$, an a.c. or dynamic load line is to be drawn. At the signal frequency C_{C2} will act as short circuit, the effective load resistance at the collector will be the parallel combination of R_L and R_C. The a.c. load line must be drawn through the operating point Q and the slope should correspond to the effective load resistance at the collector (figure 8.1b).

Figure 8.2 illustrates that the operating point Q is chosen in the middle of the load line. It gives the variation of the sinusoidal collector current and collector voltage corresponding to the input signal (base current). It is clear from this figure that the output signal is without distortion. If the operating shifts nearer to Q_2 the output voltage and current get clipped at the positive peaks. If on the other hand the operating point shifts nearer to Q_1 the clipping will at the negative peaks of the output voltage and the current. This results the distortion in the output signal which is undesirable. Thus the maximum signal that can be handled by an amplifier will be decided by the choice of the operating point. In other words we may say that the variation of the base current (signal current) should be such that operating point will not go either to the saturation region or to the cutoff region. The collector current will therefore, flow in the outer circuit for the whole of the input cycle.

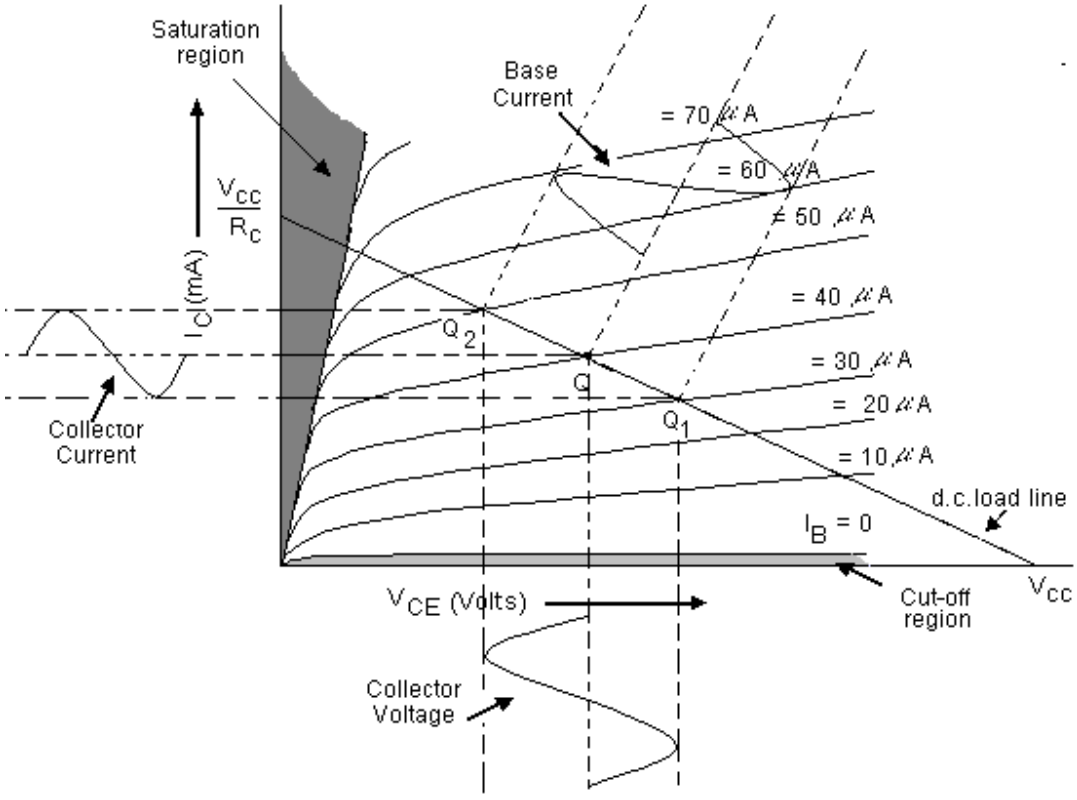


Fig. 8.2

8.2 Operating Point Stability: The operating point selected in the middle of load line should remain fixed in the amplifier circuit. But the operating point shift due to following two reasons.

(i) **Unit to unit variation**: When a transistor is replaced by another transistor of the same type (or number), the current amplification β of transistor may be three times larger or smaller than the other; and if the biasing current is not capable of balancing this, the collector current as a result the collector voltage of the circuit will become different causing shift in the operating point.

(ii) **Thermal variation**: Another important factor is the temperature variation which leads the operating point instability in the circuit. When the device temperature changes there is a variation in reverse saturation collector current I_{CO}, base to emitter voltage (V_{BE}) and β. As a result of which collector current will change affecting the operating point stability. It has been observed that I_{CO} doubles for every 10 ^{o}C rise in temperature, V_{BE} decreases at a rate of about 2.5 mV/deg. C for both Si and Ge and also there is an increase in β with the increase in temperature.

There are two techniques for maintaining the operating point stability in the transistor circuits. One is called as the Bias Stabilization and the other is known as the Compensation technique. Both these techniques will be discussed in detail in following sections. The resistive biasing circuits will be used in the bias stabilization techniques that will maintain the collector current stable in spite of variation in I_{CO}, β and V_{BE} with temperature or unit to unit variation. However, in the Compensation technique, the temperature sensitive devices such as diodes, transistors, thermistors etc. are used for the operating point stability.

8.3 The Stability Factors: Since the collector current I_C is a function of I_{CO}, β and V_{BE} , so we define the three partial derivatives of I_C with respect to these variables. These derivatives are called the stability factors S, S', S'' and are defined as:

The stability S is defined as the rate of change of collector current with respect to the reverse saturation current, keeping β and V_{BE} constant.

$$S = \frac{\partial I_C}{\partial I_{CO}} \approx \frac{\Delta I_C}{\Delta I_{CO}}$$

Here the smaller the value of S better is the stability against I_{CO}.

The stability S' is defined as the rate of change of collector current with respect to base to emitter voltage V_{BE}, keeping I_{CO} and β constant.

$$S' = \frac{\partial I_C}{\partial V_{BE}} \approx \frac{\Delta I_C}{\Delta V_{BE}}$$

Similarly the variation of I_C with respect to β is known as stability factor S'' given by $S'' = \frac{\partial I_C}{\partial \beta} \approx \frac{\Delta I_C}{\Delta \beta}$.

We shall now discuss the different biasing circuit and study the stability factor of each case.

8.4 Fixed Base Bias: The circuit diagram for the fixed base bias is shown in figure (8.4). The required voltage at the base is obtained from V_{CC} by having an excess voltage drop across R_B.

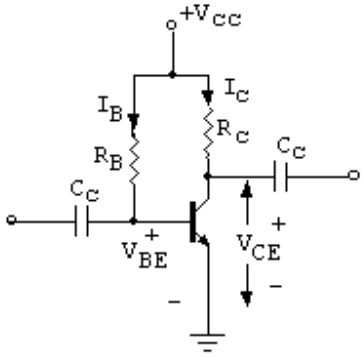


Fig. 8.4

For a CE configuration, we have the collector current as:

$$I_C = (1+\beta)I_{CO} + \beta I_B \qquad \text{------- (8.1)}$$

Applying the KVL to the input circuit, we have:

$$V_{CC} = I_B R_B + V_{BE} \qquad \text{------- (8.2)}$$

or
$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \approx \frac{V_{CC}}{R_B} \qquad \text{------- (8.3)}$$

Since $V_{CC} >> V_{BE}$ and the change in V_{BE} will have little effect on base bias I_B. For this reason it is called as fixed base bias.

From the equations (8.1) and (8.2) we have:

$$I_C = \frac{V_{CC}}{R_B}\beta + (1+\beta)I_{CO} \qquad \text{------ (8.4)}$$

Differentiating this equation with respect to I_{CO} we get:

$$S = \frac{\partial I_C}{\partial I_{CO}} = (1+\beta) \qquad \text{------- (8.5)}$$

As β is always sufficiently a larger quantity, the circuit has thus a poor thermal stability. In other words it may be discussed that the fixed bias circuit is very simple to design as it has only a few components and the operating point may be fixed at the desired place just by varying the value of R_B. But with the rise of temperature the collector current increases which further rises the temperature of the device. A cumulative action takes place and the collector current goes on increasing which leads the thermal instability. The circuit provides no check on the increase in the collector current.

Further this circuit does not provide the stability against unit to unit variation. Since $I_C = \beta I_B$, the base current is fixed by R_B so the collector current depends only on β. This biasing method is thus not of practical use.

8.5 Collector to Base Bias: An improvement in operating point stability is possible by connecting resistance R_B between collector and base as shown in figure (8.5). This method of biasing is known as collector to base bias.

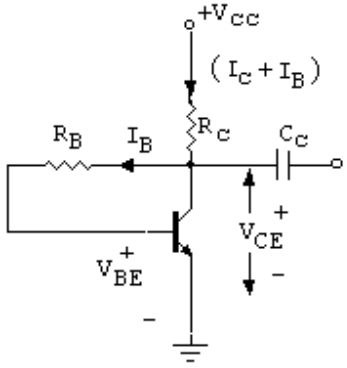


Fig. 8.5

Here, as I_C increases because of β or temperature, V_{CE} decreases as there is a larger drop across R_C . Hence I_B decreases which leads a decrease in I_C. Thus it gives the stability in the operating point. Let us calculate the stability factor in this case.

We have $I_C = \beta I_B + (!+\beta) I_{CO}$

or $$I_B = \frac{[I_C - (1+\beta) I_{CO}]}{\beta}$$ ------- (8.6)

Applying the KVL to the input circuit, we get

$$V_{CC} = R_C (I_C + I_B) + I_B R_B + V_{BE}$$ ------ (8.7)

From equations (8.6) and (8.7) we have

$$V_{CC} - V_{BE} = I_C R_C + \frac{(R_B + R_C)}{\beta}[I_C - (1+\beta) I_{CO}]$$ ------ (8.8)

Differentiating this equation with respect to I_{CO}, we have

$$\frac{\partial I_C}{\partial I_{CO}} R_C + \left[\frac{R_B + R_C}{\beta}\right]\left[\frac{\partial I_C}{\partial I_{CO}} - (1+\beta)\right] = 0$$

$$\frac{\partial I_C}{\partial I_{CO}}\left[R_C + \frac{(R_B + R_C)}{\beta}\right] - (1+\beta)\left[\frac{R_B + R_C}{\beta}\right] = 0$$

or $$S = \frac{\partial I_C}{\partial I_{CO}} = \frac{[(1+\beta)(R_B + R_C)]/\beta}{[(1+\beta)R_C + R_B]/\beta}$$

$$S = \frac{[(1+\beta)(R_B + R_C)]}{[(1+\beta)R_C + R_B]} \qquad \text{------ (8.9)}$$

or $$S = \frac{(1+\beta)}{\left[1 + \frac{\beta R_C}{R_B + R_C}\right]} \qquad \text{------ (8.10)}$$

The stability factor *S* is smaller than its value for fixed base bias circuit. The improvement in operating point stability in this circuit is due to the fact that a part of the signal output is coupled back to the base through R_B, reducing thereby the voltage gain of the amplifier. The operating point stability is obtained in this case at the cost of the voltage gain.

Differentiating the equation (8.8) with respect to V_{BE}, we get the expression for stability factor S' as

$$-1 = \left[\frac{R_B + (1+\beta)R_C}{\beta}\right]\frac{\partial I_C}{\partial V_{BE}}$$

or $$S' = \frac{\partial I_C}{\partial V_{BE}} = \frac{-\beta}{[R_B + (1+\beta)R_C]}$$

This equation may further be manipulated as

$$S' = \frac{-\beta(1+\beta)(R_B + R_C)}{[R_B + (1+\beta)R_C]x(1+\beta)(R_B + R_C)}$$

or $$S' = \frac{-\beta.S}{(1+\beta)(R_B + R_C)} \qquad \text{------ (8.11)}$$

where $\frac{[(1+\beta)(R_B + R_C)]}{[(1+\beta)R_C + R_B]} = S$.

Further differentiating the equation (8.8) with respect to β, we may obtain the expression for stability factor S''. Rewriting the equation (8.8) we have:

$$\left[\frac{R_B + (1+\beta)R_C}{\beta}\right]I_C = \frac{(V_{CC} - V_{BE})\beta + (R_C + R_B)(1+\beta)I_{CO}}{\beta}$$

If $(1 + \beta) I_{CO} \approx \beta I_{CO}$ then

$$I_C = \frac{[V_{CC} - V_{BE} + (R_C + R_B)I_{CO}]\beta}{[R_B + (1+\beta)R_C]} \qquad \text{------- (8.12)}$$

Now differentiating equation (8.12) with respect to β we may have:

$$S'' = \frac{\partial I_C}{\partial \beta} = \frac{[R_B + (1+\beta)R_C][V_{CC} - V_{BE} + (R_C + R_B)I_{CO}] - [V_{CC} - V_{BE} + (R_C + R_B)I_{CO}]\beta R_C}{[R_B + (1+\beta)R_C]^2}$$

$$= \frac{[V_{CC} - V_{BE} + (R_C + R_B)I_{CO}][R_B + R_C + \beta R_C - \beta R_C]}{[R_B + (1+\beta)R_C]^2}$$

$$= \frac{[V_{CC} - V_{BE} + (R_C + R_B)I_{CO}][R_B + R_C]}{[R_B + (1+\beta)R_C]^2} \qquad \text{------ (8.13)}$$

Using the equations (8.12) and (8.13) we get:

$$S'' = \frac{I_C[R_B + R_C]}{[R_B + (1+\beta)R_C]\beta} x \frac{(1+\beta)}{(1+\beta)} \qquad \text{------ (8.14)}$$

By using the equations (8.13) and (8.9) we may get the expression of S'' in terms of *S* as:

$$S'' = \frac{I_C}{\beta} \cdot \frac{S}{(1+\beta)}$$

8.6 Self Bias or Emitter Bias: Another circuit that is very commonly employed is the Self bias or Emitter bias. This arrangement is shown in figure (8.6), which comprises of resistance R_E in series with the emitter across which a voltage equal to $R_E I_E$ is dropped,

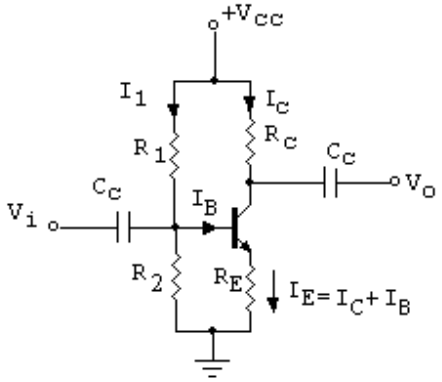


Fig. 8.6

and a potential divider consisting of R_1 & R_2. The improvement in the stability of the operating point in this circuit may be explained as given below.

As the temperature is increased, I_{CO} as well I_C increases. The increase in I_C will in turn increases the voltage drop across R_E, reducing thereby the base current and hence reduction in the collector current.

The circuit may be analyzed to find the stability factors S, S', S'' as follows:

In the self biasing circuit, the values of resistances R_1 and R_2 are so chosen so that the current I_1 flowing through the resistance R_1 is large enough compared with I_B. The current flowing through the resistance R_2 will therefore be equal to I_1. The potential drop V_{BB} across R_2 will be given by:

$$V_{BB} = \frac{V_{CC}.R_C}{R_1 + R_2} \qquad \text{------ (8.15)}$$

The circuit can be redrawn by its Thevenin's equivalent as shown in figure (8.7).

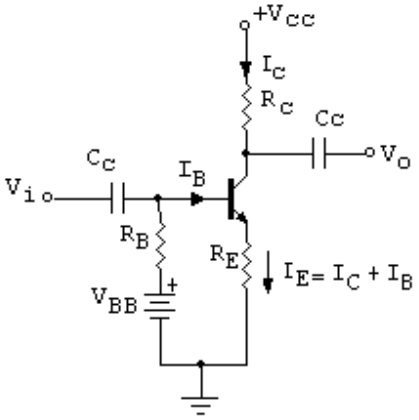


Fig. 8.7

Thevenin's resistance R_B may be obtained by shorting the supply voltage V_{CC} to ground. This resistance is given by:

$$R_B = \frac{R_1 R_2}{R_1 + R_2} \qquad \text{------ (8.16)}$$

Applying the KVL to the input circuit we get:

$$V_{BB} = I_B R_B + V_{BE} + (I_B + I_C) R_E$$

or
$$V_{BB} - V_{BE} = (R_B + R_E) I_B + I_C R_E \qquad \text{----- (8.17)}$$

The collector current is given by:

$$I_C = \beta I_B + (1+\beta) I_{CO}$$

or
$$I_B = \frac{[I_C - (1+\beta) I_{CO}]}{\beta} \qquad \text{------ (8.18)}$$

Using the equations (8.17) & (8.18) we get:

$$V_{BB} - V_{BE} = (R_B + R_E)\left[\frac{I_C - (1+\beta) I_{CO}}{\beta}\right] + I_C R_E \qquad \text{------(8.19)}$$

Differentiating the equation (8.19) with respect to I_{CO} , we may obtain the expression for the stability factor *S*.

$$\left[\frac{R_B + R_E}{\beta}\right]\left[\frac{\partial I_C}{\partial I_{CO}} - (1+\beta)\right] + \frac{\partial I_C}{\partial I_{CO}} R_E = 0$$

or $$\left[\frac{R_B + R_E}{\beta} + R_E\right]\frac{\partial I_C}{\partial I_{CO}} = (1+\beta)\frac{R_B + R_E}{\beta}$$

or $$S = \frac{\partial I_C}{\partial I_{CO}} = \frac{[(1+\beta)(R_B + R_E)]/\beta}{[(1+\beta)R_E + R_B]/\beta}$$

$$S = \frac{[(1+\beta)(R_B + R_E)]}{[(1+\beta)R_E + R_B]} \qquad \text{------ (8.20)}$$

or $$S = \frac{(1+\beta)\left[1+\frac{R_B}{R_E}\right]}{\left[1+\beta+\frac{R_B}{R_E}\right]} \qquad \text{------ (8.21)}$$

From this equation it is clear that the stability factor S will be equal to unity if $\frac{R_B}{R_E}$ is small and it will be equal to $(1+\beta)$ if $\frac{R_B}{R_E}$ is ∞. So smaller the value of R_B, better will be the stabilization. But if the operating is fixed for low values of R_B, the current drawn from the d.c. source V_{CC} will be large enough. That means power dissipation of the source will be larger. If R_E is increased keeping R_B constant, V_{CC} has to be increased to maintain the same value of quiescent current otherwise there will be loss of gain due to the negative feedback. So there must be compromise between the good stability and low loss of power. The resistance may also be bye passed by a large capacitance to reduce the negative feedback and to improve the stability.

Differentiating the equation (8.19) with respect to V_{BE} , we may obtain the expression for the stability factor S'.

$$-1 = \left[\frac{R_B + (1+\beta)R_E}{\beta}\right]\frac{\partial I_C}{\partial V_{BE}}$$

or $$S' = \frac{\partial I_C}{\partial V_{BE}} = \frac{-\beta}{[R_B + (1+\beta)R_E]}$$

This equation may further be manipulated as

$$S' = \frac{-\beta(1+\beta)(R_B + R_E)}{[R_B + (1+\beta)R_E]x(1+\beta)(R_B + R_E)}$$

or $$S' = \frac{-\beta.S}{(1+\beta)(R_B + R_E)} \qquad \text{------ (8.22)}$$

where $\frac{[(1+\beta)(R_B + R_E)]}{[(1+\beta)R_C + R_B]} = S$.

From this equation it is clear that as S is reduced, S' also reduces.

Further differentiating the equation (8.19) with respect to β, we may obtain the expression for stability factor S''. Rewriting the equation (8.19) we have:

$$\left[\frac{R_B+(1+\beta)R_E}{\beta}\right]I_C=\frac{(V_{BB}-V_{BE})\beta+(R_E+R_B)(1+\beta)I_{CO}}{\beta}$$

If $(1+\beta)I_{CO}\approx\beta I_{CO}$ then

$$I_C=\frac{[V_{BB}-V_{BE}+(R_E+R_B)I_{CO}]\beta}{[R_B+(1+\beta)R_E]} \qquad \text{------- (8.23)}$$

Now differentiating equation (8.23) with respect to β we may have:

$$S''=\frac{\partial I_C}{\partial\beta}=\frac{[R_B+(1+\beta)R_E][V_{BB}-V_{BE}+(R_E+R_B)I_{CO}]-[V_{BB}-V_{BE}+(R_E+R_B)I_{CO}]\beta R_E}{[R_B+(1+\beta)R_E]^2}$$

$$=\frac{[V_{BB}-V_{BE}+(R_E+R_B)I_{CO}][R_B+R_E+\beta R_E-\beta R_E]}{[R_B+(1+\beta)R_E]^2}$$

$$=\frac{[V_{BB}-V_{BE}+(R_E+R_B)I_{CO}][R_B+R_E]}{[R_B+(1+\beta)R_E]^2} \qquad \text{------ (8.24)}$$

Using the equations (8.23) and (8.24) we get:

$$S''=\frac{I_C[R_B+R_E]}{[R_B+(1+\beta)R_E]\beta}x\frac{(1+\beta)}{(1+\beta)}$$

By using the equations (8.24) and (8.20), we may get the expression of S'' in terms of *S* as:

$$S''=\frac{I_C}{\beta}.\frac{S}{(1+\beta)} \qquad \text{------ (8.25)}$$

One can observe from this equation that the stability S'' is a function of β. So if there is change in β due to some reason then what value of β should be taken in to account while calculating the stability factor S'', the initial value, final value or the average value of β. This type of problem does not occur in S or S'. To sort out this problem, let us calculate the change in collector current I_C due to change in β given by:

$$\Delta I_C=S''.\Delta\beta=\frac{S.I_C}{\beta(1+\beta)}\Delta\beta$$

Now we calculate S'', by taking the ratio of the difference of I_C and the difference of β as:

$$S''=\frac{I_{C2}-I_{C1}}{\beta_2-\beta_1}=\frac{\Delta I_C}{\Delta\beta}$$

From equation (8.23), we have

$$\frac{I_{C2}}{I_{C1}} = \left(\frac{\beta_2}{\beta_1}\right)\left(\frac{R_B + (1+\beta_1)R_E}{R_B + (1+\beta_2)R_E}\right)$$

or $$\frac{I_{C2}}{I_{C1}} - 1 = \left(\frac{\beta_2}{\beta_1}\right)\left(\frac{R_B + (1+\beta_1)R_E}{R_B + (1+\beta_2)R_E}\right) - 1$$

or $$\frac{I_{C2} - I_{C1}}{I_{C1}} = \left(\frac{\beta_2 - \beta_1}{\beta_1}\right)\left(\frac{R_B + R_E}{R_B + (1+\beta_2)R_E}\right)$$

or $$S'' = \frac{\Delta I_C}{\Delta \beta} = \left(\frac{I_{C1}}{\beta_1}\right)\left(\frac{R_B + R_E}{R_B + (1+\beta_2)R_E}\right)$$

or $$S'' = \left(\frac{I_{C1}}{\beta_1}\right)\left(\frac{S_2}{(1+\beta_2)}\right) \qquad \text{------ (8.26)}$$

Where S_2 is the value of S for $\beta = \beta_2$ given by $S_2 = \dfrac{[(1+\beta)(R_B + R_E)]}{[(1+\beta)R_E + R_B]}$

If $\Delta\beta \to 0$ so that $I_{C1} \cong I_{C2}$, then equation (8.26) reduces to

$S'' = \dfrac{I_C}{\beta}.\dfrac{S}{(1+\beta)}$, which is the same as equation (8.25).

Equation (8.26) signifies that the maximum value of S_2 can be determined, for a given span of β and a given value of I_{C1}. The variation in β may be due to the temperature variation or unit to unit variation of the transistor.

Example 8.1: Show that S for self biasing circuit may be put in the form

$$S = \frac{G_E + G_1 + G_2}{\left[\dfrac{G_E}{(1+\beta)} + G_1 + G_2\right]}$$ where the G's are conductances corresponding to the R's used in self biasing circuit.

Solution: The stability factor S for self biasing circuit is given by equation (8.20), which is rewritten here.

$$S = \frac{[(1+\beta)(R_B + R_E)]}{[(1+\beta)R_E + R_B]} \qquad \text{------ (8.27)}$$

where $R_B = \dfrac{R_1 R_2}{R_1 + R_2}$ or $\dfrac{1}{R_B} = \dfrac{1}{R_1} + \dfrac{1}{R_2} = G_1 + G_2$ ------ (8.28)

and $G_E = \dfrac{1}{R_E}$ ------ (8.29)

Using the equations (8.27) to (8.29) we have:

$$S = \frac{[(1+\beta)R_B(1+\frac{R_E}{R_B})]}{[(1+\beta)\frac{R_E}{R_B}+1]R_B}$$

$$S = \frac{(1+\beta)[1+\frac{1}{G_E}(G_1+G_2)]}{[1+\frac{(1+\beta)}{G_E}(G_1+G_2)]}$$

or $$= \frac{(1+\beta)[G_E+G_1+G_2)]}{[G_E+(1+\beta)(G_1+G_2)]}$$

or $$S = \frac{G_E+G_1+G_2}{\left[\frac{G_E}{(1+\beta)}+G_1+G_2\right]}$$ Hence proved.

Example 8.2: For the two battery transistor circuit shown in figure (8.8), prove that the stabilization factor *S* is given by:

$$S = \frac{1+\beta}{1+\beta\frac{R_E}{(R_E+R_B)}}$$

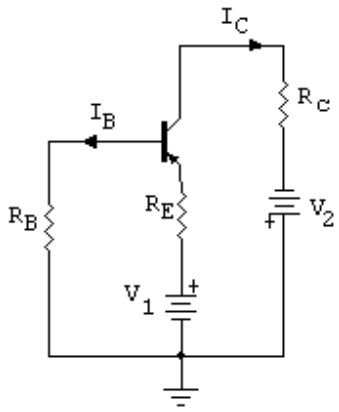


Fig. 8.8

Solution: Applying the KVL to the input circuit we get:

$$V_1 - V_{BE} = I_B R_B + I_B R_E + R_E I_C$$

or $$V_1 - V_{BE} = I_B(R_B+R_E) + R_E I_C$$ ----- (8.30)

The collector current is given by:

$$I_C = \beta I_B + (1+\beta)I_{CO}$$

or $$I_B = \frac{[I_C - (1+\beta)I_{CO}]}{\beta}$$ ------ (8.31)

From equations (8.30) & (8.31), we have

$$V_1 - V_{BE} = \left[\frac{I_C - (1+\beta)I_{CO}}{\beta}\right](R_B + R_E) + R_E I_C \qquad \text{------ (8.32)}$$

Differentiating the equation (8.32) with respect to I_{CO} , we may obtain the expression for the stability factor *S*.

$$\frac{\partial I_C}{\partial I_{CO}} R_E - \frac{(R_E + R_B)(1+\beta)}{\beta} + \frac{(R_B + R_E)}{\beta}\frac{\partial I_C}{\partial I_{CO}} = 0$$

$$\frac{(1+\beta)R_E + R_B}{\beta} S = \frac{(1+\beta)(R_E + R_B)}{\beta}$$

or $$S = \frac{(1+\beta)(R_E + R_B)}{[R_B + R_E + \beta R_E]}$$

or $$S = \frac{1+\beta}{1+\beta\dfrac{R_E}{(R_E + R_B)}}$$ Hence proved.

Example 8.3: In the transformer coupled amplifier shown in figure (8.9),

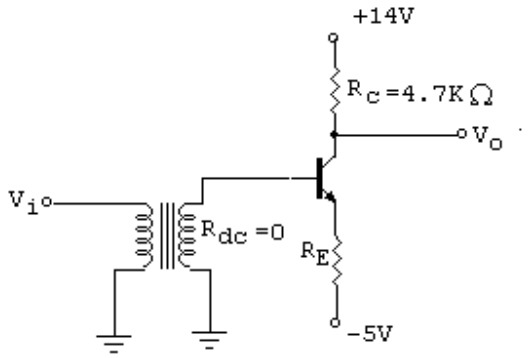


Fig. 8.9

V_{BE} = 0.7 volt, β = 45 and V_{CE} = 4 volts. Determine R_E and stability factor *S*.

Solution: Applying the KVL to the output circuit we get:

$$14 + 5 \approx 4.7K.I_C + 4 + R_E.I_C$$

or $$(4.7K + R_E)I_C = 15V$$

Applying KVL to the input circuit we get:

$$V_{BE} + R_E I_C - 5V = 0$$

or $$0.7 + I_C R_E = 5V$$

or $$I_C R_E = 4.3V$$

Solving for R_E and I_C, we have $I_C = 2.28\ mA$ and $R_E = 1.9\ K\Omega$.

The given circuit is the Thevenin's equivalent of the self biasing circuit with $R_B = 0$.

The stability factor S is given by:

$$S = \frac{1+\beta}{1+\beta\dfrac{R_E}{(R_E + R_B)}}$$

Putting the value of R_B =0, we get $S = 1$.

Example 8.4: Assume that a Silicon transistor with $\beta = 45$, $V_{BE} = 0.7$ V, $V_{CC} = 24$ V and $R_C = 5\ K\Omega$ is used in self biasing *CE* amplifier. It is desired to establish a *Q* point at $V_{CE} = 14$ V and $I_C = 1.7\ mA$ and stability factor $S \le 4$. Find the values of R_E, R_1 and R_2.

Solution: The current in R_E is $I_E \approx I_C = 1.7mA$

Applying the KVL to the output circuit we have,

$$24V = (5K + R_E)1.7mA + 14V$$

or $$5K + R_E = \frac{10}{1.7} K\Omega = 5.88K\Omega$$

or $$R_E = 0.88K\Omega$$

The stability factor S is given by:

$$S = \frac{(1+\beta)\left[1+\dfrac{R_B}{R_E}\right]}{\left[1+\beta+\dfrac{R_B}{R_E}\right]}$$

$$4 = 46\frac{1+\dfrac{R_B}{R_E}}{46+\dfrac{R_B}{R_E}}$$

or $$4x46 + 4\frac{R_B}{R_E} = 46 + 46\frac{R_B}{R_E}$$

or $$\frac{R_B}{R_E} = 3.29$$

or $R_B = 3.29x0.88K\Omega = 2.9K\Omega$

Applying the KVL to the input circuit, we get

$$V_{BB} = R_B I_B + V_{BE} + R_E I_C$$

$$I_B = \frac{I_C}{\beta} = \frac{1.7mA}{45} = 37.8\mu A$$

or $V_{BB} = 2.9x10^3 x37.8x10^{-6} + 0.7 + 0.88x10^3 x1.7x10^{-3}$

$= 2.31V$

Further $V_{BB} = \frac{V_{CC} x R_2}{R_1 + R_2}$ and $R_B = \frac{R_1 x R_2}{R_1 + R_2}$

$$2.31 = \frac{14xR_2}{R_1 + R_2} \quad \text{or} \quad \frac{R_2}{R_1 + R_2} = \frac{2.31}{14} = 0.165$$

$$R_1 = \frac{2.9K}{0.165} = 17.6K\Omega \text{ and } R_2 = 3.48K\Omega$$

Variation of Operating Point Stability with Simultaneous Variation of I_{CO}, V_{BE} and β: We have defined the stability factors S, S', and S'' as the partial derivatives of I_C with I_{CO}, V_{BE} and β respectively. Each partial derivatives is calculated when all other parameters are held constant. The collector current is a function of I_{CO}, V_{BE} and β as

$$I_C = f(I_{C0}, V_{BE}, \beta)$$

The total change in collector current due to the simultaneous variation in I_{CO}, V_{BE} and β may be given as

$$\Delta I_C = \frac{\partial I_C}{\partial I_{CO}}.\Delta I_{CO} + \frac{\partial I_C}{\partial V_{BE}}.\Delta V_{BE} + \frac{\partial IC}{\partial \beta}.\Delta\beta$$

$$= S.\Delta I_{CO} + S'.\Delta V_{BE} + S''\Delta\beta \qquad \text{------ (8.33)}$$

The stability factor S, S', and S'' may be expressed in terms of parameter M defined by $M = \frac{1}{1 + \frac{R_B}{[R_E(1+\beta)]}}$

$$S = \frac{[(1+\beta)(R_B + R_E)]}{[(1+\beta)R_E + R_B]} = \frac{(1+\beta)R_E(1+\frac{R_B}{R_E})}{(1+\beta)R_E[1+\frac{R_B}{R_E(1+\beta)}]} = (1+\frac{R_B}{R_E})M \qquad \text{------ (8.34)}$$

$$S' = \frac{-\beta.S}{(1+\beta)(R_B + R_E)} = \frac{-\beta M (R_B + R_E)}{(1+\beta)(R_B + R_E)R_E} = \frac{-M}{R_E} \qquad \text{------ (8.35)}$$

(Provided $\beta >>1$)

and $$S'' = \left(\frac{I_{C1}}{\beta_1}\right)\left(\frac{S_2}{(1+\beta_2)}\right) = \frac{I_{C1}}{\beta_1}\frac{M_2(1+\frac{R_B}{R_E})}{(1+\beta_2)} \approx (1+\frac{R_B}{R_E})\frac{I_{C1}M_2}{\beta_1\beta_2}$$ ------- (8.36)

(Provided $\beta >>1$ and M_2 is the value of M for $\beta = \beta_2$)

Putting the values of S, S', and S'' from equations (8.34) to (8.35) in equation (8.33), we get the fractional change in collector current as:

$$\frac{\Delta I_C}{I_{C1}} = \left(1+\frac{R_B}{R_E}\right)\frac{M_1\Delta I_{CO}}{I_{C1}} - \frac{M_1\Delta V_{BE}}{I_{C1}R_E} + \left(1+\frac{R_B}{R_E}\right)\frac{M_2\Delta\beta}{\beta_1\beta_2}$$ ------ (8.37)

Once ΔI_C is known, ΔV_{CE} is obtained from the d.c. load line.

The range of temperature over which a transistor is operated is normally – 65^0C to + 75^0C for Germanium transistor and – 65^0C to + 175^0C for Silicon transistor. Table 8.1 illustrates the typical parameters for both Silicon and Germanium transistors for their respective temperature range of operation. It is of significance to study the order of ΔI_C due to the change in I_{CO}, V_{BE} and β over the temperature range of operation.

Table : 8.1

Parameters	Silicon Transistor			Germanium Transistor		
	$-\ 65^0C$	$+25^0C$	$+175^0C$	$-\ 65^0C$	$+25^0C$	$+\ 75^0C$
I_{CO}	$1.95x10^{-3}nA$	$1.0nA$	$33000nA$	$1.95x10^{-3}\mu A$	$1.0\mu A$	$32\mu A$
β	25	55	100	20	55	90
$V_{BE}(volts)$	0.78	0.6	0.225	0.38	0.2	0.1

It may clearly be seen from this table that at room temperature (25^0C) both Si and Ge transistors have the same value of β. For Si I_{CO} is much smaller than Ge. However, I_{CO} approximately doubles for every 10^0C and V_{BE} decreases by approximately 2.5 mV/^{0}C rise in temperature.

Making use of the parameters given in table (8.1) and equation (8.37), one can easily find that the change in collector current for both Si and Ge transistors for their respective range of temperature. The study reveals that the collector current variation for 240^0C change of temperature for Si transistor is approximately same for 140^0C change of temperature for Ge transistor. This means that the variation of collector current due to temperature is more for Ge transistor than that of Si. Hence the Si transistor is superior. However, for Si the influence of V_{BE} is more significant than that of I_{CO} on collector current.

Example 8.5 : For the self bias CE transistor (Ge) amplifier has the following values R_E = 4.5 KΩ, R_1 = 90 KΩ, R_2 =10 KΩ. The collector supply voltage and collector resistance R_C are adjusted so as to give the collector current of 1.5 *mA* at 25 ^{0}C. Determine the variation of collector current in the temperature range +25 ^{0}C to + 75 ^{0}C, when the Ge transistor of table 8.1 is used.

Solution: The resistance R_B is given by: $R_B = \dfrac{R_1 x R_2}{R_1 + R_2} = \dfrac{90x10}{100} = 9K\Omega$

The ratio $\dfrac{R_B}{R_E} = \dfrac{9K\Omega}{4.5K\Omega} = 2$

At room temperature $M_1 = \dfrac{1}{1 + \dfrac{R_B}{[R_E(1+\beta)]}} = \dfrac{1}{1 + \dfrac{2}{56}} \cong 1$

The stability factor S at room temperature (+25 ^{0}C) is given by:

$$S \text{ (at } +25\ ^0\text{C)} = (1 + \frac{R_B}{R_E})M_1 = 3$$

The stability factor S' is given by:

$$S' \text{ (at } +25\ ^0\text{C)} = \frac{-M}{R_E} = -\frac{1}{4.5} mA/V = -0.22 mA/V$$

The stability factor S'' is given by

$$S'' \text{ (at } +75\ ^0\text{C)} = (1 + \frac{R_B}{R_E})\frac{I_{C1}M_2}{\beta_1\beta_2} = \frac{(1+2)x1.5x10^{-3}x1}{55x90} = 0.91x10^{-6}\, mA$$

where M_2 is also approximately equal to unity.

The change in I_C in the temperature range from (+25 ^{0}C to + 75 ^{0}C) is obtained as:

$$\begin{aligned}\Delta I_C &= S.\Delta I_{CO} + S'.\Delta V_{BE} + S''\Delta\beta \\ &= 3x31x10^{-6} + (-0.22x10^{-3})(-0.1) + (0.91x10^{-6})(35) \\ &= 0.147mA\end{aligned}$$

8.8 Bias Compensation: For the proper use of transistor as amplifier it has earlier been discussed that it is to be biased in the active region. The operating point should be made stable against the variation of temperature. The resistive biasing circuits have been used for stabilization techniques that maintain the collector current stable in spite of variation in I_{CO}, β and V_{BE} with temperature or unit to unit variation. However, in the Compensation technique, the temperature sensitive devices such as diodes, transistors, thermistors etc. are introduced that compensate the change of V_{BE} or I_{CO} in the circuits.

8.8.1 Diode Compensation for V_{BE}: Figure (8.10) illustrates the self biasing circuit having the diode *D* connected in the emitter circuit as the compensating element. This diode *D* is biased in the forward direction by a voltage source V_{DD} through a resistance r_d.

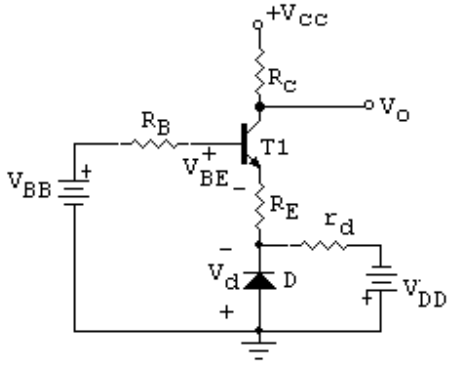


Fig.8.10

Further it is assumed that the diode *D* and the transistor T_1 are made up of the same material and same type. The voltage drop across the diode V_d and the voltage across the emitter base junction V_{BE} will therefore have the same temperature coefficient.

Applying the KVL to the input circuit, we have:

$$-V_{BB} + R_B I_B + V_{BE} + R_E (I_C + I_B) - V_d = 0$$

or $$(V_{BB} - V_{BE} + V_d) = R_E I_C + (R_B + R_E) I_B \qquad \text{------ (8.38)}$$

But $$I_C = \beta I_B + (1+\beta) I_{CO}$$

or $$I_B = \frac{I_C - (1+\beta) I_{CO}}{\beta} \qquad \text{------ (8.39)}$$

Substituting the value of I_B from equation (8.38) in equation (8.39), we get

$$(V_{BB} - V_{BE} + V_d) = R_E I_C + (R_B + R_E)\left[\frac{I_C - (1+\beta) I_{CO}}{\beta}\right]$$

$$\beta[V_{BB} - (V_{BE} - V_d)] + I_{CO}(R_B + R_E)(1+\beta) = I_C[R_B + R_E + \beta R_E]$$

or $$I_C = \frac{\beta[V_{BB} - (V_{BE} - V_d)] + I_{CO}(R_B + R_E)(1+\beta)}{R_B + R_E(1+\beta)} \qquad \text{------ (8.40)}$$

From the equation (8.40), it is clear that since the variation in V_{BE} and V_d is the same as both are made of the same material and type, the factor $(V_{BE} - V_d)$ will remain to be constant. That is the change in V_{BE} is compensated by the change in V_d by almost the same amount. Hence I_C will be insensitive to the variations in V_{BE}. Although this compensation is not perfect but it is sufficiently effective to take care the transistor drift due to the variations in V_{BE}. This compensation technique is generally used in Silicon transistors as the change of V_{BE} has dominating effect in the variation of I_C.

8.8.2 Diode Compensation for I_{CO}: The diode compensation for I_{CO} is useful for stabilizing the Germanium transistors as the variation of I_{CO} with temperature is quite large in these transistors. Figure (8.11) illustrates the circuit of Germanium transistor

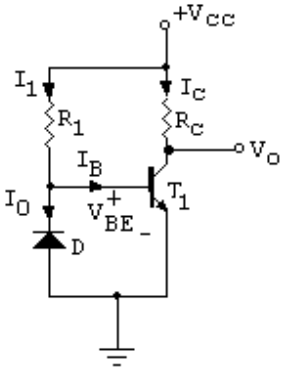


Fig. 8.11

amplifier in which a diode D is connected between emitter and the base of the transistor. If the diode and the transistor are made up of the same material and the type, the reverse saturation current I_0 of the diode will increase with temperature at the same rate as the transistor collector saturation current I_{CO}.

From the figure (8.11), we get

$$I_1 = \frac{V_{CC} - V_{BE}}{R_1}$$

Since the diode is reverse biased by an amount $V_{BE} \approx 0.2$ volt for Germanium devices, so

$$I_1 \approx \frac{V_{CC}}{R_1} = \text{Constant}$$

The base current is given by:

$$I_B = I_1 - I_0 \qquad \text{------ (8.41)}$$

But we have the collector current I_C as:

$$I_C = \beta I_B + (1+\beta) I_{CO} \qquad \text{------ (8.42)}$$

Substituting the value of I_B from equation (8.41) in equation (8.42) we have:

$$I_C = \beta I_1 - \beta I_0 + (1+\beta) I_{CO}$$

If $\beta >> 1$, then

$$I_C \approx \beta I_1 - \beta I_0 + \beta I_{CO}$$

From this equation it is clear that if the diode and the transistor T1 are made up of same material and type, the variation of I_0 and I_{CO} will be nullified and the collector current will remain to be essentially constant over the entire range of temperature.

Example 8.6: For the biasing arrangement shown in figure (8.11) and assuming that the reverse saturation currents of the diode and transistor are equal, prove that

$$S = 1$$

$$S' = -\frac{\beta}{R_1}$$

$$S'' = \frac{\Delta(I_C - I_{CO})}{\Delta\beta} = \frac{I_{C1} - I_{CO1}}{\beta_1}$$

Solution: We have

$$I_1 = \frac{V_{CC} - V_{BE}}{R_1} \approx \frac{V_{CC}}{R_1} = \text{constant}$$

and

$$I_C = \beta I_B + (1 + \beta) I_{CO}$$

$$I_B = I_1 - I_0$$

where I_0 is the reverse saturation current of the diode.

or

$$I_C = \beta I_1 - \beta I_0 + (1 + \beta) I_{CO}$$

$$= \beta I_1 - \beta I_0 + I_{CO} + \beta I_{CO}$$

Since $I_{CO} = I_0$ so we have

$$I_C = \beta I_1 + I_{CO} \qquad \text{------ (8.43)}$$

Differentiating this equation with respect to I_{CO}, we get

$$\frac{\partial I_C}{\partial I_{CO}} = 0 + 1$$

or

$$S = \frac{\partial I_C}{\partial I_{CO}} = 1$$ Proved part I

Equation (8.43) may be rewritten:

$$I_C = \frac{\beta}{R_1}(V_{CC} - V_{BE}) + I_{CO}$$

Differentiating this equation with respect to V_{BE}, we get

$$\frac{\partial I_C}{\partial V_{BE}} = -\frac{\beta}{R_1}$$

or

$$S' = \frac{\partial I_C}{\partial V_{BE}} = -\frac{\beta}{R_1}$$ Proved part II

Equation (8.43) may be rewritten as

$$I_C - I_{CO} = \beta I_1$$

or $$I_{C2} - I_{CO2} = \beta_2 I_1$$

and $$I_{C1} - I_{CO1} = \beta_1 I_1$$

or $$\frac{I_{C2} - I_{CO2}}{I_{C1} - I_{CO1}} - 1 = \frac{\beta_2}{\beta_1} - 1$$

or $$\frac{\Delta I_C - \Delta I_{CO}}{I_{C1} - I_{CO1}} = \frac{\beta_2 - \beta_1}{\beta_1}$$

or $$\frac{\Delta(I_C - I_{CO})}{\Delta\beta} = \frac{I_{C1} - I_{CO1}}{\beta_1}$$

$$S'' = \frac{\Delta(I_C - I_{CO})}{\Delta\beta} = \frac{I_{C1} - I_{CO1}}{\beta_1}$$ Proved part III

8.9 Thermistor and Sensistor Compensation: In this compensation technique, the temperature sensitive element such as a thermistor or a sensistor is used rather than diode.

8.9.1 Thermistor Compensation: Figure (8.12) shows the self-bias CE transistor amplifier with thermistor compensation. Thermistor has a negative temperature

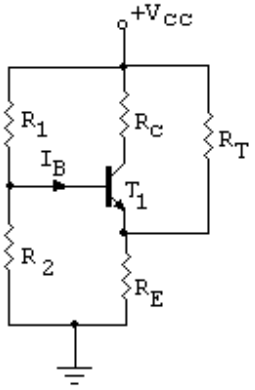


Fig. 8.12

coefficient of resistance i.e. its resistance decreases with the increase of temperature. Thermistor R_T connected between emitter and the positive supply compensate the variation of I_C with I_{CO}, V_{BE} or β caused due to the variation in temperature. As the temperature increases the resistance of R_T decreases, the current flowing through R_T in to R_E increases. Since voltage across R_E is in the direction to reverse bias the emitter base junction of the transistor, the increase in temperature will reduce the net forward bias of the emitter junction and as a result the collector current will remain fairly constant.

An alternative configuration using thermistor compensation is shown in Figure (8.13), in which the thermistor R_T is placed in parallel with R_2. As the temperature

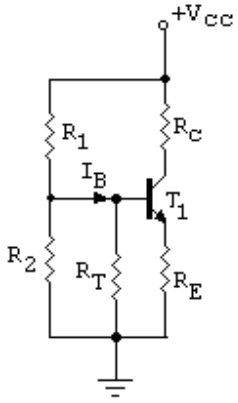


Fig. 8.13

increases, R_T decreases also the drop across R_T decreases. This results the decrease in collector current. This reduced I_C tends to compensate for the increased collector current caused by the rise in temperature.

8.9.2 Sensistor Compensation: For the operating point stability of the transistor, sensistor which has the positive temperature coefficient of resistance may be used as the compensating element instead of thermistor. The sensistor is placed either in parallel with R_1 or in parallel with R_E or in place of R_E in the self bias circuit. This may be explained that the net voltage drop across R_2 increases reducing thereby the collector current.

8.10 Thermal Runaway: In transistor amplifier, one must control the biasing of the transistor in such a way as to keep the average dissipation of the device below its maximum value. The amplifier design leads the operating point to shift with temperature in such a way as to burn out the transistor. The biasing problem is aggravated by an effect known as Thermal Runaway, which may be explained as follows. The collector-base junction of the transistor, where all of the dissipation occurs, is not in perfect contact with the transistor case. That is, there is some thermal resistance between the junction and the case, and between the case and ambient. Thus the power dissipated in the transistor will heat the junction to a temperature considerably above the ambient. The internal increase in temperature will cause a change in the transistor characteristics which in turn increases the power dissipation and the internal temperature. This cyclic chain of events will cause the rapid rise in transistor temperature and finally result the destruction of the transistor, known as thermal runaway. The power dissipation can be controlled and hence the thermal runaway be prevented, if the biasing arrangement is designed to keep the operating point approximately fixed in the $V_{CE} - I_C$ plane.

8.10.1 Thermal Resistance: Let T_J is the temperature of the collector base junction and T_A is the ambient temperature of the transistor. Here $T_J > T_A$ due to heating within the transistor. It has been observed that:

$$(T_J - T_A) \propto P_D$$

Where P_D is the power dissipated in the transistor.

or $(T_J - T_A) = \Theta P_D$ ------ (8.44)

Θ is the proportionality constant and is called as thermal resistance given in ^{0}C/watt.

or $T_J = T_A + \Theta P_D$

The collector junction temperature T_J is higher than the ambient temperature T_A by an amount equal to the product of thermal resistance Θ and power dissipation P_D in the transistor.

The value of the thermal resistance Θ depends upon

1. the size of the transistor,
2. convection and radiation,
3. on the nature of the cooling like forced air colling, and
4. the thermal connection of the device to the metal chassis or a heat sink.

Typical values for various transistor designs vary from 0.2 ^{0}C/watt for high power transistor fitted with a suitable heat sink to 100 ^{0}C/watt for a low power transistor which has no cooling arrangement.

The maximum collector power P_C permitted for a transistor for safe operation at 25 ^{0}C (room temperature) is specified by the manufacturer. For ambient temperatures above this value, the maximum permissible value reduces. The P_C reduces to zero at the temperature at which the transistor may operate, that is this value is 100 ^{0}C for Ge transistor and 225 ^{0}C for Si transistor as shown in figure (8.14).

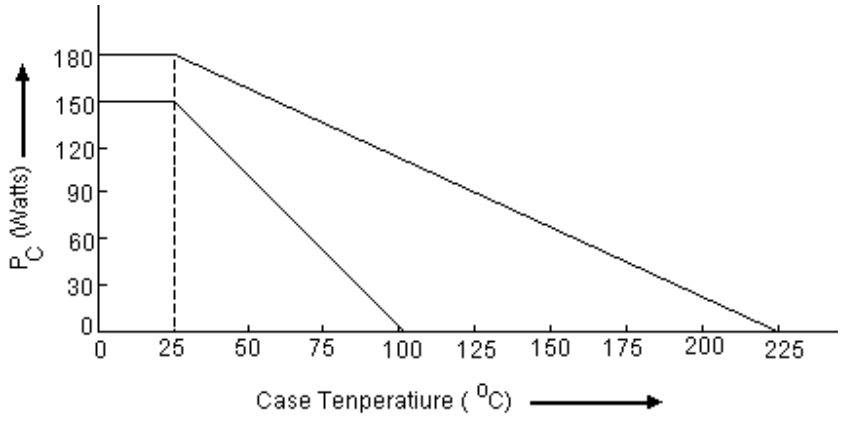


Fig. 8.14

8.10.2 Condition to Prevent Thermal Runaway: The condition to avoid the thermal runaway is that the rate at which heat is released at the collector junction must not exceed the rate at which heat can be dissipated i.e.

$$\frac{\partial P_C}{\partial T_J} < \frac{\partial P_D}{\partial T_J} \qquad \text{------ (8.45)}$$

Differentiating equation (8.44) with respect to T_J, we have

$$1 = \Theta \frac{\partial P_D}{\partial T_J}$$

or $$\frac{\partial P_D}{\partial T_J} = \frac{1}{\Theta} \qquad \text{------ (8.46)}$$

From equations (8.45) and (8.46), we get

$$\frac{\partial P_C}{\partial T_J} < \frac{1}{\Theta} \qquad \text{------ (8.47)}$$

To prevent the thermal runaway in the transistor this condition must be satisfied.

8.10.3 Thermal Stability: It is well known that the transistor is to be biased in the active region for the transistor to work as an amplifier. The power generated at the collector junction with no signal is given by:

$$P_C = I_C V_{CE} \approx I_C V_{CE} \qquad \text{------ (8.48)}$$

If we assume that he quiescent collector current almost equals the emitter current, then equation (8.48) can be written as:

$$P_C = I_C \left[V_{CC} - I_C (R_E + R_C) \right]$$

$$= I_C V_{CC} - I_C^2 (R_E + R_C) \qquad \text{------ (8.49)}$$

The condition (equation 8.47) to prevent the thermal runaway is rewritten as:

$$\frac{\partial P_C}{\partial I_C} \cdot \frac{\partial I_C}{\partial T_J} < \frac{1}{\Theta} \qquad \text{------ (8.50)}$$

Here Θ and $\frac{\partial I_C}{\partial T_J}$ are positive, hence equation (8.50) is always satisfied when $\frac{\partial P_C}{\partial I_C}$ is negative. Differentiating equation (8.49) with respect to I_C we get,

$$\frac{\partial P_C}{\partial I_C} = V_{CC} - 2 I_C (R_E + R_C) \qquad \text{------ (8.51)}$$

Hence to prevent the thermal runaway the right hand side of the equation (8.51) should be negative. That is

$$I_C > \frac{V_{CC}}{2(R_E + R_C)} \qquad \text{------ (8.52)}$$

But $V_{CE} = V_{CC} - I_C(R_E + R_C)$

So the equation (8.52) implies that

$$V_{CE} < \frac{V_{CC}}{2} \qquad \text{----- (8.53)}$$

If I_C is not greater than $\frac{V_{CC}}{2(R_E + R_C)}$ and $V_{CE} > \frac{V_{CC}}{2}$ then $\frac{\partial P_C}{\partial I_C}$ is positive; and thus the thermal runaway is not prevented. To ensure that the thermal runaway does not occur, the equation (8.50) should be satisfied.

It is well known that I_C depends on the variation of I_{CO}, V_{BE} and β due to the junction temperature T_J .

Hence

$$\Delta I_C = \frac{\partial I_C}{\partial I_{CO}}\Delta I_{CO} + \frac{\partial I_C}{\partial V_{BE}}\Delta V_{BE} + \frac{\partial I_C}{\partial \beta}\Delta\beta$$

$$\frac{\partial I_C}{\partial T_J} = \frac{\partial I_C}{\partial I_{CO}}.\frac{\partial I_{CO}}{\partial T_J} + \frac{\partial I_C}{\partial V_{BE}}.\frac{\partial V_{BE}}{\partial T_J} + \frac{\partial I_C}{\partial \beta}.\frac{\partial \beta}{\partial T_J}$$

$$\frac{\partial I_C}{\partial T_J} = S.\frac{\partial I_{CO}}{\partial T_J} + S^{'}.\frac{\partial V_{BE}}{\partial T_J} + S^{''}.\frac{\partial \beta}{\partial T_J} \qquad \text{----- (8.54)}$$

For a given transistor, $\frac{\partial I_{CO}}{\partial T_J}$, $\frac{\partial V_{BE}}{\partial T_J}$ and $\frac{\partial \beta}{\partial T_J}$ are known and thus by selecting the values of S, $S^{'}$ and $S^{''}$ equation (8.50) may be satisfied.

In some practical problems the effect of I_{CO} dominates, the term $\frac{\partial I_{CO}}{\partial T_J}$ in equation (8.54) also dominates and the problem of thermal runaway may be analyzed as follows. From the equations (8.50) and (8.54) we have:

$$\frac{\partial P_C}{\partial I_C}.\left[S\frac{\partial I_{CO}}{\partial T_J}\right] < \frac{1}{\Theta} \qquad \text{------ (8.55)}$$

Here we neglected all the terms on right hand side of equation (8.54) except first.

It is well known that for both Ge and Si transistors the reverse saturation current I_{CO} increases by 7 percent/ ^{0}C i.e.,

$$\frac{\partial I_{CO}}{\partial T_J} = 0.07 I_{CO} \qquad \text{------ (8.56)}$$

Putting the value of $\frac{\partial P_C}{\partial I_C}$ from (8.51) and the value of $\frac{\partial I_{CO}}{\partial T_J}$ from equation (8.56) in equation (8.55) we have:

$$[V_{CC} - 2I_C(R_E + R_C)].[S(0.07I_{CO})] < \frac{1}{\Theta} \qquad \text{------ (8.57)}$$

This equation gives the necessary condition for avoiding thermal runaway and is valid for any NPN transistor. It can also be valid for any PNP transistor if I_C and I_{CO} are used to represent the magnitudes of the current.

Equation (8.57) shows that amplifiers operated at low currents and used for stability factor S<10 are almost free from thermal runaway. On the other hand, power amplifiers operating at high power levels use R_E of low value, resulting high value of stability factor S. Thus in power amplifiers, thermal runaway is the common problem and care must be taken in the design of power amplifiers to prevent the thermal runaway.

Example 8.7: What will be the temperature of the junction if the transistor dissipates 2.5 watt of power? Given for a transistor Θ = 12 ^{0}C/watt and working in an ambient temperature of 25 ^{0}C.

Solution: We have

$$T_J = T_A + \Theta P_D$$
$$= 25 + 12x2.5 = 55\ ^0\text{C}$$

Example 8.8 : Find the maximum permissible value of thermal resistance Θ required for a transistor so that the circuit is thermally stable. Given that V_{CC} = 24 volts, V_{CE} = 15 volts R_C = 5 KΩ R_E = 1.5 KΩ, I_C = 1.5 mA and stability factor S = 8. Assume I_{CO} = 2.0 nA at 25 ^{0}C.

Solution: $\frac{V_{CC}}{2} = \frac{24}{2} = 12$ volts

and V_{CE} = 15 volts

Since V_{CE} > (V_{CC} / 2), the circuit is not stable. The condition to prevent thermal runaway is

$$[V_{CC} - 2I_C(R_E + R_C)].[S(0.07I_{CO})] < \frac{1}{\Theta}$$

or $$[24 - 2x1.5mA(1.5 + 5)K\Omega].[8(0.07x2x10^{-9})] < \frac{1}{\Theta}$$

or $$4.5x1.12x10^{-9} < \frac{1}{\Theta}$$

or $$\Theta < \frac{10}{5.04}x10^8$$

or $$\Theta < 1.98x10^8\ ^0\text{C/Watt}$$

Example 8.9: Figure (8.15) shows a power amplifier using a PNP

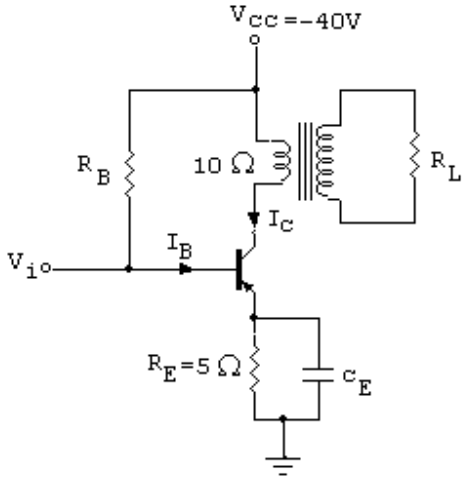


Fig. 8.15

Ge transistor with β = 120 and I_{CO} = – 6 mA . The quiescent collector current I_C = – 1A. Find

(i) the value of R_B

(ii) the largest value of Θ that can result the thermal stable circuit . Assume that the effect of I_{CO} dominates.

Solution: (i) The collector current is given by:

$$I_C = \beta I_B + (1+\beta) I_{CO} \approx \beta (I_B + I_{CO})$$

or $$I_B = \frac{I_C - \beta I_{CO}}{\beta} = -\frac{1 - 120\ x 6 x 10^{-3}}{120} A = -2.8 \text{ mA}$$

Neglecting V_{BE}, we get

$$R_B = \frac{V_{CC} - V_E}{I_B} = \frac{40-5}{2.8} k\Omega$$

$$= 12.5 k\Omega$$

(ii) $$|V_{CE}| = 40 - (10+5)1A = 25$$

and $$\frac{V_{CC}}{2} = \frac{40}{2} = 20V$$

Since $V_{CE} > V_{CC}/2$, so the circuit is not inherently stable. The stability factor is given by:

$$S = \frac{(1+\beta)\left[1+\frac{R_B}{R_E}\right]}{\left[1+\beta+\frac{R_B}{R_E}\right]} = 121x\frac{1+\frac{12500}{5}}{121+\frac{12500}{5}} = 115.5$$

The condition to prevent thermal runaway is

$$[V_{CC} - 2I_C(R_E + R_C)][S(0.07I_{CO})] < \frac{1}{\Theta}$$

or $$[40 - 2x1(5+10)][115.5(0.07x6x10^{-3})] < \frac{1}{\Theta}$$

or $$10x115.5x0.42x10^{-3} < \frac{1}{\Theta}$$

or $$\Theta < 2.06\ ^0\text{C/Watt}$$

Problems:

1. What do you understand by transistor biasing? Why it is needed?
2. What do you mean by the stabilization of operating point? Define the Stability factor. Derive the expression for the stability factor for fixed base bias.
3. Draw the self bias circuit. Find the expression for the stability factor for the self bias circuit. Explain qualitatively why such a circuit is an improvement on the fixed bias circuit.
4. List the three sources of instability of collector current. Define the three stability factor.
5. Show that the stability factor for the collector to base bias is $S = \frac{(1+\beta)}{\left[1+\frac{\beta R_C}{R_B+R_C}\right]}$, the symbols have their meaning.
6. Find the expression for the stability factor S' for self bias circuit.
7. Find the expression for the stability factor S'' for self bias circuit.
8. Define Stabilization and Compensation techniques.
9. Show that the stability factor for the self bias is , $S = \frac{(1+\beta)\left[1+\frac{R_B}{R_E}\right]}{\left[1+\beta+\frac{R_B}{R_E}\right]}$, the symbols have their meaning.

10. Prove that the stability factor S' for self bias circuit is $S' = \frac{-\beta.S}{(1+\beta)(R_B + R_E)}$. The symbols have their usual meaning.

11. Prove that the stability factor S'' for self bias circuit is $S'' = \frac{I_C}{\beta}.\frac{S}{(1+\beta)}$. The symbols have their usual meaning.

12. Prove that the stability factor S'' for self bias circuit is $S'' = \left(\frac{I_{C1}}{\beta_1}\right)\left(\frac{S_2}{(1+\beta_2)}\right)$

Where S_2 is the value of S for $\beta = \beta_2$ given by $S_2 = \frac{[(1+\beta)(R_B + R_E)]}{[(1+\beta)R_E + R_B]}$. The other symbols have their usual meaning.

13. Prove that the stability factor S' for collector to base bias circuit is $S' = \frac{-\beta.S}{(1+\beta)(R_B + R_C)}$. The symbols have their usual meaning.

14. Prove that the stability factor S'' for the collector to base bias circuit is $S'' = \frac{I_C}{\beta}.\frac{S}{(1+\beta)}$. Where S is the stability factor for this circuit and other symbols have their meaning.

15. Draw and explain the circuit for diode compensation for the changes in V_{BE}.

16. Draw and explain the circuit for diode compensation for the changes in I_{CO}.

17. Draw and explain the circuit employing thermistor compensation and sesistor compensation.

18. Discuss thermal runaway.

19. Define thermal resistance. What is the condition for thermal stability? Explain.

20. Show that the thermal runaway cannot take place if $V_{CE} < \frac{V_{CC}}{2}$.

21. Determine the operating point for a silicon transistor with $\beta = 50$ used in the self biasing circuit. The circuit components are $V_{CC} = 20$ V, $R_C = 2$ *KΩ*, $R_E = 100$ *Ω*, $R_1 = 100$ *KΩ* and $R_2 = 5$ *KΩ*. Find also the stability factor for this circuit.

22. Determine the operating point for a Germanium transistor with $\beta = 50$ used in the self biasing circuit. The circuit components are $V_{CC} = 20$ V, $R_C = 2$ *KΩ*, $R_E = 100$ *Ω*, $R_1 = 100$ *KΩ* and $R_2 = 5$ *KΩ*. Find also the stability factor for this circuit.

23. In self bias *CE* transistor (NPN) amplifier circuit $R_1 = 90$ *KΩ*, $R_2 = 10$ *KΩ* and $\alpha = 0.98$. Find the value of *S* when emitter resistance is :

(i) 1 *KΩ* and (ii) 2 *KΩ*. (Ans. 8.46, 5.0)

24. An NPN transistor with β = 100 is used in *CE* circuit with V_{CC} =15 V, R_C = 4.7 *KΩ*. Bias is obtained by connecting a 220 *KΩ* resistance from collector to base (collector to base bias). Find Q – point and the stability factor *S*.

(Ans. I_B = 21μA, I_C = 2.1 *mA*, V_{CE} = 5.13*V* & *S* = 32.7)

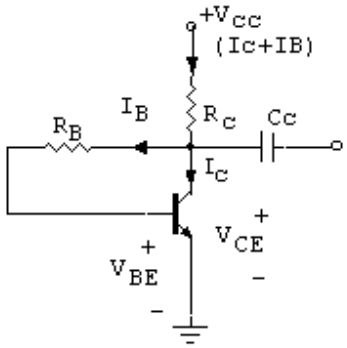
25. A Si NPN transistor is used in self bias CE amplifier, with V_{CC} = 24 V, R_C = 5 *KΩ*, R_E = 1 *KΩ*, R_1 = 81 *KΩ* and R_2 = 9 *KΩ*. The transistor has β = 60. Find the Q – point and the value of stability factor *S*.

(Ans. I_B = 25μA, I_C =1.5 *mA*, V_{CE} = 15*V* & *S* = 8.03)

26. A Si NPN transistor is used in the self bias *CE* amplifier, having β = 50 at room temperature. The circuit has R_1 = 90 *KΩ* and R_2 = 10 *KΩ* and R_E = 1.5 *KΩ*. The values of other components are adjusted to have the collector current I_C equal to 2 *mA*. Calculate the values of *S*, S', S''.

(Ans. *S* = 6.3, S' = –5.85x 10 $^{-4}$ amp/volt, S'' = 4.9x10^{-6} amp.)

27. For the circuit given below



(i) Calculate the values of I_B, I_C & V_{CE} if Si transistor with β = 45 is used. Provided V_{CC} = 12 V, R_C = 2.7 *KΩ*, R_B = 120 *KΩ* and V_{BE} = 0.7 V

(ii) Find the value of R_B so that V_{CE} = 8 Volts.

(Ans. (i) I_B = 46.3 μA, I_C = 2.08 mA V_{CE} = 6.23 V (ii) R_B = 207 KΩ)

28. For the self bias CE transistor (Si) amplifier has the following values R_E = 4.5 KΩ, R_1 = 90 KΩ, R_2 =10 KΩ. The collector supply voltage and collector resistance R_C are adjusted so as to give the collector current of 1.5 *mA* at 25 ^{0}C. Determine the variation of collector current in the temperature range +25 ^{0}C to + 175 ^{0}C, when the Si transistor of table 8.1 is used.
(Ans. 0.21*mA*)

9

Field Effect Transistors

The field effect transistor (FET) is a three terminal semiconductor device in which the current is controlled by an electric field and used for variety of applications in electronics. Unlike the usual transistor, its operation depends upon the flow of majority carriers and the minority charge carriers play no significant role in the operation of the device. It is, therefore, known as a unipolar device. In usual transistors the current flow due to both types of charge carriers (electrons and holes) hence known as bipolar junction transistors (BJT). In the present chapter classification, operation and characteristics of the field effect transistors will be discussed. In addition the biasing and applications of the field effect transistors will also be illustrated.

9.1 Field Effect Transistors: The field effect transistors may broadly be classified in to the following two categories:

(1) Junction Field Effect Transistors (JFET)

(2) Metal Oxide Semiconductor Field Effect Transistors (MOSFET)

It is also called Insulated Gate Field Effect Transistors (IGFET)

MOSFET's are further subdivided in two categories:

(i) Enhancement type MOSFET

(ii) Depletion type MOSFET

Each of the above categories may be either N – channel or P - channel.

The field effect transistors differ from the bipolar junction transistors in the following important characteristics:

FET'S	BJT'S
1. Its operation depends upon the flow of majority charge carriers only. It is therefore a Unipolar device.	Its operation depends upon the flow of both the two types of charge carriers i.e. electrons and holes hence it is named as Bipolar device.
2. It exhibits very high input resistance, typically many megohms.	It exhibits low input resistance. It ranges from 10^2 to 10^6 ohms.
3. It is voltage controlled device i.e. the output (drain) current is controlled by the input (gate) voltage.	It is current controlled device i.e. the output current is controlled by the input current.
4. Less noisy.	More noisy.
5. Thermal stability is good.	Thermal stability is poor.
6. Immune to radiations.	Sensitive to radiations.
7. It is simpler to fabricate and occupies less space in the integrated form.	It occupies more space in the integrated form.
8. It exhibits no offset voltage at zero drain current, and hence makes an excellent signal chopper.	It has offset voltage at zero collector current, hence not used for signal chopper.
9. It has small band width product. (This is the only drawback).	It has high band width product.

9.2 Junction Field Effect Transistor: Figure 9.1(a) shows the structure of N – channel junction field effect transistor. It consists of an *N* – type substrate (bar or channel) in which two P^+ – regions (heavily doped *P* – regions) are diffused. One end of the *N* – type substrate is called source and other end is called drain. The two heavily doped *P* – regions are connected to a third terminal called gate. The detailed sketch of the *N* – channel JFET is shown in figure 9.1(b).

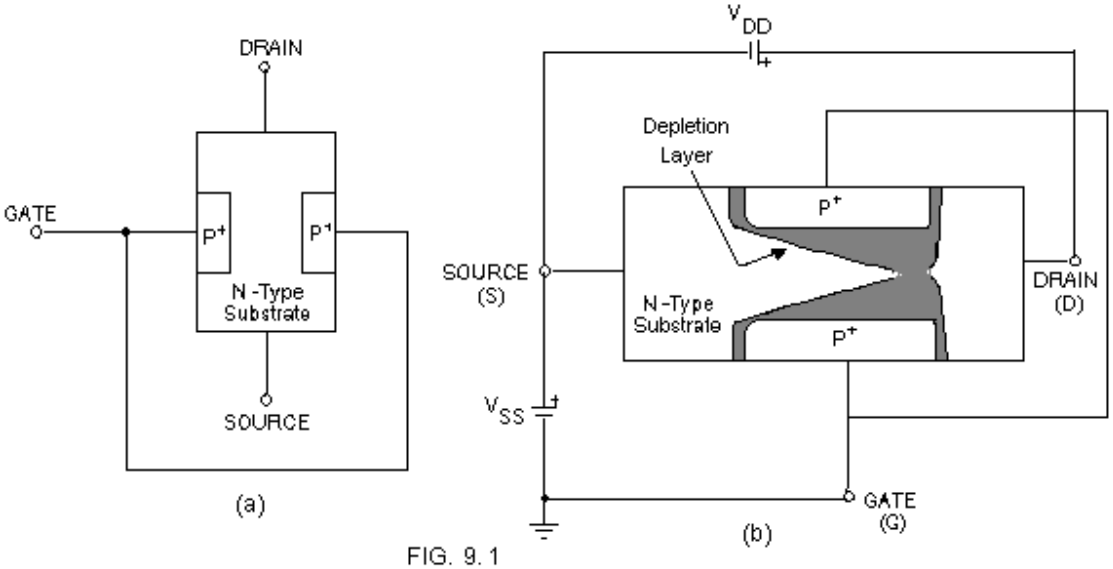


FIG. 9.1

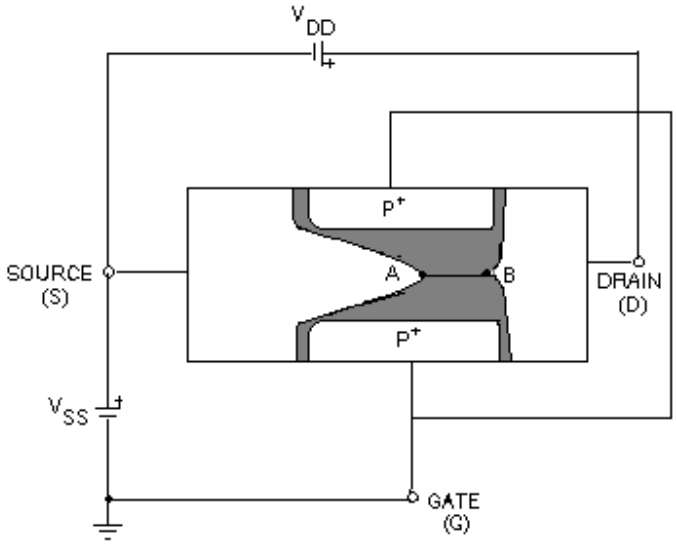


Fig. 9.1(c)

In the normal operation of JFET, the gate is kept at a negative potential with respect to the source. The two *P – N* junctions are in reverse bias. The drain is operated at a positive potential with respect to the source. There exist space charge region (depletion region) on either side of junction leaving uncovered positive ions on the *N* side and uncovered negative ions on the *P* side. The electric field extends from the positive ions on one side to the negative ions on the other side. As the magnitude of the reverse bias across the junction increases, the thickness of the uncovered region also increases. The conductivity of the uncovered region is zero since there are no current carriers. Since the two *P* regions are heavily doped compared to the *N* region (bar), the bulk of the depletion region may assume to be in the *N* channel. It is well known that the width of the depletion region increases with increasing reverse bias. So when we move from the source end to the drain end the reverse bias across the *P – N* junction increases. The width of the depletion region will also increase from the source end to the drain end. The width of depletion region in the *N* – channel is, therefore, like wedge shaped as shown by dark shadow in figure 9.1(b)and (c). Thus for a fixed drain to source voltage, it becomes possible to control the drain current by varying the reverse bias voltage across the gate junction. The field effect transistor is named so because the current control is the effect of the extension of the field associated with depletion region as caused by the increasing reverse bias.

9.2.1 Static characteristics of JFET: The symbols used for *N* – channel *JFET* and *P* – channel JFET are shown in Figure (9.2). The direction of

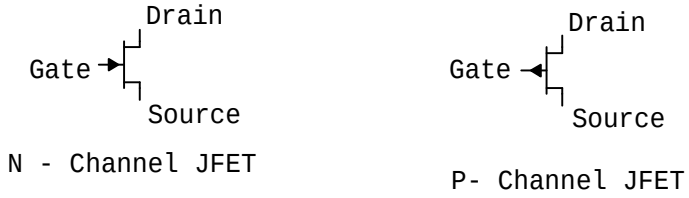


Figure 9.2

arrow at the gate of the JFET indicates the direction in which the gate current would flow if the gate junction were forward biased. Figure 9.3 shows the circuit diagram, to plot the

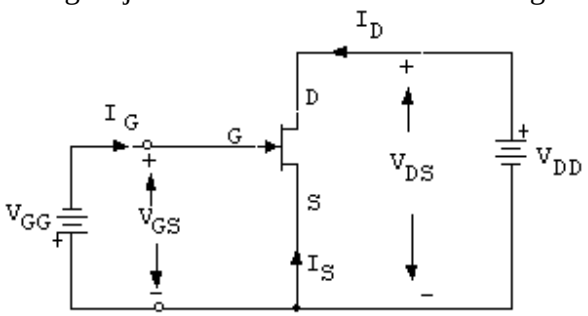


Figure 9.3

characteristic curves of *JFET* in common source mode. Figure (9.4) shows the characteristic curves of the *FET*, where the drain current I_D is plotted against V_{DS} (drain

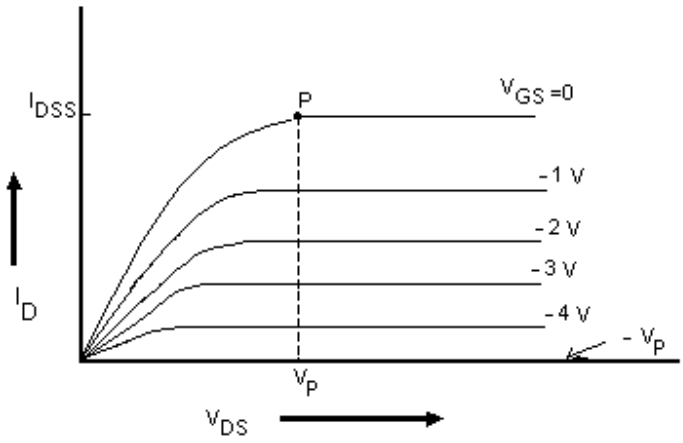


Figure 9.4

to source voltage) with V_{GS} (gate to source voltage) as the parameter. To understand the nature of these characteristic curves, let us set V_{GS} equal to zero. As we increase V_{DS} the drain current I_D increases until V_{DS} equals V_P. At the voltage V_P the two depletion regions meet at a point in the channel and we say channel is pinched off. Thus, if we increase the V_{DS} beyond V_P, the drain current I_D saturates and becomes a constant value denoted by

I_{DSS}. In fact when V_{DS} is increased beyond V_P, the potential at two points A and B (ref. 9.1c) is responsible for the drain current to flow. The electric field so generated helps in sweeping the carriers from the source end to drain end. The increase of V_{DS} (beyond V_P) corresponds the proportional increase in the distance AB. This in turn gives the constant electric field $\frac{V_{DS}}{AB}$. The constant electric field does not allow the drain current to increase further. The saturated drain current (I_{DSS}) is thus obtained.

If now the magnitude of the gate potential is increased in the direction to provide the additional reverse bias, pinch off will take place at a lower voltage. Further increase in V_{DS} does not result in any increase in I_D and the current saturates at a lower value than I_{DSS}. Curves corresponding to different negative values of V_{GS} are shown in figure (9.4). If V_{DS} is increased beyond a certain limit, the JFET enters in the break down region where the drain current increases drastically. This happens because the reverse biased $P - N$ junctions undergo avalanche breakdown when small changes in V_{DS} produce very large change in drain current.

It is worthwhile to mention that the characteristic curves beyond the pinch off voltage are parallel to each other (having the saturated values) and the saturated drain currents for this region are denoted by I_{DS}. In amplifier applications the FET is always used in this region. The value of I_{DS} with gate shorted to the source (V_{GS} = 0) is denoted by I_{DSS}.

The transfer characteristic of the JFET (I_D versus V_{GS} with V_{DS} kept constant at a value greater than V_P), is shown in figure (9.5). Note that for V_{GS} = 0, I_{DS} = I_{DSS}

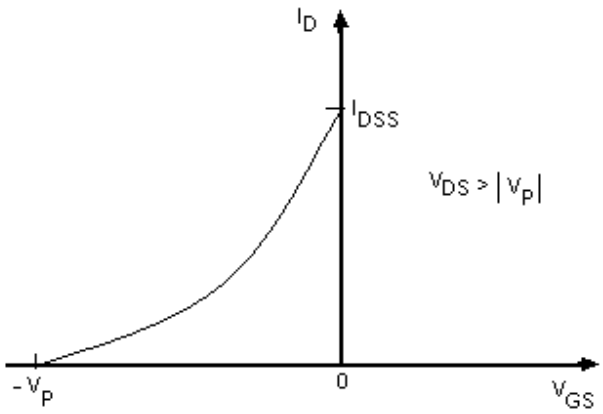


Fig. 9.5

and for V_{GS} = - V_p, I_{DS} = 0. The dependence of I_{DS} on V_{GS} can be approximated by the parabola given by:

$$I_{DS} = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \qquad \text{------ (9.1)}$$

This simple parabolic approximation gives an excellent fit, with the experimentally determined transfer characteristics for FET's made by diffusion process.

9.3 Metal Oxide Semiconductor (MOS) FET: There are two types of Metal Oxide Semiconductor field Effect Transistors. One is called as Enhancement type MOSFET, while the other is called as Depletion type MOSFET. Both the two types of MOSFET's will be discussed below.

9.3.1 Enhance type MOSFET: Figure (9.6) shows the *P* – channel enhancement type MOSFET which consists of a lightly doped N – type substrate into which two heavily doped P^+ regions are diffused. These two P^+ regions act as the source and the drain. The source and drain are 10 to 20 μm apart. Then a thin layer of insulating silicon dioxide (SiO_2) is grown over the entire surface of the structure. Holes are cut in the SiO_2 layer above the P^+ regions (source and drain). The metal film is evaporated between the source and the drain and also through the cut in the SiO_2 layer. The metallic layer between the source and the drain forms the gate. The contacts to the source and the drain are also taken through the metallic layers in the cuts over the two P^+ regions.

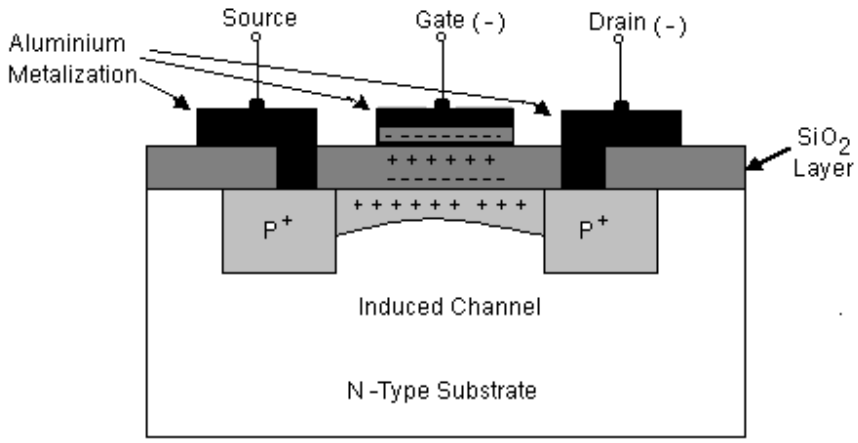


Fig. 9.6

The metal area of the gate, in addition with the insulating dielectric oxide layer and the semiconductor channel, form a parallel plate capacitor. The insulating layer of silicon dioxide is the reason why this device is called the insulated gate field effect transistor.

When a negative voltage is applied at the gate, it induces positive charges in insulating layer and correspondingly positive charges in semiconductor. The positive charge region in semiconductor increases as the negative voltage at the gate increases. When the drain is kept at the negative potential with respect to source, the positive charges in the semiconductor help in conducting between source and the drain. In this way the drain current is enhanced by the negative gate voltage. Figure 9.7 (a) and 9.7(b)

show the drain characteristics and transfer characteristics of P – channel enhancement type MOSFET.

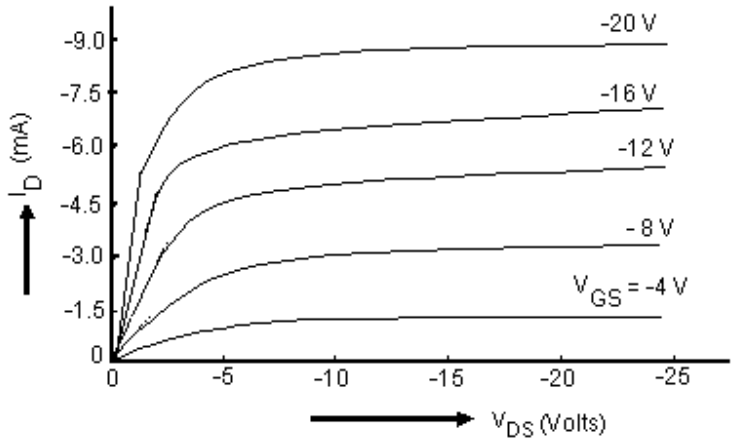


Fig. 9.7(a)

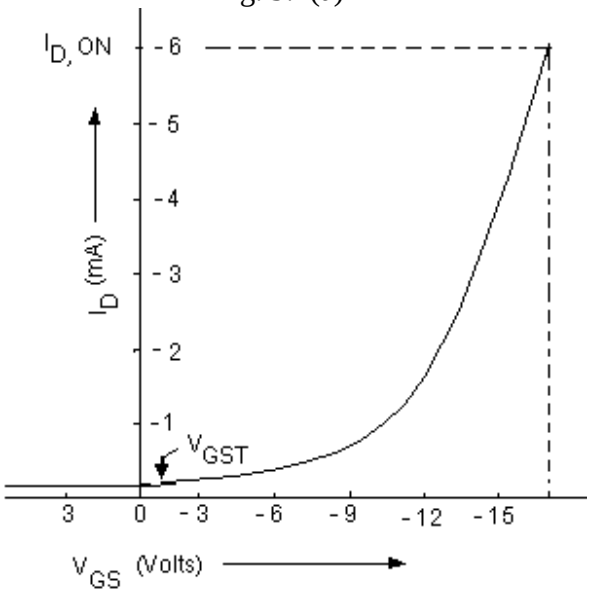


Fig. 9.7(b)

It may be noted from figure 9.7(a), that the drain current for V_{GS} greater than zero is very small and as V_{GS} is made more negative the current increases slowly at first and then increases drastically. The manufacturer often indicates the gate source threshold voltage V_{GST} or V_T at which the drain current $|I_D|$ attains some defined small value say about 10 μA. The current $I_{D,ON}$, the maximum permissible current on the drain characteristics, is also specified by the manufacturer.

9.3.2 Depletion type MOSFET: The basic structure of N – channel depletion type MOSFET is shown in figure 9.8(a). It consists of a lightly doped P – type substrate into which two heavily doped N^+ regions are diffused. These two N^+ regions act as the source and the drain. In depletion type MOSFET a lightly doped N – channel is diffused between

the source and the drain. Then a thin layer of insulating silicon dioxide (SiO_2) is grown over the entire surface of the structure. Holes are cut in the SiO_2 layer above the N^+ regions (source and drain). The metal film is evaporated between the source and the drain and also through the cut in the SiO_2 layer. The metallic layer between the source and the drain forms the gate.

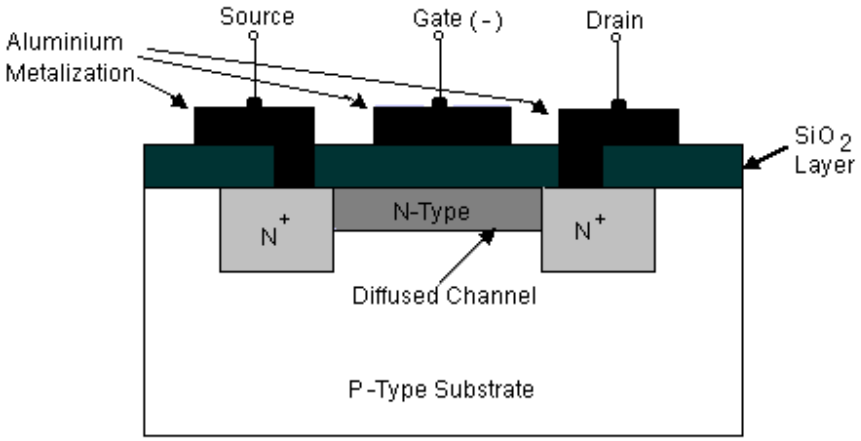


Fig. 9.8(a)

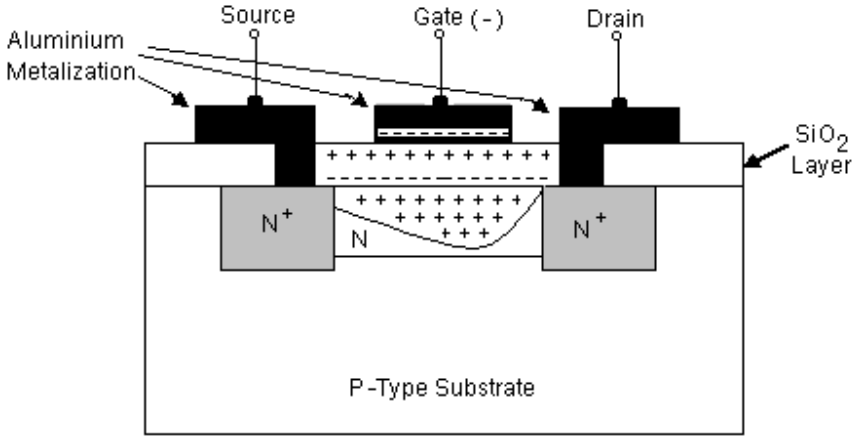


Fig 9.8(b)

When negative voltage is applied at the gate, positive charges are induced in the *N* – channel through the insulating layer of SiO_2, as shown in figure 9.8(b). Since the conduction of current through the channel is by means of majority carriers (electrons in *N* – channel), the conductivity of the channel reduces as the gate voltage is made more negative. It is so because the channel is depleted of majority carriers due to induction of positive charge in it by application of negative gate voltage. The drain current I_D is decreased as the gate voltage V_{GS} is made more negative. This phenomenon is analogous

to that of pinch off occurring in the JFET at the drain end of the channel. Figures 9.9(a) and 9.9(b) show the drain and transfer characteristics of the depletion type MOSFET. The drain characteristics of a depletion type MOSFET are similar to that of a JFET.

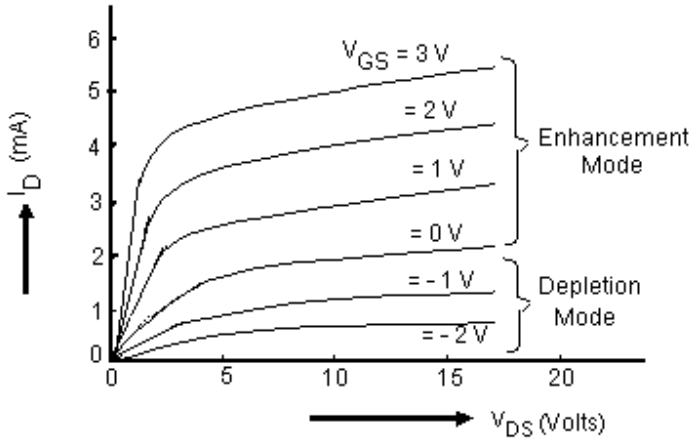


Fig. 9.9(a)

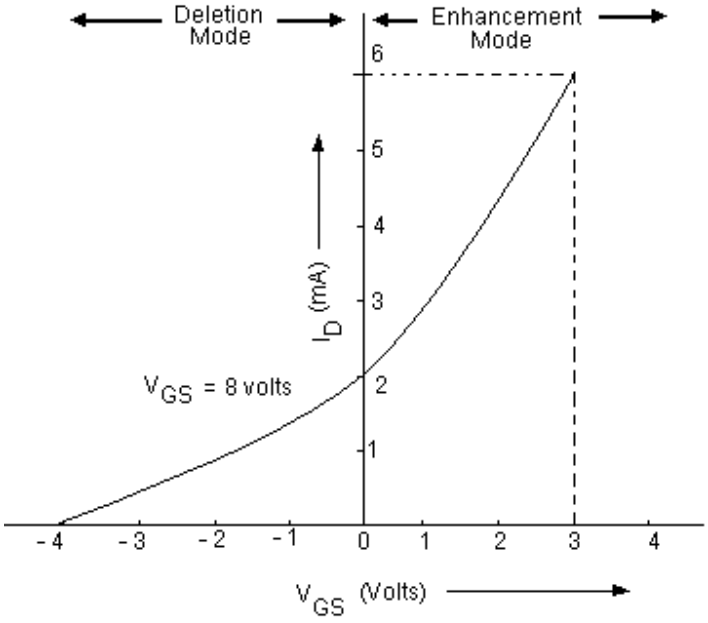


Fig. 9.9(b)

A depletion MOSFET may also be used in enhancement mode, by applying positive voltage at the gate. In this case the negative charges are induced in the channel and its conductivity increases. The drain current I_D thus increases or enhances as more positive potential is applied. Figures 9.9(a) and 9.9(b) also show the drain and transfer characteristics of the MOSFET operated in both depletion and enhancement mode.

9.3.3 Circuit Symbols: The graphical symbols for an N – channel and P – channel depletion type MOSFET's are shown in figure 9.10(a) and 9.10(b) respectively. Two

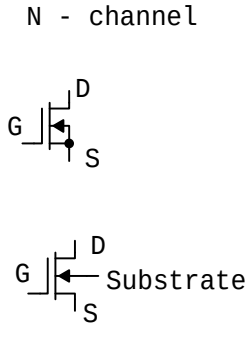


Figure 9.10(a)

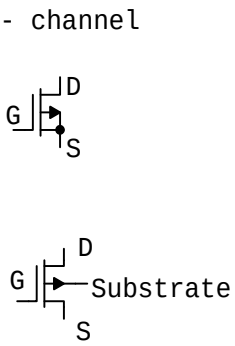


Figure 9.10(b)

symbols are provided for each type of channel. Similarly the symbols for N – channel and P – channel enhancement type MOSFET's are shown in figure 9.11(a) and 9.11(b) respectively.

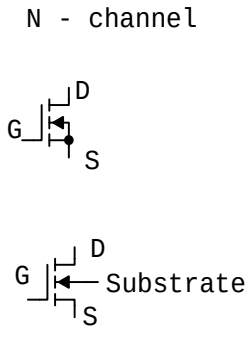


Figure 9.11(a)

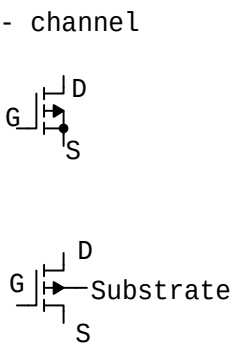


Figure 9.11(b)

9.4 Parameters of FET: The linear small signal model for the field effect transistor can be obtained in the same fashion as for bipolar junction transistor. As is well known that the drain current i_D may be expressed as a function of gate voltage v_{GS} and v_{DS} as given below:

$$i_D = f(v_{GS}, v_{DS}) \qquad \text{------ (9.2)}$$

The three parameters namely the drain resistance r_d, transconductance g_m and amplification factor μ may be defined as follows:

9.4.1 Drain Resistance r_d : The parameter r_d known as dynamic drain resistance or the output resistance is defined as the ratio of the change in the drain to source voltage to the change in drain current at a constant gate to source voltage v_{GS}.

$$r_d = \left.\frac{\partial v_{DS}}{\partial i_D}\right|_{v_{GS}} \approx \left.\frac{\Delta v_{DS}}{\Delta i_D}\right|_{v_{GS}} \qquad \text{------ (9.3)}$$

The reciprocal of the drain resistance is known as the drain conductance g_d.

9.4.2 Transconductance g_m: The mutual conductance or transconductance g_m is defined as the ratio of the change in the drain current to the change in gate to source voltage at a constant drain to source voltage.

$$g_m = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{DS}} \approx \left.\frac{\Delta i_D}{\Delta v_{GS}}\right|_{v_{DS}} \qquad \text{------ (9.4)}$$

9.4.3 Amplification Factor μ: Third important parameter of the FET is the amplification factor μ defined as the ratio of the change in drain to source voltage to the change in gate to source voltage when the drain current is kept constant.

$$\mu = -\left.\frac{\partial v_{DS}}{\partial v_{GS}}\right|_{i_D} \approx -\left.\frac{\Delta v_{DS}}{\Delta v_{GS}}\right|_{i_D} \qquad \text{------ (9.5)}$$

The negative sign in this expression represents that when the drain to source voltage is increased; the drain current increases now to keep the drain current constant the gate to voltage has to be decreased. In other words to the keep the drain current constant the drain to source voltage and the gate to source voltage should of opposite sign.

The amplification factor μ can also be written in the following form

$$|\mu| = \frac{\partial v_{DS}}{\partial i_D} \cdot \frac{\partial i_D}{\partial v_{GS}} = r_d \cdot g_m \qquad \text{------ (9.6)}$$

9.4.4 Relation between Transconductance g_m and Drain Current I_{DS} of the FET: We have the drain current given by equation (9.1) which is reproduced here:

$$I_{DS} = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \qquad \text{------- (9.7)}$$

Differentiating this equation with respect to v_{GS} we get:

$$g_m = \frac{\partial I_{DS}}{\partial V_{GS}} = 2.I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)x\left(-\frac{1}{V_P}\right)$$

$$= -\frac{2.I_{DSS}}{V_P}\left(1-\frac{V_{GS}}{V_P}\right)$$

$$= g_{m0}\left(1-\frac{V_{GS}}{V_P}\right) \qquad \text{------ (9.8)}$$

Where g_{m0} is the value of gm for $V_{GS} = 0$ and given by:

$$g_{m0} = -\frac{2I_{DSS}}{V_P}$$

Combining the equations (9.7) and (9.8), we get:

$$g_m = -\frac{2}{V_P}\sqrt{I_{DSS}.I_{DS}} \qquad \text{------ (9.9)}$$

From the equation (9.9) it is clear that the transconductance gm varies as the square root of the drain current.

9.5 Small Signal Model of Field Effect Transistor: The drain current i_D may be expressed as a function of gate voltage v_{GS} and v_{DS} as given below:

$$i_D = f(v_{GS}, v_{DS})$$

The incremental change in the drain current i_D may be given by the Taylor series expansion as:

$$\Delta i_D = \frac{\partial.i_D}{\partial v_{GS}}.\Delta v_{GS} + \frac{\partial.i_D}{\partial v_{DS}}.\Delta v_{DS} \qquad \text{------ (9.10)}$$

In the small signal notation, $\Delta.i_D = i_d$, $\Delta.v_{GS} = v_{gs}$ and $\Delta.v_{DS} = v_{ds}$, the equation (9.10) may be written in the form:

$$i_D = g_m.v_{gS} + \frac{1}{r_d}.v_{dS} \qquad \text{------ (9.11)}$$

A circuit satisfying the equation (9.11) may be drawn as shown in the figure (9.12).

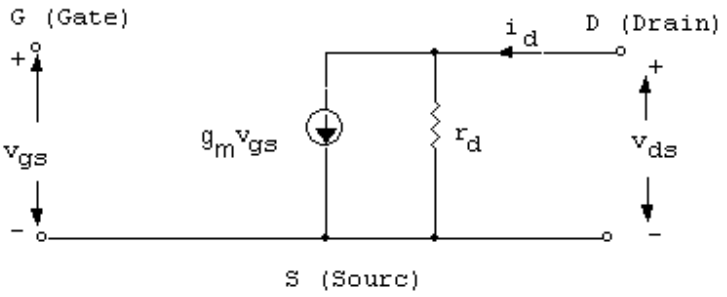


Fig. 9.12

This circuit is known as the small signal model of the field effect transistor. This small signal model has a Norton's output (current source) whose current is proportional to the gate to source voltage. The proportionality factor is the transconductance g_m. The input resistance between gate and source is infinite since reverse biased gate current is assumed to be zero. Similarly the resistance between the gate and the drain is infinite. The output resistance is drain resistance r_d , whose value is infinitely large when FET is operated in the linear region where the characteristic curves are parallel and equidistance from each other. In that case the output will behave like a constant current source whose value will depend upon the input voltage v_{gs}. Hence the FET is known as the voltage operated device or Voltage Controlled Current Source (VCCS).

The small signal model discussed above is suitable for low frequency signal only. The capacitances between the terminals of the field effect transistors may gets added at high frequencies as these will produce finite reactance at the frequency. So the small signal model at the high frequencies will be as given in figure (9.13).

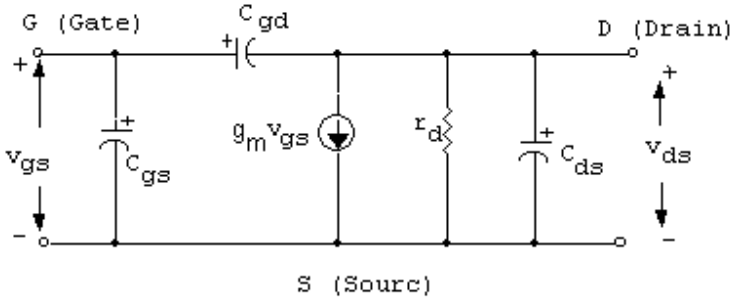


Fig. 9.13

In this circuit the capacitance C_{gs} is the barrier capacitance between the gate and the source, C_{gd} is the barrier capacitance between the gate and the drain and the capacitance C_{ds} is similarly the drain to source capacitance of the channel. These capacitances produce the feedback from the output to the input and the voltage gain drops rapidly as the frequency increases.

Example 9.1 Show that for small values of V_{GS} compared with V_P, the drain current is approximately given by:

$$I_D \cong I_{DSS} + g_{m0} V_{GS}$$

Solution: We have

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right)$$

We also have the expression for the drain current as:

$$I_D = g_m . V_{GS} + \frac{1}{r_d} . V_{DS}$$

Putting the value of gm in this expression we get:

$$I_D = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right) + \frac{1}{r_d} . V_{DS}$$

or $$I_D = g_{m0} V_{GS} - g_{m0} \frac{V_{GS}}{V_P} V_{GS} + \frac{1}{r_d} . V_{DS}$$

If V_{GS} is smaller than the magnitude of Vp then we have

$$I_D \cong g_{m0} V_{GS} + \frac{1}{r_d} . V_{DS}$$

If VGS = 0 then $I_D = I_{DSS} = \frac{V_{DS}}{r_d}$, so

$$I_D \cong I_{DSS} + g_{m0} V_{GS}$$ Proved.

Example 9.2 If the two FETs (which are not identical) are connected in parallel then prove that the effective transconductance and drain resistance of this combination are:

$$g_m = g_{m1} + g_{m2}$$

and $$\frac{1}{r_d} = \frac{1}{r_{d1}} + \frac{1}{r_{d2}}$$ respectively.

The amplification factor of the combination is given by:

$$\mu = \frac{\mu_1 r_{d2} + \mu_2 r_{d1}}{r_{d1} + r_{d2}}$$

Solution: Let two FETs, whose drain resistances are rd1 and rd2 and their transconductance are gm1 and gm2, are connected in parallel as shown in figure 9.14. The combination of these FETs behaves like a single FET.

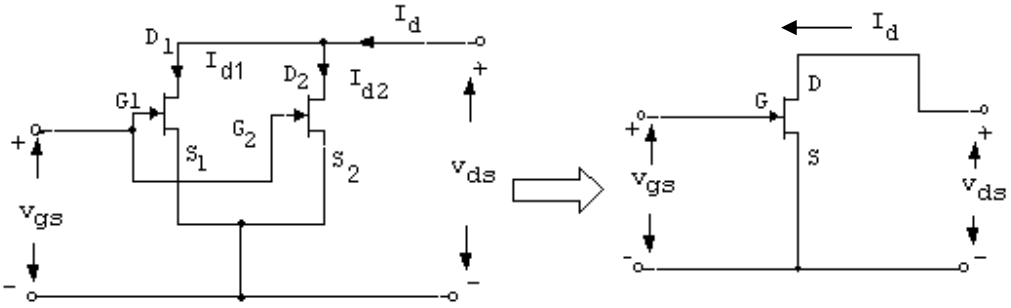


Fig. 9.14

Let I_{d1} and I_{d2} are the two drain currents of the individual FET given by:

$$I_{d1} = g_{m1}.V_{GS} + \frac{1}{r_{d1}}V_{DS}$$

$$I_{d2} = g_{m2}.V_{GS} + \frac{1}{r_{d2}}V_{DS}$$

and $$I_d = I_{d1} + I_{d2}$$

or $$I_d = (g_{m1} + g_{m2})V_{GS} + (\frac{1}{r_{d1}} + \frac{1}{r_{d2}})V_{DS}$$

This implies $$g_m = g_{m1} + g_{m2}$$

and $$\frac{1}{r_d} = \frac{1}{r_{d1}} + \frac{1}{r_{d2}}$$

Amplification factor μ is given by:

$$\mu = r_d.g_m = (\frac{r_{d1}.r_{d2}}{r_{d1} + r_{d2}}).(g_{m1} + g_{m2})$$

or $$= \frac{g_{m1}.r_{d1}.r_{d2} + g_{m2}.r_{d1}.r_{d2}}{r_{d1} + r_{d2}}$$

$$= \frac{\mu_1.r_{d2} + \mu_2.r_{d1}}{r_{d1} + r_{d2}}$$ Proved.

9.6 Low Frequency FET Amplifiers: As is well known that the bipolar transistors can be used in three amplifiers configurations CE, CB and CC. The field effect transistors can also be used in three amplifier configurations namely common source (CS), common gate (CG) and common drain (CD). The common source amplifier is

analogous to the common emitter amplifier of the bipolar transistor. The common drain (CD) amplifier is analogous to the common collector amplifier. The common drain amplifier is also called source follower as the common collector amplifier is known as emitter follower. We shall analyze these amplifiers in detail.

9.6.1 Common Source Amplifier: The basic circuit diagram of common source amplifier is shown in figure 9.15, in which the source is common between input and output circuit. The input signal V_i to be amplified is applied to the gate terminal of the FET. The output is taken at the drain and across the load resistance R_L. We shall analyze the circuit by calculating the voltage gain and output resistance of the amplifier.

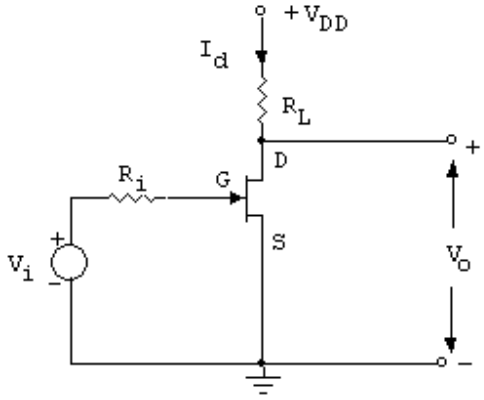


Fig. 9.15

The small signal model for the purpose the above mentioned quantities of this circuit may be drawn as given in figure 9.16.

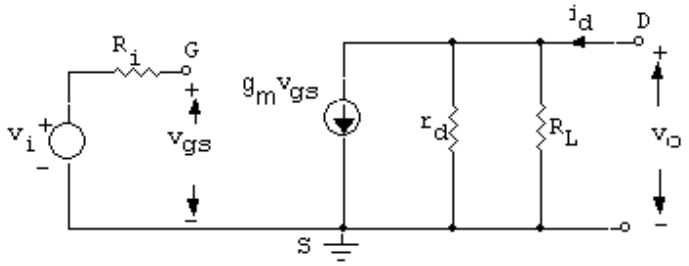


Fig. 9.16

The Thevenin's equivalent model of this circuit may also be drawn as shown in figure 9.17.

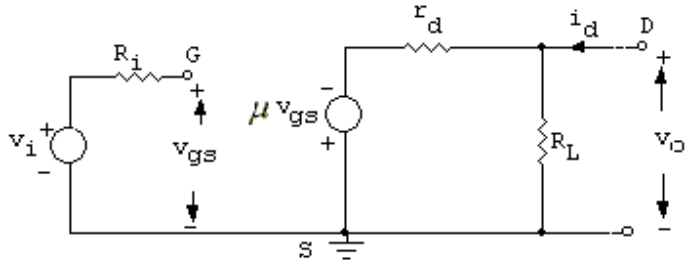


Fig. 9.17

Voltage Gain: The voltage gain $A_v = \frac{V_O}{V_i}$ of the amplifier can be calculated from the figure 9.17, as:

$$A_v = \frac{V_O}{V_i} = \frac{-\mu}{r_d + R_L} x R_L \qquad \text{------ (9.12)}$$

$$= \frac{-\left(\frac{\mu}{r_d}\right)}{\left(1 + \frac{R_L}{r_d}\right)} x R_L = \frac{-g_m}{\left(1 + \frac{R_L}{r_d}\right)} x R_L$$

If $r_d >> R_L$, then the voltage gain of the amplifier is given as:

$$A_v = -g_m . R_L \qquad \text{------ (9.13)}$$

The negative sign in this expression indicates that there is a phase reversal of 180 ^{0} between input and output signal.

From the equivalent circuit the output resistance is:

$$R_O = r_d \qquad \text{------ (9.14)}$$

9.6.2 Common Drain Amplifier: The basic circuit of a common drain amplifier using field effect transistor is shown in figure (9.18).

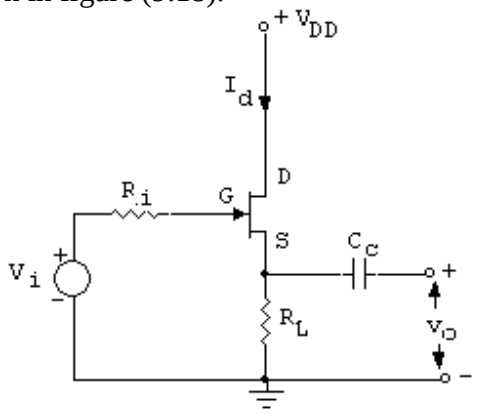


Fig. 9.18

In this circuit the drain is connected to the d.c. supply and for the a.c. signal analysis the d.c. supply is shorted to ground. Hence the signal applied between the gate and the ground is basically between gate and the drain. The output is being taken between the source and the ground (means drain). Now we shall find the voltage gain and output resistance of this circuit. The equivalent circuit is therefore, drawn as shown in figure (9.19).

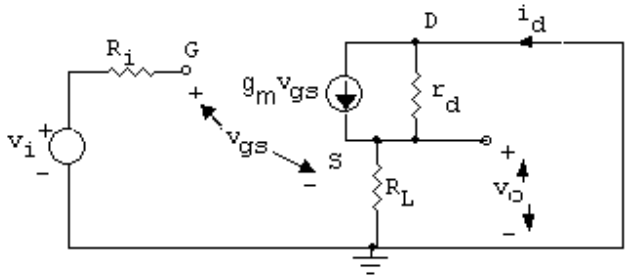


Fig. 9.19

Applying the KVL to the output circuit, we get:

$$i_d.R_L + (i_d - g_m.V_{GS})r_d = 0 \quad \text{------ (9.15)}$$

and $$V_{GS} = V_i - i_d.R_L \quad \text{------ (9.16)}$$

or $$i_d.(R_L + r_d) - g_m.r_d(V_i - i_d R_L) = 0$$

or $$i_d = \frac{g_m.r_d.V_i}{[R_L(1+\mu) + r_d]} = \frac{\mu.V_i}{[r_d + (1+\mu)R_L]} \quad \text{------ (9.17)}$$

The output voltage is given by:

$$V_O = i_d.R_L = \frac{\mu.V_i.R_L}{[r_d + (1+\mu)R_L]} \quad \text{------ (9.18)}$$

The voltage gain of the amplifier can be obtained as

$$A_V = \frac{V_O}{V_i} = \frac{\mu.R_L}{[r_d + (1+\mu)R_L]} \quad \text{------ (9.19)}$$

If $(\mu+1)R_L >> r_d$ then $A_V \approx \frac{\mu.}{(1+\mu)}$

Generally $\mu >> 1$ for the field effect transistors, so the gain of the amplifier will become almost equal to unity ($A_V \approx 1$).

The voltage of unity means that the output (Source) follows the input (gate) signal. Hence the Common Drain configuration may be called as the Source Follower circuit, similar to the emitter follower of bipolar transistor amplifiers.

The equation (9.18) may be rewritten of the following form:

$$V_O = \frac{\frac{\mu.V_i}{(1+\mu)}.R_L}{[\frac{r_d}{(1+\mu)} + R_L]} \quad \text{------ (9.20)}$$

Thevenin equivalent circuit, for the amplifier whose output voltage V_O is given by equation (9.20), is shown in figure (9.20).

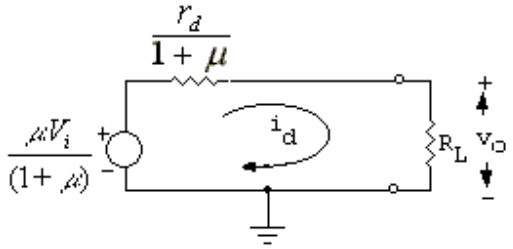


Fig. 9.20

The output resistance R_0 of the source follower circuit is obtained from the figure (9.19) as:

$$R_0 = \frac{r_d}{1+\mu} \approx \frac{r_d}{\mu} \approx \frac{1}{g_m} \qquad \text{------ (9.21)}$$

Provided $\mu >> 1$.

9.6.3 Common Gate Amplifier: The circuit in figure (9.21) is the common gate amplifier using field effect transistor. In his circuit the gate is common between input and output terminals. The input signal is applied at source terminal with respect to gate and output is taken across the drain and the gate.

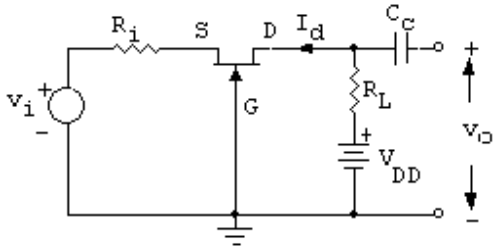


Fig. 9.21

The voltage gain and output resistance of this circuit may be obtained by drawing its equivalent circuit as shown in figure (9.22).

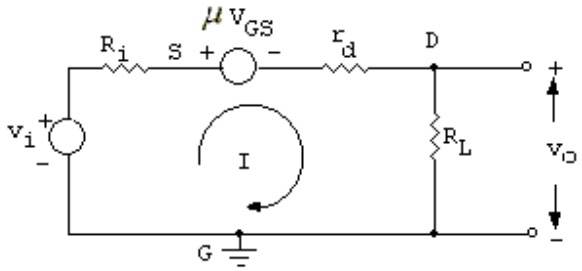


Fig. 9.22

Applying the KVL to this circuit:

$$R_i.I + \mu V_{GS} + (r_d + R_L)I - V_i = 0$$

or $$I(R_i + r_d + R_L) = V_i - \mu V_{GS}$$

or $$I = \frac{V_i - \mu.V_{GS}}{(R_i + R_L + r_d)} = \frac{V_i - \mu(-V_i + R_i I)}{(R_i + R_L + r_d)}$$

or $$I\left[1 + \frac{\mu.R_i}{R_i + r_d + R_L}\right] = \frac{(1+\mu)V_i}{(R_i + r_d + R_L)}$$

or $$I = \frac{(1+\mu)V_i}{[(1+\mu)R_i + r_d + R_L]} \qquad \text{------ (9.22)}$$

the output voltage V_O is therefore calculated as:

$$V_O = R_L I = \frac{(1+\mu)V_i.R_L}{[(1+\mu)R_i + r_d + R_L]} \qquad \text{------ (9.23)}$$

The voltage gain of the common gate amplifier is given by:

$$A_V = \frac{V_O}{V_i} = \frac{(1+\mu).R_L}{[(1+\mu)R_i + r_d + R_L]} \qquad \text{------ (9.24)}$$

The output resistance of the common gate amplifier can be obtained by drawing the Thevenin's equivalent circuit for the equation (9.23) as shown in figure (9.23).

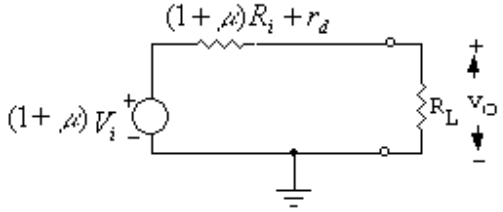


Fig. 9.23

The output resistance from this figure is therefore given by:

$$R_O = (1+\mu)R_i + r_d \qquad \text{------- (9.25)}$$

Example 9.3: The figure (9.24) shows the circuit diagram for common source amplifier with unbypassed source resistance. Prove that

(i) the expression for the voltage gain $A_v = \frac{V_O}{V_i} = \frac{-\mu}{r_d + R_L + (1+\mu)R_S} xR_L$, and

(ii) the expression for output resistance $R_O = r_d + (1+\mu)R_S$.

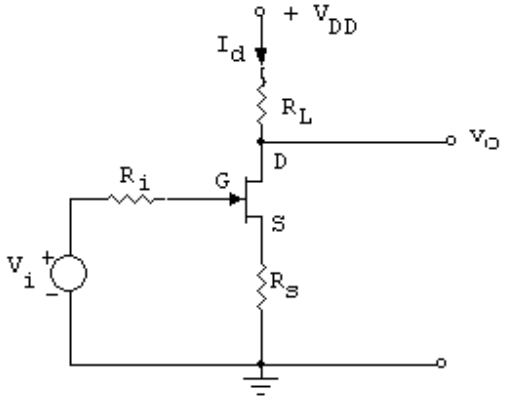


Fig. 9.24

Solution: To find the voltage gain and the output resistance of the circuit we draw small signal model of the given circuit as shown in figure (9.25).

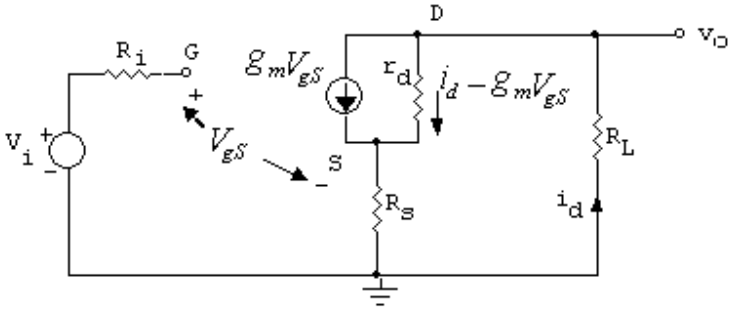


Fig. 9.25

Applying the KVL to the output circuit, we get:

$$i_d.R_L + (i_d - g_m.V_{gS})r_d + R_S.i_d = 0 \qquad \text{------ (9.26)}$$

and $$V_{gS} = V_i - i_d.R_S \qquad \text{------ (9.27)}$$

or $$i_d.(R_L + R_S + r_d) - g_m.r_d(V_i - i_d R_S) = 0$$

or $$i_d = \frac{g_m.r_d.V_i}{[R_S(1+\mu) + r_d + R_L]} = \frac{\mu.V_i}{[r_d + (\mu+1)R_S + R_L]}$$

The output voltage is given by:

$$V_O = -i_d.R_L = \frac{-\mu.V_i.R_L}{[r_d + (1+\mu)R_S + R_L]} \qquad \text{------ (9.28)}$$

The voltage gain of the amplifier can be obtained as

$$A_v = \frac{V_O}{V_i} = \frac{-\mu}{r_d + R_L + (1+\mu)R_S} xR_L \qquad \text{------ (9.29)}$$

Thevenin's equivalent of the equation (9.29) is given by (fig. 9.26):

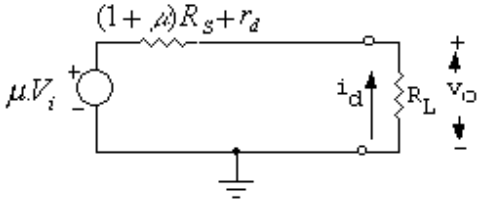


Fig. 9.26

The output resistance R_O is therefore given by:

$$R_O = r_d + (1+\mu)R_S$$

Example 9.4: In Common Source amplifier, a FET has r_d =200 $K\Omega$ and μ= 18. The load resistance R_L = 120 $K\Omega$. Compute the voltage gain and output resistance of this amplifier.

Solution: We know that the voltage gain of the Common Source amplifier is given by:

$$A_v = \frac{-\mu}{r_d + R_L} xR_L$$

$$= -\frac{18x10K\Omega}{(200+120)K\Omega} = -\frac{18}{21} = -6.7$$

The output resistance is given by:

$$R_O = r_d = 200\ K\Omega$$

Example 9.5: A Common Source FET amplifier uses load resistance R_L = 100 kΩ and an unbypassed resistance R_S = 10KΩ connected between the source and the ground. The drain resistance of the FET is 300 KΩ and μ=15. Compute the voltage gain and the output resistance of the amplifier.

Solution: The voltage gain of the common source amplifier with a source resistance R_s connected between source and the ground is given by:

$$A_v = \frac{-\mu}{r_d + R_L + (1+\mu)R_S} xR_L$$

$$= \frac{-15x100K\Omega}{(300+100+16x10)K\Omega} = \frac{-150}{56} = -2.67$$

The output resistance of the amplifier is given by :

$$R_O = r_d + (1+\mu)R_S = (300 + 16x10)K\Omega \quad = 460K\Omega$$

9.7 Biasing the FET: The amplifiers using FET have so far been discussed without showing the biasing arrangement as the d.c. biasing does not produce any significant role for the a.c. signal analysis. However, for the practical point of view the biasing of the FET' is very necessary. Like the bipolar transistors, the FET's have also to be biased properly. The consideration must be given to operate the field effect transistor in the linear region of its characteristic curves. Following are the biasing arrangements generally used in field effect transistors.

9.7.1 Source Self Bias: Figure (9.27) shows the self biasing arrangement for the field effect transistor. This type of biasing is generally used in JFET or depletion type MOS

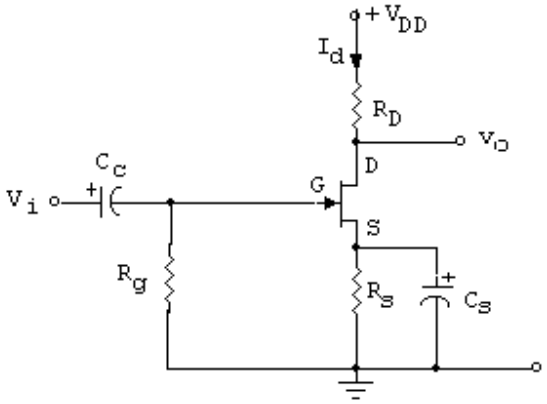


Fig. 9.27

FET's. In this circuit a resistance R_g is connected between the gate and the ground to bias the gate source junction in the reverse bias. The gate is supposed to be at ground potential as the voltage drop across R_g is negligibly small due to the very small gate current. To keep the gate at the negative potential with respect to the source, a resistance R_s is connected between the source and the ground. The voltage drop across Rs will be equal to $I_d.R_s$ i.e. the source is at the positive potential ($I_d.R_s$) with respect to ground. The gate is, therefore, at the negative potential with respect to the source since the gate is at the ground potential and source is at the positive potential. However the resistance R_s will produce a degenerative feedback for the a.c. signal to be applied to the input terminals. To avoid this problem a capacitance C_E is connected in parallel with the resistance R_s. The capacitance C_E bye passes the signal available at the source. Source is therefore called at the signal ground.

9.7.2 Voltage Divider Biasing: The voltage divider biasing arrangement can also be applied to the FET amplifiers as shown in figure (9.28).

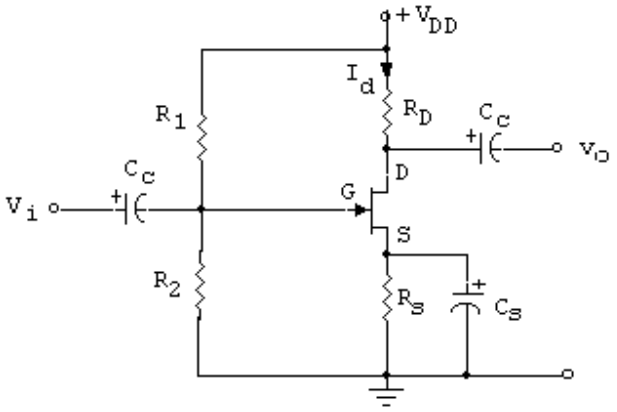


Fig. 9.28

In this case the gate to source voltage VGS is given by:

$$V_{GS} = V_{GG} - I_d.R_S$$

where V_{GG} is the voltage between gate and the ground given by:

$$V_{GG} = \frac{V_{DD}xR_2}{R_1 + R_2}$$

9.8 Common Source Amplifier at High Frequencies: The common source amplifier at low frequencies has been discussed in the earlier section of this chapter. At high frequencies the inter electrode capacitance will influence the amplifier. So we shall discuss the common source amplifier at high frequencies taking into account the effect of inter electrode capacitances. Figure (9.29) shows the CS amplifier without biasing.

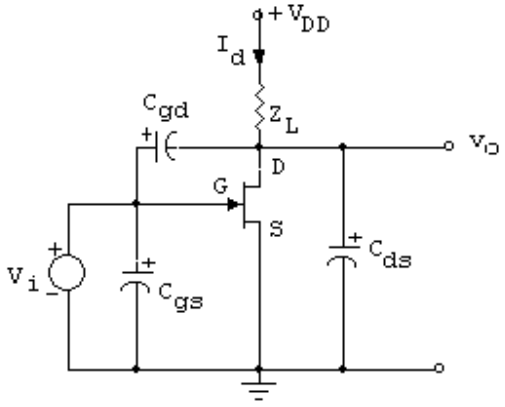


Fig. 9.29

In this circuit the capacitance C_{gs} is the barrier capacitance between the gate and the source, C_{gd} is the barrier capacitance between the gate and the drain and the capacitance C_{ds} is similarly the drain to source capacitance of the channel. The magnitude of these capacitances being very low produces finite reactance at high frequencies. We shall calculate voltage gain, input and output admittances and input capacitance, by

drawing the equivalent circuit of the CS amplifier. Such an equivalent circuit is shown in figure (9.30).

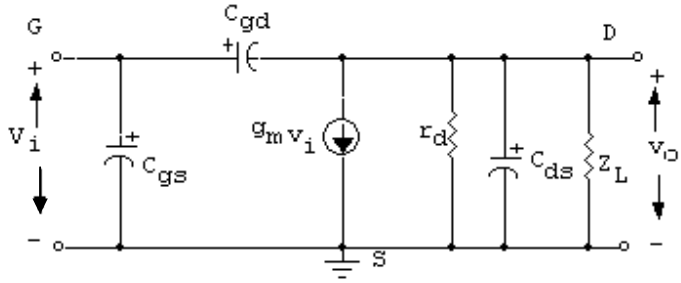


Fig. 9.30

Voltage Gain: The output voltage of the circuit shown in figure (9.30) can easily be obtained by applying Norton's theorem. The output voltage V_o at the drain source terminals is given by: $V_O = I.Z$, where I is the short circuit current at the output terminals and the impedance Z is impedance at the output terminals when all the sources are replaced by their equivalent internal impedances (i.e. the voltage sources are shorted and current sources are open circuited).

So to find the impedance Z , we short circuit the independent source V_i (i.e. $V_i = 0$) and open circuit the dependent current source $g_m.V_i$ (i.e. the current source $g_m.V_i$ is removed). So the impedance Z will be the parallel combination of impedances corresponding to Z_L, C_{ds}, r_d and C_{gd}. Hence

$$\frac{1}{Z} = \frac{1}{Z_L} + \frac{1}{r_d} + \frac{1}{(1/\, j\omega.C_{ds})} + \frac{1}{(j\omega.C_{gd})}$$

or
$$Y = \frac{1}{Z} = Y_L + g_d + Y_{ds} + Y_{gd} \qquad \text{------ (9.30)}$$

Where $Y_L = 1/Z_L =$ admittance corresponding to Z_L.

$g_d = 1/r_d =$ admittance corresponding to r_d.

$Y_{ds} = 1/C_{ds} =$ admittance corresponding to C_{ds}.

$Y_{gd} = 1/C_{gd} =$ admittance corresponding to C_{gd}.

The short circuit current I is obtained by short circuiting the drain to source terminals as:

$$I = -g_m.V_i + V_{i.}(1/\, j\omega.C_{ds}) = (-g_m + Y_{ds}).V_i \qquad \text{------ (9.31)}$$

The voltage gain is therefore given by:

$$A_V = \frac{V_O}{V_i} = \frac{Z.I}{V_i} = \frac{I}{V_i..Y}$$
$$= \frac{(-g_m + Y_{ds})V_i}{(Y_L + g_d + Y_{dS} + Y_{gd})V_i}$$

$$A_V = \frac{-g_m + Y_{ds}}{Y_L + g_d + Y_{dS} + Y_{gd}} \qquad \text{------- (9.32)}$$

Input Admittance : To obtain the input admittance of the figure (9.30), we can apply the Miller's theorem as a capacitance Cgd is connected between the gate and the drain. This capacitance can be replaced by impedance $Z_1 = \frac{1}{j\omega C_{gd}.(1-A_V)}$ connected between the gate and the source and another impedance $Z_2 = \frac{A_V}{j\omega C_{gd}.(A_V - 1)}$ connected between the drain and the source. Hence the input admittance is given by :

$$Y_i = Y_{gS} + \frac{1}{Z_1} = Y_{gS} + j\omega.C_{gd}.(1 - A_V)$$

or $$Y_i = Y_{gS} + Y_{gd}.(1 - A_V) \qquad \text{------ (9.33)}$$

Input Capacitance: Input capacitance is important to find since it is needed in cascaded amplifiers. In such an amplifier the output of one stage is used as the input of the second stage. The input of the second stage acts as shunt across the output of the first stage. So the input capacitance C_i of the second stage is shunted by the output load resistance of the first stage.

Let the drain circuit impedance Z_L be a pure resistance and for the low frequency, the gain of the amplifier is given by:

$$A_v = \frac{-\mu}{r_d + Z_L} xZ_L = -gm.Z_L^{'}$$

where $Z_L^{'} = \frac{r_d.Z_L}{r_d + Z_L}$

In this case the equation (9.33) becomes:

$$\frac{Y_i}{j.\omega} \equiv C_i = C_{gS} + (1 + g_m.Z_L^{'})C_{gd} \qquad \text{------(9.34)}$$

From this equation it is clear that there is an increase in the input capacitance C_i due to the inter electrode capacitance between the gate and the drain. This increase in the input capacitance of the amplifier is called as the Miller effect.

Output Admittance : The output admittance is obtained by looking back from output terminals into drain with Vi is kept zero. If the input source is shorted, then r_d, C_{dS} and C_{gd} will seem to be in parallel. Thus the output admittance will be given by:

$$Y_O = Y_{dS} + Y_{gd} \qquad \text{------ (9.35)}$$

9.9 Common Drain Amplifier (Source Follower) at High Frequencies:

Figure (9.31) shows the CD amplifier without biasing and having the inter electrode capacitances.

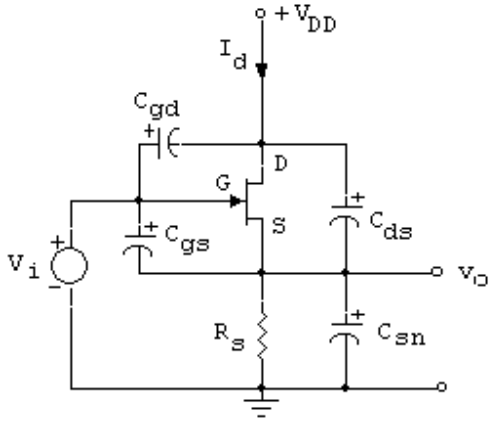


Fig. 9.31

Now we shall calculate voltage gain, input and output admittances and input capacitance, by drawing the equivalent circuit of the CS amplifier. Such an equivalent circuit is shown in figure (9.32).

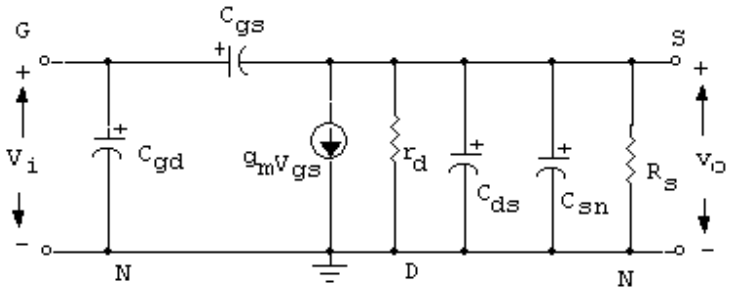


Fig. 9.32

Voltage Gain: In this circuit C_{sn} is the capacitance between the source and the ground terminal. The output voltage can be obtained as:

$$V_O = Z.I \qquad \text{------ (9.36)}$$

where I is the short circuit current across the terminals S and N; and Z is the impedance at the output terminals (S and N) when all the source have been replaced by their internal impedances (Norton's resistance).

The short circuit current is given by:

$$I = g_m.V_{gS} + V_i.(1/\ j\omega C_{gS}) = g_m.V_{gS} + V_i.Y_{gS}$$

But $V_{gS} = V_i - V_O = V_i - Z.I$

or $I(1 + g_m.Z) = g_m.V_i + V_i.Y_{gS}$

or $I = \dfrac{(g_m + Y_{gS}).V_i}{(1 + g_m.Z)}$

or $V_O = I.Z = \dfrac{(g_m + Y_{gS}).V_i}{(Y + g_m)}$ ------ (9.37)

The value of admittance Y is given by:

$$Y = \frac{1}{Z} = \frac{1}{R_S} + g_d + Y_{Sn} + Y_{dS} + Y_{gS}$$ ------ (9.38)

From equations (9.37) & (9.38) we have:

$$A_V = \frac{V_O}{V_i} = \frac{(g_m + Y_{gS})R_S}{1 + (g_m + g_d + Y_{Sn} + Y_{dS} + Y_{gS})R_S}$$ ------(9.39)

Input Admittance : The input admittance can be obtained by applying the Miller's theorem to the capacitance C_{gS}. Thus the input admittance is given by:

$$Y_i = j.\omega.C_{gd} + j\omega.C_{gS}(1 - A_V) \approx j.\omega.C_{gd}$$ ------ (9.40)

as $A_V \approx 1$ for source follower.

Input Capacitance: The input capacitance C_i is given by:

$$C_i = C_{gd} + (1 - A_V)C_{gS}$$ ------ (9.41)

Output Admittance: The output admittance without considering the load resistance R_S is given by:

$$Y_O = g_m + g_d + Y_{dS} + Y_{gS} + Y_{Sn}$$ ------ (9.42)

Example 9.6: Using the high frequency model of the field effect transistor, show that for common gate amplifier with $R_s = 0$ and $C_{dS} = 0$.

(i) Voltage gain $A_V = \dfrac{(g_m + g_d)R_L}{1 + R_L(g_d + j\omega.C_{gS})}$

(ii) Input admittance $Y_i = g_m + g_d(1 - A_V) + j\omega C_{gS}$

Solution: To find the voltage gain and input admittance of the common gate amplifier we draw the circuit of CG amplifier with its high frequency model as shown in figure (9.33).

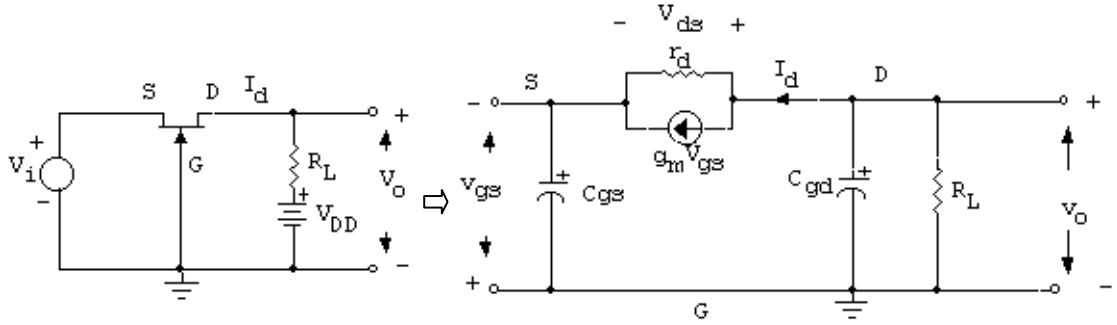


Fig. 9.33

(i) From this circuit, the drain current I_d is given by:

$$I_d = g_m.V_{gS} + \frac{V_{dS}}{r_d} = g_m.V_{gS} + g_d(V_{dS} + V_{gS})$$

Since $V_{dS} = V_{dg} + V_{gS}$

or $$I_d = (g_m + g_d).V_{gS} + g_d.V_{dg} \qquad \text{------ (9.43)}$$

The output voltage is given by:

$$V_{dg} = -I_d.\left(\frac{R_L.(1/j\omega.C_{gd})}{R_L + (1/j.\omega.C_{gd})}\right)$$

or $$I_d = \frac{-V_{dg}(1 + j.\omega.C_{gd}.R_L)}{R_L} \qquad \text{------ (9.44)}$$

From equations (9.43) & (9.44), we have:

$$(g_m + g_d).V_{gS} + g_d.V_{dg} = \frac{-V_{dg}(1 + j.\omega.C_{gd}.R_L)}{R_L}$$

or $$(g_m + g_d).R_L.V_{gS} = -V_{dg}(1 + j.\omega.C_{gd}.R_L + g_d.R_L)$$

or $$A_V = \frac{V_{dS}}{-V_{gS}} = \frac{(g_m + g_d).R_L}{1 + (g_d + j\omega.C_{gd})R_L}$$

(ii) Input admittance will be obtained by applying Miller's theorem to the resistance r_d . In this case a resistance $r_d/(1-A_V)$ will be in parallel with C_{gS}.

So $$Y_i = g_m + \frac{1}{[r_d/(1-A_V)\}} + \frac{1}{1/j.\omega.C_{gS}}$$

or $$Y_i = g_m + g_d(1-A_V) + j\omega C_{gS}$$

Problems:

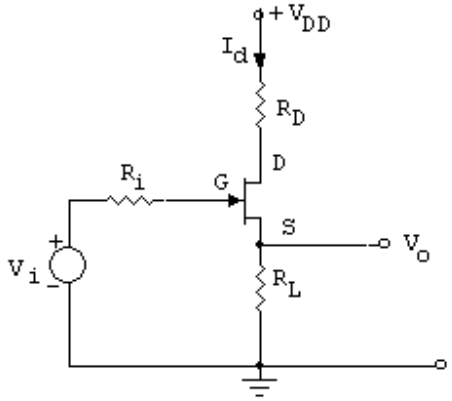
1. What are the advantages of the FET over the conventional transistors? What do the terms Unipolar and Bipolar refer to?
2. Define transconductance g_m, drain resistance r_d and amplification factor μ of a FET. What is the relation between these three constants?
3. Draw the drain characteristics of an N – channel JFET. Explain the shape of these curves qualitatively.
4. Sketch the basic structure of N – channel Junction Field Effect Transistor and explain its working. Give its circuit symbol also.
5. Give the cross section of a P – channel enhancement MOSFET and explain its working. Give the two circuit symbols of this MOSFET.
6. Sketch the basic structure of N – channel depletion type MOSFET and explain its working. Give its two circuit symbols also.
7. Draw and explain the drain and transfer characteristics of a P – channel enhancement MOSFET.
8. Draw and explain the drain and transfer characteristics of an N – channel depletion MOSFET.
9. Draw the small signal model of a field effect transistor.
10. Draw the circuit of common source amplifier and find the expression for the voltage gain and output resistance of this amplifier.
11. Draw the circuit of common drain amplifier and find the expression for the voltage gain and output resistance of this amplifier.
12. Draw the circuit of common gate amplifier and find the expression for the voltage gain and output resistance of this amplifier.
13. What is source follower circuit? Draw the circuit of source follower and show that its voltage gain is almost unity. Find also the expression for the output resistance of the source follower.
14. If the two identical FETs are connected in parallel then prove that the effective transconductance is doubled and drain resistance is halved. The amplification factor remains unchanged.

15. Draw the small signal model of a FET and show that the FET behaves as a voltage controlled current source (VCCS).

16. The figure given below shows the circuit diagram for common drain amplifier with unbypassed drain resistance R_d. Prove that

(i) the expression for the voltage gain

$A_v = \frac{V_O}{V_i} = \frac{\mu . V_i}{r_d + R_D + (1+\mu)R_L} x R_L$, and (ii) the expression for output resistance $R_O = \frac{r_d + R_D}{\mu + 1}$.



17. A Common Source amplifier uses FET having drain resistance r_d = 10 *KΩ* and μ = 16. Calculate the voltage and the output resistance of the amplifier for load resistance equal to (i) 100 *KΩ* (ii) 200 *KΩ* and (iii) 1 *MΩ.*

Ans.: (i) -8, 100 *KΩ* (ii) – 10.67, 100 *KΩ* (iii)- 14.5, *100 KΩ.*

18. A Common Source FET amplifier uses load resistance R_L = 100 k*Ω* and an unbypassed resistance R_S connected between the source and the ground. The drain resistance of the FET is 200 K*Ω* and transconductance g_m = 0.1 millimhos. Compute the voltage gain and the output resistance of the amplifier for R_S equal to :

(i) 2 K*Ω* (ii) 10 K *Ω* and (iii) 20 K*Ω*.

Ans.: (i) – 5.85, 242 *KΩ*, (ii) – 3.9, 410 *KΩ* (iii) – 2.78, 620 *KΩ*

19. A source follower amplifier uses FET having r_d = 200 *KΩ* and μ = 20. The load resistance connected between the source and the ground is 120 *KΩ*. Calculate the voltage gain and output resistance of the amplifier.

(Ans.: 0.88, 9.5 *KΩ*)

10

Multistage Amplifiers

In many applications of electronic circuits, the gain of the single stage transistor amplifier is not sufficient to the desired level, so more stages of amplification are generally used. In such cases the output of one stage is connected to the second stage of the amplifier through some coupling network. In this chapter different types of coupling in addition to the classification of amplifiers will be discussed. High frequency amplifiers, power amplifiers as well as distortion in amplifiers will also be discussed.

10.1 Classification of amplifiers: Amplifiers may be classified into the following ways:

(i) According to frequency range
(ii) According to the method of operation
(iii) According to their applications
(iv) According to the method of inter stage coupling

(i) According to frequency range : The amplifiers are further classified according to frequency range as follows:

(a) **D.C. amplifiers:** These amplifiers are capable of amplifying even zero frequency signals, i.e. D.C. signals.

(b) **Audio frequency amplifiers:** The amplifiers capable of amplifying the signal of frequency range 20 Hz to 20 KHz are known as Audio frequency amplifiers.

(c) **Video amplifiers:** These amplifiers amplify the signal of frequency up to a few megahertz.

(d) **Radio frequency amplifiers:** Its frequency range operation is a few kilohertz to hundreds of megahertz.

(e) **Ultrahigh – frequency amplifiers:** The frequency range of operation of these amplifiers is up to hundreds or thousands of megahertz.

(ii) According to the method of operation: The method of operation means the position of the operating point. The amplifiers are thus classified according to the method of operation as follows:

(a) **Class *A* amplifiers:** In class *A* amplifiers the operating point is chosen in the middle of the load line such that the output current follows for the entire input signal.

(b) **Class *B* amplifiers:** In class *B* amplifiers the operating point is fixed at an extreme end of the characteristics, so that quiescent power is very small. In this case the output current flows only for half of the input sinusoidal signal.

(c) **Class *AB* amplifiers:** The operating point, in this class *AB* amplifier, is fixed between the two extremes defined for class *A* and class *B* amplifiers. Hence in this class of amplifiers the output current is zero for a small part but less than half of the input sinusoidal signal.

(d) **Class *C* amplifiers:** The operating point in this class of amplifiers is fixed such that the output current flows only for less than half of the input sinusoidal.

(iii) According to their applications: The classification according to use includes voltage, power, current or general purpose amplifiers.

(iv) According to the method of inter - stage coupling: The amplifiers according to the method of inter stage coupling may be classified as follows:

(a) **Resistance – capacitance coupling** : When the coupling network, comprising resistance and capacitance is used to couple the output of one stage to the input of second stage then the amplifiers of this type are called as R – C coupled amplifiers.

(b) **Transformer coupling**: Amplifiers using this coupling are known as transformer coupled amplifiers. In these amplifiers transformers are used for inter stage coupling. The transformer coupled amplifiers saves the power dissipation.

(c) **Direct coupling**: For applications where the signal frequency is very low (may be below 10 Hz), the coupling networks discussed above are not used. In such cases output of the one stage is directly coupled to the input of the succeeding stages. The amplifiers using direct coupling are known as direct coupled amplifiers.

10.2 R – C Coupled Amplifier :

The circuit diagram of two stage R – C coupled amplifier is shown in figure (10.1). In this circuit two single stage common emitter amplifiers are coupled through R_{C1} and C_C network.

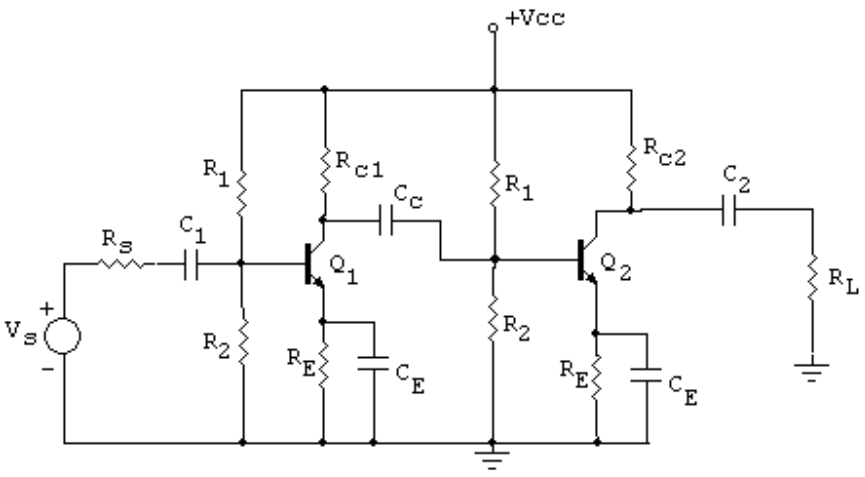
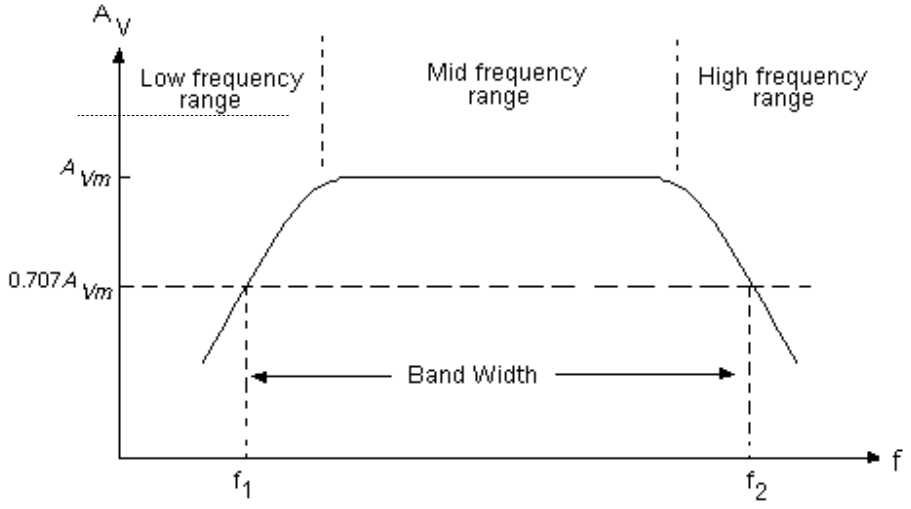


Fig. 10.1

In this circuit the resistances R_1, R_2, R_{C1}, R_{C2} and R_E are used to provide the self biasing and bias stabilization to the two transistors Q_1 and Q_2. The coupling capacitor C_C blocks the quiescent *d.c.* current and prevents it from appearing at the input of the second stage. However, C_C is so chosen that it offers negligible reactance for signal frequencies. Thus it couples the signal effectively to the second stage.

As is well known, the emitter resistance R_E is used to provide good stabilization to the transistor circuits; and larger the value of this resistance better will be biasing stability. But the larger value of R_E introduces larger amount of negative feedback in the circuit which on the contrary reduces the gain of the amplifier. To overcome this difficulty a bye pass capacitor C_E is connected in parallel with the emitter resistance R_E. The choice of a larger value for C_E makes the emitter at the signal ground.

10.2.1 Frequency Response: The curve representing the variation of gain of the amplifier with frequency is known as frequency response curve. It is shown for RC coupled amplifier in figure (10.2). The frequency response of



the *RC* coupled amplifier will be studied over a band of frequencies. The entire frequency range is divided into three regions namely low frequency region, mid frequency region and high frequency region. The coupling and the emitter bye pass capacitances are assumed to be large in the mid and high frequency region so that at the signal frequency these can be considered as short circuited. But at lower frequencies, their reactances will not be negligible. Thus the gain of an *RC* coupled amplifier will fall off at lower frequencies. Generally the frequency response of this amplifier will be governed by the two time constants.

(i) The coupling capacitor C_C together with the input impedance of the second stage of the amplifier will form the simple *RC* high pass filter. The effect of this time constant will be that the gain of the amplifier will reduce at lower frequencies.

(ii) The second time constant is due to the emitter bye pass capacitor C_E. The parallel combination of R_E and C_E offers the finite impedance in the emitter circuit at lower frequencies. As a result of which the gain of the amplifier decreases at lower frequencies.

Now the effect of these two time constants on the frequency response of this amplifier will be discussed.

10.2.2 Effect of coupling capacitor: To study the effect of time constant due to the coupling capacitor C_C together with input impedance of second stage of the amplifier, the effect of second time constant due to R_E and C_E is assumed to be negligible .As a result the emitter is assumed to be shorted to ground. Also the parallel combination of resistances R_1 and R_2 is assumed to be much larger than R_i. So the biasing resistances are not taking into account in the equivalent circuit. The a.c. equivalent circuit and the approximate *h* – parameter model of the amplifier are shown in figures (10.3) and (10.4) respectively. In the approximate *h* – parameter model, only the parameters h_{ie} and h_{fe} are considered and h_{oe} and h_{re} are neglected.

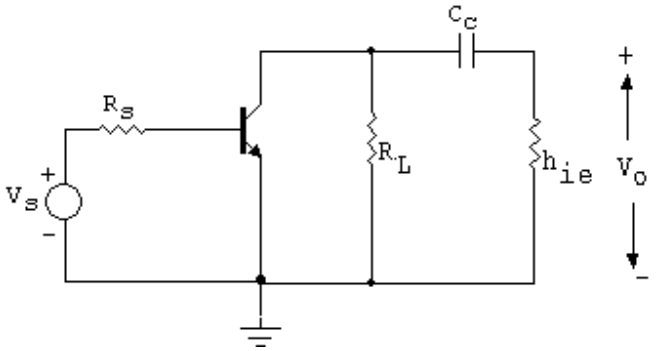


Fig. 10.3

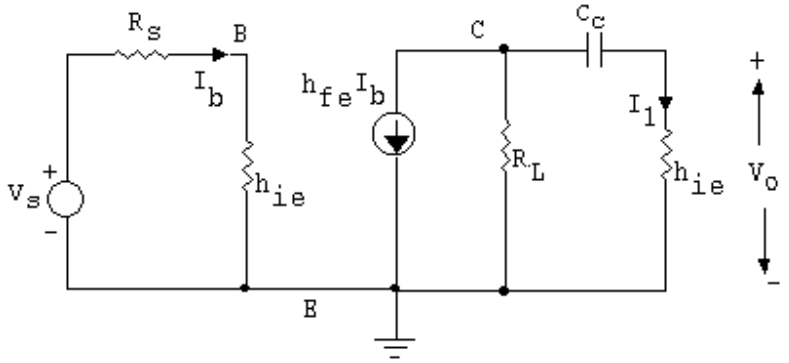


Fig. 10.4

Considering the figure (10.4), the effective load impedance Z_L at the collector of the first stage of the amplifier is the parallel combination of R_L and $(h_{ie} + \frac{1}{j\omega.C_C})$.

$$Z_L = \frac{R_L(h_{ie} + \frac{1}{j\omega.C_C})}{R_L + h_{ie} + \frac{1}{j\omega.C_C}} \qquad \text{------ (10.1)}$$

From the figure (10.4), we have:

$$I_1(h_{ie} + \frac{1}{j\omega.C_C}) = h_{fe} I_b.Z_L$$

or
$$I_1 = \frac{h_{fe} I_b Z_L}{(h_{ie} + \frac{1}{j\omega.C_C})} \qquad \text{------(10.2)}$$

The output voltage V_o is given by:

$$V_o = -I_1.h_{ie}$$

or
$$V_o = -\frac{h_{fe}.h_{ie}.I_b.Z_L}{(h_{ie} + \frac{1}{j\omega C_C})} \qquad \text{------ (10.3)}$$

The input voltage V_s is given by:

$$V_S = (R_S + h_{ie}).I_b \qquad \text{------ (10.4)}$$

Using the equations (10.3) & (10.4), the voltage gain A_{VL1} is given as :

$$A_{VL1} = \frac{V_o}{V_S} = \frac{-h_{fe}.h_{ie}.I_b.Z_L}{(h_{ie} + \frac{1}{j\omega C_C}).I_b.(R_S + h_{ie})}$$

or $$= \frac{-h_{fe}.h_{ie}.Z_L}{(h_{ie} + \frac{1}{j\omega C_C}).(R_S + h_{ie})}$$

Putting the value of Z_L from equation (10.1), A_{VL1} is given as:

$$A_{VL1} = \frac{-h_{fe}.h_{ie}.R_L(h_{ie} + \frac{1}{j\omega.C_C})}{(h_{ie} + \frac{1}{j\omega C_C}).(R_S + h_{ie})(R_L + h_{ie} + \frac{1}{j\omega C_C})}$$

$$= \frac{-h_{fe}.h_{ie}.R_L}{(R_S + h_{ie})(R_L + h_{ie} + \frac{1}{j\omega C_C})} \quad \text{------ (10.5)}$$

or $$A_{VL1} = \frac{-h_{fe}.h_{ie}.R_L}{(R_S + h_{ie})(R_L + h_{ie})(1 + \frac{1}{j\omega.(R_L + h_{ie})C_C})} \quad \text{------(10.6)}$$

This is the low frequency gain when only the effect of coupling capacitor is considered. The mid frequency gain of the amplifier is obtained by neglecting the effect of C_C . So by putting the factor $(\frac{1}{j\omega.C_C})$ equal to zero in equation (10.5), the mid frequency gain of the amplifier may be obtained as:

$$A_{Vm} = \frac{-h_{fe}.h_{ie}.R_L}{(R_S + h_{ie})(R_L + h_{ie})} \quad \text{------ (10.7)}$$

From equations (10.6) & (10.7), we have:

$$A_{VL1} = \frac{A_{Vm}}{(1 + \frac{1}{j\omega.(R_L + h_{ie})C_C})}$$

or $$|A_{VL1}| = \frac{|A_{Vm}|}{\sqrt{1 + \left[\frac{1}{\omega.(R_L + h_{ie})C_C}\right]^2}} \quad \text{------ (10.8)}$$

At $\omega = \frac{1}{(R_L + h_{ie}).C_C}$, the voltage gain of the amplifier reduces to $\frac{1}{\sqrt{2}}$ (or – 3 *db*) of the mid frequency gain of the amplifier, i.e.,

$$|A_{VL1}| = \frac{|A_{Vm}|}{\sqrt{2}}$$

This frequency is known as the lower cutoff frequency,

$$\omega_{L1} = \frac{1}{(R_L + h_{ie}).C_C}$$

or

$$f_{L1} = \frac{1}{2\pi(R_L + h_{ie}).C_C}$$

And the low frequency gain is reduced to

$$|A_{VL1}| = \frac{|A_{Vm}|}{\sqrt{1+\left[\frac{f_{L1}}{f}\right]^2}}$$

10.2.3 Effect of Emitter Bye - pass Capacitor: The effect of second time constant due to R_E and C_E will be studied and the coupling capacitor C_C is assumed to be large enough so that it reactance is negligibly small and it has no effect on the response. The biasing resistances are also not being taking into account in the equivalent circuit. The a.c. equivalent circuit and the approximate *h* – parameter model of the amplifier are shown in figures (10.5) and (10.6) respectively. In the approximate *h* – parameter model only the parameters h_{ie} and h_{fe} are considered and h_{oe} and h_{re} are neglected.

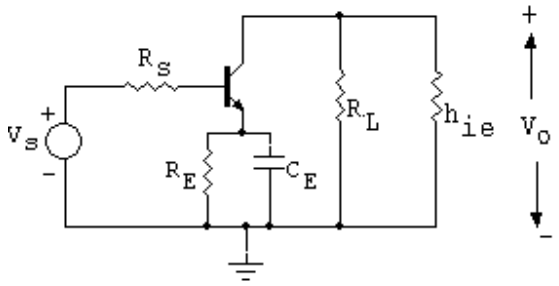


Fig. 10.5

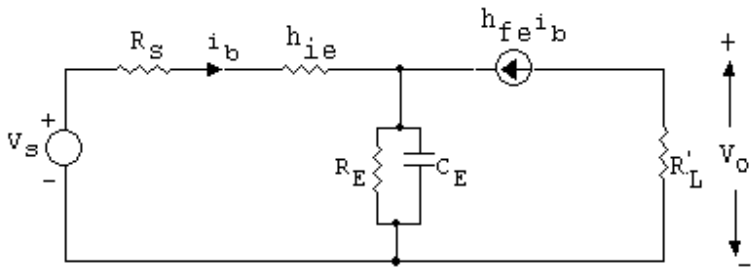


Fig. 10.6

With reference to the figure (10.6) the output voltage V_o is given by:

$$V_o = -h_{fe}.i_b.R_L^{'} \qquad \text{------(10.9)}$$

where $.R_L^{'}$ is the parallel combination of R_L and h_{ie}.

Applying the KVL to the input circuit we get:

$$V_S = (R_S + h_{ie}).i_b + Z_E(1+.h_{fe}).i_b \qquad \text{------ (10.10)}$$

or
$$i_b = \frac{V_S}{R_S + h_{ie} + (1+h_{fe})Z_E} \qquad \text{------ (10.11)}$$

where Z_E is the parallel combination of R_E and C_E given by:

$$Z_E = \frac{(R_E)(\frac{1}{j\omega.C_E})}{(R_E + \frac{1}{j\omega.C_E})} = \frac{R_E}{1+ j.\omega.R_E.C_E} \qquad \text{----- (10.12)}$$

From equations (10.9) and (10.11), V_o is given by:

$$V_o = \frac{-V_S.h_{fe}.R_L^{'}}{R_S + h_{ie} + (1+h_{fe})Z_E}$$

The voltage gain at low frequency is given by:

$$A_{VL.2} = \frac{V_o}{V_S} = \frac{-.h_{fe}.R_L^{'}}{R_S + h_{ie} + (1+h_{fe})Z_E} \qquad \text{------(10.13)}$$

Note that when ω is large (in the mid frequency region), the voltage gain known as mid band gain is given by:

$$A_{VM} = \frac{-h_{fe}.R_L^{'}}{R_S + h_{ie}} \qquad \text{------- (10.14)}$$

Since Z_E will behave as short circuit at the mid and high frequency region.
Put the value of Z_E from equation (10.12) into equation (10.13) we have:

$$A_{VL.2} = \frac{-.h_{fe}.R_L^{'}}{R_S + h_{ie} + \frac{(1+h_{fe})R_E}{1+ j.\omega.R_E.C_E}}$$

or
$$= \frac{-h_{fe}.R_L^{'}.(1+ j.\omega.R_E.C_E)}{(R_S + h_{ie})(1+ j.\omega.R_E C_E) + (1+h_{fe}).R_E}$$

or $$= \frac{-h_{fe}.R_L^{'}.(1+ j.\omega.R_E.C_E)}{\{R_S + h_{ie} + (1+h_{fe}).R_E\} + \{j.\omega.R_E.C_E(R_S + h_{ie})\}}$$

$$= \left(\frac{-h_{fe}.R_L^{'}}{R_S + h_{ie} + (1+h_{fe}).R_E}\right)\left(\frac{1+ j.\omega.R_E.C_E}{1+ j.\omega.\left\{\frac{R_E.C_E(R_S + hie)}{R_S + h_{ie} + (1+h_{fe})R_E}\right\}}\right)$$

$$= \left(\frac{-h_{fe}.R_L^{'}}{R_S + h_{ie} + (1+h_{fe}).R_E}\right)\left(\frac{1+ j.\omega.\tau_1}{1+ j.\omega.\tau_2}\right)$$

where $\tau_1 = R_E.C_E$

and $\tau_2 = \frac{R_E.C_E(R_S + h_{ie})}{R_S + h_{ie} + (1+h_{fe})R_E}$

Generally $(1+h_{fe})R_E >> (R_S + h_{ie})$, so $\tau_2 << \tau_1$.

The voltage gain at $\omega = \frac{1}{\tau_2}$ is, therefore, given by:

$$A_{VL.2} = \left(\frac{-h_{fe}.R_L^{'}}{R_S + h_{ie} + (1+h_{fe}).R_E}\right)\left(\frac{1+ j.\left(\frac{\tau_1}{\tau_2}\right)}{1+ j}\right) \quad \text{------ (10.15)}$$

For $\tau_2 << \tau_1$, the voltage gain is approximated as:

$$A_{VL.2} = \left(\frac{-h_{fe}.R_L^{'}}{R_S + h_{ie} + (1+h_{fe}).R_E}\right)\left(\frac{j.\left(\frac{\tau_1}{\tau_2}\right)}{1+ j}\right)$$

0r $$\left|A_{VL.2}\right| = \frac{1}{\sqrt{2}}\left(\frac{h_{fe}.R_L^{'}(\frac{\tau_1}{\tau_2})}{R_S + h_{ie} + (1+h_{fe}).R_E}\right)$$

Putting the values of τ_1 and τ_2, in the above equation the voltage gain is given by:

or $$|A_{VL.2}| = \frac{1}{\sqrt{2}} \left(\frac{h_{fe}.R_L^{'} \left(\frac{R_E.C_E}{\frac{R_E.C_E(R_S + h_{ie})}{R_S + h_{ie} + (1+h_{fe})R_E}} \right)}{R_S + h_{ie} + (1+h_{fe}).R_E} \right)$$

$$|A_{VL.2}| = \frac{1}{\sqrt{2}} \left(\frac{h_{fe}.R_L^{'}}{R_S + h_{ie}} \right) \qquad \text{------ (10.16)}$$

or $$\left| \frac{A_{VL.2}}{A_{VM}} \right| = \frac{1}{\sqrt{2}} \qquad \text{------ (10.17)}$$

From equation (10.14) $$|A_{VM}| = \frac{h_{fe}.R_L^{'}}{R_S + h_{ie}} \qquad \text{------ (10.18)}$$

Further it is clear from the equation (10.17) that the voltage gain of the amplifier has dropped by 3 *db* from the gain at the mid frequency region. In other words, at $\omega = \frac{1}{\tau_2}$ the magnitude of the gain is $\frac{1}{\sqrt{2}}$ times the mid frequency gain. This frequency is called the lower cut off frequency, which may be given by:

$$\omega_{L2} = \frac{1}{\tau_2} = \frac{R_S + h_{ie} + (1+h_{fe})R_E}{R_E.C_E(R_S + h_{ie})}$$

If $(1+h_{fe})R_E >> (R_S + h_{ie})$ then the lower cut off frequency is given by:

$$\omega_{L2} = \frac{(1+h_{fe})}{C_E(R_S + h_{ie})} \qquad \text{------ (10.19)}$$

or $$f_{L2} = \frac{1+h_{fe}}{2.\pi.(R_S + h_{ie}).C_E} \qquad \text{------ (10.20)}$$

Note that the expression for the lower cut off frequency $f_{C.2}$ does not contain the emitter resistance, so the lower 3 *db* frequency is dependent on the transistor parameters and the source resistance.

10.2.4 High Frequency Response : The variation of voltage gain of the RC coupled amplifier in the high frequency region will now be studied. In this region the coupling capacitor and emitter bye pass capacitor will offer negligibly low reactance and these

capacitances will behave as short circuited. However, in this range of frequencies, effective shunt capacitance C_O due to the input and output capacitances of the devices along with the associated stray wiring capacitance plays important role in the variation of the voltage gain of the amplifier. The detailed discussion of these capacitances will be illustrated in the following section using the high frequency π model of the transistor. The values of these capacitances are very low, so at high frequencies its reactance will considerably be low. The effective load impedance at the collector of the first stage will be the parallel combination of the R_L, h_{ie} and the reactance of C_O. As a result the gain of the amplifier will decrease with the increase in frequency.

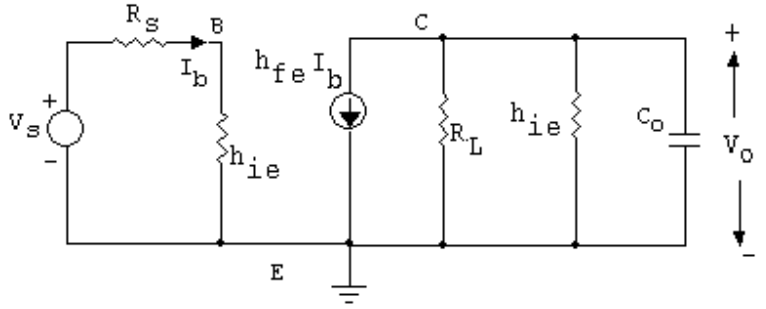


Fig. 10.7

Considering the figure (10.7), the effective load impedance Z_L is given by:

$$\frac{1}{Z_L} = \frac{1}{R_L} + \frac{1}{h_{ie}} + j\omega.C_o$$

$$Z_L = \frac{R_L h_{ie}}{R_L + h_{ie} + j\omega.C_o h_{ie}.R_L} \qquad \text{------ (10.21)}$$

The output voltage V_o is given by:

$$V_o = -h_{fe} I_b . Z_L$$

The input voltage V_s is given by:

$$V_S = (R_S + h_{ie}).I_b \qquad \text{------ (10.22)}$$

Using the equations (10.21) & (10.22), the voltage gain A_{VH} is given as :

$$A_{VH} = \frac{V_o}{V_S} = \frac{-h_{fe}.I_b.Z_L}{I_b.(R_S + h_{ie})} = \frac{-h_{fe}.Z_L}{R_S + h_{ie}}$$

Putting the value of Z_L from equation (10.21), A_{VH} is given as:

$$A_{VH} = \frac{-h_{fe}.h_{ie}.R_L}{(R_S + h_{ie})(R_L + h_{ie} + j.\omega.C_o.h_{ie}.R_L)} \qquad \text{------- (10.23)}$$

The mid frequency gain of the amplifier is obtained by neglecting the effect of C_o . So the mid frequency gain of the amplifier may be obtained as:

$$A_{Vm} = \frac{-h_{fe}.h_{ie}.R_L}{(R_S + h_{ie})(R_L + h_{ie})} \qquad \text{------ (10.24)}$$

From equations (10.23) & (10.24), we have:

$$A_{VH} = \frac{A_{Vm}}{(1 + \frac{j.\omega.C_o.h_{ie}R_L}{R_L + h_{ie}})}$$

or

$$|A_{VH}| = \frac{|A_{Vm}|}{\sqrt{1+\left[\frac{\omega.C_{o.}h_{ie}.R_L}{(R_L + h_{ie})}\right]^2}} \qquad \text{------ (10.25)}$$

At $\omega = \frac{R_L + h_{ie}}{R_L h_{ie}.C_o}$, the voltage gain of the amplifier reduces to $\frac{1}{\sqrt{2}}$ (or –3 *db*) of the mid frequency gain of the amplifier, i.e.,

$$|A_{VH}| = \frac{|A_{Vm}|}{\sqrt{2}}$$

This frequency is known as the higher cutoff frequency,

$$\omega_H = \frac{R_L + h_{ie}}{R_L h_{ie}.C_o}$$

$$f_H = \frac{R_L + h_{ie}}{2\pi R_L h_{ie}.C_o}$$

And the gain at high frequencies is reduced to

$$|A_{VL1}| = \frac{|A_{Vm}|}{\sqrt{1+\left[\frac{f}{f_H}\right]^2}}$$

Example 10.1 The individual voltage gains of the three stages amplifier are 50, 60, 70. Calculate the overall gain in *db* of this three stage amplifier.

Solution: Overall gain = 50 x 60 x 70
= 210000
Gain in *db* = 20 *log* (210000) = 106.4 *db*

Example 10.2 The mid frequency gain of RC coupled amplifier is 100. If the gain falls by 3 *db* at the lower cut off frequency, calculate the gin at the cutoff frequency in *db*.

Solution: Mid frequency gain = 100
Mid frequency gain in *db* = 20 *log* (100) = 40 *db*

Gain at cut off frequency = 40 – 3 = 37 *db*

Example 10.3 The mid frequency gain of a RC coupled amplifier is 100. The values of lower and higher cut off frequencies are 100 *Hz* and 100 *KHz*. Find the frequency at which the gain reduces to 90.

Solition: The gain at lower frequency region is given by:

$$|A_{VL}| = \frac{|A_{Vm}|}{\sqrt{1+\left[\frac{f_L}{f}\right]^2}}$$

According to the problem $A_{Vm} = 100$ $A_{VL} = 90$ $f_L = 100$ *Hz*

So the frequency *f* at which the gain reduces to 90 is given by:

$$\sqrt{1+\left[\frac{100}{f}\right]^2} = \frac{100}{90}$$

or

$$f = \frac{100}{\sqrt{\left[\frac{100}{81}-1\right]}} = 206 \quad Hz$$

The gain at higher frequency region is given by:

$$|A_{VH}| = \frac{|A_{Vm}|}{\sqrt{1+\left[\frac{f'}{f_H}\right]^2}}$$

According to the problem $A_{Vm} = 100$ $A_{VH} = 90$ $f_H = 100$ *KHz*

So the frequency *f*' at which the gain reduces to 90 is given by:

$$\sqrt{1+\left[\frac{f'}{100}\right]^2} = \frac{100}{90}$$

or

$$\frac{f'}{100} = \sqrt{\frac{100}{81}-1} = 0.484$$

or

$$f' = 0.484x100 = 48.4 \ KHz$$

Example 10.4 The *h* parameters of the transistors used in two stage RC coupled amplifier are $h_{fe} = 600$, $h_{ie} = 10\ K\Omega$. If the shunt capacitance at high frequency is 400 *pf*, coupling capacitance is 0.5 μF and $R_L = 10\ K\Omega$, calculate the lower and higher cutoff frequencies of the amplifier. The source may assume to be negligibly small.

Solution: The lower cut off frequency is given by:

$$f_L = \frac{1}{2\pi(R_L + h_{ie}).C_C}$$

$$= \frac{1}{2x3.14x(10+10)x10^3\,x0.5x10^{-6}}$$

$$= \frac{1000}{62.8} = 15.9\ Hz$$

The higher cut off frequency is given by:

$$f_H = \frac{R_L + h_{ie}}{2\pi R_L h_{ie}.C_o}$$

$$= \frac{(10+10)x10^3}{2x3.14x10^4\,x10^4\,x400x10^{-12}}$$

$$= \frac{10^6}{12.56} = 79.6\,KHz$$

10.3 Hybrid π - model For the CE Transistor Amplifier: Hybrid π model of the transistor will be considered to discuss the performance of a transistor amplifier at high frequencies. The capacitive effect of the PN junctions of the transistor is taken into account. This model helps in determining the common emitter short circuit current gain and its dependence on frequency. The hybrid π model also known as the Giacoletto model of the transistor is shown in figure 10.8.

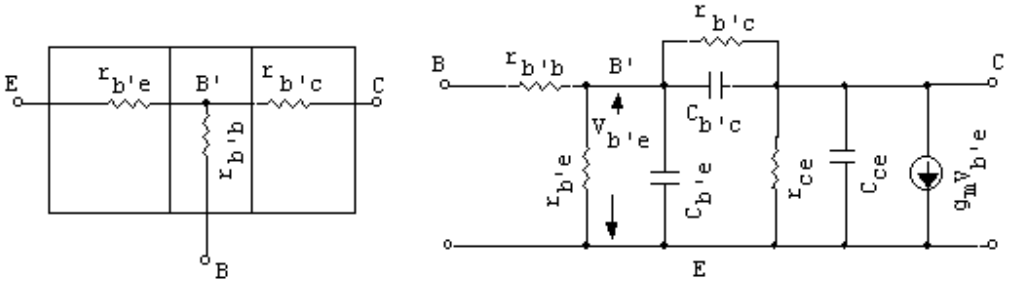


Fig. 10.8

In a transistor forward bias, emitter base junction has a capacitance associated with it. This arises from the diffusion across the base emitter junction. This is represented by $C_{b'e}$. The node *B'* is an imaginary node within the base of the transistor. The resistance $r_{b'.b}$ is the bulk resistance of the base region through which current has to flow from base lead *B*. The resistance $r_{b'C}$ is the incremental resistance; $C_{b'C}$ is the transition capacitance of the reverse biased collector base junction. The resistance and capacitance between the output terminals (collector and emitter) are r_{ce} and c_{ce}.

$V_{b'e}$ is the voltage across the emitter base junction. For small changes in $V_{b'e}$ across the emitter junction, the extra minority carrier concentration injected into the base

region is proportional to $V_{b'e}$. Now the small signal collector current with collector shorted to emitter will be proportional to the voltage $V_{b'e}$. It will act as the constant current source across the collector and emitter having the magnitude equal to $g_m.V_{b'e}$.

The common emitter short circuit current gain may be calculated using this hybrid π model. The following assumptions are made while calculating the short circuit current gain of the common emitter amplifier.

(i) In the equivalent circuit, $r_{b'c}$ is neglected as $r_{b'c} >> r_{b'e}$.

(ii) The resistance r_{ce} disappears as output is short circuited.

(iii) The load resistance $R_L = R_C = 0$.

The equivalent circuit of the common emitter amplifier will be as shown in figure 10.9.

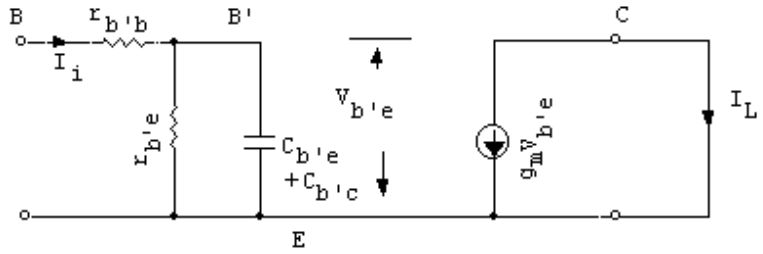


Fig. 10.9

From this equivalent circuit the short circuit current I_L is given by:

$$I_L = -g_m.V_{b'e} \qquad \text{------ (10.26)}$$

The voltage $V_{b'e}$ is given by:

$$V_{b'e} = I_i.Z_{eq} \qquad \text{----- (10.27)}$$

where Z_{eq} is the parallel combination of the resistance $r_{b'e}$ and the capacitive reactance $j.\omega.(C_{b'c} + C_{b'e})$ given by:

$$Z_{eq} = \frac{r_{b'e}.\dfrac{1}{j.\omega.(C_{b'e} + C_{b'c})}}{r_{b'e} + \dfrac{1}{j.\omega.(C_{b'e} + C_{b'c})}}$$

$$= \frac{1}{\dfrac{1}{r_{b'e}} + j.\omega.(C_{b'e} + C_{b'c})} \qquad \text{------ (10.28)}$$

Using the equations (10.26) to (10.28), the short circuit current gain is given as:

$$I_L = \frac{-g_m.I_i}{\dfrac{1}{r_{b'e}} + j.\omega.(C_{b'e} + C_{b'c})}$$

The short circuit current gain A_I is obtained as:

$$A_I = \frac{I_L}{I_i} = \frac{-g_m}{\frac{1}{r_{b'e}} + j.\omega.(C_{b'e} + C_{b'c})} \qquad \text{------ (10.29)}$$

Put $r_{b'e} = \frac{h_{fe}}{g_m}$ in equation (10.24) we have:

$$A_I = \frac{-g_m h_{fe}}{g_m + j.\omega.h_{fe}.(C_{b'e} + C_{b'c})}$$

$$= \frac{-h_{fe}}{1 + j.\omega.r_{b'e}.(C_{b'e} + C_{b'c})}$$

$$= \frac{-h_{fe}}{1 + j.\left(\frac{f}{f_h}\right)} \qquad \text{------- (10.30)}$$

where $f_h = \frac{1}{2.\pi.r_{b'e}.(C_{b'e} + C_{b'c})}$, is the cutoff frequency.

When $f = 0$, $A_I = -h_{fe}$ (low frequency current gain with output short circuited).

At $f = f_H$, $A_I = \frac{-h_{fe}}{\sqrt{2}}$ i.e. the current gain falls by $\frac{1}{\sqrt{2}}$ times the low frequency gain (or –3db fall of gain). The gain variation is shown in figure (10.10).

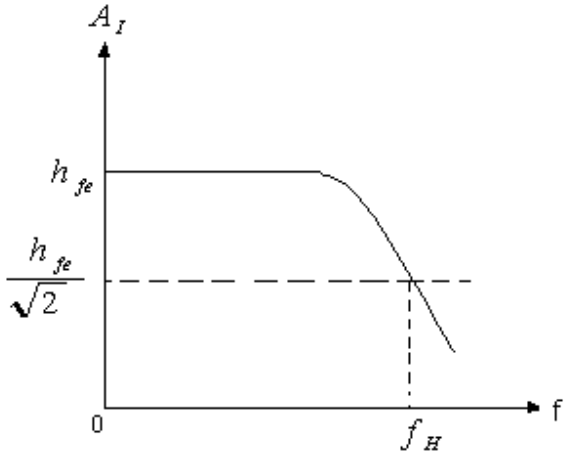


Fig. 10.10

10.4 Class A power Amplifier: The power amplifier also called large signal amplifier is used to obtain large power at the output of the amplifier. To have the signal up to the desired level, before the signal is applied to the power amplifier the signal is to be amplified by the multistage amplifiers. In many electronic systems, such as public address system, audio amplifier of the television receiver or transistor receiver, power amplifiers are used. The power amplifier provides a large voltage swing and also large current swing, so as to deliver the maximum power to the load with minimum distortion. Figure 10.11 shows the class *A* power amplifier in the common emitter configuration. Here the load resistance R_L is directly coupled to the amplifier.

The theoretical efficiency of such an amplifier will be calculated which is defined as the ratio of the output power delivered to the load to the d.c. input power from the power supply.

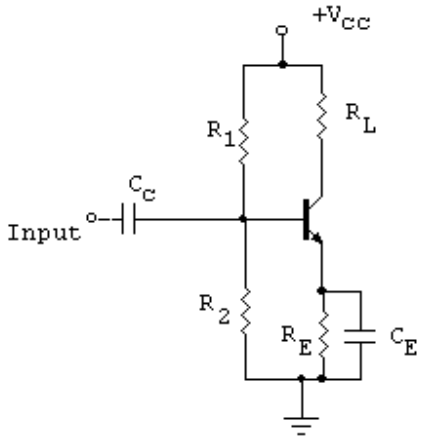


Fig. 10.11

To calculate the efficiency of this transistor amplifier circuit, it is assumed that the emitter is at the signal ground and the $V_{CE,sat.}$ is equal to zero, so that the signal swing from 0 to V_{CC}. The quiescent voltage V_{CQ} will be equal to peak swing $V_m = \frac{V_{CC}}{2}$. The maximum collector current is $I_m = \frac{V_{CC}}{2R_L}$.

The output power $P_o = V_{rms} . I_{rms}$

$$= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} = \frac{V_m . I_m}{2}$$

$$= \frac{1}{2} \cdot \frac{V_{CC}}{2} \cdot \frac{V_{CC}}{2R_L}$$

$$= \frac{V_{CC}^2}{8R_L}$$

The input power from the supply is given by:

$$P_{d.c}. = V_{CC}.I_{CQ}$$

where I_{CQ} is the average current from the supply which is equal to the maximum collector current I_m. So the input power from the supply is given by:

$$P_{d.c.} = \frac{(V_{CC})^2}{2R_L}$$

The maximum efficiency is given by:

$$\eta = \frac{P_o}{P_{d.c.}} = \frac{\{V_{CC}^2 / 8R_L\}}{\{V_{CC}^2 / 2R_L\}}$$

$$= \frac{1}{4} = 25\ \%$$

This is the maximum theoretical efficiency; however, the efficiency of the practical amplifier will always be less than this efficiency, because the bias resistances also consume some power which has not been considered in the above calculations.

10.5 Transformer Coupled Amplifier: The load for the power amplifier is generally in the form of the voice coil of the loudspeaker having low output impedance. However, the ordinary class *A* amplifier has the high impedance in the collector circuit. So the maximum power will not be transferred from the output of the power amplifier to the voice coil of the loudspeaker. In order to transfer the maximum power to the load impedance, the transistor collector circuit should be coupled by means of a transformer. The circuit of a transformer coupled amplifier is shown in figure (10.12).

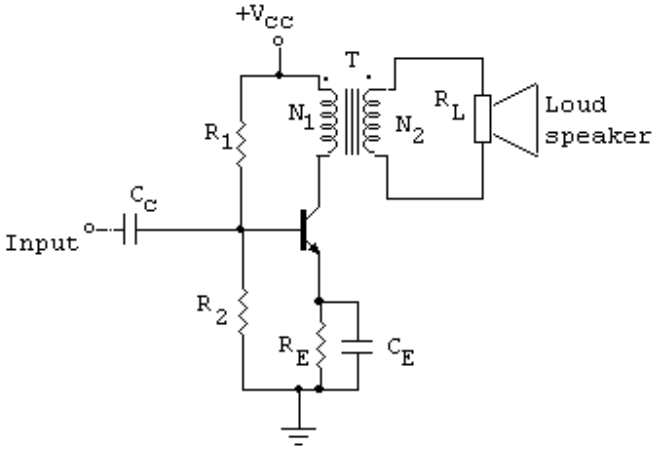


Fig. 10.12

In this circuit there is no coupling capacitor, the d.c. isolation is provided by the transformer itself. There exist no d.c. path between the primary and the secondary windings of a transformer. The a.c. signal across the primary winding is transformed to

the secondary windings. The function of the resistance R_1 and R_2 are used to bias the transistor in class A mode. The emitter resistance R_E provides the bias stabilization and the bye pass capacitor C_E is used to prevent the amplified signal to appear at the input.

From the figure 10.12, it is clear that the effective resistance in the collector circuit is approximately equal to d.c. resistance of the primary of the transformer which is negligibly small. The d.c. load line will, therefore, be vertical straight line passing through the operating point as shown in figure 10.13.

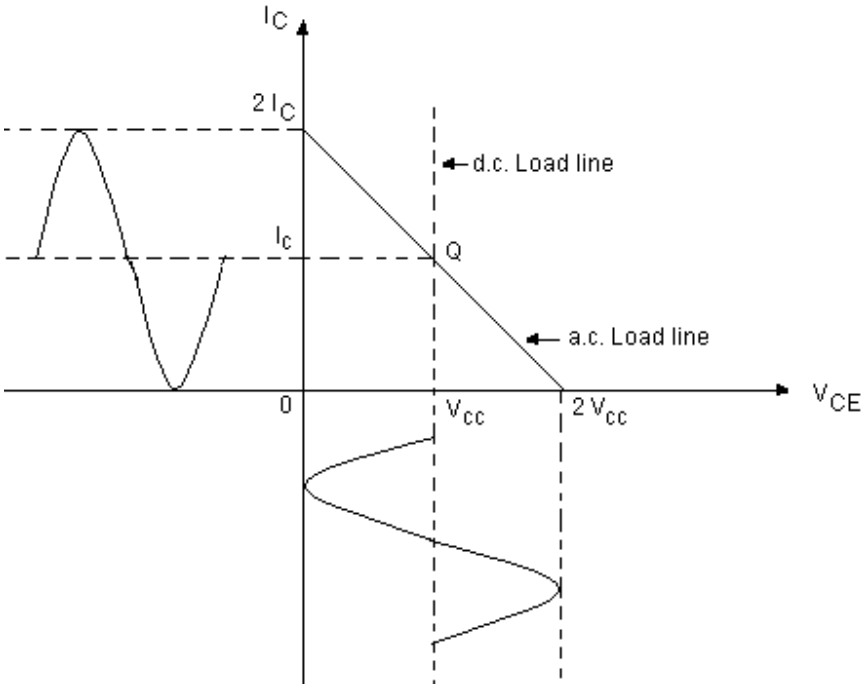


Fig. 10.13

When a.c. signal is applied to the input of the amplifier, the resistance in the collector circuit is formed by the reflected resistance of the load. Thus the effective load R'_L as seen by the collector of the amplifier is given by:

$$\frac{R'_L}{R_L} = \frac{V_1/I_1}{V_2/I_2} = \frac{V_1}{V_2}\frac{I_2}{I_1}$$

$$= \frac{N_1}{N_2}\frac{N_1}{N_2} = \left(\frac{N_1}{N_2}\right)^2$$

Where V_1 and V_2 are the voltages across the primary and the secondary windings of a transformer respectively and I_1 and I_2 are the corresponding current in the primary and the secondary windings of the transformer.

or $$R'_L = n^2.R_L$$

where $n = \left(\frac{N_1}{N_2}\right)$ is the turn ratio.

The operating point moves along the a.c. load line whose slope is $\left(-\frac{1}{R_L^{'}}\right)$, after the application of the large signal to the input of the amplifier. During the peak of the positive half cycle of the signal, the collector current is $2I_C$ and $V_{CE} = 0$. However during the negative half cycle, the collector current is zero and $V_{CE} = 2V_{CC}$.

Thus peak to peak collector emitter voltage is given by:

$$v_c = 2.V_{CC}$$

The r.m.s. value of the collector voltage is $V_{rms} = \frac{1}{2}\frac{v_C}{\sqrt{2}} = \frac{V_{CC}}{\sqrt{2}}$

The peak to peak collector current is given by:

$$i_C = \frac{v_C}{R_L^{'}} = \frac{2.V_{CC}}{R_L^{'}}$$

The r.m.s. value of the collector current is :

$$I_{rms} = \frac{1}{2}\frac{i_C}{\sqrt{2}} = \frac{V_{CC}}{\sqrt{2}R_L^{'}}$$

The output power $P_o = V_{rms} . I_{rms}$

$$= \frac{V_{CC}}{\sqrt{2}} . \frac{V_{CC}}{\sqrt{2}\, R_L^{'}}$$

$$= \frac{V_{CC}^2}{2\, R_L^{'}}$$

The d.c. input power from the supply is

$$P_{d.c.} = V_{CC} . I_C = \frac{V_{CC}^2}{R_L^{'}}$$

The maximum efficiency is given by:

$$\eta = \frac{P_o}{P_{d.c.}} = \frac{\{V_{CC}^2 / 2R_L^{'}\}}{\{V_{CC}^2 / R_L^{'}\}}$$

$$= \frac{1}{2} = 50\ \%$$

The transformer coupled transistor amplifier has the advantage that its theoretical efficiency is 50% which is twice the power available in the RC coupled amplifier. Secondly the transformer provides the proper impedance matching to the output impedance of the collector circuit with the load resistance of the few ohms (resistance of the voice coil of the loud speaker). However, this amplifier has the disadvantage that the output transformer saturates (because of the core saturation), when the large d.c. current flows through the primary of the transformer. This leads the distortion at the extreme points of the output signal.

Example 10.5: A 12 volt battery is connected to a power transistor used in class A mode. If the maximum change in collector current is 120 *mA*, find the power transferred to the load, when a loudspeaker of 8 Ω is:

(i) connected directly in the collector circuit.

(ii) transformer coupled.

Solution: (i) Voltage across the loudspeaker when directly connected to the collector circuit V_C is given by:

$$V_C = \Delta I_C x R_L$$

$$= 120 \quad x8 = 960 \quad mV \quad = 0.96 \; Volt$$

Required power transferred to the loudspeaker is given by:

$$P_L = V_C \; xI_C$$

$$= 0.96 \; Vx \; 120 \quad mA \quad = 115 \quad .2 \; mW$$

(ii) The output impedance looking at the primary of the transformer is equal to the output impedance at the collector is given by:

$$R_L^{'} = \frac{V_{CC}}{\Delta I_C} = \frac{12V}{120 \; mA} = 100 \; \Omega$$

Now $R_L^{'} = n^2 . R_L$

$100 = n^2 x 8$

or $n = \sqrt{12.5} = 3.54$

This is the turn ratio of the transformer.

Voltage at the secondary of the transformer $= \frac{12V}{3.54} = 3.39V$

Now the load current $I_L = \frac{3.39\,V}{8\Omega} = 0.424 \; A$

The power delivered to the loudspeaker

$$= I_L^2 . R_L = (0.424)^2 x8$$

$$= 1.44 \quad Watt$$

Example 10.6 If the load connected to the secondary of the transformer in a transformer coupled power amplifier is 15 Ω and zero signal current is 120 *mA*. Find the maximum power output, if the turn ratio of the transformer is 10.

Solution: The effective load resistance at the collector of the transistor is:

$$R_L^{'} = n^2 R_L$$

$$= (10)^2 15 = 1500 \; \Omega$$

Power output $P_O = I_{rms}^2 . R_L^{'}$

And $I_{rms} = \frac{I_C}{\sqrt{2}}$

So $P_O = \frac{I_C^2}{2} . R_L^{'} = \frac{(120)^2 x1500}{2} \mu W$

$= 10.8W$

10.6 Class B Push – Pull Amplifier: The most frequently used power amplifier in the output stage of electronic circuits is the push pull amplifier. The push pull amplifier consists of two transistors and each transistor conducts in class *B* operation i.e. transistors are biased in the cutoff region so that the current flows only for half cycle in each transistor. In push pull amplifiers the transistors can also be used in class *A* mode but in this case the amplifier efficiency is not more than 50% and distortion is high. However the class *B* push pull amplifier provides low distortion and has the efficiency about 78.5%. The class *B* push pull amplifier is, therefore, used in the practical circuits. Figure 10.14 shows the circuit diagram of this class *B* push pull amplifier.

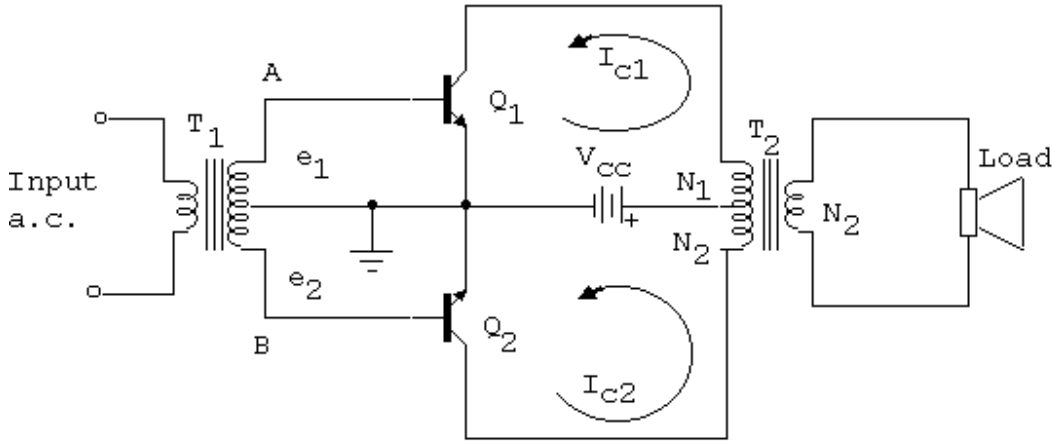


Fig. 10.14

This circuit consists of two transistors *Q1* and *Q2* whose emitters are coupled together and two centre tapped transformers T_1 and T_2. The two transistors are biased in the cutoff region so that the collector current flows only for half of the input cycle. The input transformer T_1 converts the input signal into two waves e_1 and e_2 which are 180^0 out of phase. When the point *A* is positive with respect to ground, point *B* will be negative with respect to ground and vice versa.

During the positive half of the input signal, the point *A* is positive and point *B* is negative with respect to the common point. The transistor Q_1 will conduct and the collector current Ic_1 will flow in the collector circuit of the transistor Q_1. The transistor Q_2 will, of course, be in the cutoff stage and the collector current I_{c2} in the collector circuit of transistor T_2 will not flow. During the next cycle of the input signal the case is reversed that the collector current I_{c2} in transistor Q_2 will be flowing and the collector current I_{c1} in the transistor Q_1 will be zero. These two currents flow in opposite directions in the two halves of the primary windings of the output transformer T_2. The centre tapped primary of this transformer T_2 combines two collector currents to form a sine wave output at the secondary. The effective load as seen by the amplifier is given by:

$$R_L^{'} = n^2.R_L$$

Where R_L is load resistance at the secondary of the transformer T_2 and *n* is the turn ratio of this transformer which is given by:

$$n = \left(\frac{2N_1}{N_2} \right).$$

It is worth to mention that in the first half cycle of the input signal the transistor Q_1 is pushed up for conduction and Q_2 is pulled down.; and in second cycle the case is reversed. Hence it is named as push pull amplifier.

Analysis: Let the base currents of the two transistors are sinusoidal in nature which are given as :

$$I_{B1} = I_B \sin \omega.t$$

and $$I_{B2} = I_B \sin(\omega.t + \pi)$$

Where ω is frequency of the input signal in radians and π is phase difference which occurs in the two signals at the base of the two transistors due to secondary windings of the input transformer.

Due to the nonlinear characteristics of the transistors the collector currents will not be the function of frequency ω of their signals at the base but will have the higher harmonics of this frequency ω in addition to the d.c. currents. The collector current I_{c1} and I_{c2} are given as:

$$I_{c1} = I_0 + I_1 \sin \omega.t + I_2 \sin 2\omega.t + I_3 \sin 3\omega.t + ...$$

$$I_{c2} = I_0 + I_1 \sin(\omega.t + \pi) + I_2 \sin 2(\omega.t + \pi) + I_3 \sin 3(\omega.t + \pi) + ...$$

$$= I_0 - I_1 \sin \omega.t + I_2 \sin 2\omega.t - I_3 \sin 3\omega.t + ...$$

The coefficients of harmonic terms in the two equations are same since the transistors are assumed to be identical. The matched pairs of transistors for push pull amplifiers are available in the market.

The voltage induces in the secondary of the output transformer T_2 which is proportional to the difference of collector currents $(I_{c1} \sim I_{c2})$, which is given by:

$$V_0 = k(I_{c1} \sim I_{c2})$$

$$= 2k\left(I_1 \sin \omega.t + I_3 \sin 3\omega.t + I_5 \sin 5\omega.t + ...\right)$$

where k is the proportionality constant.

It is important to note from the above equation that the d.c. components (which causes the saturation in the core of the output transformer) and all even harmonic terms are eliminated leaving the third harmonic term as the only source of distortion. Fifth and other odd harmonic terms will have negligible magnitude since in the power amplifiers all harmonic terms have the magnitudes in the decreasing order not equal. So the net distortion in the output of the push pull amplifier is very less than that of the single ended amplifier.

Theoretical Efficiency: The current drawn in the load is the sum of currents I_{c1} and I_{c2} and will have the full wave rectified output as shown in figure 10.15.

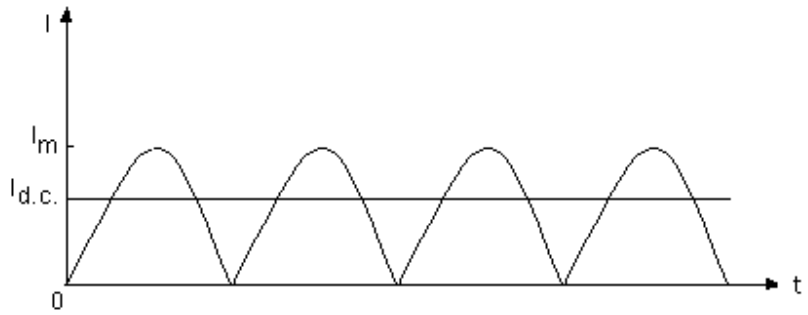


Fig. 10.15

The average value of this full wave rectified current is given by:

$$I_{dc} = \frac{2I_m}{\pi}$$

I_m is the peak value of current given by:

$$I_m = \frac{V_{CC}}{R_L^{'}}$$

so

$$I_{dc} = \frac{2.V_{CC}}{\pi.R_L^{'}}$$

$$P_{dc} = V_{CC}.I_{dc} = \frac{2.V_{CC}^2}{\pi.R_L^{'}}$$

The maximum a.c. power in the load is given by:

$$P_o = V_{rms}.I_{rms}$$
$$= \frac{V_{CC}}{\sqrt{2}} \cdot \frac{V_{CC}}{\sqrt{2}\,R_L^{'}}$$
$$= \frac{V_{CC}^2}{2\,R_L^{'}}$$

Thus the maximum efficiency is

$$\eta = \frac{P_o}{P_{d.c.}} = \frac{\{V_{CC}^2 / 2R_L^{'}\}}{\{2V_{CC}^2 / \pi.R_L^{'}\}}$$
$$= \frac{\pi}{4} = 78.5\%$$

A**dvantages of Push Pull Amplifier**:

1. The magnetic saturation of the core of the output transformer by the d.c. current does not occur, as it opposes each other in the transformer. Small size transformers can, therefore, be used in the push pull amplifier which will reduce its cost.
2. The harmonic distortion in the output is very less due to the cancellation of all the even harmonic components.
3. The maximum theoretical efficiency of this amplifier is 78.5% which is much larger than that of the single ended amplifier.
4. The ripple content in the d.c. power supply does not affect the output because these components get cancelled, as the ripple currents flow in the opposite direction in the two halves of the primary windings of the output transformer.

Disadvantages of Push Pull Amplifier:

1. Two identical transistors must be used in this circuit otherwise harmonic terms or harmonic distortion will not be reduced.
2. The push pull amplifier becomes very bulky because of the use of two centre tapped transformers.
3. The frequency response of this amplifier becomes poor because of the stray inter-winding capacitances.

10.7 More About Properties of Amplifiers: One must study the following properties about the amplifiers.

10.7.1 Distortion: When a sinusoidal signal is applied to the input of an amplifier, the output should be the exact replica of the input wave shape with, of course, having power gain. But due to some reasons, output wave form differs from the input signal wave shape, this change in output wave form is known as distortion. There are generally of the following three types of distortions present individually or simultaneously in the amplifiers.

1. Non – linear distortion
2. Frequency distortion
3. Phase – shift distortion

1. Non – linear distortion: Due to the non – linearity of the dynamic transfer characteristics of the active device in the operating range, the wave form of the output voltage differs from that of the input signal. Such a distortion is called non –linear or amplitude distortion. Further it is observed that the output signal does not only contain the input signal frequency but also the new frequencies which are higher harmonics of the frequency of the input signal. This type of distortion is called as harmonic distortion. Quantitatively harmonic distortion is defined as the ratio of the total power in all the harmonics to the power contained in the fundamental frequency at the output.

Consider a sinusoidal signal $v_i = v_o \sin \omega.t$ is applied to the input of the power amplifier, and then the output current wave form may be expressed as:

$$i_0 = I_0 + I_1 \sin \omega.t + I_2 \sin 2\omega.t + I_3 \sin 3\omega.t + ...$$

where I_0 is the d.c. component, I_1 the peak value of fundamental frequency (original frequency or first harmonic), I_2 the peak value of the second harmonic and so on.

The harmonic distortion for the second harmonic known as second harmonic distortion is given by:

$$D_2 = \frac{I_2}{I_1}$$

Similarly, third harmonic distortion $D_3 = \frac{I_3}{I_1}$

and so on.

When distortion is present that power delivered to the fundamental frequency is :

$$P_1 = \frac{I_1^2 R_L}{2}$$

The total power delivered to all the harmonic components is given by:

$$P_T = \left[I_1^2 + I_2^2 + I_3^2 + ...\right]\frac{R_L}{2}$$

$$= \frac{I_1^2 R_L}{2}\left[1 + \left(\frac{I_2}{I_1}\right)^2 + \left(\frac{I_3}{I_1}\right)^2 + ...\right]$$

$$= P_1\left[1 + D_2^2 + D_3^2 + ...\right]$$

$$= P_1\left[1 + D^2\right]$$

where D is called as the total distortion or distortion factor given by:

$$D = \sqrt{D_2^2 + D_3^2 + ...}$$

Example 10.7 When a sinusoidal signal of fundamental frequency of 500 radians/ sec. is fed to a transistor amplifier, the resulting output collector current is of the form:

$$I_C = 12.Sin500t + 1.1.Sin1000t + Sin1500t + 0.6.Sin2000t + ...$$

Calculate (i) Second, third and fourth harmonic distortions.

(ii) Percentage increase in power due to harmonic distortion.

Solution: (i) Second harmonic distortion:

$$D_2 = \frac{I_2}{I_1} = \frac{1.1}{12} = 0.0917 = 9.17\%$$

Third Second harmonic distortion:

$$D_3 = \frac{I_3}{I_1} = \frac{1.0}{12} = 0.083 = 8.33\%$$

Fourth harmonic distortion:

$$D_4 = \frac{I_4}{I_1} = \frac{.6}{12} = 0.05 = 5\%$$

(ii) Distortion Factor

$$D = \sqrt{D_2^2 + D_3^2 + D_4^2}$$

$$= \sqrt{(0.0917)^2 + (0.083)^2 + (0.05)^2} = 0.133 = 13.3\%$$

Power delivered to all the harmonic components:

$$= P_1(1 + D^2) = P_1\left(1 + (.133)^2\right)$$

$$= 1.0177 \ \ P_1$$

Percentage increase in power due to distortion:

$$= \frac{(1.0177 - 1)P_1}{P_1} x100 = 1.77\%$$

2. Frequency Distortion: When the signals of different frequencies are amplified by an amplifier than the amplification will be different at different frequencies. The frequency distortion is, therefore, said to exist. At different frequencies the different behavior of the amplifier is due to coupling network or due to the device inter–capacitances. The graph plotted between the gains of the amplifier versus the frequency of the input signal is called as amplitude frequency response characteristic. When the frequency response characteristic is not horizontal over the range of frequencies under consideration, the frequency distortion is said to exist.

3. Phase – Shift Distortion: It has been observed that all amplifiers behave as low pass filter at high frequencies. In the region of frequency response curve where the gain falls, the signals of different frequencies suffer unequal phase shift. The phase shift distortion is, therefore, said to exist. The plot of the phase shift as a function of frequency is called phase response. There will be no phase shit distortion if there is linear increase in phase shift with frequency. The phase shift distortion has no consequence in audio amplifiers since human ear is not sensitive to the phase relationship between the Fourier signal components. Video amplifiers used in television system suffer problem due to phase shift distortion, and highly distorted pictures will be observed. Video amplifiers are thus designed to have linear phase – shift characteristics.

10.7.2 Noise in Amplifiers: It has been observed that even when no signal is applied to the input of an amplifier, some voltage variation of appreciable magnitude is available at its output. This voltage variation is referred to as noise. Sometimes it is impossible to distinguish between the signal and the background noise. The background hiss in a radio receiver is an example of noise. Snow like appearance on the television screen is due to the noise in the video amplifier of the television system. Noise in the amplifiers is caused by ransom movement of charge carriers in transistors and resistors used in amplifiers. Various sources of noise in an amplifier are being discussed.

10.7.3 Thermal Noise or Johnson Noise: The random movement of electrons in a conductor is due to the thermal energy possessed by the electrons. If there a small fluctuation in the energy, it will produce small noise potential in the conductor. This type

of noise is known as thermal noise or Johnson noise. The mean square value of the thermally varying voltage in a resistor R at T^0K is given by:

$$E_n^2 = 4RKT(BW)$$

where k is the Boltzmann's constant (Joules/^{0}K),
T is the absolute temperature of the resistor, and
BW is the bandwidth (Hz).

This thermal voltage available across the resistance will be amplified and appear as the noise at the output terminals, when the resistance is connected across the input terminals of the amplifier. This noise is also called the white noise since it gives almost the same noise per unit band width over a wide frequency spectrum.

10.7.4 Shot Noise: The current in a transistor or field effect transistor is normally assumed to be constant under d.c. conditions. But in practice there are fluctuations in these currents due to the random movements of charge carriers in the semiconductor. In other words the d.c. current is the average value of this current. Microscopically one can say that the random component of the current known as shot noise is superimposed on the average value.

The mean square value of the shot noise current is proportional to the d.c. current I_{dc} given by:

$$I_N^2 = 2qI_{dc}(BW)$$

where q is the magnitude of the electronic charge.

If R_L is the value of load resistance than the noise voltage $I_N R_L$ will appear across the load resistance.

10.7.5 Noise Figure: To know quantitatively how noisy a device is, a term known as noise figure has been introduced. Before discussing the noise figure one more quantity known as signal to noise power ratio, which is defined as the ratio of the input signal power to the noise power at the input signal. The signal noise power ratio may be defined for both the input as well as for the output. Thus the input signal to noise power ratio (*SNPI*) and output signal to noise power ratio (*SNPO*) are given by:

$$SNPI = \frac{S_{Pi}}{N_{Pi}}$$

$$SNPO = \frac{S_{Po}}{N_{Po}}$$

where S_{Pi} = signal power input
N_{Pi} = Noise power input due to source resistance
S_{Po} = signal power output
N_{Po} = noise power output

The ratio of the input signal to noise power ratio (SNPI) and output signal to noise power ratio (SNPO) is called as the noise figure given as:

$$NF = \frac{S_{Pi}/N_{Pi}}{S_{Po}/N_{Po}} = \frac{S_{Pi}.N_{Po}}{S_{Po}.N_{Pi}}$$

Noise figure is often expressed in decibels as :

$$NF\,(db) = 10\log\left(\frac{S_{Pi}.N_{Po}}{S_{Po}.N_{Pi}}\right)$$
$$= 10\ \log\left(\frac{S_{Pi}}{N_{Pi}}\right) - 10\ \log\left(\frac{S_{Po}}{N_{Po}}\right)$$

Noise figure in decibels is given by the input signal to noise power ratio in decibels minus output signal to noise power ratio in decibels.

The noise figure in decibels can also be given in form of signal to noise voltage ratio as:

$$NF\,(db) = 20\log\left(\frac{S_{Vi}.N_{Vo}}{S_{Vo}.N_{Vi}}\right)$$
$$= 20\log\left(\frac{S_{Vi}}{N_{Vi}}\right) - 20\log\left(\frac{S_{Vo}}{N_{Vo}}\right)$$

where (S_{Vi}/N_{Vi}) is called input signal to noise voltage ratio, and

(S_{Vo}/N_{Vo}) is called output signal to noise voltage ratio.

Example 10. 8 The signal input to a small signal amplifier consists of 60 μW of signal power and 0.8 μW of noise power. The amplifier generates an internal noise of power of 50 μW and has power of 20 *db*. Compute for this amplifier:

(i) Signal to Noise power input (SNPI)

(ii) Signal to Noise power output (SNPO)

(iii) Noise Figure and its value in *db* also.

Solution: (i) Signal to Noise power input

$$SNPI = \frac{S_{Pi}}{N_{Pi}} = \frac{60\,\mu W}{0.8\mu W} = 75$$

(ii) Power gain is given as 20 db

i.e. $20 = 10\log_{10}(power.gain)$

So power gain $= 100$

Input signal power = 60 μW

Output signal power (S_{Po}) = (input signal power)x(power gain)

$= 60 \; x \; 100 = 6000 \; \mu W$

Noise power output (N_{Po}) =

(power gain).(noise power input)+ internal noise of the amplifier

$= 100 \text{ x } 0.8 \; \mu W + 50 \; \mu W = 130 \; \mu W$

Signal to noise power output $SNPO = \frac{S_{Po}}{N_{Po}} = \frac{6000}{130} = 46.2$

Noise Figure $NF = \frac{SNPI}{SPNO} = \frac{75}{46.2} = 1.63$

Noise figure (db) $= 10 \log_{10} (1.63) = 2.12 \; db$

Example 10. 9 A 1 MΩ resistance is connected between the base and the ground in CE transistor amplifier whose input impedance is infinitely large. The amplifier has a voltage gain of 100 and band width 100 *KHz*. Find the amount of noise voltage at the output of amplifier at room temperature (300 ^oK) when no source is connected to the amplifier. The amplifier itself does not provide any noise.

Solution: (i) The noise voltage will be developed across 1 MΩ input resistance whose root mean square value is given by:

$$E_n = \sqrt{4RKT\ (BW\ )}$$

$$= \sqrt{4(10^{\ 6})(1.38\ x10^{\ -23})(300\ )(100\ x10^{\ 3})}$$

$$= 40.7\ \mu V$$

The noise voltage at the output is given by:

$E_{no} = E_n \; x$ (gian of the amplifier)

$= 100 \; x 40.7 = 4.7 mV$

Problems:

1. Discuss the classifications of amplifiers.
2. Draw the circuit diagram of R – C coupled transistor amplifier. Discuss the working of this circuit explaining the function of each component used in the circuit.
3. Explain the frequency response curve of R – C coupled amplifier for the low and mid frequency region.
4. Drive the expression for the lower cutoff frequency of the R – C coupled transistor amplifier due to emitter bye pass capacitance alone and assume other capacitances to have zero impedance.

5. Discuss hybrid $\pi-$model of transistor. Find the expression for higher cut off frequency and show that the current gain falls by $\frac{1}{\sqrt{2}}$ times the low frequency gain (or –3db fall of gain).

6. What is power amplifier? Draw the circuit diagram of a Class A power Amplifier and show that its maximum theoretical efficiency is only 25%.

7. Draw the circuit diagram of a single ended transformer coupled class *A* power amplifier. Explain the working of this amplifier. What are its advantages and disadvantages?

8. Draw the circuit diagram of a single ended transformer coupled class *A* power amplifier and show that its maximum theoretical efficiency is about 50%.

9. Draw the circuit diagram of class B push - pull amplifier and explain its working. Give the advantages and disadvantages of this amplifier.

10. Explain the circuit diagram of class B push – pull amplifier. Show that its maximum theoretical efficiency is about 78.5%.

11. Draw the circuit diagram of class B push - pull amplifier and explain the harmonic distortion in class B push – pull amplifier.

12. Discuss various types of distortions in transistor amplifiers.

13. Discuss various types of noise in transistor amplifiers.

14. Define following modes of operation of an amplifier (i) Class A, (ii) Class B, (iii) Class AB and (iv) Class C.

15. Draw the circuit diagram of two stage R – C coupled CE transistor amplifier. By drawing the low frequency model of the first stage of the amplifier, find the expression of lower cutoff frequency of the amplifier assuming only the emitter bye pass capacitance.

16. What is meat by amplifier noise? Define Johnson noise and shot noise.

17. Define noise figure and signal to noise ratio. Drive the expression for the noise figure in terms of the input and output signal to noise ratio.

18. Discuss the following types of distortion (i) non-linear distortion (ii) frequency distortion and (iii) phase-shift distortion.

19. Show that the maximum theoretical efficiency of a single ended class A power amplifier is 25%. Also show that if the output is transformer coupled, then by proper matching, efficiency can be increased to 50%.

20. In a transformer coupled class A power amplifier, the turn ratio of the transformer is 10 and load resistance connected to the secondary is 50Ω. The zero signal collector current is *80 mA*. (Ans.16 *W*)

21. The individual gains of the three stage amplifier are 30, 40, 50. Calculate the overall gain of the amplifier and express it in *db* also.

(Ans. 60000, 95.6 *db*)

22. The voltage gain of an RC coupled amplifier in the mid frequency range is 90. If the gain falls by 3 *db* at the lower cut off frequency, calculate the gin at the cutoff frequency in *db*. (Ans. 36.08 *db*)

23. The mid frequency gain of a RC coupled amplifier is 120. The values of lower and higher cut off frequencies are 50 *Hz* and 75 *KHz*. Find the frequency at which the gain reduces to 100.

(Ans. 75.4 *Hz*, 49.75 *KHz*)

24. The *h* parameters of the transistors used in two stage RC coupled amplifier are h_{fe} = 400, h_{ie} = 8 $K\Omega$. If the shunt capacitance at high frequency is 500 *pf*, coupling capacitance is 0.5 μF and R_L = 15 $K\Omega$, calculate the lower and higher cutoff frequencies of the amplifier. The source may assume to be negligibly small.

(Ans. 13.85*Hz*, 61*KHz*)

25. A 10 volt battery is connected to a power transistor used in class A mode. If the maximum change in collector current is 100 *mA*, find the power transferred to the load, when a loudspeaker of 16 Ω is:

(i) connected directly in the collector circuit.

(ii) transformer coupled. (Ans. 160 *mW*, 1*W*)

26. If the load connected to the secondary of the transformer in a transformer coupled power amplifier is 10 Ω and zero signal current is 100 mA. Find the maximum power output, if the turn ratio of the transformer is 12.

(Ans. 7.2 *W*)

27. When a sinusoidal signal of fundamental frequency of 300 radians/ sec. is fed to a transistor amplifier, the resulting output collector current is of the form:

$$I_C = 10.Sin300t + 1.1.Sin600t + 0.9Sin900t + 0.6.Sin1200t + ...$$

Calculate (i) Second, third and fourth harmonic distortions.

(ii) Percentage increase in power due to harmonic distortion.

(Ans. (i) 11%, 9%, 6% : (ii)2.38%)

———

11

Electronic Instruments

With the advent of Nanotechnology, now a days very sophisticated, modern and sensitive electronic instrument are available for the use of human beings in every walks of life. In this chapter only the very basic electronic instruments for the use in laboratories will be discussed. The purpose of this chapter is to acquaint the students with the basic principle, working and applications of these instruments. The most commonly used instruments in laboratories are Multimeters, Cathode Ray Oscilloscope, Digital Frequency Meter and Function Generators etc.

11.1 Multimeters: While working with the electronic circuits, it is sometimes necessary to measure the basic electronic quantities namely current, voltage (for both a.c. and d.c.), and also the resistances. With the use of an important electronic instrument known as multimeter, these quantities can easily be measured. Multimeter is a very versatile, rugged and general purpose instrument for use in laboratories. It is also known as AVO meter (Ampere Volt Ohm meter). The multimeters are basically of two types namely Analog multimeters and digital multimeters.

11.1.1 Analog Multimeters: In analog multimeters, the reading of the measuring quantity is given by the deflection of the needle on the indicating meter. The heart of this

meter is a pivoted type moving coil galvanometer having a coil on jeweled bearings between the poles of a permanent magnet. The indicating needle is fastened to the coil. The zero is marked on the extreme left of the galvanometer and not in the middle as is normally marked in the galvanometer. The needle is, therefore, rests on the zero of the meter. With this galvanometer, the arrangements are made so that it can measure the voltage, current and resistances.

Measurement of Voltage: To use the multimeter for the measurement of voltage, a high resistance *R* is connected in series with the galvanometer of the multimeter as shown in figure (11.1). Let I_g is the current sensitivity for the full scale deflection of the galvanometer and *G* is the galvanometer resistance. The maximum voltage that can be measured by this meter will be $I_g.G$, which is known as the voltage rating of the

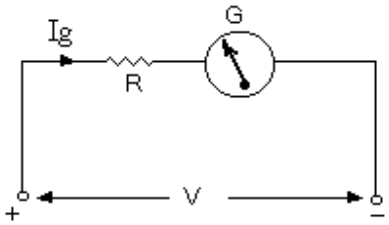


Fig. 11.1

instrument. The high resistance *R* connected in series with the galvanometer will increase the voltage ratings of the instrument. The value of the series resistance *R* for the measurement of voltage from 0 to *V* volt is given by:

$$V = I_g (R + G)$$

or
$$R = \frac{V}{I_G} - G \qquad \text{------ (11.1)}$$

From this equation it is clear that to have the larger voltage range, larger series resistance is required.

A multi - range voltmeter can, therefore, be constructed by providing a number of high resistance in series with the galvanometer and a rotary switch for the selection of the proper voltage range as shown in figure (11.2).

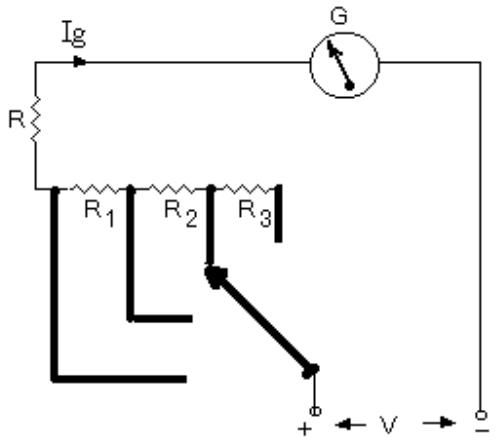


Fig. 11.2

The a.c. voltage can also be measured in the similar fashion, simply by connecting the bridge rectifier with the galvanometer as shown in figure (11.3).

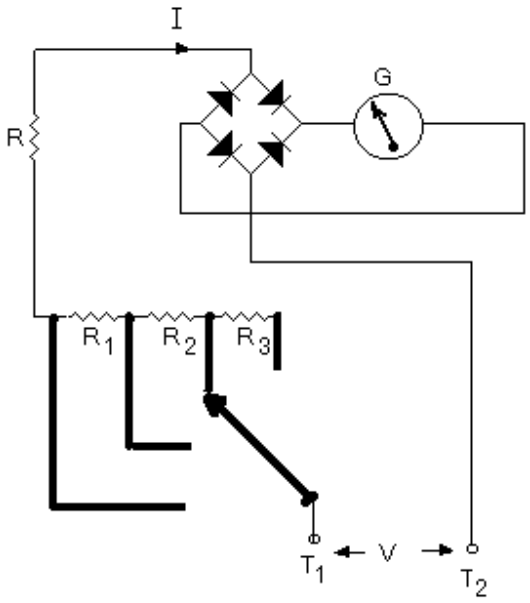


Fig. 11.3

Measurement of Current: For the measurement of current, using the analog multimeter, a low resistance R is connected in shunt (parallel) with the galvanometer as shown in figure (11.4). Let Ig is the sensitivity of the galvanometer and G is the resistance of the

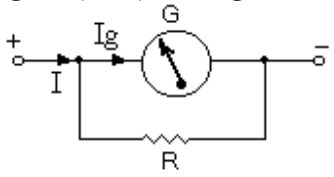


Fig. 11.4

galvanometer. The current I flowing through the terminals of the galvanometer is given by:

$$\frac{R.G}{(R+G)}.I = G.I_g$$

or $$\frac{R.I}{(R+G)} = I_g$$

or $$R = \frac{G.I_g}{(I - I_g)} \qquad \text{------ (11.2)}$$

From this equation, it is clear that for the different current ranges, the different values of shunt resistances are required. A multi-range current meter, is therefore, constructed by providing a number of shunt resistances and a rotary switch for the selection of proper current range as shown in figure (11.5).

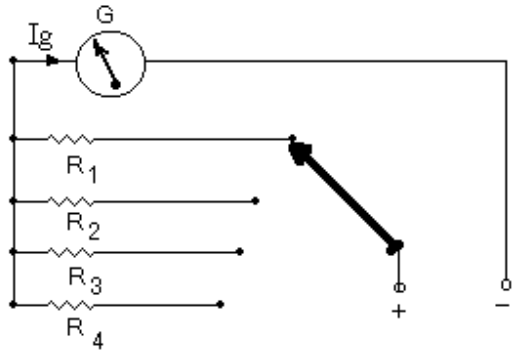


Fig.11.5

Measurement of Resistance: The measurement of resistance is possible with the help of multimeter if its galvanometer is connected with a battery and a current limiting resistance R_S as shown in figure (11.6).

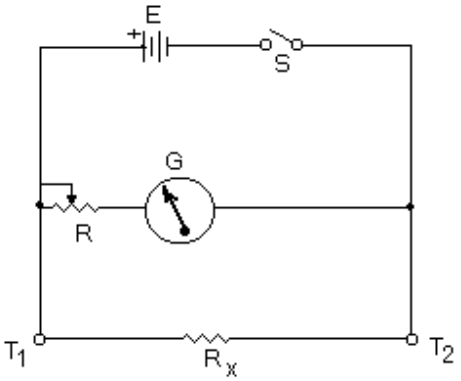


Fig. 11.6

The resistor whose resistance is to be measured is connected across the output terminal of the arrangement shown in figure (11.6). In this case the amount of current flowing through the meter will depend on the value of the measuring resistance. So the scale of the meter can be calibrated in ohms. When the leads of the multimeter are shorted together, the variable resistance *R* should be adjusted such that the galvanometer show the full scale deflection to the right side of the galvanometer where zero for the ohm scale is marked.

The galvanometer current is given by:

$$\frac{E}{R+G} = I_g$$

When the leads are open, the resistance between the leads is infinite and needle rests to the left side of the scale where infinite for the resistance is marked. It is worth mentioning that the scale for the ohmmeter is not linear. The value of the resistances at the left side of the meter is crowded. The different ranges of resistances may also be provided with the help of selector switch and by using the different values of resistance *R*. The block diagram of a basic analog multimeter is shown in figure 11.7.

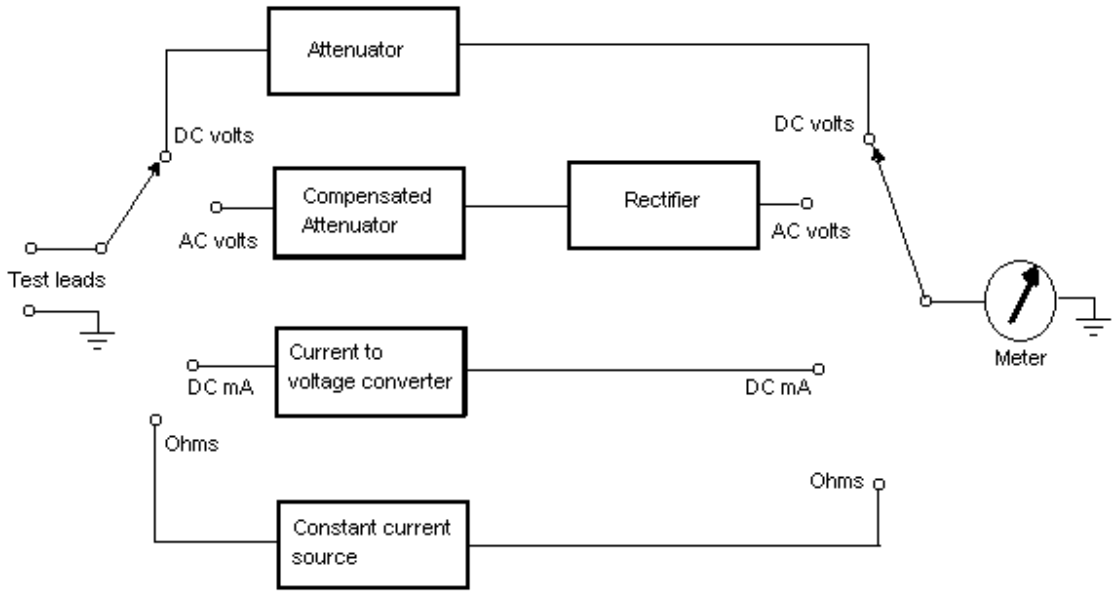


Fig. 11.7

Commercially available Analog multimeters is shown in figure 11.8.



Fig. 11.8

11.1.2 Electronic Voltmeter: The simple multimeter discussed above have the input impedance in the range of 10 KΩ to 20 KΩ/ volt. These are, therefore, low impedance voltmeter especially when the voltage in the range of milli - volts or micro - volts is measured. Further, these multimeters have poor sensitivity and low bandwidth. Thus these multimeters are not suitable for the measurement of low voltage from high impedance source. In such cases the electronic voltmeter is used.

The electronic voltmeter has very high input impedance, good sensitivity and larger bandwidth. The electronic voltmeter consists of amplifier, rectifier and other circuit so as to give the current proportional to the voltage to be measured. This current is then passed through a conventional analog voltmeter. The block diagram of such an electronic voltmeter is shown in figure (11.9). These voltmeters have the input impedance in the range of 1 MΩ to 10 MΩ and bandwidth of several hundred megahertz and the sensitivity lies in the range of milli-volts.

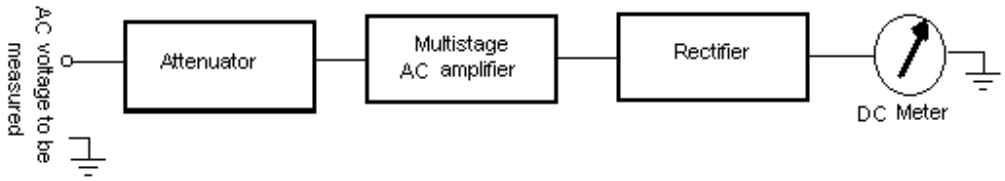


Fig. 11.9

11.1.3 Digital Multimeters: In digital multimeters, the reading of the measuring quantity is given in the form of digital readout on the panel of the meter; and not by the deflection of the needle on the indicating meter as is done in the analog multimeters. The digital multimeters are also used to measure voltage, current and resistance. The measuring quantities are first converted to d.c. voltage by some device, which is measured and displayed on the digital panel meter. For the measurement of a.c. voltage, the input voltage is converted to d.c. voltage by means of a rectifier. For the measurement of current, it is passed through a precision resistor in the meter and voltage across that resistor will be processed to have its digital readout on the panel meter. Similarly, for the resistance measurement a constant current supplied by a constant current source is passed through the measuring resistance and voltage drop across it is measured and displayed in ohms on the panel meter. The functional block diagram of the digital multimeter is shown in the figure (11.10).

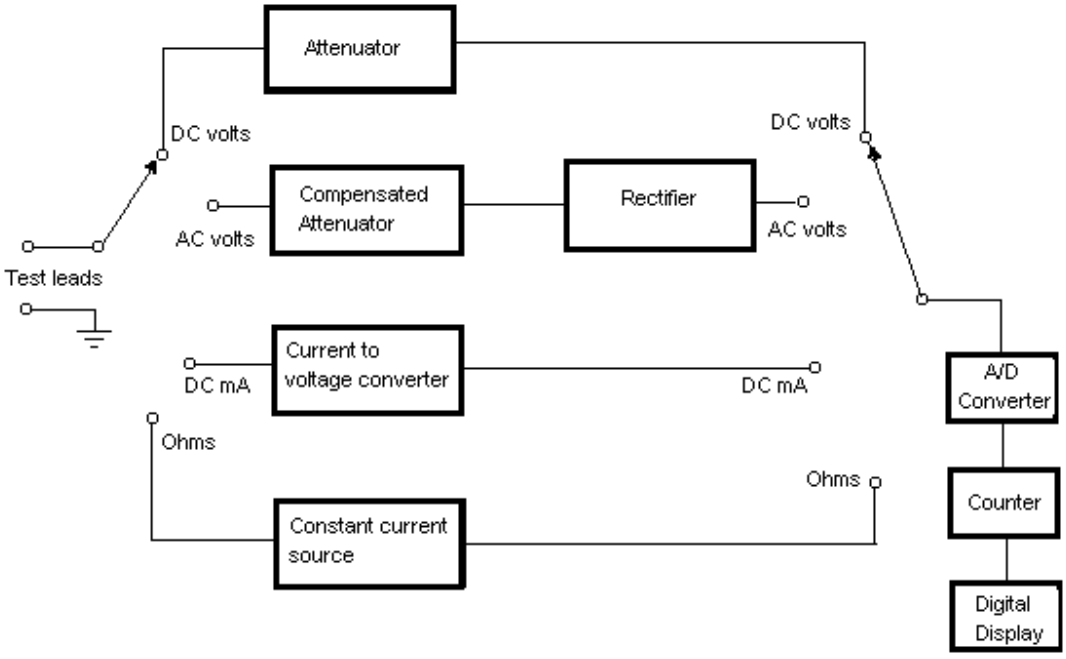


Fig. 11.10

Commercially available digital multimeter is shown in figure 11.11.



Fig. 11.11

Advantages of the Digital multimeters:

(1) The digital multimeters provide better resolution than the conventional analog multimeters. A typical resolution of 1 part in 10^6 is easily observed.
(2) The digital multimeters are more accurate than the analog multimeters.
(3) Input impedance on both a.c. and d.c. is much larger than the analog multimeters.
(4) These are operated on low voltage cells which are kept inside the multimeters.
(5) Since the result of the measuring quantity is displayed on the penal meter in the form of digits so it avoids errors due to reading, interpolation and parallax.

11.2 Cathode Ray Oscilloscope: The cathode ray oscilloscope (CRO) is another measuring electronic instrument used in laboratories. It is capable of displaying the signal wave shapes on the screen of the CRO, so it is widely used for the trouble shooting in the electronic circuits in the laboratories. It is very versatile instrument and can also be used to measure the voltage, frequency and the phase shift. The heart of the cathode ray oscilloscope is the cathode ray tube (CRT).

11.2.1 Cathode Ray tube: The schematic diagram of a cathode ray tube (CRT) is shown in figure (11.12). It has the following four major parts.

1. Electron gun – an arrangement for producing and focusing the electron beam.

2. Deflecting system – a system for deflecting the electron beam librated from the electron gun.

3. Fluorescent screen – for producing bright spot.

4. Evacuated glass enclosure – all assemblies fitted in evacuated glass enclosure.

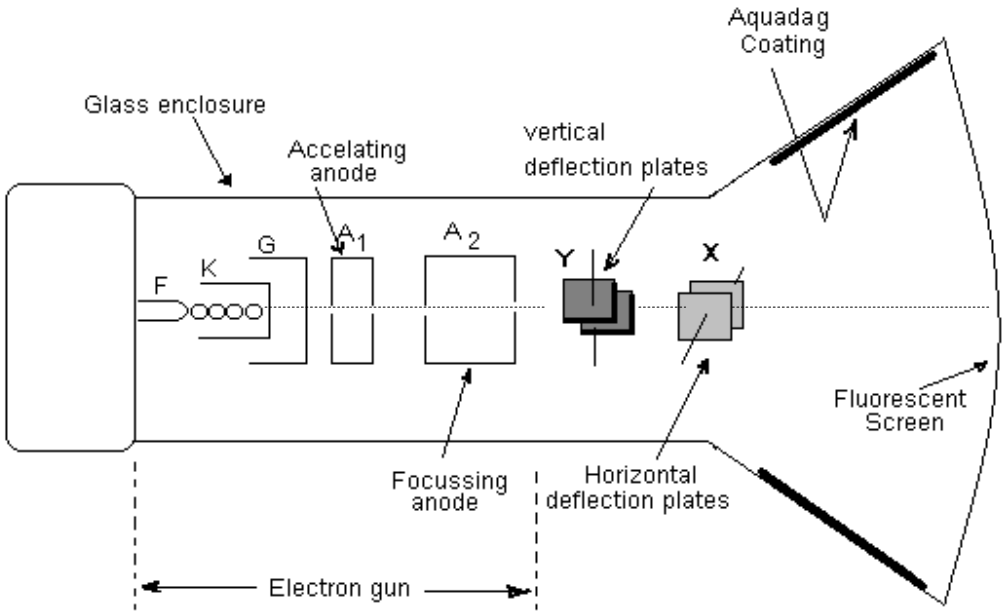


Fig. 11.12

1. Electron gun: The arrangement of electrodes that produces a focused beam of electrons is known as electron gun. It consists of an indirectly heated cathode K, a control grid G, a focusing anode A_1 and accelerating anode A_2. The control grid is held at a negative potential with respect to cathode and focusing and accelerating grids are maintained at successively higher potential with respect to cathode.

The cathode consists of a nickel cylinder coated with oxides of barium and strontium for liberation plenty of electrons. The control grid enclosing the cathode consists of metallic cylinder with very small aperture. This small opening helps to keep the electron beam of very small in size. By controlling the positive potential of the focusing anode A_1, the electron beam is focused in to a sharp pin – dot. The accelerating anode A_2 which is at higher potential than the focusing anode accelerates the electron beam to a very high velocity. The electron gun thus librates a narrow, accelerated beam of electrons which produces a sharp spot of light when strikes on the fluorescent screen.

2. Deflecting system: For deflecting the narrow accelerated electron beam librated from the electron gun are deflected in the vertical and horizontal directions by the two sets of deflecting plates. One set marked as Y is known vertical deflecting plates and the other set marked as X is known as horizontal deflecting plates. By the application of proper

potential to the two sets plates, the electron beam deflects in the vertical and horizontal direction. If no potential is applied to the two sets of plates, the electron beam without deflection strikes the screen at the centre producing a bright spot. If on the other hand, a positive potential is applied to the upper vertical deflecting plate with respect to the lower vertical plate, the electron beam deflect upwards. The height of the deflecting beam will be proportional to the applied potential. If an alternating voltage is applied to the set of Y plates, the spot will be moving continuously upward and down ward thereby producing a luminous trace in the vertical direction on the fluorescent screen due to the persistence of vision. The beam will be deflecting in the horizontal direction if the similar potential is applied to the set of horizontal plates.

3. Fluorescent Screen: The inside face of the tube is coated with some fluorescent material such as Zinc orthosilicate, Zinc oxide etc which works as fluorescent screen. It makes the visible. Various kinds of phosphors or their combinations are used to obtain spots of variety of colors. The green spot is produced if Zinc orthosilicate is used as the fluorescent material.

4. Evacuated glass enclosure: The all parts discussed above are enclosed inside a funnel shaped evacuated glass envelope. The vacuum created inside the tube helps the electron beam to transverse the tube easily without any collision. The flared part of the tube is coated from inside with a conducting graphite layer called aquadag. This coating is kept at a positive potential with respect to the cathode and collets the primary as well as secondary electrons returned from the screen.

11.2.2 Construction: The simplified block diagram of a cathode ray oscilloscope is shown in figure 11.13. The filament of the electron gun is heated with help of the a.c. supply. The electrons are therefore, emitted. The intensity of the beam is controlled by the control grid supply. The electron beam after the control of the control grid is influenced by the focusing and the accelerating grid. These two electrodes are maintained at high positive potential with respect to cathode. The electrostatic field that exists between the anodes provides the necessary focusing of the electron beam and the system of electrodes is known as electron lens. The focusing of the beam that strikes on the fluorescent screen can be corrected by varying voltage on the two electrodes. The focus control is provided on the front panel of the oscilloscope.

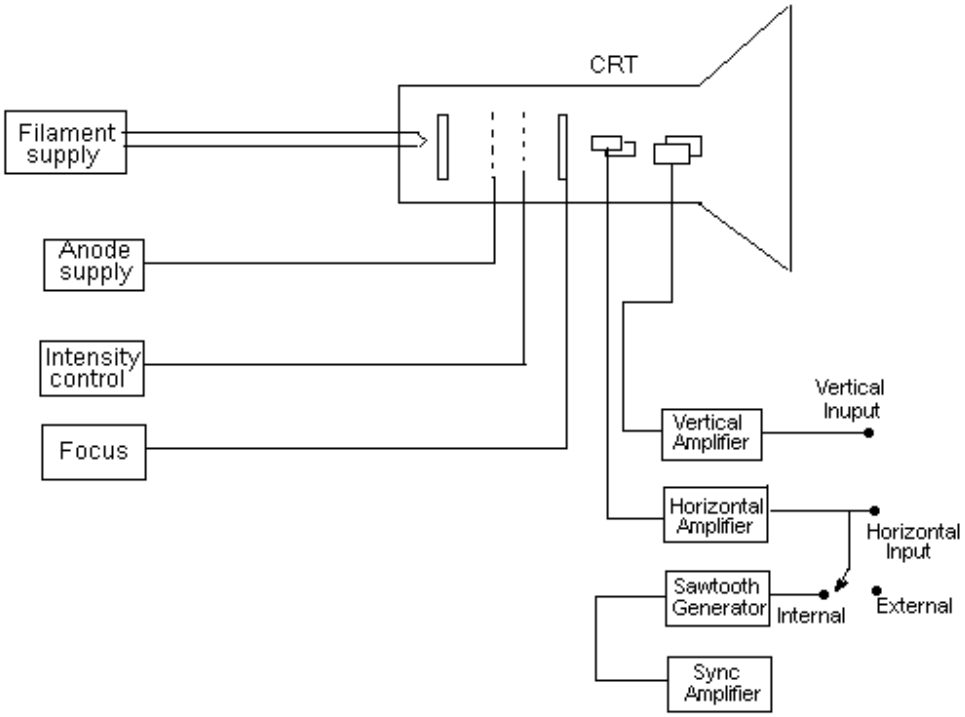


Fig. 11.13

Now after the electron beam leaves the accelerating anode, it comes under the influence of vertical and horizontal plates. If no voltage is applied to the deflection plates, the electron beam will produce spot of light at the centre of the screen (point P). The electron beam will be deflected upwards (point Q), if certain voltage is applied to the vertical plates (figure 11.14). The height of the deflection will be proportional to the applied voltage to the vertical plates. The electron beam will be deflected downwards (point R) if the polarity of the voltage on the vertical plates is reversed. Similarly, the spot can be moved in the horizontal direction if the proper voltage is applied across the horizontal plates. If the sinusoidal voltage is applied to the vertical plates, the spot will

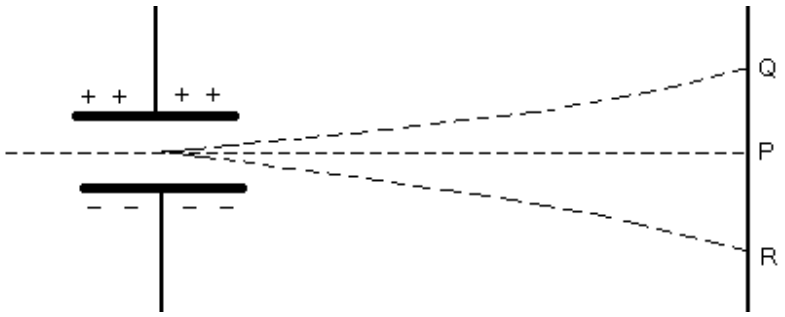


Fig. 11.14

deflect upward and downward direction continuously and due to the persistence of vision a vertical line will be observed on the fluorescent screen. However, to see the variation of the sinusoidal signal with time on the CRO screen, the spot is simultaneously to be moved in the horizontal direction uniform speed. This is possible if suitable waveform (saw tooth) is applied to the horizontal plates. So a saw tooth is generated internally in the CRO with the help of saw tooth generator or linear time base generator. During the trace path of the saw tooth wave (fig. 11.15) the complete cycle is traced on the screen, while during the fly back or retrace path of the saw tooth wave the spot is blanked out and not visible on the screen. The fly back time of the wave is kept very small.

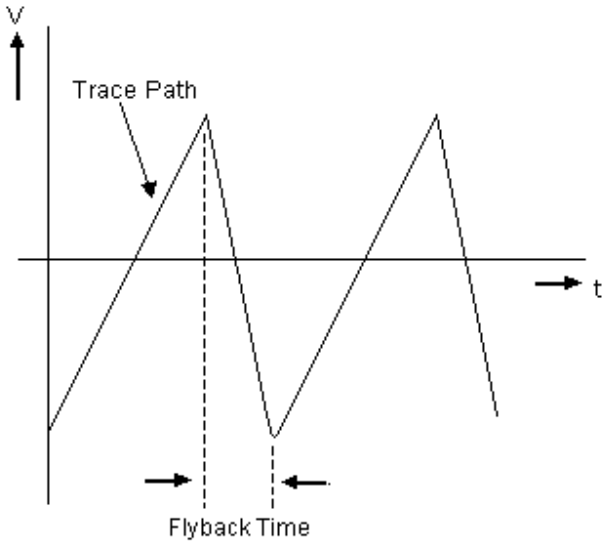


Fig. 11.15

Figure 11.16 clearly illustrates the synchronization between the input sinusoidal wave applied across the vertical plates of the CR tube and saw tooth wave applied internally across the horizontal plates of the tube.

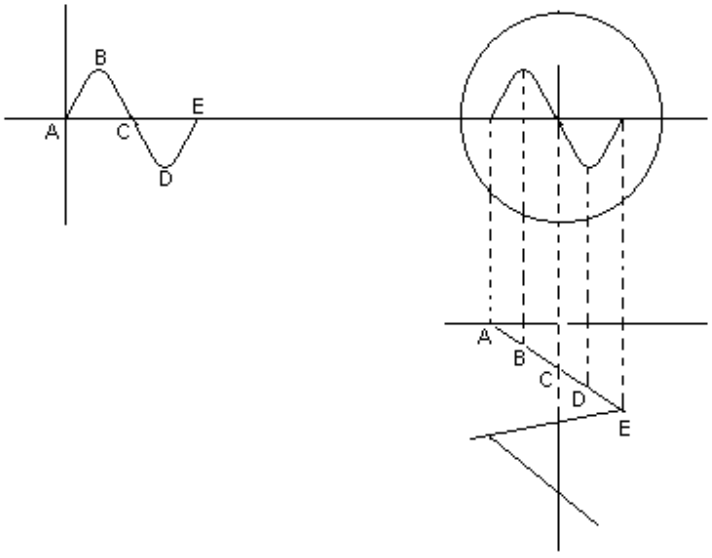


Fig. 11.16

On the screen of the CRO scale (equidistant horizontal and vertical lines as in graph sheet) is provided as shown in figure 11.17. The size of the input wave can be seen on this scale.

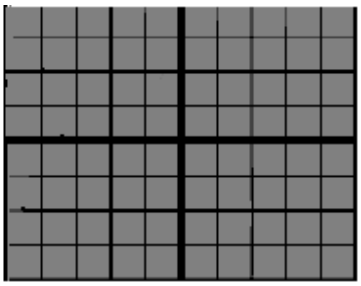


Fig. 11.17

Commercially available cathode ray oscilloscope for use in electronics laboratory is shown in figure 11.18.

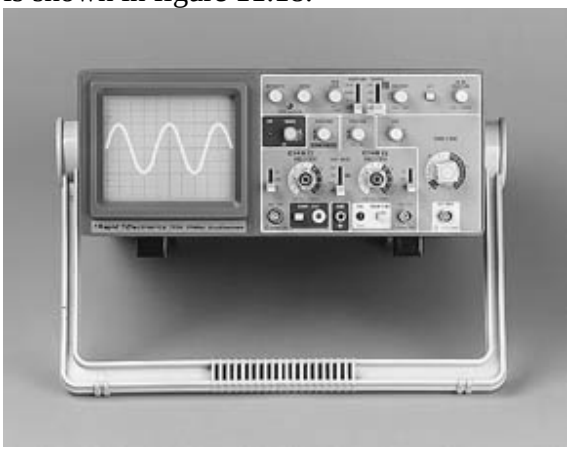


Fig. 11.18

11.3 Application of CRO: The CRO finds many applications in electronics laboratory. Here some very common and important applications of CRO will be discussed.

11.3.1 Measurement of Voltage and Time Period: For the measurement of voltage the signal whose voltage is to be measured, is applied to the Y –axis and internal time base is used. The scale is determined by the Y – amplifier (volts/cm) control provided on the front panel of the oscilloscope. The height of the wave observed on the screen of CRO is adjusted with the help of Y –amplifier such that it fits on the screen of the CRO. Further the wave is deflected so that it coincides with the scale of the screen as shown in figure 11.19. From this figure it is clear that the length of the trace (in cm) measured on the scale will give the peak to peak voltage of the a.c. signal applied on the vertical input of CRO. The peak to peak voltage of the signal is calculated by multiplying the length of the

trace with the deflection sensitivity of the vertical amplifier. The rms value of this signal is obtained by the dividing the peak to peak value by $2\sqrt{2}$.

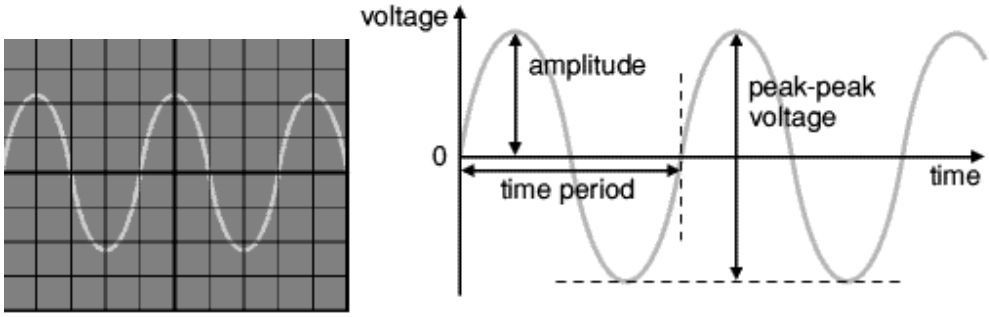


Fig. 11.19

To measure the d.c. voltage on the CRO, the position of the zero volt (normally halfway up the screen) is first checked. The AC/DC/GND switch is kept to GND (0V) and Y – shift is used to adjust the position of the trace to middle line on the CRO screen. The AC/DC/GND switch is thrown to the DC position and d.c. voltage is applied to the Y –input. The vertical displacement (in cm) of the horizontal line is multiplied by the deflection sensitivity of the vertical amplifier. This results the magnitude of the d.c. voltage.

The time period of the signal can also be measured with help of CRO. The time period is the time for one cycle of the signal. The frequency of the signal is the number of cycles per second i.e.

$$Frequency = \frac{1}{time\ period}$$

The horizontal distance (in cm) of the complete one cycle of the wave (fig. 11.19) is obtained from the CRO screen. This distance when multiplied by the scale of the time base control (Time/cm) gives the time period of the wave. The time period may be converted to frequency from the above formula.

11.3.2 Measurement of Phase Difference: The cathode ray oscilloscope can be used for the measurement of phase difference between two sinusoidal signals of the same frequency. One signal is applied to the Y – input of the CRO and other signal is applied to the X – input of CRO. The time base of the oscilloscope is in the external mode. On the screen of the CRO, a Lissajous figure (an ellipse) will be formed as shown in figure 11.20. From this figure on the CRO screen the maximum displacement Y2 in the vertical direction and intercept y_1 on the Y – axis are measured. The phase difference θ of the two signals is given by the formula:

$$\theta = \sin^{-1}(\frac{y_1}{y_2})$$

This phase difference can also be given by the formula:

$$\theta = \sin^{-1}(\frac{x_1}{x_2}),$$

where X_1 is the intercept on the X – axis and X2 is the maximum horizontal displacement.

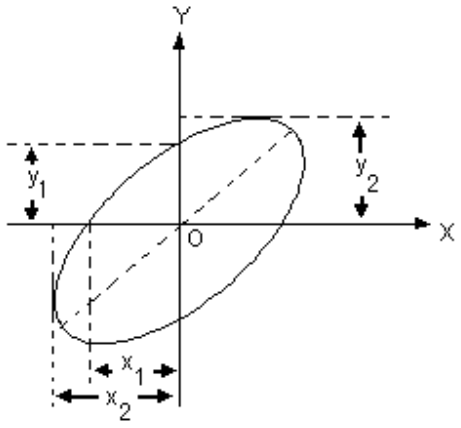


Fig. 11.20

11.4 Function Generator: A function generator is an electronic instruments used in laboratories. It provides different wave forms sine, square and triangular of adjustable frequencies and amplitude. The frequency of a wave may be adjusted from a fraction of an Hz to several hundred KHz.

In LC or RC oscillators the frequency is controlled by varying capacitor. However, in the function generators various types of wave shapes are to be generated, so the saw tooth wave is first generated with the help of an integrator. In the integrator circuit a capacitor is charged by the constant current source. The frequency of the saw tooth will depend upon the magnitude of the constant current source. The saw tooth wave is smoothened by the resistance and diode shaping circuit to get the sinusoidal wave. Further, square or triangular wave is obtained by simultaneously connecting the saw tooth wave to a comparator circuit. The comparator circuit limits the height of the square wave and time of integration.

Figure 11.21 shows the block diagram of such a function generator. It consists of two constant current sources namely positive and negative current source, the magnitude of these current sources is controlled by the frequency controlled network. Its control is done by a variable resistance provided on the front panel of the function generator. The outputs of these current sources are applied to an integrator circuit through a switching circuit. The switching circuit allows the positive constant current source to charge the capacitor of the integrated circuit. The output voltage of the integrator increases linearly with time by the relation given as:

$$V_o = -\frac{1}{C}\int_0^t I.dt$$

where I is the magnitude of the constant current source.

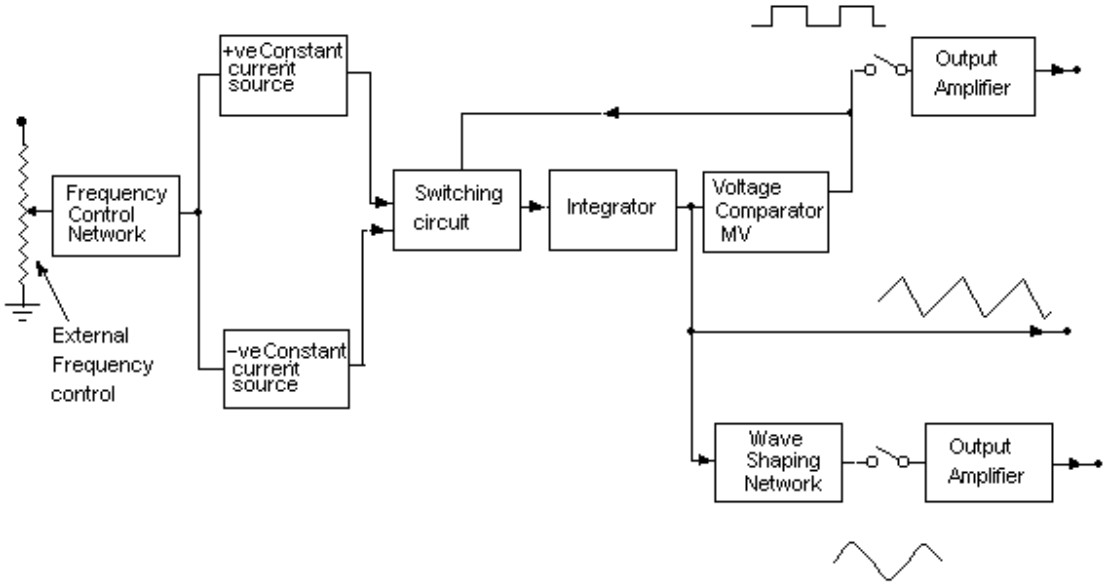


Fig. 11.21

Rising slope of the integrator output will depend upon the magnitude of the current source. As soon as the output voltage of the integrator reaches a predetermined level, the voltage comparator multivibrator changes its state and the switching circuit connects the input of the integrator to the negative constant current source.

The negative constant current source supplies the reverse current to the capacitor C. Thus the output voltage of the integrator decreases linearly with time. Further, as the voltage decreases to a predetermined level, the voltage comparator multivibrator changes its state. The switching circuit again connects back the positive constant current source to the integrator. The triangular wave form is therefore obtained at the output of the integrator, whose frequency is determined by the magnitude of the current supplied by the constant current source. The output of the comparator is the square wave; its frequency is the same as that of the triangular waves. The resistance and diode shaping network connected to the triangular wave, gives the sinusoidal wave. The output amplifiers are used to get the waves.

11.5 Digital Frequency Meter: The digital frequency meter is an electronic instrument used to measure the frequency of a periodic waveform. The basic principle for the precise determination of frequency of an unknown signal is illustrated in figure 11.22. The unknown signal is applied to amplifier/attenuator, where the signal is amplified if it is a weak signal and attenuated if the signal of high amplitude. The amplified signal is then connected to a Schmitt trigger circuit where the signal is converted to a square wave. The square is differentiated to get the narrow pulse train. The number of pulses in the pulse train is equal to the frequency of the input unknown signal. This narrow pulse train is then applied to one input of a two input AND gate. The second input of this AND gate is connected another standard sample pulse of constant width. The sample pulse controls

for how long the pulse train is allowed to pass through the AND gate to the digital counter. If the width of this sample pulse is kept as 1 second, then the AND gate will allow the pulse train to go to the input of the counter for 1 second. The counter will display the counts on the display devices (in digital form) counted by it for 1 second. The number displayed on the display devices will show the frequency of the input signal directly in Hz, since the number of pulses in the pulse train is equal to the frequency of the input unknown signal. The digital counter basically contains BCD counter, decoder and display unit (seven segment display).

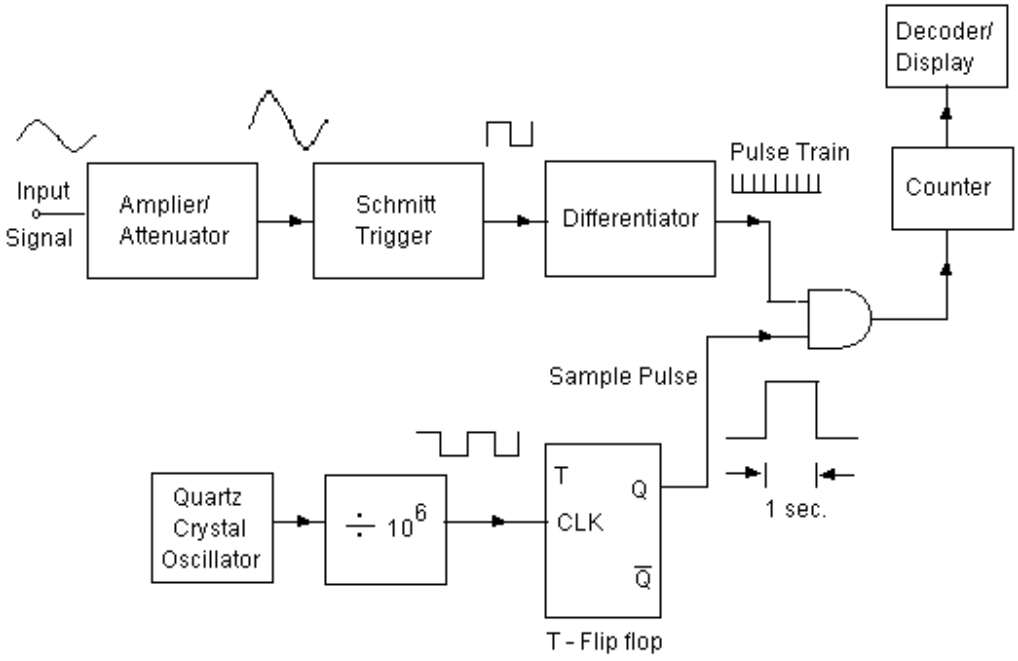


Fig. 11.22

The accuracy of the counter will depend on the accuracy of the width of the sample pulse. The sample pulse of standard time period is therefore, obtained from a high frequency quartz crystal oscillator, say, 1 MHz. The frequency of this crystal oscillator is divided by a factor of 10^6 using frequency divider circuit, which gives a square wave of 1 Hz frequency. Finally the 1 Hz frequency is divided a factor of 2, to obtain a square whose pulse width is 1 second. If the width of the sample pulse is taken as 1 msec, then the counter will display the frequency directly in KHz, if it is taken as 1 μsec, the counter will display the frequency in MHz.

Problem:

1. Draw the functional block diagram of the function generator. Explain the working of each bock.
2. Discuss the principle and working of a digital frequency meter.
3. Draw the block diagram of a cathode ray tube and explain its main components.

4. Draw the basic block diagram of a cathode ray oscilloscope and explain the function of each block.
5. Explain the use of CRO for the measurement of frequency of a signal.
6. How the voltage of a signal is measured with the help of CRO?
7. Explain how the phase difference of two signals is measured with the help of CRO.
8. What is the need for time base generator in CRO?
9. Draw the functional block diagram of an electronic multimeter. Explain the function of each block. What are its advantages over the conventional multimeter.
10. Draw and explain the functional block diagram of a digital multimeter. What are the advantages of digital multimeter.
11. What is an analog multimeter? Discuss the principle of measuring a.c./d.c. voltage with an analog multimeter.
12. Discuss the principle of measuring the current with an analog multimeter.
13. Discuss the principle of measuring resistance with an analog multimeter.

Appendix - I

Decibel:

To discuss the term decibel, consider an amplifier whose output power is P_2 and input power is P_1. The ratio $\frac{P_2}{P_1}$ is known as the power gain of the amplifier. This ratio may be more than unity or less than unity. If this ratio is more than unity the amplifier is said to have been gained, and if, $\frac{P_2}{P_1}$ is less than unity, there is attenuation in the power of the amplifier.

When a number of power amplifiers are connected in tandem, then the overall power gain of such a cascaded amplifier is the product of power gains of the individual amplifier. It is, therefore, customary to define the power ratio on the logarithmic scale since it an established fact that the power (or audio level) is related on the logarithmic basis. The overall gain will easily be obtained by adding the logarithmic units rather than multiplying.

The power gain in logarithmic scale is defined as :

$$\text{Power gain} = \log_{10}\left(\frac{P_2}{P_1}\right) \quad bel$$

The unit *bel* was found to be quite larger unit for practical purposes, a smaller unit known as *decibel* (*db*) is defined which is one - tenth of a *bel*.

$$1\ bel = 10\ db$$

So power gain in *db* is defined as:

$$\text{Power gain (in } db) = 10\log_{10}\left(\frac{P_2}{P_1}\right) \quad db$$

It is interesting to note that there is a power gain of o *db* if output power is equal to the input power ($P_2 = P_1$). If $P_2 = 2P_1$ i.e. output power is twice the input power than there a gain a 3db as:

$$\text{Power gain (in } db) = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\frac{2P_1}{P_1}\ db$$

$$= 10\log_{10}(2) = 10x0.3010 = 3db$$

If on the other hand the power of the amplifier is attenuated by a factor of two i.e. $P_2 = \frac{P_1}{2}$, there is the loss of power (reduced half); however, in *db* it will denoted by – 3*db* (negative quantity) as:

$$\text{Power gain (in } db) = 10 \log_{10}\left(\frac{P_2}{P_1}\right) = 10 \log_{10} \frac{P_1}{2P_1} \; db$$

$$= 10\log_{10}\left(\frac{1}{2}\right) = -10\log_{10}(2) = -10x0.3010 = -3db$$

Further, if V_1 and I_1 are the voltage and current of the input of the amplifier; and V_2 and I_2 are the voltage and current of the output of the amplifier, the ratio $\frac{P_2}{P_1}$ is given as:

$$\frac{P_2}{P_1} = \frac{V_2^2/R_2}{V_1^2/R_1} = \frac{I_2^2 R_2}{I_1^2 R_1}$$

If $R_1 = R_2$, then

$$\frac{P_2}{P_1} = \frac{V_2^2}{V_1^2} = \frac{I_2^2}{I_1^2}$$

Thus power gain in *db*

$$= 10\log_1\left(\frac{P_2}{P_1}\right) = 20\log_{10}\left(\frac{V_2}{V_1}\right) = 20\log_{10}\left(\frac{I_2}{I_1}\right)$$

Generally, the input and output impedances are not equal, so if the effect of different impedances is ignored, the voltage gain or the current gain of the amplifier can conveniently be expressed in *decibels* as:

$$\text{Voltage gain in } db = 20\log_{10}\left(\frac{V_2}{V_1}\right) db = 20\log_{10}(A_V) \; db$$

$$\text{Current gain in } db = 20\log_{10}\left(\frac{I_2}{I_1}\right) db = 20\log_{10}(A_I) \; db$$

Appendix – II

Switches

Switch is a device used for opening and closing a circuit. Switches are designed in variety of sizes, types and shapes. Some most commonly available switches are being discussed.

(1) Single Pole Single Throw (SPST) Switches: The SPST switch also called toggle switch, has one switch contact set and one conducting position. It can connect or disconnect of a single wire or line. It is in fact a simple on – off switch. This type of switch can be used to switch the power supply to a circuit. Figure 1(a) shows the symbolic representation of SPST switch and figure 1(b) shows the commercially available switch.



Fig. 1(a)



Fig. 1(b)

(2) Push – to – On Momentary SPST Switch: Figure 2(a) shows the symbolic representation of this switch and its physical appearance is shown in figure 2(b). This switch returns to its normally open (OFF) position, when the push button is released. It momentarily makes the connection. It is like the door bell switch.

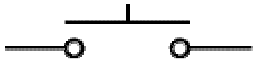


Fig. 2(a)



Fig. 2(b)

(3) Push – to – Off Momentary SPST switch: This type of switch is shown in figure 3. It performs the reverse function of push to on momentary switch, that is, the switch returns to its normally close (ON) position, when the push button is released. It momentarily breaks the circuit.

Fig. 3(b)



Fig. 3(a)

(4) Single Pole Double Throw Switch (SPDT): Such a switch has two ON positions. It has three terminals, when the switching position is in the centre the two circuits will be in the off positions. When the switch is thrown to either of the two positions it closes the corresponding one circuit. Figure 4 shows its physical appearance and schematic diagram.



Fig. 4(a)

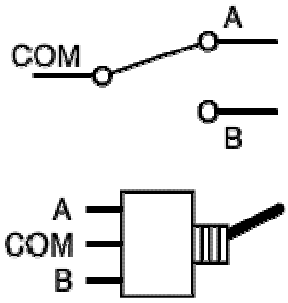


Fig. 4(b)

(5) Double Pole Double Throw (DPDT) Switch: It has a pair of On – Off switches which operate together as shown by dotted lines of its schematic diagram (fig. 5a). Figure 5(b) shows its physical appearance.

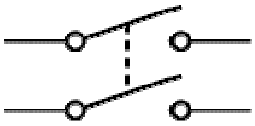


Fig. 5(a)



Fig. 5(b)

(6) Push – Push Switch: Physical appearance of this type of switch is shown in figure 6. It looks like a momentary action push switch but it is standard on – off switch. It is switched to on when pushed once and it is pushed off when pushed again. This is called a latching action.



Fig. 6

(7) Reed Switch: The reed switches have a glass body inside which micro switches usually SPST are there. It is shown in figure 7. Its contacts are closed when a small magnet is brought near the switch. Such switches may be used in security circuits.



Fig. 7

(8) DIP (Dual – in – line parallel) Switch: This switch is a set of miniature SPST on – off switches is a dual in line integrated circuit package. Figure 8 shows such switch containing 8 miniature switches.



Fig. 8

(9) Rotary Switch: Rotary switches have 3 or more conduction positions, by rotating the shaft connections to different poles may be made. Figure 9 shows the schematic and physical appearance of such switch.

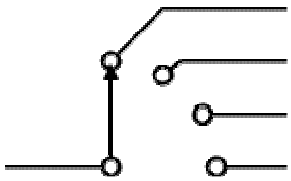


Fig. 9(a)



Fig. 9(b)

Appendix – III

Resistances

Resistances are the passive elements used in electronics. They are measured in ohms (Ω). The resistances are available in different variety. While using the resistance one should know the following specifications of the resistances.

	Specifications		Remarks
1.	Nominal Value	in ohms	Value indicated on the resistor usually by color code.
2.	Tolerance	in $\pm$ X %	Real value will be with in $\pm$ X % of the nominal value.
3.	Wattage	in watts	Maximum power that can be dissipated in a resistor.
4.	Temperature coefficient	ppm/ ^{0}C	Variation of resistance with temperature.
5.	Stability	in %	Variation of resistance with ageing and ambient conditions.

The most widely used resistances are carbon composition, carbon film, metal film and wire wound. The carbon composition resistors are inexpensive and therefore, generally used in entertainment purposes. They are not very stable and their tolerances are also not very good. Metal films resistances are, however, very stable and its tolerances are also very good. They are costly. Such resistances are used in the circuits in which accuracy is the important factor. The wire wound resistances are available in high wattages; the size of such resistors increases as the wattage is increased. Such resistances are never used in high frequency circuits.

The nominal value and the tolerance of the resistances are obtained from the color bands indicated on the resistances. Each resistance has four color bands and each color band represents a number shown in tabe1.

Table 1

The Resistor Color Code	
Color	Number
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Grey	8
White	9

- The first band gives the first digit.
- The second band gives the second digit.
- The third band indicates the number of zeros.
- The fourth band is used to shows the tolerance (precision) of the resistor, usually fourth band is silver or gold. If the fourth band is silver then it tolerance is 10%, and for gold its tolerance is 5%.

Figure 1 shows the color code of one resistance. This resistor has red (2), violet (7), yellow (4 zeros) and gold bands 5%. So its value is 270000 Ω ± 5%= 270 K Ω ± 5%.

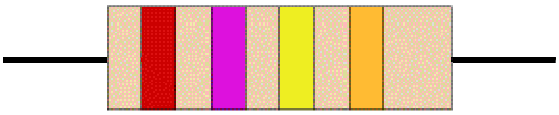


Fig. 1

Small value resistors (less than 10 ohm): The standard color code cannot show values of less than 10 Ω. To show these small values two special colors are used for the third band: gold which means × 0.1 and silver which means × 0.01. The first and second bands represent the digits as normal.

For example: The color code of 2.7 Ω resistance is red, violet and gold.

Variable resistances: Variable resistors consist of a resistance track with connections at both ends and a wiper which moves along the track as the spindle is turned. The track may be made from carbon or a coil of wire (for low resistances). Figure 2 shows the physical appearance of this type of variable resistance also called potentiometer.

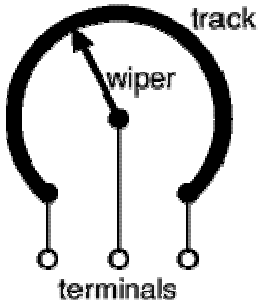




Fig. 2

The potentiometers are available in linear and logarithmic tracks. The linear track means that the resistance changes at a constant rate as the wiper is moves. In most of the electronic circuits LIN potentiometers are use. However, in log potentiometers, the resistance changes slowly at one end of the track and changes rapidly at the other end, so halfway along the track is not half the total resistance. These types of potentiometers are used for volume controls in audio equipments.

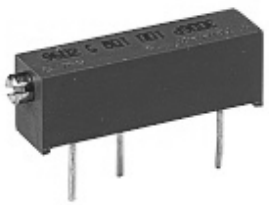
Preset potentiometers: Preset pots are used where precise variation of resistances is required. Preset pots are available in single turn and multi-turn pot. The screw is provided in multi-turn pots. The screw moves the slider from one end of the track to the other end giving the very fine control. Figure 3 shows such presets.



Single turn preset (open)



single turn preset (closed)



Multi - turn

Fig. 3

Appendix – IV

Capacitors

Capacitors store electric charge. The capacitors are also used in filter circuits because it easily pass AC (changing) signals but they block DC (constant) signals. The capacity of a capacitor is its ability to store charge. A large capacitance means that more charge can be stored. Capacitance is measured in farads, and its symbol is F. However 1 F is very large, so prefixes are used to show the smaller values.

Three prefixes (multipliers) used with Farad F are μ (micro), n (nano) and p (pico):

$1\ \mu F = 10^{-6}\ F$

$1\ nF = 10^{-9}\ F$

$1\ pF = 10^{-12}\ F$

All capacitors are generally classified in to two categories, namely:

(1) Fixed Capacitors and (2) Variable Capacitors.

(1) Fixed Capacitors: The fixed capacitors can be sub-divided into two groups.

(a) Electrolytic capacitors: They are large value capacitors (more than 1 μF). These capacitors use electrolyte (borax or carbon salt) as the negative plate. An Aluminium anode acts as the positive plate, while a thin film of Aluminium oxide on the anodes acts as the dielectric. The electrolytic capacitors are used in circuits where a dc voltage is present, because it requires the dc polarizing voltage. Since Electrolytic capacitors are polarized so they must be connected to the correct polarity. At least one lead of the electrolytic capacitor is marked as + or -. They are not damaged by heat when soldering.

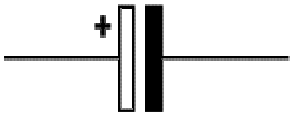


Fig. 1(a)

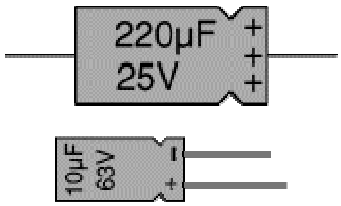


Fig. 1(b)

The value of electrolytic capacitors is printed on the capacitor with their voltage ratings (fig.1). The voltage rating of the capacitors should always be checked while selecting an electrolytic capacitor. In the circuits, one should use the capacitors with a voltage rating greater than the circuit's power supply voltage; otherwise the capacitor may be damaged. The symbolic representation of the electrolytic capacitor is shown in figure 1(a).

(b) Non electrolytic capacitors: It includes paper, mica and ceramic capacitors. These capacitors have no polarity requirement, i.e., they can be connected in either direction in the capacitors. They are low value capacitors (less than 1µF) with small size. The circuit symbol and its physical appearance of these capacitors are shown in figure 2.

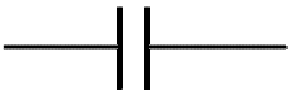


Fig. 2(a)

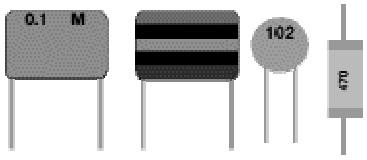


Fig. 2(b)

To find the value of these capacitors a number code is often used on small capacitors where printing is difficult:

- 1st number is the 1st digit,
- 2nd number is the 2nd digit,
- 3rd number is the number of zeros to give the capacitance in pF.
- Ignore any letters - they just indicate tolerance and voltage rating.

For example: **102** means 1000 pF = 1nF *(not 102pF)*

472J means 4700 pF = 4.7nF (J means 5% tolerance).

222 means 2200 pF = 2.2 nF

(2) Variable Capacitors: As the name indicates its capacitance can be varied by rotating the small spindle connected with the capacitors. These capacitances are also called gang capacitors. The gang capacitor is shown in figure 3 with its symbol. Variable capacitors are mostly used in radio tuning circuits and they are sometimes called 'tuning capacitors'. They have very small capacitance values, typically between 100 pF and 500 pF.



Fig. 3(a)



Fig. 3(b)

Trimmer capacitors: Trimmer capacitors (trimmers) are miniature variable capacitors. They are designed to be mounted directly onto the circuit board and adjusted only when the circuit is built. A small screwdriver or similar tool is required to adjust trimmers. The process of adjusting them requires patience because slight variation in the screw driver position will not give the required capacitance in the circuit. Trimmer capacitors are only available with very small capacitances, normally less than 100pF. It is usually specified by their minimum and maximum values, for example 2-10 pF. The Trimmer capacitor is shown in figure 4.



Fig. 4(a)



Fig. 4(b)

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